faktor

SPECIFICATION · REVISION 8

IDE experience & implementation

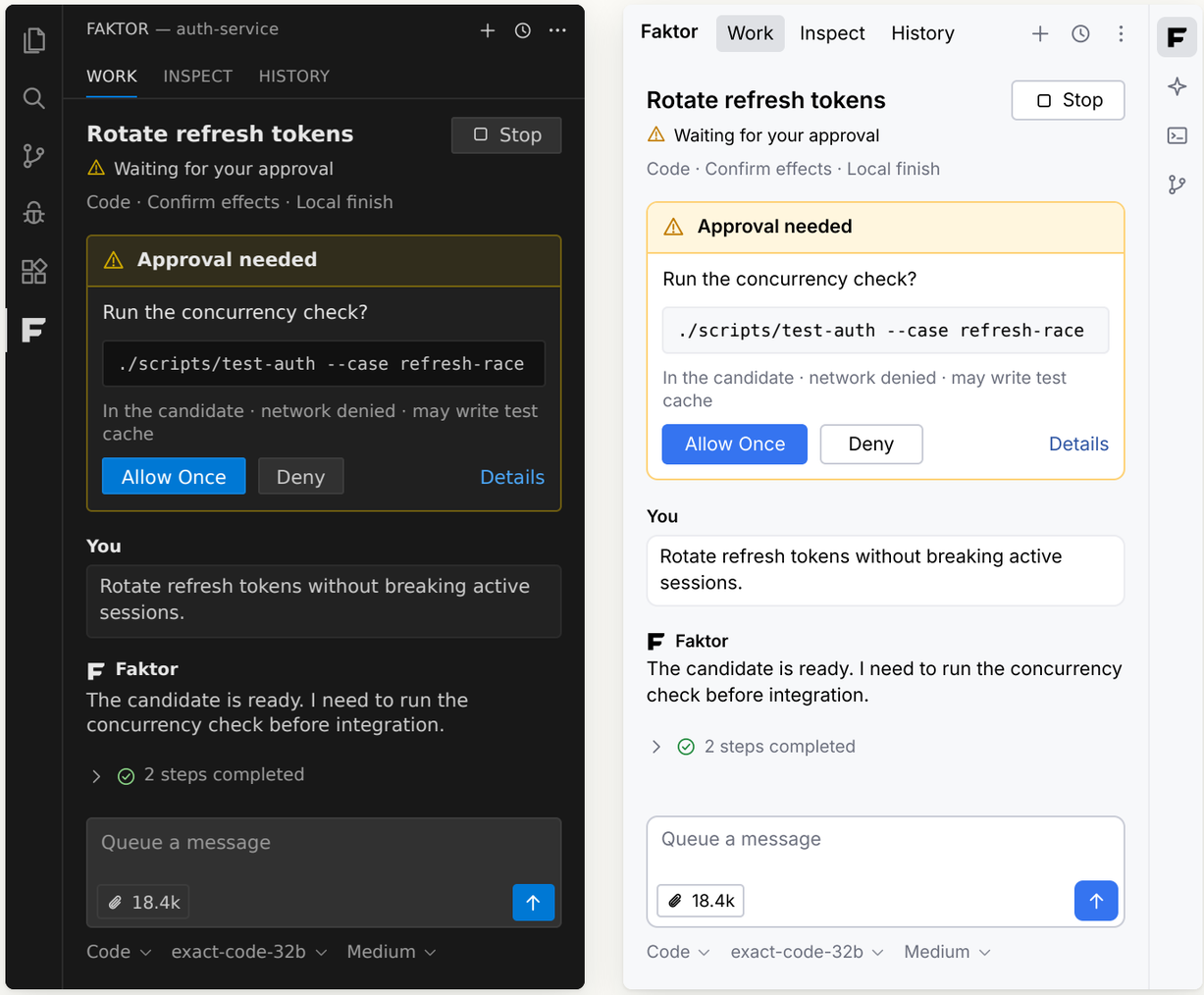
A complete specification for an agent workspace in VS Code and JetBrains: quieter by default, native in each host, with every feature and all technical documentation retained.

200
controls

72
surfaces, two hosts

838
plate-copy changes recorded, none lost

Providers and models. Context and permissions.
Tools, customization, run control and engineering proof.



Work view, revision 8. VS Code Dark Modern (left), JetBrains Light (right). Reference renderings, not product screenshots.

Revision 8 / 17 October 2026
Source baseline: a89c00dbd612
Design targets and required backend construction are identified explicitly.

NAVIGATE THE MANUAL

Contents

Front matter and Part 1 define the corrected components. Part 2 shows every screen in both IDEs. Part 3 defines the validation still required. Parts 4 and 5 retain all revision 7 documentation.

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FRONT MATTER

Revision 8 at a glance

The everyday interface is quieter and native in each host. Shared components were corrected first; every screen was then redrawn on them. Nothing was removed from the product or the documentation.

	PROBLEM	REVISION 7	REVISION 8	SEE
1	**Button labels sat high**	Measured on all 288 R7 buttons: cap height centred 4.6 px above the midpoint (range 4.4–4.8).	Labels are trimmed to cap height and baseline and centred from font metrics. Reference renderer: median +0.0 px across 1036 labels.	p. 10
2	**Too much explanatory copy**	5,372 words across the 72 VS Code plates; subtitles, helper paragraphs and status rows on every surface.	3,898 words (window titles excluded in both counts). One status line per header, help only where a decision needs it, details behind disclosures. All 838 changes are recorded per screen.	p. 15
3	**Implementation instructions in the UI**	R7 page 038 told users how developers should distinguish empty history from failed searches.	Moved to the S61 contract and drawn as three distinct states. Same treatment for S62 (null context usage), S28 (config key alias), S72 and S56.	p. 95
4	**Backend detail beside fields**	R7 page 068 explained the micro-USD conversion under the money cap.	The conversion lives in control B-02. The field now says what happens at the limit: Faktor won't start a call that would go over it.	p. 161
5	**Weak prioritisation in Work**	Conversation, activity, approval, configuration, usage and composer had similar weight.	Fixed order: pinned attention, conversation, composer, configuration, usage. The approval keeps its decision facts visible; Stop moves to the task header.	p. 40
6	**Native fidelity was cosmetic**	Colours and radii changed; type, sizing, capitalisation, placement and surfaces did not.	Each host gets its own type, metrics, label style, button order, validation pattern and native surface (Settings, Quick Pick, DialogWrapper, diff, terminal).	p. 9

Retained

Every feature, all 200 control contracts, all 72 screen contracts, the 16 workflows, 40 acceptance scenarios, code outlines, restored first-report content, sources, identity and website. Wording of retained documentation is unchanged; amendments are marked.

Not yet proven

Plates are reference renderings in Chromium with substitute fonts. Optical, contrast and retention checks were run here; real task flows in both IDEs with assistive technology were not. Part 3 defines that work.

FRONT MATTER

Acceptance criteria for this revision

Drawn from the official VS Code and JetBrains guidance, WCAG 2.2 and established usability principles. A status of “Specified” means the design and contract define it; only an IDE test can confirm it.

#	GUIDELINE	WHAT FAKTOR SHOULD SATISFY	HOW REVISION 8 APPLIES IT	STATUS	SEE
1	**Native platform conventions**	Follow VS Code’s extension UX and JetBrains’ component rules; inherit host fonts, themes and interaction behavior.	Host foundations table; native surfaces per screen (Settings editor, Configurable, Quick Pick, popup, DialogWrapper, modal message, diff, terminal); host capitalisation, label style and validation.	Specified	p. 9
2	**Optical alignment**	Consistent text baselines, centered button contents and balanced icon–label groups.	Metric-trimmed labels; label/value baseline rows; 16 px icon boxes on the cap centre with 6 px gaps.	Measured in reference renderer	p. 10
3	**Meaningful spacing and grouping**	Related elements sit closer together; section spacing communicates structure. Arbitrary whitespace does not fix hierarchy.	4-step scale; within-group 4–8 px, between groups 14–16 px, sections titled; actions placed next to the content they act on.	Specified	p. 14
4	**Action hierarchy**	A clearly identifiable primary action, with secondary and destructive actions treated appropriately.	One primary per surface, named on every spread; destructive actions name verb and object and confirm; no colour-only danger.	Specified per screen	p. 11
5	**Progressive disclosure**	Show common controls first; reveal advanced tuning and diagnostic detail when requested.	Four tiers: glance, on demand, inspector, documentation. Advanced parameters, request details and proof records sit behind labelled disclosures.	Specified	p. 16
6	**Concise, task-oriented language**	Explain the user’s decision and consequences. Keep backend terminology and implementation instructions in appropriate detail views.	Glossary of replaced terms; 838 copy changes recorded with destinations; implementation rules moved to contracts.	Specified	p. 15
7	**Accessible contrast**	WCAG 2.2 AA: generally 4.5:1 for normal text and 3:1 for large text; evaluate essential graphical controls separately.	Measured 80 token pairs in five themes. Faktor-owned tokens pass; 9 host defaults fall short and are listed for decision.	Measured; decisions open	p. 17
8	**Usable targets**	WCAG AA specifies 24 × 24 CSS-pixel pointer targets, subject to exceptions. The enhanced AAA criterion uses 44 × 44.	Faktor controls are ≥ 24 px tall (buttons 26/28 px); host-drawn toolbar icons are noted where smaller.	Specified	p. 17
9	**Keyboard and focus behavior**	Logical navigation, visible focus, appropriate activation keys, focus restoration and unobscured focused controls.	Host focus rings; initial focus rules (never on Allow Once); Escape returns to trigger; sticky footers never cover focus.	Specified; IDE test required	p. 17
10	**Resizing and text enlargement**	Preserve readable text and reachable actions across narrow panels, enlarged fonts and host zoom.	Stacked VS Code forms; 320 px Work panel; 200% text plates; actions stay reachable.	Rendered; IDE test required	p. 194
11	**Clear status and recovery**	Distinguish running, waiting, failed, cancelled and completed states; preserve input and provide useful recovery actions.	One status vocabulary (icon + word); every problem row carries its fix; drafts are always kept.	Specified	p. 13
12	**Semantic accessibility**	Correct accessible names, roles, states and announcements, verified with assistive technology.	Role and name table for webview and Swing; announcement rules; AT verification protocol.	Specified; AT test required	p. 182

FRONT MATTER

Nielsen’s ten heuristics, checked explicitly

Each heuristic, where revision 7 fell short, and the specific change in revision 8.

#	HEURISTIC	REVISION 7	REVISION 8	SCREENS
1	**Visibility of system status**	Status repeated in badges, subtitles and rows; pending states looked like cards.	One status line in each header (icon + word); pinned attention; Starting… / Stopping… states.	S01 S38 S22
2	**Match with the real world (familiar language)**	Backend vocabulary: admission, receipt, envelope, generation, owner checkout, micro-USD, VCSC.	Plain words with a glossary; acronyms spelled out; consequences stated.	S21 S53 S44
3	**User control and freedom**	Stop scopes and queue edits were present but dense.	Scoped stop dialog; Move Up/Down; Cancel adjacent to primary; drafts kept on every failure.	S70 S59 S39
4	**Consistency and standards**	One custom chrome for both IDEs.	Host chrome, capitalisation, label style, button order and validation per platform; one status vocabulary.	S03 S29 S02
5	**Error prevention**	Good guards, explained in prose.	Guards expressed by the control: disabled with reason, exact command shown before approval, confirmation names the object.	S34 S14 S66
6	**Recognition rather than recall**	Settings meaning lived in helper paragraphs.	Labels that describe outcomes; pickers show capabilities; breadcrumbs and selection persist.	S02 S06 S13
7	**Flexibility and efficiency of use**	Composer controls competed with content.	Composer pickers in one row; Ctrl/Cmd+Enter; Quick Pick and popup search; slash commands.	S36 S17
8	**Aesthetic and minimalist design**	Every surface carried subtitle, helper paragraph, metadata and disclosure description.	5,372 → 3,898 words in VS Code plates; one primary action; quiet secondary rows.	All
9	**Help users recover from errors**	Errors explained the system rather than the fix.	Every problem row carries its fix (Retry, Sign In…, Configure Remote…); host validation patterns.	S37 S30 S48
10	**Help and documentation**	Documentation text sat inside controls.	Help links (Setup Guide), tooltips, details disclosures and the contracts in this document.	S34 S61

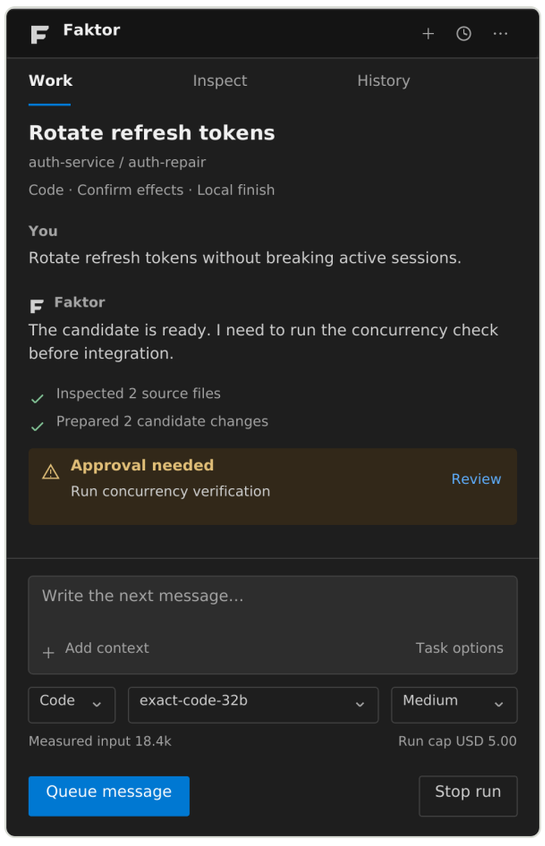
Revision 7 underperformed most on minimalism (8) and familiar language (2). Both are addressed on every screen; each spread lists what left the interface and where it now lives.

FRONT MATTER

Before and after

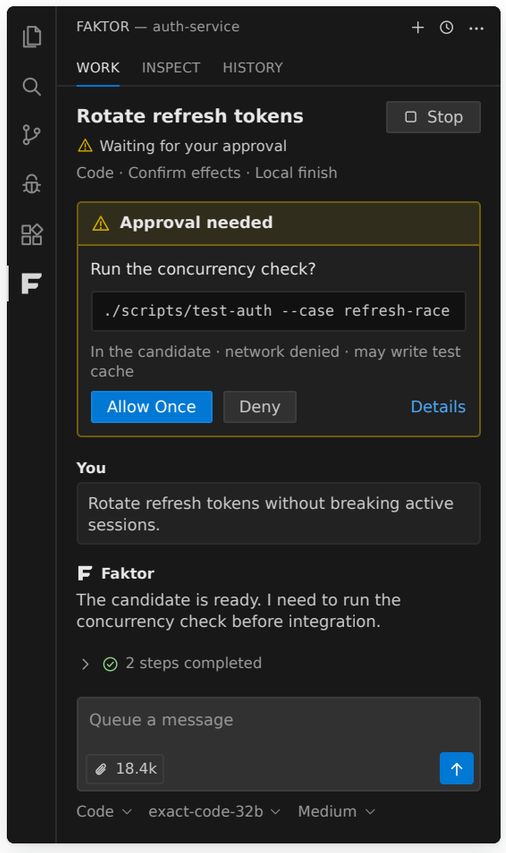
Revision 7 pages beside their revision 8 replacements. The measurements come from rasterised R7 pages and the R8 reference renderer.

R7 Work (page 014)



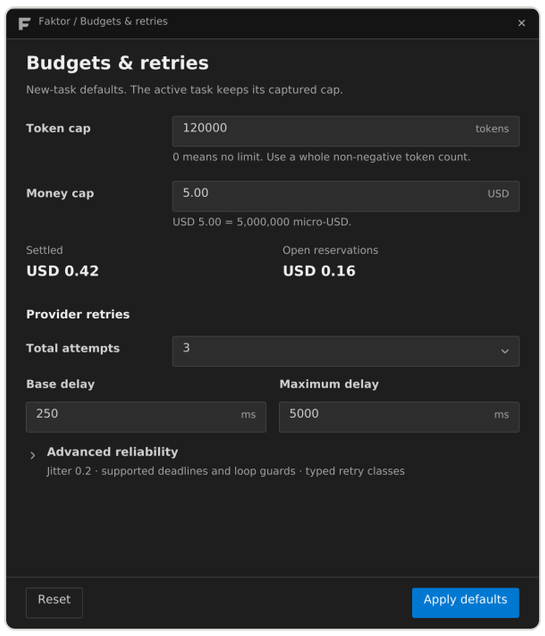
Six regions of similar weight: header, transcript, steps, approval, composer, configuration and usage.

R8 Work (S01)



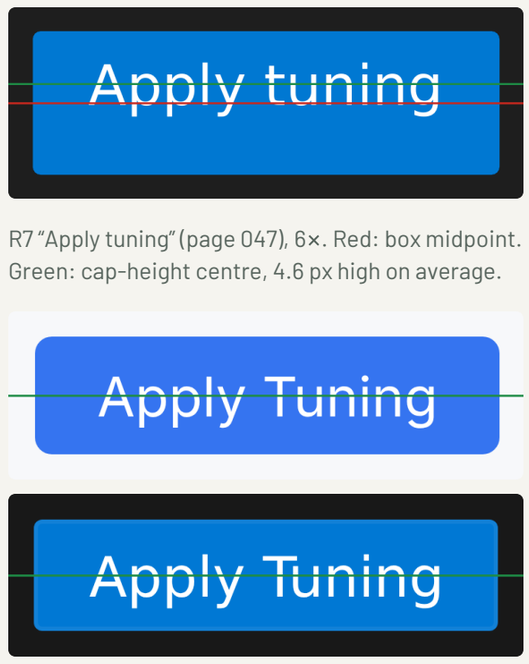
Attention pinned, conversation, composer; configuration in one quiet row; usage is the context count.

R7 Budgets (page 068)



"USD 5.00 = 5,000,000 micro-USD" beside the field; R7 page 038 likewise explained developer rules to users.

Button centring



R7 "Apply tuning" (page 047), 6×. Red: box midpoint. Green: cap-height centre, 4.6 px high on average.

R8 JetBrains and VS Code, 6×. The lines coincide (offset +0.0 px median).

FRONT MATTER

How to read a screen spread

Design spread

Left page of each screen. VS Code Dark Modern on the left, JetBrains Light on the right, each in its native surface (named above the plate). The header gives the original job statement, the primary action, the control IDs and the R7 page it replaces.

Contract page

Right page. The revision 7 behaviour contract, verbatim, plus what changed and every line of R7 plate copy that left the default interface with its new home.

Plates

Reference renderings drawn at 1:1 logical pixels and scaled to fit (typically 106–120%). JetBrains uses Inter 13 px, its default UI font. VS Code uses the platform font; plates show the Linux reference (DejaVu Sans 13 px), the widest common case. Code uses DejaVu Sans Mono in both.

Themes

Every surface is drawn in four host themes in the build. Spreads show Dark Modern and JetBrains Light; Part 3 shows VS Code Light, JetBrains Dark and High Contrast.

Button order

Plates show the Windows and Linux order (primary first). On macOS both hosts reverse dialog buttons; implementations call the host dialog APIs instead of ordering buttons by hand.

What a plate is not

Not a product screenshot and not evidence of IDE behaviour. Focus, announcements, zoom and real fonts are verified only by the Part 3 protocol.

How to review (retained from revision 7)

1. Follow a real journey

Start with setup, admission, work, plan, evidence, integration and outcome. Settings are subordinate.

2. Inspect the exact behavior

Each screen has scope, action, recovery, focus, state, layout and implementation contracts.

3. Verify retained content

Both original IDE reports are archived. Original journeys, states, file targets, identity and website stay included.

4. Build and validate

Use all control/build rows, detailed UI rules, code outlines and adverse-content fixtures.

The source snapshot and existing/new implementation classification are retained; the interface and document layout are fully redrawn.

PART 1

1 Shared components, corrected first

Every screen in Part 2 is built from these components. They were fixed before any screen was redrawn: host typography and sizing, optical centring, action hierarchy, validation, status, spacing, language and accessibility.

	SECTION	PAGE
1.1	Host foundations	p. 9
1.2	Optical alignment	p. 10
1.3	Buttons and action hierarchy	p. 11
1.4	Fields, forms and validation	p. 12
1.5	Choices	p. 13
1.6	Status, attention and recovery	p. 13
1.7	Lists, rows, disclosure and tabs	p. 14
1.8	Spacing and grouping	p. 14
1.9	Language	p. 15
1.10	Progressive disclosure	p. 16
1.11	Accessibility	p. 17
1.12	Revision 7 construction standard (retained)	p. 19

PART 1 · SHARED COMPONENTS

1.1 Host foundations

Native fidelity is typography, sizing, label style, action placement, validation and choice of surface, not colour and radius. Rules were checked against the official guidance on 17 October 2026; pixel values are reference values read from the host at runtime.

	VS CODE	JETBRAINS	FAKTOR RULE
UI font	Platform UI font from `--vscode-font-family`; size from `--vscode-font-size`.	Inter on all platforms in the New UI; size from Appearance settings.	Never set a font family or size. Plates: DejaVu Sans 13 px (Linux reference) and Inter 13 px.
Code font	`--vscode-editor-font-family` and size.	Editor colour-scheme font.	Monospace only for commands, paths, IDs and output. Plates use DejaVu Sans Mono.
Control size	Reference 26 px buttons and inputs in webviews.	Reference 28 px; JBUI-scaled.	Use host components where they exist; webview controls never below 24 px.
Corners	2 px controls.	4 px controls; 8–10 px popups and dialogs.	Follow the host; no custom radii.
Field labels	Stacked: label, one-line description, control (Settings editor pattern).	Left label, a noun ending with a colon, sentence case.	One label component, two host layouts.
Button labels	Title case, as in VS Code's own commands and buttons.	Title case, imperative verb, five words or fewer; ellipsis when more input follows.	Same label text on both hosts; ellipsis only for Locate..., Configure... and similar.
Primary action	Sidebar: full-width at the end of the content. Editor tab: left-aligned row under the form.	Tool window: left-aligned row. Dialog: right-aligned, default first (Windows/Linux). Settings: OK, Cancel, Apply.	Exactly one primary per surface, named on every spread.
Dialog buttons	Modal message through the API; the host orders buttons.	DialogWrapper; one default button; Help bottom-left.	Never hand-order dialog buttons; macOS reverses them.
Validation	Message box under the input (Settings editor); webviews mirror the input-validation tokens.	Tooltip above the field on focus; inline error above the buttons after submit; complex forms keep OK enabled.	Validate format on input, rules on blur or submit, never empty required fields on input.
Inline help	Description line under the setting title.	Under the control, up to five lines and 70 characters wide; never repeat the label.	One line by default, stating a consequence.
Persistent settings	Native Settings editor; webviews must not repeat Settings.	Configurable pages under Settings › Tools › Faktor.	Editor-tab forms only where atomic multi-field apply is required.
Simple choices	Quick Pick (single, multi, with separators and item buttons).	Popup chooser with search, groups and an ad line.	Pickers for model, agent and context (S02, S06, S09).
Views	One view container; 3–5 views at most; no sidebar content that could be a command.	One tool window with Work, Inspect and History content tabs.	Three clusters only; Settings, Usage and Diagnostics open from commands.
Theming	All elements themeable from `--vscode-*` tokens.	JBColor and UIManager keys; JBUI.scale for metrics.	No Faktor colours in reading controls; brand stays on the marketing site.
Diff and terminal	`vscode.diff`; integrated terminal in the Panel.	DiffManager; console tool window.	Never redraw diffs or terminals in a webview.
Focus	1 px `focusBorder` outline.	2 px accent focus ring.	Host focus only; never remove outlines.

PART 1 · SHARED COMPONENTS

1.2 Optical alignment

Button contents appear centred; body copy, field values and form labels share left edges and baselines.

What was wrong

Revision 7 placed every button label with a fixed text offset. Measured on all 288 plate buttons, the cap-height centre sat 4.6 px above the button midpoint (logical pixels; range 4.4–4.8). Horizontal centring was already correct.

The rule

Centre the cap-height box, not the font's line box. A single-line label is trimmed to its cap height and alphabetic baseline, then centred by the layout engine. Mixed-case labels with descenders ("Apply") look centred because the eye reads the cap and x-height band, not the descenders.

Icon and label groups

A 16 px icon box is centred on the label's cap box with a 6 px gap. Trailing chevrons are 12–14 px. Icons never set the row height.

Baselines in forms

In JetBrains label-left rows the label and the field text share one baseline (first baseline for wrapped labels). In VS Code stacked rows, label, description and control share one left edge. Values, units and help keep the same left edge as their field.

Webview (VS Code)

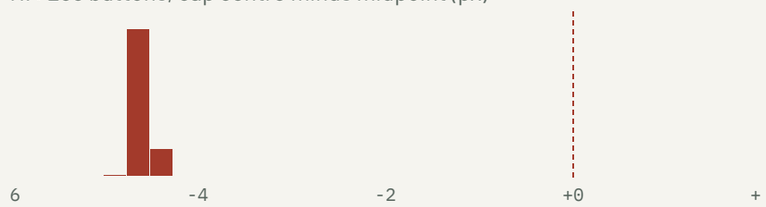
```
.btn {
  display: inline-flex;
  align-items: center;
  justify-content: center;
  height: var(--control-h);
  padding-inline: var(--btn-px);
}
/* cap height to baseline */
.btn > .label {
  text-box: trim-both cap alphabetic;
}
@supports not (text-box: trim-both cap
alphabetic) {
  .btn > .label {
    line-height: var(--control-h);
  }
}
```

Swing (JetBrains)

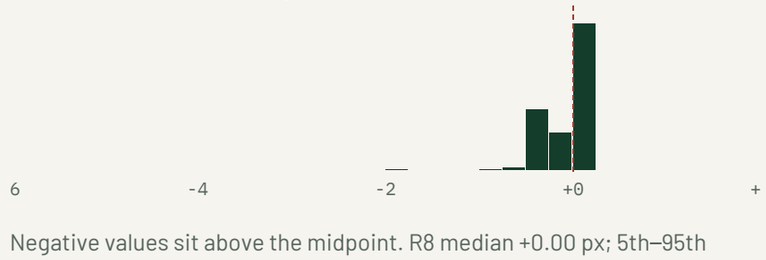
```
// Custom painting only; prefer JButton
val cap = font
  .createGlyphVector(g.fontRenderContext, "H")
  .visualBounds.height
// baseline that centres the cap height
val y = ((height + cap) / 2.0).roundToInt()
g.drawString(label, x, y) // no fixed offset
```

Measured offsets

R7: 288 buttons, cap centre minus midpoint (px)



R8: 1036 labels (buttons, segments, tabs)



Negative values sit above the midpoint. R8 median +0.00 px; 5th–95th percentile -0.33 to +0.17 px. The few larger R8 readings are labels whose ascenders or bowls ("l", "P") bias the ink-based measurement, not misplaced text.



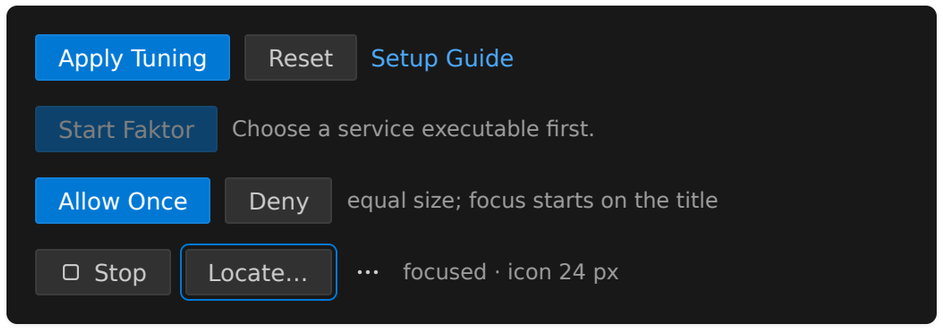
Red: box midpoint. Green: cap-height centre. Left: R7. Right: R8, where the lines coincide.

PART 1 · SHARED COMPONENTS

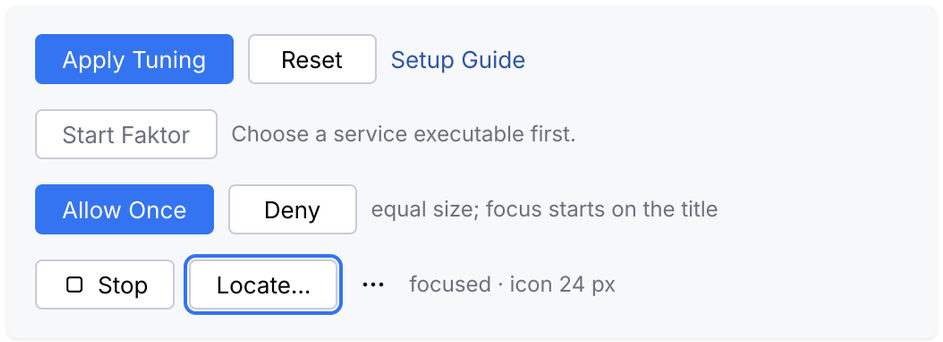
1.3 Buttons and action hierarchy

A clearly identifiable primary action, with secondary and destructive actions treated appropriately.

VS CODE · DARK MODERN



JETBRAINS · LIGHT



Placement by surface

SURFACE	VS CODE	JETBRAINS
Sidebar / tool window	Full-width primary at the end of the content; secondary below as a button or link.	Left-aligned row: primary, then secondary; links right.
Composer	Send button carries the verb as its name (Run Task / Queue Message); Ctrl/Cmd+Enter.	Same.
Editor tab form	Left-aligned row under the form: primary, then Cancel or Reset.	Configurable page: OK, Cancel, Apply; Reset link top-right; page actions inside the page.
Confirmation	Native modal message: primary, other options, Cancel.	DialogWrapper: default button, Cancel; Help bottom-left.
Picker	Enter selects; multi-select shows an OK button.	Enter selects; ad line states timing.

Rules

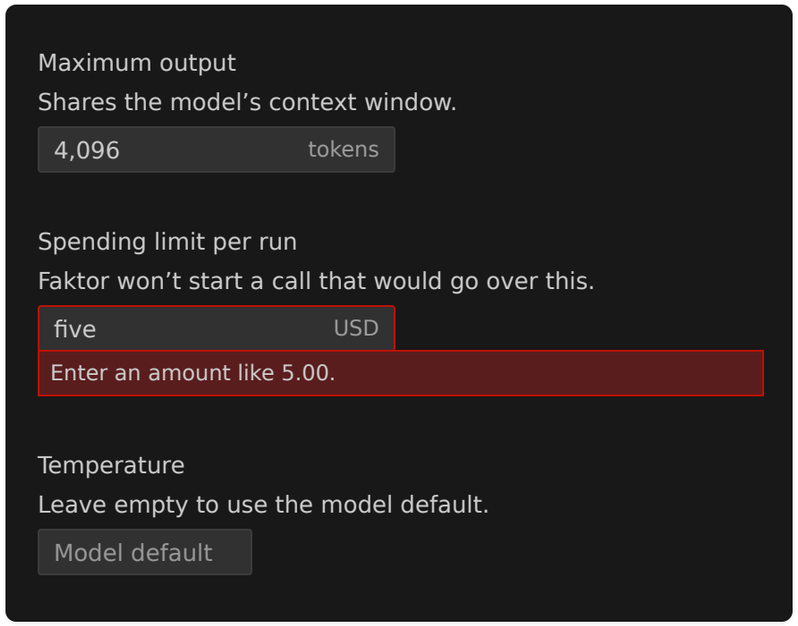
- One primary action per surface. It advances the current job and is named on every spread in Part 2.
- Label the verb and the object when there is any doubt: Stop Run, Delete Memory, Check Task Status, Open AC-03.
- Cancel sits next to the primary action. Reset is separate (a link on JetBrains Settings pages).
- Neither host has a red “danger” button. Destructive actions name their target, confirm through the host dialog and are never the default when a safe action exists.
- A disabled primary shows its reason next to it, associated for assistive technology.
- Approvals: Allow Once and Deny have equal size; initial focus is on the question, never on Allow Once.
- Ellipsis only when the action asks for more input (Locate..., Configure..., Review Restore...).
- A background refresh, an optional account and Stop never compete with the composer for emphasis.

PART 1 · SHARED COMPONENTS

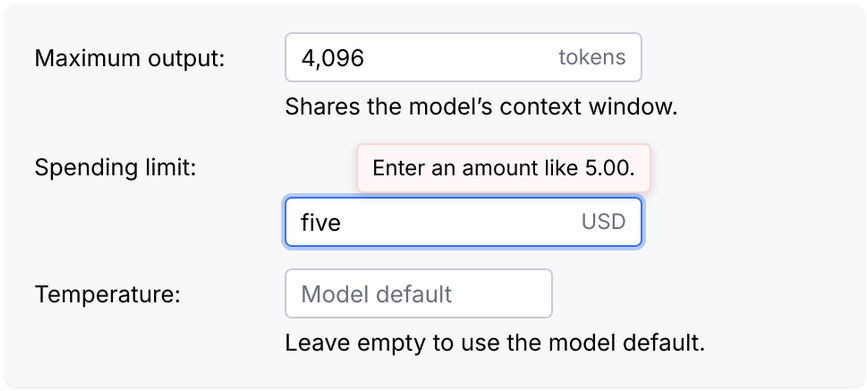
1.4 Fields, forms and validation

One field component; two host layouts. Help explains consequences; validation follows each platform's timing and placement.

VS CODE · DARK MODERN



JETBRAINS · LIGHT



Validation timing

WHAT IS VALIDATED	VS CODE	JETBRAINS
Characters a field can never hold	On input	On input
Format and range (5.00, 0–100)	On blur or submit	On blur or submit; tooltip on focus
Empty required field	On submit only	Simple form: disable OK. Complex form: highlight on OK
Remote checks (connection, model)	Explicit Test action	Explicit Test action
Whole-form rules (atomic apply)	Native Settings saves only valid values; Faktor reports conflicts (S60)	Inline error above the buttons; focus first invalid field

Rules

- Help is one line by default and states a consequence ("Faktor won't start a call that would go over this"), not a mechanism ("USD 5.00 = 5,000,000 micro-USD").
- Units sit inside the field as a suffix; numbers use the locale's grouping while exact values are kept for the backend.
- Unset is not zero: an empty field shows the inherited value as placeholder; "0 means no limit" appears only where the backend uses zero that way.
- Errors describe the problem and the fix in user terms and keep the user's text. Editing hides the error.
- Locked values show who set them ("Set by your organization") next to a lock icon.

PART 1 · SHARED COMPONENTS

1.5 Choices

Pick the control by the shape of the decision.

CONTROL	USE WHEN	EXAMPLES
Segmented	2–4 short, exclusive options (≤ 12 characters each).	Reasoning effort, Finish, Restore scope
Radio group	Exclusive options that each need one line of explanation.	Shell isolation, Stop scope, hook failure
Checkbox	Independent on/off settings.	Efficiency switches, instruction sources
Select	Five or more options, or options loaded at runtime.	API, adapter, keep loaded
Picker	Searchable choice with capabilities or groups.	Model, agent, context

JETBRAINS · LIGHT

Reasoning effort

Off | Low | Medium | High

- (•) Fail-closed (recommended)
  Blocks shell commands when unsafe.
- ( ) Network-capable
  Weaker isolation.
- [x] Compress tool output
  Full output stays available.

PART 1 · SHARED COMPONENTS

1.6 Status, attention and recovery

Running, waiting, failed, cancelled and completed are distinct, with a recovery action.

	WORDS USED	RECOVERY OFFERED
✓	Done, Passed, Verified, Ready, Kept	None needed
◌	Running, Starting…, Checking AC-03	Stop (scoped)
⚠	Waiting for your approval, Needs input, Out of date, Conflict	Open, Review, Verify current candidate
⊗	Failed, Action needed, Blocked by an error	Retry, Sign In…, Configure Remote…
◌	Pending, Needs evidence, Incomplete	Open the criterion or step
⊘	Blocked, Denied by preset, Can't restore	Reason is shown; fix where it exists
○	Not started, Queued, Unavailable, Stopped, Cancelled	Quiet; no colour emphasis
🔒	Locked (organization)	Who set it

VS CODE · DARK MODERN

⚠ **Approval needed**

Run the concurrency check?

```
./scripts/test-auth --case refresh-race
```

In the candidate · network denied · may write test cache

Allow Once | Deny | Details

⊗ **Faktor and the extension are out of sync**
Nothing was skipped; restart the service to continue.

Restart Service…

trace.json ⊗ Failed
Upload interrupted Retry

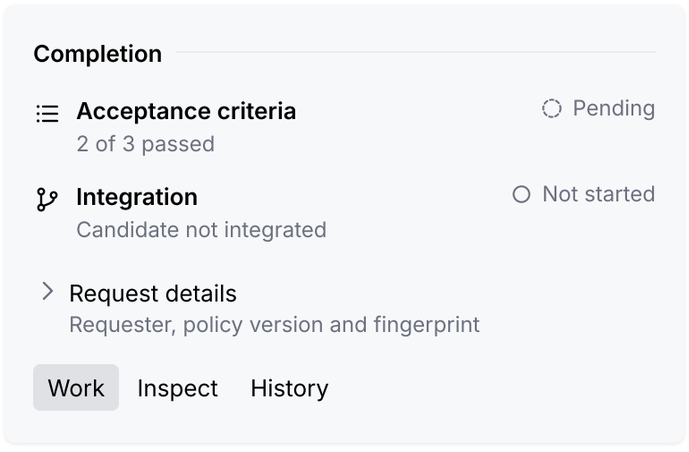
- Status is always an icon and a word; colour is never the only signal.
- Use one status line per header. Do not repeat the same state in a badge, a subtitle and a row.
- Banners only for conditions that affect the whole screen, one at a time.
- Attention that needs a decision (approval, question) is pinned above the transcript and keeps its facts visible.
- Every problem offers its own recovery; drafts and selections are always kept.
- Pending, stale, unavailable, failed and effect-unknown never render as success.

PART 1 · SHARED COMPONENTS

1.7 Lists, rows, disclosure and tabs

Shallow lists followed by one selected detail.

JETBRAINS · LIGHT



- A row has a 16 px icon, a title, at most one meta line and at most one trailing status or action (both stack on separate baselines).
- Rows bind to entity IDs; refresh keeps selection and scroll anchor.
- A disclosure has a chevron, a title and a one-line summary of what it hides.
- Tabs use the host idiom: VS Code panel-style uppercase tabs inside the webview; JetBrains tool-window content tabs.
- Read-only facts are text, never disabled inputs.

PART 1 · SHARED COMPONENTS

1.8 Spacing and grouping

Related elements sit closer together; sections are titled.

PX	USE
2	Icon to its own label (inside a chip)
4	Title to meta line; label to control in a stacked field
6	Icon to label; status icon to word
8	Between buttons; rows in a list (padding)
12	Between fields and between blocks in a group
14–16	Between groups; section title above its rows
24	Between unrelated regions; header to first group in editors

- Spacing communicates structure: the space between groups is at least twice the space within them.
- Actions sit next to the content they act on. Sticky footers are used only where content scrolls (Work, long forms).
- Whitespace never stands in for hierarchy: headers, sections and a single primary action do that work.
- All values scale with host zoom (JBUI.scale in JetBrains; CSS px in webviews).

PART 1 · SHARED COMPONENTS

1.9 Language

Concise, task-oriented copy. Backend terminology stays available in details, inspectors and this specification.

Rules

- Write for the decision in front of the user and its consequence.
- One header status line; no subtitle that repeats the title.
- Keep implementation instructions in this specification, not in the interface.
- Name objects exactly (file, command, model); spell out acronyms.
- Buttons: Title Case verbs. Labels and help: sentence case.
- Errors: the problem, then the fix. No “invalid”, “illegal” or “error” alone.
- Avoid “please”, “successfully” and exclamation marks.
- Numbers: locale grouping and a unit; exact values for copy and export.

> Plate copy across the 72 VS Code surfaces fell from 5,372 to 3,898 words. Each of the 838 changes is listed on its screen’s contract page.

BACKEND TERM	INTERFACE WORDING	WHERE THE EXACT TERM LIVES
admission, admitted	start, started	Contracts; Inspect › Activity
admission receipt	confirmation (“Faktor confirmed the start”)	Inspector details
submission ID, body digest, generation, epoch, cursor	not shown	Tooltips and details
owner checkout	your workspace	S44 contract
landed, landing	integrated, integration	V-controls
authority / tool envelope	permissions; “what it can do”	Q- and A-controls
daemon, daemon host	Faktor; the machine running Faktor	P- and O-controls
managed policy	organization policy	Q-12
VCSC	cost per verified change	O-06, K-12
micro-USD	not shown	B-02
settled / open reservations	spent / held for calls in progress	B-04
stale	out of date	V-09
effect unknown	not confirmed; may have happened	X-05, D-07
child agent	agent (Verifier), “agent 3 of run 047”	A-controls
FailClosed / FailOpen	Block the tool / Allow the tool	S-12
lexical fallback	text search	C-10
semantic_context alias	not shown	K-08
capability authority	capabilities (info button)	M-07
protocol / schema refused	out of sync	O-04, D-13
provenance	where this value comes from	O-01, D-11

PART 1 · SHARED COMPONENTS

1.10 Progressive disclosure

Common controls first; advanced tuning and diagnostic detail when requested. Counts show where the revision 7 plate copy went.

TIER 1 · ALWAYS VISIBLE

Glance

State, decision-critical facts, the primary action. Approval command, target and network stay here.

711

Rewritten in place, moved to host chrome or removed as redundant

TIER 2 · ONE STEP AWAY, SAME PLACE

On demand

Tooltips, info buttons and labelled disclosures (Request details, Advanced parameters).

61

Tooltips and disclosures

TIER 3 · A DEDICATED VIEW

Inspector

Activity, Context, Verification, Evidence, Usage and Diagnostics in the Inspect cluster.

7

Inspector views

TIER 4 · OUTSIDE THE PRODUCT UI

Documentation

This specification (contracts, controls, rules) and the product documentation.

59

Specification and documentation

Rules

- Show common controls first: model, effort, permissions, finish and caps are reachable from the composer or Task options.
- Advanced tuning (thinking budget, penalties, seeds), request fingerprints, protocol versions and raw identities sit behind disclosures.
- Decision-critical approval information is never disclosed lazily: the exact command, where it runs, network and effect are visible.
- A disclosure names what it hides; collapsed state is announced (aria-expanded / AccessibleState.EXPANDED).
- Nothing is removed from the product: each relocation is recorded with its destination on the screen's contract page.

PART 1 · SHARED COMPONENTS

1.11 Accessibility: contrast

WCAG 2.2 AA: 4.5:1 for normal text, 3:1 for large text and for graphical objects and component boundaries needed to identify them (1.4.3, 1.4.11). Ratios computed from the plate tokens.

PAIR	NEEDS	DARK MODERN	LIGHT MODERN	JB LIGHT	JB DARK	HIGH CONTRAST
Body text	4.5:1	11.06	10.55	19.76	10.55	21.00
Body text on editor	4.5:1	10.26	11.20	21.00	12.59	21.00
Secondary text	4.5:1	6.55	5.83	4.64	5.28	21.00
Secondary text on editor	4.5:1	6.08	6.19	4.93	6.30	21.00
Link	4.5:1	7.16	5.94	6.71	4.18	7.96
Placeholder	4.5:1	4.51	4.54	4.93	4.26	11.42
Primary button label	4.5:1	4.53	6.31	4.28	4.28	21.00
Secondary button label	4.5:1	8.10	8.89	21.00	10.55	21.00
Primary button vs view	3.0:1	3.92	5.94	4.03	3.23	10.55
Focus indicator vs view	3.0:1	3.92	5.94	4.03	3.23	8.18
Checkbox / radio outline	3.0:1	4.84	4.54	3.67	4.75	10.55
Input border vs editor	3.0:1	1.49	1.57	1.60	2.07	10.55
Warning icon	3.0:1	7.69	2.93	3.17	8.49	14.97
Error icon	3.0:1	5.30	4.46	4.17	3.87	8.55
Success icon	3.0:1	9.74	4.07	4.15	5.61	11.52
Info / running icon	3.0:1	5.79	3.38	4.03	4.18	6.84

Below threshold

- VS Code Dark Modern: input border vs editor 1.49:1(#3C3C3C on #1F1F1F)
- VS Code Light Modern: input border vs editor 1.57:1(#CECECE on #FFFFFF)
- VS Code Light Modern: warning icon 2.93:1(#BF8803 on #F8F8F8)
- JetBrains Light: primary button label 4.28:1(#FFFFFF on #3574F0)
- JetBrains Light: input border vs editor 1.60:1(#C9CCD6 on #FFFFFF)
- JetBrains Dark: link 4.18:1(#548AF7 on #2B2D30)
- JetBrains Dark: placeholder 4.26:1(#8C8F96 on #2B2D30)
- JetBrains Dark: primary button label 4.28:1(#FFFFFF on #3574F0)
- JetBrains Dark: input border vs editor 2.07:1(#4E5157 on #1E1F22)

Decision needed

All failing pairs are host defaults that native components inherit. Overriding them would break native fidelity (criterion 1) to meet contrast (criterion 7). Recommended: keep native components native; for Faktor-drawn elements (status icons, segmented controls, custom checkboxes, focus) require ≥ 3:1 using the strongest available host token; report the JetBrains primary-button and link values upstream; verify again in each host's high-contrast theme.

PART 1 · SHARED COMPONENTS

1.11 Accessibility: semantics, keyboard, targets and zoom

Accessible names, roles, states and announcements must be verified with assistive technology in each host (Part 3).

Roles and names

ELEMENT	VS CODE WEBVIEW	JETBRAINS SWING
Work / Inspect / History	tablist + tab + tabpanel; arrow keys	Tool-window content tabs (native)
Header status line	role=status (polite), text + icon (aria-hidden)	AccessibleContext name/description on the label
Approval card	region named “Approval needed”; announce once	Notification group or panel with accessible name; announce once
Composer	textarea labelled “Message”; send button named by its verb	JBTextArea with accessible name; action button with text
Lists and rows	listbox/option or list/listitem with selected state	JBList; AccessibleState.SELECTED
Disclosure	button with aria-expanded	Collapsible group with expanded state
Segmented / radio	radiogroup + radio with checked state	JRadioButton group / SegmentedButton
Errors	aria-invalid + aria-describedby to the message	Validation tooltip linked to the field
Pickers	Native Quick Pick (host accessibility)	Native popup (host accessibility)

Keyboard and focus

- Tab order follows reading order; the composer is reachable in one step from the transcript.
- Visible focus is the host’s indicator (2.4.7); sticky footers and pinned cards never cover the focused control (2.4.11).
- Escape closes transient detail and returns focus to its trigger; it never cancels a run.
- Enter is newline in the composer; Ctrl/Cmd+Enter sends; IME composition never submits.
- Opening a detail moves focus to its heading; background events never steal focus.
- New approvals and final outcomes announce once (4.1.3); streaming text is not announced token by token.

Targets

- Faktor-drawn controls are at least 24 × 24 CSS px (2.5.8); buttons are 26 px (VS Code) and 28 px (JetBrains) tall.
- Text links in rows get a 24 px tall hit area; inline links in sentences are exempt.
- Host-drawn toolbar icons in VS Code view titles are 22 px with no spacing margin; they belong to the host and are noted, not overridden.
- The enhanced 44 × 44 target (2.5.5) is not adopted, because it would break host density.

Zoom and reflow

- Content reflows at 320 CSS px without horizontal scrolling (1.4.10); Part 3 shows 320 px and 200% text.
- Text resizes to 200% through host zoom without loss (1.4.4); never shrink text to fit a box.
- Labels wrap; values that cannot wrap (paths, IDs) elide in the middle with the full value available to copy.

PART 1 · SHARED COMPONENTS

1.12 Revision 7 construction standard (retained)

One grid, intrinsic sizing and a clear distinction between controls, status and supporting text. Kept verbatim from R7 pages 003–005, with each value's revision 8 status.

	RULE	R7 WORDING	R8 STATUS
01	**24 px panel insets**	The title, section headings, fields, help, errors and action bar share the same content edges.	Superseded by host insets (16 px sidebar content, 24 px editor forms, JetBrains Settings page insets).
02	**156 px labels + 20 px gutter**	At wide widths, control values and helper text use one fixed value column. Labels wrap before touching controls.	Retained for JetBrains label-left rows (128–160 px by page); VS Code stacks labels.
03	**36 px controls + 16 px row gap**	Inputs and primary actions have a consistent height. Longer help or validation expands the row and moves following content.	Height superseded by host metrics (26 / 28 px); intrinsic-height rule retained.
04	**8 px helper gap**	Helper and error text stay beneath their own input. They never drift below another field or cover its label.	Retained (4–6 px in host metrics). JetBrains errors use the platform tooltip.
05	**One action bar**	The footer is reserved independently. Complex forms keep Apply reachable to identify problems; unsupported effects remain disabled with a reason.	Retained where content scrolls; short surfaces place actions after content.

States and actions have different jobs

An input edits a value. A status reports a fact. A button performs a named action.

Control states: Default, focused, staged, invalid, disabled, locked and inherited each have an explicit treatment. Operational states: Pending, stale, unavailable, failed and effect-unknown never become success through optimistic rendering.

Narrow views reflow instead of shrinking

Stack fields, wrap supporting text and scroll the content region while keeping the action bar separate.

Reflow at measured width: below 560 logical px the label moves above the input; multi-choice rows wrap into a deliberate grid when needed. Preserve readable text: keep the host font and zoom; never reduce font size to make a fixed-height box appear to fit. Bound the correct region: use header / minmax(0, 1fr) content / footer; scroll long forms or transcripts inside the content region. Test adverse content: 240/320/480 px panels, 200% text, long paths, translations, errors, keyboard focus and IME are required fixtures.

R7 example plates on these pages, retained as text: "Agent defaults / Project scope · three examples on a single shared grid / Default model exact-code-32b / Maximum output 4096 tokens / Input and reserves must fit the same context window. / Money cap five USD / Enter a non-negative decimal such as 5.00. / Ask before external writes / A style instruction cannot widen execution authority. / Reset / Apply changes"; "State treatment / Values, errors and outcomes remain visually distinct. / Endpoint https://inference.example.invalid/v1 / A normal editable value with an attributed execution host. / Run cap five USD / Use a decimal amount such as 5.00. / Connection status Ready / Observed authentication, not a writable field. / Managed network deny / Locked constraints show their source and explanation. / An acknowledgement is still pending / Keep the previous coherent state. Do not label the operation applied yet."; "Connection / Stacked-field layout / Endpoint / The complete value remains available for copy and inspection. / Credential reference TEAM_API_KEY / No secret value is placed in the project configuration. / Ask before external effects / A wrapped checkbox keeps its checkbox and help text together. / Connection note / Use the team endpoint for this project. Keep local-only tasks on the local route. / Cancel / Save". Footnotes: "Logical dimensions scale with the host. The

PART 2

2 Screens: 72 surfaces in both IDEs

Each screen keeps its ID, job and control IDs. The left page shows it natively in VS Code and JetBrains; the right page keeps its full behaviour contract and lists every line of copy that moved.

	GROUP	SCREENS	PAGE
2.1	**Setup and start**	8	p. 21
2.2	**Work**	6	p. 39
2.3	**Inspect and proof**	13	p. 54
2.4	**Agents**	4	p. 81
2.5	**History**	4	p. 90
2.6	**Usage**	3	p. 100
2.7	**Models and providers**	7	p. 107
2.8	**Context**	4	p. 122
2.9	**Instructions and memory**	4	p. 132
2.10	**Permissions and tools**	9	p. 141
2.11	**Settings and diagnostics**	10	p. 160

PART 2 · SCREENS

2.1 Setup and start

8 screens. Each has a design spread for both IDEs and a contract page.

ID	SCREEN	JOB	PAGE
S34	**First open**	Open a project, review prerequisites and start the service explicitly while preserving the draft.	p. 22
S35	**Service executable trust**	Inspect the exact machine executable and its separate checksum and publisher trust basis before any start.	p. 25
S57	**Shell isolation setup**	Choose supported shell isolation explicitly and show platform-established filesystem and network guarantees.	p. 27
S36	**Ready conversation**	Write one task, choose an exact model and intent, and review the captured authority, criteria, finish and run caps.	p. 29
S33	**Task options**	Edit the goal, required criteria, completion contract, finish level and exact run caps in one draft sheet. The active task keeps its captured options.	p. 31
S37	**Attachment admission**	Resolve file, image and upload admission per source; failed or unsupported sources remain visible in the draft.	p. 33
S38	**Task admission pending**	Show one captured logical submission while the admission receipt is pending; suppress duplicate starts.	p. 35
S39	**Ambiguous task start**	Reconcile an ambiguous admission against the original request identity and body before any permitted retry.	p. 37

S34 SETUP AND START

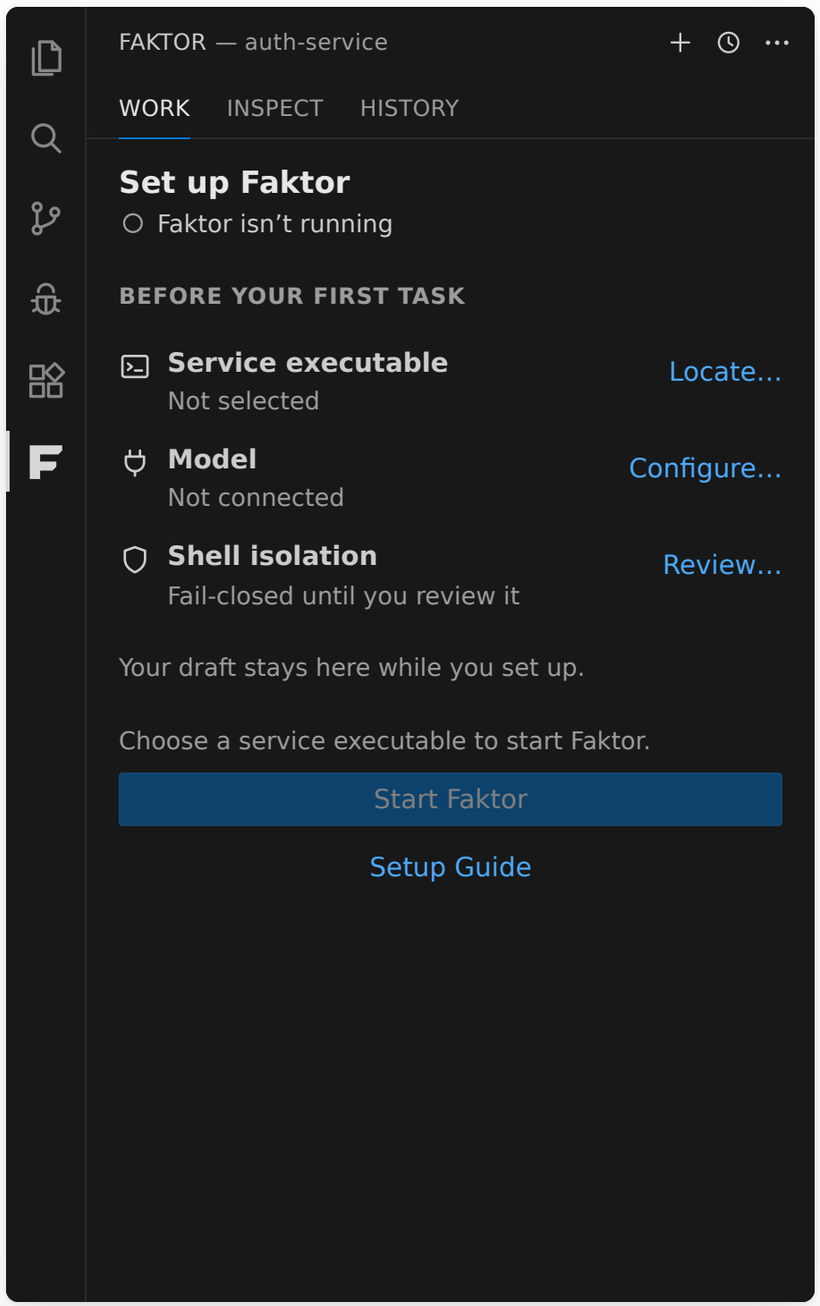
First open

Open a project, review prerequisites and start the service explicitly while preserving the draft.

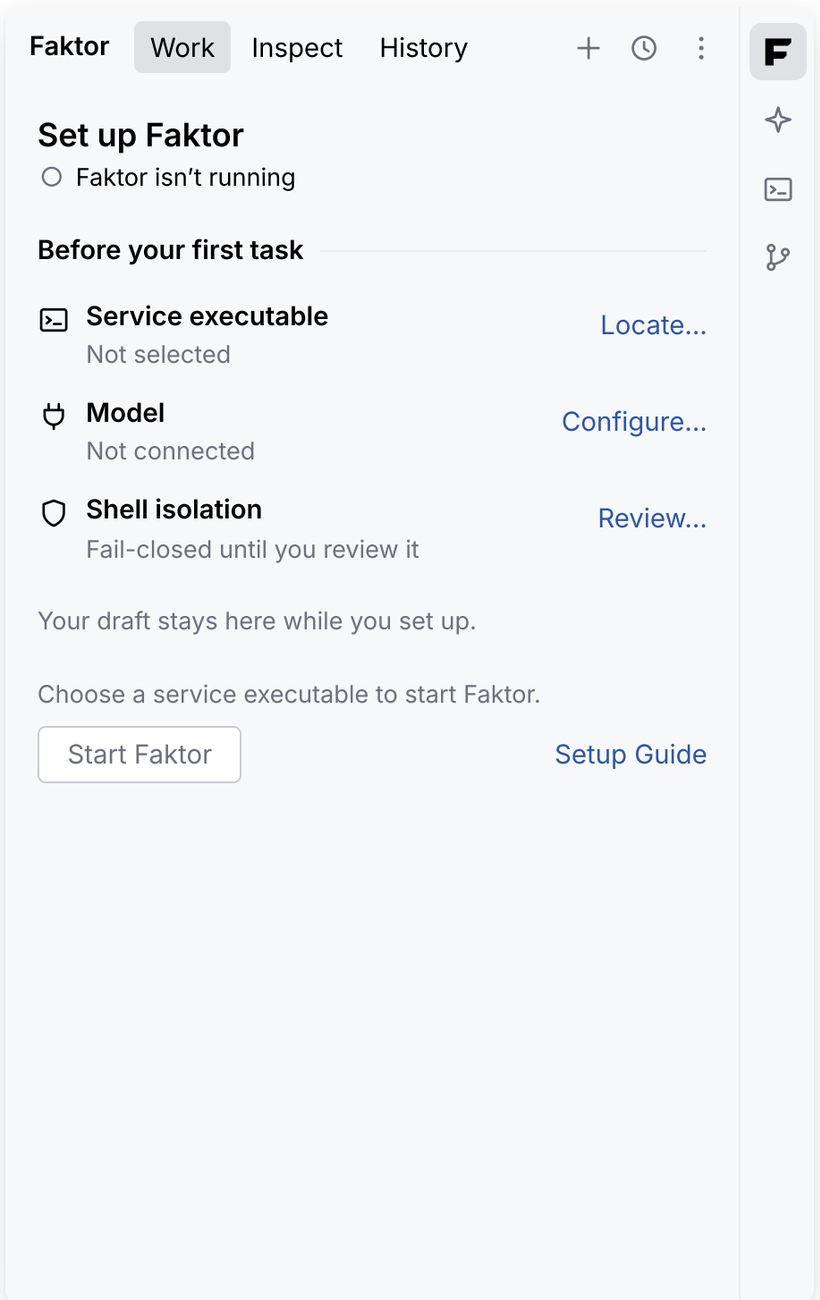
PRIMARY ACTION

Start Faktor

VS CODE · DARK MODERN — SIDEBAR VIEW · WEBVIEW



JETBRAINS · LIGHT — TOOL WINDOW · SWING / KOTLIN UI DSL



CONTROLS O-04 O-05 P-01 Q-11

REPLACES R7 page 006 · R4-S01

CONTRACT p. 23

S34 · SETUP AND START · BEHAVIOUR CONTRACT

First open: contract and revision notes

Original contract retained verbatim (R4-S01 / controls: O-04 O-05 P-01 Q-11). Design on p. 22.

What changed in revision 8

- One job: complete three prerequisites, then start. Start is the only primary action and states why it is disabled.
- VS Code renders this as a welcome-view style checklist with full-width actions; JetBrains uses its tool-window empty-state pattern with links.
- Workspace identity moves to the host frame instead of a repeated card.

Entry and return

More / Diagnostics, the setup walkthrough, or the exact blocked operation. Work / Inspect / History remain the only top-level clusters. Settings, Usage and Diagnostics use explicit Back. Return to the originating object, not the first row.

Scope and captured state

Execution host + service instance/generation + protocol version/cursor + trust source; account bindings stay distinct. Starting, restarting and stopping the service are explicit owned operations. Rebind after recovery; opening a view is not a start command.

Primary action and recovery

Start service. Show a bounded human reason and actionable recovery. Missing checksum, checksum mismatch, signature status, schema refusal and local/cloud boundary are separate facts.

Loading, empty and stale

Keep last coherent data stale, retain event cursors and distinguish automatic transient reconnect from a protocol-blocked stop.

Keyboard and focus

A background disconnect never steals typing focus. Recovery opens in a native command/dialog; Back returns to the blocked task. Copy diagnostics confirms once.

Layout and adverse content

Show task-facing recovery first, technical detail behind disclosure. Wrap exact material paths; redact secrets before display and before copy/export.

Starting point and required construction

Starting classifications: E, N, R. E existing operation; R runtime-backed adapter; N new contract; H host-only work. Exact per-control classifications remain in the catalog.

Reuse the reviewed domain operations and preserve all served fields. Build every missing native read/write adapter, capability gate and state renderer identified by the matching control/build rows.

BK10 owns readiness/discovery/diagnostics; BK08 fences stream and operation owners; BK05 reports actual isolation and policy.

Required interaction fixtures

Exercise first-read, empty/no-match, ready, refreshing/stale, error/refusal, action pending and acknowledged result. Also switch scope while the request is in flight.

Moved out of the interface

Every line of revision 7 plate copy that left the default view, and where it lives now.

"Faktor / Work (R7 window title)"
→ Native host chrome (VS Code view title, tabs or editor tab; JetBrains tool-window tabs, dialog title or Settings breadcrumb)

"Faktor is stopped. Opening this view does not start the service."
→ Status line "Faktor isn't running"; start rule in Specification: screen contract

"Workspace / auth-service / Open / /work/auth-service · current project"
→ VS Code view description and JetBrains project frame; path in Tooltip / hover help

"Machine-level path · never from project settings"
→ Help text in the Locate dialog (S35) and Specification: screen contract

"Use local inference or a configured provider"
→ Model setup pages (S03, S04)

"Choose the host-supported sandbox deliberately"
→ Shell isolation page (S57)

"No account or sibling product is required."
→ Product documentation › Setup Guide

"Start in your workspace"
→ Title "Set up Faktor"

"Before the first task"
→ Section "Before your first task"

"Choose a service executable / Locate"
→ Row "Service executable — Locate…"

"Connect a model / Configure"
→ Row "Model — Configure…"

"Review shell isolation / Review"
→ Row "Shell isolation — Review…"

"Your task draft stays here while you complete setup."
→ Note "Your draft stays here while you set up."

S34 · FIRST OPEN · CONTRACT, CONTINUED

“Setup guide / Start service”
→ Setup Guide link / Start Faktor (disabled with its reason)

S35 SETUP AND START

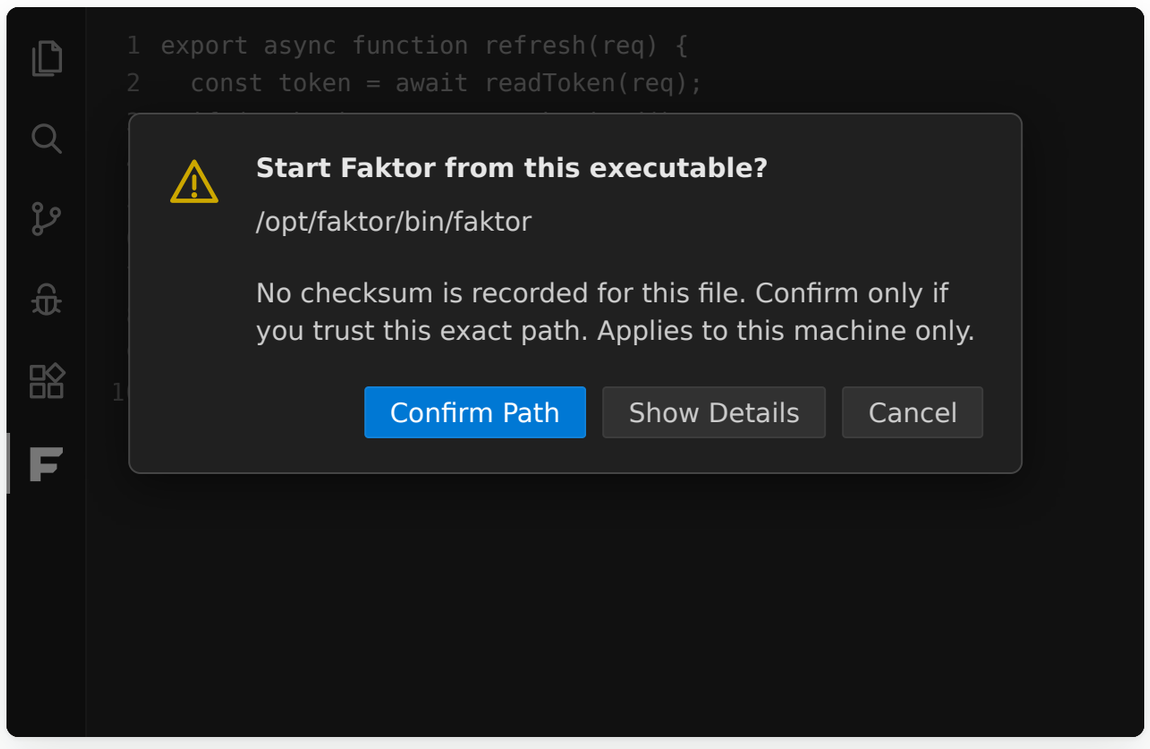
Service executable trust

Inspect the exact machine executable and its separate checksum and publisher trust basis before any start.

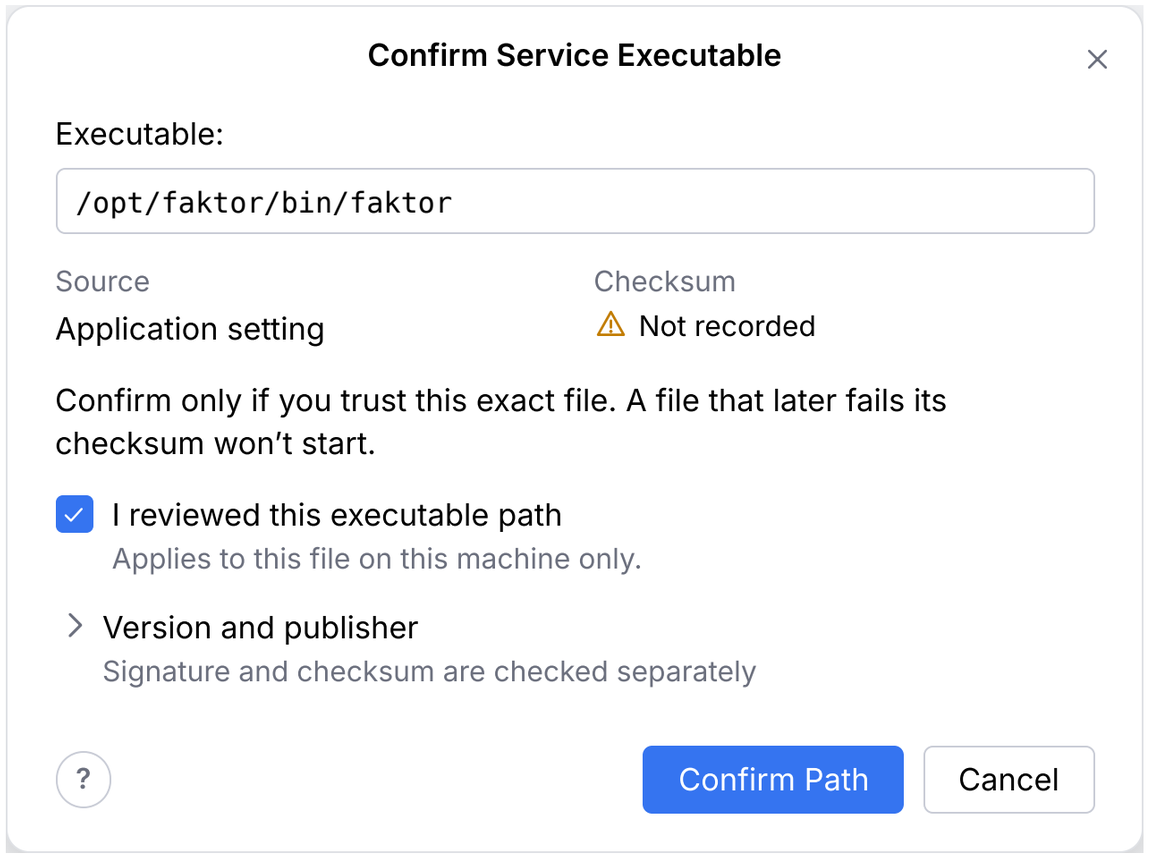
PRIMARY ACTION

Confirm Path

VS CODE · DARK MODERN — MODAL MESSAGE · NATIVE



JETBRAINS · LIGHT — DIALOGWRAPPER



CONTROLS O-04 O-05

REPLACES R7 page 007 · R4-S02

CONTRACT p. 26

S35 · SETUP AND START · BEHAVIOUR CONTRACT

Service executable trust: contract and revision notes

Original contract retained verbatim (R4-S02 / controls: O-04 O-05). Design on p. 25.

What changed in revision 8

- VS Code uses the native modal message (the extension API has no checkbox), so clicking Confirm Path is the explicit confirmation.
- JetBrains uses DialogWrapper with the checkbox; Confirm Path stays disabled until it is checked.
- Trust facts (source, checksum) stay visible; provenance detail moves behind a disclosure.

Entry and return

More / Diagnostics, the setup walkthrough, or the exact blocked operation. Work / Inspect / History remain the only top-level clusters. Settings, Usage and Diagnostics use explicit Back. Return to the originating object, not the first row.

Scope and captured state

Execution host + service instance/generation + protocol version/cursor + trust source; account bindings stay distinct. Starting, restarting and stopping the service are explicit owned operations. Rebind after recovery; opening a view is not a start command.

Primary action and recovery

Confirm path. Show a bounded human reason and actionable recovery. Missing checksum, checksum mismatch, signature status, schema refusal and local/cloud boundary are separate facts.

Loading, empty and stale

Keep last coherent data stale, retain event cursors and distinguish automatic transient reconnect from a protocol-blocked stop.

Keyboard and focus

A background disconnect never steals typing focus. Recovery opens in a native command/dialog; Back returns to the blocked task. Copy diagnostics confirms once.

Layout and adverse content

Show task-facing recovery first, technical detail behind disclosure. Wrap exact material paths; redact secrets before display and before copy/export.

Starting point and required construction

Starting classifications: E. E existing operation; R runtime-backed adapter; N new contract; H host-only work. Exact per-control classifications remain in the catalog.

Reuse the reviewed domain operations and preserve all served fields. Build every missing native read/write adapter, capability gate and state renderer identified by the matching control/build rows.

BK10 owns readiness/discovery/diagnostics; BK08 fences stream and operation owners; BK05 reports actual isolation and policy.

Required interaction fixtures

Exercise first-read, empty/no-match, ready, refreshing/stale, error/refusal, action pending and acknowledged result. Also switch scope while the request is in flight.

Moved out of the interface

Every line of revision 7 plate copy that left the default view, and where it lives now.

"Faktor / Diagnostics / Runtime (R7 window title)"
→ Native host chrome (VS Code view title, tabs or editor tab; JetBrains tool-window tabs, dialog title or Settings breadcrumb)

"Application setting · dev-box · exact selected path"
→ Source field; host name in Tooltip / hover help

"A checksum mismatch is a refusal, with no override in this dialog."
→ Refusal state R4-T04 and Specification: screen contract

"Publisher signature and recorded checksum are separate checks. Review version and provenance before starting."
→ Version and publisher disclosure

"This does not trust project-supplied binaries or change future machine policy."
→ Checkbox help "Applies to this file on this machine only"; full rule in Specification: screen contract

"Review the service executable"
→ Dialog title "Confirm Service Executable"

"Executable to start"
→ Field "Executable"

"Source / Checksum record / Unavailable"
→ Facts "Source" and "Checksum — Not recorded"

"This executable has no recorded checksum"
→ Warning text / VS Code modal detail

"Confirm only this exact path."
→ Body text "Confirm only if you trust this exact file."

"Cancel / Confirm path"
→ Cancel / Confirm Path

S57 SETUP AND START

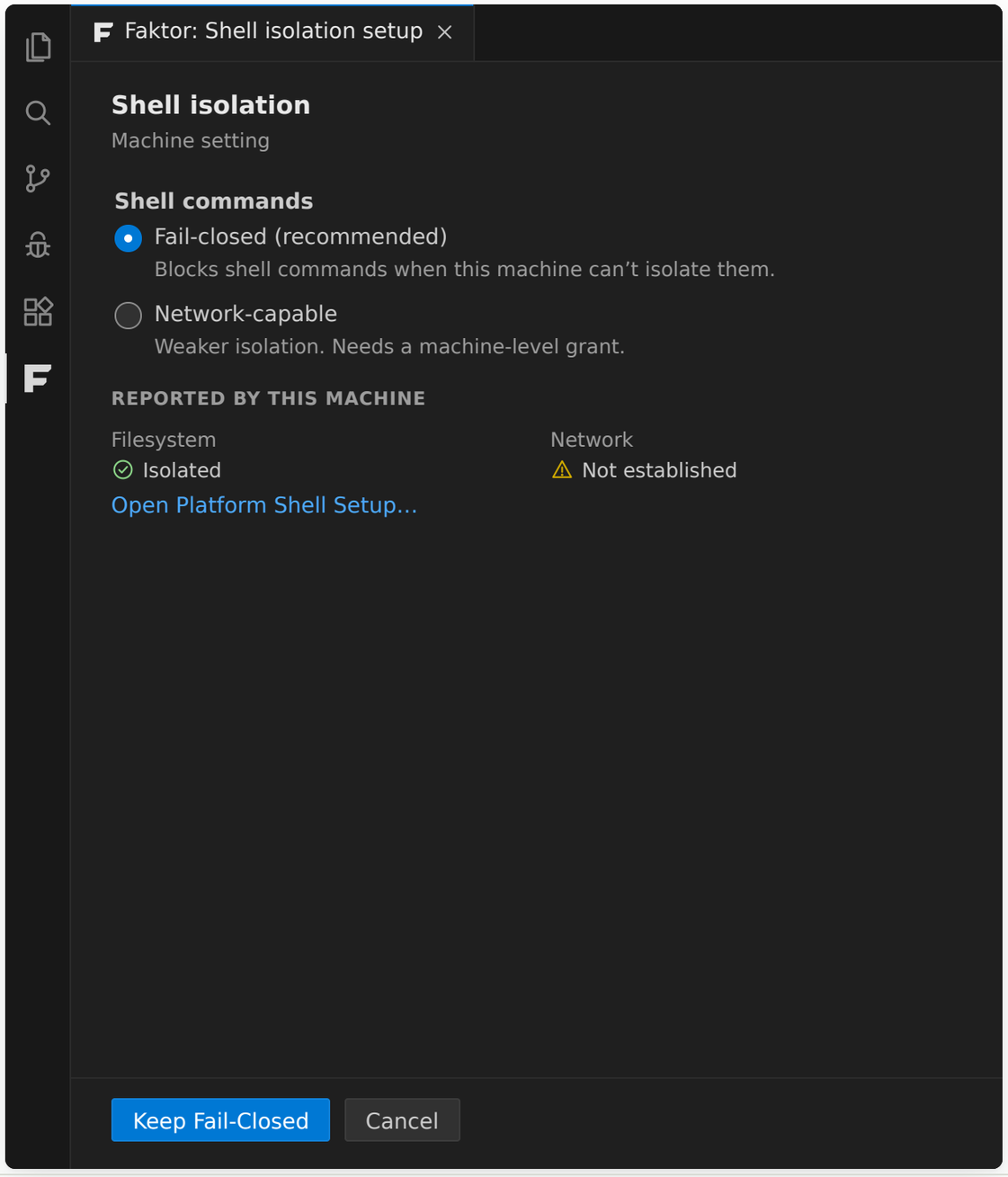
Shell isolation setup

Choose supported shell isolation explicitly and show platform-established filesystem and network guarantees.

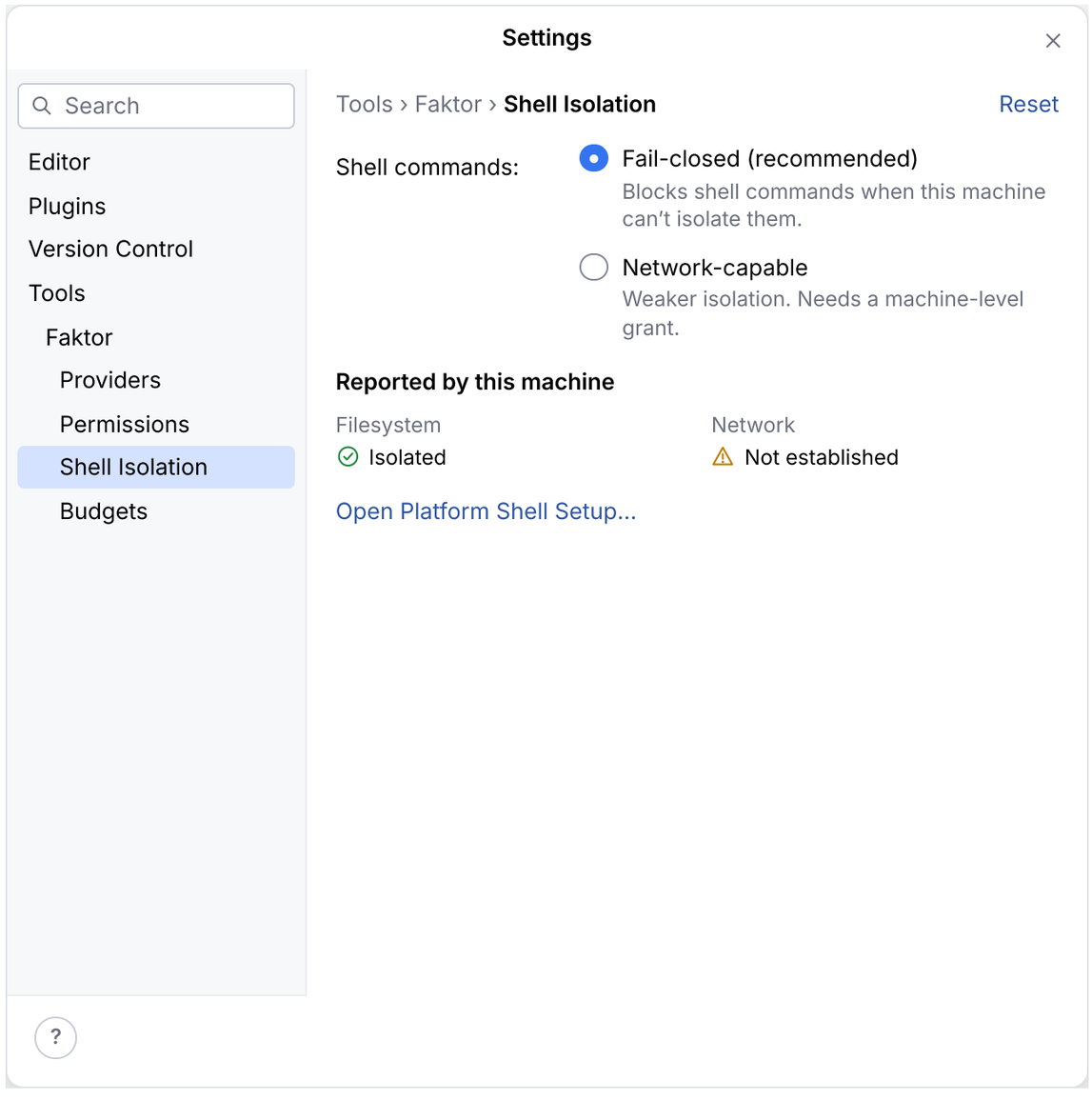
PRIMARY ACTION

Keep Fail-Closed · JetBrains: OK / Apply

VS CODE · DARK MODERN — EDITOR TAB · WEBVIEW SETTINGS



JETBRAINS · LIGHT — SETTINGS DIALOG · CONFIGURABLE



CONTROLS Q-10 Q-11

REPLACES R7 page 008 · R4-S34

CONTRACT p. 28

S57 · SETUP AND START · BEHAVIOUR CONTRACT

Shell isolation setup: contract and revision notes

Original contract retained verbatim (R4-S34 / controls: Q-10 Q-11). Design on p. 27.

What changed in revision 8

- Persistent machine configuration moves to JetBrains Settings (Configurable) and a VS Code settings editor.
- A radio group makes the exclusive choice explicit; the reported guarantees are read-only facts with icon and text.

Entry and return

More / Diagnostics, the setup walkthrough, or the exact blocked operation. Work / Inspect / History remain the only top-level clusters. Settings, Usage and Diagnostics use explicit Back. Return to the originating object, not the first row.

Scope and captured state

Execution host + service instance/generation + protocol version/cursor + trust source; account bindings stay distinct. Starting, restarting and stopping the service are explicit owned operations. Rebind after recovery; opening a view is not a start command.

Primary action and recovery

Open native shell setup. Show a bounded human reason and actionable recovery. Missing checksum, checksum mismatch, signature status, schema refusal and local/cloud boundary are separate facts.

Loading, empty and stale

Keep last coherent data stale, retain event cursors and distinguish automatic transient reconnect from a protocol-blocked stop.

Keyboard and focus

A background disconnect never steals typing focus. Recovery opens in a native command/dialog; Back returns to the blocked task. Copy diagnostics confirms once.

Layout and adverse content

Show task-facing recovery first, technical detail behind disclosure. Wrap exact material paths; redact secrets before display and before copy/export.

Starting point and required construction

Starting classifications: R. E existing operation; R runtime-backed adapter; N new contract; H host-only work. Exact per-control classifications remain in the catalog.

Existing CLI shell setup is real; add native invocation/status adapters. A new policy editor needs a supported write capability.

BK10 owns readiness/discovery/diagnostics; BK08 fences stream and operation owners; BK05 reports actual isolation and policy.

Required interaction fixtures

Exercise first-read, empty/no-match, ready, refreshing/stale, error/refusal, action pending and acknowledged result. Also switch scope while the request is in flight.

Moved out of the interface

Every line of revision 7 plate copy that left the default view, and where it lives now.

"Faktor / Setup / Shell (R7 window title)"
→ Native host chrome (VS Code view title, tabs or editor tab; JetBrains tool-window tabs, dialog title or Settings breadcrumb)

"Explicit platform setup · machine scope"
→ Header "Machine setting"; JetBrains Settings location

"Default"
→ Label "(recommended)"

"A label is not an isolation guarantee"
→ Specification: screen contract (rendering rule)

"A checkbox cannot create network isolation. Reuse the real first-run setup and report the platform result."
→ Specification: screen contract — implementation instruction; the UI shows only platform-reported results

"Platform-reported"
→ Section title "Reported by this machine"

"Choose shell isolation"
→ Title "Shell isolation"

"Fail-closed shell / Deny shell work when the platform cannot provide required isolation."
→ Option "Fail-closed (recommended)" with help

"Network-capable shell / Weaker isolation. Requires an explicit supported machine grant."
→ Option "Network-capable" with help

"This host / Filesystem boundary / Network boundary / Not yet established"
→ Section "Reported by this machine": Filesystem, Network — Not established

"Keep fail-closed / Open native shell setup"
→ Keep Fail-Closed / Open Platform Shell Setup…

S36 SETUP AND START

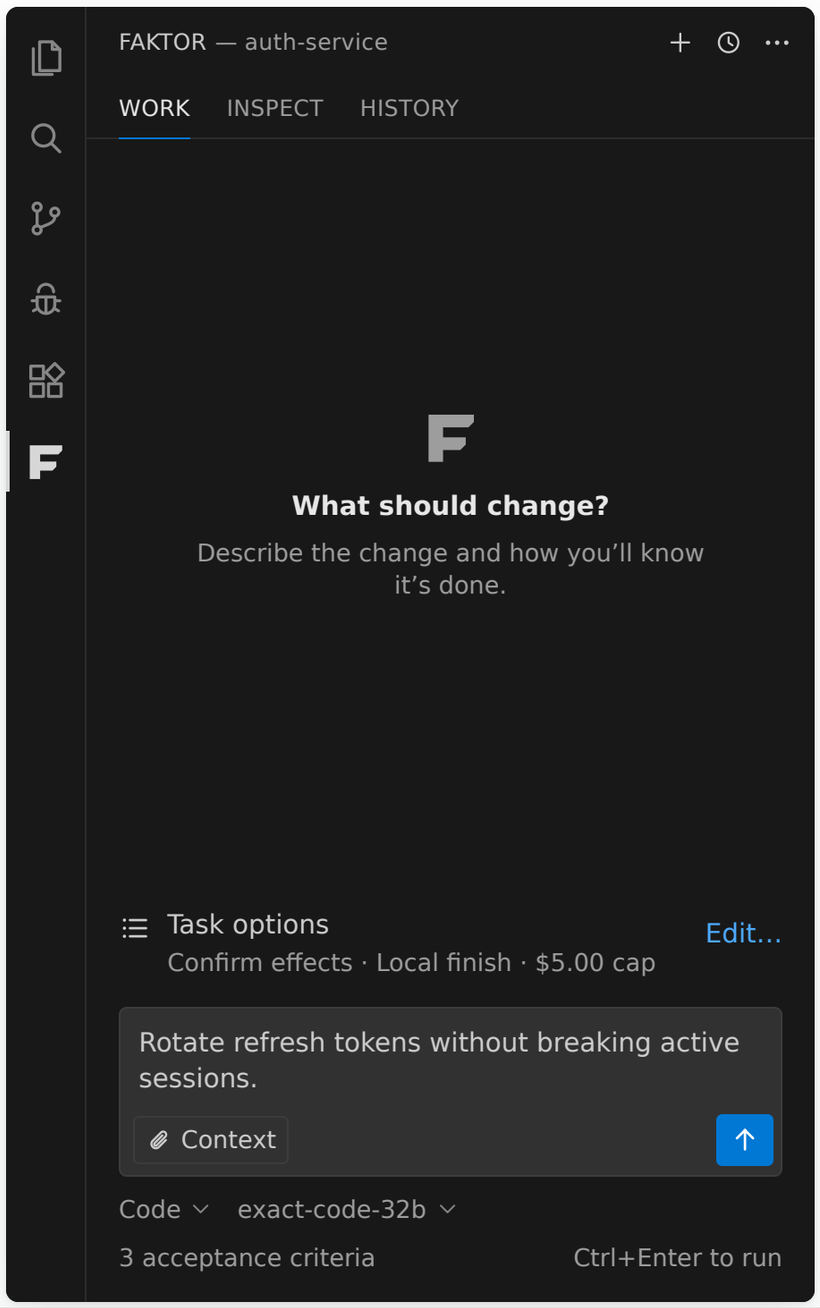
Ready conversation

Write one task, choose an exact model and intent, and review the captured authority, criteria, finish and run caps.

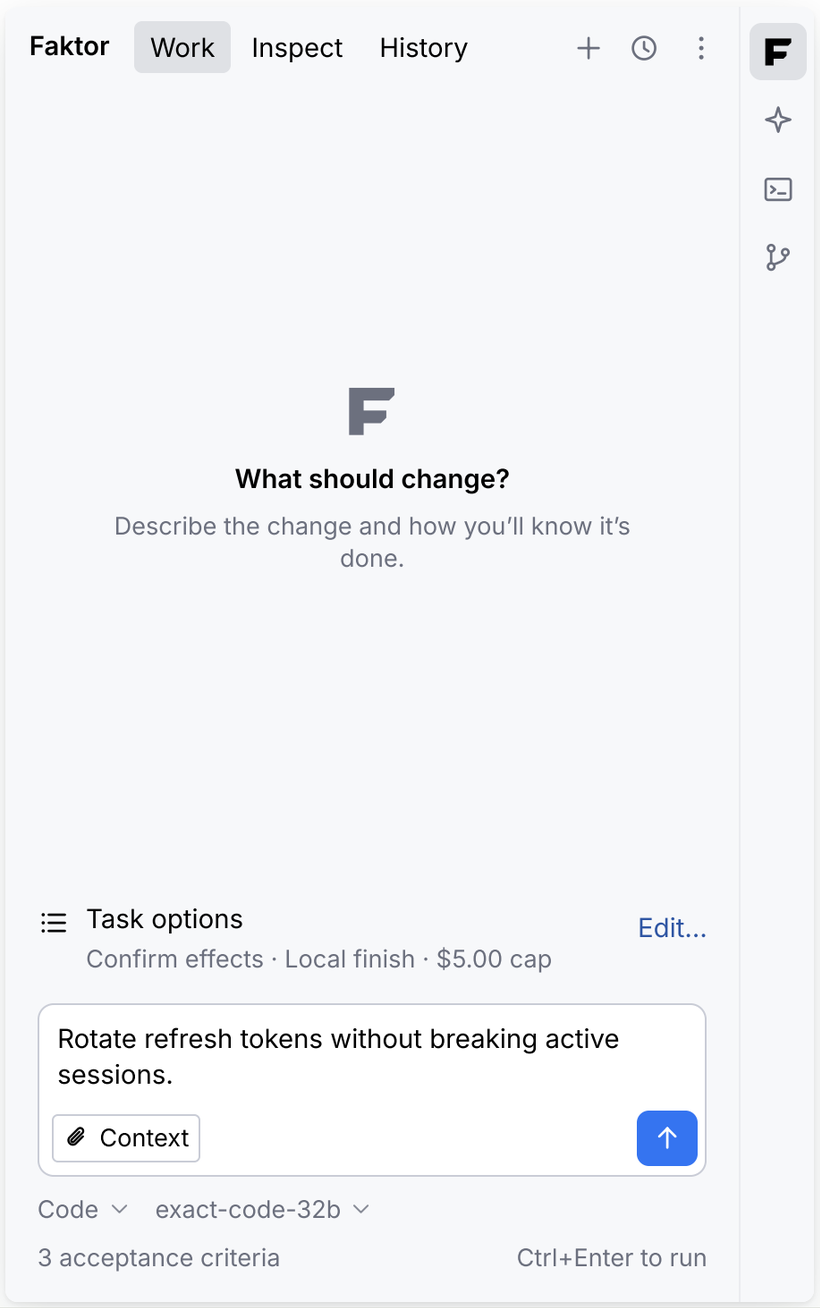
PRIMARY ACTION

Run Task (composer)

VS CODE · DARK MODERN — SIDEBAR VIEW · WEBVIEW



JETBRAINS · LIGHT — TOOL WINDOW · SWING / KOTLIN UI DSL



CONTROLS A-01 M-02 Q-01 V-07 V-10 B-01 B-02

REPLACES R7 page 009 · R4-S03

CONTRACT p. 30

S36 · SETUP AND START · BEHAVIOUR CONTRACT

Ready conversation: contract and revision notes

Original contract retained verbatim (R4-S03 / controls: A-01 M-02 Q-01 V-07 V-10 B-01 B-02). Design on p. 29.

What changed in revision 8

- The composer is the single place to start work; pickers live in its toolbar instead of a separate form.
- Captured authority, finish and cap are one quiet summary line with an Edit action.
- Run Task is the send button; Ctrl/Cmd+Enter is shown once, Enter stays newline.

Entry and return

Work, a captured submission receipt, or a task-specific attention item. Work / Inspect / History remain the only top-level clusters. Settings, Usage and Diagnostics use explicit Back. Return to the originating object, not the first row.

Scope and captured state

Daemon/workspace/worktree/conversation/task/request generation. Draft scope is separate from an admitted run. Capture one immutable logical start. Clear only its acknowledged draft version; preserve text typed after capture. Queue items acquire their own receipt and revision.

Primary action and recovery

Run task. Retain the captured body, options and source handles. Unknown admission/effect requires receipt reconciliation before replay. A changed body needs a different request identity.

Loading, empty and stale

Show the named operation and current owner. Keep the composer available for an independent draft. No fake percentages, fabricated tool steps or optimistic completion.

Keyboard and focus

Enter inserts a newline by default. The host send command respects IME composition. New approvals announce once without taking focus. Escape closes a detail; it never stops a run.

Layout and adverse content

Header, transcript scroll and composer are siblings. Collapse secondary metadata first. Keep a scoped Stop action and the active task reachable at short heights.

Starting point and required construction

Starting classifications: E, N. E existing operation; R runtime-backed adapter; N new contract; H host-only work. Exact per-control classifications remain in the catalog.

Reuse the reviewed domain operations and preserve all served fields. Build every missing native read/write adapter, capability gate and state renderer identified by the matching control/build rows.

BK08 owns admission, receipts, queue, control handles and reconciliation; BK05 owns authority and BK07 owns a shared exact budget ledger.

Required interaction fixtures

Exercise first-read, empty/no-match, ready, refreshing/stale, error/refusal, action pending and acknowledged result. Also switch scope while the request is in flight.

Moved out of the interface

Every line of revision 7 plate copy that left the default view, and where it lives now.

"Faktor / Work (R7 window title)"
→ Native host chrome (VS Code view title, tabs or editor tab; JetBrains tool-window tabs, dialog title or Settings breadcrumb)

"auth-service · isolated candidate · Local finish"
→ View description (workspace) and Task options summary (finish)

"Define the change. Keep its acceptance criteria and constraints with the task."
→ Empty-state hint "Describe the change and how you'll know it's done."

"Agent / Code; Model / exact-code-32b"
→ Composer pickers

"Authority / Confirm effects; Run cap / USD 5.00 / 120k tokens"
→ Task options summary; token cap inside Task options (S33)

"Task options / Review / 3 required criteria · Local finish"
→ Task options row and composer hint

"Add context / Run task"
→ Composer context chip and send button (Run Task)

"Run task"
→ Send button "Run Task"

S33 SETUP AND START

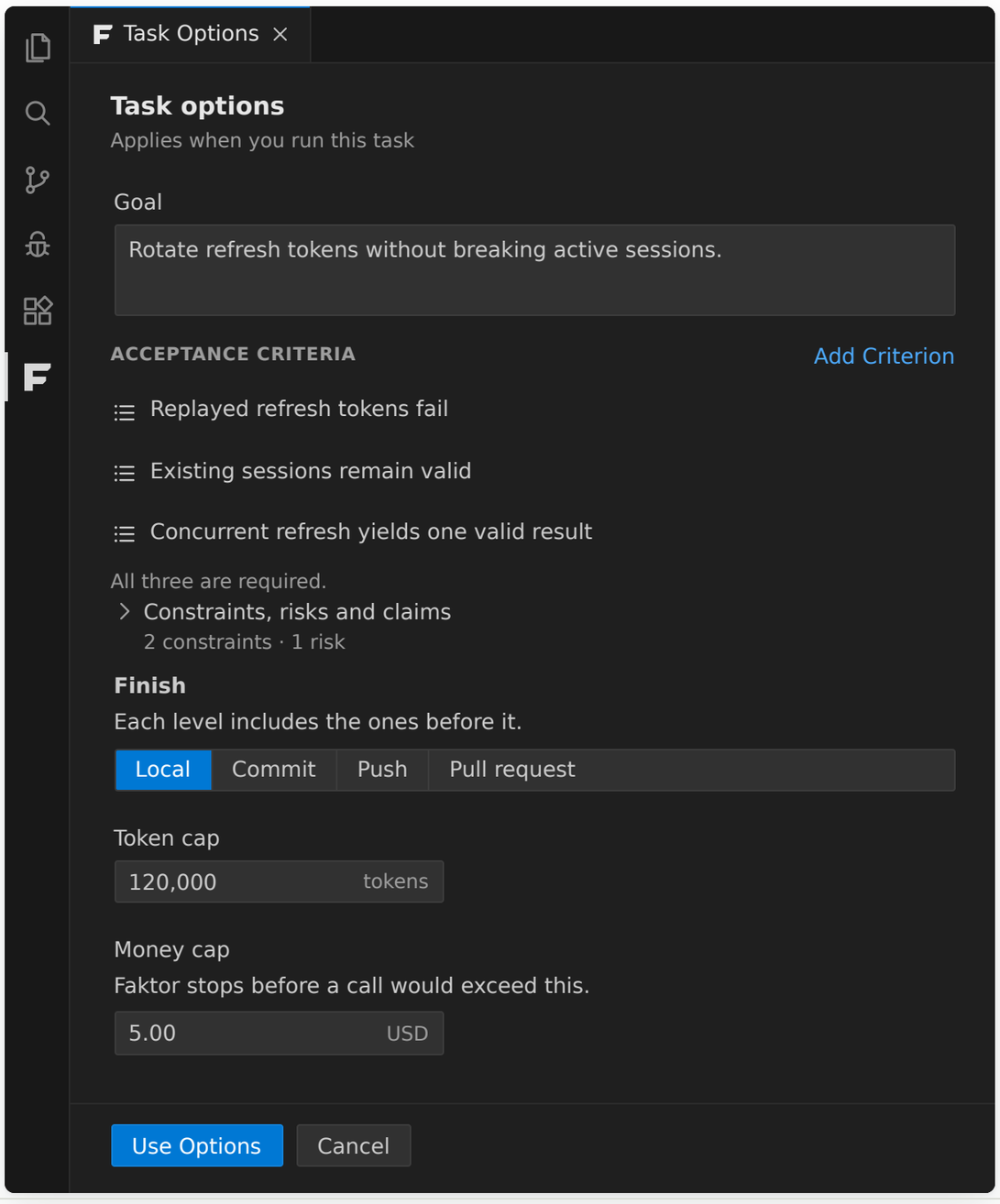
PRIMARY ACTION

Use Options

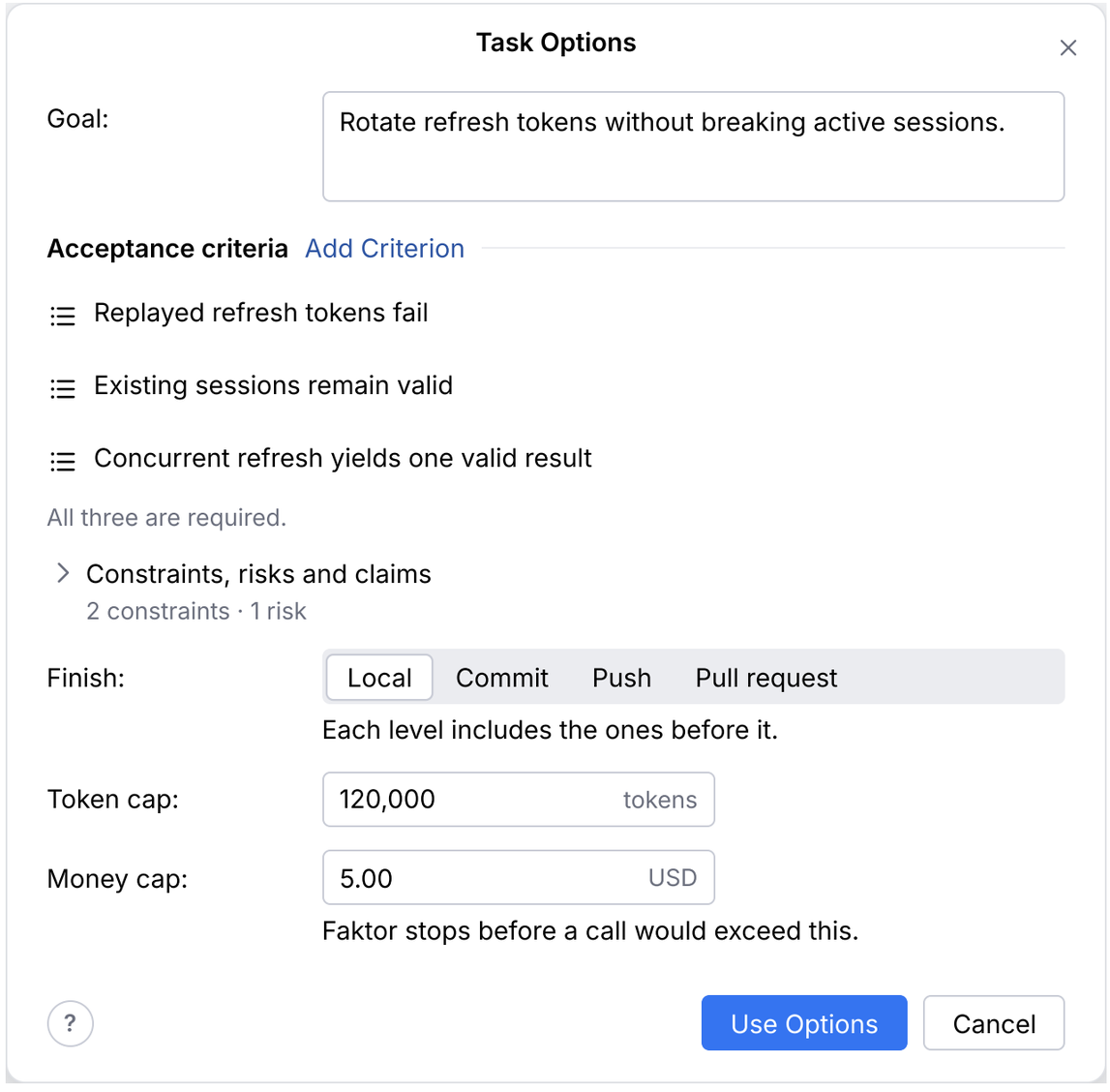
Task options

Edit the goal, required criteria, completion contract, finish level and exact run caps in one draft sheet. The active task keeps its captured options.

VS CODE · DARK MODERN — EDITOR TAB · WEBVIEW



JETBRAINS · LIGHT — DIALOGWRAPPER



CONTROLS V-07 V-10 B-01 B-02

REPLACES R7 page 010 · R4-S04

CONTRACT p. 32

S33 · SETUP AND START · BEHAVIOUR CONTRACT

Task options: contract and revision notes

Original contract retained verbatim (R4-S04 / controls: V-07 V-10 B-01 B-02). Design on p. 31.

What changed in revision 8

- JetBrains opens a DialogWrapper; VS Code opens an editor tab so the composer stays reachable.
- Money cap explains the consequence instead of the storage unit.

Entry and return

Work, a captured submission receipt, or a task-specific attention item. Work / Inspect / History remain the only top-level clusters. Settings, Usage and Diagnostics use explicit Back. Return to the originating object, not the first row.

Scope and captured state

Daemon/workspace/worktree/conversation/task/request generation. Draft scope is separate from an admitted run. Capture one immutable logical start. Clear only its acknowledged draft version; preserve text typed after capture. Queue items acquire their own receipt and revision.

Primary action and recovery

Use options returns the validated values to the composer. Run task later captures one immutable submission body and the requested finish hierarchy.. Invalid criteria or caps preserve the staged draft and block Use options. An Ask-only prompt does not silently acquire a task contract or finish action.

Loading, empty and stale

Show the named operation and current owner. Keep the composer available for an independent draft. No fake percentages, fabricated tool steps or optimistic completion.

Keyboard and focus

Enter inserts a newline by default. The host send command respects IME composition. New approvals announce once without taking focus. Escape closes a detail; it never stops a run.

Layout and adverse content

Header, transcript scroll and composer are siblings. Collapse secondary metadata first. Keep a scoped Stop action and the active task reachable at short heights.

Starting point and required construction

Starting classifications: E. E existing operation; R runtime-backed adapter; N new contract; H host-only work. Exact per-control classifications remain in the catalog.

Reuse the reviewed domain operations and preserve all served fields. Build every missing native read/write adapter, capability gate and state renderer identified by the matching control/build rows.

BK08 owns admission, receipts, queue, control handles and reconciliation; BK05 owns authority and BK07 owns a shared exact budget ledger.

Required interaction fixtures

Exercise first-read, empty/no-match, ready, refreshing/stale, error/refusal, action pending and acknowledged result. Also switch scope while the request is in flight.

Moved out of the interface

Every line of revision 7 plate copy that left the default view, and where it lives now.

"Faktor / Work / Task options (R7 window title)"
→ Native host chrome (VS Code view title, tabs or editor tab; JetBrains tool-window tabs, dialog title or Settings breadcrumb)

"New task draft · captured only when Run task is admitted"
→ Subtitle "Applies when you run this task"; capture rule in Specification: screen contract

"AC-01 · required / AC-02 · required / AC-03 · required"
→ "Required" status; criterion IDs in Tooltip / hover help and the Inspector

"Edit list / Add"
→ Add Criterion; editing is in-place on each row

"Completion contract / Constraints, invariants, risks and claims"
→ Disclosure "Constraints, risks and claims"

"Finish level / Local / Commit / Push / PR"
→ Finish segmented control with help "Each level includes the ones before it."

"Run token cap / Run money cap"
→ Token cap and Money cap fields with consequence help

"120000 tokens / 5.00 USD"
→ Formatted as 120,000 tokens / 5.00 USD

"Cancel / Use options"
→ Cancel / Use Options

S37 SETUP AND START

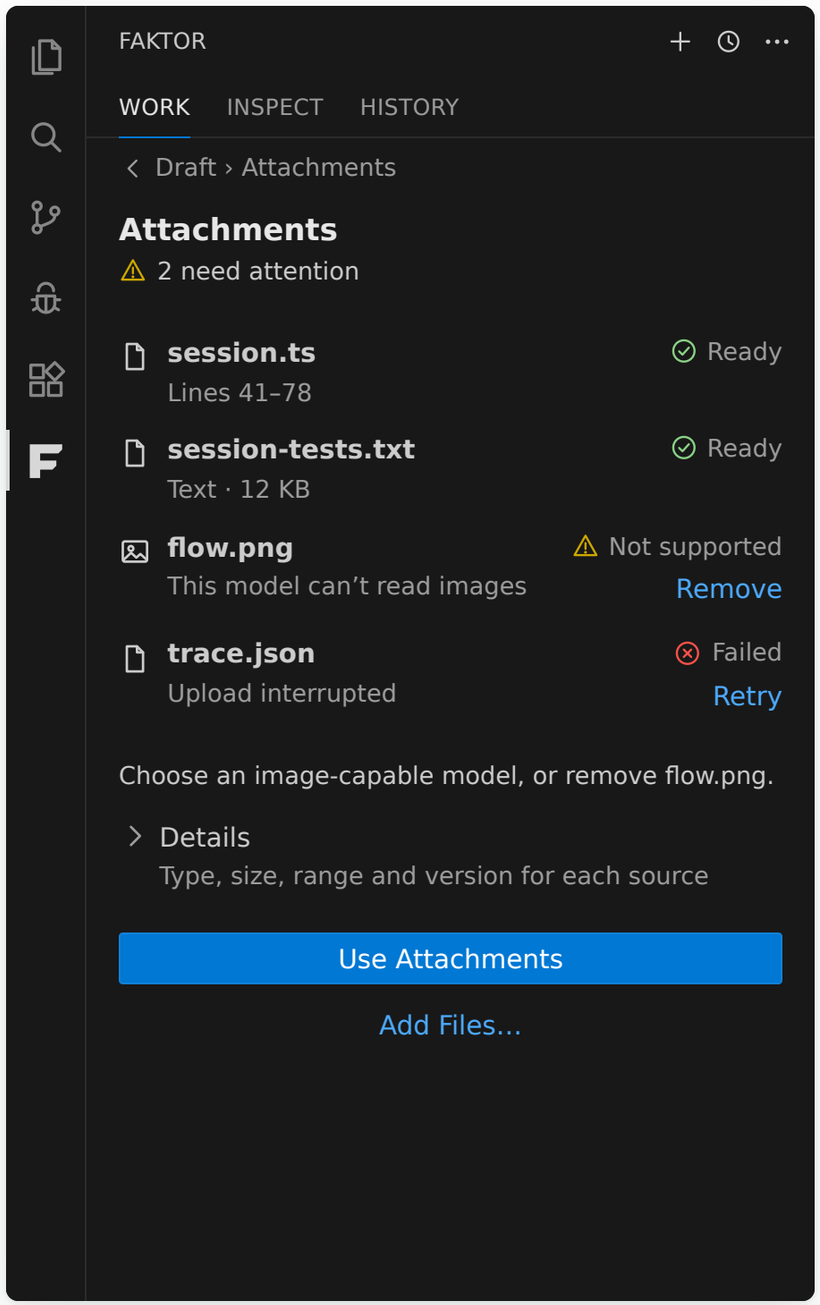
Attachment admission

Resolve file, image and upload admission per source; failed or unsupported sources remain visible in the draft.

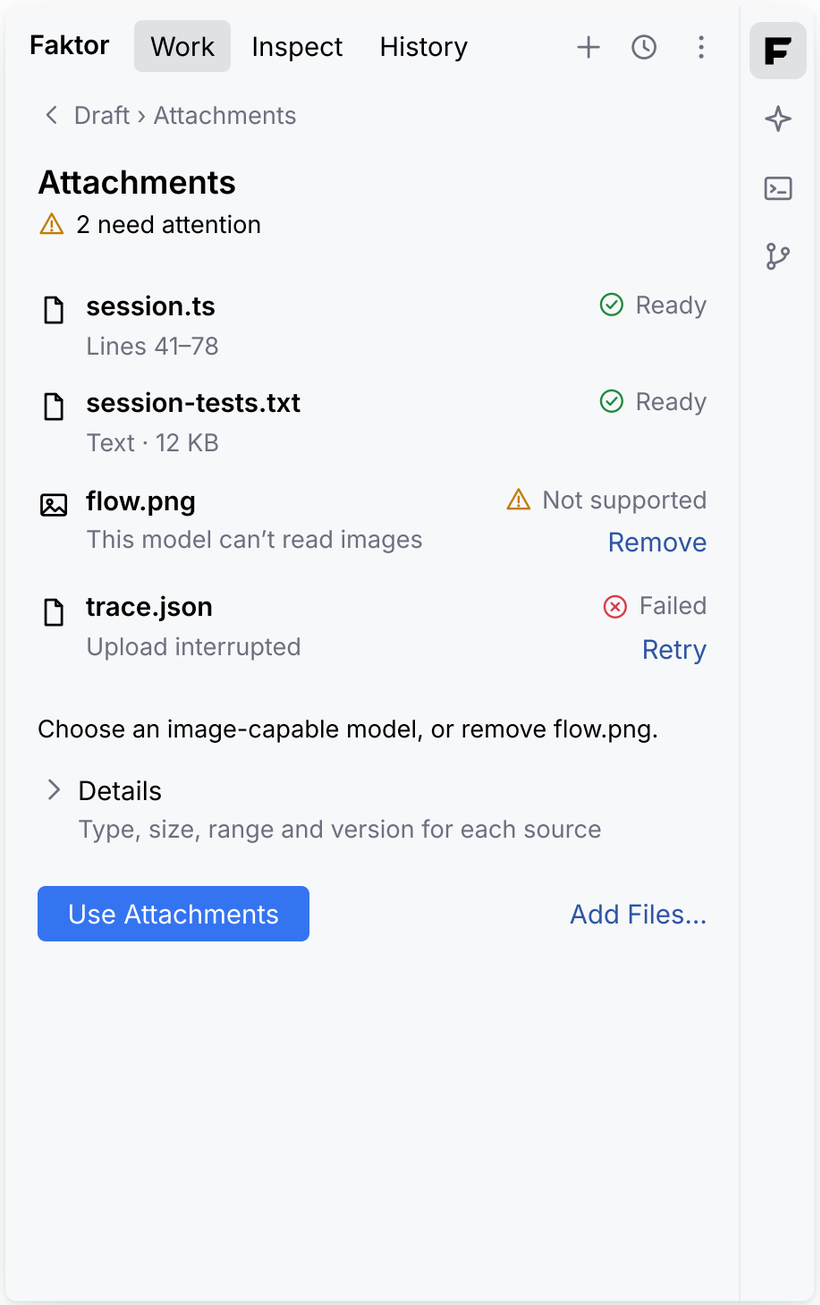
PRIMARY ACTION

Use Attachments

VS CODE · DARK MODERN — SIDEBAR VIEW · WEBVIEW



JETBRAINS · LIGHT — TOOL WINDOW · SWING / KOTLIN UI DSL



CONTROLS C-03 C-05 C-06

REPLACES R7 page 011 · R4-S05

CONTRACT p. 34

S37 · SETUP AND START · BEHAVIOUR CONTRACT

Attachment admission: contract and revision notes

Original contract retained verbatim (R4-S05 / controls: C-03 C-05 C-06). Design on p. 33.

What changed in revision 8

- Each problem is fixed on its own row, so the global banner is gone.
- Statuses use icon and text: Ready, Not supported, Failed.

Entry and return

Composer context picker/meter, attachment staging, memory or compaction command. Work / Inspect / History remain the only top-level clusters. Settings, Usage and Diagnostics use explicit Back. Return to the originating object, not the first row.

Scope and captured state

Draft/call receipt + exact source URI/range/version + token accounting basis; memory has origin and retention scope. Explicit inclusion acquires a versioned source snapshot. Do not replace an admitted source with the current editor or silently drop an unsupported attachment.

Primary action and recovery

Use attachments. Show unsupported type, oversize, upload error, capacity refusal and stale source separately. Remove only explicitly; failed compaction keeps the prior valid accounting basis.

Loading, empty and stale

Measured input, draft estimate, model capacity and output/safety reserve use separate labels. null contextUsage is unavailable. Upload and compaction each retain original inputs.

Keyboard and focus

Picker arrows select sources; Enter stages the selected identity; Escape returns to the same composer caret. New usage observations do not interrupt typing.

Layout and adverse content

Wrap source labels and exact rejection reasons. Keep previews bounded with full artifact/copy actions. Do not fit a long source list between several competing scroll panes.

Starting point and required construction

Starting classifications: E, H. E existing operation; R runtime-backed adapter; N new contract; H host-only work. Exact per-control classifications remain in the catalog.

Reuse the reviewed domain operations and preserve all served fields. Build every missing native read/write adapter, capability gate and state renderer identified by the matching control/build rows.

BK04 owns accounting and compaction; BK03 owns versioned memory/customization; BK02 supplies model capabilities and BK08 owns upload handles.

Required interaction fixtures

Exercise first-read, empty/no-match, ready, refreshing/stale, error/refusal, action pending and acknowledged result. Also switch scope while the request is in flight.

Moved out of the interface

Every line of revision 7 plate copy that left the default view, and where it lives now.

"Faktor / Work / Context (R7 window title)"	→ Native host chrome (VS Code view title, tabs or editor tab; JetBrains tool-window tabs, dialog title or Settings breadcrumb)
"The next task receives only admitted, explicitly included sources."	→ Product documentation and Specification: screen contract
"Selected lines 41-78 · editor version 27"	→ "Lines 41–78"; editor version in Details
"Text · 12 KiB · local upload handle"	→ "Text · 12 KB"; upload handle in Details
"Image · 84 KiB · kept in the draft"	→ Row stays visible with its fix; size in Details
"Upload was interrupted. Original selection is retained."	→ Row meta "Upload interrupted" + Retry; retention rule in Specification: screen contract
"Resolve two attachments before sending"	→ Header status "2 need attention"
"Retry trace.json or remove it explicitly."	→ Per-row Retry action
"Included source details / Inspect / MIME, bytes, source range, version and omission reason"	→ Details disclosure
"Choose files"	→ Add Files…
"Review attachments"	→ Title "Attachments"
"Choose an image-capable model or remove flow.png."	→ Help line under the list
"Use attachments"	→ Use Attachments

S38 SETUP AND START

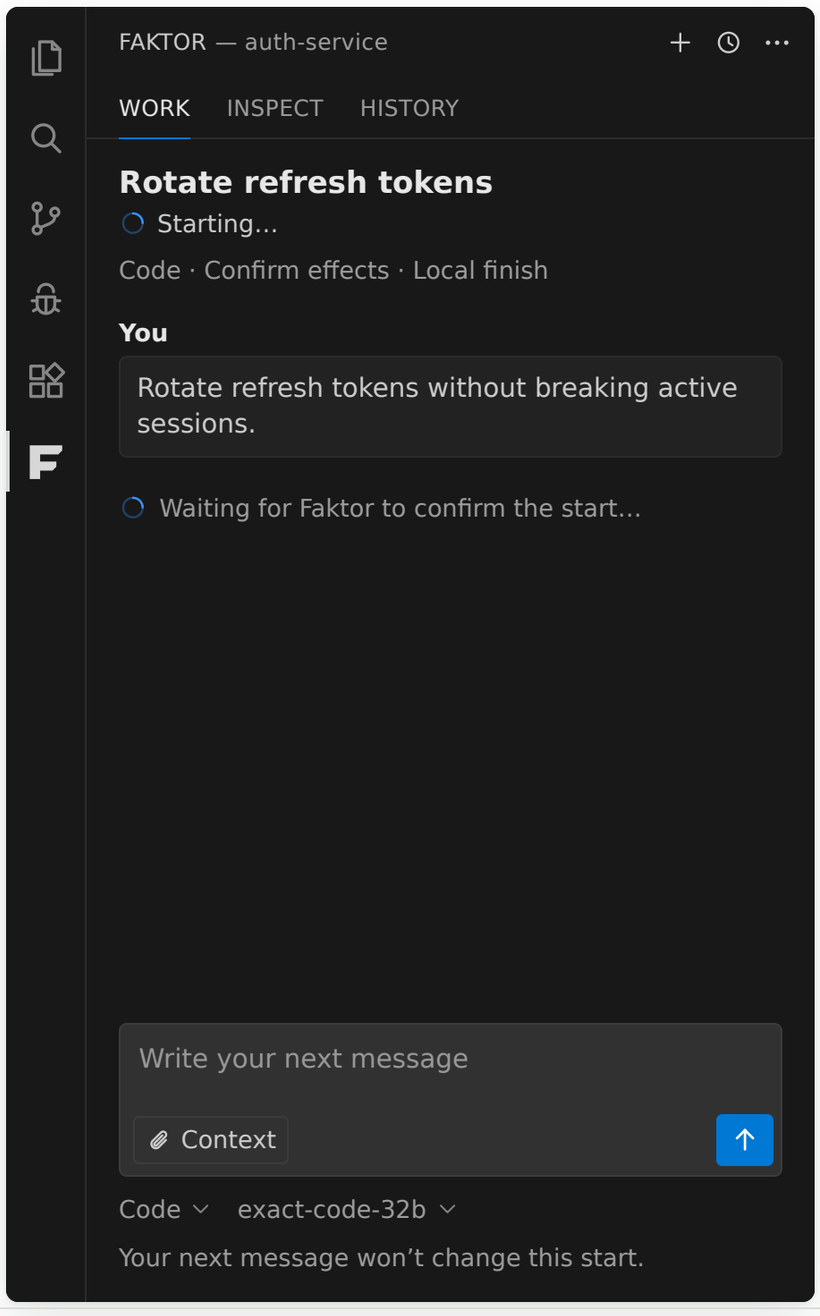
Task admission pending

Show one captured logical submission while the admission receipt is pending; suppress duplicate starts.

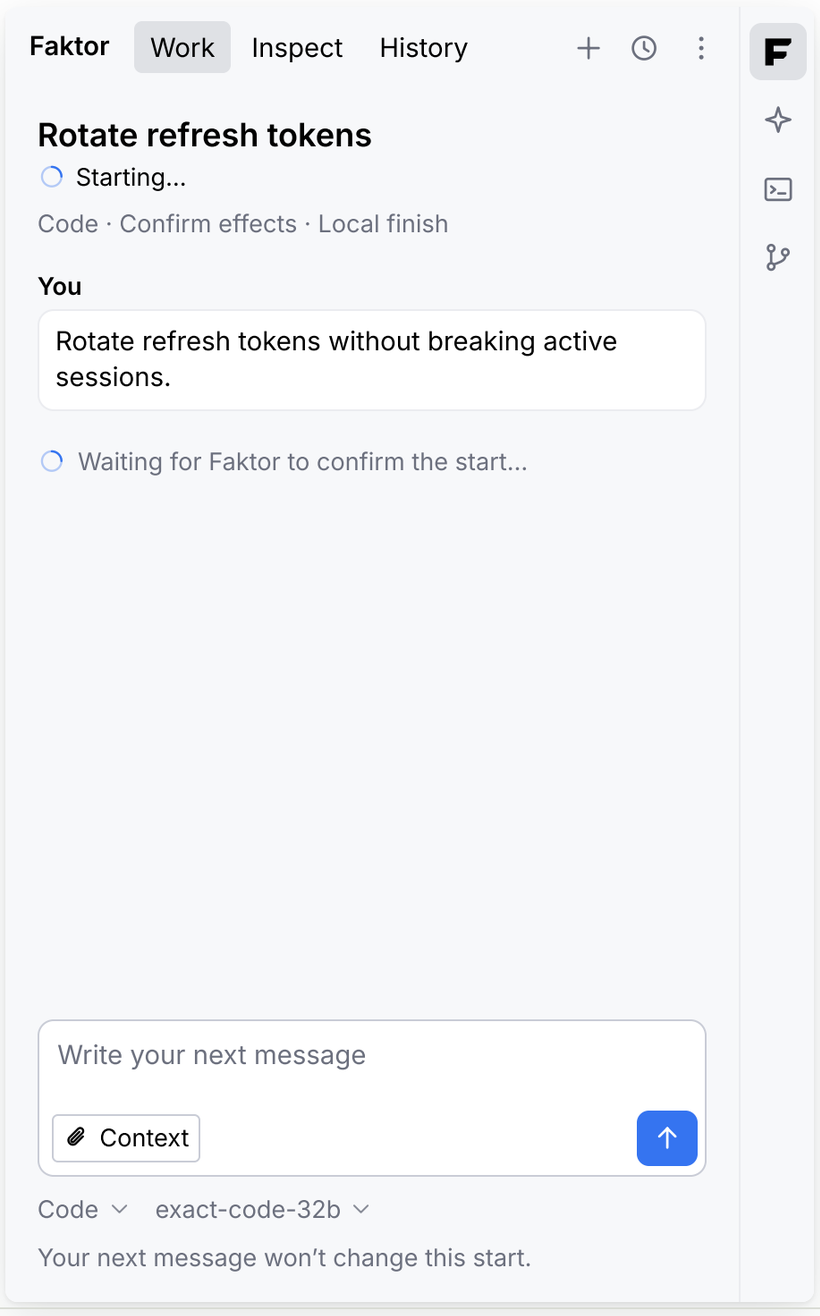
PRIMARY ACTION

Queue Message (composer)

VS CODE · DARK MODERN — SIDEBAR VIEW · WEBVIEW



JETBRAINS · LIGHT — TOOL WINDOW · SWING / KOTLIN UI DSL



CONTROLS D-05 V-07

REPLACES R7 page 012 · R4-S06

CONTRACT p. 36

S38 · SETUP AND START · BEHAVIOUR CONTRACT

Task admission pending: contract and revision notes

Original contract retained verbatim (R4-S06 / controls: D-05 V-07). Design on p. 35.

What changed in revision 8

- Pending start is a header state plus one status line instead of three cards.
- The composer stays usable for the next message, which is the only useful action.

Entry and return

Work, a captured submission receipt, or a task-specific attention item. Work / Inspect / History remain the only top-level clusters. Settings, Usage and Diagnostics use explicit Back. Return to the originating object, not the first row.

Scope and captured state

Daemon/workspace/worktree/conversation/task/request generation. Draft scope is separate from an admitted run. Capture one immutable logical start. Clear only its acknowledged draft version; preserve text typed after capture. Queue items acquire their own receipt and revision.

Primary action and recovery

Starting task…. Retain the captured body, options and source handles. Unknown admission/effect requires receipt reconciliation before replay. A changed body needs a different request identity.

Loading, empty and stale

Show the named operation and current owner. Keep the composer available for an independent draft. No fake percentages, fabricated tool steps or optimistic completion.

Keyboard and focus

Enter inserts a newline by default. The host send command respects IME composition. New approvals announce once without taking focus. Escape closes a detail; it never stops a run.

Layout and adverse content

Header, transcript scroll and composer are siblings. Collapse secondary metadata first. Keep a scoped Stop action and the active task reachable at short heights.

Starting point and required construction

Starting classifications: E, R. E existing operation; R runtime-backed adapter; N new contract; H host-only work. Exact per-control classifications remain in the catalog.

Reuse the reviewed domain operations and preserve all served fields. Build every missing native read/write adapter, capability gate and state renderer identified by the matching control/build rows.

BK08 owns admission, receipts, queue, control handles and reconciliation; BK05 owns authority and BK07 owns a shared exact budget ledger.

Required interaction fixtures

Exercise first-read, empty/no-match, ready, refreshing/stale, error/refusal, action pending and acknowledged result. Also switch scope while the request is in flight.

Moved out of the interface

Every line of revision 7 plate copy that left the default view, and where it lives now.

"Faktor / Work (R7 window title)"
→ Native host chrome (VS Code view title, tabs or editor tab; JetBrains tool-window tabs, dialog title or Settings breadcrumb)

"Submission start-047 · captured once · response pending"
→ Inspector › Activity (request ID)

"The task and its options are captured. Repeated clicks do not create another run."
→ Run button is replaced by the Starting status; duplicate-start rule in Specification: screen contract

"Captured / 3 required criteria · Local finish"
→ Task header line and Task options

"Model / exact-code-32b; Run cap / USD 5.00 / 120k tokens"
→ Composer picker and Task details

"While this start is pending / Draft for the next task / You can write independently. It does not change this start."
→ Composer hint

"Admission receipt / Pending / Waiting for the service. No second start has been sent."
→ Status line "Waiting for Faktor to confirm the start…"

"Review captured task / Starting task…"
→ Task header (Starting…); captured task is the first message

"Starting the task"
→ Header status "Starting…"

S39 SETUP AND START

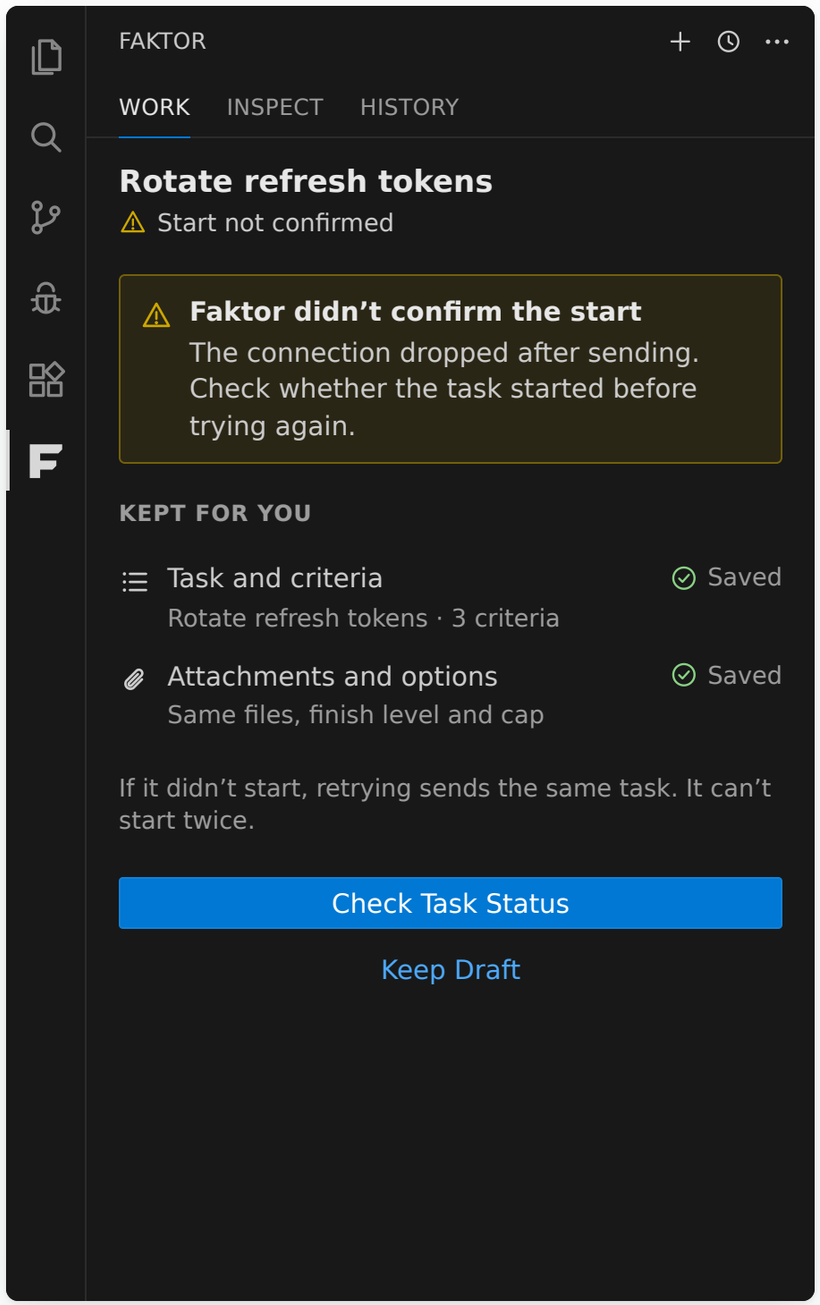
Ambiguous task start

Reconcile an ambiguous admission against the original request identity and body before any permitted retry.

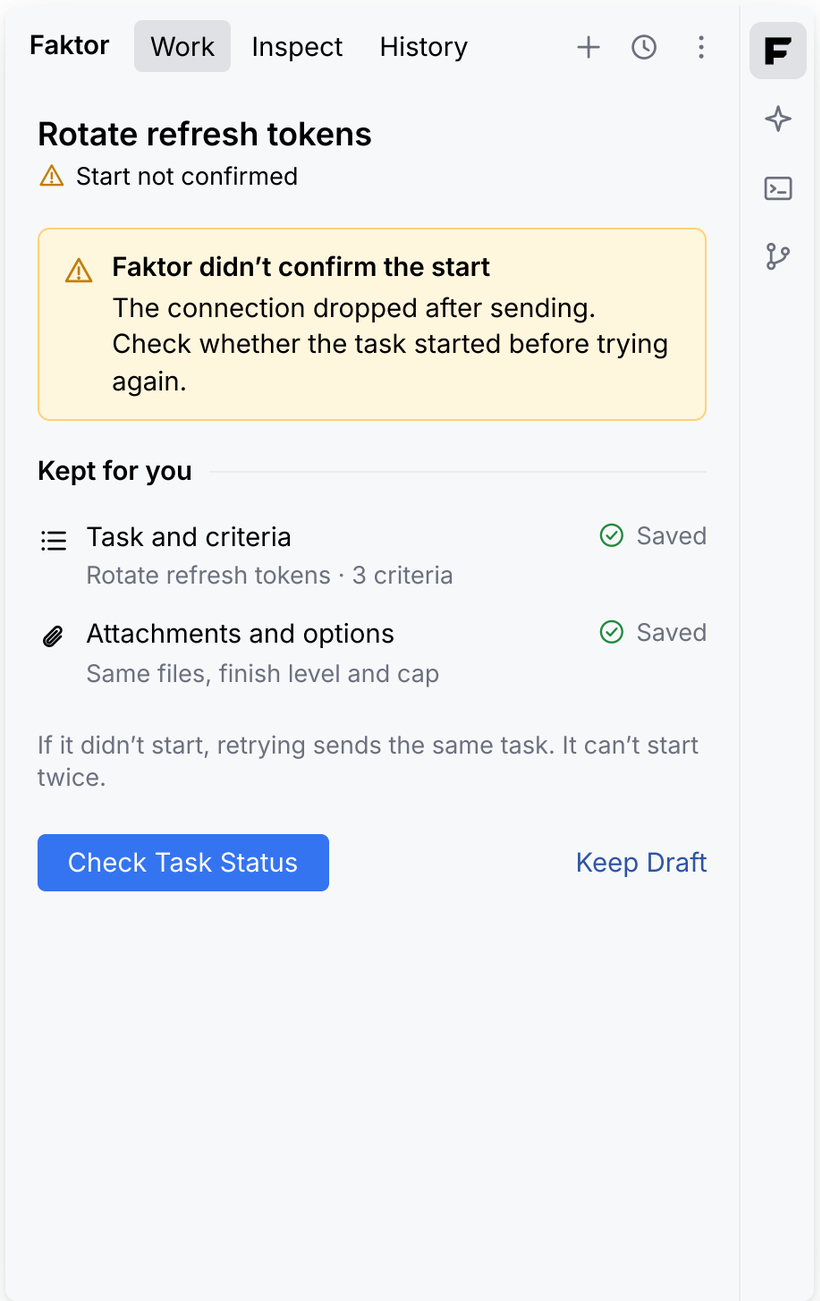
PRIMARY ACTION

Check Task Status

VS CODE · DARK MODERN — SIDEBAR VIEW · WEBVIEW



JETBRAINS · LIGHT — TOOL WINDOW · SWING / KOTLIN UI DSL



CONTROLS B-12 D-05

REPLACES R7 page 013 · R4-S07

CONTRACT p. 38

S39 · SETUP AND START · BEHAVIOUR CONTRACT

Ambiguous task start: contract and revision notes

Original contract retained verbatim (R4-S07 / controls: B-12 D-05). Design on p. 37.

What changed in revision 8

- Plain-language recovery: what happened, what is kept, what the button does.

Entry and return

Work, a captured submission receipt, or a task-specific attention item. Work / Inspect / History remain the only top-level clusters. Settings, Usage and Diagnostics use explicit Back. Return to the originating object, not the first row.

Scope and captured state

Daemon/workspace/worktree/conversation/task/request generation. Draft scope is separate from an admitted run. Capture one immutable logical start. Clear only its acknowledged draft version; preserve text typed after capture. Queue items acquire their own receipt and revision.

Primary action and recovery

Check receipt. Retain the captured body, options and source handles. Unknown admission/effect requires receipt reconciliation before replay. A changed body needs a different request identity.

Loading, empty and stale

Show the named operation and current owner. Keep the composer available for an independent draft. No fake percentages, fabricated tool steps or optimistic completion.

Keyboard and focus

Enter inserts a newline by default. The host send command respects IME composition. New approvals announce once without taking focus. Escape closes a detail; it never stops a run.

Layout and adverse content

Header, transcript scroll and composer are siblings. Collapse secondary metadata first. Keep a scoped Stop action and the active task reachable at short heights.

Starting point and required construction

Starting classifications: E, R. E existing operation; R runtime-backed adapter; N new contract; H host-only work. Exact per-control classifications remain in the catalog.

Reuse the reviewed domain operations and preserve all served fields. Build every missing native read/write adapter, capability gate and state renderer identified by the matching control/build rows.

BK08 owns admission, receipts, queue, control handles and reconciliation; BK05 owns authority and BK07 owns a shared exact budget ledger.

Required interaction fixtures

Exercise first-read, empty/no-match, ready, refreshing/stale, error/refusal, action pending and acknowledged result. Also switch scope while the request is in flight.

Moved out of the interface

Every line of revision 7 plate copy that left the default view, and where it lives now.

"Faktor / Work / Recovery (R7 window title)"
→ Native host chrome (VS Code view title, tabs or editor tab; JetBrains tool-window tabs, dialog title or Settings breadcrumb)

"The response was lost after submission start-047."
→ Banner text; request ID in Inspector

"Admission is not known yet"
→ Banner title "Faktor didn't confirm the start"

"The service may have accepted this exact captured submission. Check its receipt before starting different work."
→ Banner text

"Preserved"
→ Status "Saved"

"Receipt lookup uses start-047 and its captured body digest. If absent, an allowed retry uses the same identity. A changed task becomes a new request."
→ Specification: screen contract (B-12); UI keeps the consequence: "retrying sends the same task"

"Check receipt"
→ Check Task Status

"Check the task start"
→ Header status "Start not confirmed"

"Captured task / Rotate refresh tokens · 3 required criteria"
→ Row "Task and criteria"

"Attachments and options / Same handles, finish level and budget"
→ Row "Attachments and options — Same files, finish level and cap"

"Safe recovery"
→ Recovery sentence and Check Task Status

"Keep draft"
→ Keep Draft

PART 2 · SCREENS

2.2 Work

6 screens. Each has a design spread for both IDEs and a contract page.

ID	SCREEN	JOB	PAGE
S01	**Work**	The compact tool window exposes intent, model, effort, authority and context beside the draft. Running work keeps the active configuration and pending attention visible.	p. 40
S14	**Approve an action**	The approval dock shows the requesting session/child, operation, target, reason and effect boundary without losing the running transcript.	p. 42
S40	**Answer a task question**	Answer the exact current task question and record the response without changing the original task contract.	p. 44
S22	**Live activity**	A live timeline explains what is active, waiting, queued, failed or uncertain. Critical attention remains visible when details are collapsed.	p. 46
S59	**Queued messages**	Edit, reorder or remove only unadmitted queued inputs through an acknowledged queue revision.	p. 49
S70	**Scoped stop actions**	Choose run cancellation, turn abort or owned-process stop and acknowledge the requested operation separately.	p. 51

S01 WORK

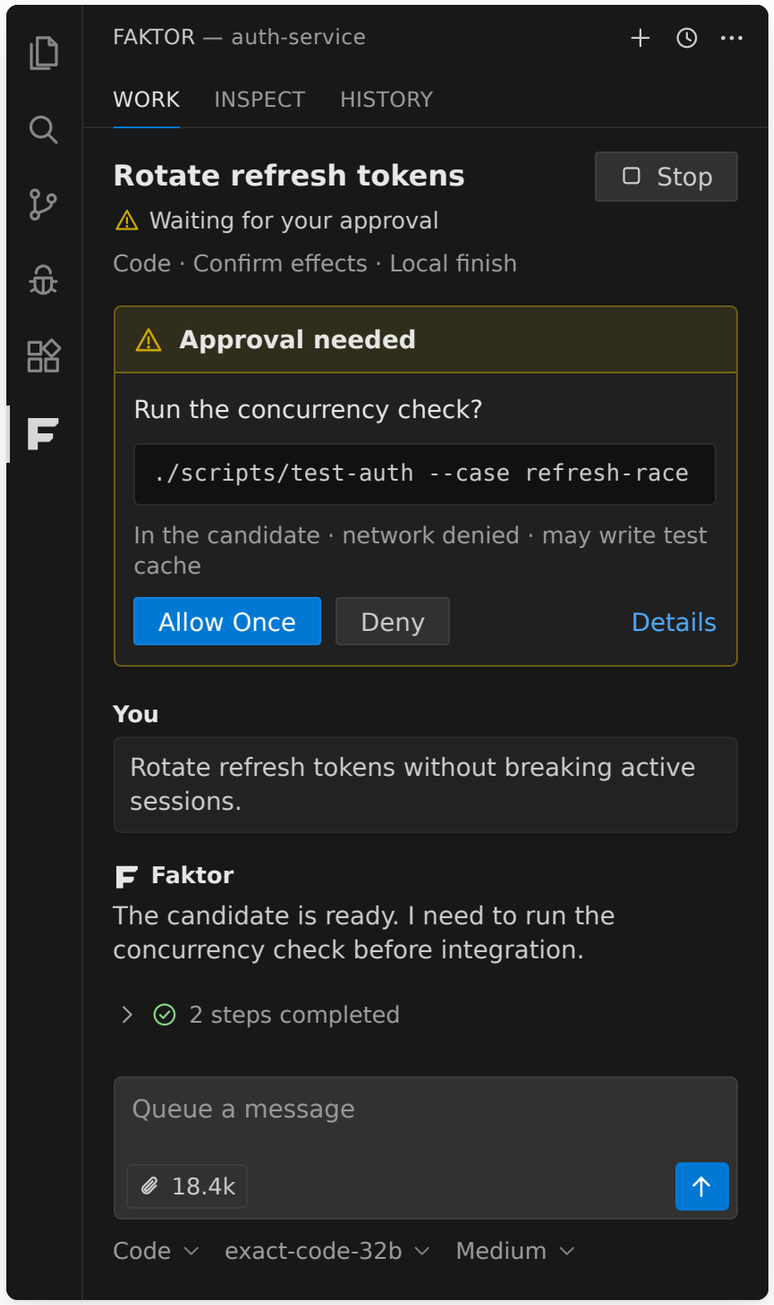
Work

PRIMARY ACTION

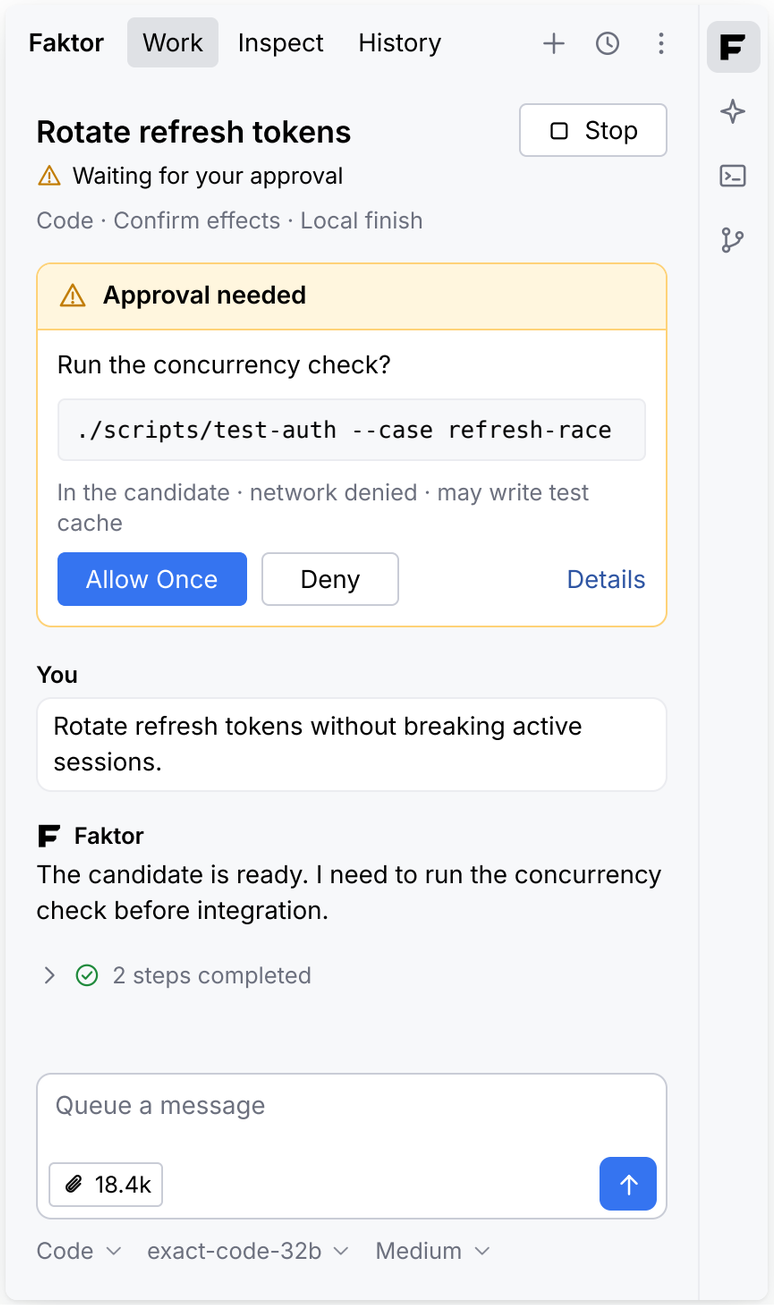
Queue Message (composer)

The compact tool window exposes intent, model, effort, authority and context beside the draft. Running work keeps the active configuration and pending attention visible.

VS CODE · DARK MODERN — SIDEBAR VIEW · WEBVIEW



JETBRAINS · LIGHT — TOOL WINDOW · SWING / KOTLIN UI DSL



CONTROLS A-01 M-01 M-02 M-04 G-01 C-01 Q-01 B-01 B-02 D-05 D-07 V-07 V-10

REPLACES R7 page 014 · R4-S08

CONTRACT p. 41

S01 · WORK · BEHAVIOUR CONTRACT

Work: contract and revision notes

Original contract retained verbatim (R4-S08 / controls: A-01 M-01 M-02 M-04 G-01 C-01 Q-01 B-01 B-02 D-05 D-07 V-07 V-10). Design on p. 40.

What changed in revision 8

- Priority order: attention (approval) › conversation › composer › configuration › usage.
- Decision-critical approval facts stay visible inline: command, where it runs, network and effect.
- Configuration collapses into the composer toolbar; usage becomes the context chip count.
- Stop moves to the task header so it never competes with sending.

Entry and return

Work, a captured submission receipt, or a task-specific attention item. Work / Inspect / History remain the only top-level clusters. Settings, Usage and Diagnostics use explicit Back. Return to the originating object, not the first row.

Scope and captured state

Daemon/workspace/worktree/conversation/task/request generation. Draft scope is separate from an admitted run. Capture one immutable logical start. Clear only its acknowledged draft version; preserve text typed after capture. Queue items acquire their own receipt and revision.

Primary action and recovery

Open Task options without losing the draft; Send captures an immutable configuration summary and task contract.. While disconnected, retain the draft and last coherent transcript. Mark observations stale and disable admission until the session is reconciled.

Loading, empty and stale

Show the named operation and current owner. Keep the composer available for an independent draft. No fake percentages, fabricated tool steps or optimistic completion.

Keyboard and focus

Enter inserts a newline by default. The host send command respects IME composition. New approvals announce once without taking focus. Escape closes a detail; it never stops a run.

Layout and adverse content

Header, transcript scroll and composer are siblings. Collapse secondary metadata first. Keep a scoped Stop action and the active task reachable at short heights.

Starting point and required construction

Starting classifications: E, N, R. E existing operation; R runtime-backed adapter; N new contract; H host-only work. Exact per-control classifications remain in the catalog.

Reuse the reviewed domain operations and preserve all served fields. Build every missing native read/write adapter, capability gate and state renderer identified by the matching control/build rows.

BK08 owns admission, receipts, queue, control handles and reconciliation; BK05 owns authority and BK07 owns a shared exact budget ledger.

Required interaction fixtures

Exercise first-read, empty/no-match, ready, refreshing/stale, error/refusal, action pending and acknowledged result. Also switch scope while the request is in flight.

Moved out of the interface

Every line of revision 7 plate copy that left the default view, and where it lives now.

"Faktor (R7 window title)"
→ Native host chrome (VS Code view title, tabs or editor tab; JetBrains tool-window tabs, dialog title or Settings breadcrumb)

"auth-service / auth-repair"
→ View description (workspace); branch in Task details

"Inspected 2 source files / Prepared 2 candidate changes"
→ Collapsed "2 steps completed"; full list in Inspect › Activity

"Approval needed / Run concurrency verification / Review"
→ Approval card with the exact command, scope and Allow Once / Deny

"Write the next message…"
→ Placeholder "Queue a message" (the run is active)

"Add context"
→ Context chip showing the measured input (18.4k)

"Task options"
→ Task header › Task details

"Measured input 18.4k"
→ Context chip; breakdown in Inspect › Context

"Run cap USD 5.00"
→ Task details (header) and Inspect › Usage

"Queue message / Stop run"
→ Send button (Queue Message) and header Stop

S14 WORK

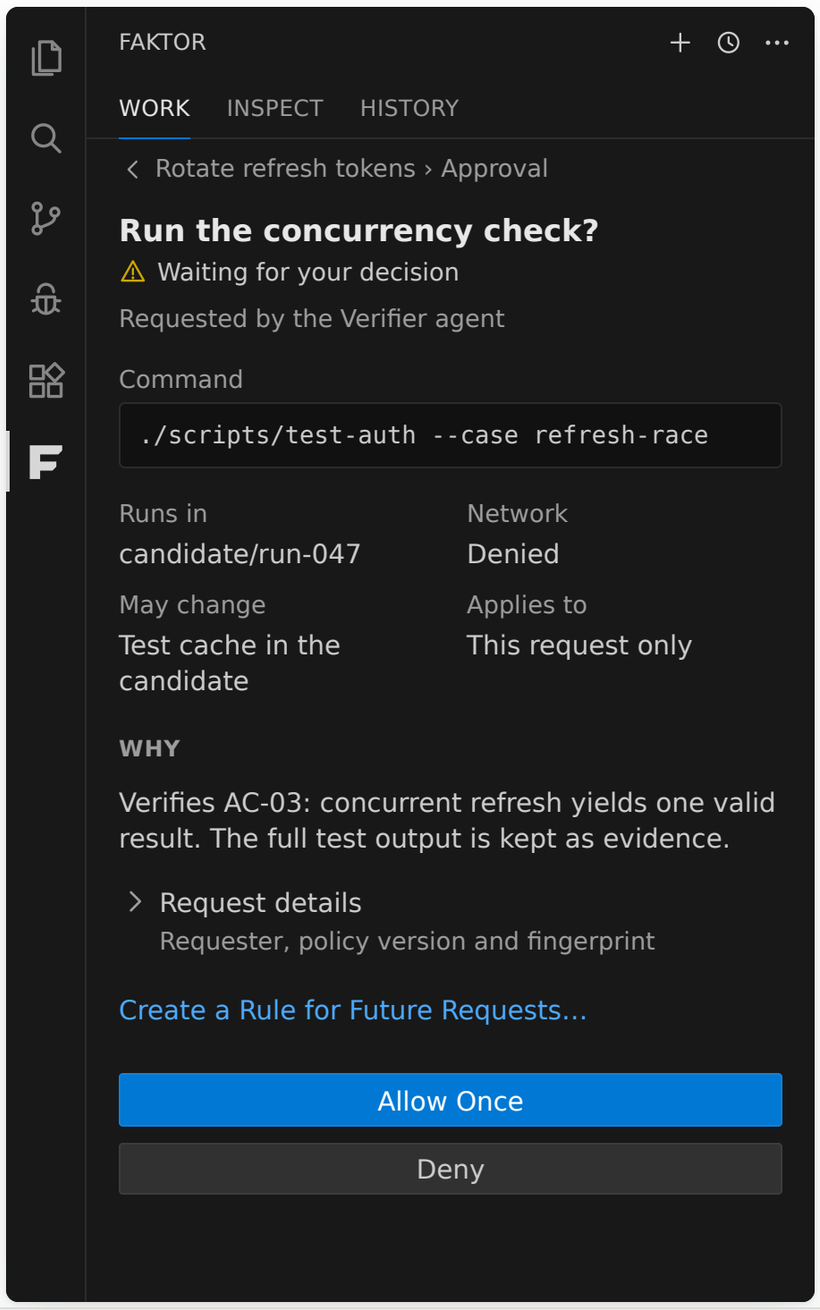
PRIMARY ACTION

Allow Once

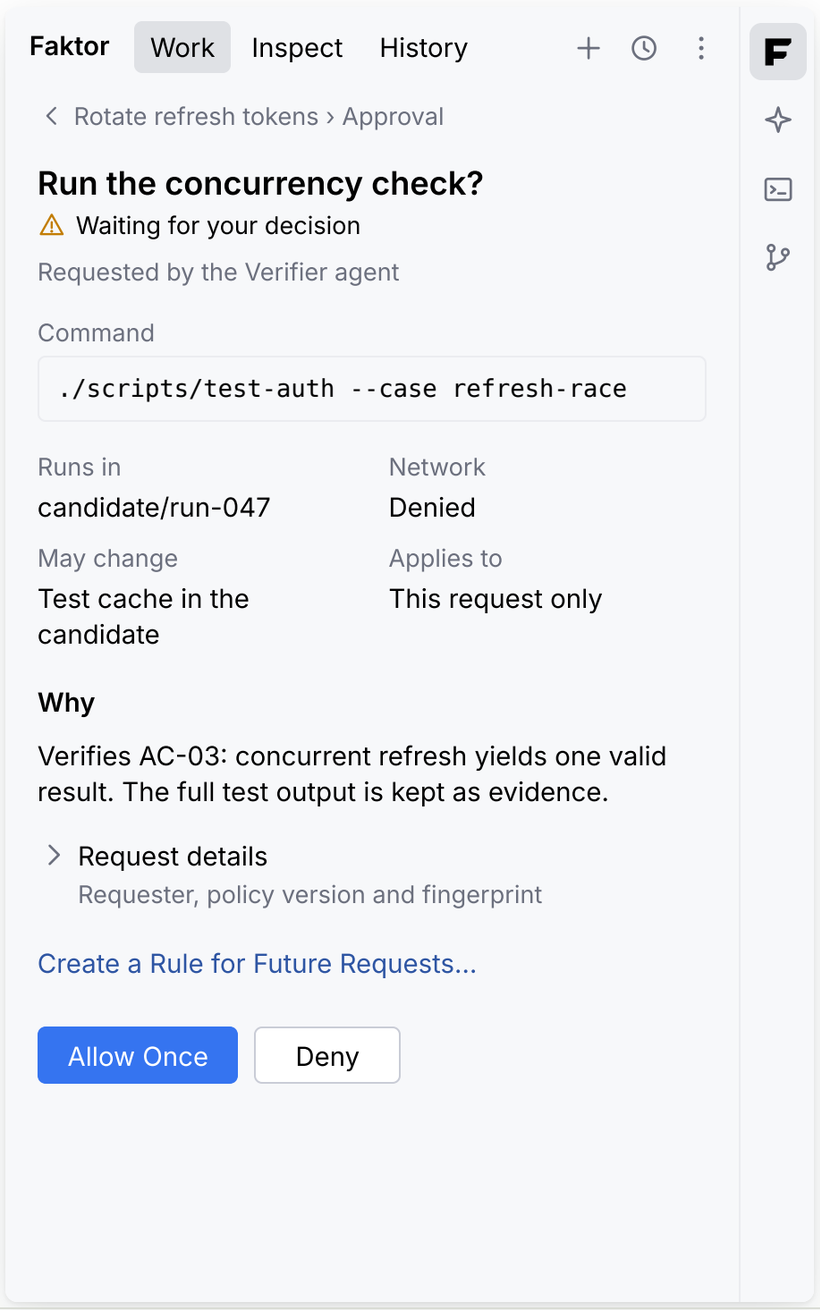
Approve an action

The approval dock shows the requesting session/child, operation, target, reason and effect boundary without losing the running transcript.

VS CODE · DARK MODERN — SIDEBAR VIEW · WEBVIEW



JETBRAINS · LIGHT — TOOL WINDOW · SWING / KOTLIN UI DSL



CONTROLS Q-06 Q-07 Q-09 Q-13 X-12 B-12 D-12

REPLACES R7 page 015 · R4-S09

CONTRACT p. 43

S14 · WORK · BEHAVIOUR CONTRACT

Approve an action: contract and revision notes

Original contract retained verbatim (R4-S09 / controls: Q-06 Q-07 Q-09 Q-13 X-12 B-12 D-12). Design on p. 42.

What changed in revision 8

- All decision-critical facts are visible without expanding anything.
- Deny and Allow Once are adjacent; initial focus goes to the title, never to Allow Once.
- The future-rule action is visually separate so it cannot be mistaken for answering.

Entry and return

Work, a captured submission receipt, or a task-specific attention item. Work / Inspect / History remain the only top-level clusters. Settings, Usage and Diagnostics use explicit Back. Return to the originating object, not the first row.

Scope and captured state

Daemon/workspace/worktree/conversation/task/request generation. Draft scope is separate from an admitted run. Capture one immutable logical start. Clear only its acknowledged draft version; preserve text typed after capture. Queue items acquire their own receipt and revision.

Primary action and recovery

Allow once or Deny addresses one exact waiter generation. Remember rule opens a separate persistent-policy draft only when supported.. Expired, already-replied or wrong-session requests disable response and offer Refresh. A timeout preserves unknown state until reconciled.

Loading, empty and stale

Show the named operation and current owner. Keep the composer available for an independent draft. No fake percentages, fabricated tool steps or optimistic completion.

Keyboard and focus

Enter inserts a newline by default. The host send command respects IME composition. New approvals announce once without taking focus. Escape closes a detail; it never stops a run.

Layout and adverse content

Header, transcript scroll and composer are siblings. Collapse secondary metadata first. Keep a scoped Stop action and the active task reachable at short heights.

Starting point and required construction

Starting classifications: E, H, N, R. E existing operation; R runtime-backed adapter; N new contract; H host-only work. Exact per-control classifications remain in the catalog.

Reuse the reviewed domain operations and preserve all served fields. Build every missing native read/write adapter, capability gate and state renderer identified by the matching control/build rows.

BK08 owns admission, receipts, queue, control handles and reconciliation; BK05 owns authority and BK07 owns a shared exact budget ledger.

Required interaction fixtures

Exercise first-read, empty/no-match, ready, refreshing/stale, error/refusal, action pending and acknowledged result. Also switch scope while the request is in flight.

Moved out of the interface

Every line of revision 7 plate copy that left the default view, and where it lives now.

"Faktor / Work / Approval (R7 window title)"
→ Native host chrome (VS Code view title, tabs or editor tab; JetBrains tool-window tabs, dialog title or Settings breadcrumb)

"Run 047 / verifier 3 · current request only"
→ Subtitle "Requested by the Verifier agent" and fact "Applies to: This request only"

"Run concurrency verification?"
→ Title

"This request may write test cache inside the isolated candidate."
→ Fact "May change: Test cache in the candidate"

"Exact command"
→ Code block label "Command"

"Working directory / candidate/run-047; Network / Denied"
→ Facts "Runs in" and "Network"

"Why it is needed / Verify acceptance criterion AC-03 and retain the full test artifact."
→ "Why" section

"Create a rule for future requests / Review / A separate policy draft does not answer this request."
→ Separate link; the rule never answers this request (Specification: screen contract)

"Owner, generation, action fingerprint and effective policy version"
→ Request details disclosure

"Approve an action"
→ Breadcrumb "Approval"; title is the question

"Deny / Allow once"
→ Deny / Allow Once

S40 WORK

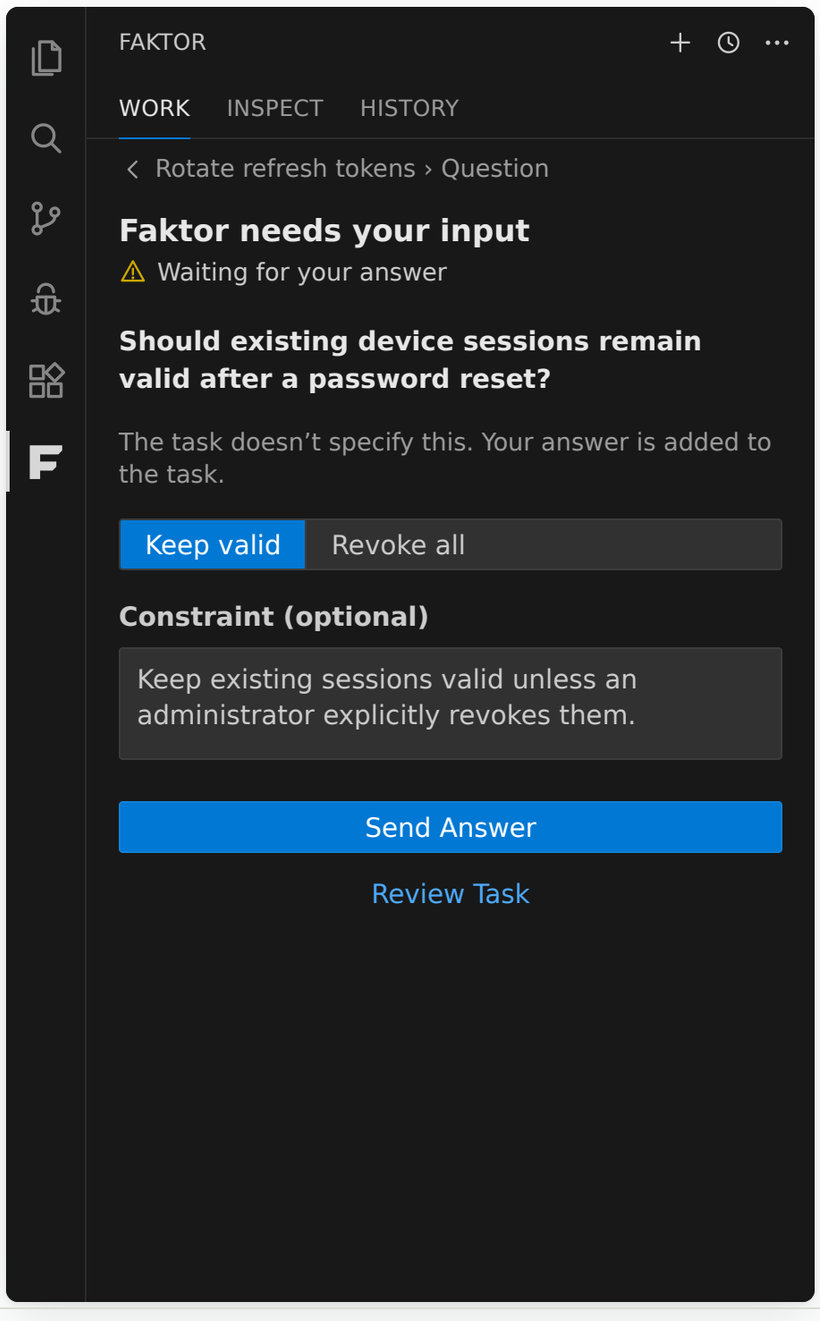
Answer a task question

Answer the exact current task question and record the response without changing the original task contract.

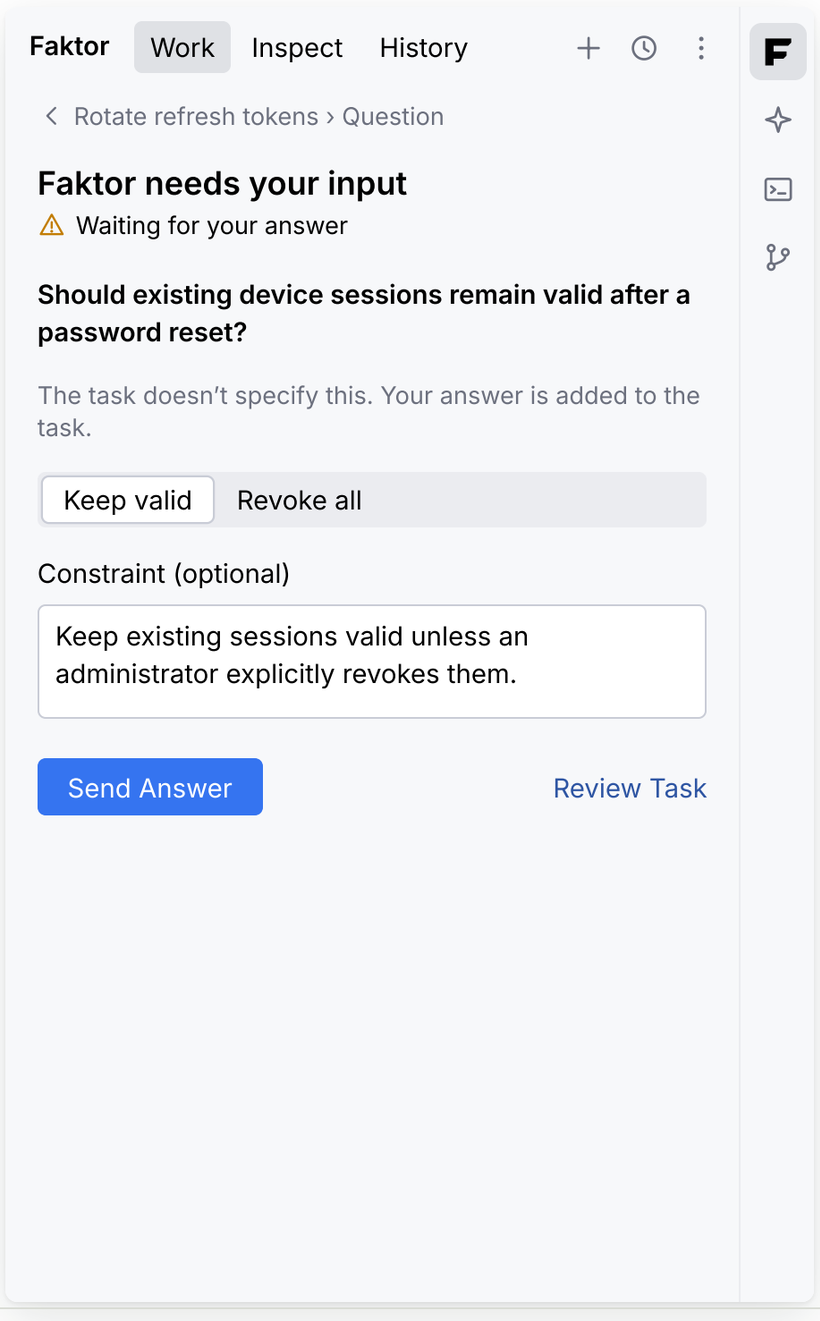
PRIMARY ACTION

Send Answer

VS CODE · DARK MODERN — SIDEBAR VIEW · WEBVIEW



JETBRAINS · LIGHT — TOOL WINDOW · SWING / KOTLIN UI DSL



CONTROLS D-07 V-07

REPLACES R7 page 016 · R4-S10

CONTRACT p. 45

S40 · WORK · BEHAVIOUR CONTRACT

Answer a task question: contract and revision notes

Original contract retained verbatim (R4-S10 / controls: D-07 V-07). Design on p. 44.

What changed in revision 8

- The question is the headline; the choice and optional constraint follow directly.

Entry and return

Work, a captured submission receipt, or a task-specific attention item. Work / Inspect / History remain the only top-level clusters. Settings, Usage and Diagnostics use explicit Back. Return to the originating object, not the first row.

Scope and captured state

Daemon/workspace/worktree/conversation/task/request generation. Draft scope is separate from an admitted run. Capture one immutable logical start. Clear only its acknowledged draft version; preserve text typed after capture. Queue items acquire their own receipt and revision.

Primary action and recovery

Send answer. Retain the captured body, options and source handles. Unknown admission/effect requires receipt reconciliation before replay. A changed body needs a different request identity.

Loading, empty and stale

Show the named operation and current owner. Keep the composer available for an independent draft. No fake percentages, fabricated tool steps or optimistic completion.

Keyboard and focus

Enter inserts a newline by default. The host send command respects IME composition. New approvals announce once without taking focus. Escape closes a detail; it never stops a run.

Layout and adverse content

Header, transcript scroll and composer are siblings. Collapse secondary metadata first. Keep a scoped Stop action and the active task reachable at short heights.

Starting point and required construction

Starting classifications: E. E existing operation; R runtime-backed adapter; N new contract; H host-only work. Exact per-control classifications remain in the catalog.

Reuse the reviewed domain operations and preserve all served fields. Build every missing native read/write adapter, capability gate and state renderer identified by the matching control/build rows.

BK08 owns admission, receipts, queue, control handles and reconciliation; BK05 owns authority and BK07 owns a shared exact budget ledger.

Required interaction fixtures

Exercise first-read, empty/no-match, ready, refreshing/stale, error/refusal, action pending and acknowledged result. Also switch scope while the request is in flight.

Moved out of the interface

Every line of revision 7 plate copy that left the default view, and where it lives now.

"Faktor / Work / Question (R7 window title)"
→ Native host chrome (VS Code view title, tabs or editor tab; JetBrains tool-window tabs, dialog title or Settings breadcrumb)

"Run 047 · question q-12 · waiting for your answer"
→ Status "Waiting for your answer"; IDs in Inspector

"The current requirement does not decide this behavior. Your answer will be recorded against this task."
→ Short help "The task doesn't specify this. Your answer is added to the task."

"Session policy"
→ Implied by the question; label kept as the accessible name of the choice

"Additional constraint"
→ Field "Constraint (optional)"

"Captured question / Current / Run 047 / q-12 · still open"
→ Inspector › Activity

"Keep valid / Revoke all"
→ Segmented choice

"Review task / Send answer"
→ Review Task / Send Answer

S22 WORK

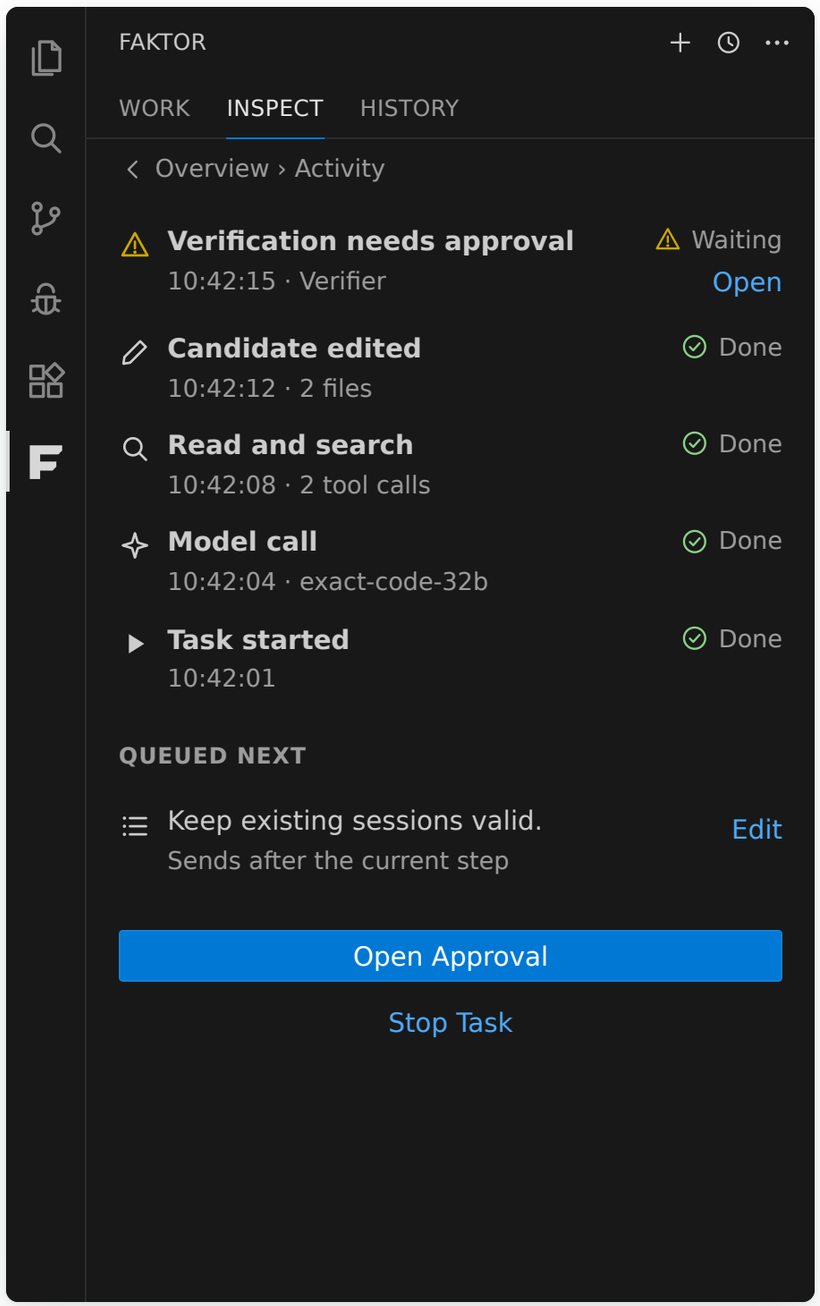
PRIMARY ACTION

Open Approval

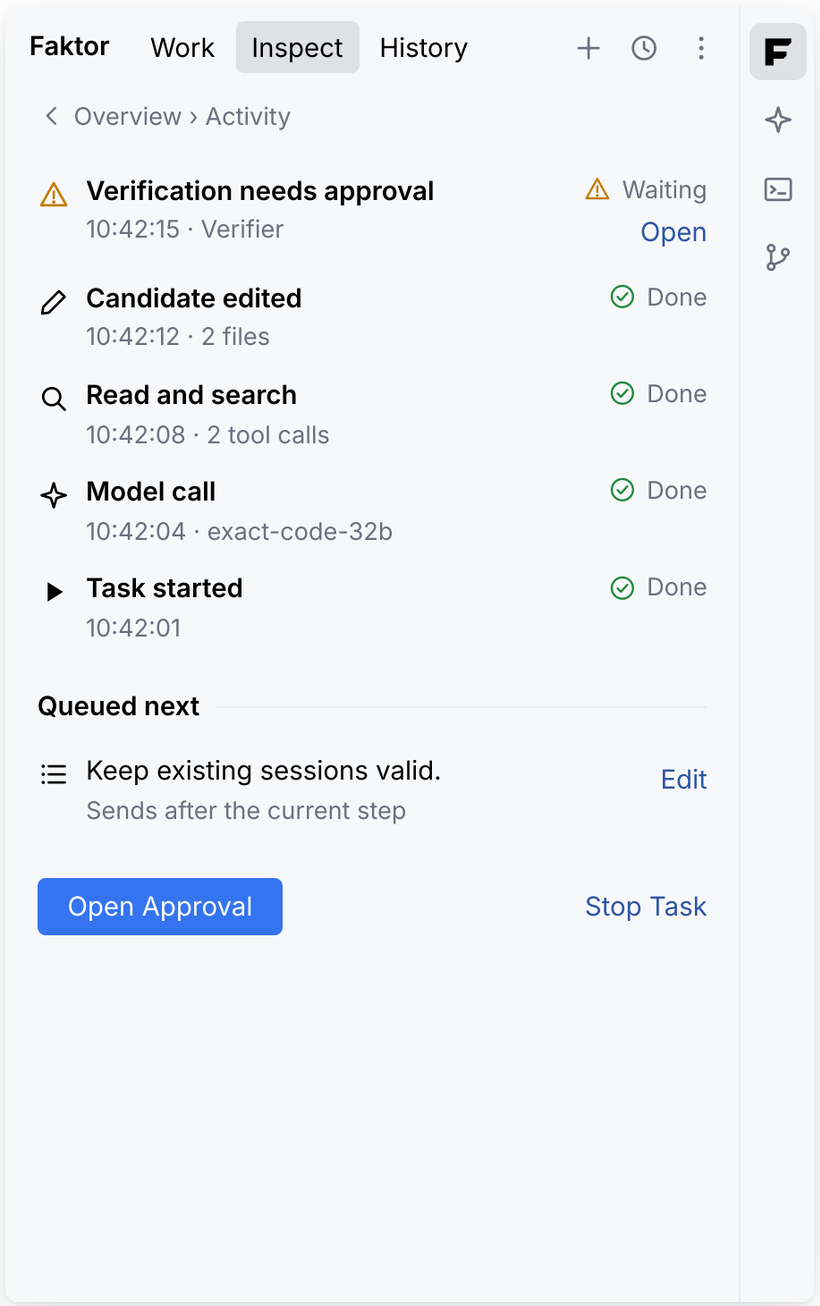
Live activity

A live timeline explains what is active, waiting, queued, failed or uncertain. Critical attention remains visible when details are collapsed.

VS CODE · DARK MODERN — SIDEBAR VIEW · WEBVIEW



JETBRAINS · LIGHT — TOOL WINDOW · SWING / KOTLIN UI DSL



CONTROLS D-05 D-07 D-08 D-12 D-13 M-11 G-14 Q-06 T-09 B-04

REPLACES R7 page 017 · Expanded standard-agent surface

CONTRACT p. 47

S22 · WORK · BEHAVIOUR CONTRACT

Live activity: contract and revision notes

Original contract retained verbatim (Expanded standard-agent surface / controls: D-05 D-07 D-08 D-12 D-13 M-11 G-14 Q-06 T-09 B-04). Design on p. 46.

What changed in revision 8

- Attention is pinned to the top of the timeline; completed steps use the quiet Done state.

Entry and return

Work, a captured submission receipt, or a task-specific attention item. Work / Inspect / History remain the only top-level clusters. Settings, Usage and Diagnostics use explicit Back. Return to the originating object, not the first row.

Scope and captured state

Daemon/workspace/worktree/conversation/task/request generation. Draft scope is separate from an admitted run. Capture one immutable logical start. Clear only its acknowledged draft version; preserve text typed after capture. Queue items acquire their own receipt and revision.

Primary action and recovery

Inspect a tool call, edit the owned next-message queue or stop the current run. A steer action states its delivery point.. Lost stream marks the projection stale without turning the task idle. Reconnect from durable cursor and reconcile before enabling duplicate effects.

Loading, empty and stale

Show the named operation and current owner. Keep the composer available for an independent draft. No fake percentages, fabricated tool steps or optimistic completion.

Keyboard and focus

Enter inserts a newline by default. The host send command respects IME composition. New approvals announce once without taking focus. Escape closes a detail; it never stops a run.

Layout and adverse content

Header, transcript scroll and composer are siblings. Collapse secondary metadata first. Keep a scoped Stop action and the active task reachable at short heights.

Starting point and required construction

Starting classifications: E, H, N, R. E existing operation; R runtime-backed adapter; N new contract; H host-only work. Exact per-control classifications remain in the catalog.

Reuse the reviewed domain operations and preserve all served fields. Build every missing native read/write adapter, capability gate and state renderer identified by the matching control/build rows.

BK08 owns admission, receipts, queue, control handles and reconciliation; BK05 owns authority and BK07 owns a shared exact budget ledger.

Required interaction fixtures

Exercise first-read, empty/no-match, ready, refreshing/stale, error/refusal, action pending and acknowledged result. Also switch scope while the request is in flight.

Moved out of the interface

Every line of revision 7 plate copy that left the default view, and where it lives now.

"Faktor / Inspect / Activity (R7 window title)"
→ Native host chrome (VS Code view title, tabs or editor tab; JetBrains tool-window tabs, dialog title or Settings breadcrumb)

"Run 047 · durable activity in time order"
→ Removed: restated by the layout or label (newest first, attention pinned on top)

"Task admitted / Pinned local-code · contract version 4"
→ Row "Task started"; route and contract version in the row detail

"Actual model exact-code-32b · usage observed"
→ Row meta; usage in row detail

"full artifacts retained"
→ Row detail

"candidate identity advanced"
→ Row detail

"Verifier 3 · exact request available"
→ Row meta "Verifier" + Open

"Edit / Remove"
→ Edit on the row; Remove inside Queued messages (S59)

"Not admitted yet. Editing the queue never interrupts the active task."
→ Row meta "Sends after the current step"; rule in Specification: screen contract

"Live activity"
→ Breadcrumb "Activity"

"Task admitted / Done"
→ Row "Task started — Done"

"Model call completed"
→ Row "Model call"

"10:42:01 · Pinned local-code · contract version 4"
→ Row meta "10:42:01"; details on expand

"10:42:04 · Actual model exact-code-32b · usage observed"
→ Row meta "10:42:04 · exact-code-32b"

"Read and search completed"
→ Row "Read and search"

"10:42:08 · 2 tool calls · full artifacts retained"
→ Row meta "10:42:08 · 2 tool calls"

S22 · LIVE ACTIVITY · CONTRACT, CONTINUED

"Candidate edited / 10:42:12 · 2 files · candidate identity advanced"
→ Row meta "10:42:12 · 2 files"

"Verification needs approval / Waiting / 10:42:15 · Verifier 3 · exact request available"
→ Pinned row with Open

"Queued next / Keep existing sessions valid."
→ Section "Queued next"

"Stop task / Open approval"
→ Stop Task / Open Approval

S59 WORK

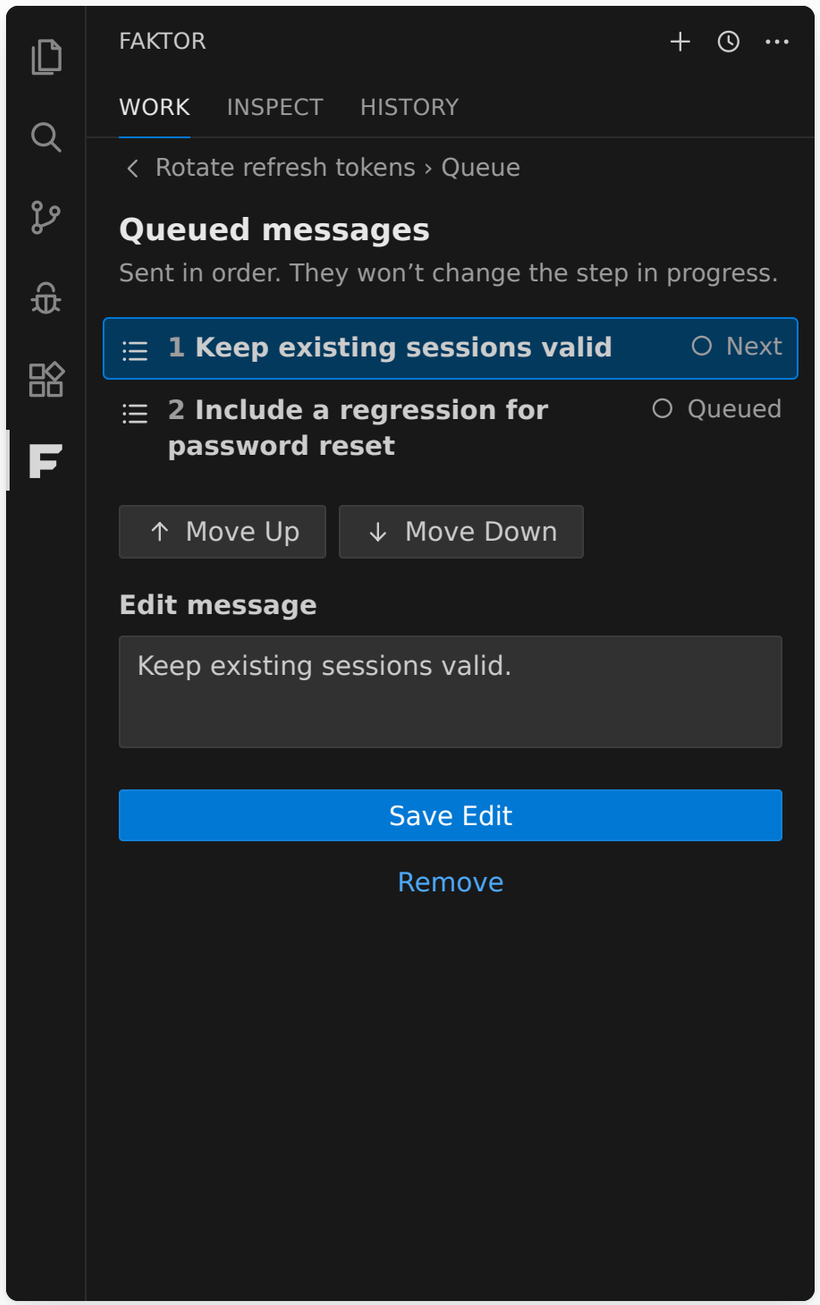
Queued messages

Edit, reorder or remove only unadmitted queued inputs through an acknowledged queue revision.

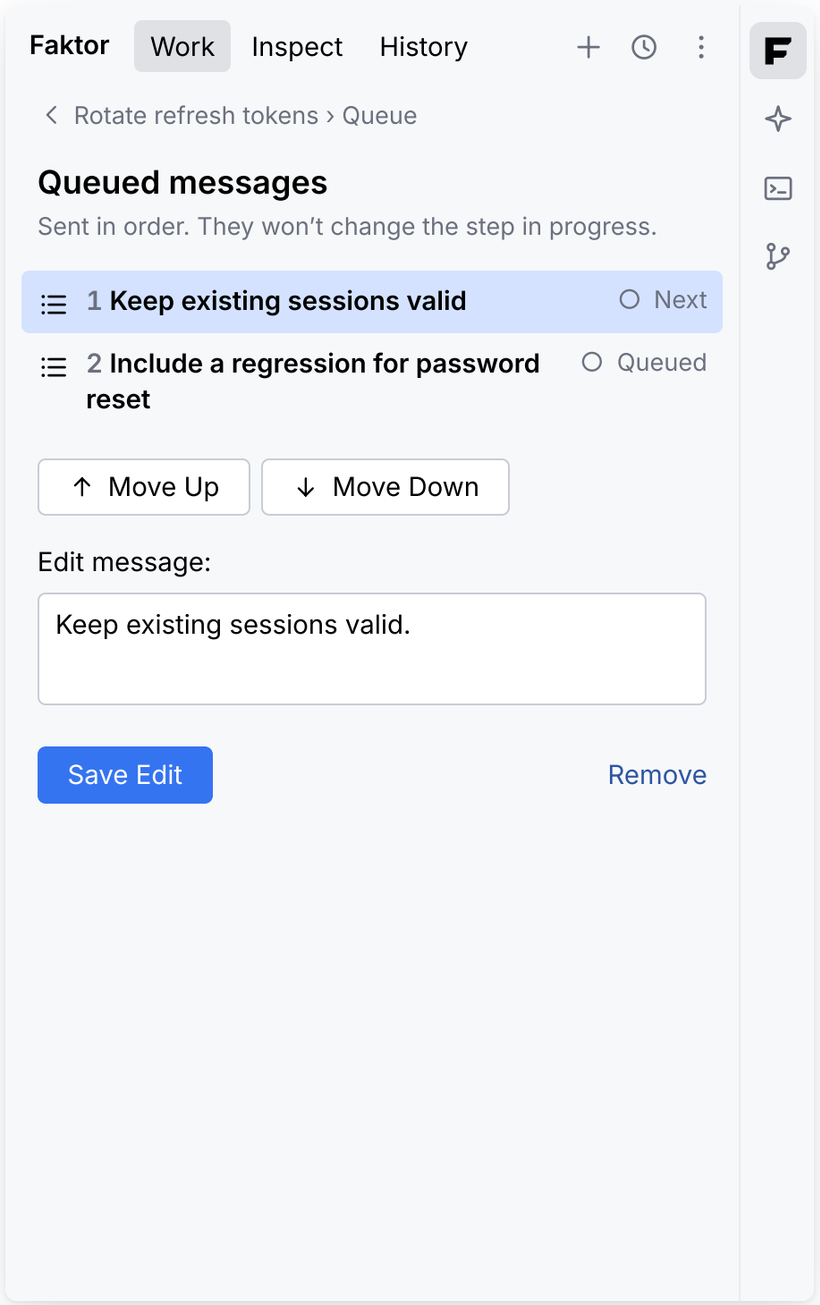
PRIMARY ACTION

Save Edit

VS CODE · DARK MODERN — SIDEBAR VIEW · WEBVIEW



JETBRAINS · LIGHT — TOOL WINDOW · SWING / KOTLIN UI DSL



CONTROLS D-05

REPLACES R7 page 018 · Expanded standard-agent surface

CONTRACT p. 50

S59 · WORK · BEHAVIOUR CONTRACT

Queued messages: contract and revision notes

Original contract retained verbatim (Expanded standard-agent surface / controls: D-05). Design on p. 49.

What changed in revision 8

- Reordering uses labelled Move Up / Move Down buttons (keyboard-operable), not drag only.

Entry and return

Work, a captured submission receipt, or a task-specific attention item. Work / Inspect / History remain the only top-level clusters. Settings, Usage and Diagnostics use explicit Back. Return to the originating object, not the first row.

Scope and captured state

Daemon/workspace/worktree/conversation/task/request generation. Draft scope is separate from an admitted run. Capture one immutable logical start. Clear only its acknowledged draft version; preserve text typed after capture. Queue items acquire their own receipt and revision.

Primary action and recovery

Save queued edit. Retain the captured body, options and source handles. Unknown admission/effect requires receipt reconciliation before replay. A changed body needs a different request identity.

Loading, empty and stale

Show the named operation and current owner. Keep the composer available for an independent draft. No fake percentages, fabricated tool steps or optimistic completion.

Keyboard and focus

Enter inserts a newline by default. The host send command respects IME composition. New approvals announce once without taking focus. Escape closes a detail; it never stops a run.

Layout and adverse content

Header, transcript scroll and composer are siblings. Collapse secondary metadata first. Keep a scoped Stop action and the active task reachable at short heights.

Starting point and required construction

Starting classifications: R. E existing operation; R runtime-backed adapter; N new contract; H host-only work. Exact per-control classifications remain in the catalog.

NEW: build a durable session queue with revisioned edit/reorder/remove, captured admission envelopes, owner checks and receipts.

BK08 owns admission, receipts, queue, control handles and reconciliation; BK05 owns authority and BK07 owns a shared exact budget ledger.

Required interaction fixtures

Exercise first-read, empty/no-match, ready, refreshing/stale, error/refusal, action pending and acknowledged result. Also switch scope while the request is in flight.

Moved out of the interface

Every line of revision 7 plate copy that left the default view, and where it lives now.

"Faktor / Work / Queue (R7 window title)"
→ Native host chrome (VS Code view title, tabs or editor tab; JetBrains tool-window tabs, dialog title or Settings breadcrumb)

"Run 047 · two queued inputs · none changes the active call"
→ Subtitle

"Queue item q-19 · waits for a safe admission boundary"
→ "Next" status; queue IDs in Tooltip / hover help

"Queue item q-20 · draft snapshot retained"
→ "Queued" status

"Selected message / Edit queued text"
→ Field "Edit message"

"Reordering uses an acknowledged queue revision. A message already admitted cannot be edited or recalled."
→ Specification: screen contract; sent messages simply have no Edit action

"Remove selected / Save queued edit"
→ Remove link and Save Edit

"Next messages"
→ Title "Queued messages"

S70 WORK

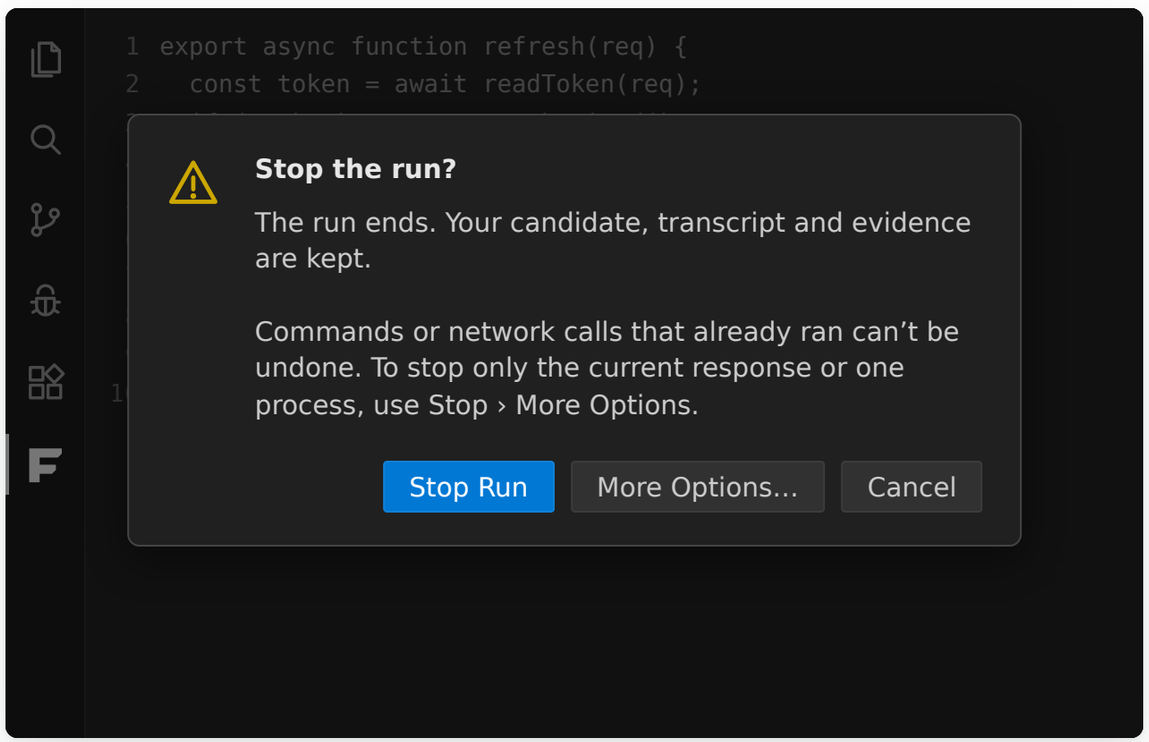
Scoped stop actions

Choose run cancellation, turn abort or owned-process stop and acknowledge the requested operation separately.

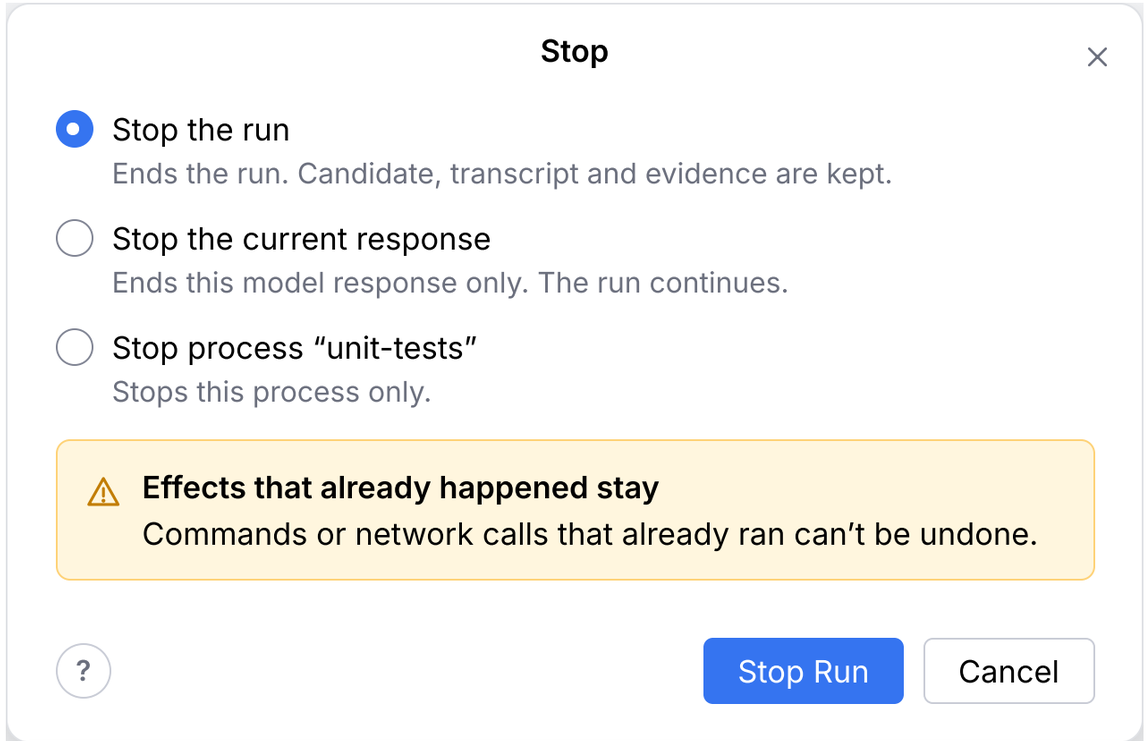
PRIMARY ACTION

Stop Run

VS CODE · DARK MODERN — MODAL MESSAGE · NATIVE



JETBRAINS · LIGHT — DIALOGWRAPPER



CONTROLS A-10 X-05 D-07

REPLACES R7 page 019 · Expanded standard-agent surface

CONTRACT p. 52

S70 · WORK · BEHAVIOUR CONTRACT

Scoped stop actions: contract and revision notes

Original contract retained verbatim (Expanded standard-agent surface / controls: A-10 X-05 D-07). Design on p. 51.

What changed in revision 8

- The button names the exact operation; neither host has a red destructive button, so clarity comes from the verb and object.
- VS Code shows the common case as a native modal; the other scopes are one click away.

Entry and return

Work, a captured submission receipt, or a task-specific attention item. Work / Inspect / History remain the only top-level clusters. Settings, Usage and Diagnostics use explicit Back. Return to the originating object, not the first row.

Scope and captured state

Daemon/workspace/worktree/conversation/task/request generation. Draft scope is separate from an admitted run. Capture one immutable logical start. Clear only its acknowledged draft version; preserve text typed after capture. Queue items acquire their own receipt and revision.

Primary action and recovery

Request run stop. Retain the captured body, options and source handles. Unknown admission/effect requires receipt reconciliation before replay. A changed body needs a different request identity.

Loading, empty and stale

Show the named operation and current owner. Keep the composer available for an independent draft. No fake percentages, fabricated tool steps or optimistic completion.

Keyboard and focus

Enter inserts a newline by default. The host send command respects IME composition. New approvals announce once without taking focus. Escape closes a detail; it never stops a run.

Layout and adverse content

Header, transcript scroll and composer are siblings. Collapse secondary metadata first. Keep a scoped Stop action and the active task reachable at short heights.

Starting point and required construction

Starting classifications: E. E existing operation; R runtime-backed adapter; N new contract; H host-only work. Exact per-control classifications remain in the catalog.

Keep existing run/turn controls separate; build current-generation owned process handles and unknown-effect reconciliation wherever absent.

BK08 owns admission, receipts, queue, control handles and reconciliation; BK05 owns authority and BK07 owns a shared exact budget ledger.

Required interaction fixtures

Exercise first-read, empty/no-match, ready, refreshing/stale, error/refusal, action pending and acknowledged result. Also switch scope while the request is in flight.

Moved out of the interface

Every line of revision 7 plate copy that left the default view, and where it lives now.

"Faktor / Work / Stop (R7 window title)"
→ Native host chrome (VS Code view title, tabs or editor tab; JetBrains tool-window tabs, dialog title or Settings breadcrumb)

"Run 047 owns model turn t-18 and process p-6."
→ Option labels name the response and the process; IDs in Tooltip / hover help

"Selected"
→ Radio selection

"Request durable run cancellation; retain candidate and proof"
→ Option help "Ends the run. Candidate, transcript and evidence are kept."

"End this turn only; do not mark the run cancelled"
→ Option help "Ends this model response only. The run continues."

"Stop the exact current handle; never a stale PID"
→ Option help; stale-handle rule in Specification: screen contract

"A request is not a stopped receipt"
→ Stopping… state after confirmation; rule in Specification: screen contract

"If the connection is lost, show effect unknown and reconcile. An external effect may already have happened."
→ Warning "Effects that already happened stay"; reconciliation in Specification: screen contract

"Request run stop"
→ Stop Run (label follows the selected option)

"Stop which operation?"
→ Dialog title "Stop" and the radio group

"Stop run"
→ Option "Stop the run"

"Abort current model turn"
→ Option "Stop the current response"

"Stop owned process p-6"
→ Option "Stop process "unit-tests""

S70 · SCOPED STOP ACTIONS · CONTRACT, CONTINUED

“Cancel”
→ Cancel

PART 2 · SCREENS

2.3 Inspect and proof

13 screens. Each has a design spread for both IDEs and a contract page.

ID	SCREEN	JOB	PAGE
S41	**Inspect overview**	Read all four engineering boundaries and open the selected plan, criterion, artifact or candidate from one overview.	p. 55
S42	**Plan and dependencies**	Inspect the dependency-ordered plan and selected step without confusing a planned checkmark with observed proof.	p. 57
S24	**Review changes**	Use each host's native diff editor. Keep base/candidate identity, acceptance criteria and remaining verification adjacent to code review.	p. 59
S43	**Native diff review**	Review the immutable base and selected candidate in the native editor with exact hunk and claim navigation.	p. 61
S45	**Required verification**	Inspect every required criterion, its current evidence and explicit pending, failed, refused or unavailable result.	p. 63
S46	**Evidence detail**	Follow one criterion to its exact artifact, source, claim and candidate; open bounded excerpts and full content.	p. 65
S47	**Proof identity mismatch**	Remove stale verified presentation when proof belongs to a different candidate; retain the historical record separately.	p. 67
S44	**Integration conflict**	Preserve user edits when landing preconditions change; review a new candidate and rerun affected verification.	p. 69
S48	**Finish actions**	Inspect the hierarchical Local / Commit / Push / PR receipts and resolve only the blocked native finish step.	p. 71
S32	**Task outcome**	Finish with explicit engineering status: required acceptance, claim evidence, matching integration and requested finish. Conversation completion alone proves none of them.	p. 73
S65	**Verified outcome**	Show a separate complete example only when all required acceptance, claims, matching integration and finish agree.	p. 75
S25	**Restore & fork**	Show recorded checkpoint identity, content coverage and branching separately. Preview the exact files and conversation range affected by any supported restore.	p. 77
S49	**Cancelled run**	Retain candidate work, evidence, recorded spend and drafts after a durable cancellation; completion remains separate.	p. 79

S41 INSPECT AND PROOF

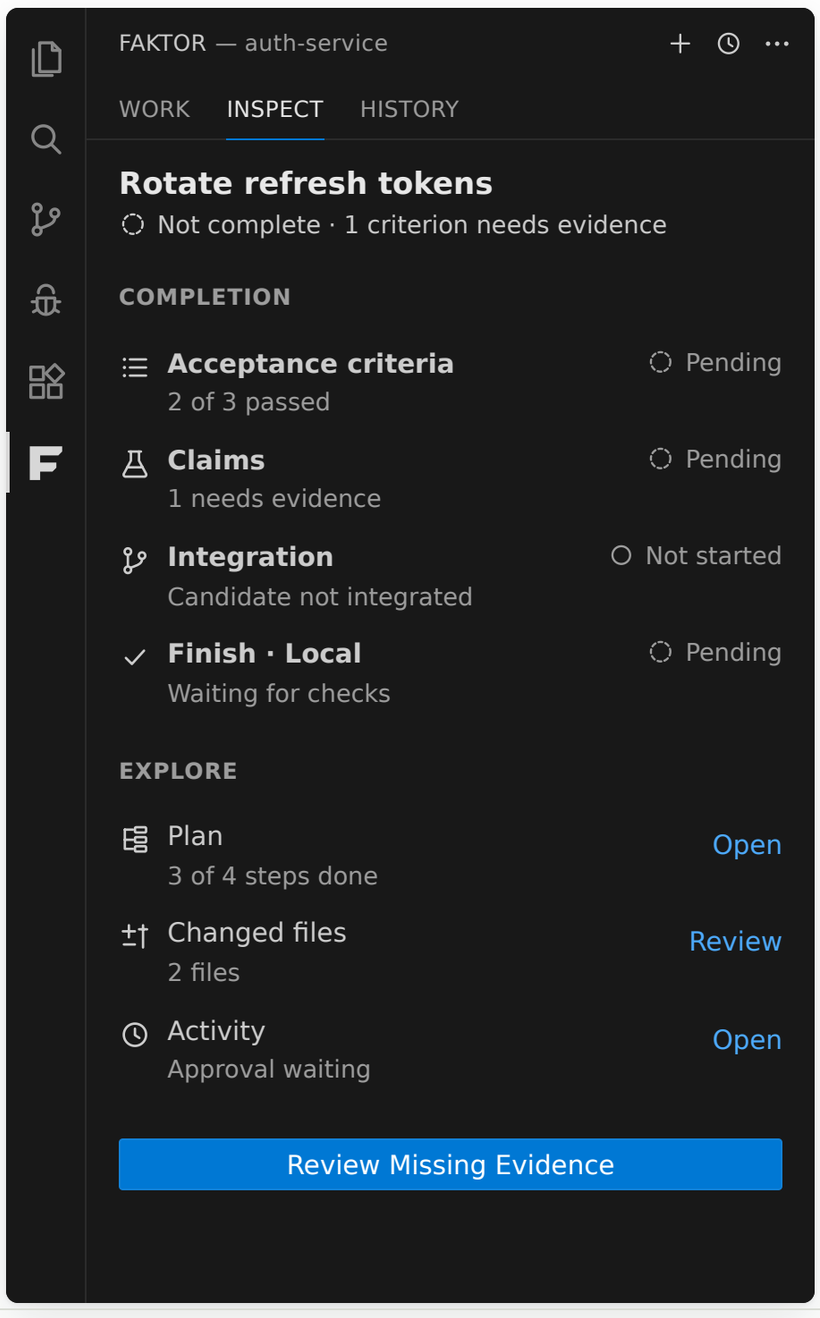
Inspect overview

Read all four engineering boundaries and open the selected plan, criterion, artifact or candidate from one overview.

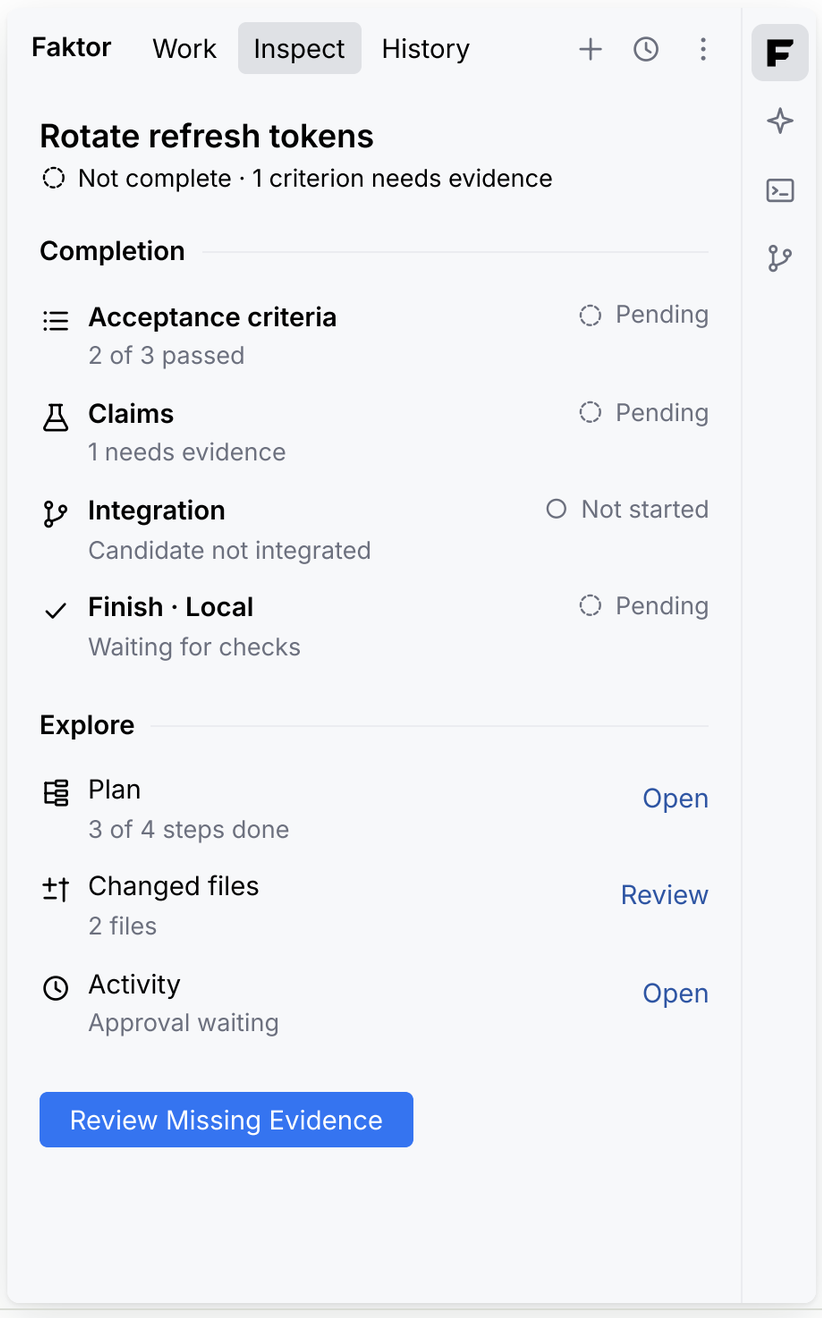
PRIMARY ACTION

Review Missing Evidence

VS CODE · DARK MODERN — SIDEBAR VIEW · WEBVIEW



JETBRAINS · LIGHT — TOOL WINDOW · SWING / KOTLIN UI DSL



CONTROLS V-07 V-08 V-09 V-10

REPLACES R7 page 020 · R4-S11

CONTRACT p. 56

S41 · INSPECT AND PROOF · BEHAVIOUR CONTRACT

Inspect overview: contract and revision notes

Original contract retained verbatim (R4-S11 / controls: V-07 V-08 V-09 V-10). Design on p. 55.

What changed in revision 8

- The four completion domains read as one checklist; navigation rows are visually quieter.
- Return to Work becomes the tab itself (no duplicate navigation button).

Entry and return

Inspect overview, a changed file, a criterion or a recorded proof link. Work / Inspect / History remain the only top-level clusters. Settings, Usage and Diagnostics use explicit Back. Return to the originating object, not the first row.

Scope and captured state

Task + criterion/claim + immutable candidate + artifact/proof version; landing has its own identity. Selection is a stable entity key, not a row index. Evidence and proof remain bound to the exact snapshot that produced them.

Primary action and recovery

Review missing evidence. Distinguish missing, failed, refused, unavailable and mismatched proof. Never fill a missing required result with a quick check, plan state or completed conversation.

Loading, empty and stale

Keep previous facts clearly stale during refresh. If a proof read fails, remove its verified styling and show the typed unavailable reason.

Keyboard and focus

Arrow keys move the native list/tree; Enter opens detail. Back returns to the exact selected file or criterion. Native diff actions preserve editor focus conventions.

Layout and adverse content

Use a shallow master/detail hierarchy. At narrow widths show one selected detail with Back; open real diffs in the editor instead of squeezing two panes into the tool window.

Starting point and required construction

Starting classifications: E, R. E existing operation; R runtime-backed adapter; N new contract; H host-only work. Exact per-control classifications remain in the catalog.

Reuse the reviewed domain operations and preserve all served fields. Build every missing native read/write adapter, capability gate and state renderer identified by the matching control/build rows.

BK09 owns immutable content, preconditions, claim/candidate binding, integration and finish receipts. Existing read DTOs must retain all proof fields.

Required interaction fixtures

Exercise first-read, empty/no-match, ready, refreshing/stale, error/refusal, action pending and acknowledged result. Also switch scope while the request is in flight.

Moved out of the interface

Every line of revision 7 plate copy that left the default view, and where it lives now.

"Faktor / Inspect / Overview (R7 window title)"
→ Native host chrome (VS Code view title, tabs or editor tab; JetBrains tool-window tabs, dialog title or Settings breadcrumb)

"Run 047 · candidate c-08 · auth-repair"
→ Tooltip / hover help on the title; IDs in Inspector details

"2 passed · 1 needs evidence"
→ "2 of 3 passed"

"One claim needs a current artifact"
→ "1 needs evidence"

"Candidate and integration / Not integrated / c-08 · isolated · not landed"
→ Row "Integration — Candidate not integrated"

"Requested finish / Local · waiting on required proof"
→ Row "Finish · Local — Waiting for checks"

"Continue inspection"
→ Section "Explore"

"3 completed steps · 1 waiting on verification"
→ "3 of 4 steps done"

"2 files in this candidate · native diff"
→ "2 files"; native diff opens from Review

"Return to Work"
→ Work tab

"Inspect the change"
→ Header (task title) on the Inspect tab

"Acceptance criteria / Pending"
→ Row

"Named claims"
→ Row "Claims"

"Plan and dependencies / Open"
→ Row "Plan — Open"

"Changed files / Review"
→ Row "Changed files — Review"

"Review missing evidence"
→ Review Missing Evidence

S42 INSPECT AND PROOF

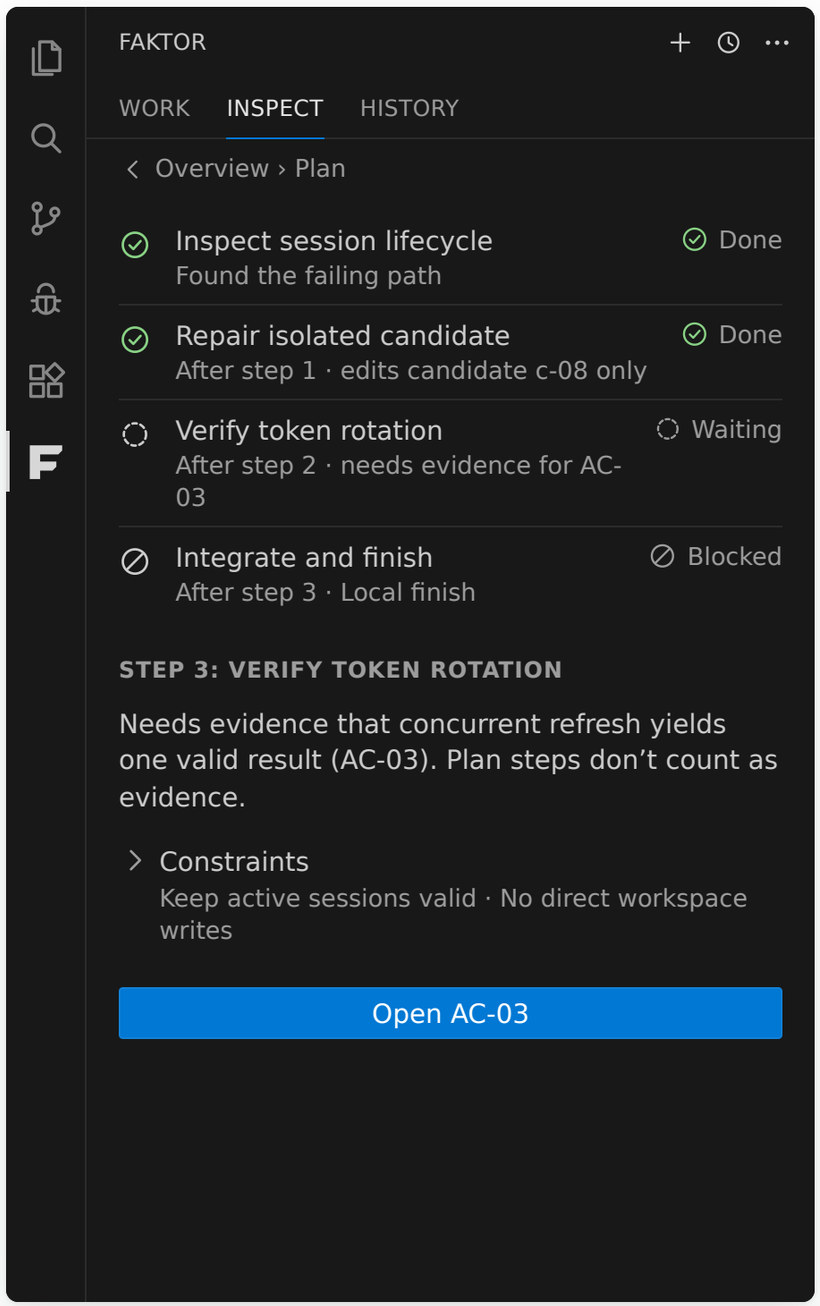
Plan and dependencies

PRIMARY ACTION

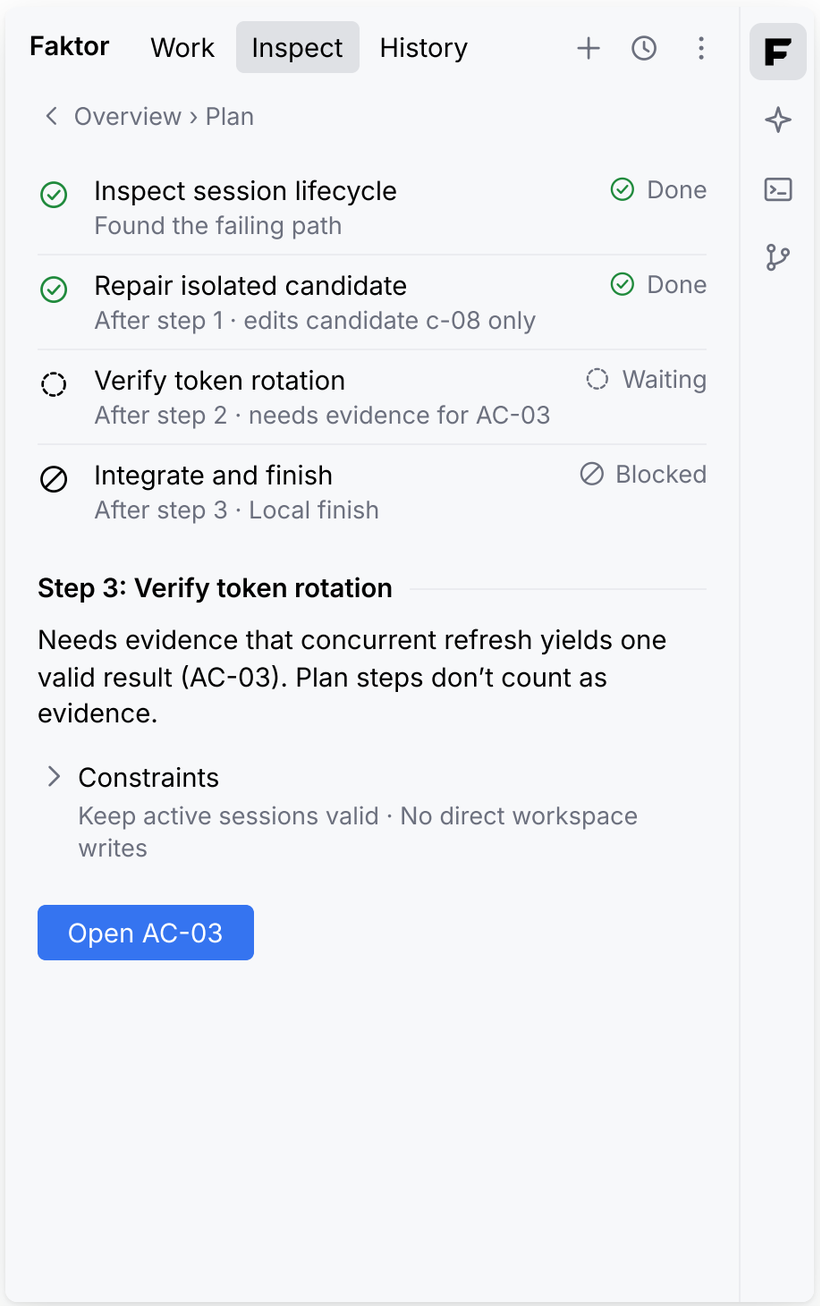
Open AC-03

Inspect the dependency-ordered plan and selected step without confusing a planned checkmark with observed proof.

VS CODE · DARK MODERN — SIDEBAR VIEW · WEBVIEW



JETBRAINS · LIGHT — TOOL WINDOW · SWING / KOTLIN UI DSL



CONTROLS V-07 V-08

REPLACES R7 page 021 · R4-S12

CONTRACT p. 58

S42 · INSPECT AND PROOF · BEHAVIOUR CONTRACT

Plan and dependencies: contract and revision notes

Original contract retained verbatim (R4-S12 / controls: V-07 V-08). Design on p. 57.

What changed in revision 8

- Plan steps use a vertical stepper; Done/Waiting/Blocked carry icon and text.

Entry and return

Inspect overview, a changed file, a criterion or a recorded proof link. Work / Inspect / History remain the only top-level clusters. Settings, Usage and Diagnostics use explicit Back. Return to the originating object, not the first row.

Scope and captured state

Task + criterion/claim + immutable candidate + artifact/proof version; landing has its own identity. Selection is a stable entity key, not a row index. Evidence and proof remain bound to the exact snapshot that produced them.

Primary action and recovery

Open AC-03. Distinguish missing, failed, refused, unavailable and mismatched proof. Never fill a missing required result with a quick check, plan state or completed conversation.

Loading, empty and stale

Keep previous facts clearly stale during refresh. If a proof read fails, remove its verified styling and show the typed unavailable reason.

Keyboard and focus

Arrow keys move the native list/tree; Enter opens detail. Back returns to the exact selected file or criterion. Native diff actions preserve editor focus conventions.

Layout and adverse content

Use a shallow master/detail hierarchy. At narrow widths show one selected detail with Back; open real diffs in the editor instead of squeezing two panes into the tool window.

Starting point and required construction

Starting classifications: E, R. E existing operation; R runtime-backed adapter; N new contract; H host-only work. Exact per-control classifications remain in the catalog.

Reuse the reviewed domain operations and preserve all served fields. Build every missing native read/write adapter, capability gate and state renderer identified by the matching control/build rows.

BK09 owns immutable content, preconditions, claim/candidate binding, integration and finish receipts. Existing read DTOs must retain all proof fields.

Required interaction fixtures

Exercise first-read, empty/no-match, ready, refreshing/stale, error/refusal, action pending and acknowledged result. Also switch scope while the request is in flight.

Moved out of the interface

Every line of revision 7 plate copy that left the default view, and where it lives now.

"Faktor / Inspect / Plan (R7 window title)"
→ Native host chrome (VS Code view title, tabs or editor tab; JetBrains tool-window tabs, dialog title or Settings breadcrumb)

"Run 047 · plan version 6 · captured task contract"
→ Inspector details

"Read current behavior and identify the failure"
→ Step meta "Found the failing path"

"Depends on 1 · edit only candidate c-08"
→ "After step 1 · edits candidate c-08 only"

"Depends on 2 · AC-03 still has no current artifact"
→ "After step 2 · needs evidence for AC-03"

"Depends on 3 · Local finish"
→ "After step 3 · Local finish"

"Required: concurrent refresh yields one valid result. A plan checkmark cannot satisfy this criterion."
→ Step detail (shortened); rule in Specification: screen contract

"Contract and constraints / Inspect / Keep active sessions valid · no owner-checkout writes"
→ Constraints disclosure ("No direct workspace writes")

"Return to run"
→ Breadcrumb back

"Plan and dependencies"
→ Breadcrumb "Plan"

"1 Inspect session lifecycle / Complete"
→ Step 1 — Done

"2 Repair isolated candidate / Complete"
→ Step 2 — Done

"3 Verify token rotation / Waiting"
→ Step 3 — Waiting

"4 Integrate and finish / Blocked"
→ Step 4 — Blocked

"Selected step: verify token rotation"
→ Section "Step 3: Verify token rotation"

"Open AC-03"
→ Open AC-03

S24 INSPECT AND PROOF

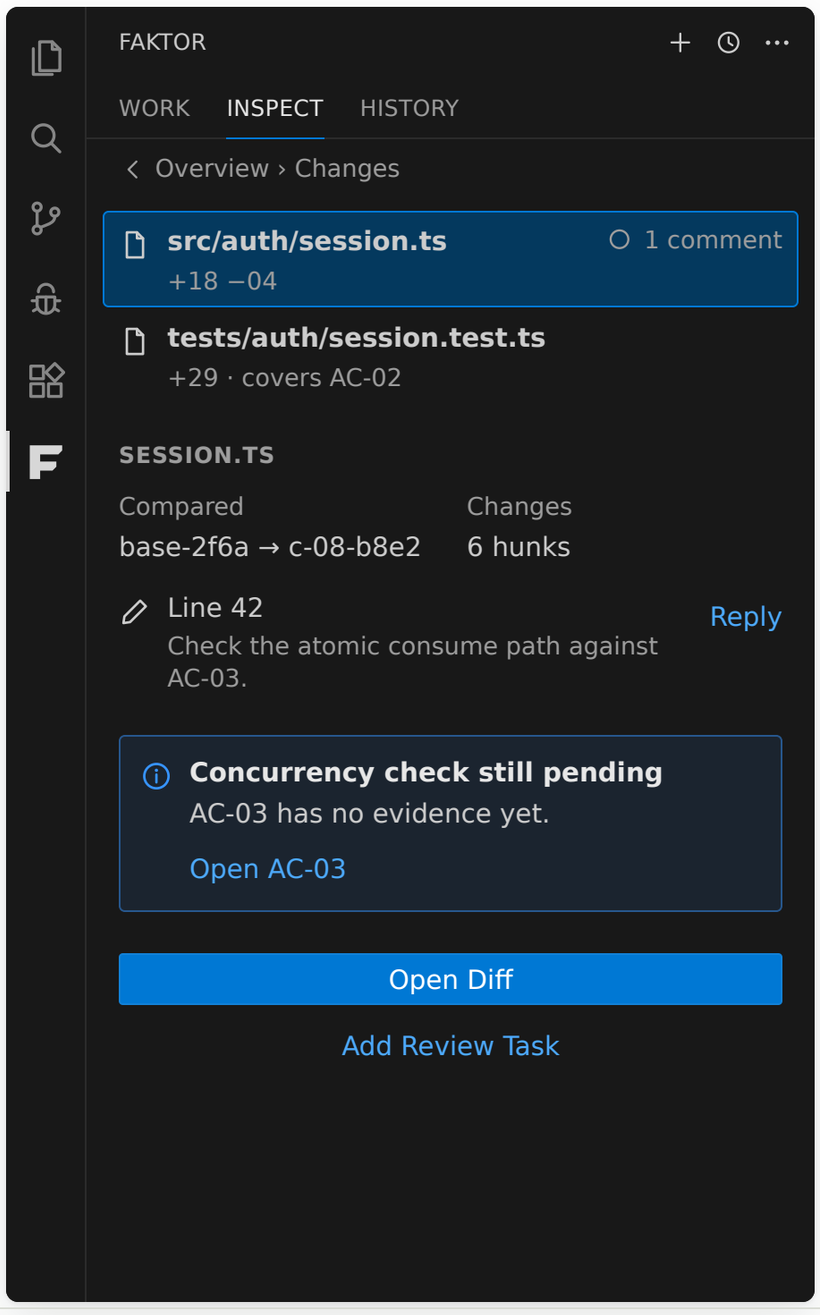
Review changes

PRIMARY ACTION

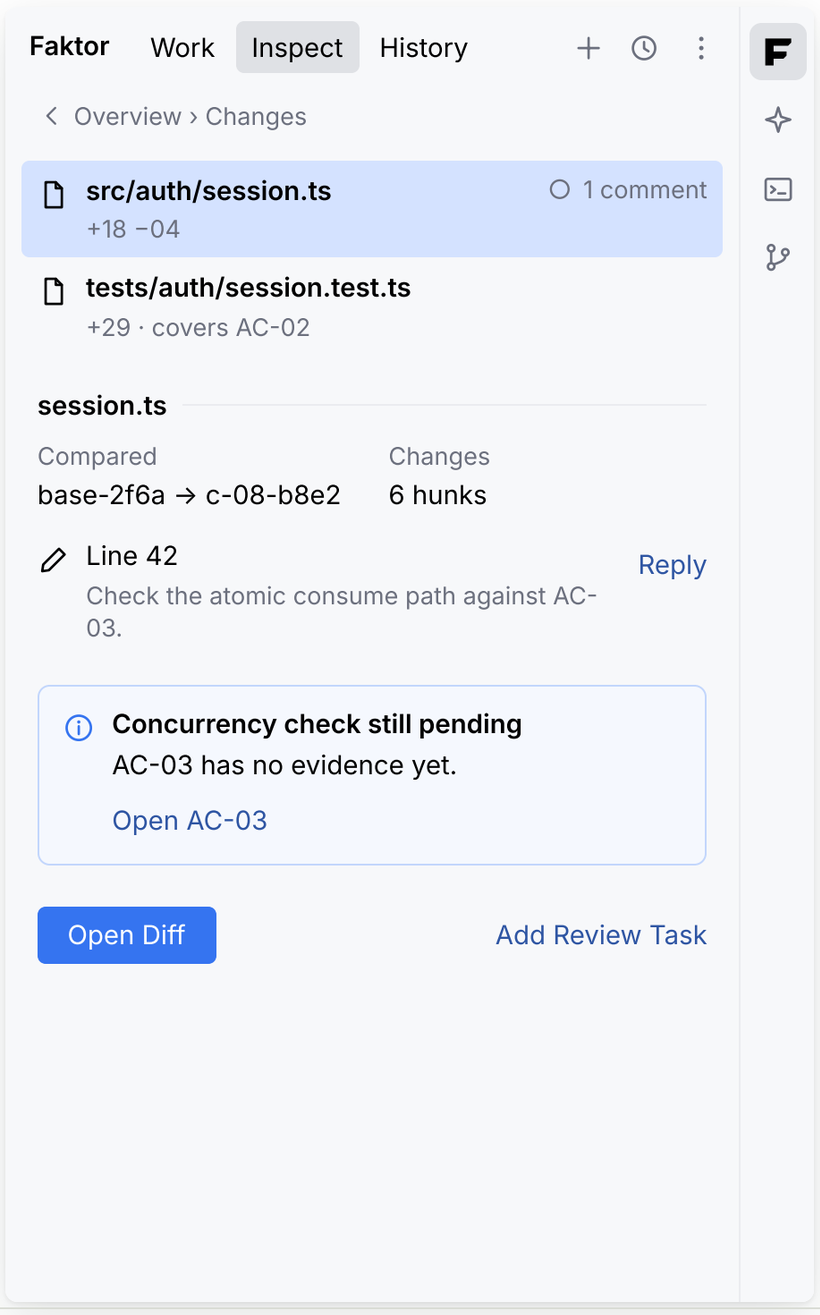
Open Diff

Use each host's native diff editor. Keep base/candidate identity, acceptance criteria and remaining verification adjacent to code review.

VS CODE · DARK MODERN — SIDEBAR VIEW · WEBVIEW



JETBRAINS · LIGHT — TOOL WINDOW · SWING / KOTLIN UI DSL



CONTROLS V-01 V-02 V-03 V-07 V-09 V-11 V-12 A-11

REPLACES R7 page 022 · R4-S13

CONTRACT p. 60

S24 · INSPECT AND PROOF · BEHAVIOUR CONTRACT

Review changes: contract and revision notes

Original contract retained verbatim (R4-S13 / controls: V-01 V-02 V-03 V-07 V-09 V-11 V-12 A-11). Design on p. 59.

What changed in revision 8

- The file list and the selected file read as one master-detail; the diff itself opens in the native editor.

Entry and return

Inspect overview, a changed file, a criterion or a recorded proof link. Work / Inspect / History remain the only top-level clusters. Settings, Usage and Diagnostics use explicit Back. Return to the originating object, not the first row.

Scope and captured state

Task + criterion/claim + immutable candidate + artifact/proof version; landing has its own identity. Selection is a stable entity key, not a row index. Evidence and proof remain bound to the exact snapshot that produced them.

Primary action and recovery

Open a file/hunk, attach a versioned comment or prepare a new bounded task. Review marks alone do not integrate or publish.. Manual owner edits trigger compare-and-swap refusal. A changed or partially selected candidate needs matching new evidence.

Loading, empty and stale

Keep previous facts clearly stale during refresh. If a proof read fails, remove its verified styling and show the typed unavailable reason.

Keyboard and focus

Arrow keys move the native list/tree; Enter opens detail. Back returns to the exact selected file or criterion. Native diff actions preserve editor focus conventions.

Layout and adverse content

Use a shallow master/detail hierarchy. At narrow widths show one selected detail with Back; open real diffs in the editor instead of squeezing two panes into the tool window.

Starting point and required construction

Starting classifications: E, N, R. E existing operation; R runtime-backed adapter; N new contract; H host-only work. Exact per-control classifications remain in the catalog.

Reuse the reviewed domain operations and preserve all served fields. Build every missing native read/write adapter, capability gate and state renderer identified by the matching control/build rows.

BK09 owns immutable content, preconditions, claim/candidate binding, integration and finish receipts. Existing read DTOs must retain all proof fields.

Required interaction fixtures

Exercise first-read, empty/no-match, ready, refreshing/stale, error/refusal, action pending and acknowledged result. Also switch scope while the request is in flight.

Moved out of the interface

Every line of revision 7 plate copy that left the default view, and where it lives now.

"Faktor / Inspect / Changes (R7 window title)"
→ Native host chrome (VS Code view title, tabs or editor tab; JetBrains tool-window tabs, dialog title or Settings breadcrumb)

"Run 047 · candidate c-08 / generation 4"
→ Inspector details

"18 additions · 4 deletions"
→ "+18 −04"

"29 additions · acceptance criterion AC-02"
→ "+29 · covers AC-02"

"Selected file / Comparison basis / Change summary"
→ Section titled with the file name; "Compared" and "Changes" facts

"base-2f6a / c-08-b8e2 / 6 hunks · isolated candidate"
→ Facts

"Related acceptance / Inspect / Replay protection and active-session continuity"
→ File row meta "covers AC-02" and the info banner link

"Review comment / line 42 / 1 open"
→ Comment row "Line 42"

"Required concurrency proof is pending"
→ Info banner

"Review marks do not integrate or publish a change. Open the native editor to inspect the full immutable diff."
→ Specification: screen contract (review vs. integration); Open Diff action

"Open native diff"
→ Open Diff

"Review changes"
→ Breadcrumb "Changes"

"src/auth/session.ts / Selected"
→ Selected file row

"Add review task"
→ Add Review Task

S43 INSPECT AND PROOF

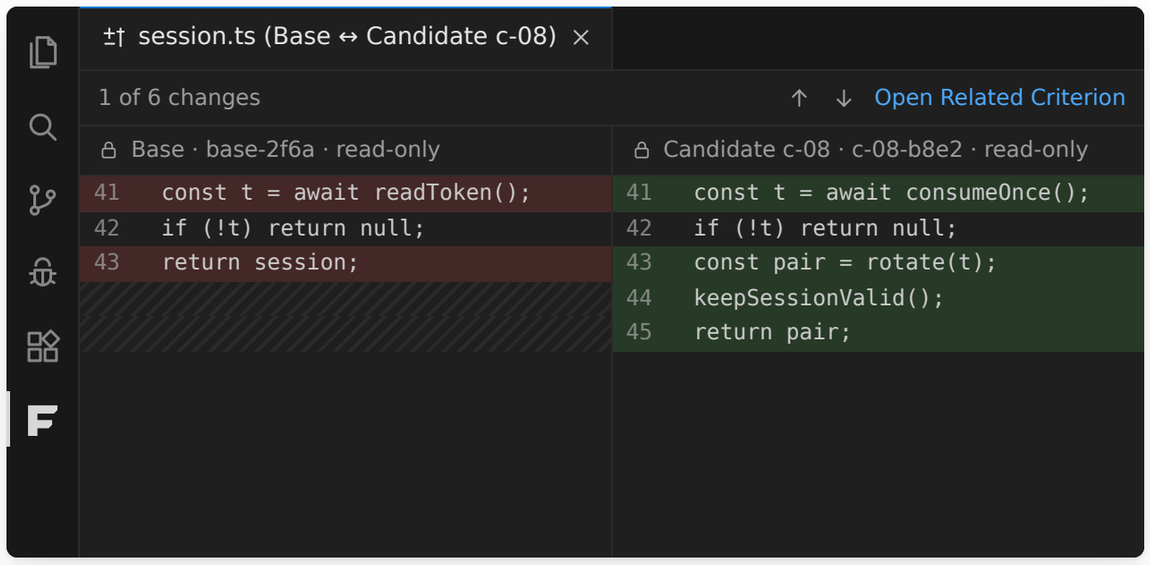
Native diff review

Review the immutable base and selected candidate in the native editor with exact hunk and claim navigation.

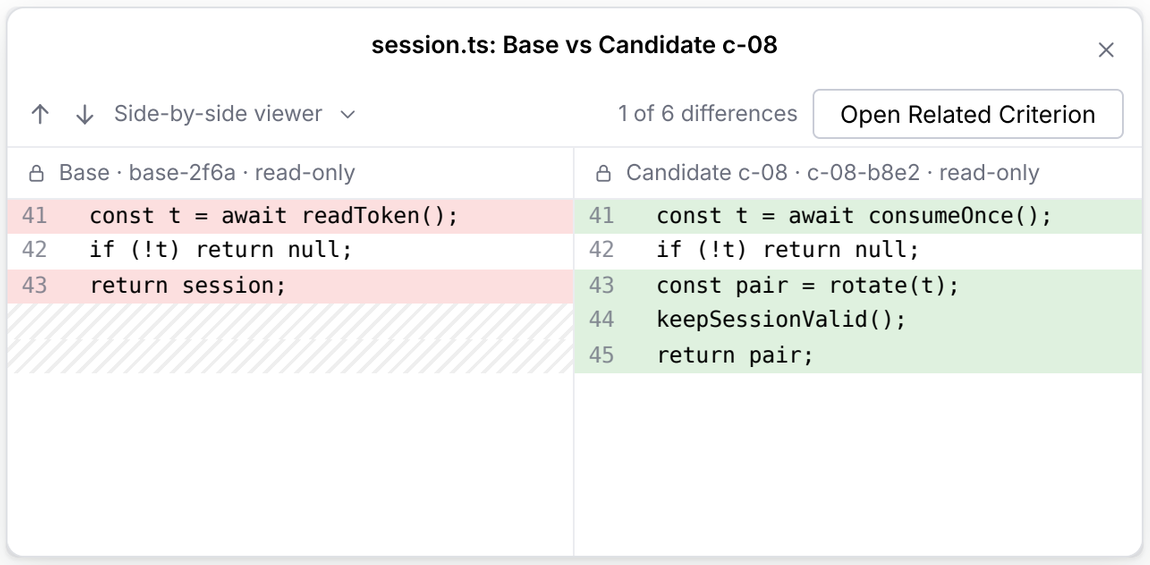
PRIMARY ACTION

Open Related Criterion

VS CODE · DARK MODERN — DIFF EDITOR · VSCODE.DIFF



JETBRAINS · LIGHT — DIFF VIEWER · DIFFMANAGER



CONTROLS V-01 V-03

REPLACES R7 page 023 · R4-S14

CONTRACT p. 62

S43 · INSPECT AND PROOF · BEHAVIOUR CONTRACT

Native diff review: contract and revision notes

Original contract retained verbatim (R4-S14 / controls: V-01 V-03). Design on p. 61.

What changed in revision 8

- The plate is the host's real diff viewer (vscode.diff / DiffManager), with one Faktor action added to its toolbar.

Entry and return

Inspect overview, a changed file, a criterion or a recorded proof link. Work / Inspect / History remain the only top-level clusters. Settings, Usage and Diagnostics use explicit Back. Return to the originating object, not the first row.

Scope and captured state

Task + criterion/claim + immutable candidate + artifact/proof version; landing has its own identity. Selection is a stable entity key, not a row index. Evidence and proof remain bound to the exact snapshot that produced them.

Primary action and recovery

Open related criterion. Distinguish missing, failed, refused, unavailable and mismatched proof. Never fill a missing required result with a quick check, plan state or completed conversation.

Loading, empty and stale

Keep previous facts clearly stale during refresh. If a proof read fails, remove its verified styling and show the typed unavailable reason.

Keyboard and focus

Arrow keys move the native list/tree; Enter opens detail. Back returns to the exact selected file or criterion. Native diff actions preserve editor focus conventions.

Layout and adverse content

Use a shallow master/detail hierarchy. At narrow widths show one selected detail with Back; open real diffs in the editor instead of squeezing two panes into the tool window.

Starting point and required construction

Starting classifications: E, N. E existing operation; R runtime-backed adapter; N new contract; H host-only work. Exact per-control classifications remain in the catalog.

Reuse the reviewed domain operations and preserve all served fields. Build every missing native read/write adapter, capability gate and state renderer identified by the matching control/build rows.

BK09 owns immutable content, preconditions, claim/candidate binding, integration and finish receipts. Existing read DTOs must retain all proof fields.

Required interaction fixtures

Exercise first-read, empty/no-match, ready, refreshing/stale, error/refusal, action pending and acknowledged result. Also switch scope while the request is in flight.

Moved out of the interface

Every line of revision 7 plate copy that left the default view, and where it lives now.

"Faktor / Inspect / Changes (R7 window title)"
→ Native host chrome (VS Code view title, tabs or editor tab; JetBrains tool-window tabs, dialog title or Settings breadcrumb)

"Native editor comparison · candidate c-08 · immutable snapshots"
→ Editor tab title and read-only pane titles

"Base / base-2f6a · read only; Candidate / c-08-b8e2 · read only"
→ Pane titles

"Base snapshot / Candidate snapshot"
→ Pane titles

"The full native editor provides hunk navigation, copy, accessibility and exact source identities. Unrelated working-tree edits are outside this comparison."
→ Specification: screen contract (why the native diff is required)

"Back to files / Open related criterion"
→ Editor tab close / Open Related Criterion action

"session.ts"
→ Diff editor tab title

S45 INSPECT AND PROOF

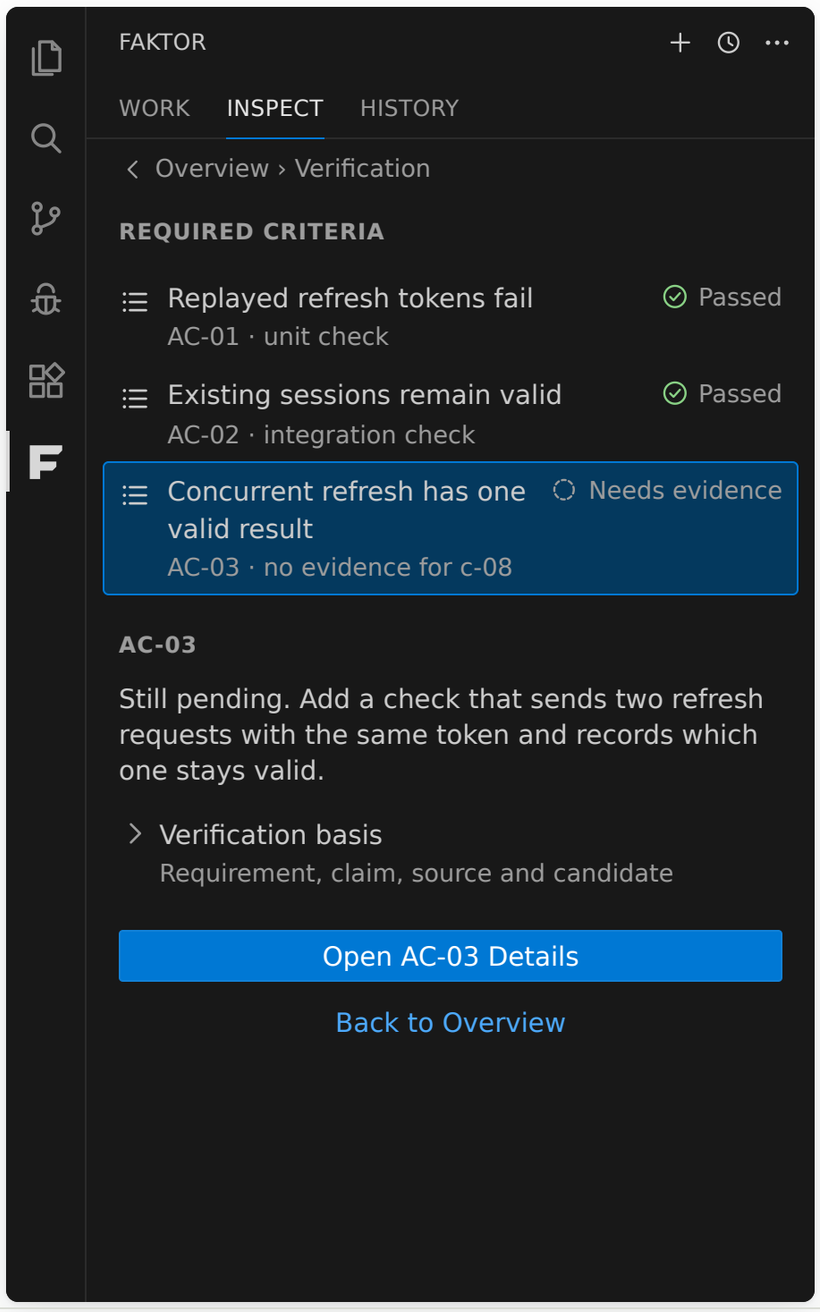
Required verification

Inspect every required criterion, its current evidence and explicit pending, failed, refused or unavailable result.

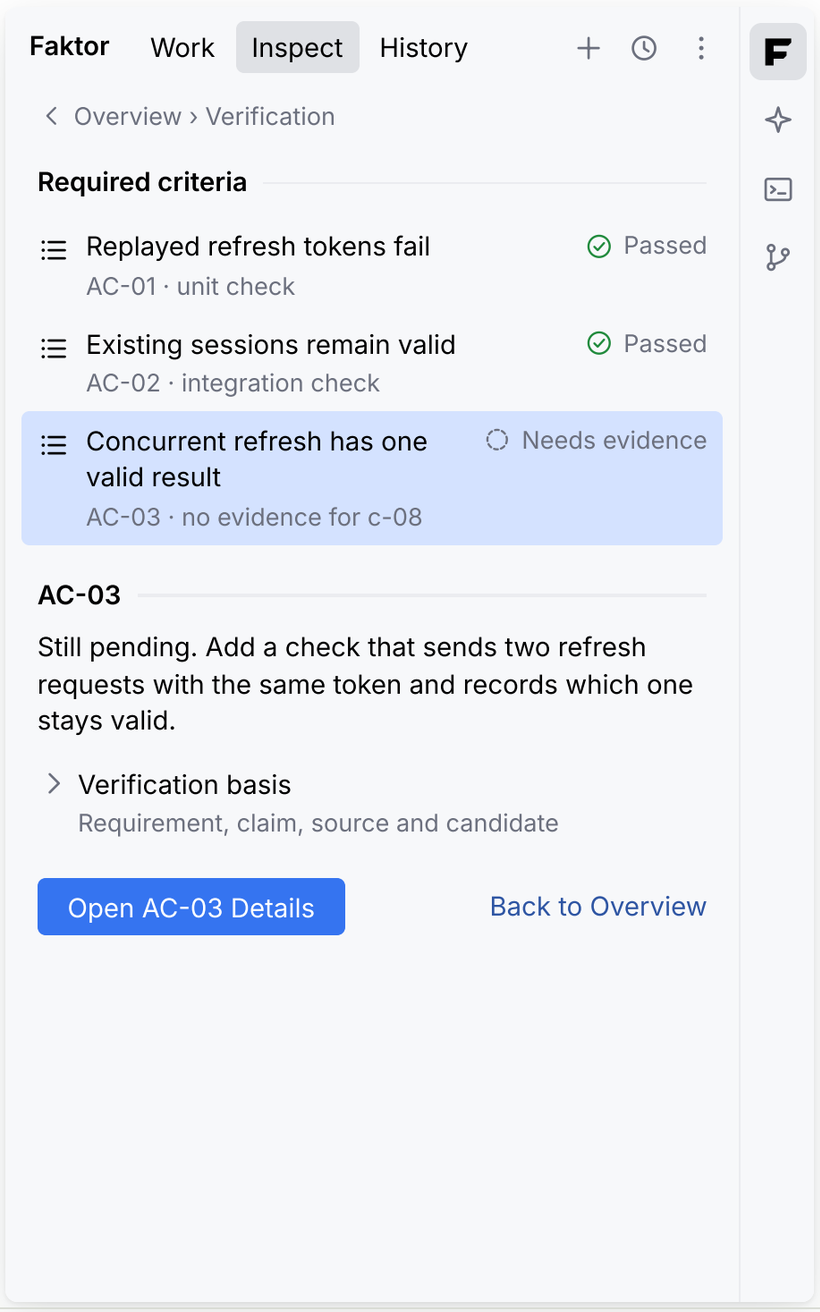
PRIMARY ACTION

Open AC-03 Details

VS CODE · DARK MODERN — SIDEBAR VIEW · WEBVIEW



JETBRAINS · LIGHT — TOOL WINDOW · SWING / KOTLIN UI DSL



CONTROLS V-07 V-08 V-09

REPLACES R7 page 024 · R4-S16

CONTRACT p. 64

S45 · INSPECT AND PROOF · BEHAVIOUR CONTRACT

Required verification: contract and revision notes

Original contract retained verbatim (R4-S16 / controls: V-07 V-08 V-09). Design on p. 63.

What changed in revision 8

- The pending criterion now says what evidence would satisfy it instead of what cannot.

Entry and return

Inspect overview, a changed file, a criterion or a recorded proof link. Work / Inspect / History remain the only top-level clusters. Settings, Usage and Diagnostics use explicit Back. Return to the originating object, not the first row.

Scope and captured state

Task + criterion/claim + immutable candidate + artifact/proof version; landing has its own identity. Selection is a stable entity key, not a row index. Evidence and proof remain bound to the exact snapshot that produced them.

Primary action and recovery

Open criterion detail. Distinguish missing, failed, refused, unavailable and mismatched proof. Never fill a missing required result with a quick check, plan state or completed conversation.

Loading, empty and stale

Keep previous facts clearly stale during refresh. If a proof read fails, remove its verified styling and show the typed unavailable reason.

Keyboard and focus

Arrow keys move the native list/tree; Enter opens detail. Back returns to the exact selected file or criterion. Native diff actions preserve editor focus conventions.

Layout and adverse content

Use a shallow master/detail hierarchy. At narrow widths show one selected detail with Back; open real diffs in the editor instead of squeezing two panes into the tool window.

Starting point and required construction

Starting classifications: E, R. E existing operation; R runtime-backed adapter; N new contract; H host-only work. Exact per-control classifications remain in the catalog.

Reuse the reviewed domain operations and preserve all served fields. Build every missing native read/write adapter, capability gate and state renderer identified by the matching control/build rows.

BK09 owns immutable content, preconditions, claim/candidate binding, integration and finish receipts. Existing read DTOs must retain all proof fields.

Required interaction fixtures

Exercise first-read, empty/no-match, ready, refreshing/stale, error/refusal, action pending and acknowledged result. Also switch scope while the request is in flight.

Moved out of the interface

Every line of revision 7 plate copy that left the default view, and where it lives now.

"Faktor / Inspect / Verification (R7 window title)"
→ Native host chrome (VS Code view title, tabs or editor tab; JetBrains tool-window tabs, dialog title or Settings breadcrumb)

"Candidate c-08 · current criterion and artifact bindings"
→ Row meta "no evidence for c-08"; bindings in Verification basis

"AC-01 · unit check · evidence e-101"
→ "AC-01 · unit check"; evidence ID in S46

"AC-02 · integration check · evidence e-102"
→ "AC-02 · integration check"

"AC-03 · no current artifact is served"
→ "AC-03 · no evidence for c-08"

"Selected criterion: AC-03"
→ Section title "AC-03"

"The requirement is still pending. An empty or quick test suite cannot fill its missing evidence."
→ Actionable next step; the rule is in Specification: screen contract

"Requiredness, claim, source and candidate details"
→ Verification basis disclosure

"Open criterion detail"
→ Open AC-03 Details

"Required verification"
→ Breadcrumb "Verification"; section "Required criteria"

"Passed / Needs evidence"
→ Statuses

"Back to overview"
→ Back to Overview

"Verification basis / Inspect"
→ Disclosure

S46 INSPECT AND PROOF

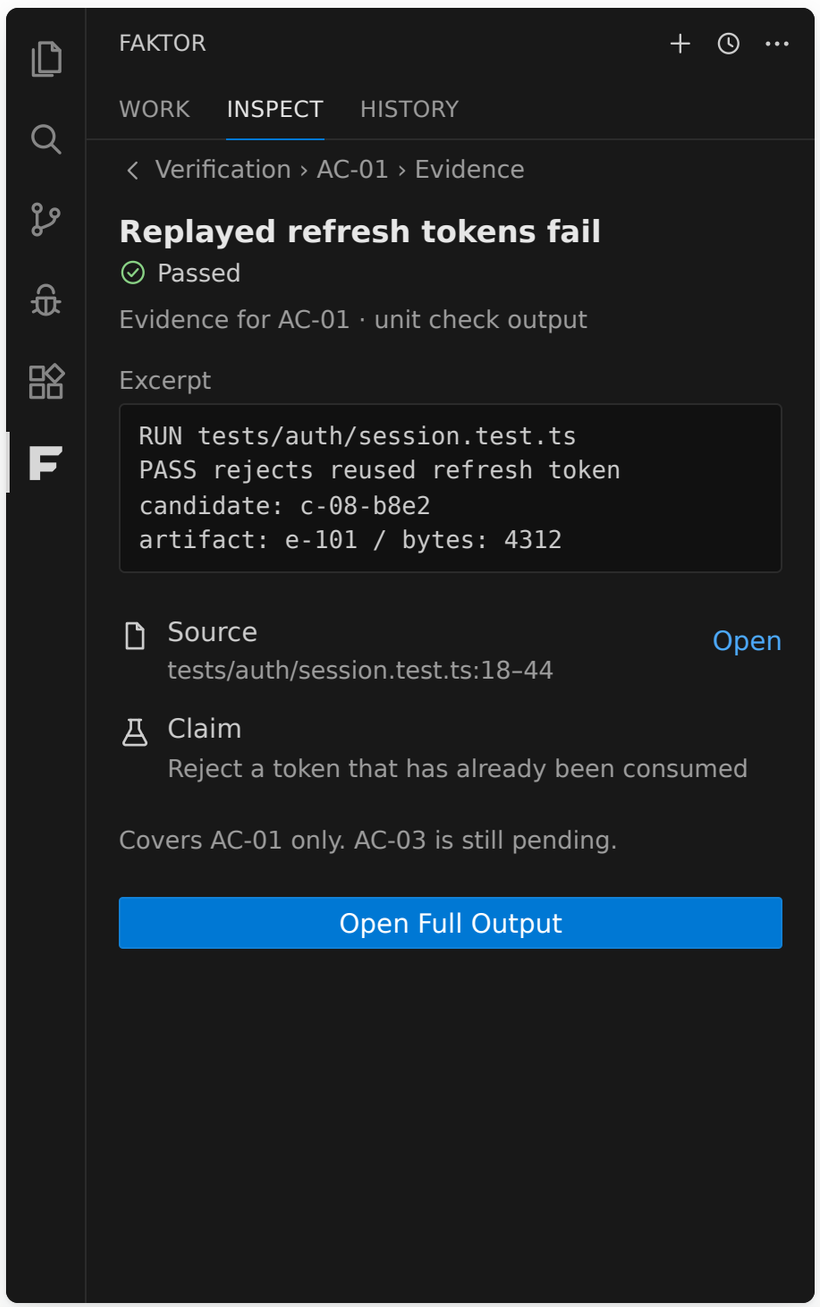
Evidence detail

Follow one criterion to its exact artifact, source, claim and candidate; open bounded excerpts and full content.

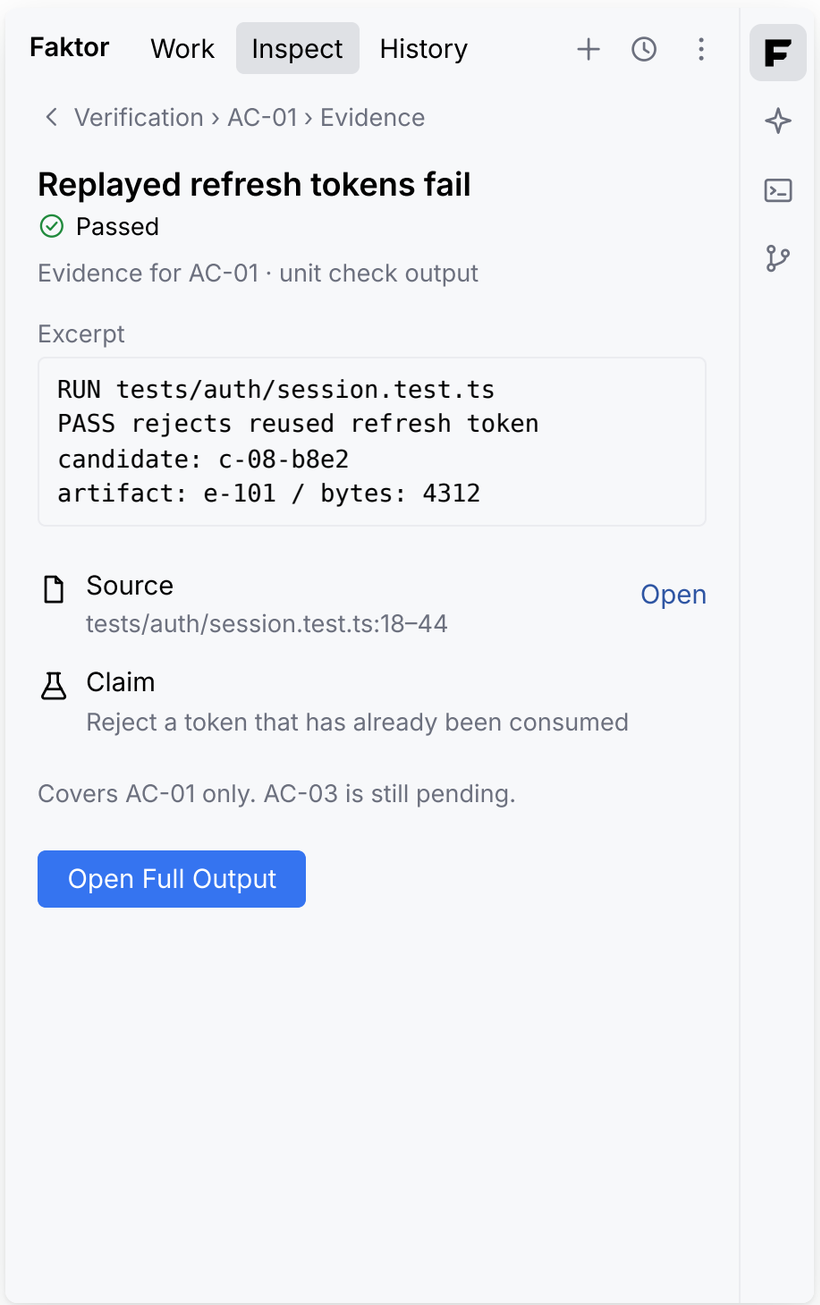
PRIMARY ACTION

Open Full Output

VS CODE · DARK MODERN — SIDEBAR VIEW · WEBVIEW



JETBRAINS · LIGHT — TOOL WINDOW · SWING / KOTLIN UI DSL



CONTROLS V-07 V-08

REPLACES R7 page 025 · R4-S17

CONTRACT p. 66

S46 · INSPECT AND PROOF · BEHAVIOUR CONTRACT

Evidence detail: contract and revision notes

Original contract retained verbatim (R4-S17 / controls: V-07 V-08). Design on p. 65.

What changed in revision 8

- The excerpt is the content; bindings are two quiet rows.

Entry and return

Inspect overview, a changed file, a criterion or a recorded proof link. Work / Inspect / History remain the only top-level clusters. Settings, Usage and Diagnostics use explicit Back. Return to the originating object, not the first row.

Scope and captured state

Task + criterion/claim + immutable candidate + artifact/proof version; landing has its own identity. Selection is a stable entity key, not a row index. Evidence and proof remain bound to the exact snapshot that produced them.

Primary action and recovery

Open full artifact. Distinguish missing, failed, refused, unavailable and mismatched proof. Never fill a missing required result with a quick check, plan state or completed conversation.

Loading, empty and stale

Keep previous facts clearly stale during refresh. If a proof read fails, remove its verified styling and show the typed unavailable reason.

Keyboard and focus

Arrow keys move the native list/tree; Enter opens detail. Back returns to the exact selected file or criterion. Native diff actions preserve editor focus conventions.

Layout and adverse content

Use a shallow master/detail hierarchy. At narrow widths show one selected detail with Back; open real diffs in the editor instead of squeezing two panes into the tool window.

Starting point and required construction

Starting classifications: E, R. E existing operation; R runtime-backed adapter; N new contract; H host-only work. Exact per-control classifications remain in the catalog.

Reuse the reviewed domain operations and preserve all served fields. Build every missing native read/write adapter, capability gate and state renderer identified by the matching control/build rows.

BK09 owns immutable content, preconditions, claim/candidate binding, integration and finish receipts. Existing read DTOs must retain all proof fields.

Required interaction fixtures

Exercise first-read, empty/no-match, ready, refreshing/stale, error/refusal, action pending and acknowledged result. Also switch scope while the request is in flight.

Moved out of the interface

Every line of revision 7 plate copy that left the default view, and where it lives now.

"Faktor / Inspect / Evidence (R7 window title)"
→ Native host chrome (VS Code view title, tabs or editor tab; JetBrains tool-window tabs, dialog title or Settings breadcrumb)

"Evidence e-101"
→ Excerpt line "artifact: e-101"

"AC-01 / replayed refresh tokens fail / candidate c-08"
→ Title and subtitle

"Artifact kind / Unit-check output; Result / Passed"
→ Subtitle and status

"Bounded excerpt · original artifact retained"
→ Label "Excerpt" + Open Full Output

"Source binding / … immutable test version"
→ Source row; version in Tooltip / hover help

"Claim binding"
→ Claim row

"Artifact coverage / This result supports AC-01 only. AC-03 remains pending."
→ Coverage note

"Return to AC-01"
→ Breadcrumb

"tests/auth/session.test.ts:18-44 · immutable test version"
→ Source row "tests/auth/session.test.ts:18–44"

"Open full artifact"
→ Open Full Output

"Reject a token that has already been consumed"
→ Claim row

S47 INSPECT AND PROOF

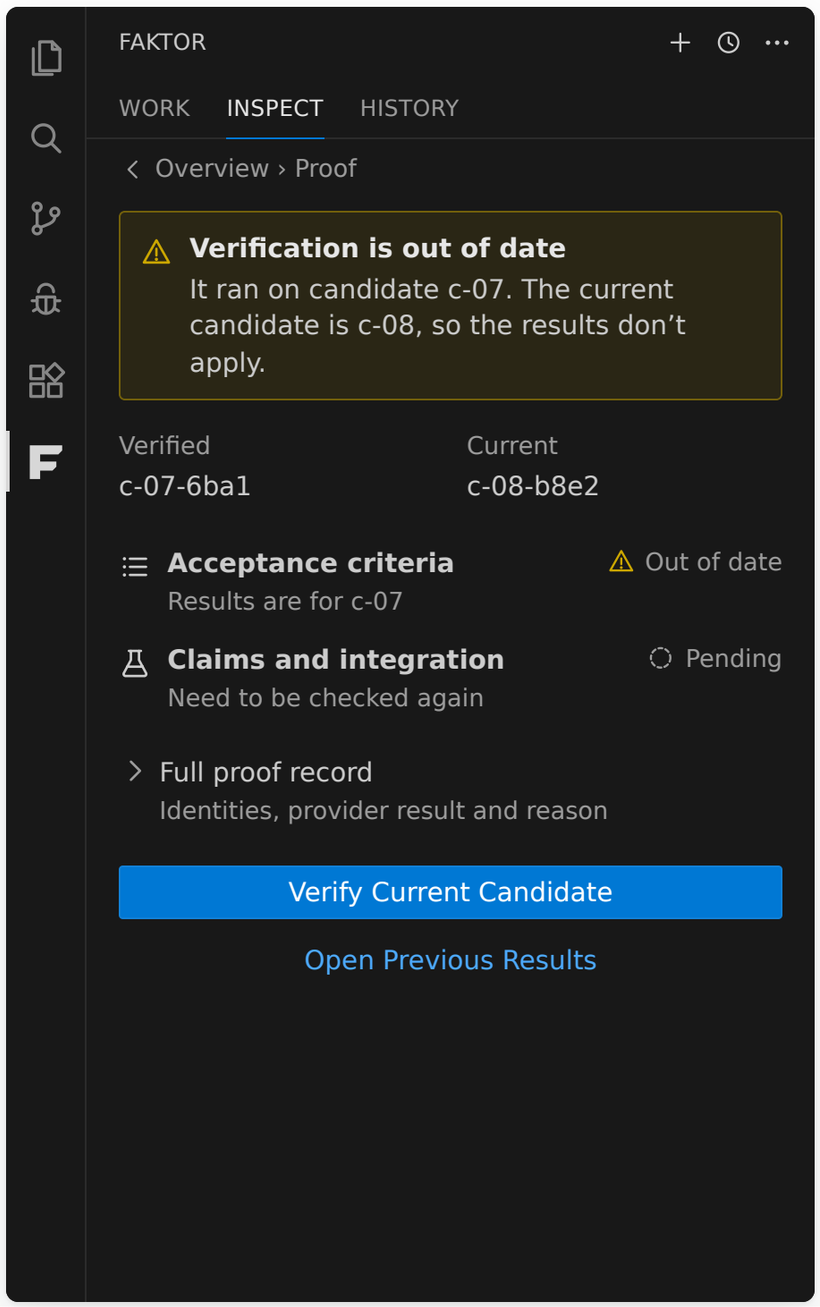
Proof identity mismatch

Remove stale verified presentation when proof belongs to a different candidate; retain the historical record separately.

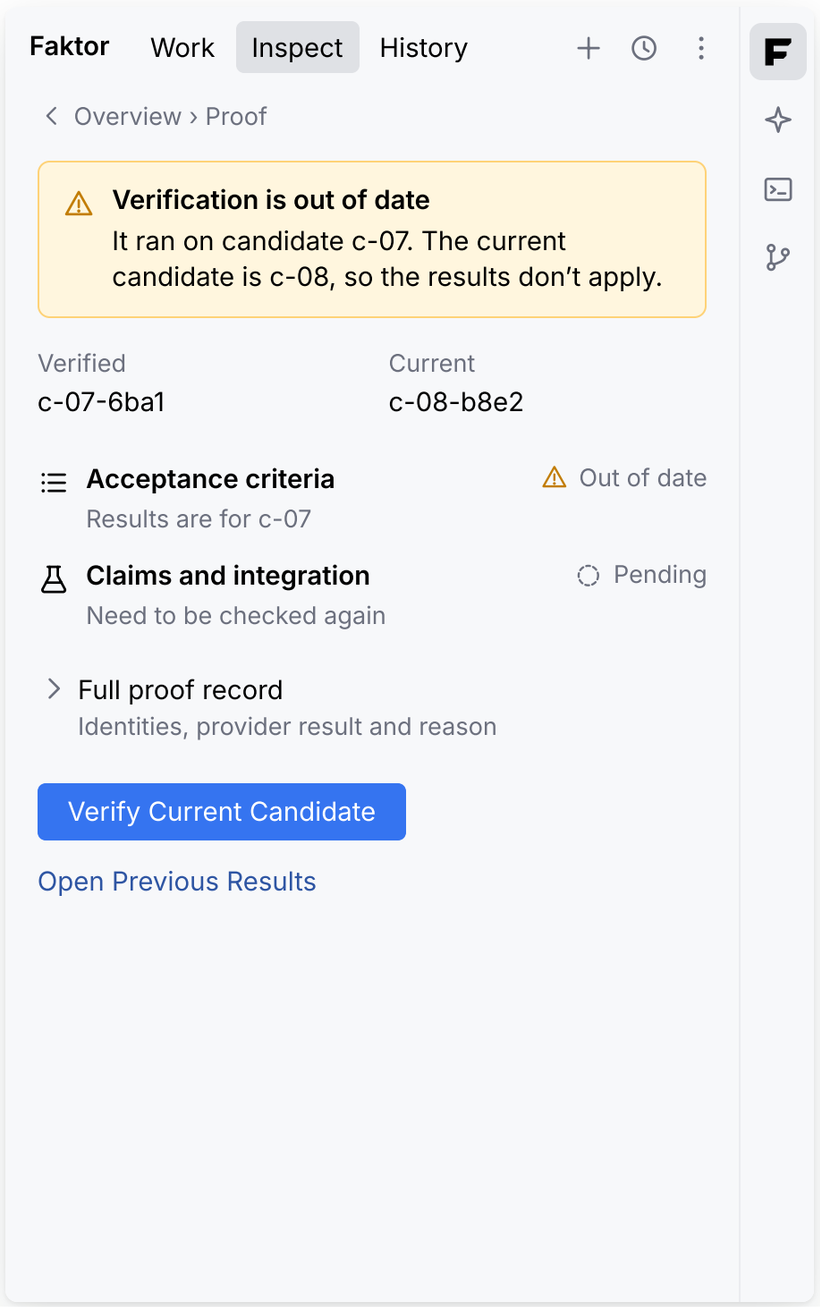
PRIMARY ACTION

Verify Current Candidate

VS CODE · DARK MODERN — SIDEBAR VIEW · WEBVIEW



JETBRAINS · LIGHT — TOOL WINDOW · SWING / KOTLIN UI DSL



CONTROLS V-09 V-12

REPLACES R7 page 026 · R4-S18

CONTRACT p. 68

S47 · INSPECT AND PROOF · BEHAVIOUR CONTRACT

Proof identity mismatch: contract and revision notes

Original contract retained verbatim (R4-S18 / controls: V-09 V-12). Design on p. 67.

What changed in revision 8

- "Stale" becomes "Out of date", with the consequence stated plainly.

Entry and return

Inspect overview, a changed file, a criterion or a recorded proof link. Work / Inspect / History remain the only top-level clusters. Settings, Usage and Diagnostics use explicit Back. Return to the originating object, not the first row.

Scope and captured state

Task + criterion/claim + immutable candidate + artifact/proof version; landing has its own identity. Selection is a stable entity key, not a row index. Evidence and proof remain bound to the exact snapshot that produced them.

Primary action and recovery

Review current verification. Distinguish missing, failed, refused, unavailable and mismatched proof. Never fill a missing required result with a quick check, plan state or completed conversation.

Loading, empty and stale

Keep previous facts clearly stale during refresh. If a proof read fails, remove its verified styling and show the typed unavailable reason.

Keyboard and focus

Arrow keys move the native list/tree; Enter opens detail. Back returns to the exact selected file or criterion. Native diff actions preserve editor focus conventions.

Layout and adverse content

Use a shallow master/detail hierarchy. At narrow widths show one selected detail with Back; open real diffs in the editor instead of squeezing two panes into the tool window.

Starting point and required construction

Starting classifications: E. E existing operation; R runtime-backed adapter; N new contract; H host-only work. Exact per-control classifications remain in the catalog.

Reuse the reviewed domain operations and preserve all served fields. Build every missing native read/write adapter, capability gate and state renderer identified by the matching control/build rows.

BK09 owns immutable content, preconditions, claim/candidate binding, integration and finish receipts. Existing read DTOs must retain all proof fields.

Required interaction fixtures

Exercise first-read, empty/no-match, ready, refreshing/stale, error/refusal, action pending and acknowledged result. Also switch scope while the request is in flight.

Moved out of the interface

Every line of revision 7 plate copy that left the default view, and where it lives now.

"Faktor / Inspect / Proof (R7 window title)"
→ Native host chrome (VS Code view title, tabs or editor tab; JetBrains tool-window tabs, dialog title or Settings breadcrumb)

"Proof no longer matches / The latest candidate differs from the verified snapshot."
→ Banner title and text

"Do not present this task as verified"
→ Specification: screen contract (the UI never shows Verified here)

"The served proof belongs to c-07. Current review is c-08. Keep the historical record, and verify the new candidate."
→ Banner text and the two actions

"Proof candidate / Selected candidate"
→ Facts "Verified" and "Current"

"Required acceptance / Stale / Historical c-07 results cannot certify c-08"
→ Row "Out of date — Results are for c-07"

"Candidate, landed identity and finish must be re-bound"
→ Row "Need to be checked again"

"Raw identities, provider result and reason"
→ Disclosure

"Open historical proof / Review current verification"
→ Open Previous Results / Verify Current Candidate

"Proof candidate / Selected candidate / c-07-6ba1 / c-08-b8e2"
→ Facts "Verified" / "Current"

S44 INSPECT AND PROOF

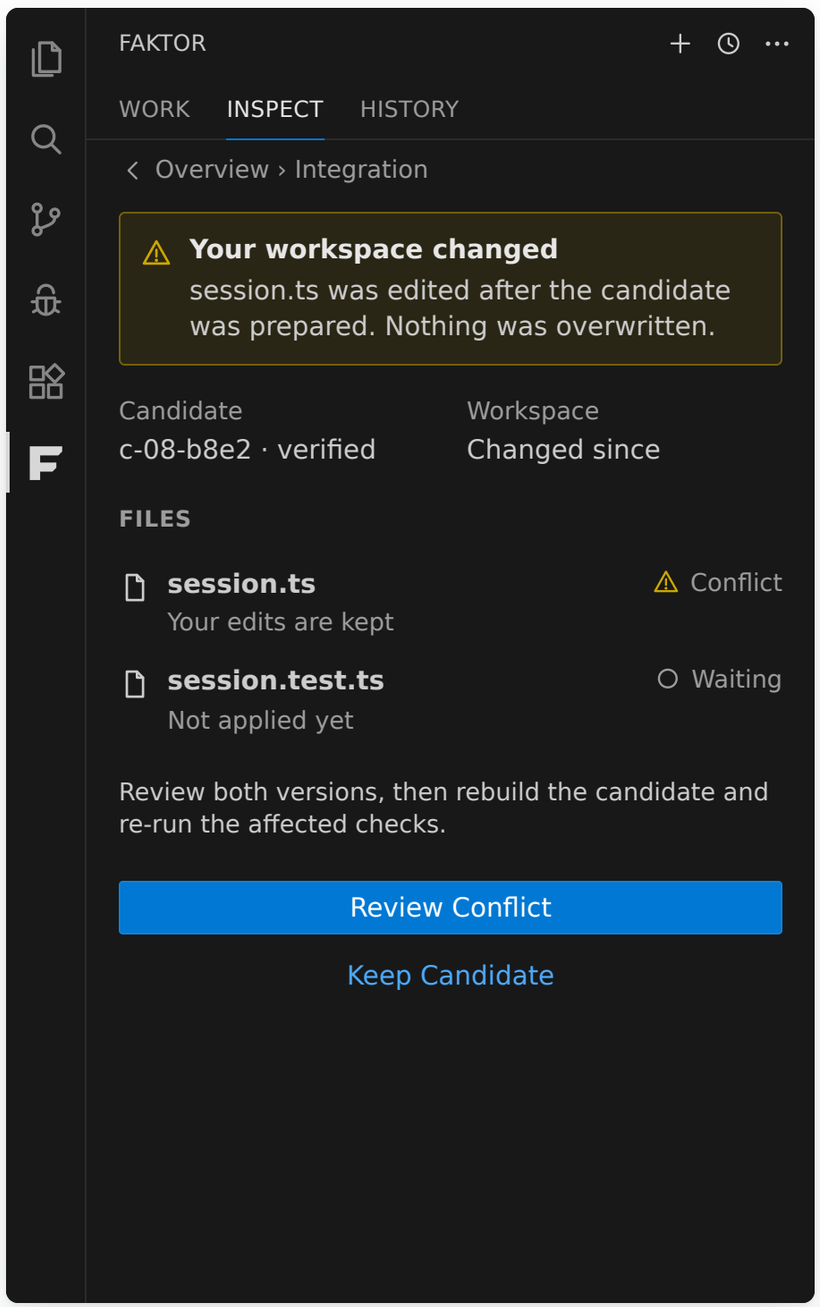
Integration conflict

Preserve user edits when landing preconditions change; review a new candidate and rerun affected verification.

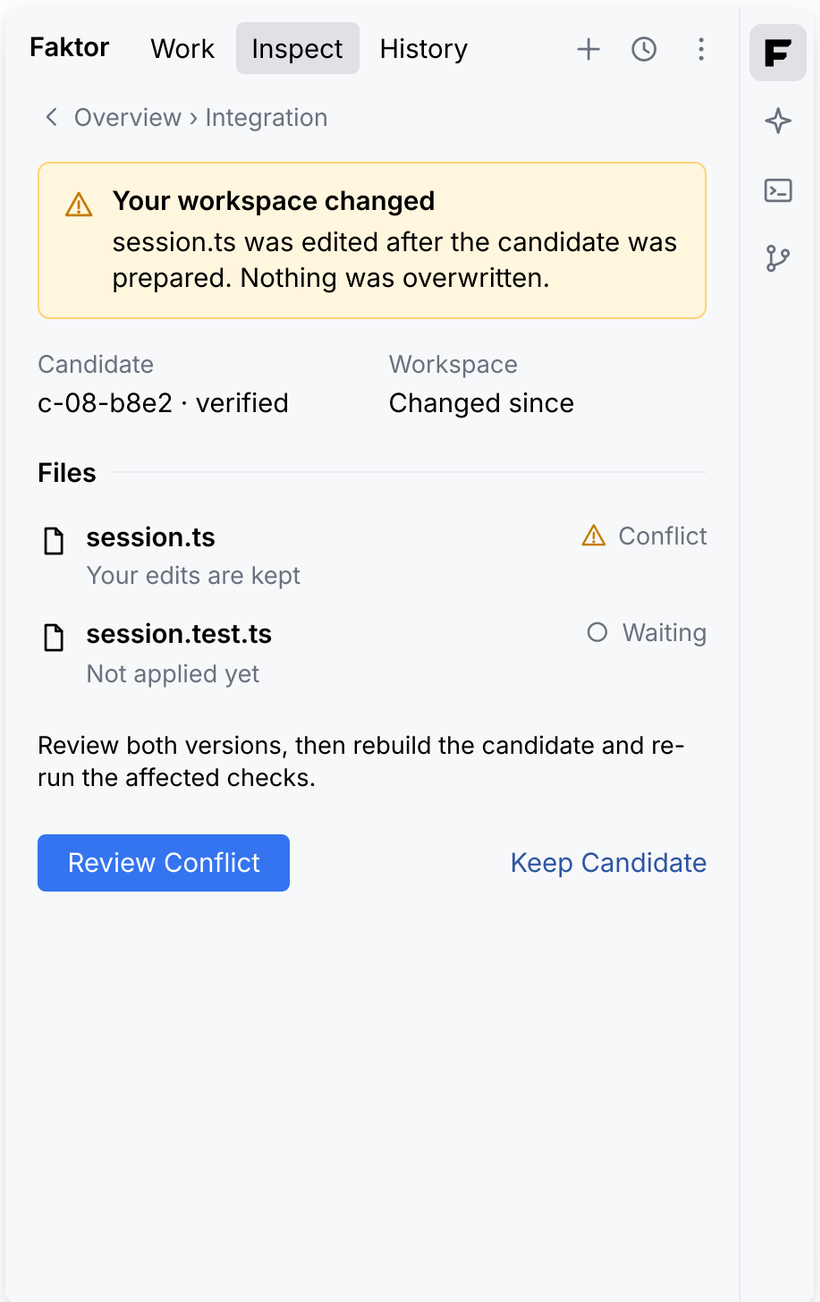
PRIMARY ACTION

Review Conflict

VS CODE · DARK MODERN — SIDEBAR VIEW · WEBVIEW



JETBRAINS · LIGHT — TOOL WINDOW · SWING / KOTLIN UI DSL



CONTROLS V-01 V-12

REPLACES R7 page 027 · R4-S15

CONTRACT p. 70

S44 · INSPECT AND PROOF · BEHAVIOUR CONTRACT

Integration conflict: contract and revision notes

Original contract retained verbatim (R4-S15 / controls: V-01 V-12). Design on p. 69.

What changed in revision 8

- Backend vocabulary (owner checkout, landing precondition) is replaced by “your workspace”.

Entry and return

Inspect overview, a changed file, a criterion or a recorded proof link. Work / Inspect / History remain the only top-level clusters. Settings, Usage and Diagnostics use explicit Back. Return to the originating object, not the first row.

Scope and captured state

Task + criterion/claim + immutable candidate + artifact/proof version; landing has its own identity. Selection is a stable entity key, not a row index. Evidence and proof remain bound to the exact snapshot that produced them.

Primary action and recovery

Review conflict. Distinguish missing, failed, refused, unavailable and mismatched proof. Never fill a missing required result with a quick check, plan state or completed conversation.

Loading, empty and stale

Keep previous facts clearly stale during refresh. If a proof read fails, remove its verified styling and show the typed unavailable reason.

Keyboard and focus

Arrow keys move the native list/tree; Enter opens detail. Back returns to the exact selected file or criterion. Native diff actions preserve editor focus conventions.

Layout and adverse content

Use a shallow master/detail hierarchy. At narrow widths show one selected detail with Back; open real diffs in the editor instead of squeezing two panes into the tool window.

Starting point and required construction

Starting classifications: E. E existing operation; R runtime-backed adapter; N new contract; H host-only work. Exact per-control classifications remain in the catalog.

Reuse the reviewed domain operations and preserve all served fields. Build every missing native read/write adapter, capability gate and state renderer identified by the matching control/build rows.

BK09 owns immutable content, preconditions, claim/candidate binding, integration and finish receipts. Existing read DTOs must retain all proof fields.

Required interaction fixtures

Exercise first-read, empty/no-match, ready, refreshing/stale, error/refusal, action pending and acknowledged result. Also switch scope while the request is in flight.

Moved out of the interface

Every line of revision 7 plate copy that left the default view, and where it lives now.

“Faktor / Inspect / Integration (R7 window title)”
→ Native host chrome (VS Code view title, tabs or editor tab; JetBrains tool-window tabs, dialog title or Settings breadcrumb)

“The workspace changed / Candidate c-08 is verified. The landing precondition no longer matches.”
→ Banner

“Integration stopped before overwrite”
→ Banner text “Nothing was overwritten.”

“session.ts changed in the owner checkout after the candidate was prepared. The existing verification belongs to c-08.”
→ Banner text; “owner checkout” becomes “your workspace”

“Verified candidate / Current owner state / owner-91ac”
→ Facts; workspace revision ID in Tooltip / hover help

“Owner content changed · your edits are preserved”
→ “Your edits are kept”

“Held / Candidate content retained · no automatic overwrite”
→ “Waiting — Not applied yet”

“Retry never silently widens the overwrite scope.”
→ Specification: screen contract

“Review both states, choose a new isolated candidate and rerun affected proof.”
→ Sentence “Review both versions, then rebuild the candidate and re-run the affected checks.”

“Keep candidate / Review conflict”
→ Keep Candidate / Review Conflict

“Conflict”
→ Status

S48 INSPECT AND PROOF

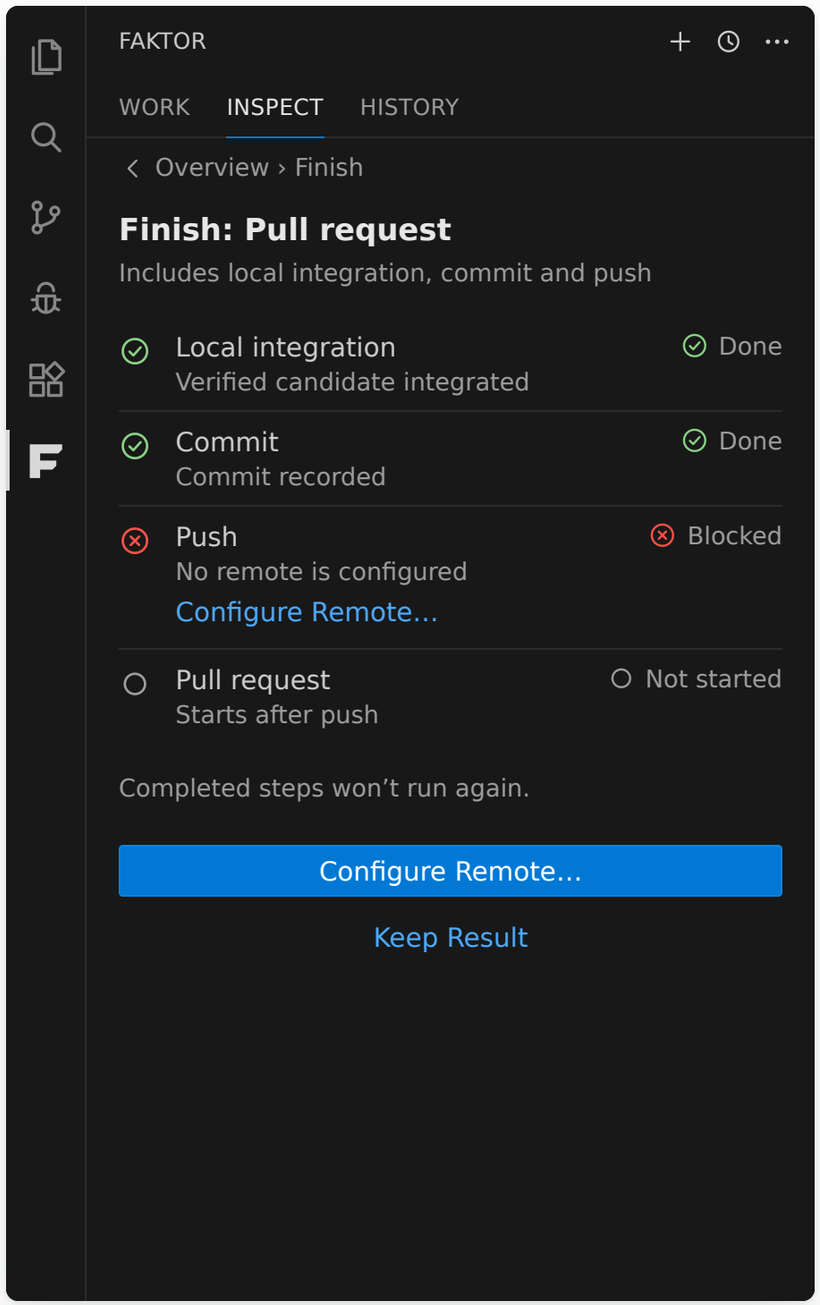
Finish actions

Inspect the hierarchical Local / Commit / Push / PR receipts and resolve only the blocked native finish step.

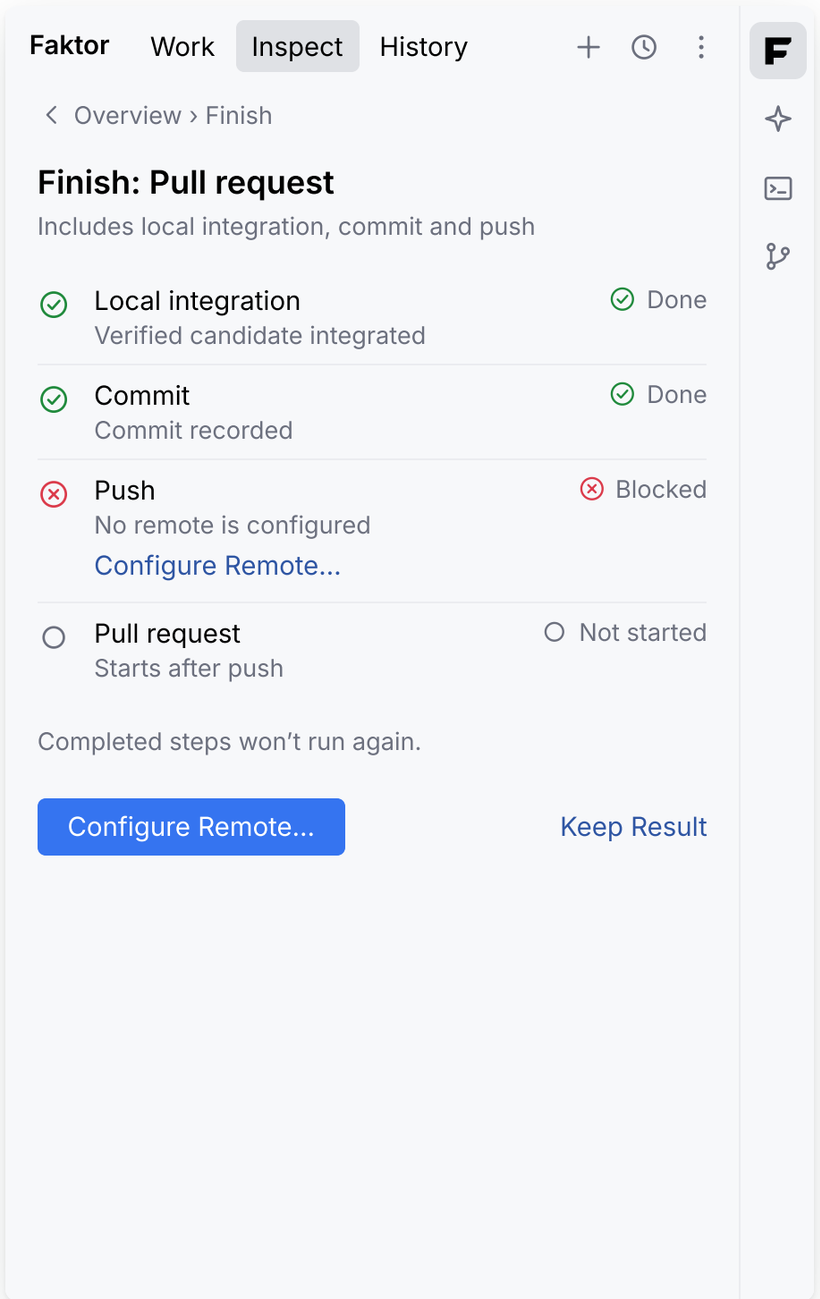
PRIMARY ACTION

Configure Remote…

VS CODE · DARK MODERN — SIDEBAR VIEW · WEBVIEW



JETBRAINS · LIGHT — TOOL WINDOW · SWING / KOTLIN UI DSL



CONTROLS V-10 V-11

REPLACES R7 page 028 · R4-S19

CONTRACT p. 72

S48 · INSPECT AND PROOF · BEHAVIOUR CONTRACT

Finish actions: contract and revision notes

Original contract retained verbatim (R4-S19 / controls: V-10 V-11). Design on p. 71.

What changed in revision 8

- A vertical stepper shows the hierarchy; the only action is on the blocked step.

Entry and return

Inspect overview, a changed file, a criterion or a recorded proof link. Work / Inspect / History remain the only top-level clusters. Settings, Usage and Diagnostics use explicit Back. Return to the originating object, not the first row.

Scope and captured state

Task + criterion/claim + immutable candidate + artifact/proof version; landing has its own identity. Selection is a stable entity key, not a row index. Evidence and proof remain bound to the exact snapshot that produced them.

Primary action and recovery

Open SCM configuration. Distinguish missing, failed, refused, unavailable and mismatched proof. Never fill a missing required result with a quick check, plan state or completed conversation.

Loading, empty and stale

Keep previous facts clearly stale during refresh. If a proof read fails, remove its verified styling and show the typed unavailable reason.

Keyboard and focus

Arrow keys move the native list/tree; Enter opens detail. Back returns to the exact selected file or criterion. Native diff actions preserve editor focus conventions.

Layout and adverse content

Use a shallow master/detail hierarchy. At narrow widths show one selected detail with Back; open real diffs in the editor instead of squeezing two panes into the tool window.

Starting point and required construction

Starting classifications: E, R. E existing operation; R runtime-backed adapter; N new contract; H host-only work. Exact per-control classifications remain in the catalog.

Reuse the reviewed domain operations and preserve all served fields. Build every missing native read/write adapter, capability gate and state renderer identified by the matching control/build rows.

BK09 owns immutable content, preconditions, claim/candidate binding, integration and finish receipts. Existing read DTOs must retain all proof fields.

Required interaction fixtures

Exercise first-read, empty/no-match, ready, refreshing/stale, error/refusal, action pending and acknowledged result. Also switch scope while the request is in flight.

Moved out of the interface

Every line of revision 7 plate copy that left the default view, and where it lives now.

"Faktor / Inspect / Finish (R7 window title)"
→ Native host chrome (VS Code view title, tabs or editor tab; JetBrains tool-window tabs, dialog title or Settings breadcrumb)

"Separate finish example · PR includes Local, Commit and Push"
→ Header "Finish: Pull request — Includes local integration, commit and push"

"Succeeded"
→ Done

"Matching verified candidate landed"
→ "Verified candidate integrated"

"Commit receipt recorded for the landed candidate"
→ "Commit recorded"

"Canonical remote is not configured"
→ "No remote is configured"

"Waits for the requested push receipt"
→ "Starts after push"

"Configure the canonical remote"
→ Step action Configure Remote…

"A successful shell command cannot replace native push or PR reconciliation. Earlier successful steps will not be repeated."
→ "Completed steps won't run again."; shell rule in Specification: screen contract

"Open SCM configuration"
→ Configure Remote… (opens the host SCM settings)

"Finish the task"
→ Header "Finish: Pull request"

"Local integration / Commit / Push / Create pull request"
→ Stepper steps ("Pull request")

"Blocked / Not started"
→ Statuses

"Keep result"
→ Keep Result

S32 INSPECT AND PROOF

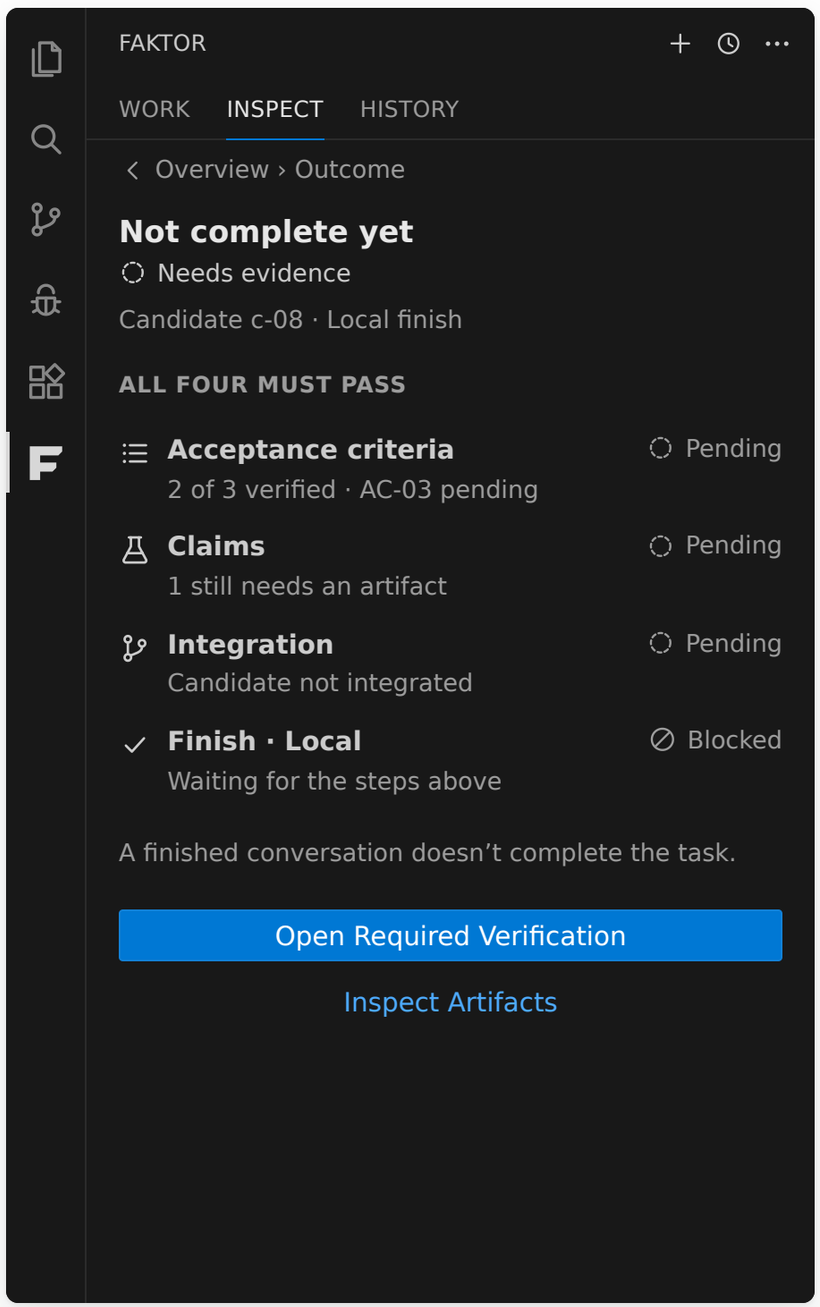
Task outcome

PRIMARY ACTION

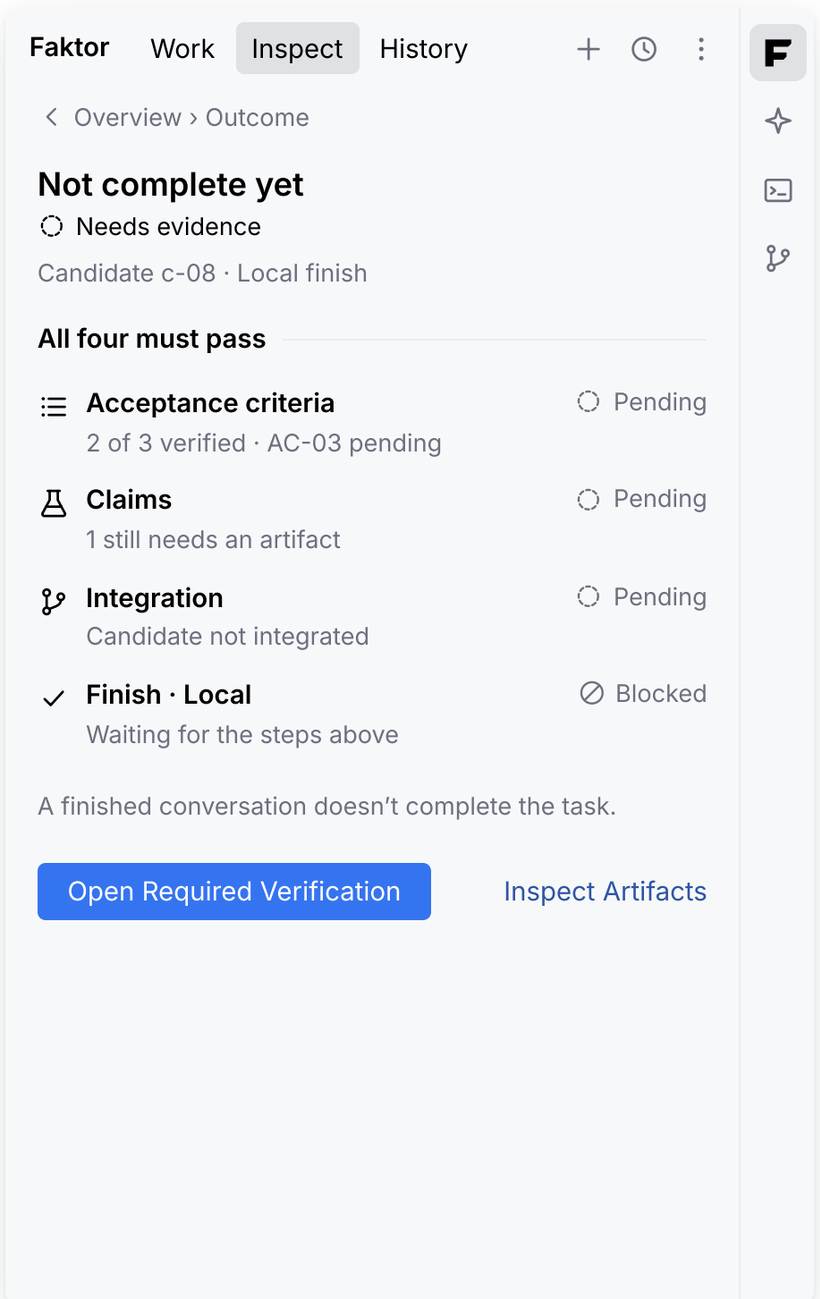
Open Required Verification

Finish with explicit engineering status: required acceptance, claim evidence, matching integration and requested finish. Conversation completion alone proves none of them.

VS CODE · DARK MODERN — SIDEBAR VIEW · WEBVIEW



JETBRAINS · LIGHT — TOOL WINDOW · SWING / KOTLIN UI DSL



CONTROLS V-07 V-08 V-09 V-10 V-11 V-12 O-06 B-04

REPLACES R7 page 029 · Expanded standard-agent surface

CONTRACT p. 74

S32 · INSPECT AND PROOF · BEHAVIOUR CONTRACT

Task outcome: contract and revision notes

Original contract retained verbatim (Expanded standard-agent surface / controls: V-07 V-08 V-09 V-10 V-11 V-12 O-06 B-04). Design on p. 73.

What changed in revision 8

- Same four-row checklist as the overview, so outcome and overview read identically.

Entry and return

Inspect overview, a changed file, a criterion or a recorded proof link. Work / Inspect / History remain the only top-level clusters. Settings, Usage and Diagnostics use explicit Back. Return to the originating object, not the first row.

Scope and captured state

Task + criterion/claim + immutable candidate + artifact/proof version; landing has its own identity. Selection is a stable entity key, not a row index. Evidence and proof remain bound to the exact snapshot that produced them.

Primary action and recovery

Open each evidence receipt, inspect matching candidate identity and resolve any missing required fact before the requested finish can be fulfilled.. Missing unit evidence, failed integration or absent SCM configuration remains a named blocker. A successful chat response cannot turn it green.

Loading, empty and stale

Keep previous facts clearly stale during refresh. If a proof read fails, remove its verified styling and show the typed unavailable reason.

Keyboard and focus

Arrow keys move the native list/tree; Enter opens detail. Back returns to the exact selected file or criterion. Native diff actions preserve editor focus conventions.

Layout and adverse content

Use a shallow master/detail hierarchy. At narrow widths show one selected detail with Back; open real diffs in the editor instead of squeezing two panes into the tool window.

Starting point and required construction

Starting classifications: E, R. E existing operation; R runtime-backed adapter; N new contract; H host-only work. Exact per-control classifications remain in the catalog.

Reuse the reviewed domain operations and preserve all served fields. Build every missing native read/write adapter, capability gate and state renderer identified by the matching control/build rows.

BK09 owns immutable content, preconditions, claim/candidate binding, integration and finish receipts. Existing read DTOs must retain all proof fields.

Required interaction fixtures

Exercise first-read, empty/no-match, ready, refreshing/stale, error/refusal, action pending and acknowledged result. Also switch scope while the request is in flight.

Moved out of the interface

Every line of revision 7 plate copy that left the default view, and where it lives now.

"Faktor / Inspect / Outcome (R7 window title)"
→ Native host chrome (VS Code view title, tabs or editor tab; JetBrains tool-window tabs, dialog title or Settings breadcrumb)

"Task outcome / Candidate c-08-b8e2 · requested finish: Local"
→ Header

"Required acceptance / 2 of 3 verified · AC-03 still pending"
→ Row

"Claims and artifacts / 2 supported · 1 artifact still required"
→ Row "1 still needs an artifact"

"Matching verification & landing / AC-03 pending · candidate c-08 not integrated"
→ Row "Integration"

"Requested finish / Local · commit/push/PR hierarchy remains explicit"
→ Row; hierarchy in S48

"Engineering outcome: pending proof"
→ Header status "Needs evidence"

"A completed conversation cannot fill missing checks, claims, matching integration or finish receipts."
→ Short note; full rule in Specification: screen contract

"Pending / Blocked"
→ Statuses

"Inspect artifacts / Open required verification"
→ Inspect Artifacts / Open Required Verification

S65 INSPECT AND PROOF

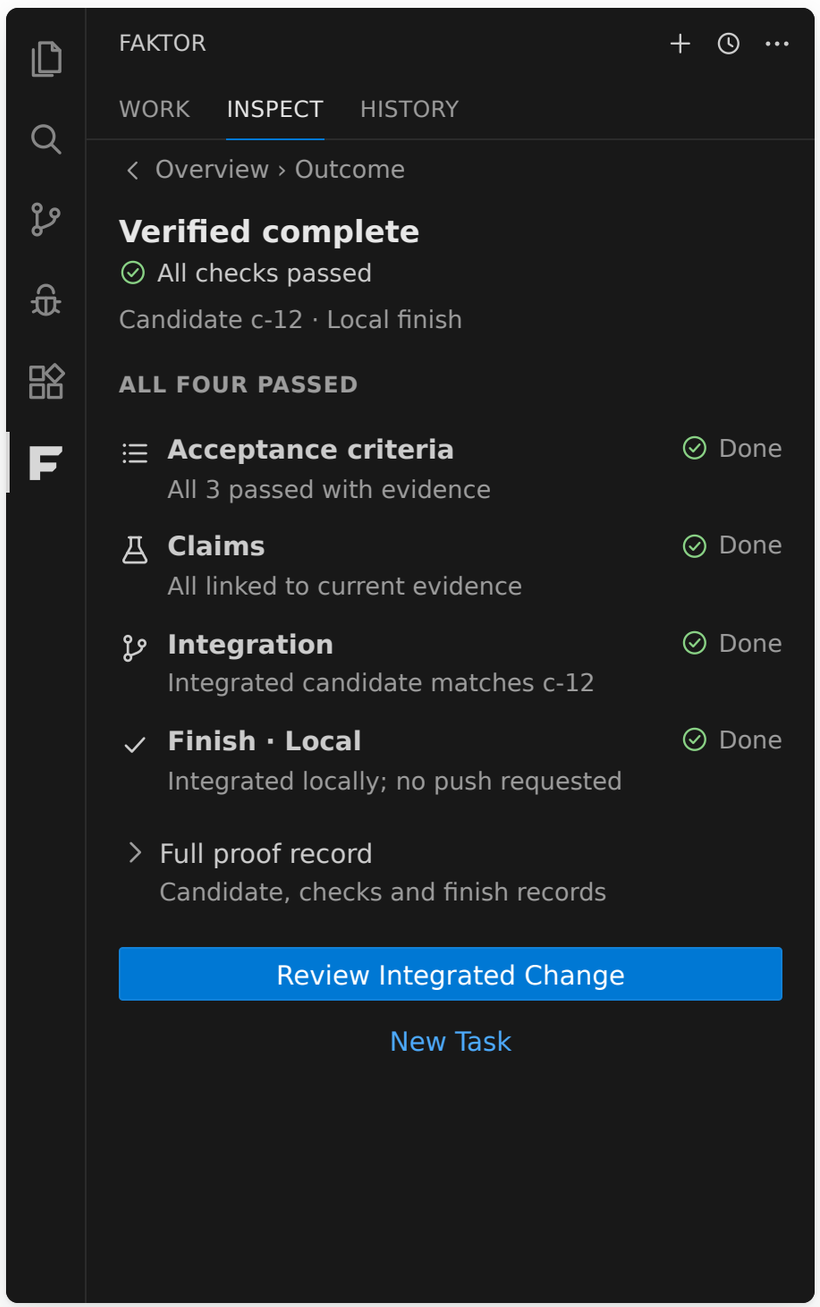
Verified outcome

PRIMARY ACTION

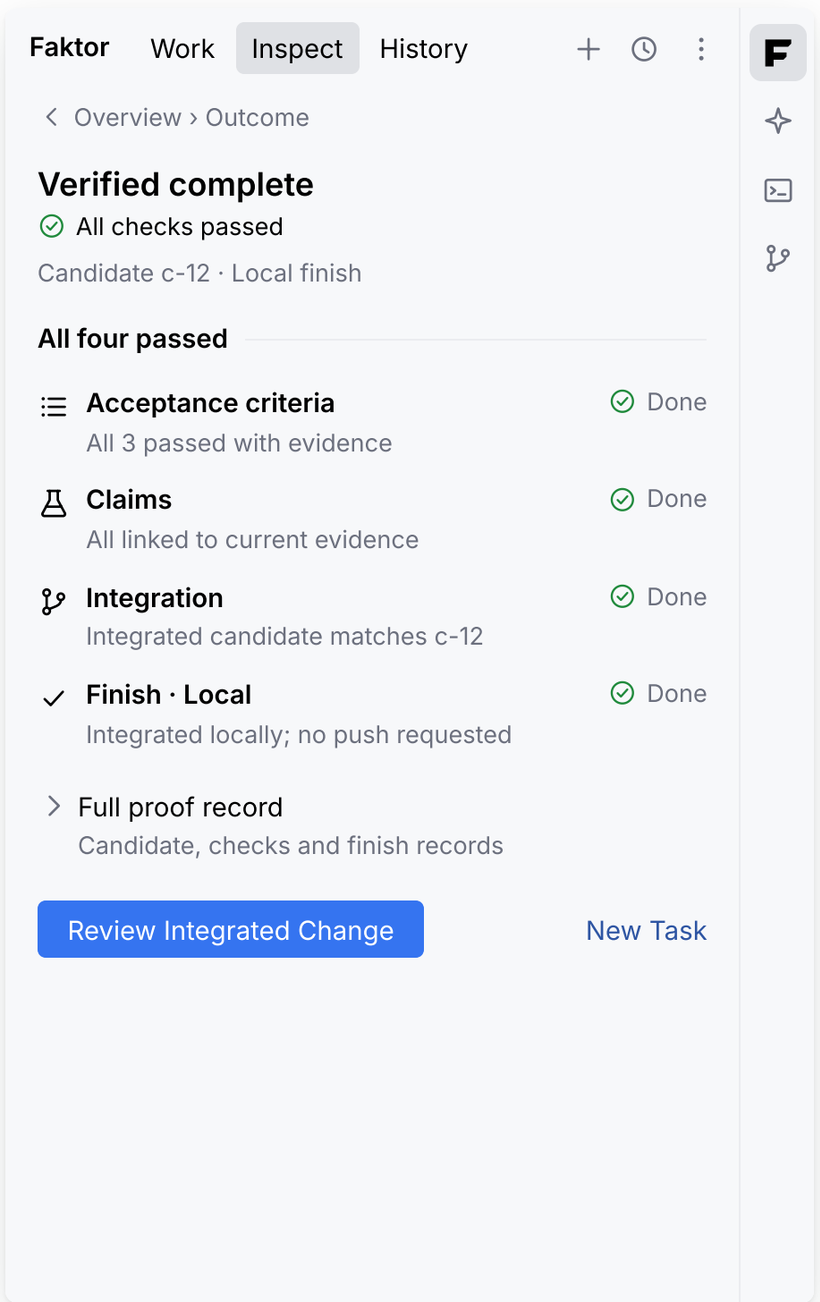
Review Integrated Change

Show a separate complete example only when all required acceptance, claims, matching integration and finish agree.

VS CODE · DARK MODERN — SIDEBAR VIEW · WEBVIEW



JETBRAINS · LIGHT — TOOL WINDOW · SWING / KOTLIN UI DSL



CONTROLS V-09 V-10

REPLACES R7 page 030 · R4-S20

CONTRACT p. 76

S65 · INSPECT AND PROOF · BEHAVIOUR CONTRACT

Verified outcome: contract and revision notes

Original contract retained verbatim (R4-S20 / controls: V-09 V-10). Design on p. 75.

What changed in revision 8

- Success uses the same layout as pending, so the difference is the state, not the decoration.

Entry and return

Inspect overview, a changed file, a criterion or a recorded proof link. Work / Inspect / History remain the only top-level clusters. Settings, Usage and Diagnostics use explicit Back. Return to the originating object, not the first row.

Scope and captured state

Task + criterion/claim + immutable candidate + artifact/proof version; landing has its own identity. Selection is a stable entity key, not a row index. Evidence and proof remain bound to the exact snapshot that produced them.

Primary action and recovery

Review landed change. Distinguish missing, failed, refused, unavailable and mismatched proof. Never fill a missing required result with a quick check, plan state or completed conversation.

Loading, empty and stale

Keep previous facts clearly stale during refresh. If a proof read fails, remove its verified styling and show the typed unavailable reason.

Keyboard and focus

Arrow keys move the native list/tree; Enter opens detail. Back returns to the exact selected file or criterion. Native diff actions preserve editor focus conventions.

Layout and adverse content

Use a shallow master/detail hierarchy. At narrow widths show one selected detail with Back; open real diffs in the editor instead of squeezing two panes into the tool window.

Starting point and required construction

Starting classifications: E. E existing operation; R runtime-backed adapter; N new contract; H host-only work. Exact per-control classifications remain in the catalog.

Reuse the reviewed domain operations and preserve all served fields. Build every missing native read/write adapter, capability gate and state renderer identified by the matching control/build rows.

BK09 owns immutable content, preconditions, claim/candidate binding, integration and finish receipts. Existing read DTOs must retain all proof fields.

Required interaction fixtures

Exercise first-read, empty/no-match, ready, refreshing/stale, error/refusal, action pending and acknowledged result. Also switch scope while the request is in flight.

Moved out of the interface

Every line of revision 7 plate copy that left the default view, and where it lives now.

"Faktor / Inspect / Outcome (R7 window title)"
→ Native host chrome (VS Code view title, tabs or editor tab; JetBrains tool-window tabs, dialog title or Settings breadcrumb)

"Separate completed example · candidate c-12 · Local finish"
→ Subtitle; "separate example" is a document caption, not UI copy

"Satisfied"
→ Done

"All 3 required criteria have matching passed evidence"
→ "All 3 passed with evidence"

"Required claims are bound to current artifacts"
→ "All linked to current evidence"

"Verified c-12 equals the recorded landed candidate"
→ "Integrated candidate matches c-12"

"Local integration succeeded; no push or PR was requested"
→ "Integrated locally; no push requested"

"Full proof record / Open / Candidate, landed identity, checks and durable finish receipts"
→ Disclosure

"Review landed change"
→ Review Integrated Change

"Verified complete"
→ Title

"Required acceptance"
→ Row "Acceptance criteria"

"Named claims"
→ Row "Claims"

"Matching integration"
→ Row "Integration"

"Requested finish"
→ Row "Finish · Local"

"New task"
→ New Task

S25 INSPECT AND PROOF

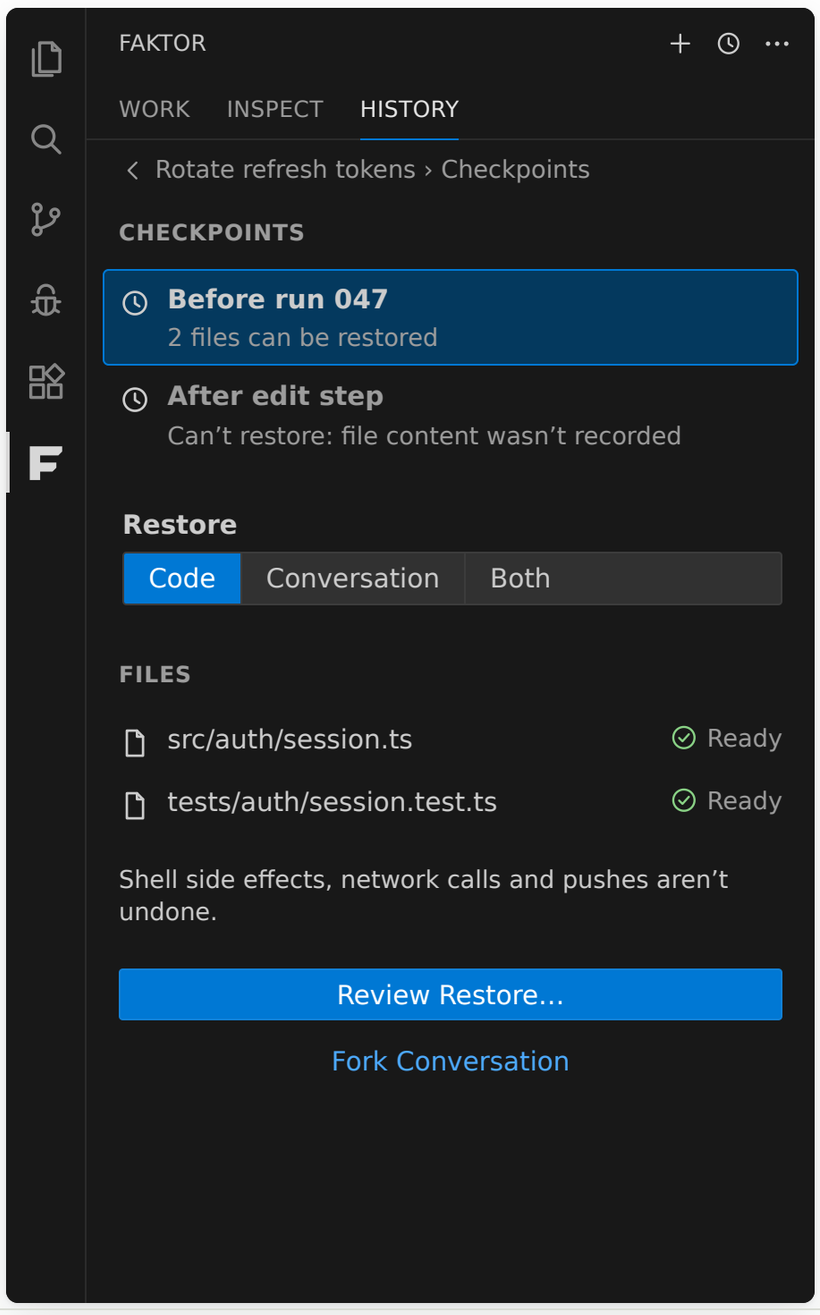
Restore & fork

PRIMARY ACTION

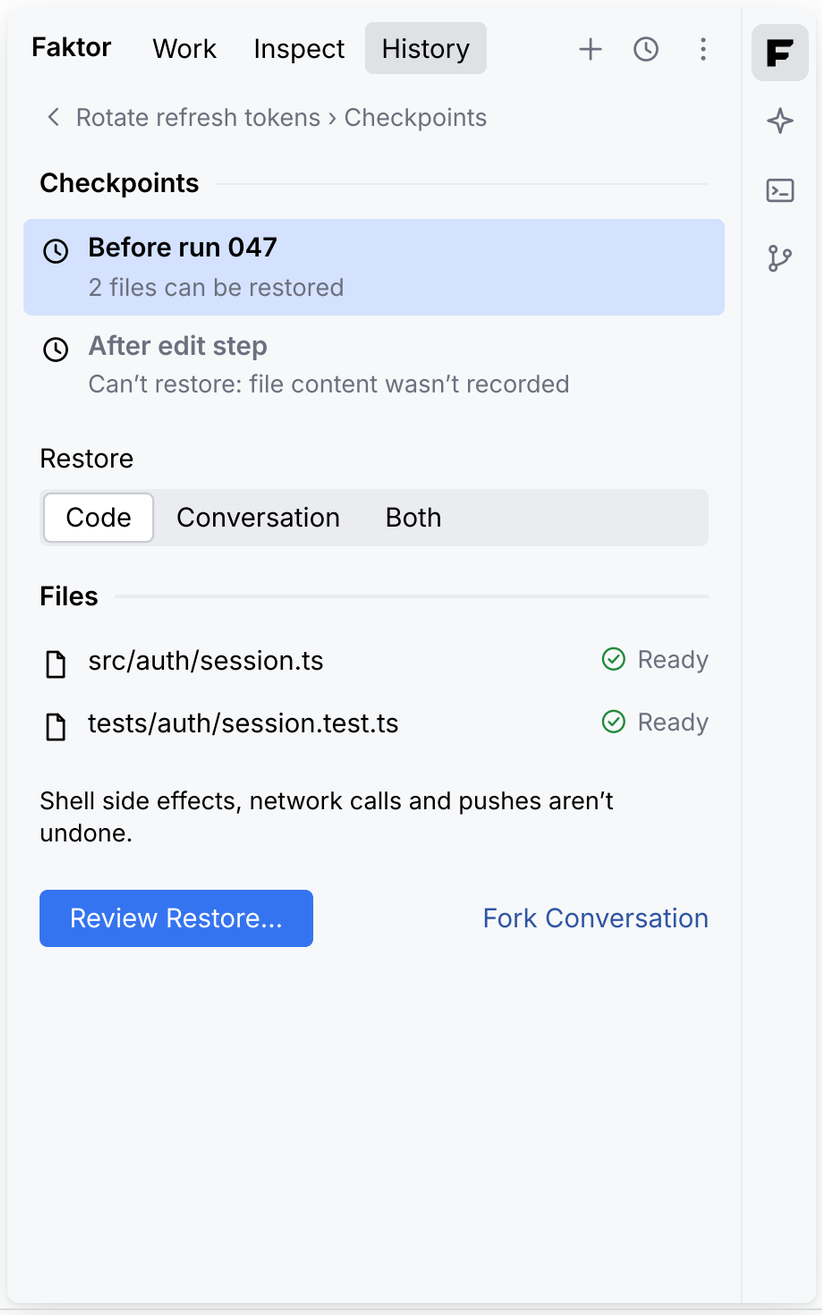
Review Restore...

Show recorded checkpoint identity, content coverage and branching separately. Preview the exact files and conversation range affected by any supported restore.

VS CODE · DARK MODERN — SIDEBAR VIEW · WEBVIEW



JETBRAINS · LIGHT — TOOL WINDOW · SWING / KOTLIN UI DSL



CONTROLS V-04 V-05 V-06 V-12 D-06 Q-09

REPLACES R7 page 031 · Expanded standard-agent surface

CONTRACT p. 78

S25 · INSPECT AND PROOF · BEHAVIOUR CONTRACT

Restore & fork: contract and revision notes

Original contract retained verbatim (Expanded standard-agent surface / controls: V-04 V-05 V-06 V-12 D-06 Q-09). Design on p. 77.

What changed in revision 8

- The disabled checkpoint states why it can't be restored instead of using a separate status label.

Entry and return

Inspect overview, a changed file, a criterion or a recorded proof link. Work / Inspect / History remain the only top-level clusters. Settings, Usage and Diagnostics use explicit Back. Return to the originating object, not the first row.

Scope and captured state

Task + criterion/claim + immutable candidate + artifact/proof version; landing has its own identity. Selection is a stable entity key, not a row index. Evidence and proof remain bound to the exact snapshot that produced them.

Primary action and recovery

Choose a stored checkpoint, inspect coverage and compare content preconditions before requesting a separate restore transaction.. Hash-only entries may not support redo/diff. Unrelated edits or missing content refuse restore; shell/network/SCM effects are disclosed as uncovered.

Loading, empty and stale

Keep previous facts clearly stale during refresh. If a proof read fails, remove its verified styling and show the typed unavailable reason.

Keyboard and focus

Arrow keys move the native list/tree; Enter opens detail. Back returns to the exact selected file or criterion. Native diff actions preserve editor focus conventions.

Layout and adverse content

Use a shallow master/detail hierarchy. At narrow widths show one selected detail with Back; open real diffs in the editor instead of squeezing two panes into the tool window.

Starting point and required construction

Starting classifications: E, N, R. E existing operation; R runtime-backed adapter; N new contract; H host-only work. Exact per-control classifications remain in the catalog.

Reuse the reviewed domain operations and preserve all served fields. Build every missing native read/write adapter, capability gate and state renderer identified by the matching control/build rows.

BK09 owns immutable content, preconditions, claim/candidate binding, integration and finish receipts. Existing read DTOs must retain all proof fields.

Required interaction fixtures

Exercise first-read, empty/no-match, ready, refreshing/stale, error/refusal, action pending and acknowledged result. Also switch scope while the request is in flight.

Moved out of the interface

Every line of revision 7 plate copy that left the default view, and where it lives now.

"Faktor / History / Checkpoints (R7 window title)"
→ Native host chrome (VS Code view title, tabs or editor tab; JetBrains tool-window tabs, dialog title or Settings breadcrumb)

"Run idle · current content preconditions checked"
→ Files show Ready only when the check passed; rule in Specification: screen contract

"Selected"
→ Selection highlight

"2 covered files · target content available"
→ "2 files can be restored"

"Redo unavailable / Hash-only record · missing after content"
→ Disabled row "Can't restore: file content wasn't recorded"

"Restore scope"
→ Choice label "Restore"

"Covered files"
→ Section "Files"

"Current hash matches preview precondition"
→ Ready status; hash detail in Tooltip / hover help

"Review 2 file changes before restore"
→ Review Restore… (preview step)

"Shell caches, network calls and pushes are not rolled back. Final content checks run again at apply."
→ Warning line; recheck rule in Specification: screen contract

"Restore & fork"
→ Breadcrumb "Checkpoints"

"Before run 047 / After edit step"
→ Checkpoint rows

"Code / Conversation / Both"
→ Segmented choice

"Fork conversation / Review restore"
→ Fork Conversation / Review Restore…

S49 INSPECT AND PROOF

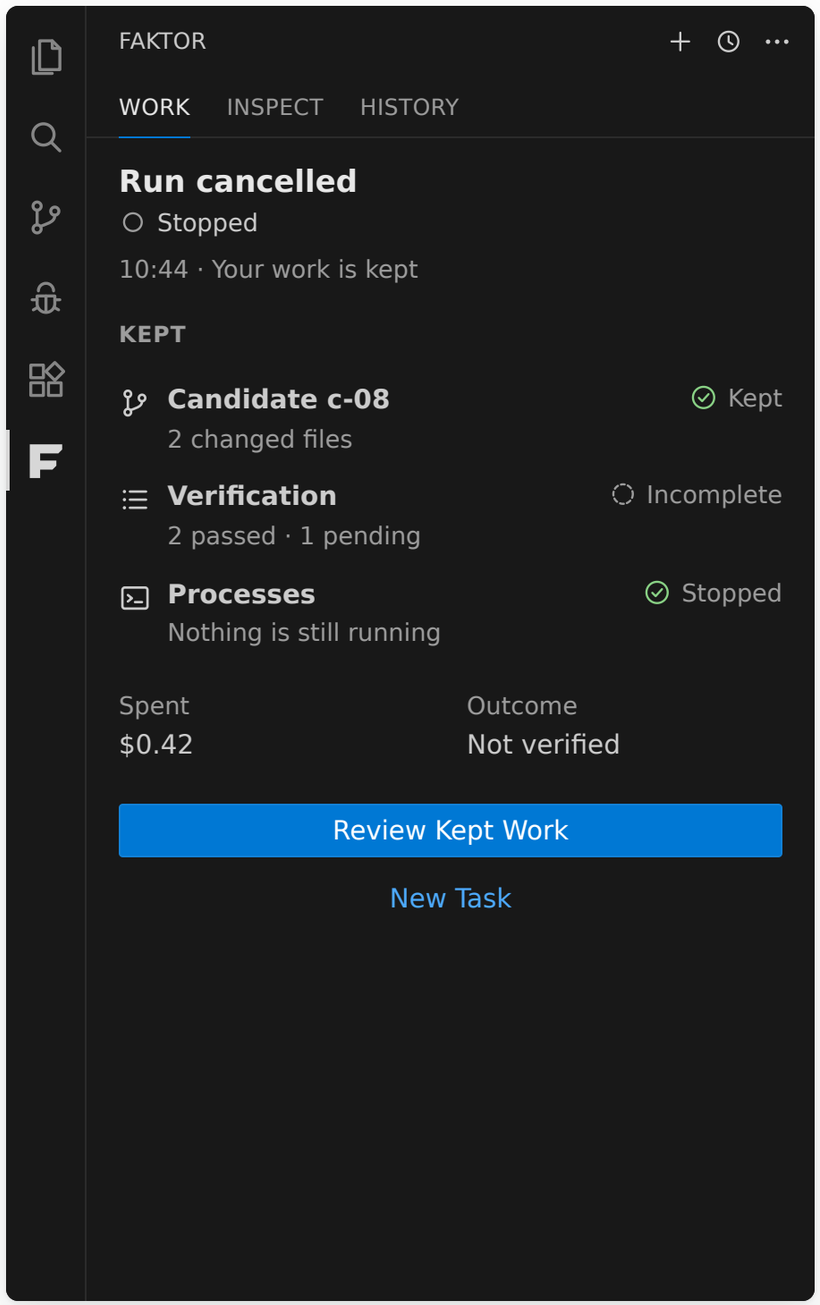
Cancelled run

PRIMARY ACTION

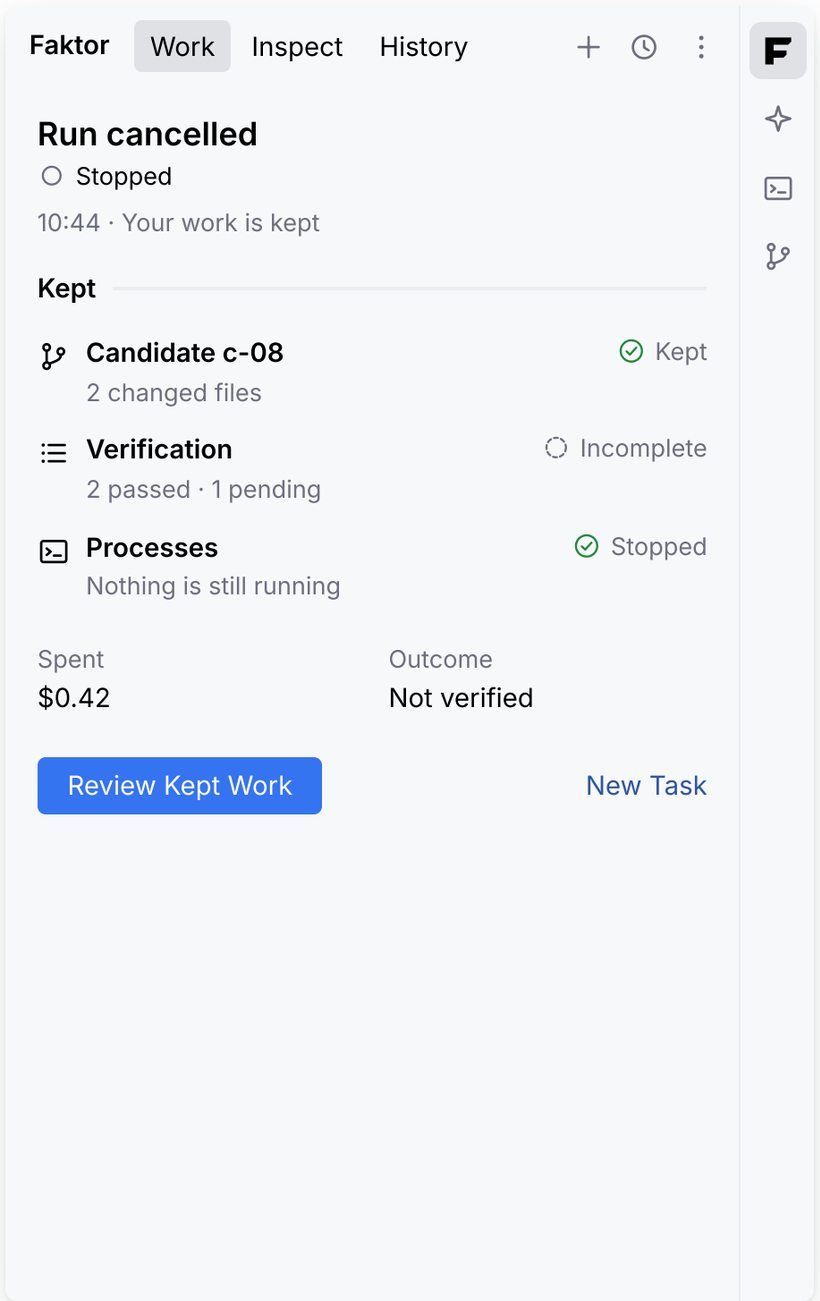
Review Kept Work

Retain candidate work, evidence, recorded spend and drafts after a durable cancellation; completion remains separate.

VS CODE · DARK MODERN — SIDEBAR VIEW · WEBVIEW



JETBRAINS · LIGHT — TOOL WINDOW · SWING / KOTLIN UI DSL



CONTROLS D-07 V-04 B-04

REPLACES R7 page 032 · R4-S21

CONTRACT p. 80

S49 · INSPECT AND PROOF · BEHAVIOUR CONTRACT

Cancelled run: contract and revision notes

Original contract retained verbatim (R4-S21 / controls: D-07 V-04 B-04). Design on p. 79.

What changed in revision 8

- Cancellation reads as a calm summary of what is kept, not as an error.

Entry and return

Work, a captured submission receipt, or a task-specific attention item. Work / Inspect / History remain the only top-level clusters. Settings, Usage and Diagnostics use explicit Back. Return to the originating object, not the first row.

Scope and captured state

Daemon/workspace/worktree/conversation/task/request generation. Draft scope is separate from an admitted run. Capture one immutable logical start. Clear only its acknowledged draft version; preserve text typed after capture. Queue items acquire their own receipt and revision.

Primary action and recovery

Review retained work. Retain the captured body, options and source handles. Unknown admission/effect requires receipt reconciliation before replay. A changed body needs a different request identity.

Loading, empty and stale

Show the named operation and current owner. Keep the composer available for an independent draft. No fake percentages, fabricated tool steps or optimistic completion.

Keyboard and focus

Enter inserts a newline by default. The host send command respects IME composition. New approvals announce once without taking focus. Escape closes a detail; it never stops a run.

Layout and adverse content

Header, transcript scroll and composer are siblings. Collapse secondary metadata first. Keep a scoped Stop action and the active task reachable at short heights.

Starting point and required construction

Starting classifications: E, R. E existing operation; R runtime-backed adapter; N new contract; H host-only work. Exact per-control classifications remain in the catalog.

Reuse the reviewed domain operations and preserve all served fields. Build every missing native read/write adapter, capability gate and state renderer identified by the matching control/build rows.

BK08 owns admission, receipts, queue, control handles and reconciliation; BK05 owns authority and BK07 owns a shared exact budget ledger.

Required interaction fixtures

Exercise first-read, empty/no-match, ready, refreshing/stale, error/refusal, action pending and acknowledged result. Also switch scope while the request is in flight.

Moved out of the interface

Every line of revision 7 plate copy that left the default view, and where it lives now.

"Faktor / Work / Outcome (R7 window title)"
→ Native host chrome (VS Code view title, tabs or editor tab; JetBrains tool-window tabs, dialog title or Settings breadcrumb)

"Run 047 · cancellation receipt recorded at 10:44"
→ Subtitle "10:44"; record ID in Inspector

"The run stopped. Its candidate, transcript and evidence remain available."
→ Subtitle "Your work is kept" and the Kept section

"Retained"
→ Kept

"2 files retained in the isolated candidate"
→ "2 changed files"

"Last verification / Pending / no completed outcome"
→ Row "Verification — Incomplete"

"Owned process / Stopped / No running owned process is reported"
→ Row "Processes — Nothing is still running"

"Attempt spend / USD 0.42 observed"
→ Fact "Spent"; observed vs. computed in Usage

"Engineering outcome / Unverified"
→ Fact "Outcome — Not verified"

"Review retained work"
→ Review Kept Work

"Run cancelled"
→ Title

"2 passed · 1 pending · no completed outcome"
→ Row meta "2 passed · 1 pending" with status "Incomplete"

"Candidate c-08"
→ Row

"New task"
→ New Task

PART 2 · SCREENS

2.4 Agents

4 screens. Each has a design spread for both IDEs and a contract page.

ID	SCREEN	JOB	PAGE
S23	**Delegated agents**	Inspect parent and child work as a tree with bounded concurrency, shared spend and candidate ownership. Child output is evidence, not automatic parent success.	p. 82
S50	**Agent detail and steer**	Inspect a child assignment, ownership, inherited authority and shared budget before steering or stopping that child.	p. 84
S51	**Compare approaches**	Compare candidate approaches against one contract and recommend without automatically integrating the winner.	p. 86
S52	**Coordination**	Read and post bounded durable coordination notes with explicit read acknowledgement and stable revision paging.	p. 88

S23 AGENTS

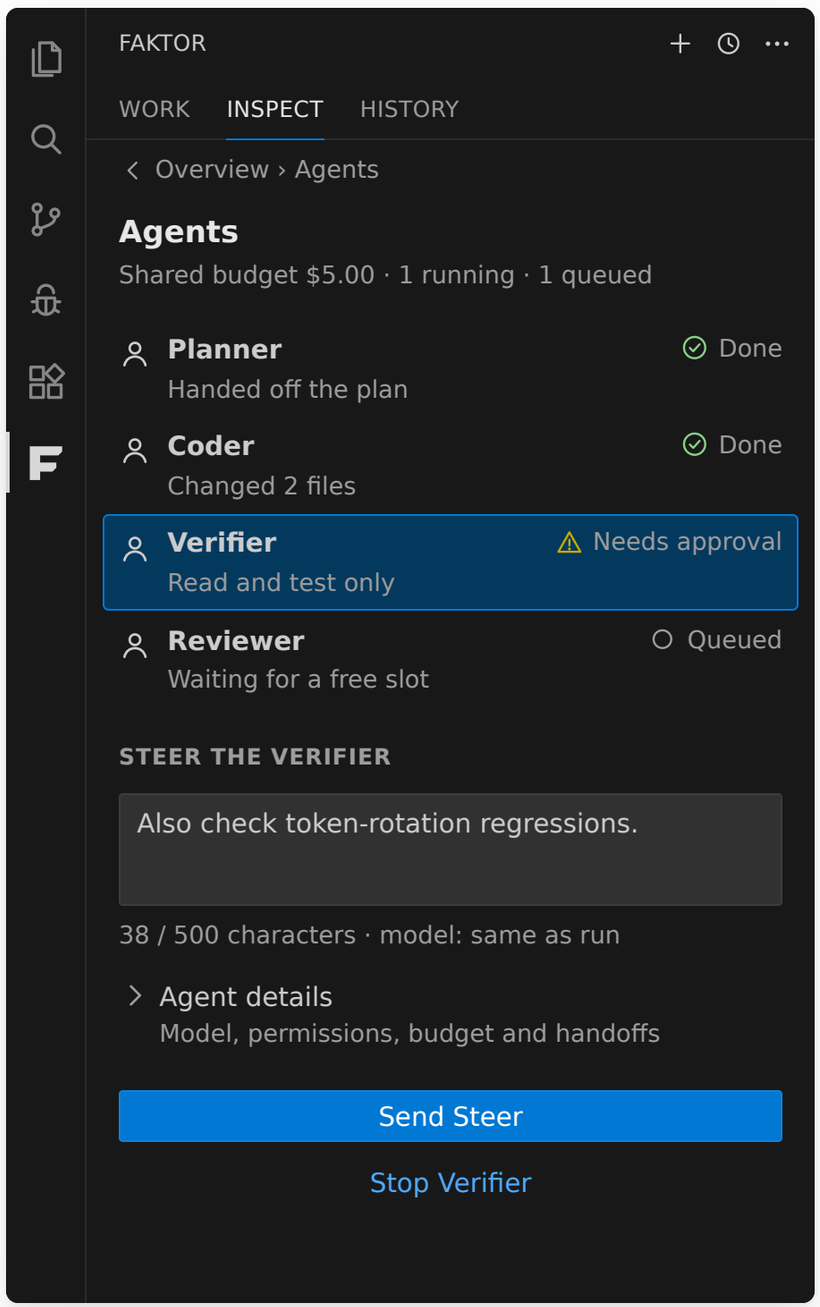
Delegated agents

PRIMARY ACTION

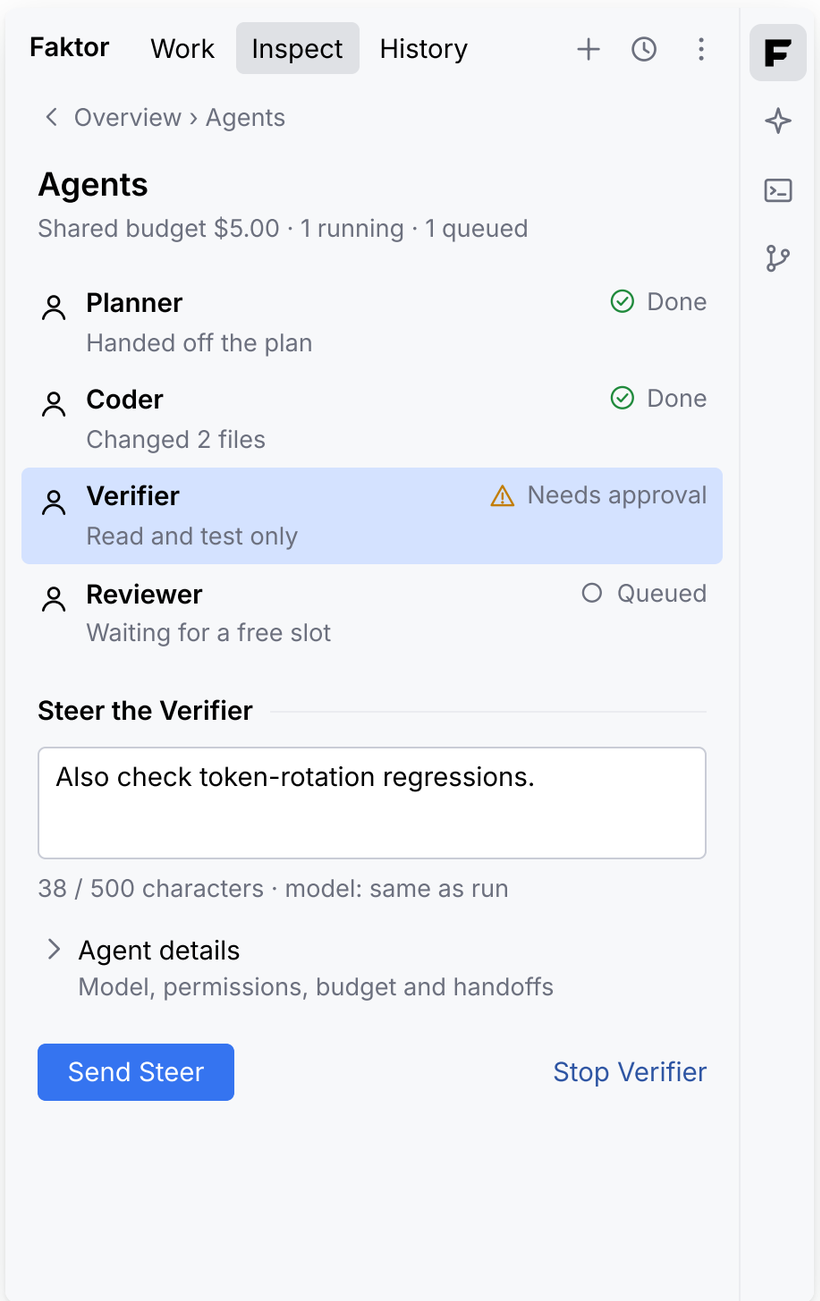
Send Steer

Inspect parent and child work as a tree with bounded concurrency, shared spend and candidate ownership. Child output is evidence, not automatic parent success.

VS CODE · DARK MODERN — SIDEBAR VIEW · WEBVIEW



JETBRAINS · LIGHT — TOOL WINDOW · SWING / KOTLIN UI DSL



CONTROLS A-06 A-07 A-08 A-09 A-10 A-11 A-12 M-11 B-04 Q-06

REPLACES R7 page 033 · R4-S22

CONTRACT p. 83

S23 · AGENTS · BEHAVIOUR CONTRACT

Delegated agents: contract and revision notes

Original contract retained verbatim (R4-S22 / controls: A-06 A-07 A-08 A-09 A-10 A-11 A-12 M-11 B-04 Q-06). Design on p. 82.

What changed in revision 8

- Child numbering and ownership vocabulary move to details; rows describe what each agent did.

Entry and return

Inspect / Agents with Roster, Approvals, Compare approaches and Coordination as subordinate destinations. Work / Inspect / History remain the only top-level clusters. Settings, Usage and Diagnostics use explicit Back. Return to the originating object, not the first row.

Scope and captured state

Parent task + current child/comparison/board generation; every write is scoped to its actual owner. Retain selected agent and the typed steer/post draft through refresh. A sent control is an immutable operation; current owner checks happen again at dispatch.

Primary action and recovery

Select a child to view its admitted role/model/tools and use supported Steer, Model or Stop controls.. Stale child controls are fenced by session and generation. A child failure preserves its artifacts and the parent's remaining obligations.

Loading, empty and stale

Show bounded queue/wait/working states. Parent and child budget reservations remain in one ledger. A comparison only recommends until explicit integration.

Keyboard and focus

Keyboard lists select by stable child/post identity. Open detail intentionally; new posts/child events announce without resetting the current selection or typed draft.

Layout and adverse content

One selected detail and one subordinate input at a time. Put raw IDs, revision cursors and provenance behind Details; keep role, assignment and outcome in the main row.

Starting point and required construction

Starting classifications: E, N, R. E existing operation; R runtime-backed adapter; N new contract; H host-only work. Exact per-control classifications remain in the catalog.

Reuse the reviewed domain operations and preserve all served fields. Build every missing native read/write adapter, capability gate and state renderer identified by the matching control/build rows.

BK03 owns intent/preset contracts; BK08 owns lifecycle/board/control adapters; BK07 owns shared budgets and BK09 owns comparison evidence.

Required interaction fixtures

Exercise first-read, empty/no-match, ready, refreshing/stale, error/refusal, action pending and acknowledged result. Also switch scope while the request is in flight.

Moved out of the interface

Every line of revision 7 plate copy that left the default view, and where it lives now.

"Faktor / Inspect / Agents (R7 window title)"
→ Native host chrome (VS Code view title, tabs or editor tab; JetBrains tool-window tabs, dialog title or Settings breadcrumb)

"Parent budget USD 5.00 · 1 active / 1 queued"
→ Subtitle

"Planner / child 1 / Handoff received · no edit ownership"
→ Row "Planner — Handed off the plan"; ownership in Agent details

"Coder / child 2 / 2 files changed · writer grant returned"
→ Row "Coder — Changed 2 files"; grant in Agent details

"Verifier / child 3 / Approval / Read/test envelope · shared parent budget"
→ Row "Verifier — Read and test only — Needs approval"

"Reviewer / child 4 / Capacity-bound queue · no model call yet"
→ Row "Reviewer — Waiting for a free slot"

"Selected model / Inherited exact model"
→ Fact "Model — Same as run"

"Steer verifier"
→ Field "Steer" in the Verifier section

"Delivered at the supported control boundary."
→ Activity entry after sending; rule in Specification: screen contract

"Child details / Model control, grants, reservations, ownership and handoff artifacts"
→ Agent details disclosure

"Stop child"
→ Stop Verifier (names the target)

"Delegated agents"
→ Title "Agents"

"Steer limit / 500 characters"
→ Counter "38 / 500 characters"

"Also check token-rotation regressions."
→ Steer field text

"Send steer"
→ Send Steer

S50 AGENTS

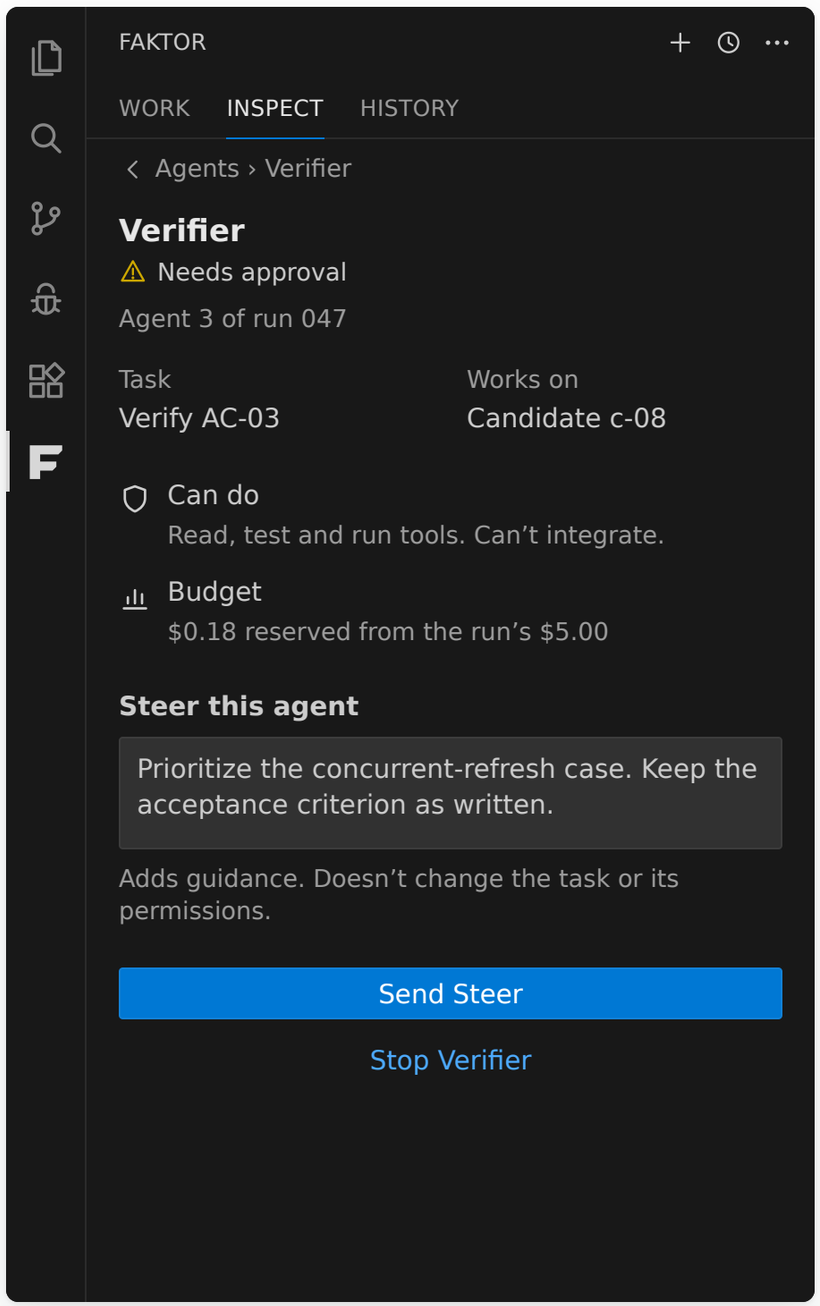
Agent detail and steer

Inspect a child assignment, ownership, inherited authority and shared budget before steering or stopping that child.

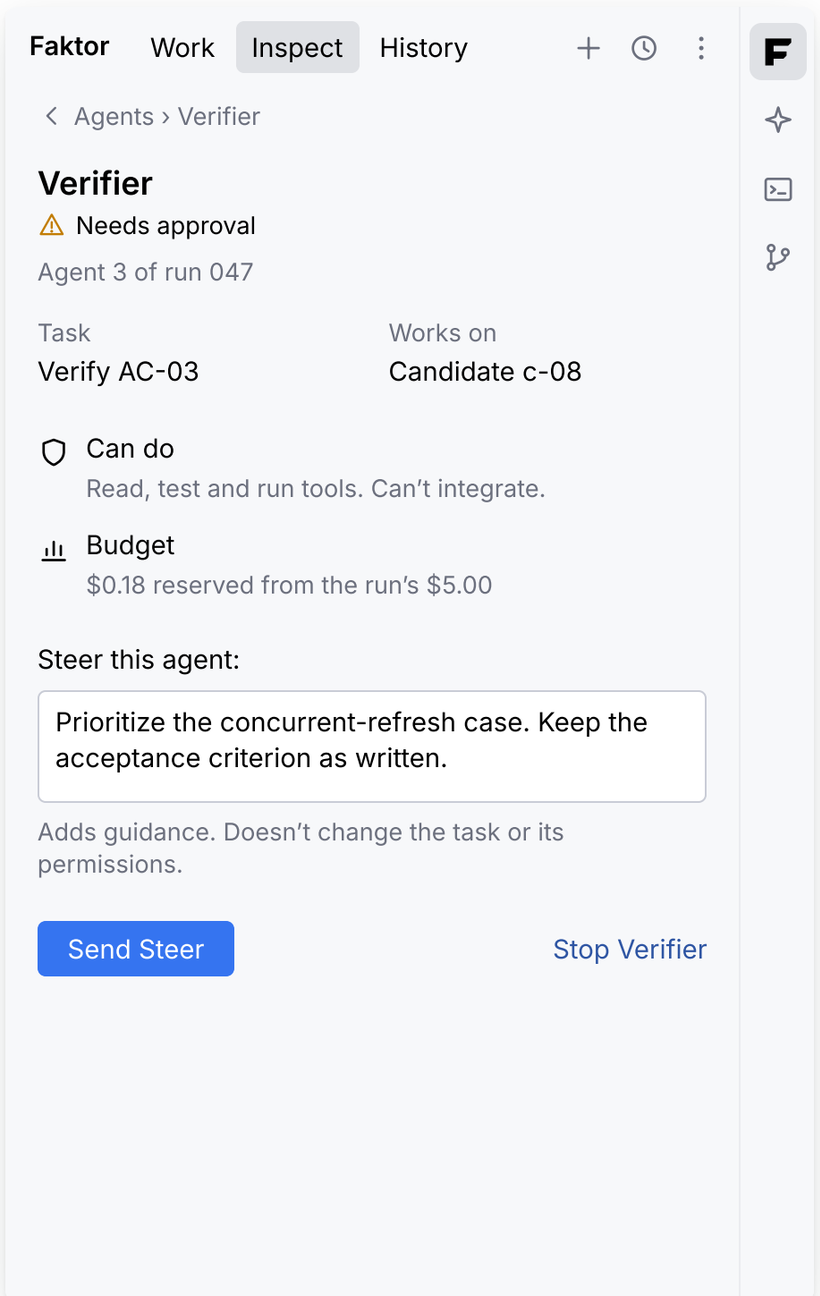
PRIMARY ACTION

Send Steer

VS CODE · DARK MODERN — SIDEBAR VIEW · WEBVIEW



JETBRAINS · LIGHT — TOOL WINDOW · SWING / KOTLIN UI DSL



CONTROLS A-09 A-10 A-11 A-12

REPLACES R7 page 034 · R4-S23

CONTRACT p. 85

S50 · AGENTS · BEHAVIOUR CONTRACT

Agent detail and steer: contract and revision notes

Original contract retained verbatim (R4-S23 / controls: A-09 A-10 A-11 A-12). Design on p. 84.

What changed in revision 8

- Authority and budget are stated as what the agent can and can't do.

Entry and return

Inspect / Agents with Roster, Approvals, Compare approaches and Coordination as subordinate destinations. Work / Inspect / History remain the only top-level clusters. Settings, Usage and Diagnostics use explicit Back. Return to the originating object, not the first row.

Scope and captured state

Parent task + current child/comparison/board generation; every write is scoped to its actual owner. Retain selected agent and the typed steer/post draft through refresh. A sent control is an immutable operation; current owner checks happen again at dispatch.

Primary action and recovery

Send steer. Reject stale child handles and duplicate controls visibly. A missing board route is unavailable, never an empty fabricated board. Recover unknown effects from receipts.

Loading, empty and stale

Show bounded queue/wait/working states. Parent and child budget reservations remain in one ledger. A comparison only recommends until explicit integration.

Keyboard and focus

Keyboard lists select by stable child/post identity. Open detail intentionally; new posts/child events announce without resetting the current selection or typed draft.

Layout and adverse content

One selected detail and one subordinate input at a time. Put raw IDs, revision cursors and provenance behind Details; keep role, assignment and outcome in the main row.

Starting point and required construction

Starting classifications: E, R. E existing operation; R runtime-backed adapter; N new contract; H host-only work. Exact per-control classifications remain in the catalog.

Reuse the reviewed domain operations and preserve all served fields. Build every missing native read/write adapter, capability gate and state renderer identified by the matching control/build rows.

BK03 owns intent/preset contracts; BK08 owns lifecycle/board/control adapters; BK07 owns shared budgets and BK09 owns comparison evidence.

Required interaction fixtures

Exercise first-read, empty/no-match, ready, refreshing/stale, error/refusal, action pending and acknowledged result. Also switch scope while the request is in flight.

Moved out of the interface

Every line of revision 7 plate copy that left the default view, and where it lives now.

"Faktor / Inspect / Agents (R7 window title)"
→ Native host chrome (VS Code view title, tabs or editor tab; JetBrains tool-window tabs, dialog title or Settings breadcrumb)

"Child 3 · parent run 047 · current generation"
→ Subtitle "Agent 3 of run 047"; generation in Tooltip / hover help

"Assignment / Verify AC-03; Owner / Candidate c-08"
→ Facts "Task" and "Works on"

"Authority envelope / Read, test and bounded tool execution · no integration"
→ Row "Can do"

"Shared budget / USD 0.18 reserved · debited from the parent ledger"
→ Row "Budget"

"Steering adds a recorded control input. It does not rewrite the original task or grant more tools."
→ Field help (shortened)

"Stop child"
→ Stop Verifier

"Verifier agent"
→ Title "Verifier"

"Prioritize the concurrent-refresh case. Preserve the captured acceptance criterion."
→ Steer text "…Keep the acceptance criterion as written."

"Steer this agent"
→ Field

"Stop child / Send steer"
→ Stop Verifier / Send Steer

S51 AGENTS

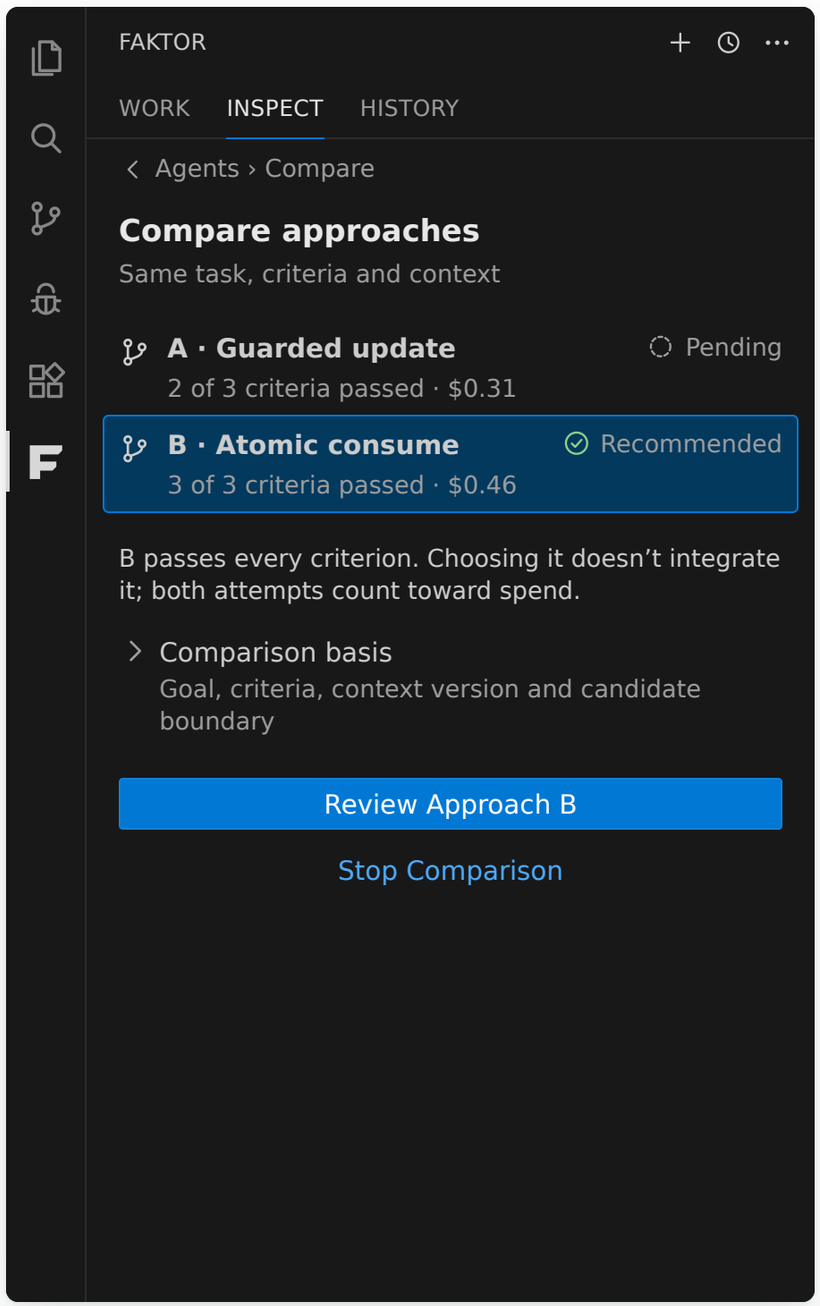
Compare approaches

Compare candidate approaches against one contract and recommend without automatically integrating the winner.

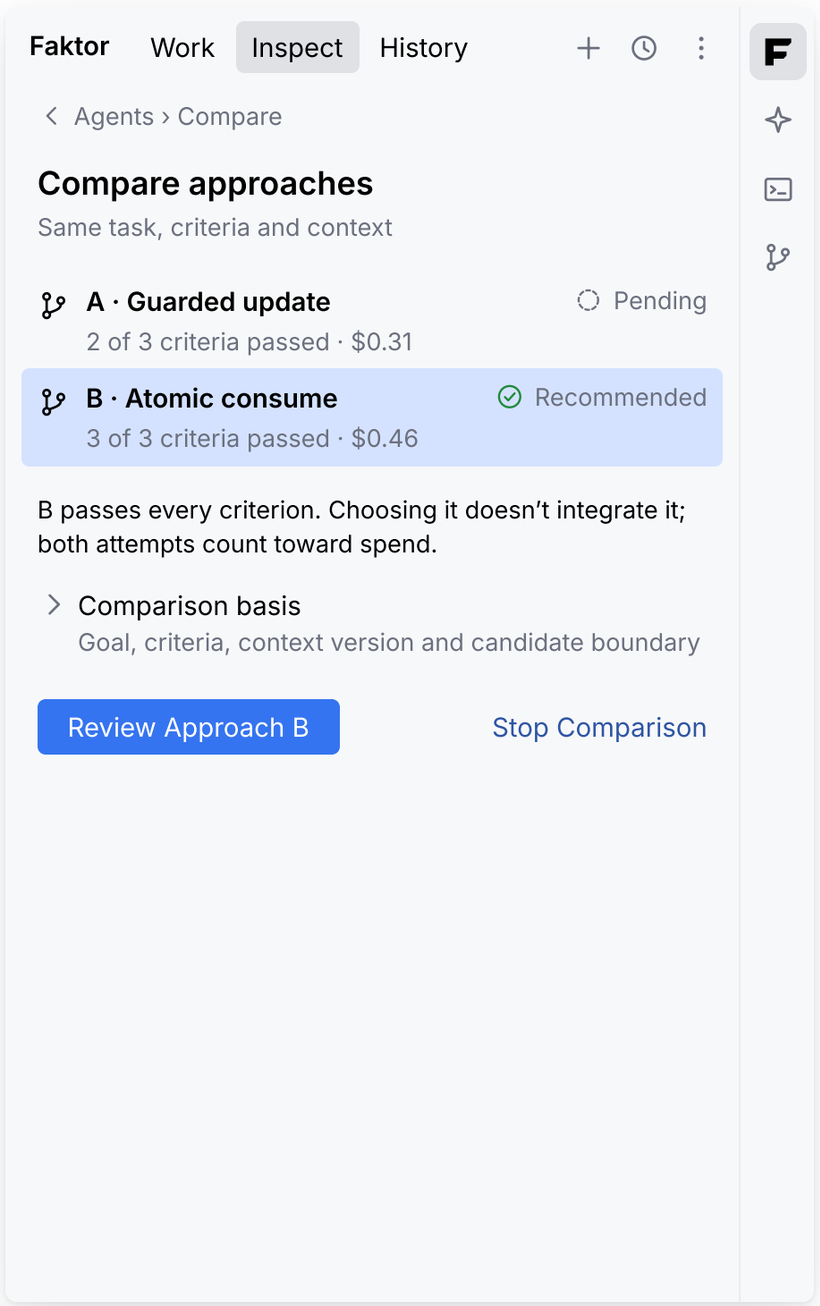
PRIMARY ACTION

Review Approach B

VS CODE · DARK MODERN — SIDEBAR VIEW · WEBVIEW



JETBRAINS · LIGHT — TOOL WINDOW · SWING / KOTLIN UI DSL



CONTROLS M-14 A-12 V-07

REPLACES R7 page 035 · R4-S24

CONTRACT p. 87

S51 · AGENTS · BEHAVIOUR CONTRACT

Compare approaches: contract and revision notes

Original contract retained verbatim (R4-S24 / controls: M-14 A-12 V-07). Design on p. 86.

What changed in revision 8

- The recommendation and its limit are one sentence; evidence opens from the primary action.

Entry and return

Inspect / Agents with Roster, Approvals, Compare approaches and Coordination as subordinate destinations. Work / Inspect / History remain the only top-level clusters. Settings, Usage and Diagnostics use explicit Back. Return to the originating object, not the first row.

Scope and captured state

Parent task + current child/comparison/board generation; every write is scoped to its actual owner. Retain selected agent and the typed steer/post draft through refresh. A sent control is an immutable operation; current owner checks happen again at dispatch.

Primary action and recovery

Review approach B. Reject stale child handles and duplicate controls visibly. A missing board route is unavailable, never an empty fabricated board. Recover unknown effects from receipts.

Loading, empty and stale

Show bounded queue/wait/working states. Parent and child budget reservations remain in one ledger. A comparison only recommends until explicit integration.

Keyboard and focus

Keyboard lists select by stable child/post identity. Open detail intentionally; new posts/child events announce without resetting the current selection or typed draft.

Layout and adverse content

One selected detail and one subordinate input at a time. Put raw IDs, revision cursors and provenance behind Details; keep role, assignment and outcome in the main row.

Starting point and required construction

Starting classifications: E, N, R. E existing operation; R runtime-backed adapter; N new contract; H host-only work. Exact per-control classifications remain in the catalog.

Reuse the reviewed domain operations and preserve all served fields. Build every missing native read/write adapter, capability gate and state renderer identified by the matching control/build rows.

BK03 owns intent/preset contracts; BK08 owns lifecycle/board/control adapters; BK07 owns shared budgets and BK09 owns comparison evidence.

Required interaction fixtures

Exercise first-read, empty/no-match, ready, refreshing/stale, error/refusal, action pending and acknowledged result. Also switch scope while the request is in flight.

Moved out of the interface

Every line of revision 7 plate copy that left the default view, and where it lives now.

"Faktor / Inspect / Agents / Compare (R7 window title)"
→ Native host chrome (VS Code view title, tabs or editor tab; JetBrains tool-window tabs, dialog title or Settings breadcrumb)

"One task contract · identical acceptance basis · separate candidates"
→ Subtitle

"Approach A / guarded update / Pending / 2 of 3 required criteria passed · USD 0.31"
→ Row A

"Approach B / atomic consume / Candidate / 3 of 3 required criteria passed · USD 0.46"
→ Row B (Recommended)

"Recommendation / Prefer B on the recorded criterion results. Both attempts contribute to eligible spend. A recommendation is not integration or completed proof."
→ Recommendation sentence (shortened)

"Comparison basis / Inspect / Same goal, criteria, context version and candidate boundary"
→ Disclosure

"Selected approach artifacts / Open / Checks, claims, cost and source identities"
→ Review Approach B opens them

"Compare approaches"
→ Title

"Stop comparison / Review approach B"
→ Stop Comparison / Review Approach B

S52 AGENTS

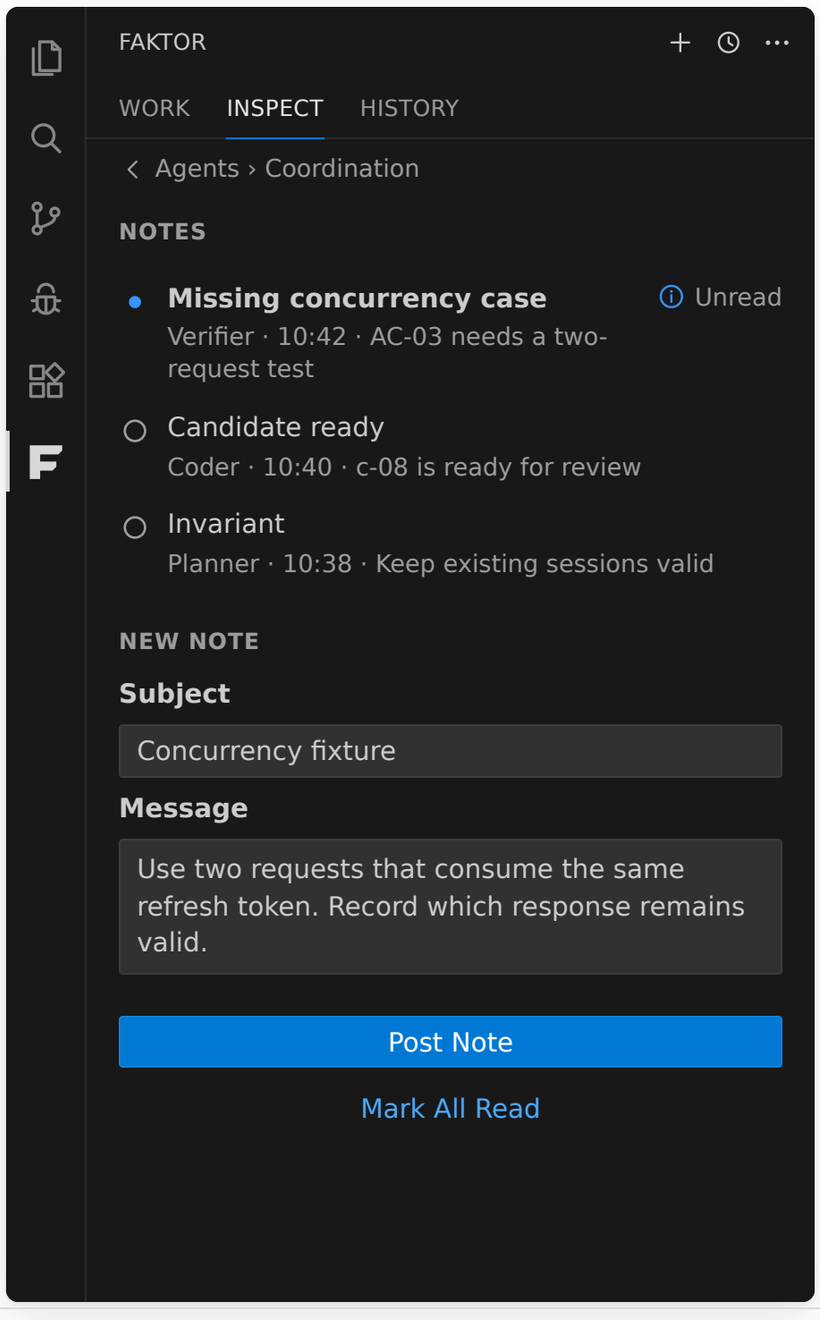
Coordination

Read and post bounded durable coordination notes with explicit read acknowledgement and stable revision paging.

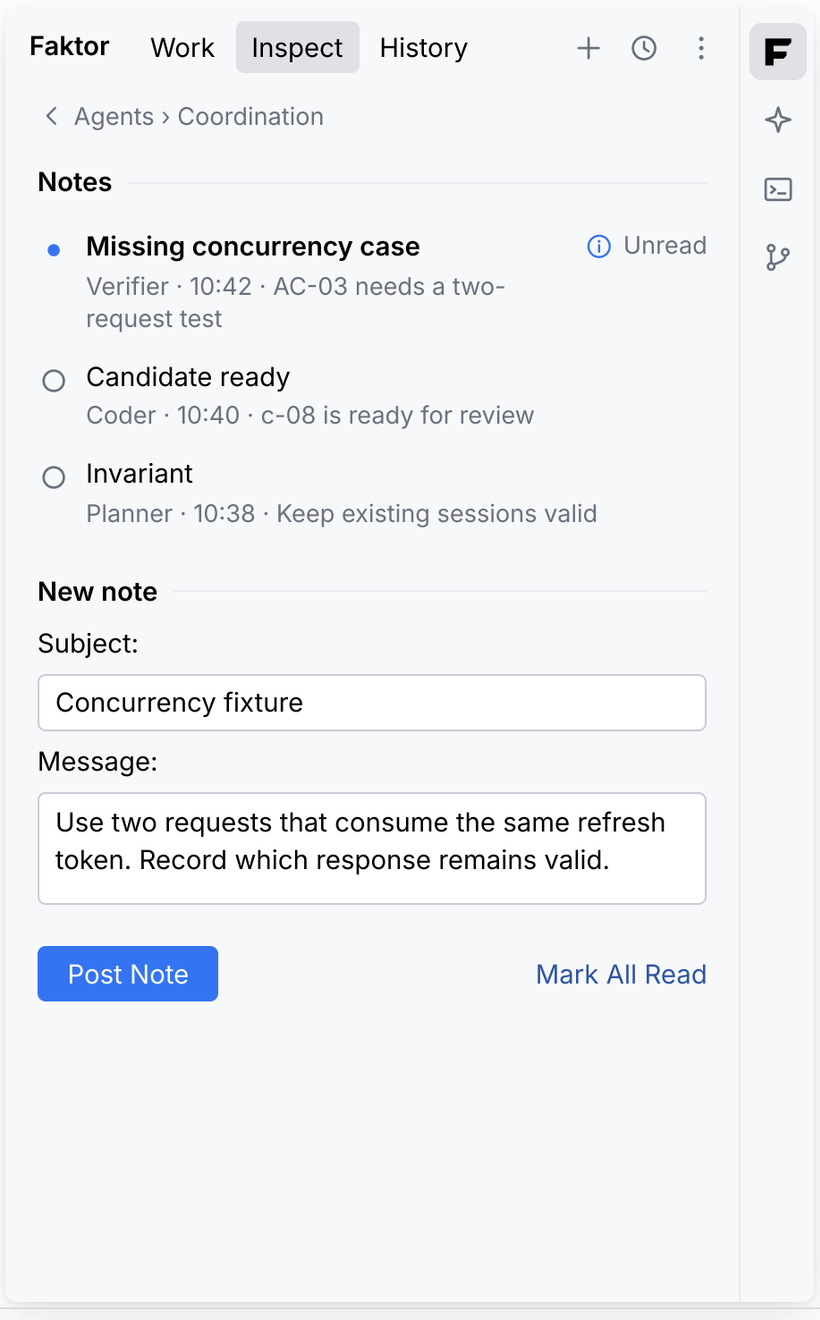
PRIMARY ACTION

Post Note

VS CODE · DARK MODERN — SIDEBAR VIEW · WEBVIEW



JETBRAINS · LIGHT — TOOL WINDOW · SWING / KOTLIN UI DSL



CONTROLS A-12 D-07

REPLACES R7 page 036 · R4-S25

CONTRACT p. 89

S52 · AGENTS · BEHAVIOUR CONTRACT

Coordination: contract and revision notes

Original contract retained verbatim (R4-S25 / controls: A-12 D-07). Design on p. 88.

What changed in revision 8

- Unread uses a dot plus the word “Unread”, never colour alone.

Entry and return

Inspect / Agents with Roster, Approvals, Compare approaches and Coordination as subordinate destinations. Work / Inspect / History remain the only top-level clusters. Settings, Usage and Diagnostics use explicit Back. Return to the originating object, not the first row.

Scope and captured state

Parent task + current child/comparison/board generation; every write is scoped to its actual owner. Retain selected agent and the typed steer/post draft through refresh. A sent control is an immutable operation; current owner checks happen again at dispatch.

Primary action and recovery

Post note. Reject stale child handles and duplicate controls visibly. A missing board route is unavailable, never an empty fabricated board. Recover unknown effects from receipts.

Loading, empty and stale

Show bounded queue/wait/working states. Parent and child budget reservations remain in one ledger. A comparison only recommends until explicit integration.

Keyboard and focus

Keyboard lists select by stable child/post identity. Open detail intentionally; new posts/child events announce without resetting the current selection or typed draft.

Layout and adverse content

One selected detail and one subordinate input at a time. Put raw IDs, revision cursors and provenance behind Details; keep role, assignment and outcome in the main row.

Starting point and required construction

Starting classifications: E, R. E existing operation; R runtime-backed adapter; N new contract; H host-only work. Exact per-control classifications remain in the catalog.

Reuse the reviewed domain operations and preserve all served fields. Build every missing native read/write adapter, capability gate and state renderer identified by the matching control/build rows.

BK03 owns intent/preset contracts; BK08 owns lifecycle/board/control adapters; BK07 owns shared budgets and BK09 owns comparison evidence.

Required interaction fixtures

Exercise first-read, empty/no-match, ready, refreshing/stale, error/refusal, action pending and acknowledged result. Also switch scope while the request is in flight.

Moved out of the interface

Every line of revision 7 plate copy that left the default view, and where it lives now.

“Faktor / Inspect / Agents / Coordination (R7 window title)”
→ Native host chrome (VS Code view title, tabs or editor tab; JetBrains tool-window tabs, dialog title or Settings breadcrumb)

“Run family 047 · three recorded posts · read state is explicit”
→ Removed: restated by the layout or label (unread marker shows read state)

“Verifier / missing concurrency case / 10:42 · AC-03 needs a two-request fixture”
→ Row

“Coder / candidate ready / 10:40 · c-08 is ready for review”
→ Row

“Planner / invariant / 10:38 · keep existing sessions valid”
→ Row

“Mark loaded posts read”
→ Mark All Read (applies to loaded notes; rule in Specification: screen contract)

“Coordination”
→ Breadcrumb

“New note / Subject / Concurrency fixture / Message”
→ Section “New note”

“Post note”
→ Post Note

PART 2 · SCREENS

2.5 History

4 screens. Each has a design spread for both IDEs and a contract page.

ID	SCREEN	JOB	PAGE
S26	**Conversation history**	History shows durable conversation identity, worktree, status and engineering outcome. Label title search separately from transcript find and indexed recall.	p. 91
S61	**Empty conversation history**	Distinguish true empty history, no search matches and failed reads; lead naturally back to a retained draft.	p. 93
S71	**Find in conversation**	Search the full durable conversation through the service cursor and return to the exact original message.	p. 96
S72	**Switch conversation safely**	Preserve draft, caret, scroll, sources and owned work while rebinding every response and action to another conversation.	p. 98

S26 HISTORY

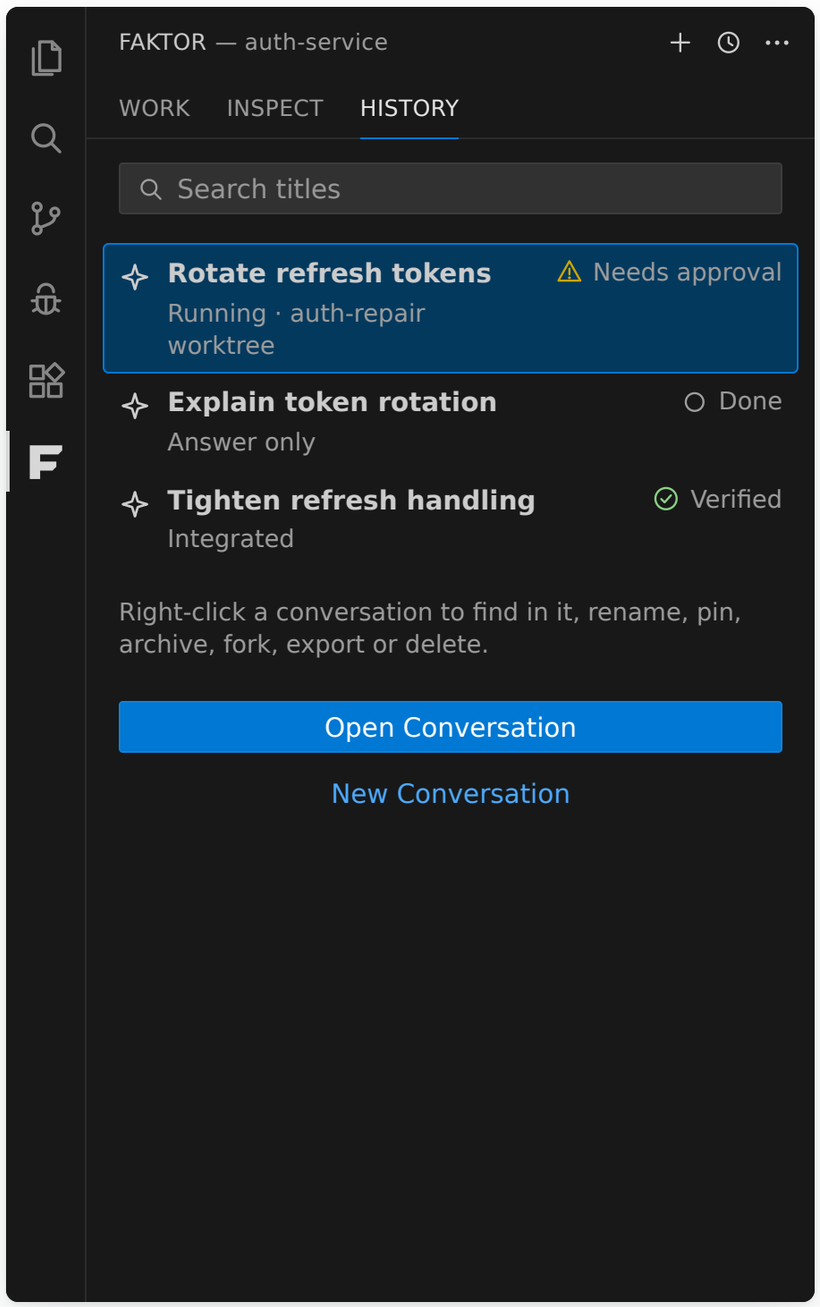
Conversation history

PRIMARY ACTION

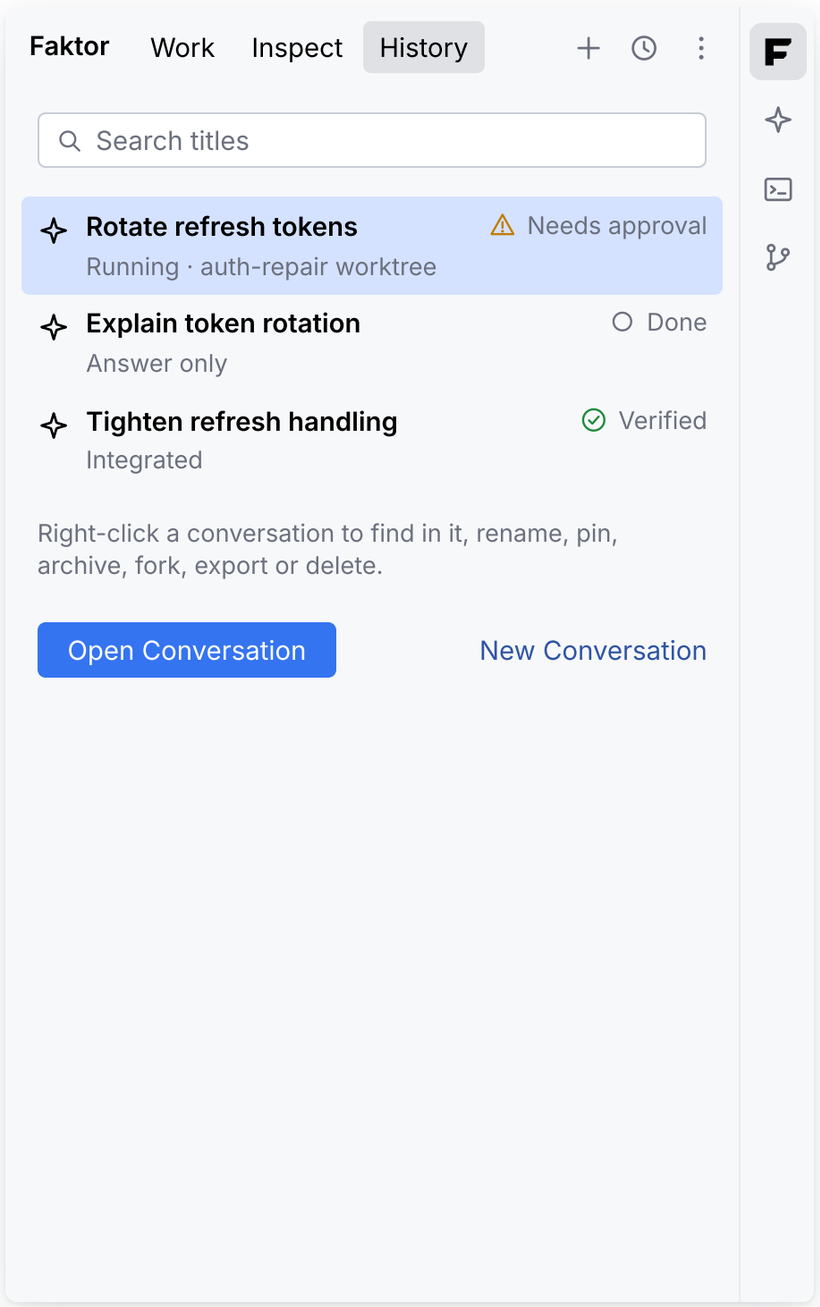
Open Conversation

History shows durable conversation identity, worktree, status and engineering outcome. Label title search separately from transcript find and indexed recall.

VS CODE · DARK MODERN — SIDEBAR VIEW · WEBVIEW



JETBRAINS · LIGHT — TOOL WINDOW · SWING / KOTLIN UI DSL



CONTROLS D-01 D-02 D-03 D-04 D-06 D-13 D-15 V-09

REPLACES R7 page 037 · R4-S26

CONTRACT p. 92

S26 · HISTORY · BEHAVIOUR CONTRACT

Conversation history: contract and revision notes

Original contract retained verbatim (R4-S26 / controls: D-01 D-02 D-03 D-04 D-06 D-13 D-15 V-09). Design on p. 91.

What changed in revision 8

- Row actions use the native context menu instead of a disclosure; status uses outcome words (Verified, Done).

Entry and return

Work, a captured submission receipt, or a task-specific attention item. Work / Inspect / History remain the only top-level clusters. Settings, Usage and Diagnostics use explicit Back. Return to the originating object, not the first row.

Scope and captured state

Daemon/workspace/worktree/conversation/task/request generation. Draft scope is separate from an admitted run. Capture one immutable logical start. Clear only its acknowledged draft version; preserve text typed after capture. Queue items acquire their own receipt and revision.

Primary action and recovery

Switch deliberately while retaining per-session drafts, scroll and pending attention; open supported session actions with precise retention scope.. Late callbacks cannot populate the wrong conversation. An unavailable history service is not an empty list; archived is not deleted.

Loading, empty and stale

Show the named operation and current owner. Keep the composer available for an independent draft. No fake percentages, fabricated tool steps or optimistic completion.

Keyboard and focus

Enter inserts a newline by default. The host send command respects IME composition. New approvals announce once without taking focus. Escape closes a detail; it never stops a run.

Layout and adverse content

Header, transcript scroll and composer are siblings. Collapse secondary metadata first. Keep a scoped Stop action and the active task reachable at short heights.

Starting point and required construction

Starting classifications: E, H, N. E existing operation; R runtime-backed adapter; N new contract; H host-only work. Exact per-control classifications remain in the catalog.

Reuse the reviewed domain operations and preserve all served fields. Build every missing native read/write adapter, capability gate and state renderer identified by the matching control/build rows.

BK08 owns admission, receipts, queue, control handles and reconciliation; BK05 owns authority and BK07 owns a shared exact budget ledger.

Required interaction fixtures

Exercise first-read, empty/no-match, ready, refreshing/stale, error/refusal, action pending and acknowledged result. Also switch scope while the request is in flight.

Moved out of the interface

Every line of revision 7 plate copy that left the default view, and where it lives now.

"Faktor / History (R7 window title)"
→ Native host chrome (VS Code view title, tabs or editor tab; JetBrains tool-window tabs, dialog title or Settings breadcrumb)

"auth-service · current project and worktree"
→ View description; worktree shown per row

"Title matches only. Find inside a transcript is a separate search."
→ Placeholder "Search titles" and the Find in Conversation menu item (S71)

"Approval / Active · run 047 · auth-repair worktree"
→ Row "Running · auth-repair worktree — Needs approval"

"Completed / Answer-only conversation · no verified change claimed"
→ Row "Answer only — Done"

"Verified / Matching evidence and landed candidate"
→ Row "Integrated — Verified"

"Other active work / 1 conversation; Session selection / Drafts stay separate"
→ Specification: screen contract (draft separation); active work is the row status

"Conversation actions / Find transcript, rename, pin, archive, fork, export and delete"
→ Native context menu, announced by one hint line

"Conversation history"
→ History tab

"Search conversation titles"
→ Placeholder "Search titles"

"Rotate refresh tokens / Explain token rotation / Tighten refresh handling"
→ Rows

"New conversation / Open conversation"
→ New Conversation / Open Conversation

S61 HISTORY

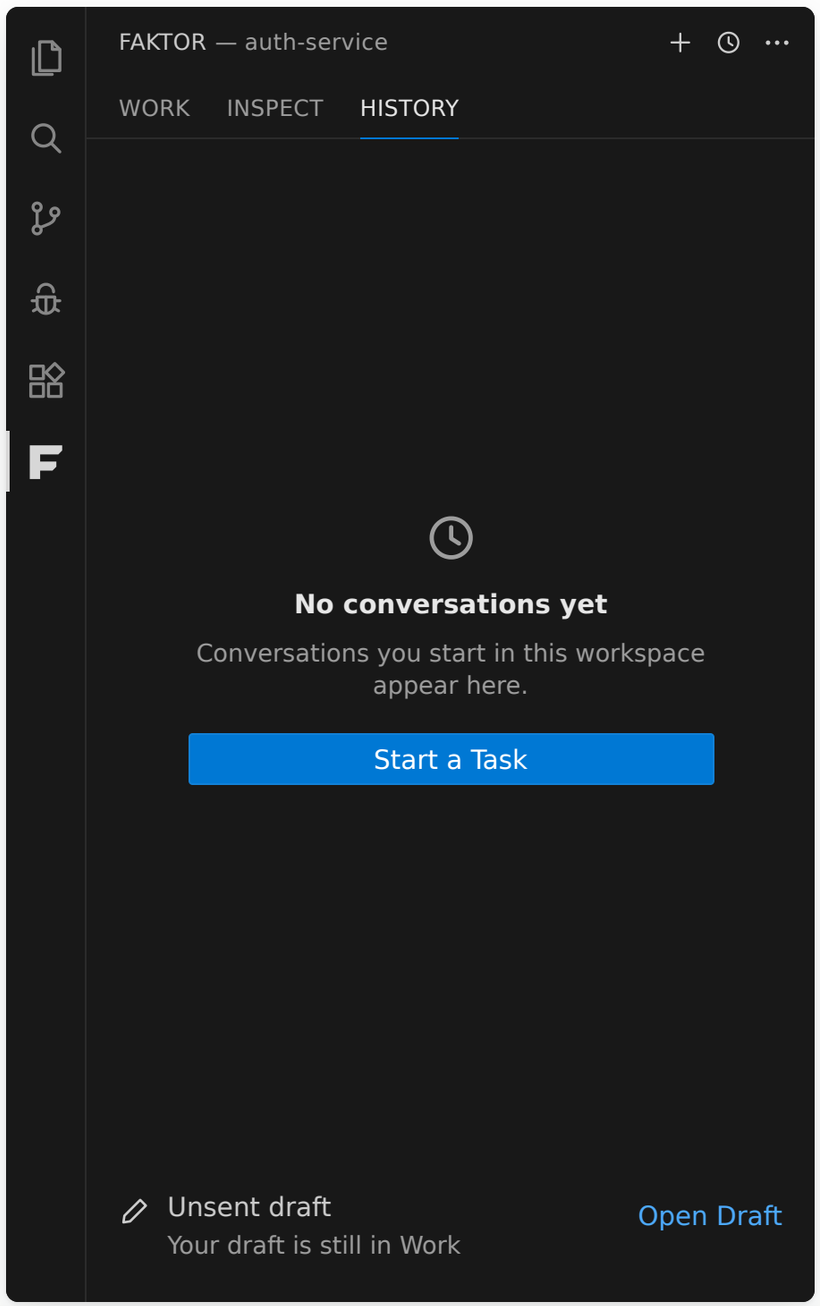
Empty conversation history

Distinguish true empty history, no search matches and failed reads; lead naturally back to a retained draft.

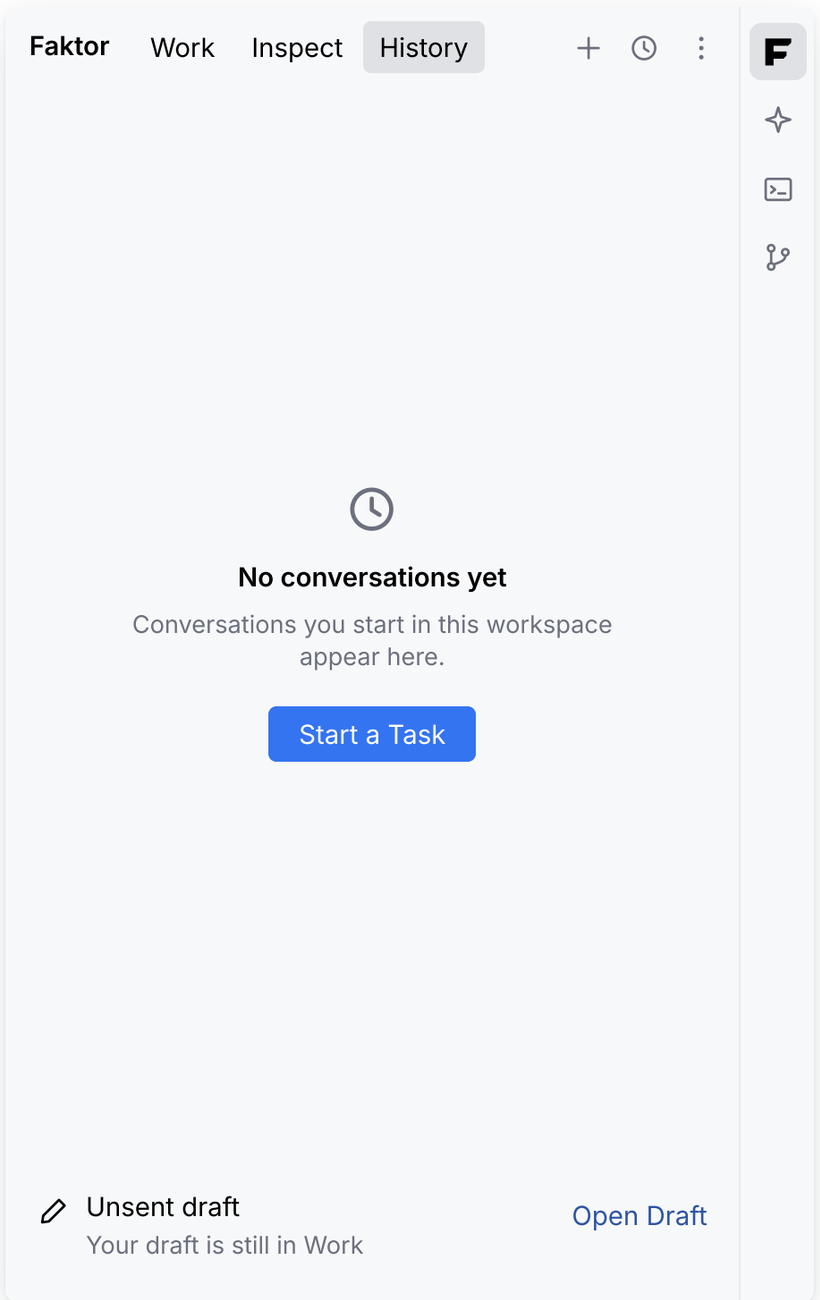
PRIMARY ACTION

Start a Task

VS CODE · DARK MODERN — SIDEBAR VIEW · WEBVIEW



JETBRAINS · LIGHT — TOOL WINDOW · SWING / KOTLIN UI DSL



CONTROLS D-01 D-02

REPLACES R7 page 038 · Expanded standard-agent surface

CONTRACT p. 94

S61 · HISTORY · BEHAVIOUR CONTRACT

Empty conversation history: contract and revision notes

Original contract retained verbatim (Expanded standard-agent surface / controls: D-01 D-02). Design on p. 93.

What changed in revision 8

- Developer guidance about other states left the interface and is now specified as three distinct states (empty, no matches, failed).

Entry and return

Work, a captured submission receipt, or a task-specific attention item. Work / Inspect / History remain the only top-level clusters. Settings, Usage and Diagnostics use explicit Back. Return to the originating object, not the first row.

Scope and captured state

Daemon/workspace/worktree/conversation/task/request generation. Draft scope is separate from an admitted run. Capture one immutable logical start. Clear only its acknowledged draft version; preserve text typed after capture. Queue items acquire their own receipt and revision.

Primary action and recovery

Start a task. Retain the captured body, options and source handles. Unknown admission/effect requires receipt reconciliation before replay. A changed body needs a different request identity.

Loading, empty and stale

Show the named operation and current owner. Keep the composer available for an independent draft. No fake percentages, fabricated tool steps or optimistic completion.

Keyboard and focus

Enter inserts a newline by default. The host send command respects IME composition. New approvals announce once without taking focus. Escape closes a detail; it never stops a run.

Layout and adverse content

Header, transcript scroll and composer are siblings. Collapse secondary metadata first. Keep a scoped Stop action and the active task reachable at short heights.

Starting point and required construction

Starting classifications: E. E existing operation; R runtime-backed adapter; N new contract; H host-only work. Exact per-control classifications remain in the catalog.

Reuse the reviewed domain operations and preserve all served fields. Build every missing native read/write adapter, capability gate and state renderer identified by the matching control/build rows.

BK08 owns admission, receipts, queue, control handles and reconciliation; BK05 owns authority and BK07 owns a shared exact budget ledger.

Required interaction fixtures

Exercise first-read, empty/no-match, ready, refreshing/stale, error/refusal, action pending and acknowledged result. Also switch scope while the request is in flight.

Moved out of the interface

Every line of revision 7 plate copy that left the default view, and where it lives now.

"Faktor / History (R7 window title)"
→ Native host chrome (VS Code view title, tabs or editor tab; JetBrains tool-window tabs, dialog title or Settings breadcrumb)

"History for auth-service · current workspace binding"
→ View description; "in this workspace"

"Your saved conversations will appear here after a task is admitted."
→ Empty text (plain language: "Conversations you start…")

"A search with no matches is a different state: keep its query and offer Clear filters. A failed history read shows Retry, rather than this empty view."
→ Specification: screen contract — implementation instruction for the History states; see the three-state reference on the contract page

"Current draft / Preserved / The task you are writing is still in Work"
→ Row "Unsent draft — Your draft is still in Work"

"Back to Work / Start a task"
→ Open Draft and Start a Task

"No conversations yet"
→ Empty-state title

S61 · STATE REFERENCE

History has three different “nothing” states

Revision 7 explained this distinction to users inside the empty view (R7 page 038). It is a specification rule, so it lives here, and each state has its own design and recovery action.

Empty

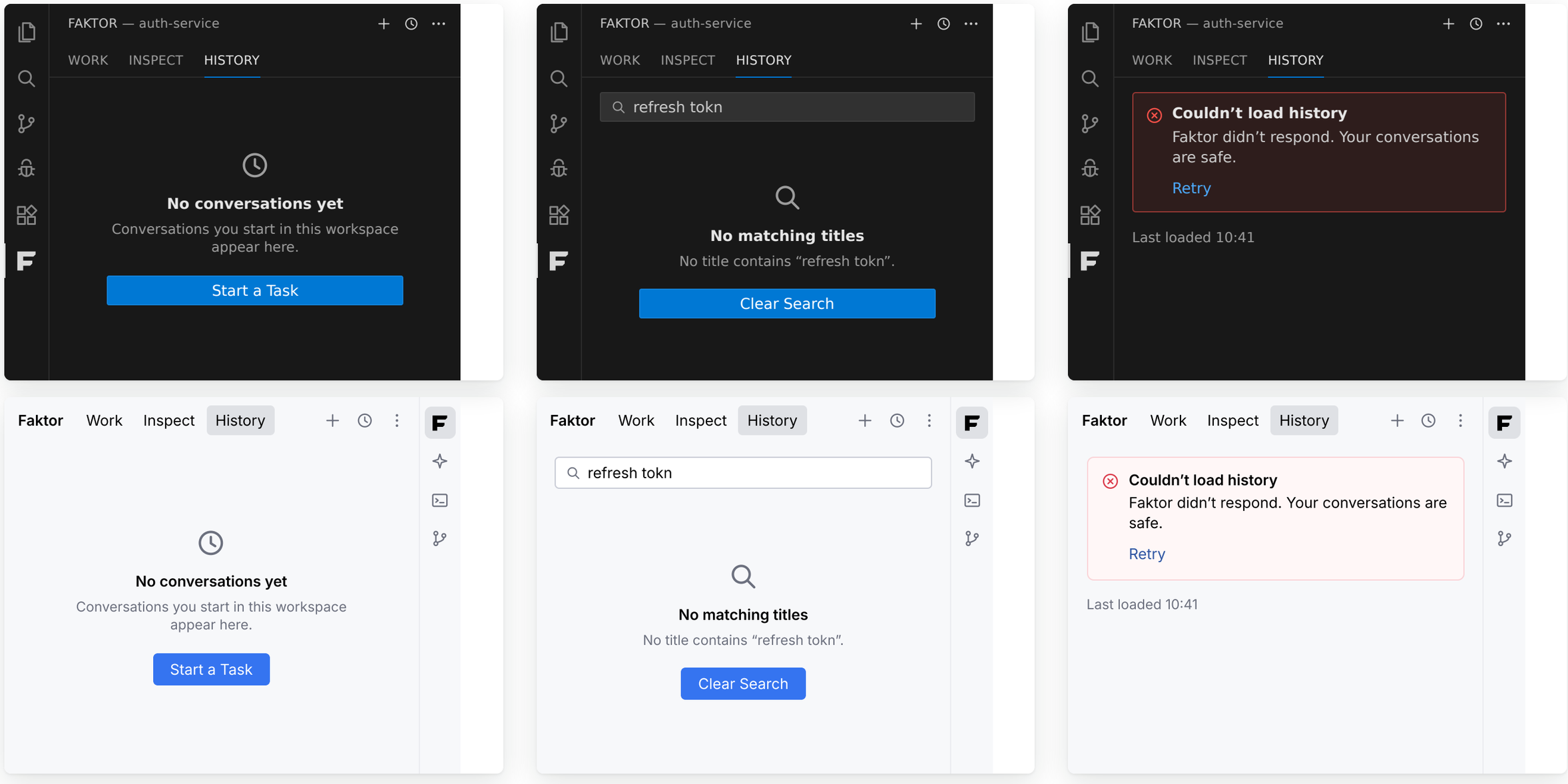
No conversations exist for this workspace. Offer Start a Task.

No matches

Conversations exist; the search matched none. Keep the query and offer Clear Search.

Couldn’t load

The read failed. Never show the empty state; keep the last list if one exists and offer Retry.



Top row VS Code Dark Modern, bottom row JetBrains Light. Announce each state once through a polite live region; the no-match state keeps focus in the search field.

S71 HISTORY

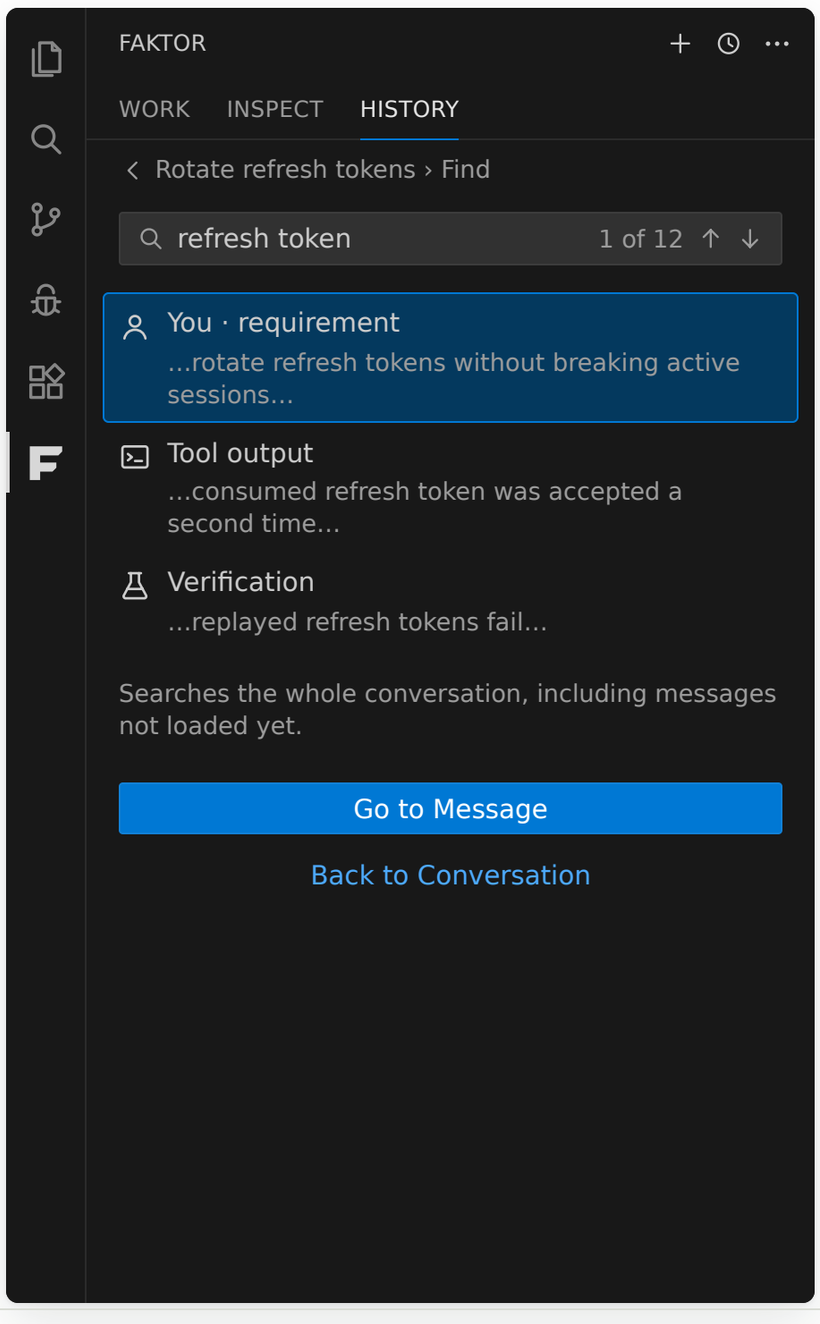
Find in conversation

Search the full durable conversation through the service cursor and return to the exact original message.

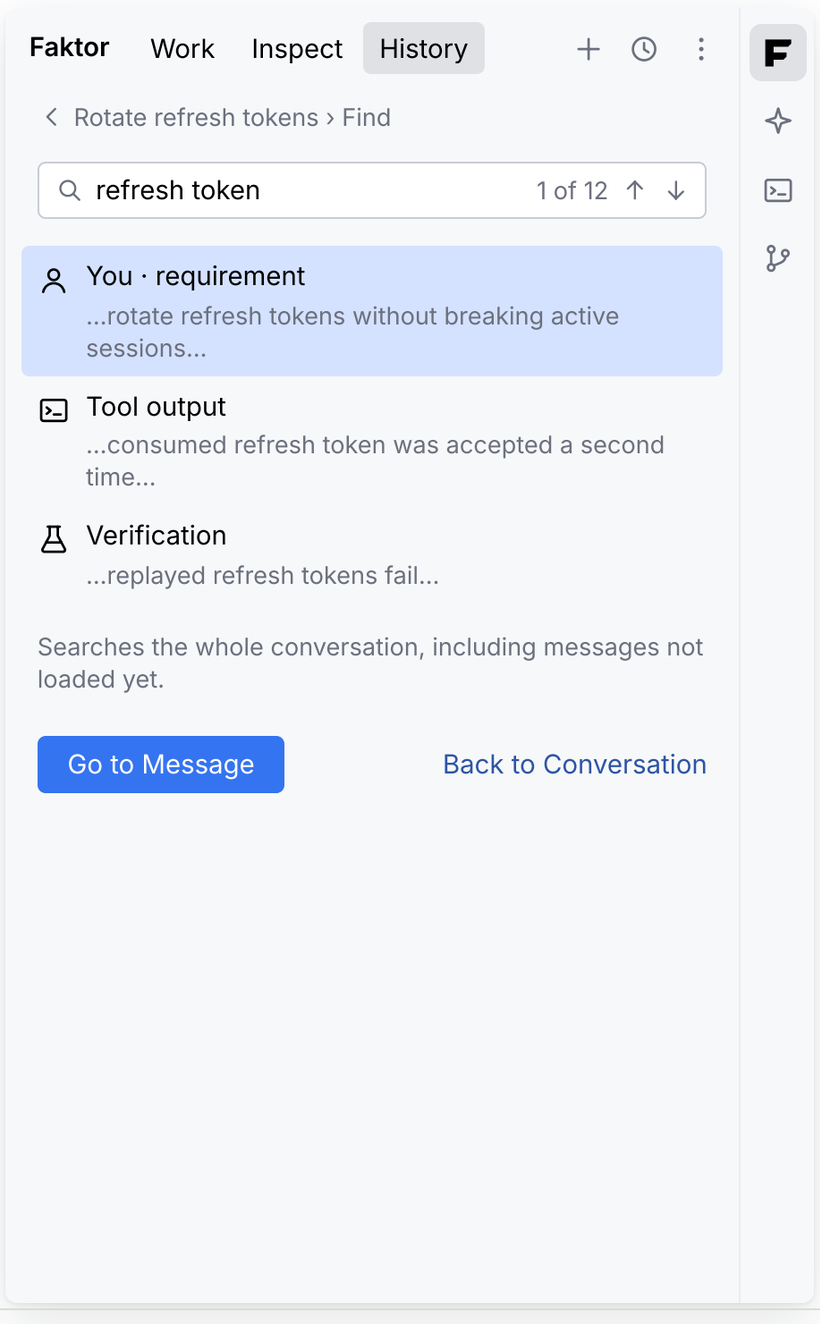
PRIMARY ACTION

Go to Message

VS CODE · DARK MODERN — SIDEBAR VIEW · WEBVIEW



JETBRAINS · LIGHT — TOOL WINDOW · SWING / KOTLIN UI DSL



CONTROLS D-03

REPLACES R7 page 039 · Expanded standard-agent surface

CONTRACT p. 97

S71 · HISTORY · BEHAVIOUR CONTRACT

Find in conversation: contract and revision notes

Original contract retained verbatim (Expanded standard-agent surface / controls: D-03). Design on p. 96.

What changed in revision 8

- The search field follows the native find-widget idiom (count plus previous/next).

Entry and return

Work, a captured submission receipt, or a task-specific attention item. Work / Inspect / History remain the only top-level clusters. Settings, Usage and Diagnostics use explicit Back. Return to the originating object, not the first row.

Scope and captured state

Daemon/workspace/worktree/conversation/task/request generation. Draft scope is separate from an admitted run. Capture one immutable logical start. Clear only its acknowledged draft version; preserve text typed after capture. Queue items acquire their own receipt and revision.

Primary action and recovery

Open selected message. Retain the captured body, options and source handles. Unknown admission/effect requires receipt reconciliation before replay. A changed body needs a different request identity.

Loading, empty and stale

Show the named operation and current owner. Keep the composer available for an independent draft. No fake percentages, fabricated tool steps or optimistic completion.

Keyboard and focus

Enter inserts a newline by default. The host send command respects IME composition. New approvals announce once without taking focus. Escape closes a detail; it never stops a run.

Layout and adverse content

Header, transcript scroll and composer are siblings. Collapse secondary metadata first. Keep a scoped Stop action and the active task reachable at short heights.

Starting point and required construction

Starting classifications: H. E existing operation; R runtime-backed adapter; N new contract; H host-only work. Exact per-control classifications remain in the catalog.

NEW: add bounded full-transcript search with a durable cursor and message identities; searching only loaded UI rows is insufficient.

BK08 owns admission, receipts, queue, control handles and reconciliation; BK05 owns authority and BK07 owns a shared exact budget ledger.

Required interaction fixtures

Exercise first-read, empty/no-match, ready, refreshing/stale, error/refusal, action pending and acknowledged result. Also switch scope while the request is in flight.

Moved out of the interface

Every line of revision 7 plate copy that left the default view, and where it lives now.

"Faktor / History / Find (R7 window title)"
→ Native host chrome (VS Code view title, tabs or editor tab; JetBrains tool-window tabs, dialog title or Settings breadcrumb)

"Run 047 · paged transcript search · original message identities"
→ Coverage note; paging rule in Specification: screen contract

"Result 1 of 12 / user requirement / Selected"
→ Counter "1 of 12" and row label

"Result 2 of 12 / tool observation"
→ Row "Tool output"

"Result 3 of 12 / verification"
→ Row "Verification"

"Search coverage / Service cursor covers the full conversation; loaded DOM is only a window."
→ Note "Searches the whole conversation, including messages not loaded yet."; DOM detail in Specification: screen contract

"Open selected message"
→ Go to Message

"Find in conversation"
→ Breadcrumb "Find"

"refresh token"
→ Search field

"Back to conversation"
→ Back to Conversation

S72 HISTORY

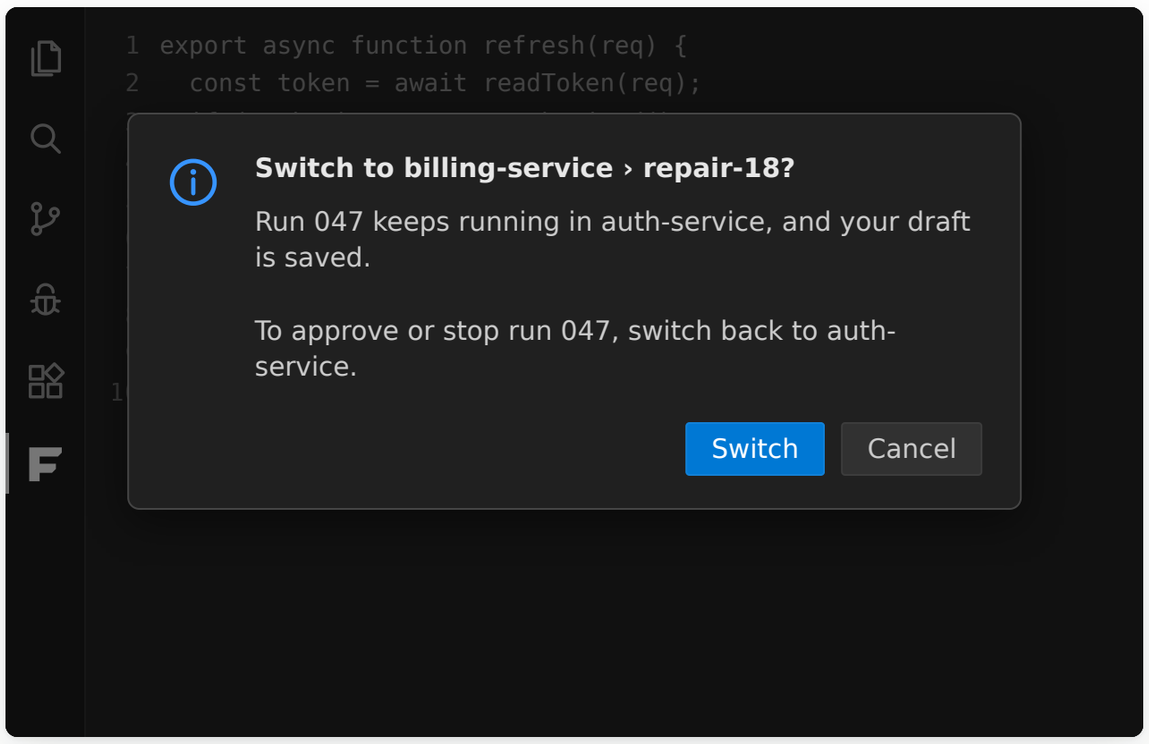
Switch conversation safely

PRIMARY ACTION

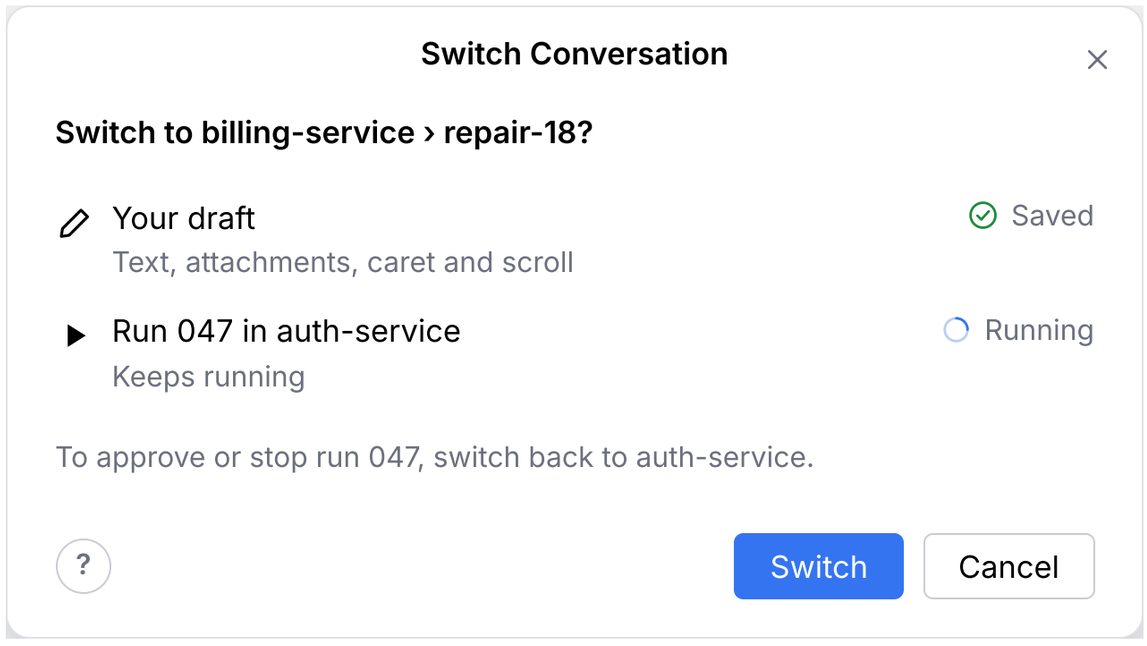
Switch

Preserve draft, caret, scroll, sources and owned work while rebinding every response and action to another conversation.

VS CODE · DARK MODERN — MODAL MESSAGE · NATIVE



JETBRAINS · LIGHT — DIALOGWRAPPER



CONTROLS D-01 D-15 X-13

REPLACES R7 page 040 · Expanded standard-agent surface

CONTRACT p. 99

S72 · HISTORY · BEHAVIOUR CONTRACT

Switch conversation safely: contract and revision notes

Original contract retained verbatim (Expanded standard-agent surface / controls: D-01 D-15 X-13). Design on p. 98.

What changed in revision 8

- Implementation rules about late responses moved to the contract; the dialog keeps only consequences.

Entry and return

More / Diagnostics, the setup walkthrough, or the exact blocked operation. Work / Inspect / History remain the only top-level clusters. Settings, Usage and Diagnostics use explicit Back. Return to the originating object, not the first row.

Scope and captured state

Execution host + service instance/generation + protocol version/cursor + trust source; account bindings stay distinct. Starting, restarting and stopping the service are explicit owned operations. Rebind after recovery; opening a view is not a start command.

Primary action and recovery

Switch to repair-18. Show a bounded human reason and actionable recovery. Missing checksum, checksum mismatch, signature status, schema refusal and local/cloud boundary are separate facts.

Loading, empty and stale

Keep last coherent data stale, retain event cursors and distinguish automatic transient reconnect from a protocol-blocked stop.

Keyboard and focus

A background disconnect never steals typing focus. Recovery opens in a native command/dialog; Back returns to the blocked task. Copy diagnostics confirms once.

Layout and adverse content

Show task-facing recovery first, technical detail behind disclosure. Wrap exact material paths; redact secrets before display and before copy/export.

Starting point and required construction

Starting classifications: E, N. E existing operation; R runtime-backed adapter; N new contract; H host-only work. Exact per-control classifications remain in the catalog.

Reuse native bindings; build per-binding draft/selection/scroll storage, request epochs and current-generation action fences.

BK10 owns readiness/discovery/diagnostics; BK08 fences stream and operation owners; BK05 reports actual isolation and policy.

Required interaction fixtures

Exercise first-read, empty/no-match, ready, refreshing/stale, error/refusal, action pending and acknowledged result. Also switch scope while the request is in flight.

Moved out of the interface

Every line of revision 7 plate copy that left the default view, and where it lives now.

"Faktor / History / Switch (R7 window title)"
→ Native host chrome (VS Code view title, tabs or editor tab; JetBrains tool-window tabs, dialog title or Settings breadcrumb)

"Switch workspace and conversation / Current draft belongs to auth-service / run 047."
→ Dialog question and draft row

"billing-service / repair-18 / Target / Different workspace and candidate boundary"
→ Question "Switch to billing-service › repair-18?"

"Preserve current draft / Saved locally / Goal, sources, options, caret and scroll position"
→ Row "Your draft — Saved"

"Active work in auth-service / Running / Run 047 keeps its existing owner. Switching does not stop it."
→ Row "Run 047 — Keeps running"

"Rebind every service view"
→ Specification: screen contract

"Late responses, approvals, process handles and proof from run 047 must not paint into repair-18. Return to the original binding to control that work."
→ Specification: screen contract (implementation rule); user-facing part kept as "To approve or stop run 047, switch back"

"Switch to repair-18"
→ Switch

"Cancel"
→ Cancel

PART 2 · SCREENS

2.6 Usage

3 screens. Each has a design spread for both IDEs and a contract page.

ID	SCREEN	JOB	PAGE
S53	**Recorded outcome cohort**	Inspect an outcome cohort with eligible attempted spend, verified landed denominator and actual local/hosted routes.	p. 101
S54	**Usage and quota**	Separate provider API spend, optional subscription quota, reservations and unmeasured local operating cost.	p. 103
S27	**Usage & outcomes**	Separate token accounting, provider/cache prices, quota, local estimates and verification-based engineering cost. Explain every denominator.	p. 105

S53 USAGE

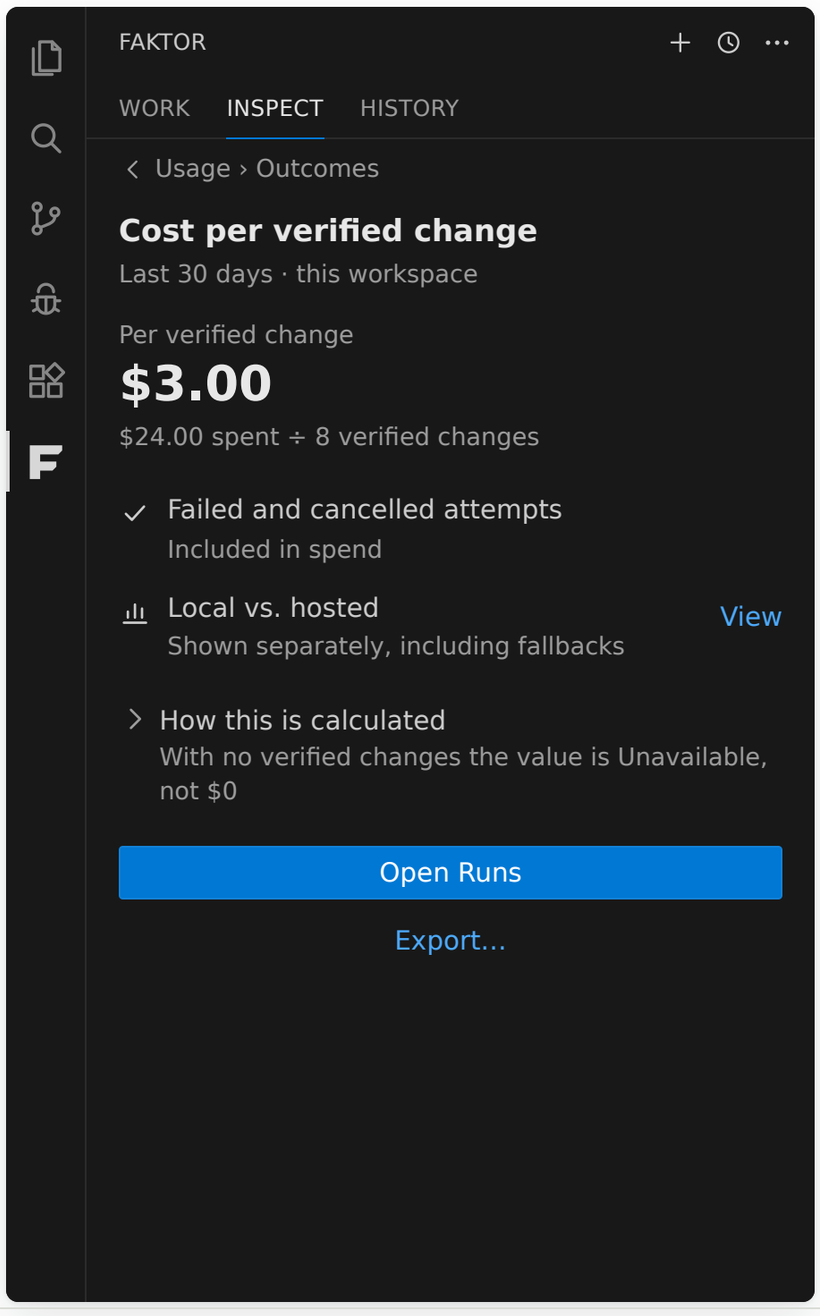
PRIMARY ACTION

Open Runs

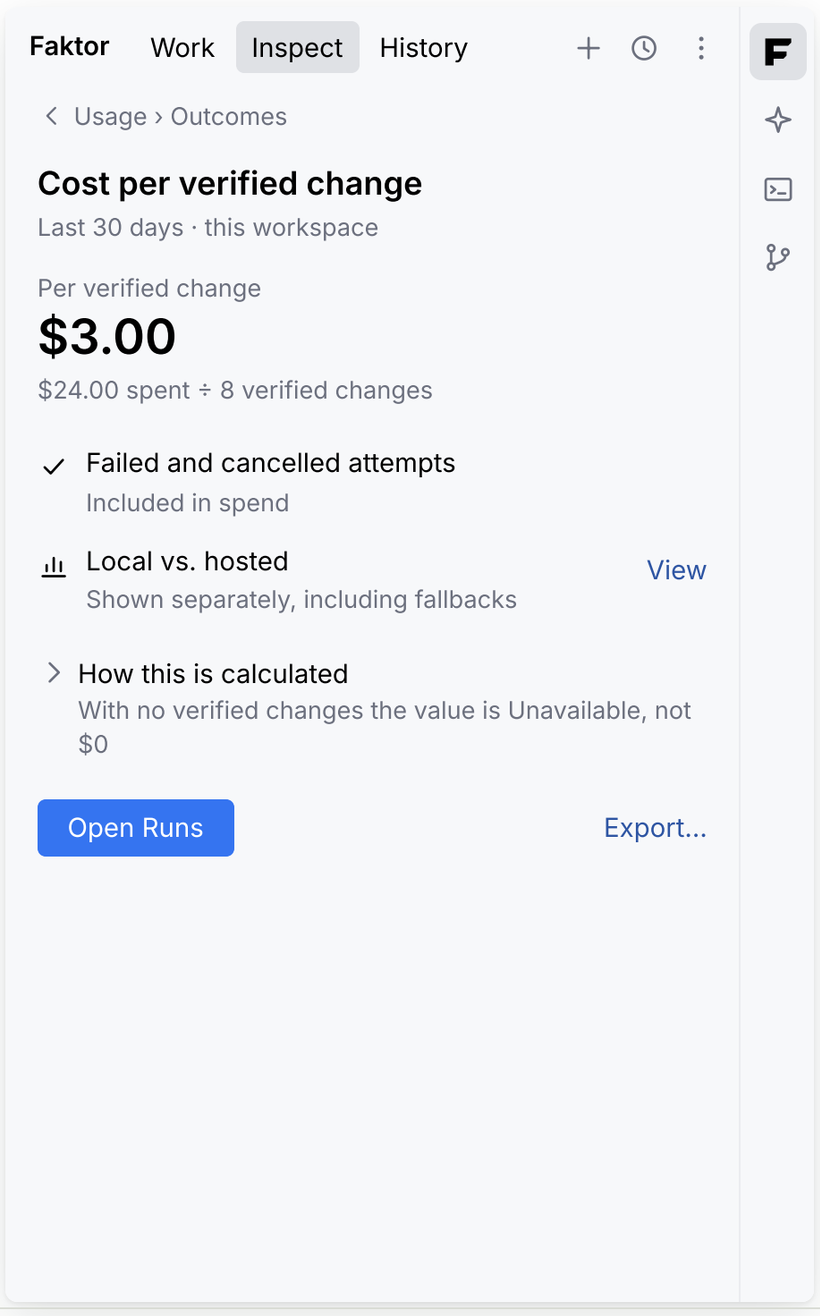
Recorded outcome cohort

Inspect an outcome cohort with eligible attempted spend, verified landed denominator and actual local/hosted routes.

VS CODE · DARK MODERN — SIDEBAR VIEW · WEBVIEW



JETBRAINS · LIGHT — TOOL WINDOW · SWING / KOTLIN UI DSL



CONTROLS O-06 K-12 M-13

REPLACES R7 page 041 · R4-S27

CONTRACT p. 102

S53 · USAGE · BEHAVIOUR CONTRACT

Recorded outcome cohort: contract and revision notes

Original contract retained verbatim (R4-S27 / controls: O-06 K-12 M-13). Design on p. 101.

What changed in revision 8

- The acronym VCSC is replaced by its meaning; the formula sits under the number.

Entry and return

More / Usage or the selected recorded run/cohort. Work / Inspect / History remain the only top-level clusters. Settings, Usage and Diagnostics use explicit Back. Return to the originating object, not the first row.

Scope and captured state

Workspace/time/filter/route cohort and service paging cursor; provider charge basis is separate from account quota. Filter changes create a new query epoch. Retain the selected run by ID and disclose the cohort denominator and eligibility rules.

Primary action and recovery

Open recorded runs. Never replace unknown with zero or average only successful attempts. Eligible failed/cancelled attempted spend remains included; exact money stays in integer micro-USD.

Loading, empty and stale

Keep prior results marked stale during refresh. First read, no records, unknown price, zero denominator and disabled local billing are distinct.

Keyboard and focus

Filters and table rows use native keyboard semantics. Refresh preserves the selected record and scroll anchor; announcements summarize changes once.

Layout and adverse content

Show definition and basis next to the metric. Tables move to a wide editor/panel area; narrow views use labelled record summaries and a selected detail.

Starting point and required construction

Starting classifications: N, R. E existing operation; R runtime-backed adapter; N new contract; H host-only work. Exact per-control classifications remain in the catalog.

Reuse the reviewed domain operations and preserve all served fields. Build every missing native read/write adapter, capability gate and state renderer identified by the matching control/build rows.

BK07 owns exact accounting/cohorts and BK10 owns optional account/service reads; native local operation cannot depend on a cloud account.

Required interaction fixtures

Exercise first-read, empty/no-match, ready, refreshing/stale, error/refusal, action pending and acknowledged result. Also switch scope while the request is in flight.

Moved out of the interface

Every line of revision 7 plate copy that left the default view, and where it lives now.

"Faktor / Usage / Outcomes (R7 window title)"
→ Native host chrome (VS Code view title, tabs or editor tab; JetBrains tool-window tabs, dialog title or Settings breadcrumb)

"Recorded outcome cohort"
→ Title "Cost per verified change" (VCSC spelled out)

"Last 30 days · exact eligible attempts · same workspace filter"
→ Subtitle; eligibility in the disclosure

"Eligible attempted spend / USD 24.00; Verified landed changes / 8"
→ Formula line under the number

"VCSC / USD 3.00 / USD 24.00 / 8 verified landed changes"
→ Big number with formula

"Failed and cancelled attempts / Included when eligible; no cherry-picked success-only cost"
→ Row "Included in spend"

"Local / hosted route / Show captured route and fallback separately"
→ Row "Local vs. hosted"

"No denominator means unavailable / A cohort with zero verified landed changes has no VCSC value. Unknown price is not zero."
→ How this is calculated disclosure

"Export cohort / Open recorded runs"
→ Export… / Open Runs

"Open recorded runs"
→ Open Runs

S54 USAGE

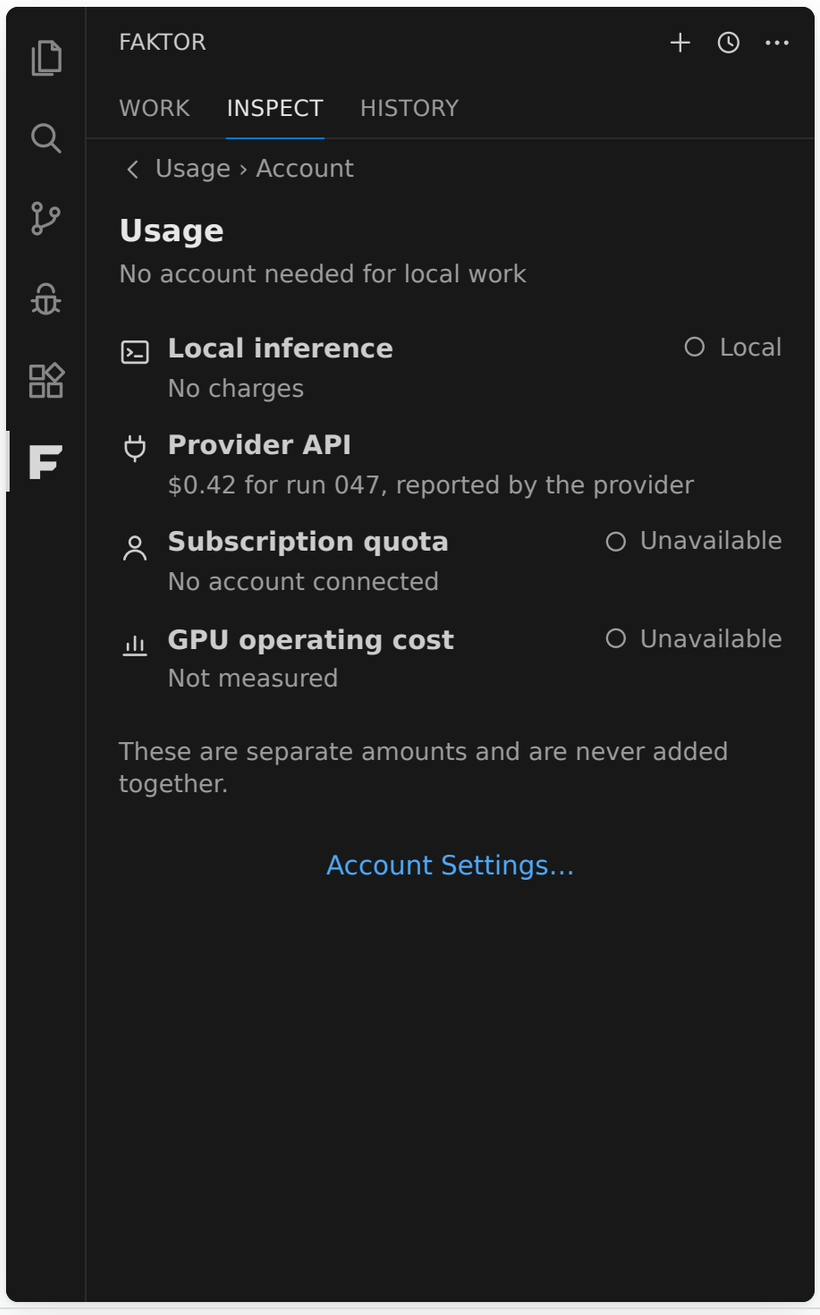
Usage and quota

Separate provider API spend, optional subscription quota, reservations and unmeasured local operating cost.

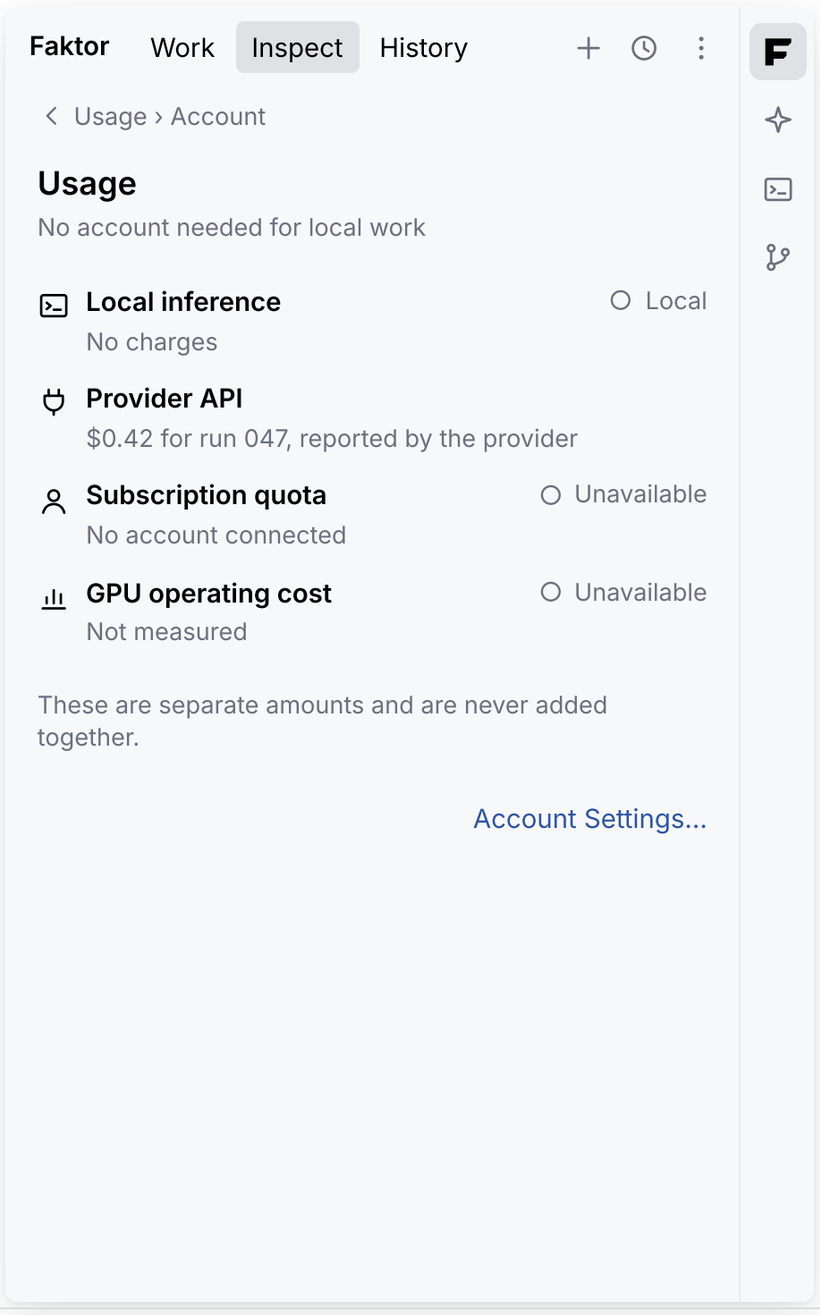
PRIMARY ACTION

None (read-only)

VS CODE · DARK MODERN — SIDEBAR VIEW · WEBVIEW



JETBRAINS · LIGHT — TOOL WINDOW · SWING / KOTLIN UI DSL



CONTROLS O-06 O-07

REPLACES R7 page 042 · R4-S28

CONTRACT p. 104

S54 · USAGE · BEHAVIOUR CONTRACT

Usage and quota: contract and revision notes

Original contract retained verbatim (R4-S28 / controls: O-06 O-07). Design on p. 103.

What changed in revision 8

- Read-only report: no primary button; the only action is a link to optional account settings.

Entry and return

More / Usage or the selected recorded run/cohort. Work / Inspect / History remain the only top-level clusters. Settings, Usage and Diagnostics use explicit Back. Return to the originating object, not the first row.

Scope and captured state

Workspace/time/filter/route cohort and service paging cursor; provider charge basis is separate from account quota. Filter changes create a new query epoch. Retain the selected run by ID and disclose the cohort denominator and eligibility rules.

Primary action and recovery

Open optional account settings. Never replace unknown with zero or average only successful attempts. Eligible failed/cancelled attempted spend remains included; exact money stays in integer micro-USD.

Loading, empty and stale

Keep prior results marked stale during refresh. First read, no records, unknown price, zero denominator and disabled local billing are distinct.

Keyboard and focus

Filters and table rows use native keyboard semantics. Refresh preserves the selected record and scroll anchor; announcements summarize changes once.

Layout and adverse content

Show definition and basis next to the metric. Tables move to a wide editor/panel area; narrow views use labelled record summaries and a selected detail.

Starting point and required construction

Starting classifications: E, R. E existing operation; R runtime-backed adapter; N new contract; H host-only work. Exact per-control classifications remain in the catalog.

Reuse the reviewed domain operations and preserve all served fields. Build every missing native read/write adapter, capability gate and state renderer identified by the matching control/build rows.

BK07 owns exact accounting/cohorts and BK10 owns optional account/service reads; native local operation cannot depend on a cloud account.

Required interaction fixtures

Exercise first-read, empty/no-match, ready, refreshing/stale, error/refusal, action pending and acknowledged result. Also switch scope while the request is in flight.

Moved out of the interface

Every line of revision 7 plate copy that left the default view, and where it lives now.

"Faktor / Usage / Account (R7 window title)"
→ Native host chrome (VS Code view title, tabs or editor tab; JetBrains tool-window tabs, dialog title or Settings breadcrumb)

"Usage and account / Native local work remains available without an account."
→ Header and subtitle

"No cloud billing service is configured"
→ Row "No charges"

"USD 0.42 observed for run 047 · provider reported"
→ Row "$0.42 for run 047, reported by the provider"

"No connected control-plane account"
→ "No account connected"

"No measured local operating-cost model configured"
→ "Not measured"

"API charges, subscription quota, reservations and estimated local operating cost are separate quantities."
→ Note (shortened)

"Back to usage / Open optional account settings"
→ Breadcrumb / Account Settings…

"Provider API spend"
→ Row "Provider API"

"Hosted subscription quota / Unavailable"
→ Row "Subscription quota — Unavailable"

"Local inference / Local"
→ Row

"GPU operating cost / Unavailable"
→ Row

S27 USAGE

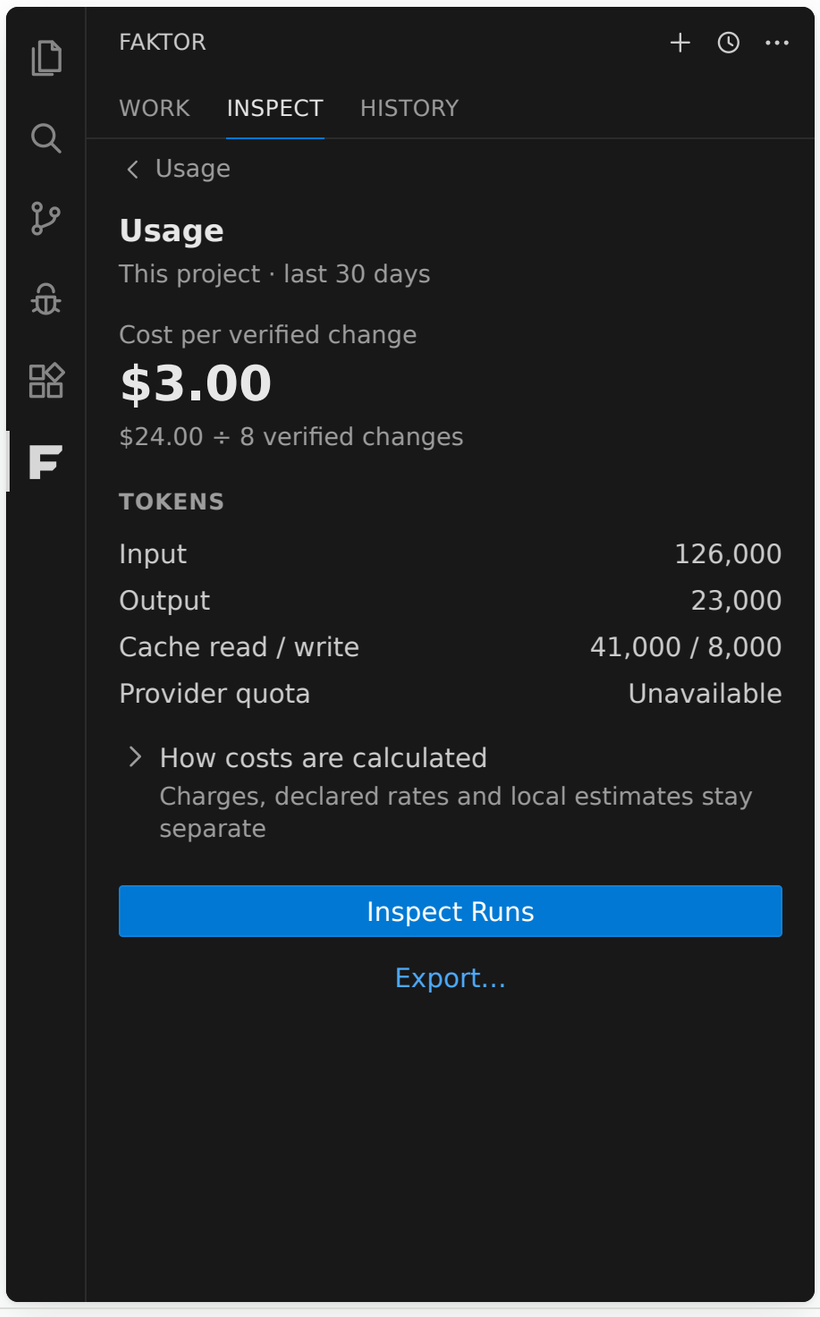
Usage & outcomes

PRIMARY ACTION

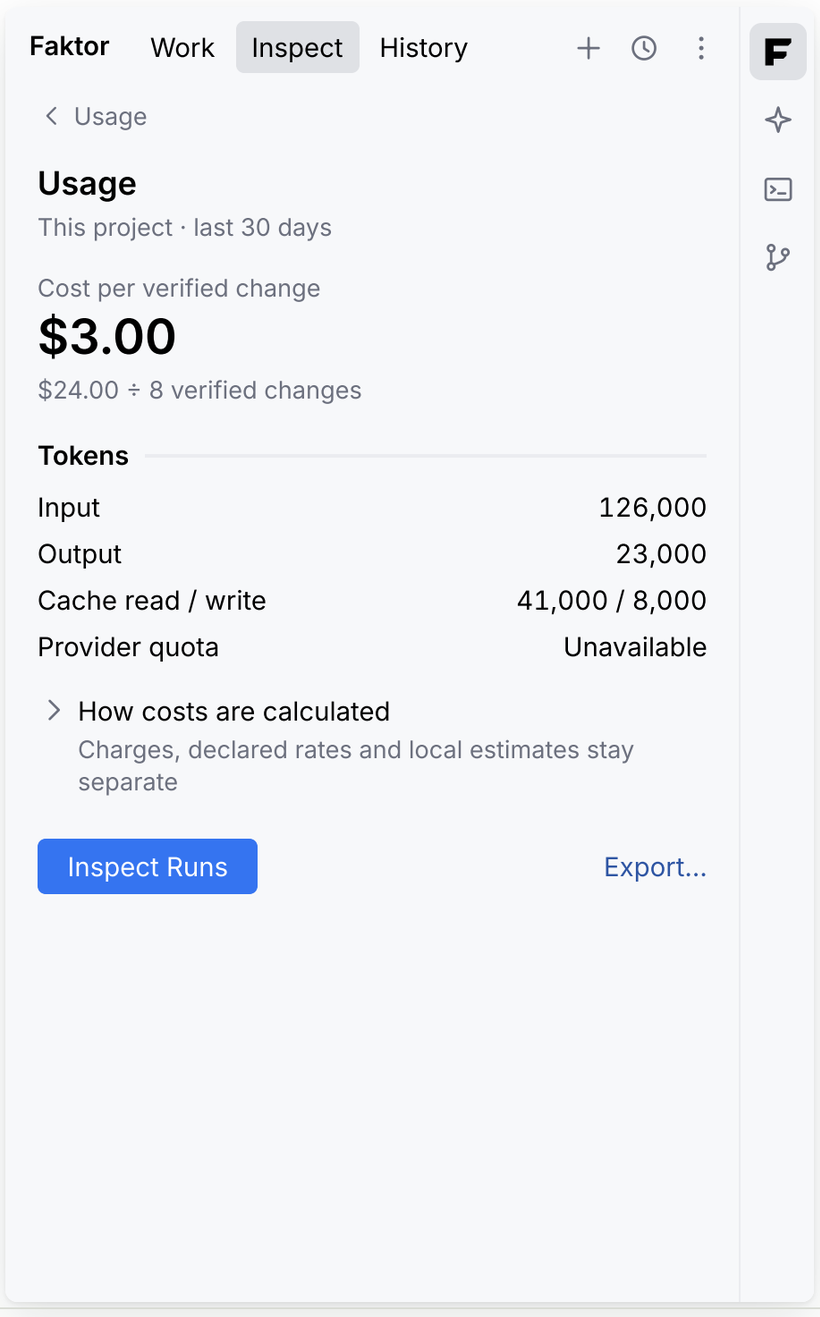
Inspect Runs

Separate token accounting, provider/cache prices, quota, local estimates and verification-based engineering cost. Explain every denominator.

VS CODE · DARK MODERN — SIDEBAR VIEW · WEBVIEW



JETBRAINS · LIGHT — TOOL WINDOW · SWING / KOTLIN UI DSL



CONTROLS B-04 M-13 K-12 O-06 V-09 D-07

REPLACES R7 page 043 · Expanded standard-agent surface

CONTRACT p. 106

S27 · USAGE · BEHAVIOUR CONTRACT

Usage & outcomes: contract and revision notes

Original contract retained verbatim (Expanded standard-agent surface / controls: B-04 M-13 K-12 O-06 V-09 D-07). Design on p. 105.

What changed in revision 8

- Numbers are right-aligned with tabular figures; explanations wait behind one disclosure.

Entry and return

More / Usage or the selected recorded run/cohort. Work / Inspect / History remain the only top-level clusters. Settings, Usage and Diagnostics use explicit Back. Return to the originating object, not the first row.

Scope and captured state

Workspace/time/filter/route cohort and service paging cursor; provider charge basis is separate from account quota. Filter changes create a new query epoch. Retain the selected run by ID and disclose the cohort denominator and eligibility rules.

Primary action and recovery

Filter by workspace, model, cohort and time basis, then inspect the underlying attempted runs and verified landed changes.. No verified changes means VCSC unavailable, not zero. Cancelled/failed attempts remain in the eligible spend numerator.

Loading, empty and stale

Keep prior results marked stale during refresh. First read, no records, unknown price, zero denominator and disabled local billing are distinct.

Keyboard and focus

Filters and table rows use native keyboard semantics. Refresh preserves the selected record and scroll anchor; announcements summarize changes once.

Layout and adverse content

Show definition and basis next to the metric. Tables move to a wide editor/panel area; narrow views use labelled record summaries and a selected detail.

Starting point and required construction

Starting classifications: E, N, R. E existing operation; R runtime-backed adapter; N new contract; H host-only work. Exact per-control classifications remain in the catalog.

Reuse the reviewed domain operations and preserve all served fields. Build every missing native read/write adapter, capability gate and state renderer identified by the matching control/build rows.

BK07 owns exact accounting/cohorts and BK10 owns optional account/service reads; native local operation cannot depend on a cloud account.

Required interaction fixtures

Exercise first-read, empty/no-match, ready, refreshing/stale, error/refusal, action pending and acknowledged result. Also switch scope while the request is in flight.

Moved out of the interface

Every line of revision 7 plate copy that left the default view, and where it lives now.

"Faktor / Usage (R7 window title)"
→ Native host chrome (VS Code view title, tabs or editor tab; JetBrains tool-window tabs, dialog title or Settings breadcrumb)

"Current project · explicit period and local-only cohort"
→ Subtitle

"Eligible attempted spend / USD 24.00; Verified landed changes / 8"
→ Formula under the number

"Verified-change cost / USD 3.00 / 24.00 / 8"
→ Big number

"Observed usage"
→ Section "Tokens"

"Failed and cancelled eligible attempts remain in spend. With no verified changes, VCSC is unavailable, never zero."
→ How costs are calculated disclosure

"Pricing and cohort basis / Observed charges, declared rates and local operating estimates stay distinct."
→ Disclosure summary

"Export usage / Inspect attempted runs"
→ Export… / Inspect Runs

"Usage & outcomes"
→ Title "Usage"

"Input tokens / 126,000"
→ Row "Input"

"Output tokens / 23,000"
→ Row "Output"

"Cache read / write / 41,000 / 8,000"
→ Row

"Provider quota / Unavailable"
→ Row

PART 2 · SCREENS

2.7 Models and providers

7 screens. Each has a design spread for both IDEs and a contract page.

ID	SCREEN	JOB	PAGE
S02	**Choose a model**	Search by exact ID, provider or capability. A selected item exposes authority, limits and availability before selection; favorite and recent lists are user-local.	p. 108
S03	**Provider connection**	Connections have a stable instance ID, API family, endpoint and credential reference. Test auth, model discovery and capability support as separate stages.	p. 110
S04	**Local inference**	Locality is relative to the daemon. Explain loopback, allocation, model readiness and resource ownership before offering local-runtime controls.	p. 112
S05	**Model tuning**	A capability-driven form exposes standard generation controls without pretending every provider supports every field. Unset is distinct from zero.	p. 114
S63	**Compare exact models**	Compare exact provider/model pairs with held-constant task and context, separate candidates and shared spend.	p. 116
S06	**Choose an agent**	Code, Ask, Plan, Debug and Review are proposed task envelopes with visible instructions, models, tool limits and delegation policy.	p. 118
S07	**Custom agent**	Build a role from bounded instructions, model settings, tools and parent-owned budget. Preview the resulting envelope before saving.	p. 120

S02 MODELS AND PROVIDERS

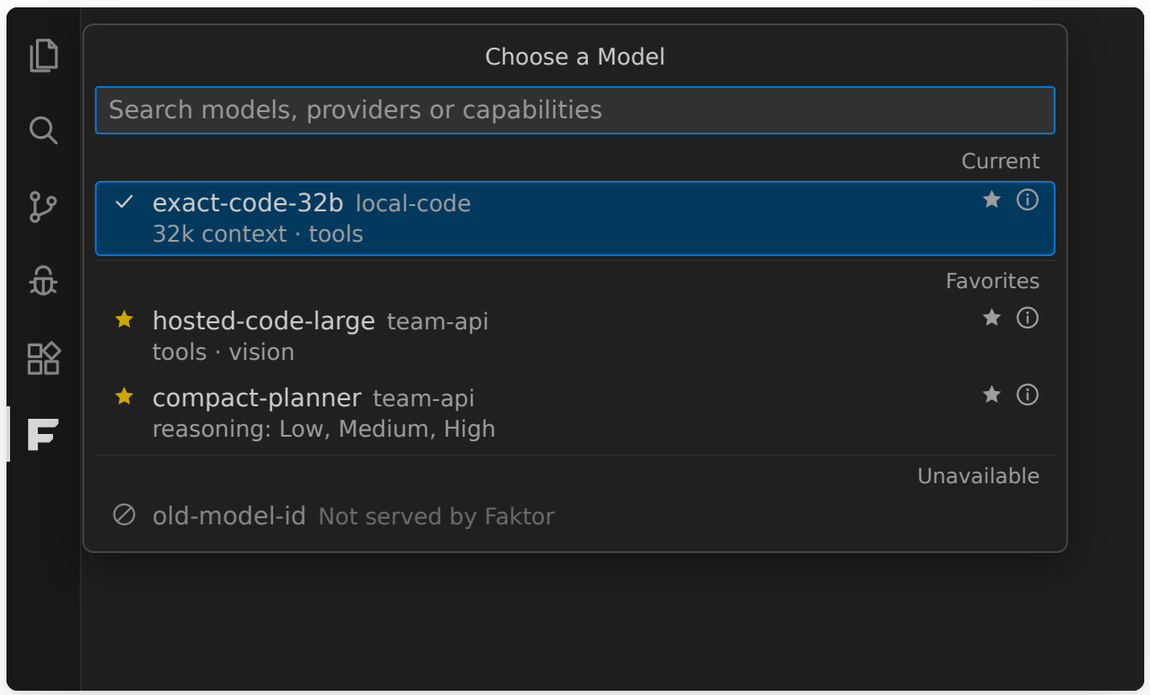
Choose a model

PRIMARY ACTION

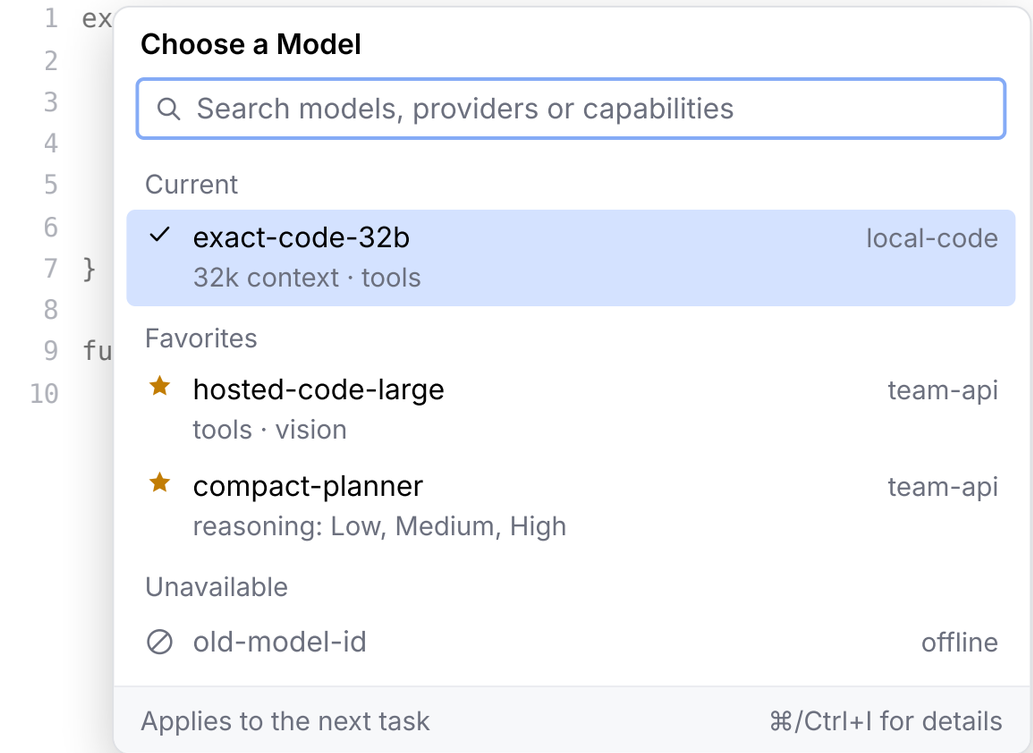
Enter selects

Search by exact ID, provider or capability. A selected item exposes authority, limits and availability before selection; favorite and recent lists are user-local.

VS CODE · DARK MODERN — QUICK PICK · NATIVE



JETBRAINS · LIGHT — POPUP CHOOSER · NATIVE



CONTROLS M-01 M-02 M-03 M-04 M-05 M-06 M-07 M-08 M-09 M-10 M-11 M-12 M-13 M-14

REPLACES R7 page 044 · R4-S29 CONTRACT p. 109

S02 · MODELS AND PROVIDERS · BEHAVIOUR CONTRACT

Choose a model: contract and revision notes

Original contract retained verbatim (R4-S29 / controls: M-01 M-02 M-03 M-04 M-05 M-06 M-07 M-08 M-09 M-10 M-11 M-12 M-13 M-14). Design on p. 108.

What changed in revision 8

- A native Quick Pick (VS Code) and popup chooser (JetBrains) replace a custom dialog.
- Capability authority and limits open from each item's info button.

Entry and return

Native Settings or the originating configuration command. Work / Inspect / History remain the only top-level clusters. Settings, Usage and Diagnostics use explicit Back. Return to the originating object, not the first row.

Scope and captured state

Application/project/connection plus effective revision; never infer task scope from an open form. Keep staged input, the effective snapshot and the acknowledged applied revision separate. Apply is atomic; Cancel preserves the previous effective configuration.

Primary action and recovery

Choose a served model and commit its provider/model pair. Show New conversation when this session cannot be changed safely.. An unknown or unavailable model remains visible with a reason. Do not silently substitute it; a blocked tool capability prevents tool-using admission.

Loading, empty and stale

Keep the last coherent values with a refresh state. A first read has no invented defaults. An absent capability remains explained and inspectable.

Keyboard and focus

Open on the triggering setting. Apply failure focuses the first invalid field on submission; async refresh never moves the caret. Escape cancels a transient editor and returns to its trigger.

Layout and adverse content

Align only related fields within their group. Long inputs use stacked labels. Native preferred sizes and zoom win over reference-art dimensions.

Starting point and required construction

Starting classifications: E, H, N, R. E existing operation; R runtime-backed adapter; N new contract; H host-only work. Exact per-control classifications remain in the catalog.

Reuse the reviewed domain operations and preserve all served fields. Build every missing native read/write adapter, capability gate and state renderer identified by the matching control/build rows.

BK01 owns validation, scope, persistence, revision comparison and acknowledgement; domain packages own parameter/capability enforcement.

Required interaction fixtures

Exercise first-read, empty/no-match, ready, refreshing/stale, error/refusal, action pending and acknowledged result. Also switch scope while the request is in flight.

Moved out of the interface

Every line of revision 7 plate copy that left the default view, and where it lives now.

"Faktor / Model picker (R7 window title)"
→ Native host chrome (VS Code view title, tabs or editor tab; JetBrains tool-window tabs, dialog title or Settings breadcrumb)

"Choose an exact model for the next task."
→ Picker footer "Applies to the next task"

"Current model / Pinned"
→ "Current" group; pin state is the check mark

"local-code · 32,768 context · tools"
→ Item description and detail

"Ready"
→ Removed: restated by the layout or label (available items are selectable; unavailable are disabled)

"team-api · tools, vision · declared limits"
→ Item detail; "declared limits" in the info button

"Offline / Not served by this daemon. Choose another model."
→ Disabled item "Not served by Faktor"

"Capability authority / Exact model; Change applies / Next task"
→ Info button (capability authority) and footer (timing)

"Cancel / Use selected model"
→ Native picker behavior: Enter selects, Escape cancels

"Search model or provider"
→ Picker placeholder "Search models, providers or capabilities"

"team-api · reasoning Low / Medium / High"
→ Item detail "reasoning: Low, Medium, High"

"exact-code-32b / hosted-code-large / compact-planner / old-model-id"
→ Picker items

"Favorites / Unavailable / Current model"
→ Picker groups

S03 MODELS AND PROVIDERS

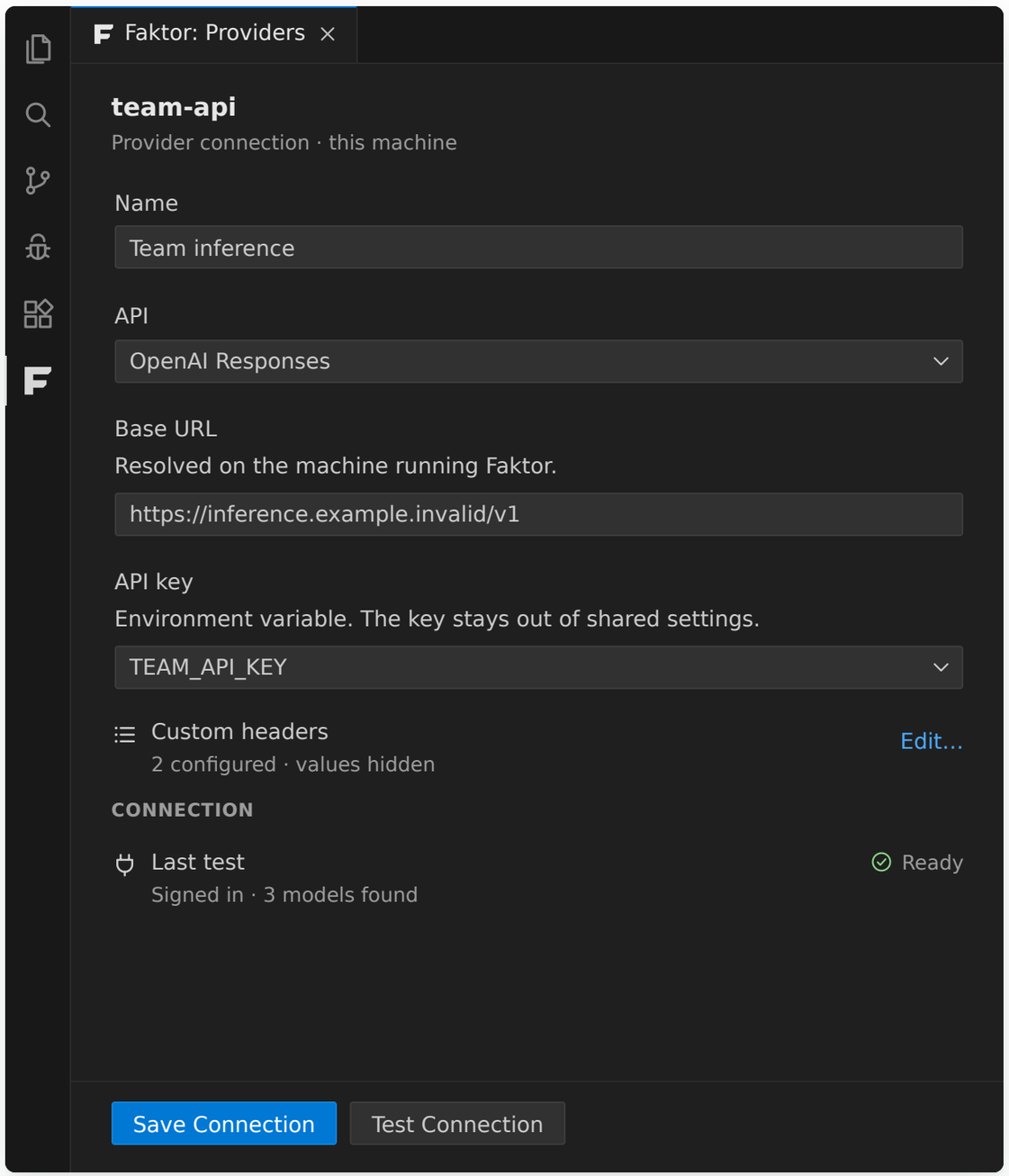
Provider connection

Connections have a stable instance ID, API family, endpoint and credential reference. Test auth, model discovery and capability support as separate stages.

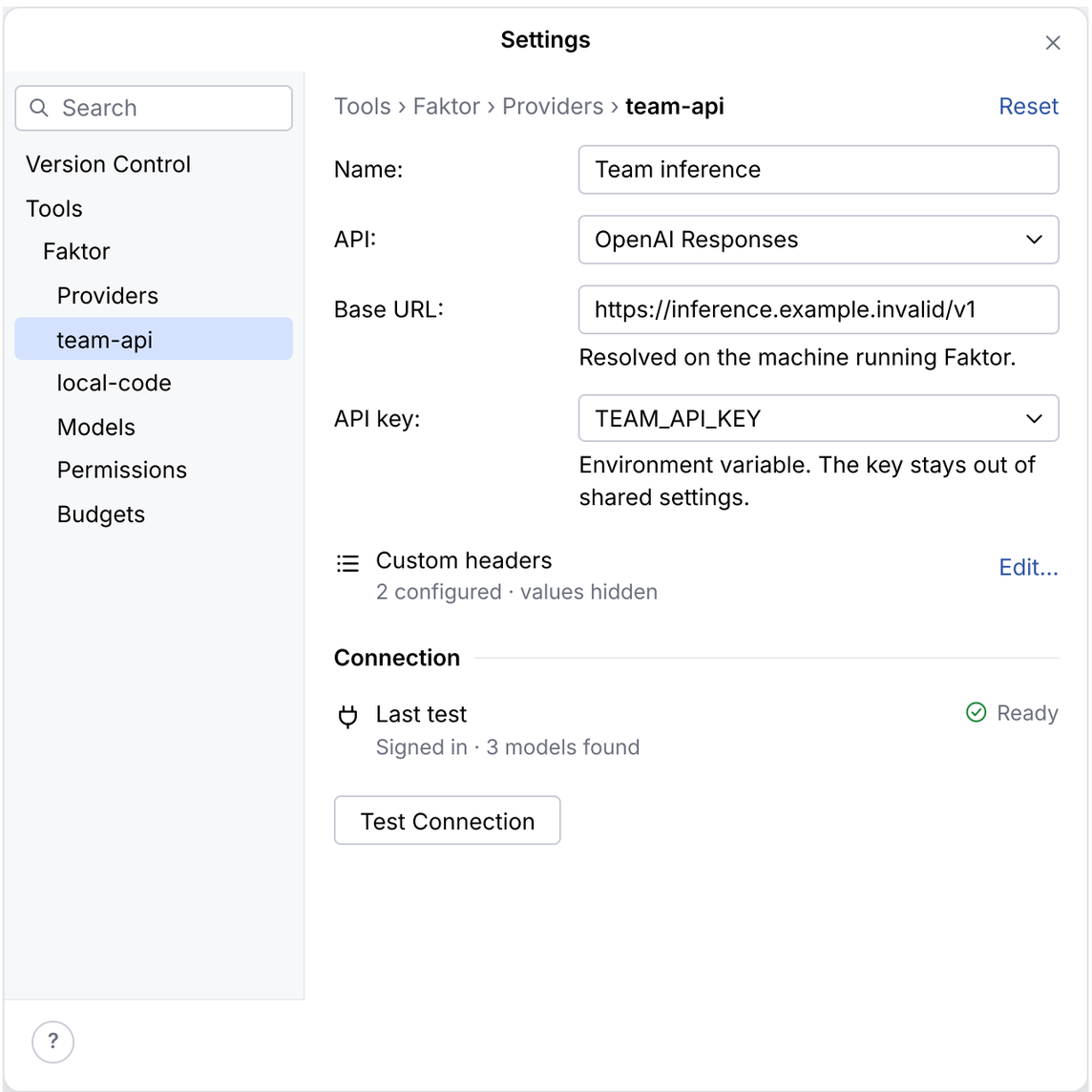
PRIMARY ACTION

Save Connection · JetBrains: OK / Apply

VS CODE · DARK MODERN — EDITOR TAB · WEBVIEW SETTINGS



JETBRAINS · LIGHT — SETTINGS DIALOG · CONFIGURABLE



CONTROLS P-01 P-02 P-03 P-04 P-05 P-06 P-10 P-11 P-12 P-14

REPLACES R7 page 045 · Expanded standard-agent surface

CONTRACT p. 111

S03 · MODELS AND PROVIDERS · BEHAVIOUR CONTRACT

Provider connection: contract and revision notes

Original contract retained verbatim (Expanded standard-agent surface / controls: P-01 P-02 P-03 P-04 P-05 P-06 P-10 P-11 P-12 P-14). Design on p. 110.

What changed in revision 8

- JetBrains: a Configurable page with left labels, colons, OK/Cancel/Apply and Reset. VS Code: a Faktor settings editor tab with stacked labels.

Entry and return

Native Settings or the originating configuration command. Work / Inspect / History remain the only top-level clusters. Settings, Usage and Diagnostics use explicit Back. Return to the originating object, not the first row.

Scope and captured state

Application/project/connection plus effective revision; never infer task scope from an open form. Keep staged input, the effective snapshot and the acknowledged applied revision separate. Apply is atomic; Cancel preserves the previous effective configuration.

Primary action and recovery

Validate a staged draft, preview the diff and save only after secure binding succeeds. Keep the last valid connection when validation fails.. A 401 offers Rotate credential; blocked destination shows endpoint policy; discovery failure keeps manually declared model IDs intact.

Loading, empty and stale

Keep the last coherent values with a refresh state. A first read has no invented defaults. An absent capability remains explained and inspectable.

Keyboard and focus

Open on the triggering setting. Apply failure focuses the first invalid field on submission; async refresh never moves the caret. Escape cancels a transient editor and returns to its trigger.

Layout and adverse content

Align only related fields within their group. Long inputs use stacked labels. Native preferred sizes and zoom win over reference-art dimensions.

Starting point and required construction

Starting classifications: N, R. E existing operation; R runtime-backed adapter; N new contract; H host-only work. Exact per-control classifications remain in the catalog.

Reuse the reviewed domain operations and preserve all served fields. Build every missing native read/write adapter, capability gate and state renderer identified by the matching control/build rows.

BK01 owns validation, scope, persistence, revision comparison and acknowledgement; domain packages own parameter/capability enforcement.

Required interaction fixtures

Exercise first-read, empty/no-match, ready, refreshing/stale, error/refusal, action pending and acknowledged result. Also switch scope while the request is in flight.

Moved out of the interface

Every line of revision 7 plate copy that left the default view, and where it lives now.

"Faktor / Provider connection (R7 window title)"
→ Native host chrome (VS Code view title, tabs or editor tab; JetBrains tool-window tabs, dialog title or Settings breadcrumb)

"Machine connection · team-api"
→ Subtitle / Settings location

"Connection name"
→ Label "Name"

"API family"
→ Label "API"

"Resolved from the daemon host."
→ Help "Resolved on the machine running Faktor."

"Credential"
→ Label "API key"

"Environment reference. The key stays out of shared settings."
→ Help (kept; consequence)

"Custom headers / 2 configured · secret values hidden"
→ Row

"Last connection test / Authentication / Ready / Models discovered / 3"
→ Row "Last test — Signed in · 3 models found"

"Test connection / Save connection"
→ Test Connection / Save Connection (JetBrains: OK)

"Provider connection"
→ Settings page subtitle

"Team inference / OpenAI Responses / TEAM_API_KEY"
→ Field values

S04 MODELS AND PROVIDERS

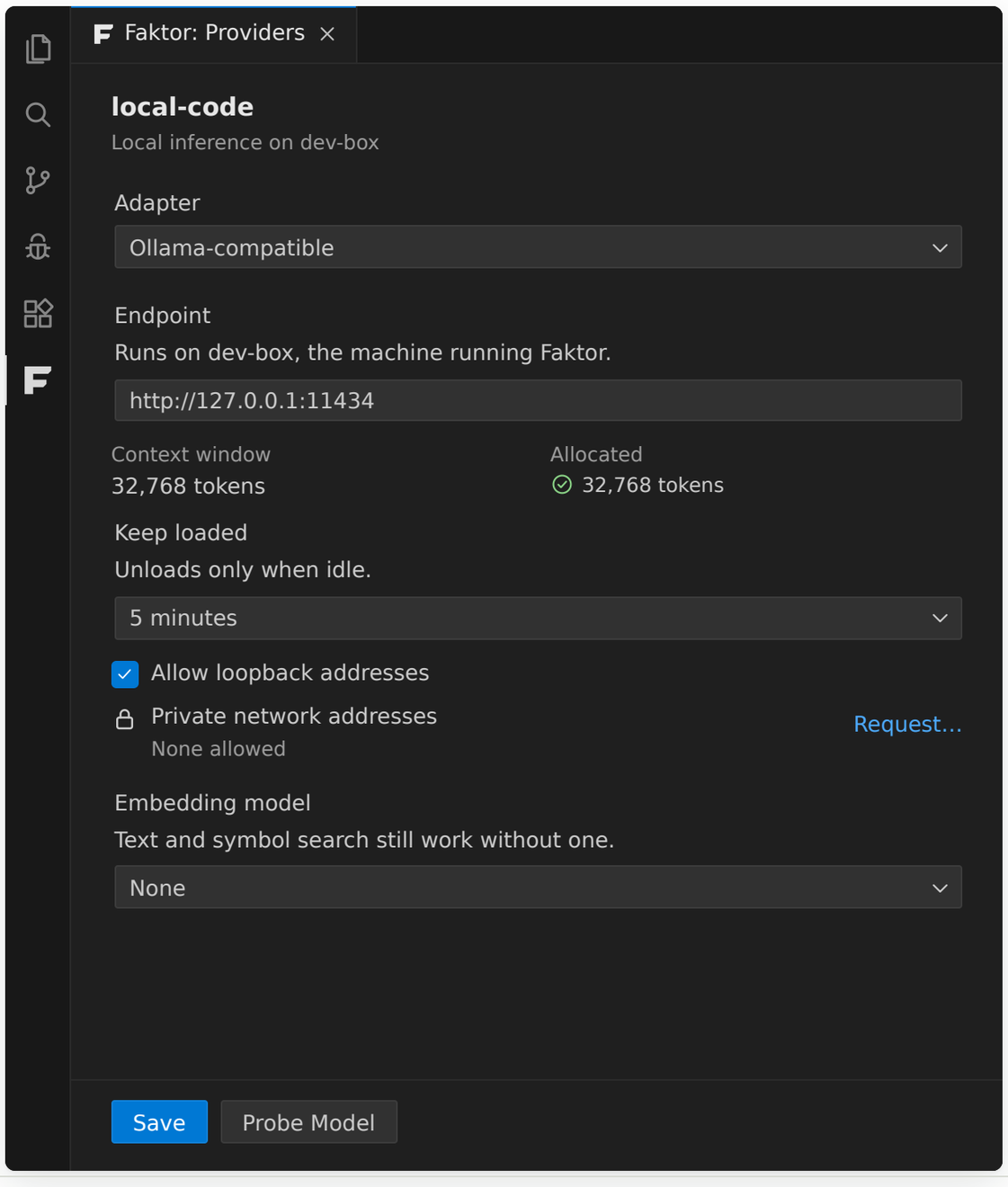
Local inference

PRIMARY ACTION

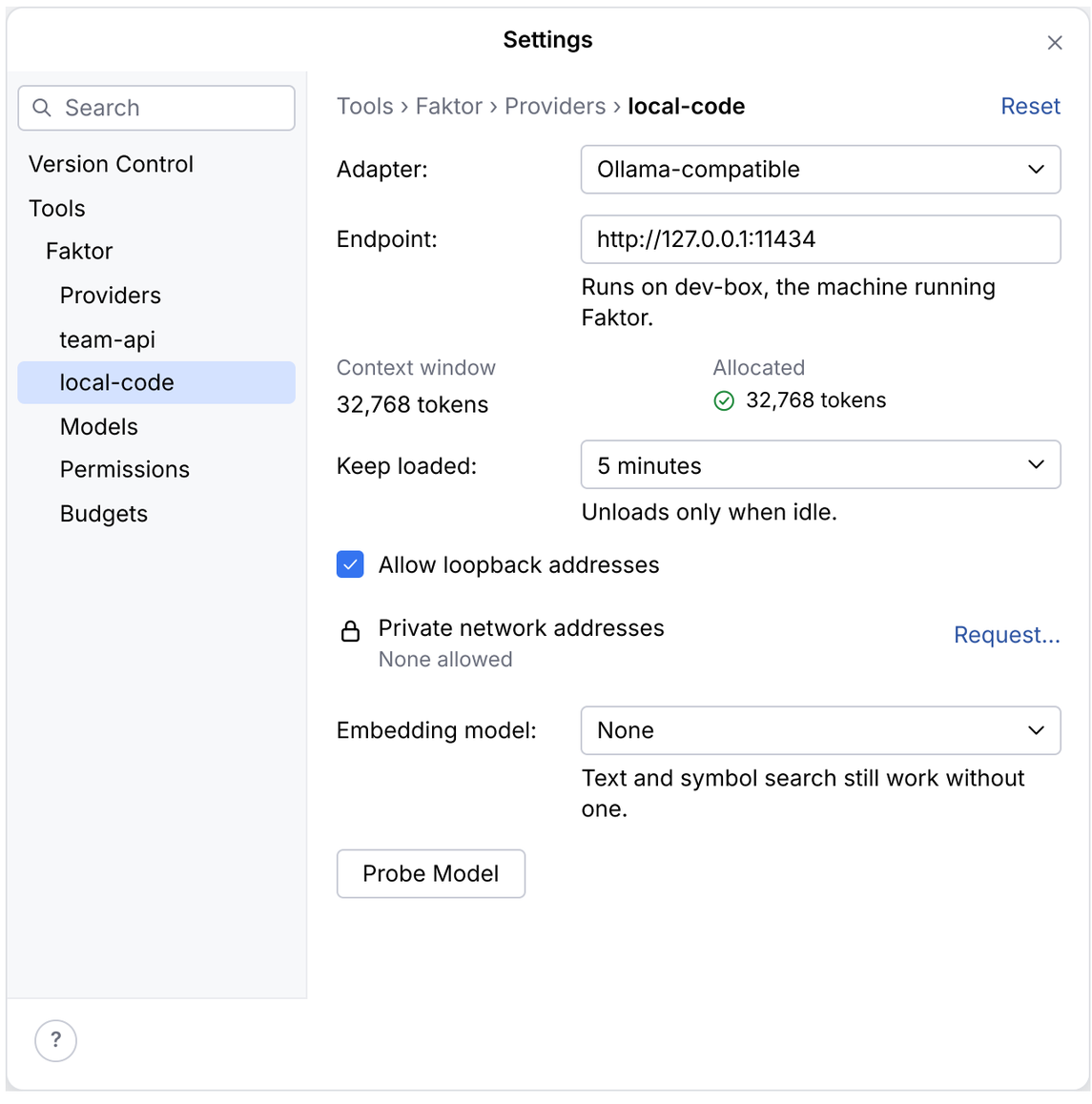
Save · JetBrains: OK / Apply

Locality is relative to the daemon. Explain loopback, allocation, model readiness and resource ownership before offering local-runtime controls.

VS CODE · DARK MODERN — EDITOR TAB · WEBVIEW SETTINGS



JETBRAINS · LIGHT — SETTINGS DIALOG · CONFIGURABLE



CONTROLS P-07 P-08 P-09 P-13 P-15 C-10 M-09

REPLACES R7 page 046 · Expanded standard-agent surface

CONTRACT p. 113

S04 · MODELS AND PROVIDERS · BEHAVIOUR CONTRACT

Local inference: contract and revision notes

Original contract retained verbatim (Expanded standard-agent surface / controls: P-07 P-08 P-09 P-13 P-15 C-10 M-09). Design on p. 112.

What changed in revision 8

- Locality is explained once ("on dev-box"); policy details move to tooltips.

Entry and return

Native Settings or the originating configuration command. Work / Inspect / History remain the only top-level clusters. Settings, Usage and Diagnostics use explicit Back. Return to the originating object, not the first row.

Scope and captured state

Application/project/connection plus effective revision; never infer task scope from an open form. Keep staged input, the effective snapshot and the acknowledged applied revision separate. Apply is atomic; Cancel preserves the previous effective configuration.

Primary action and recovery

Test the exact endpoint from the daemon host, then show configured and observed allocation side by side.. A private LAN endpoint currently receives a typed refusal; allow_loopback cannot authorize it. Unsupported unload/allocation controls stay unavailable.

Loading, empty and stale

Keep the last coherent values with a refresh state. A first read has no invented defaults. An absent capability remains explained and inspectable.

Keyboard and focus

Open on the triggering setting. Apply failure focuses the first invalid field on submission; async refresh never moves the caret. Escape cancels a transient editor and returns to its trigger.

Layout and adverse content

Align only related fields within their group. Long inputs use stacked labels. Native preferred sizes and zoom win over reference-art dimensions.

Starting point and required construction

Starting classifications: H, N, R. E existing operation; R runtime-backed adapter; N new contract; H host-only work. Exact per-control classifications remain in the catalog.

Reuse the reviewed domain operations and preserve all served fields. Build every missing native read/write adapter, capability gate and state renderer identified by the matching control/build rows.

BK01 owns validation, scope, persistence, revision comparison and acknowledgement; domain packages own parameter/capability enforcement.

Required interaction fixtures

Exercise first-read, empty/no-match, ready, refreshing/stale, error/refusal, action pending and acknowledged result. Also switch scope while the request is in flight.

Moved out of the interface

Every line of revision 7 plate copy that left the default view, and where it lives now.

"Faktor / Local inference (R7 window title)"
→ Native host chrome (VS Code view title, tabs or editor tab; JetBrains tool-window tabs, dialog title or Settings breadcrumb)

"Local means local to the daemon host: dev-box."
→ Subtitle "Local inference on dev-box" and endpoint help

"Loopback resolves inside the daemon container."
→ Tooltip / hover help on Endpoint

"Declared context / Observed allocation"
→ Facts "Context window" / "Allocated"

"Keep-alive"
→ Label "Keep loaded"

"Unload only when this runtime is idle."
→ Help "Unloads only when idle."

"Allow loopback / Enabled by this adapter's policy."
→ Checkbox; policy source in Tooltip / hover help

"Private LAN destinations / None granted / An explicit machine policy is required."
→ Row "Private network addresses — None allowed — Request…"

"Not configured"
→ "None"

"Lexical and symbol retrieval remain available."
→ Help "Text and symbol search still work without one."

"Probe model / Save local settings"
→ Probe Model / Save

"Local inference"
→ Settings page subtitle

"32,768 tokens"
→ Facts

"5 minutes / Ollama-compatible / http://127.0.0.1:11434"
→ Field values

S05 MODELS AND PROVIDERS

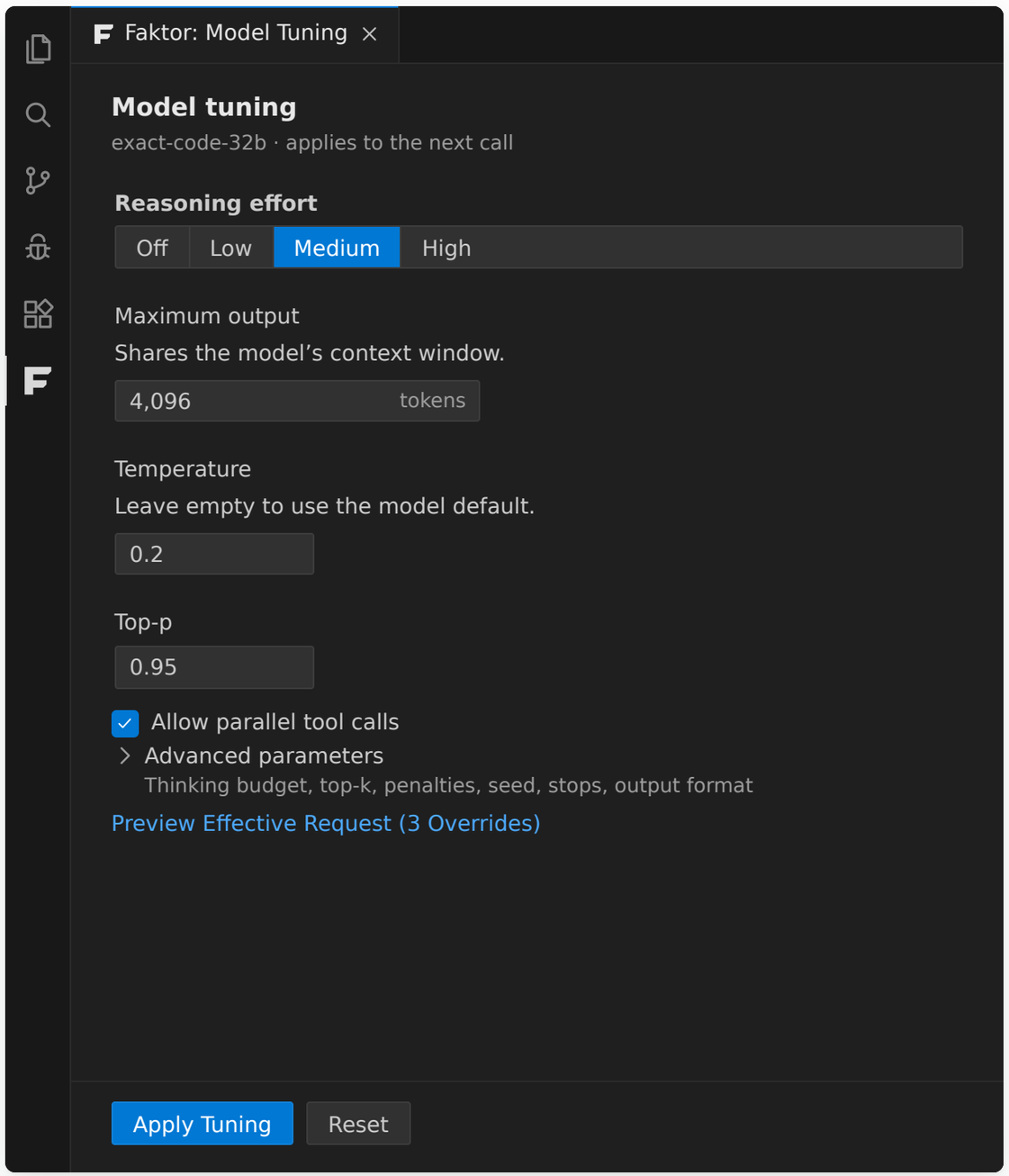
Model tuning

A capability-driven form exposes standard generation controls without pretending every provider supports every field. Unset is distinct from zero.

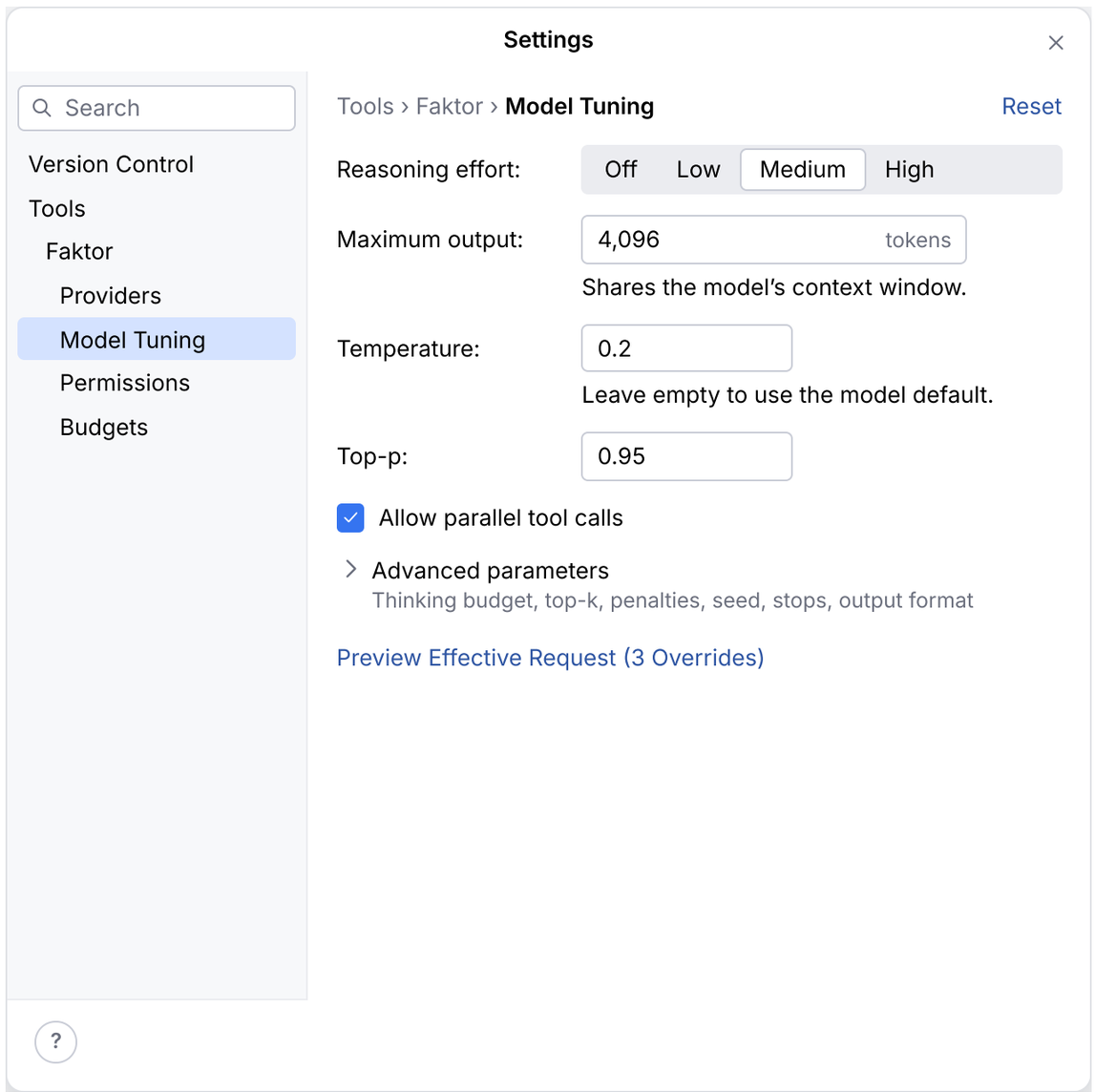
PRIMARY ACTION

Apply Tuning · JetBrains: OK / Apply

VS CODE · DARK MODERN — EDITOR TAB · WEBVIEW SETTINGS



JETBRAINS · LIGHT — SETTINGS DIALOG · CONFIGURABLE



CONTROLS G-01 G-02 G-03 G-04 G-05 G-06 G-07 G-08 G-09 G-10 G-11 G-12 G-13 G-14

REPLACES R7 page 047 · Expanded standard-agent surface

CONTRACT p. 115

S05 · MODELS AND PROVIDERS · BEHAVIOUR CONTRACT

Model tuning: contract and revision notes

Original contract retained verbatim (Expanded standard-agent surface / controls: G-01 G-02 G-03 G-04 G-05 G-06 G-07 G-08 G-09 G-10 G-11 G-12 G-13 G-14). Design on p. 114.

What changed in revision 8

- Unset vs. zero is explained by the Temperature help ("Leave empty to use the model default").
- Apply Tuning uses the corrected button component with metric-based vertical centring.

Entry and return

Native Settings or the originating configuration command. Work / Inspect / History remain the only top-level clusters. Settings, Usage and Diagnostics use explicit Back. Return to the originating object, not the first row.

Scope and captured state

Application/project/connection plus effective revision; never infer task scope from an open form. Keep staged input, the effective snapshot and the acknowledged applied revision separate. Apply is atomic; Cancel preserves the previous effective configuration.

Primary action and recovery

Reset removes an override; Preview shows the normalized request and apply timing. Changes apply only after a supported acknowledgement.. Unsupported values fail locally before admission. Changing the model revalidates the complete tuning bundle and retains incompatible edits for review.

Loading, empty and stale

Keep the last coherent values with a refresh state. A first read has no invented defaults. An absent capability remains explained and inspectable.

Keyboard and focus

Open on the triggering setting. Apply failure focuses the first invalid field on submission; async refresh never moves the caret. Escape cancels a transient editor and returns to its trigger.

Layout and adverse content

Align only related fields within their group. Long inputs use stacked labels. Native preferred sizes and zoom win over reference-art dimensions.

Starting point and required construction

Starting classifications: N. E existing operation; R runtime-backed adapter; N new contract; H host-only work. Exact per-control classifications remain in the catalog.

Reuse the reviewed domain operations and preserve all served fields. Build every missing native read/write adapter, capability gate and state renderer identified by the matching control/build rows.

BK01 owns validation, scope, persistence, revision comparison and acknowledgement; domain packages own parameter/capability enforcement.

Required interaction fixtures

Exercise first-read, empty/no-match, ready, refreshing/stale, error/refusal, action pending and acknowledged result. Also switch scope while the request is in flight.

Moved out of the interface

Every line of revision 7 plate copy that left the default view, and where it lives now.

"Faktor / Model tuning (R7 window title)"
→ Native host chrome (VS Code view title, tabs or editor tab; JetBrains tool-window tabs, dialog title or Settings breadcrumb)

"exact-code-32b · applies to the next admitted call"
→ Subtitle "exact-code-32b · applies to the next call"

"Reasoning / Effort"
→ Label "Reasoning effort"

"Output reserve shares the model context window."
→ Help "Shares the model's context window."

"Sampling"
→ Removed: restated by the layout or label (Temperature and Top-p are self-explanatory)

"The model and runtime must both support safe parallel calls."
→ Disabled-state reason, shown only when unsupported

"Thinking budget, top-k, penalties, seed, stops and output format"
→ Advanced parameters disclosure

"Effective request preview / Open / 3 overrides · requested values, resolved values and source"
→ Link "Preview Effective Request (3 Overrides)"

"Reset / Apply tuning"
→ Reset / Apply Tuning (JetBrains: Reset link + OK/Cancel/Apply)

"4096 tokens"
→ Formatted as 4,096 tokens

"Off / Low / Medium / High"
→ Segmented choice

"Temperature / Top-p / 0.2 / 0.95"
→ Fields

S63 MODELS AND PROVIDERS

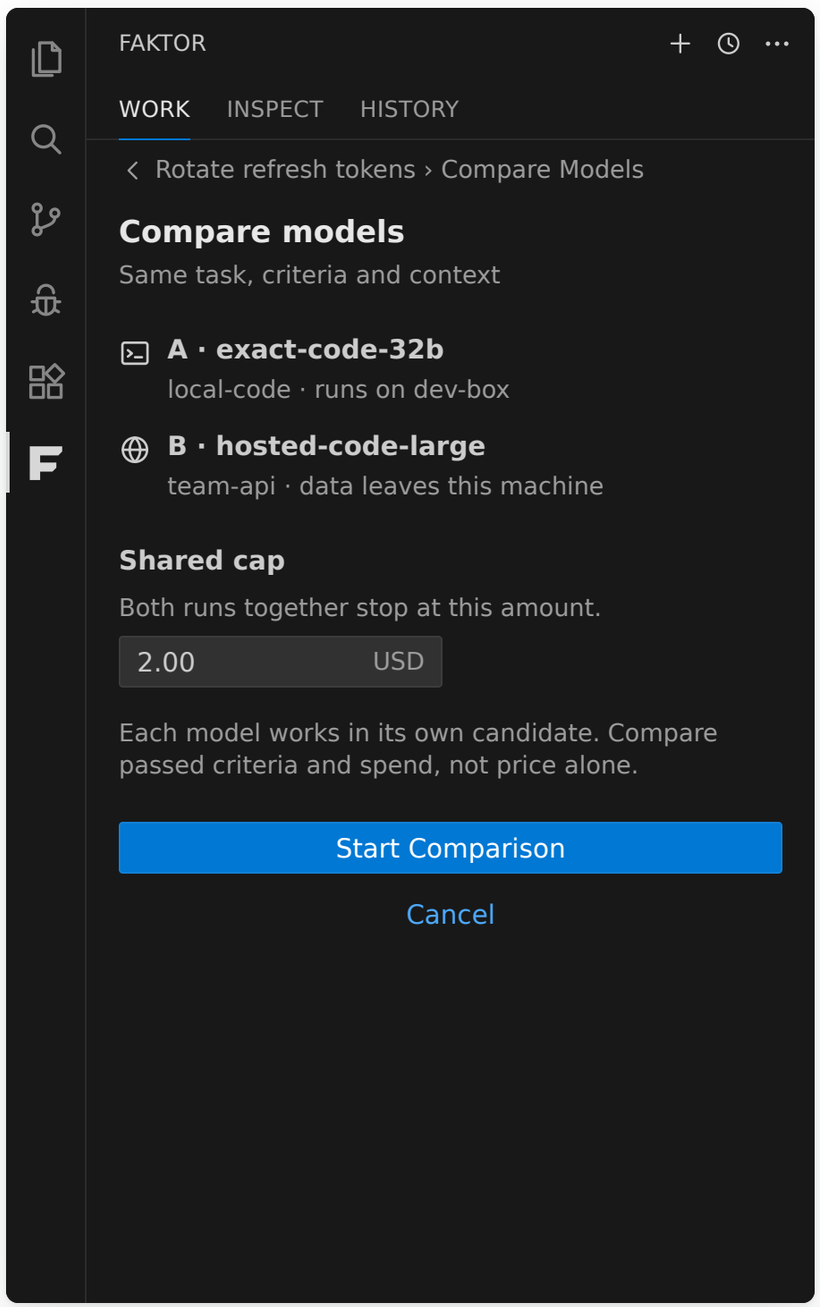
PRIMARY ACTION

Start Comparison

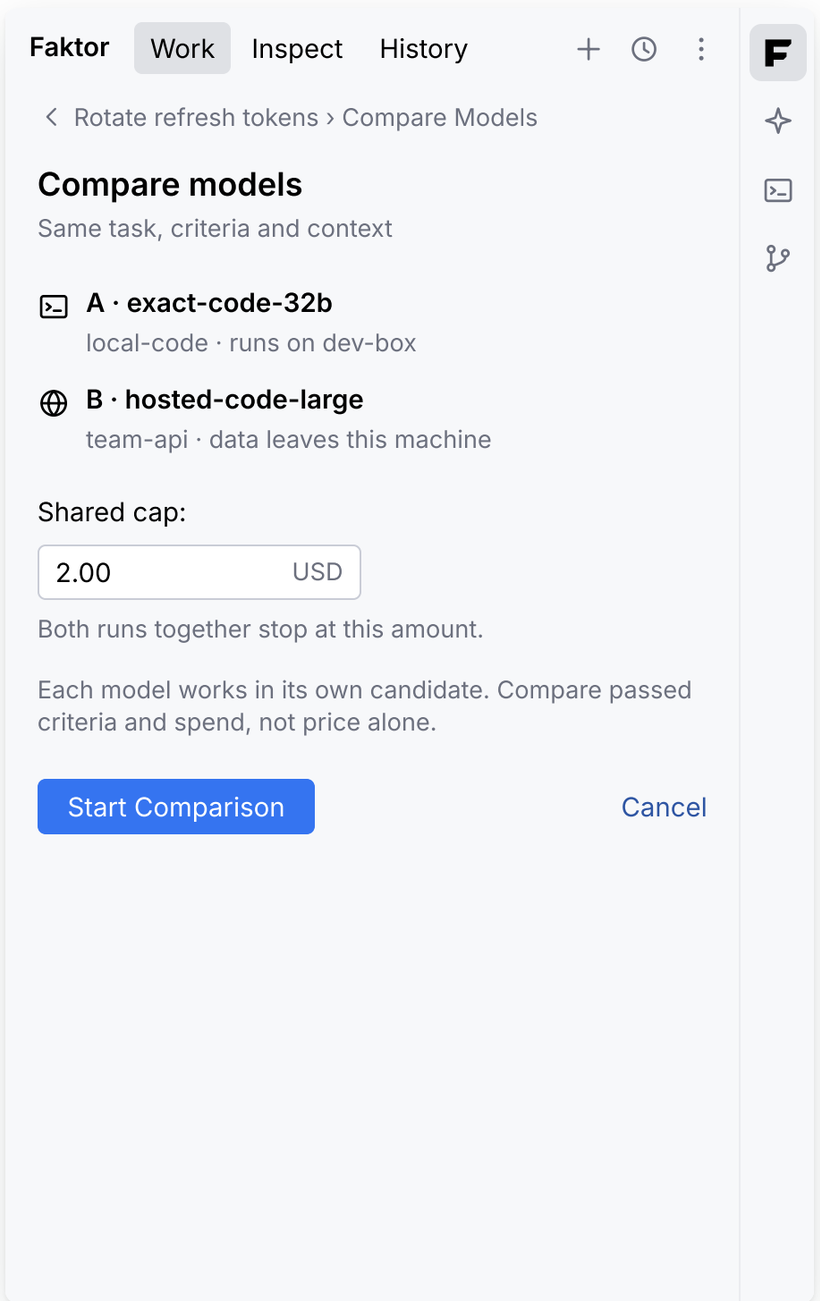
Compare exact models

Compare exact provider/model pairs with held-constant task and context, separate candidates and shared spend.

VS CODE · DARK MODERN — SIDEBAR VIEW · WEBVIEW



JETBRAINS · LIGHT — TOOL WINDOW · SWING / KOTLIN UI DSL



CONTROLS M-14 B-01 B-02

REPLACES R7 page 048 · Expanded standard-agent surface

CONTRACT p. 117

S63 · MODELS AND PROVIDERS · BEHAVIOUR CONTRACT

Compare exact models: contract and revision notes

Original contract retained verbatim (Expanded standard-agent surface / controls: M-14 B-01 B-02). Design on p. 116.

What changed in revision 8

- The data boundary is stated where it matters: on the hosted option.

Entry and return

Inspect / Agents with Roster, Approvals, Compare approaches and Coordination as subordinate destinations. Work / Inspect / History remain the only top-level clusters. Settings, Usage and Diagnostics use explicit Back. Return to the originating object, not the first row.

Scope and captured state

Parent task + current child/comparison/board generation; every write is scoped to its actual owner. Retain selected agent and the typed steer/post draft through refresh. A sent control is an immutable operation; current owner checks happen again at dispatch.

Primary action and recovery

Review comparison admission. Reject stale child handles and duplicate controls visibly. A missing board route is unavailable, never an empty fabricated board. Recover unknown effects from receipts.

Loading, empty and stale

Show bounded queue/wait/working states. Parent and child budget reservations remain in one ledger. A comparison only recommends until explicit integration.

Keyboard and focus

Keyboard lists select by stable child/post identity. Open detail intentionally; new posts/child events announce without resetting the current selection or typed draft.

Layout and adverse content

One selected detail and one subordinate input at a time. Put raw IDs, revision cursors and provenance behind Details; keep role, assignment and outcome in the main row.

Starting point and required construction

Starting classifications: E, N. E existing operation; R runtime-backed adapter; N new contract; H host-only work. Exact per-control classifications remain in the catalog.

NEW: extend the comparison admission/ledger over exact model pairs, held-constant source manifests and separately owned candidates.

BK03 owns intent/preset contracts; BK08 owns lifecycle/board/control adapters; BK07 owns shared budgets and BK09 owns comparison evidence.

Required interaction fixtures

Exercise first-read, empty/no-match, ready, refreshing/stale, error/refusal, action pending and acknowledged result. Also switch scope while the request is in flight.

Moved out of the interface

Every line of revision 7 plate copy that left the default view, and where it lives now.

"Faktor / Models / Comparison (R7 window title)"
→ Native host chrome (VS Code view title, tabs or editor tab; JetBrains tool-window tabs, dialog title or Settings breadcrumb)

"One captured task and acceptance basis · new run admission required"
→ Subtitle; admission rule in Specification: screen contract

"Exact model ID and adapter; local to dev-box"
→ Row A meta

"Exact model ID and adapter; hosted data boundary"
→ Row B meta "data leaves this machine"

"Held constant / Goal, criteria, context snapshot and requested finish"
→ Subtitle

"Shared comparison cap / USD 2.00"
→ Field with consequence help

"Candidate ownership / Separate isolated candidates"
→ Note "Each model works in its own candidate."

"Cost is not the only comparison / Keep artifact-supported quality, attempted spend and actual route visible. A cheaper declaration does not establish a better outcome."
→ Note (shortened); full rule in Product documentation

"Review comparison admission"
→ Start Comparison

"Compare exact models"
→ Title "Compare models"

"exact-code-32b / local-code"
→ Row A "exact-code-32b" with meta "local-code"

"hosted-code-large / team-api"
→ Row B "hosted-code-large" with meta "team-api"

"A / B"
→ Row prefixes

"Cancel"
→ Cancel

S06 MODELS AND PROVIDERS

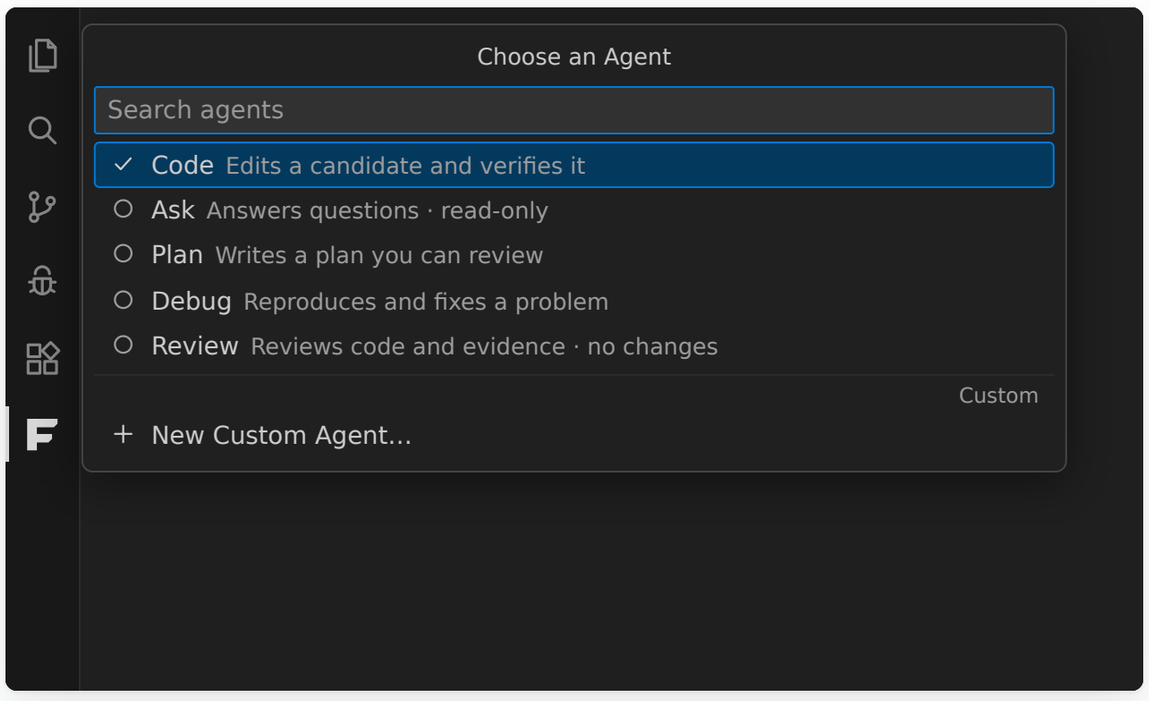
PRIMARY ACTION

Enter selects

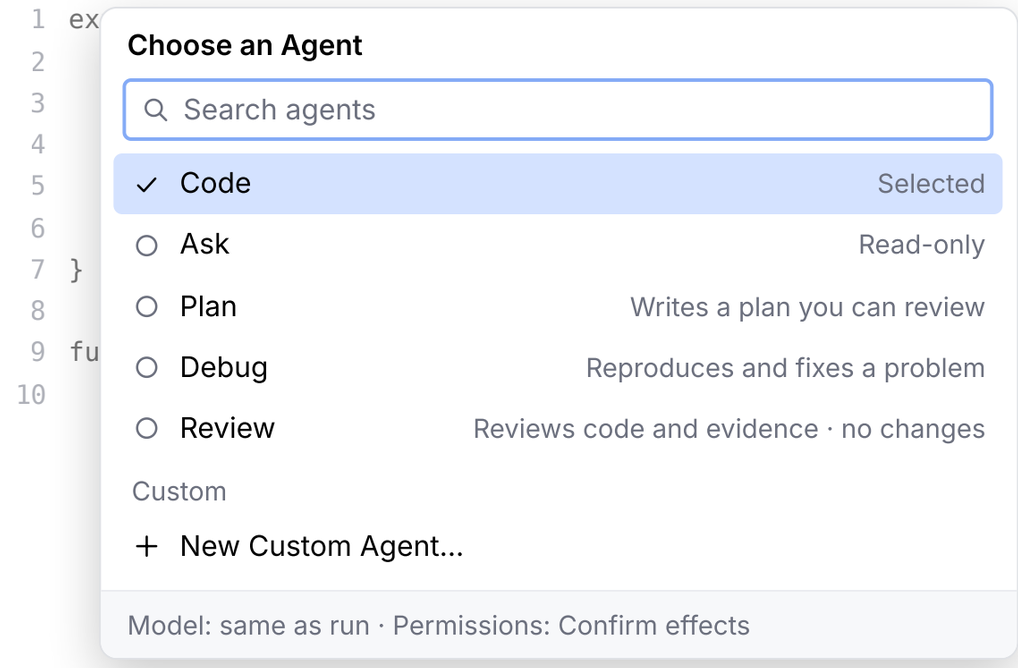
Choose an agent

Code, Ask, Plan, Debug and Review are proposed task envelopes with visible instructions, models, tool limits and delegation policy.

VS CODE · DARK MODERN — QUICK PICK · NATIVE



JETBRAINS · LIGHT — POPUP CHOOSER · NATIVE



CONTROLS A-01 A-02 A-03 A-04 A-05 A-06 A-07 A-08 A-09 A-10 A-11 A-12

REPLACES R7 page 049 · Expanded standard-agent surface

CONTRACT p. 119

S06 · MODELS AND PROVIDERS · BEHAVIOUR CONTRACT

Choose an agent: contract and revision notes

Original contract retained verbatim (Expanded standard-agent surface / controls: A-01 A-02 A-03 A-04 A-05 A-06 A-07 A-08 A-09 A-10 A-11 A-12). Design on p. 118.

What changed in revision 8

- Agent choice is a native picker; descriptions say what each agent does, not its envelope.

Entry and return

Native Settings or the originating configuration command. Work / Inspect / History remain the only top-level clusters. Settings, Usage and Diagnostics use explicit Back. Return to the originating object, not the first row.

Scope and captured state

Application/project/connection plus effective revision; never infer task scope from an open form. Keep staged input, the effective snapshot and the acknowledged applied revision separate. Apply is atomic; Cancel preserves the previous effective configuration.

Primary action and recovery

Select or duplicate a preset, then inspect its resolved permissions before applying it to a new task.. A prompt-only Ask route is labelled accordingly until read-only dispatch is enforced. A preset cannot override managed denies.

Loading, empty and stale

Keep the last coherent values with a refresh state. A first read has no invented defaults. An absent capability remains explained and inspectable.

Keyboard and focus

Open on the triggering setting. Apply failure focuses the first invalid field on submission; async refresh never moves the caret. Escape cancels a transient editor and returns to its trigger.

Layout and adverse content

Align only related fields within their group. Long inputs use stacked labels. Native preferred sizes and zoom win over reference-art dimensions.

Starting point and required construction

Starting classifications: E, N, R. E existing operation; R runtime-backed adapter; N new contract; H host-only work. Exact per-control classifications remain in the catalog.

Reuse the reviewed domain operations and preserve all served fields. Build every missing native read/write adapter, capability gate and state renderer identified by the matching control/build rows.

BK01 owns validation, scope, persistence, revision comparison and acknowledgement; domain packages own parameter/capability enforcement.

Required interaction fixtures

Exercise first-read, empty/no-match, ready, refreshing/stale, error/refusal, action pending and acknowledged result. Also switch scope while the request is in flight.

Moved out of the interface

Every line of revision 7 plate copy that left the default view, and where it lives now.

"Faktor / Agent picker (R7 window title)"
→ Native host chrome (VS Code view title, tabs or editor tab; JetBrains tool-window tabs, dialog title or Settings breadcrumb)

"Agent intent, model and authority remain separate controls."
→ Footer shows model and permissions separately; rule in Specification: screen contract

"Selected"
→ Check mark

"Edit an isolated candidate and verify the change."
→ Description

"Read only / Explore and answer with an enforced read-only envelope."
→ Description "Answers questions · read-only"

"Plan scope / Explore and write a named plan artifact."
→ Description "Writes a plan you can review"

"Reproduce, inspect and repair within granted tools."
→ Description

"Inspect code and evidence without landing changes."
→ Description

"Model / Inherited; Permission policy / Confirm effects"
→ Footer

"New custom agent / Use Code"
→ New Custom Agent… item; Enter selects

"Code / Ask / Plan / Debug / Review"
→ Picker items

S07 MODELS AND PROVIDERS

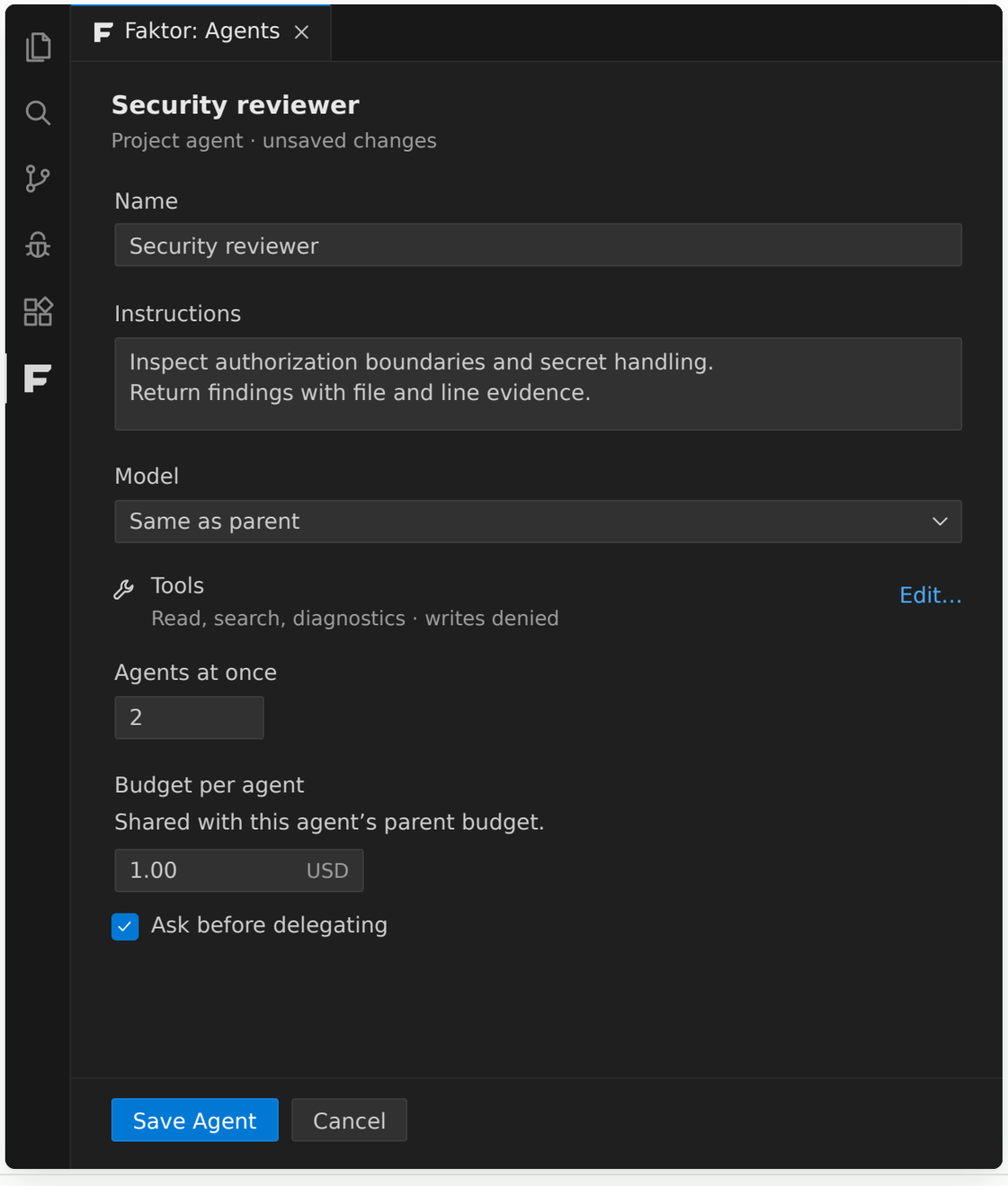
Custom agent

Build a role from bounded instructions, model settings, tools and parent-owned budget. Preview the resulting envelope before saving.

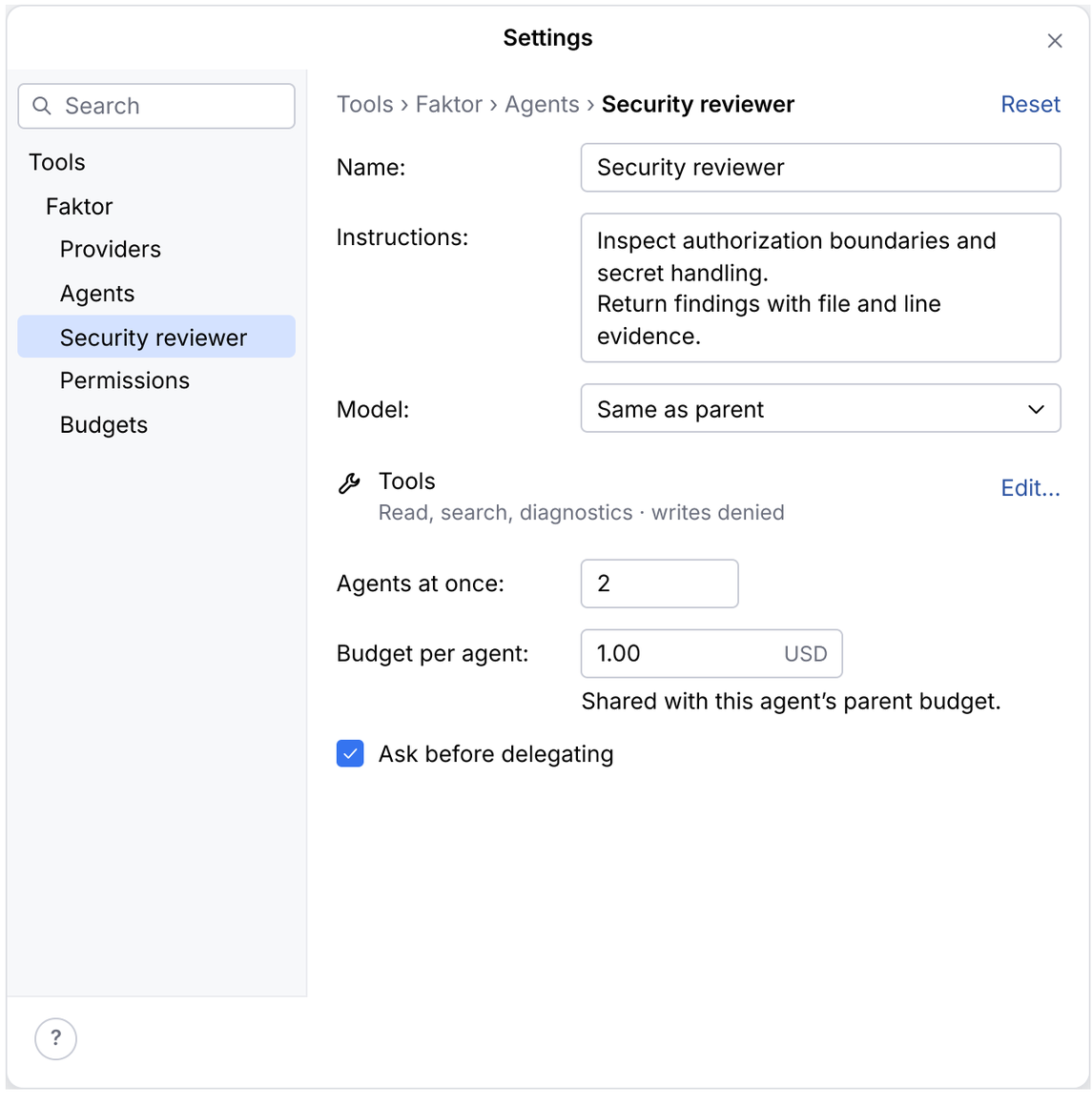
PRIMARY ACTION

Save Agent · JetBrains: OK / Apply

VS CODE · DARK MODERN — EDITOR TAB · WEBVIEW SETTINGS



JETBRAINS · LIGHT — SETTINGS DIALOG · CONFIGURABLE



CONTROLS A-04 A-05 A-06 A-07 A-08 A-09 A-11 A-12 M-05

REPLACES R7 page 050 · Expanded standard-agent surface

CONTRACT p. 121

S07 · MODELS AND PROVIDERS · BEHAVIOUR CONTRACT

Custom agent: contract and revision notes

Original contract retained verbatim (Expanded standard-agent surface / controls: A-04 A-05 A-06 A-07 A-08 A-09 A-11 A-12 M-05). Design on p. 120.

What changed in revision 8

- “Envelope”, “ceiling” and “children” become plain words.

Entry and return

Native Settings or the originating configuration command. Work / Inspect / History remain the only top-level clusters. Settings, Usage and Diagnostics use explicit Back. Return to the originating object, not the first row.

Scope and captured state

Application/project/connection plus effective revision; never infer task scope from an open form. Keep staged input, the effective snapshot and the acknowledged applied revision separate. Apply is atomic; Cancel preserves the previous effective configuration.

Primary action and recovery

Save a validated version for the next task; an active child retains its admitted version until a supported control changes it.. Duplicate names do not collide with stable IDs. Invalid capability/model combinations stay in the draft and cannot be applied.

Loading, empty and stale

Keep the last coherent values with a refresh state. A first read has no invented defaults. An absent capability remains explained and inspectable.

Keyboard and focus

Open on the triggering setting. Apply failure focuses the first invalid field on submission; async refresh never moves the caret. Escape cancels a transient editor and returns to its trigger.

Layout and adverse content

Align only related fields within their group. Long inputs use stacked labels. Native preferred sizes and zoom win over reference-art dimensions.

Starting point and required construction

Starting classifications: N, R. E existing operation; R runtime-backed adapter; N new contract; H host-only work. Exact per-control classifications remain in the catalog.

Reuse the reviewed domain operations and preserve all served fields. Build every missing native read/write adapter, capability gate and state renderer identified by the matching control/build rows.

BK01 owns validation, scope, persistence, revision comparison and acknowledgement; domain packages own parameter/capability enforcement.

Required interaction fixtures

Exercise first-read, empty/no-match, ready, refreshing/stale, error/refusal, action pending and acknowledged result. Also switch scope while the request is in flight.

Moved out of the interface

Every line of revision 7 plate copy that left the default view, and where it lives now.

“Faktor / Custom agent (R7 window title)”
→ Native host chrome (VS Code view title, tabs or editor tab; JetBrains tool-window tabs, dialog title or Settings breadcrumb)

“Project preset · version 3 · unsaved changes”
→ Subtitle; version in Tooltip / hover help

“Role instructions”
→ Label “Instructions”

“Inherit parent”
→ “Same as parent”

“Tool envelope / 3 enabled / Read, search and diagnostics. Writes are denied.”
→ Row “Tools”

“Child concurrency / 2 children”
→ Field “Agents at once”

“Child ceiling / 1.00 USD”
→ Field “Budget per agent” with help

“Ask before delegation / Children share the parent budget and capability envelope.”
→ Checkbox; budget sharing in field help, permissions rule in Specification: screen contract

“Custom agent”
→ Settings page “Agents”

“Name / Security reviewer / Model”
→ Fields

“Cancel / Save agent”
→ Cancel / Save Agent

PART 2 · SCREENS

2.8 Context

4 screens. Each has a design spread for both IDEs and a contract page.

ID	SCREEN	JOB	PAGE
S08	**Context**	Inspect what enters a call, what it costs in estimated or observed tokens, and where it came from. Capacity and input usage have different bases.	p. 123
S62	**Unavailable context usage**	Render null authoritative context usage as unavailable and keep any draft estimate on a separate basis.	p. 125
S09	**Add context**	A single discoverable picker adds exact files, ranges, diagnostics, terminal excerpts, diffs, URLs and prior-chat references to a draft.	p. 127
S10	**Compact context**	Compaction preserves task obligations and operation identity while reducing eligible history. Show a plan before dispatch and a measured result afterwards.	p. 129

S08 CONTEXT

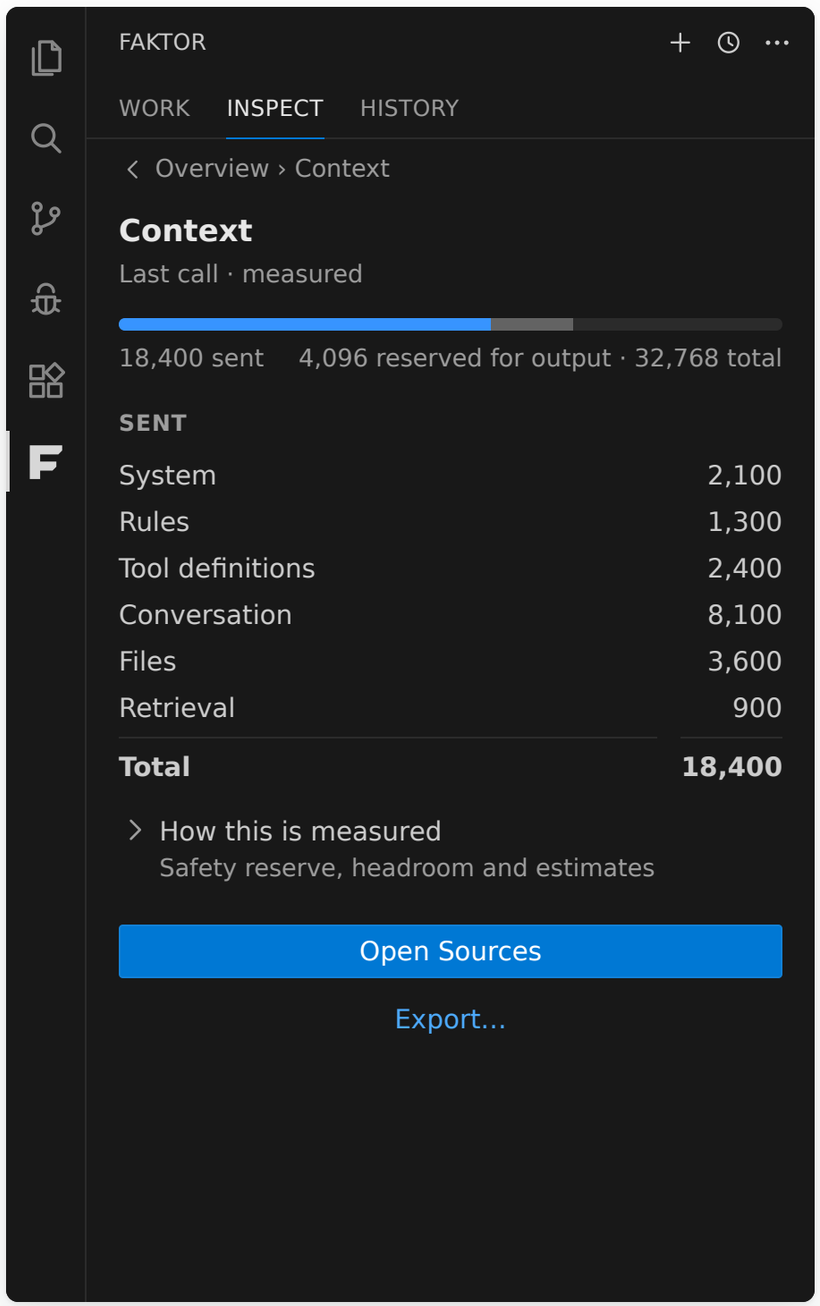
Context

PRIMARY ACTION

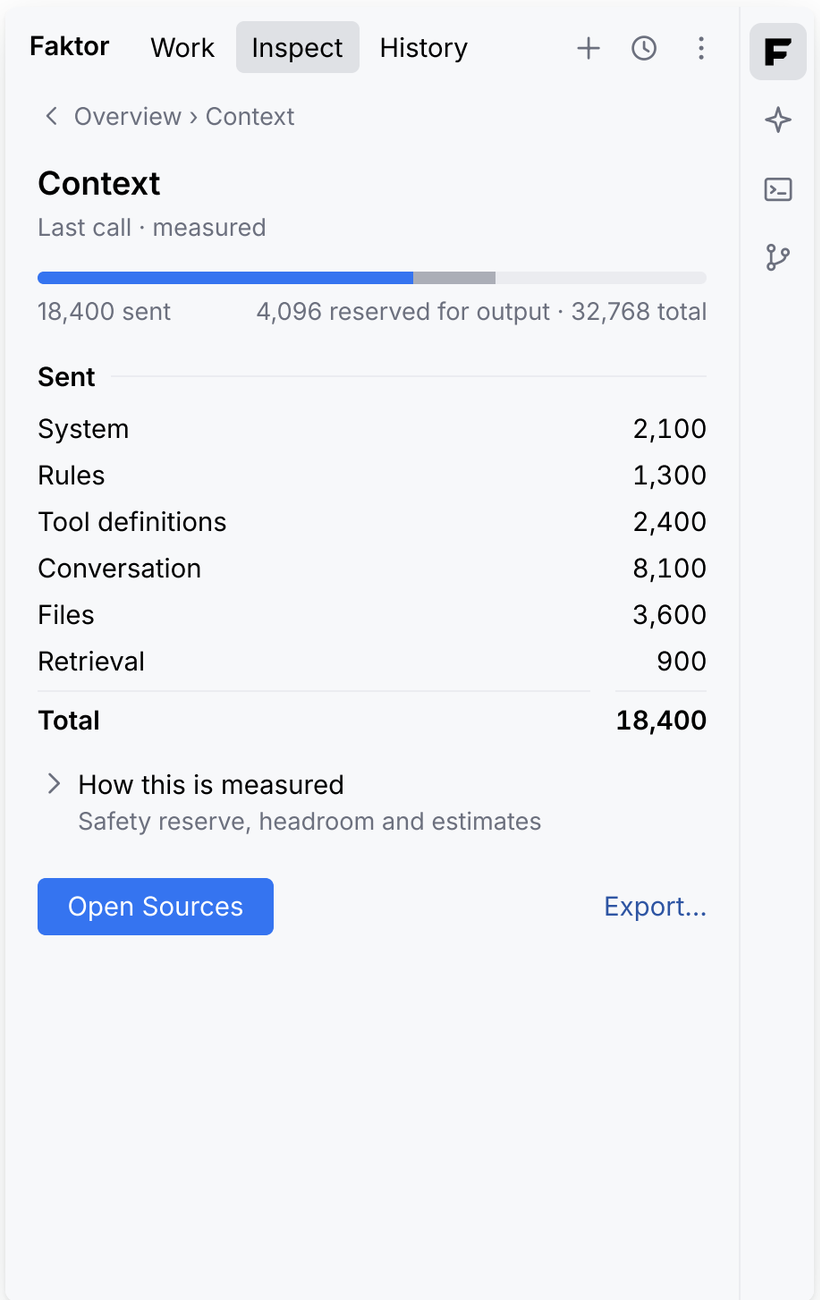
Open Sources

Inspect what enters a call, what it costs in estimated or observed tokens, and where it came from. Capacity and input usage have different bases.

VS CODE · DARK MODERN — SIDEBAR VIEW · WEBVIEW



JETBRAINS · LIGHT — TOOL WINDOW · SWING / KOTLIN UI DSL



CONTROLS C-01 C-02 C-04 C-05 C-07 C-08 C-09 C-10 C-11 C-12 C-13 C-14

REPLACES R7 page 051 · Expanded standard-agent surface

CONTRACT p. 124

S08 · CONTEXT · BEHAVIOUR CONTRACT

Context: contract and revision notes

Original contract retained verbatim (Expanded standard-agent surface / controls: C-01 C-02 C-04 C-05 C-07 C-08 C-09 C-10 C-11 C-12 C-13 C-14). Design on p. 123.

What changed in revision 8

- One meter shows sent vs. reserved vs. total; the breakdown sums to the measured total.

Entry and return

Composer context picker/meter, attachment staging, memory or compaction command. Work / Inspect / History remain the only top-level clusters. Settings, Usage and Diagnostics use explicit Back. Return to the originating object, not the first row.

Scope and captured state

Draft/call receipt + exact source URI/range/version + token accounting basis; memory has origin and retention scope. Explicit inclusion acquires a versioned source snapshot. Do not replace an admitted source with the current editor or silently drop an unsupported attachment.

Primary action and recovery

Pin, exclude or open an attributed source; resolve overflow before sending. Show the preview version that admission will capture.. With current contextUsage=null, measured usage reads Unavailable. Estimates remain useful but never masquerade as a durable runtime meter.

Loading, empty and stale

Measured input, draft estimate, model capacity and output/safety reserve use separate labels. null contextUsage is unavailable. Upload and compaction each retain original inputs.

Keyboard and focus

Picker arrows select sources; Enter stages the selected identity; Escape returns to the same composer caret. New usage observations do not interrupt typing.

Layout and adverse content

Wrap source labels and exact rejection reasons. Keep previews bounded with full artifact/copy actions. Do not fit a long source list between several competing scroll panes.

Starting point and required construction

Starting classifications: E, H, N, R. E existing operation; R runtime-backed adapter; N new contract; H host-only work. Exact per-control classifications remain in the catalog.

Reuse the reviewed domain operations and preserve all served fields. Build every missing native read/write adapter, capability gate and state renderer identified by the matching control/build rows.

BK04 owns accounting and compaction; BK03 owns versioned memory/customization; BK02 supplies model capabilities and BK08 owns upload handles.

Required interaction fixtures

Exercise first-read, empty/no-match, ready, refreshing/stale, error/refusal, action pending and acknowledged result. Also switch scope while the request is in flight.

Moved out of the interface

Every line of revision 7 plate copy that left the default view, and where it lives now.

"Faktor / Context (R7 window title)"
→ Native host chrome (VS Code view title, tabs or editor tab; JetBrains tool-window tabs, dialog title or Settings breadcrumb)

"Submitted call ctx-184-06 · measured input, explicit reserves"
→ Subtitle "Last call · measured"; call ID in Tooltip / hover help

"Submitted input / 18,400 tokens; Effective capacity / 32,768 tokens"
→ Meter label

"Output reserve / 4,096 tokens; Estimated headroom / 8,224 tokens"
→ Meter label (reserve) and disclosure (headroom)

"Included input / 18,400 total"
→ Section "Sent" with total row

"Safety reserve: 2,048 tokens. Draft estimates are shown separately from the submitted-call receipt."
→ How this is measured disclosure

"Open source manifest"
→ Open Sources

"Context"
→ Title

"System / Rules / Tool definitions / Conversation / Files / Retrieval"
→ Breakdown rows

"Export"
→ Export…

S62 CONTEXT

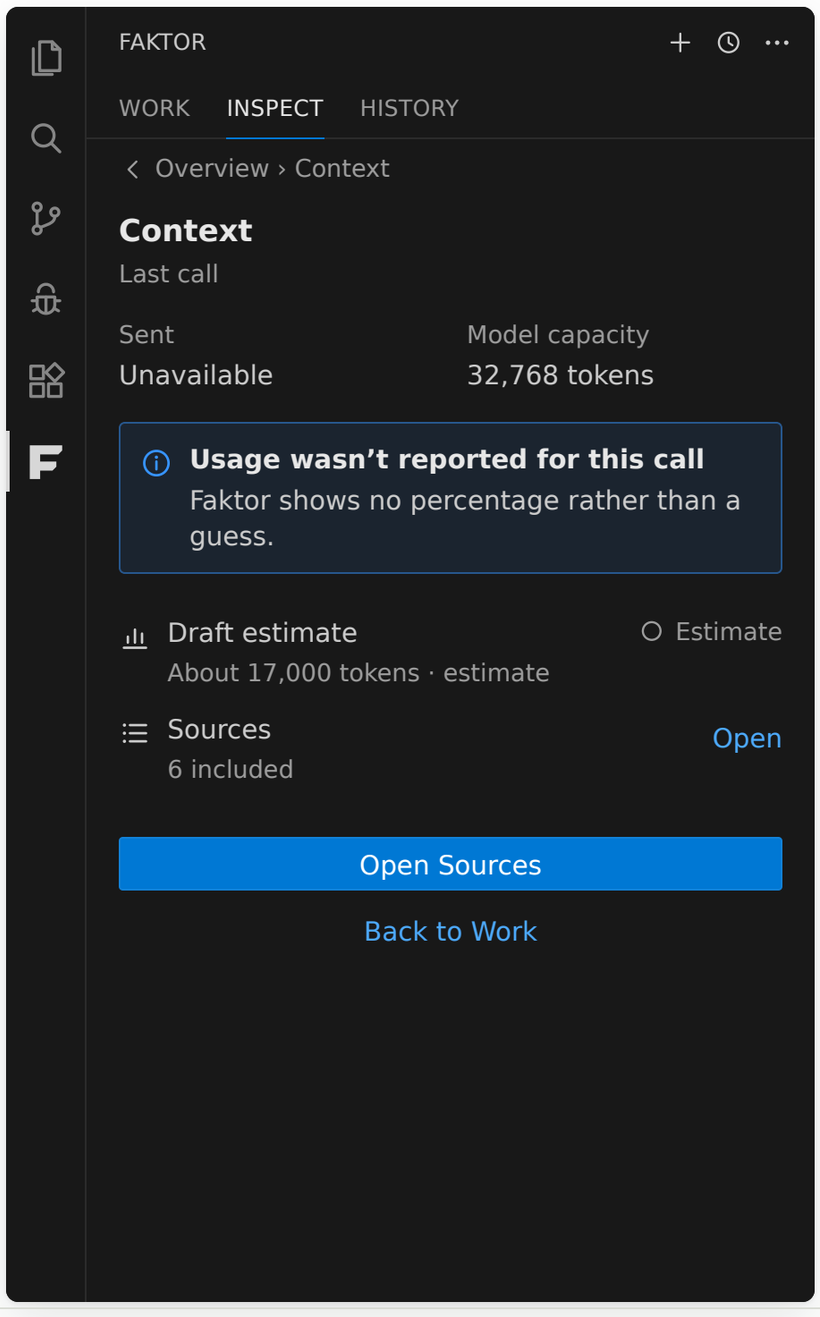
Unavailable context usage

Render null authoritative context usage as unavailable and keep any draft estimate on a separate basis.

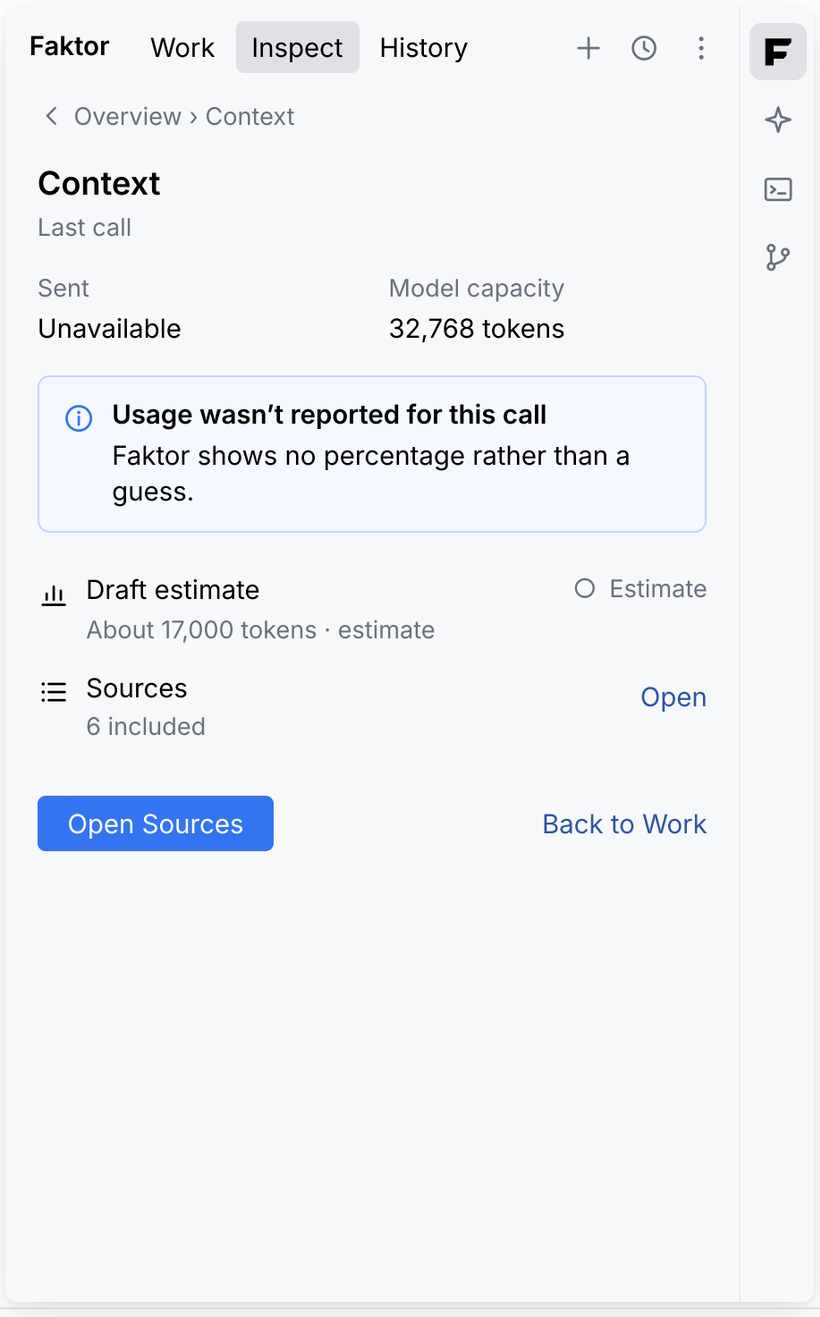
PRIMARY ACTION

Open Sources

VS CODE · DARK MODERN — SIDEBAR VIEW · WEBVIEW



JETBRAINS · LIGHT — TOOL WINDOW · SWING / KOTLIN UI DSL



CONTROLS C-01 C-02

REPLACES R7 page 052 · Expanded standard-agent surface

CONTRACT p. 126

S62 · CONTEXT · BEHAVIOUR CONTRACT

Unavailable context usage: contract and revision notes

Original contract retained verbatim (Expanded standard-agent surface / controls: C-01 C-02). Design on p. 125.

What changed in revision 8

- Like page 38, a developer instruction left the UI; users see only the fact and its meaning.

Entry and return

Composer context picker/meter, attachment staging, memory or compaction command. Work / Inspect / History remain the only top-level clusters. Settings, Usage and Diagnostics use explicit Back. Return to the originating object, not the first row.

Scope and captured state

Draft/call receipt + exact source URI/range/version + token accounting basis; memory has origin and retention scope. Explicit inclusion acquires a versioned source snapshot. Do not replace an admitted source with the current editor or silently drop an unsupported attachment.

Primary action and recovery

Open source manifest. Show unsupported type, oversize, upload error, capacity refusal and stale source separately. Remove only explicitly; failed compaction keeps the prior valid accounting basis.

Loading, empty and stale

Measured input, draft estimate, model capacity and output/safety reserve use separate labels. null contextUsage is unavailable. Upload and compaction each retain original inputs.

Keyboard and focus

Picker arrows select sources; Enter stages the selected identity; Escape returns to the same composer caret. New usage observations do not interrupt typing.

Layout and adverse content

Wrap source labels and exact rejection reasons. Keep previews bounded with full artifact/copy actions. Do not fit a long source list between several competing scroll panes.

Starting point and required construction

Starting classifications: N. E existing operation; R runtime-backed adapter; N new contract; H host-only work. Exact per-control classifications remain in the catalog.

Current contextUsage may be null. Preserve Unavailable now; build BK04 before displaying measured totals or a utilization meter.

BK04 owns accounting and compaction; BK03 owns versioned memory/customization; BK02 supplies model capabilities and BK08 owns upload handles.

Required interaction fixtures

Exercise first-read, empty/no-match, ready, refreshing/stale, error/refusal, action pending and acknowledged result. Also switch scope while the request is in flight.

Moved out of the interface

Every line of revision 7 plate copy that left the default view, and where it lives now.

"Faktor / Work / Context (R7 window title)"
→ Native host chrome (VS Code view title, tabs or editor tab; JetBrains tool-window tabs, dialog title or Settings breadcrumb)

"No authoritative accounting receipt is served for this call."
→ Banner title

"Submitted input / Unavailable; Model capacity / 32,768 declared"
→ Facts

"No usage percentage can be shown"
→ Banner text "Faktor shows no percentage rather than a guess."

"The current native contextUsage value may be null. Do not turn it into zero tokens or a fabricated utilization bar."
→ Specification: screen contract (C-01) — implementation instruction removed from the interface

"Draft estimate / Separate / Estimates may be shown with their model and tokenizer basis"
→ Row "Draft estimate — Estimate"; basis in Tooltip / hover help

"Included sources / Inspect / Selection manifest remains inspectable without a usage total"
→ Row "Sources — 6 included"

"Context usage unavailable"
→ Banner "Usage wasn't reported for this call"

"Open source manifest"
→ Open Sources

"Back to Work"
→ Back to Work

S09 CONTEXT

Add context

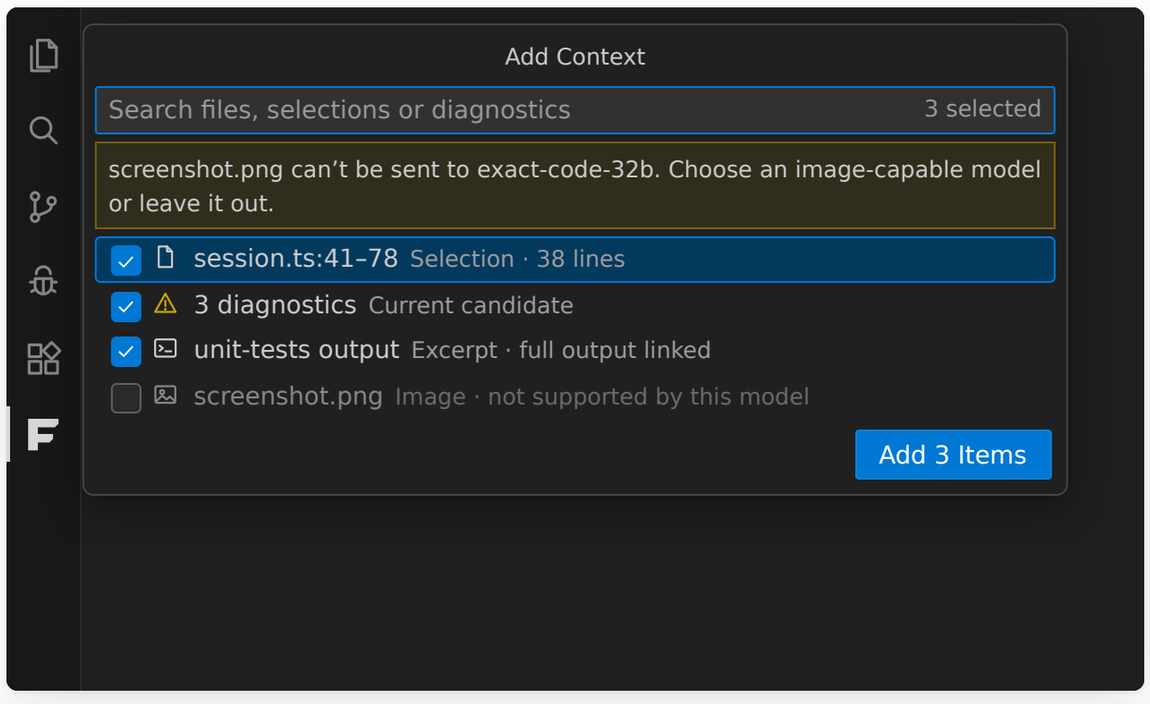
A single discoverable picker adds exact files, ranges, diagnostics, terminal excerpts, diffs, URLs and prior-chat references to a draft.

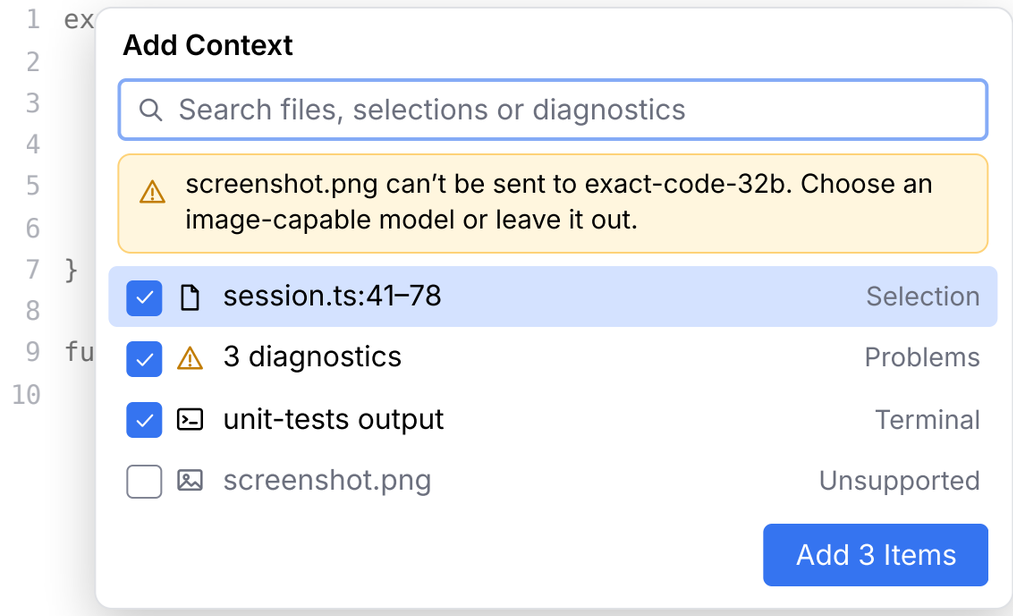
PRIMARY ACTION

Enter selects · Add 3 Items

VS CODE · DARK MODERN — QUICK PICK · NATIVE

JETBRAINS · LIGHT — POPUP CHOOSER · NATIVE





CONTROLS C-03 C-04 C-05 C-06 C-07 C-08 C-13 X-10 X-11

REPLACES R7 page 053 · Expanded standard-agent surface

CONTRACT p. 128

S09 · CONTEXT · BEHAVIOUR CONTRACT

Add context: contract and revision notes

Original contract retained verbatim (Expanded standard-agent surface / controls: C-03 C-04 C-05 C-06 C-07 C-08 C-13 X-10 X-11). Design on p. 127.

What changed in revision 8

- Multi-select Quick Pick (VS Code) and checkbox popup (JetBrains); the unsupported image stays visible with its reason.

Entry and return

Composer context picker/meter, attachment staging, memory or compaction command. Work / Inspect / History remain the only top-level clusters. Settings, Usage and Diagnostics use explicit Back. Return to the originating object, not the first row.

Scope and captured state

Draft/call receipt + exact source URI/range/version + token accounting basis; memory has origin and retention scope. Explicit inclusion acquires a versioned source snapshot. Do not replace an admitted source with the current editor or silently drop an unsupported attachment.

Primary action and recovery

Select a source to preview its resolved URI, range, version, bytes and inclusion state. Escape closes the picker without clearing text.. Oversize, stale or unsupported attachments stay visible with a reason and Remove/Retry. Do not silently omit a failed item on Send.

Loading, empty and stale

Measured input, draft estimate, model capacity and output/safety reserve use separate labels. null contextUsage is unavailable. Upload and compaction each retain original inputs.

Keyboard and focus

Picker arrows select sources; Enter stages the selected identity; Escape returns to the same composer caret. New usage observations do not interrupt typing.

Layout and adverse content

Wrap source labels and exact rejection reasons. Keep previews bounded with full artifact/copy actions. Do not fit a long source list between several competing scroll panes.

Starting point and required construction

Starting classifications: E, H, N. E existing operation; R runtime-backed adapter; N new contract; H host-only work. Exact per-control classifications remain in the catalog.

Reuse the reviewed domain operations and preserve all served fields. Build every missing native read/write adapter, capability gate and state renderer identified by the matching control/build rows.

BK04 owns accounting and compaction; BK03 owns versioned memory/customization; BK02 supplies model capabilities and BK08 owns upload handles.

Required interaction fixtures

Exercise first-read, empty/no-match, ready, refreshing/stale, error/refusal, action pending and acknowledged result. Also switch scope while the request is in flight.

Moved out of the interface

Every line of revision 7 plate copy that left the default view, and where it lives now.

"Faktor / Context picker (R7 window title)"
→ Native host chrome (VS Code view title, tabs or editor tab; JetBrains tool-window tabs, dialog title or Settings breadcrumb)

"Choose exact sources before adding them to the draft."
→ Picker placeholder and title

"Selected sources / 3 ready · 1 blocked"
→ Counter "3 selected" and the disabled item

"Selection · editor version 27 · 38 lines"
→ Item description; editor version in Tooltip / hover help

"Explicit capture · current candidate"
→ Description "Current candidate"

"Bounded excerpt · full artifact linked"
→ Description "Excerpt · full output linked"

"Unsupported / Staged image · not silently omitted"
→ Disabled item that stays visible

"This model cannot accept the image / Choose a capable model or remove screenshot.png before sending."
→ Picker validation message

"Remove image / Add selected context"
→ Uncheck / Add 3 Items

"3 selected diagnostics"
→ Item "3 diagnostics"

"Add context"
→ Picker title

"session.ts:41-78 / unit-tests output / screenshot.png"
→ Picker items

S10 CONTEXT

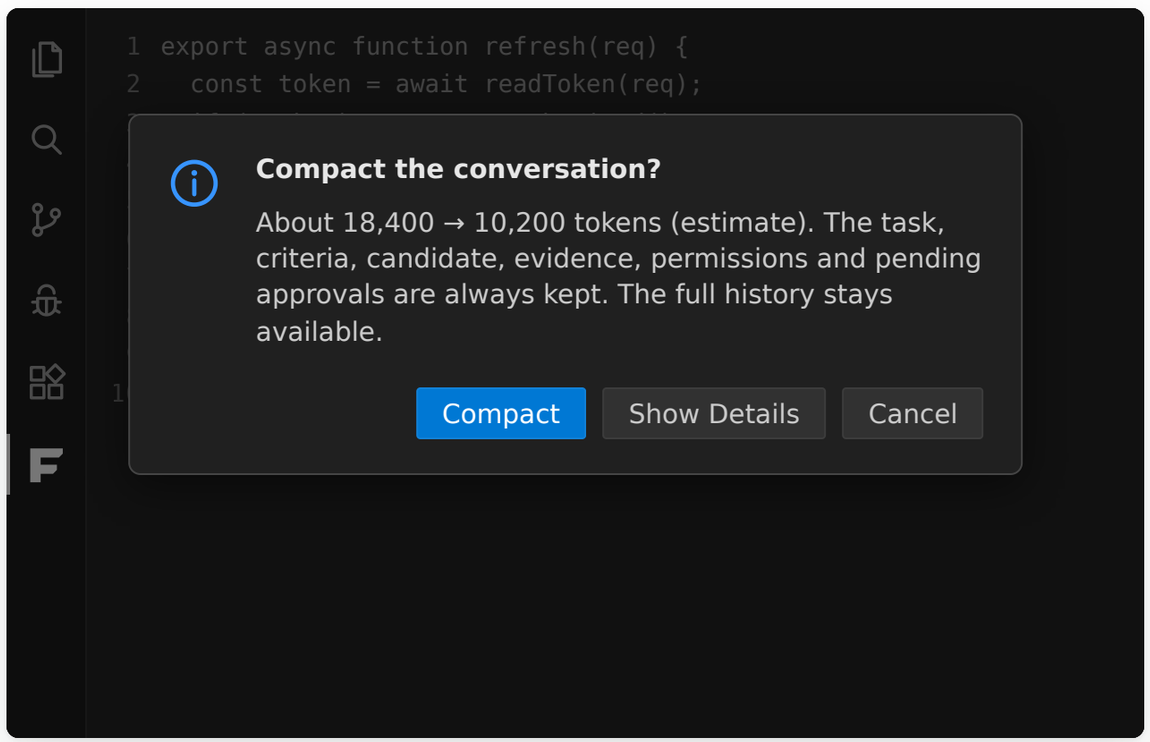
Compact context

PRIMARY ACTION

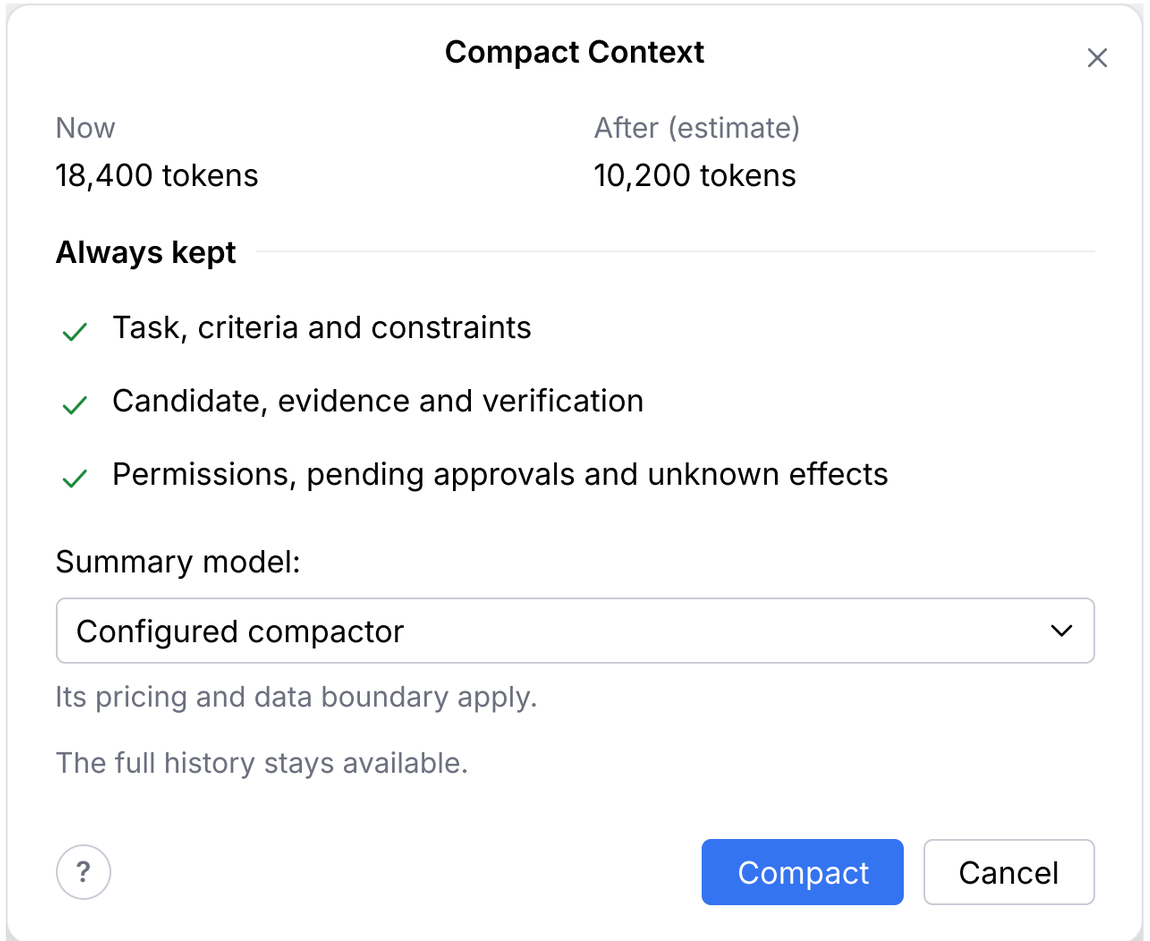
Compact

Compaction preserves task obligations and operation identity while reducing eligible history. Show a plan before dispatch and a measured result afterwards.

VS CODE · DARK MODERN — MODAL MESSAGE · NATIVE



JETBRAINS · LIGHT — DIALOGWRAPPER



CONTROLS K-01 K-02 K-03 K-04 C-12 I-11 V-07

REPLACES R7 page 054 · Expanded standard-agent surface

CONTRACT p. 130

S10 · CONTEXT · BEHAVIOUR CONTRACT

Compact context: contract and revision notes

Original contract retained verbatim (Expanded standard-agent surface / controls: K-01 K-02 K-03 K-04 C-12 I-11 V-07). Design on p. 129.

What changed in revision 8

- A confirmation dialog states the estimate and guarantees; nothing technical is needed to decide.

Entry and return

Composer context picker/meter, attachment staging, memory or compaction command. Work / Inspect / History remain the only top-level clusters. Settings, Usage and Diagnostics use explicit Back. Return to the originating object, not the first row.

Scope and captured state

Draft/call receipt + exact source URI/range/version + token accounting basis; memory has origin and retention scope. Explicit inclusion acquires a versioned source snapshot. Do not replace an admitted source with the current editor or silently drop an unsupported attachment.

Primary action and recovery

Review retained facts and the model/egress/cost implications, then Compact using a new durable operation contract.. Failure or insufficient reduction keeps the prior valid basis. Pending approvals and unknown external effects cannot be erased by summarization.

Loading, empty and stale

Measured input, draft estimate, model capacity and output/safety reserve use separate labels. null contextUsage is unavailable. Upload and compaction each retain original inputs.

Keyboard and focus

Picker arrows select sources; Enter stages the selected identity; Escape returns to the same composer caret. New usage observations do not interrupt typing.

Layout and adverse content

Wrap source labels and exact rejection reasons. Keep previews bounded with full artifact/copy actions. Do not fit a long source list between several competing scroll panes.

Starting point and required construction

Starting classifications: E, N, R. E existing operation; R runtime-backed adapter; N new contract; H host-only work. Exact per-control classifications remain in the catalog.

Reuse the reviewed domain operations and preserve all served fields. Build every missing native read/write adapter, capability gate and state renderer identified by the matching control/build rows.

BK04 owns accounting and compaction; BK03 owns versioned memory/customization; BK02 supplies model capabilities and BK08 owns upload handles.

Required interaction fixtures

Exercise first-read, empty/no-match, ready, refreshing/stale, error/refusal, action pending and acknowledged result. Also switch scope while the request is in flight.

Moved out of the interface

Every line of revision 7 plate copy that left the default view, and where it lives now.

"Faktor / Compact context (R7 window title)"
→ Native host chrome (VS Code view title, tabs or editor tab; JetBrains tool-window tabs, dialog title or Settings breadcrumb)

"Preview the retained facts and reduction before starting."
→ Dialog itself is the preview

"Current input / Estimated after"
→ Facts "Now" / "After (estimate)"

"Estimate, not a completed compaction"
→ "(estimate)" label; measured result appears after compaction

"Original history and full tool artifacts remain available by reference."
→ "The full history stays available."

"Always preserve / Retained"
→ Section "Always kept" with check icons

"Goal, required criteria, constraints and risks"
→ Row

"Candidate version, evidence and verification basis"
→ Row

"Policy version, pending approvals and unknown effects"
→ Row

"Inherit configured compactor"
→ "Configured compactor"

"Model, cost and data boundary are captured before dispatch."
→ Help "Its pricing and data boundary apply."

"Review compaction"
→ Compact

"Task contract"
→ Row "Task, criteria and constraints"

"Engineering identity"
→ Row "Candidate, evidence and verification"

"Authority and effects"
→ Row "Permissions, pending approvals and unknown effects"

"Compact context"
→ Dialog title

"Summary model"
→ Field

S10 · COMPACT CONTEXT · CONTRACT, CONTINUED

“Cancel”
→ Cancel

PART 2 · SCREENS

2.9 Instructions and memory

4 screens. Each has a design spread for both IDEs and a contract page.

ID	SCREEN	JOB	PAGE
S11	**Instructions & rules**	Inspect instructions by origin and activation scope, edit allowed sources and preview the merged bundle that the next task will capture.	p. 133
S69	**Instruction precedence**	Explain instruction precedence and managed policy conflicts without treating project text as executable authority.	p. 135
S12	**Memory**	Separate learned failure priors, user instructions and proposed personal/project memory. Each record has scope, source, date and deletion semantics.	p. 137
S66	**Memory deletion**	Delete one versioned memory from future retrieval while disclosing earlier prompts, artifacts and log copies.	p. 139

S11 INSTRUCTIONS AND MEMORY

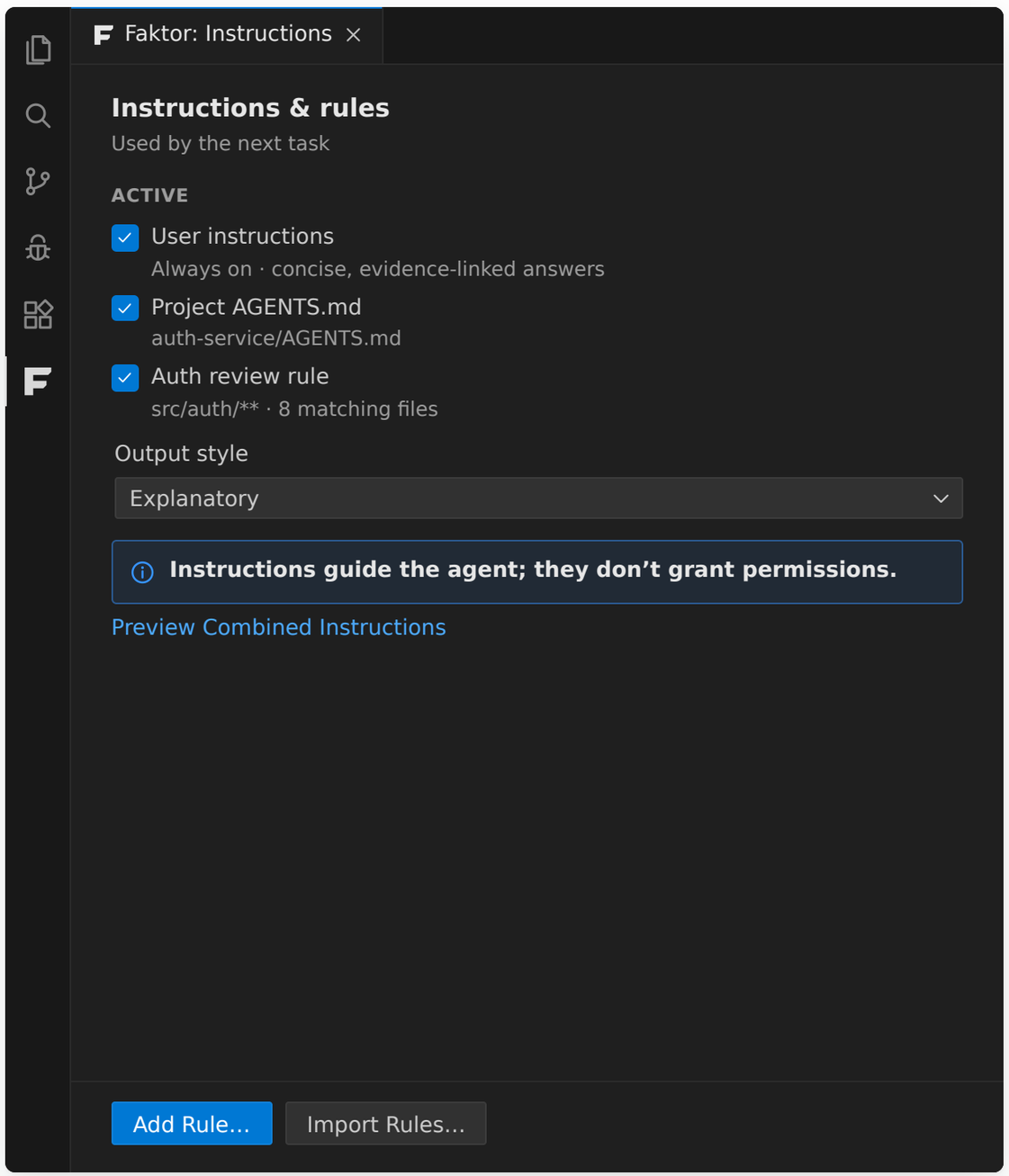
Instructions & rules

PRIMARY ACTION

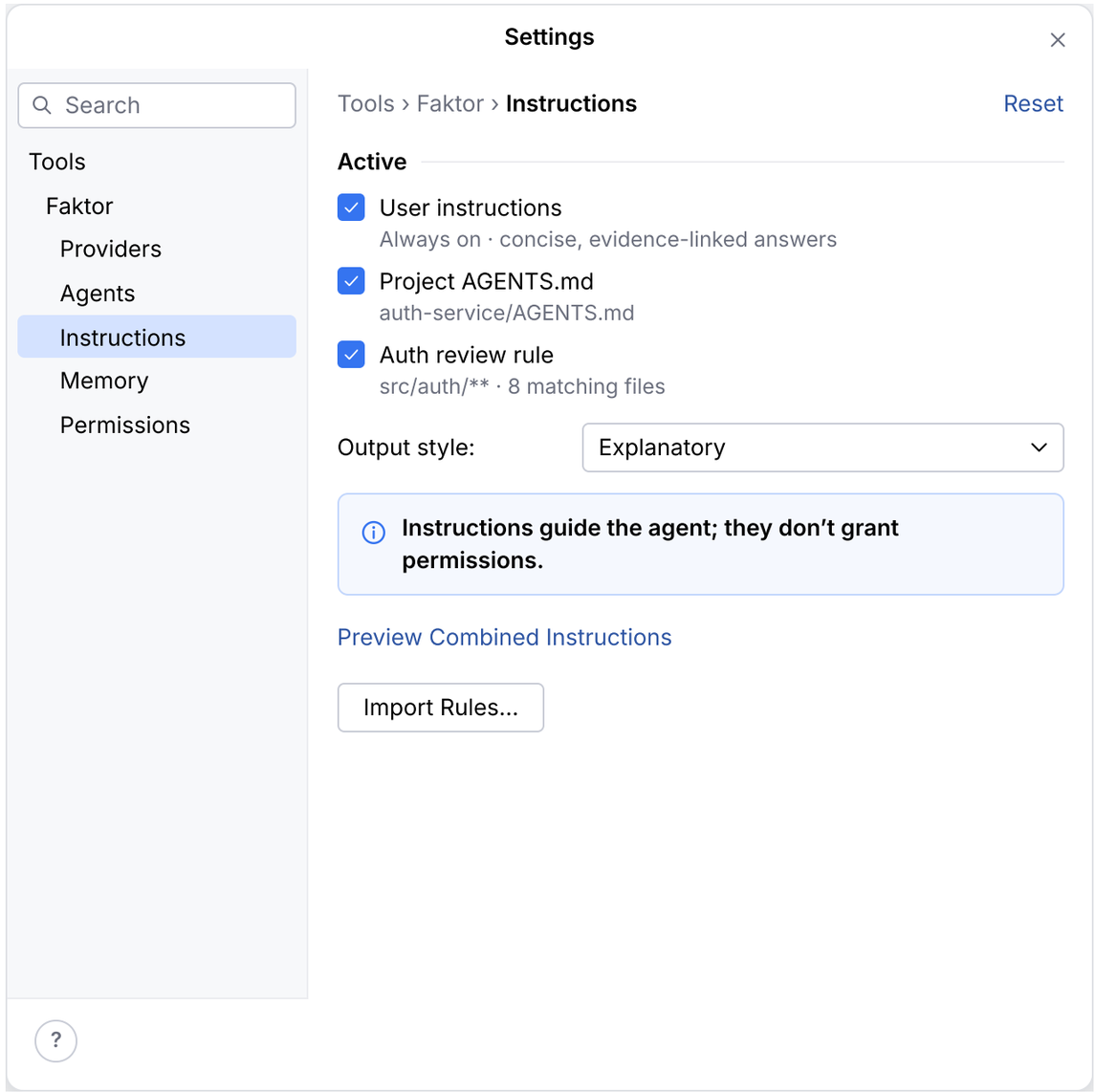
Add Rule... · JetBrains: OK / Apply

Inspect instructions by origin and activation scope, edit allowed sources and preview the merged bundle that the next task will capture.

VS CODE · DARK MODERN — EDITOR TAB · WEBVIEW SETTINGS



JETBRAINS · LIGHT — SETTINGS DIALOG · CONFIGURABLE



CONTROLS I-01 I-02 I-03 I-04 I-05 I-06 I-07 I-08 I-09 I-10 I-11

REPLACES R7 page 055 · Expanded standard-agent surface

CONTRACT p. 134

S11 · INSTRUCTIONS AND MEMORY · BEHAVIOUR CONTRACT

Instructions & rules: contract and revision notes

Original contract retained verbatim (Expanded standard-agent surface / controls: I-01 I-02 I-03 I-04 I-05 I-06 I-07 I-08 I-09 I-10 I-11). Design on p. 133.

What changed in revision 8

- Sources are checkboxes because they are independent toggles.

Entry and return

Native Settings or the originating configuration command. Work / Inspect / History remain the only top-level clusters. Settings, Usage and Diagnostics use explicit Back. Return to the originating object, not the first row.

Scope and captured state

Application/project/connection plus effective revision; never infer task scope from an open form. Keep staged input, the effective snapshot and the acknowledged applied revision separate. Apply is atomic; Cancel preserves the previous effective configuration.

Primary action and recovery

Test a scoped rule against sample files, review conflicts and save a version. Show a policy lock rather than pretending text can override it.. An invalid glob or unresolved template variable blocks Apply. Unsaved edits survive navigation and are never mistaken for applied runtime state.

Loading, empty and stale

Keep the last coherent values with a refresh state. A first read has no invented defaults. An absent capability remains explained and inspectable.

Keyboard and focus

Open on the triggering setting. Apply failure focuses the first invalid field on submission; async refresh never moves the caret. Escape cancels a transient editor and returns to its trigger.

Layout and adverse content

Align only related fields within their group. Long inputs use stacked labels. Native preferred sizes and zoom win over reference-art dimensions.

Starting point and required construction

Starting classifications: H, N, R. E existing operation; R runtime-backed adapter; N new contract; H host-only work. Exact per-control classifications remain in the catalog.

Reuse the reviewed domain operations and preserve all served fields. Build every missing native read/write adapter, capability gate and state renderer identified by the matching control/build rows.

BK01 owns validation, scope, persistence, revision comparison and acknowledgement; domain packages own parameter/capability enforcement.

Required interaction fixtures

Exercise first-read, empty/no-match, ready, refreshing/stale, error/refusal, action pending and acknowledged result. Also switch scope while the request is in flight.

Moved out of the interface

Every line of revision 7 plate copy that left the default view, and where it lives now.

"Faktor / Instructions & rules (R7 window title)"
→ Native host chrome (VS Code view title, tabs or editor tab; JetBrains tool-window tabs, dialog title or Settings breadcrumb)

"Inspect the source and activation of every instruction."
→ Subtitle "Used by the next task"

"Always active · concise, evidence-linked responses"
→ Checkbox help

"auth-service/AGENTS.md · trusted project source"
→ Checkbox help; trust state in Tooltip / hover help

"Effective instruction bundle / Preview / 3 active sources · version and hashes captured"
→ Link "Preview Combined Instructions"; hashes in the preview

"Instructions do not grant tool authority / Managed permissions remain enforced at dispatch."
→ Info banner (shortened)

"Instructions & rules"
→ Title

"Active sources"
→ Section "Active"

"User instructions / Project AGENTS.md / Auth review rule"
→ Checkboxes

"Output style / Explanatory"
→ Field

"Import rules / Add rule"
→ Import Rules... / Add Rule...

S69 INSTRUCTIONS AND MEMORY

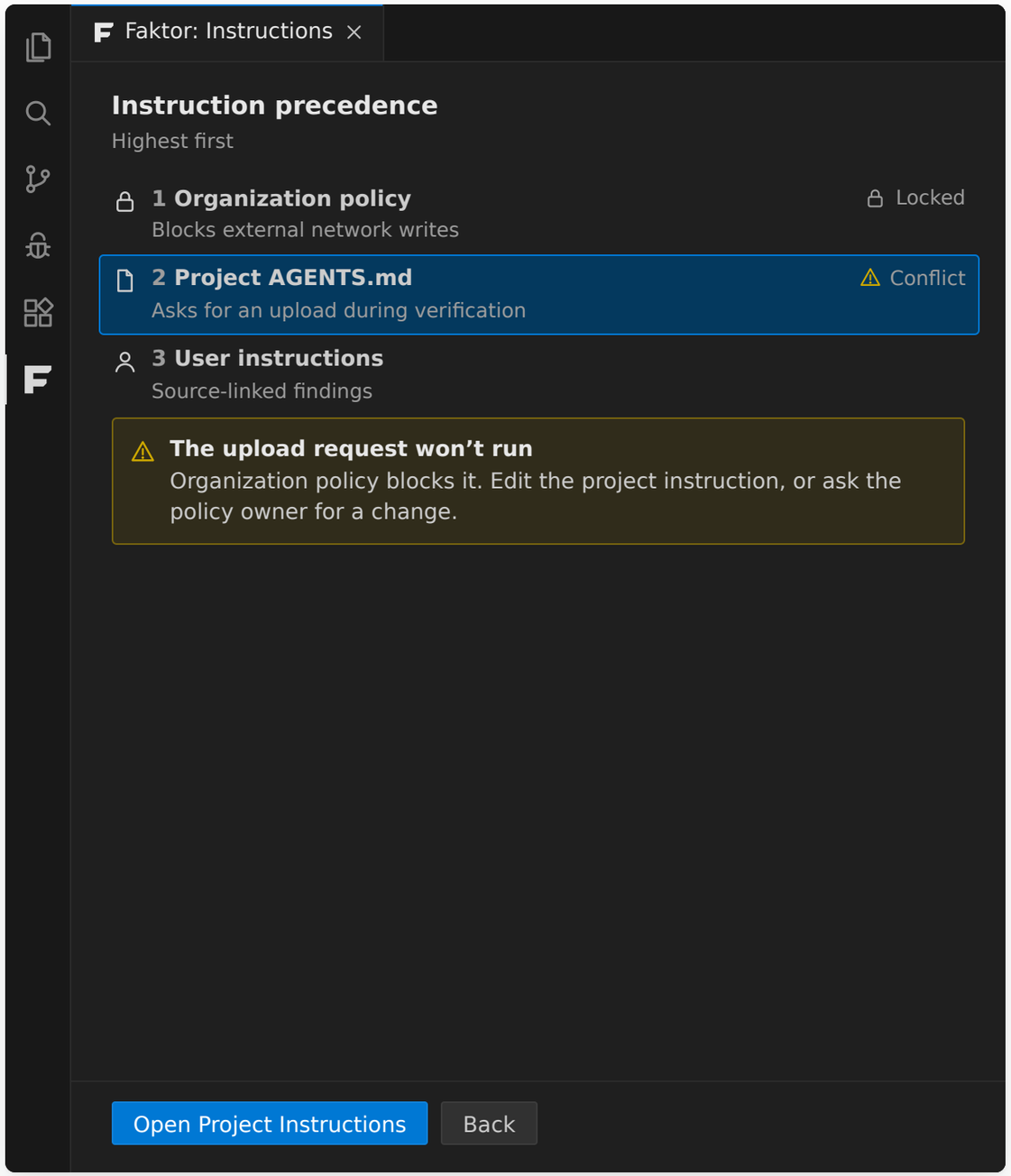
Instruction precedence

Explain instruction precedence and managed policy conflicts without treating project text as executable authority.

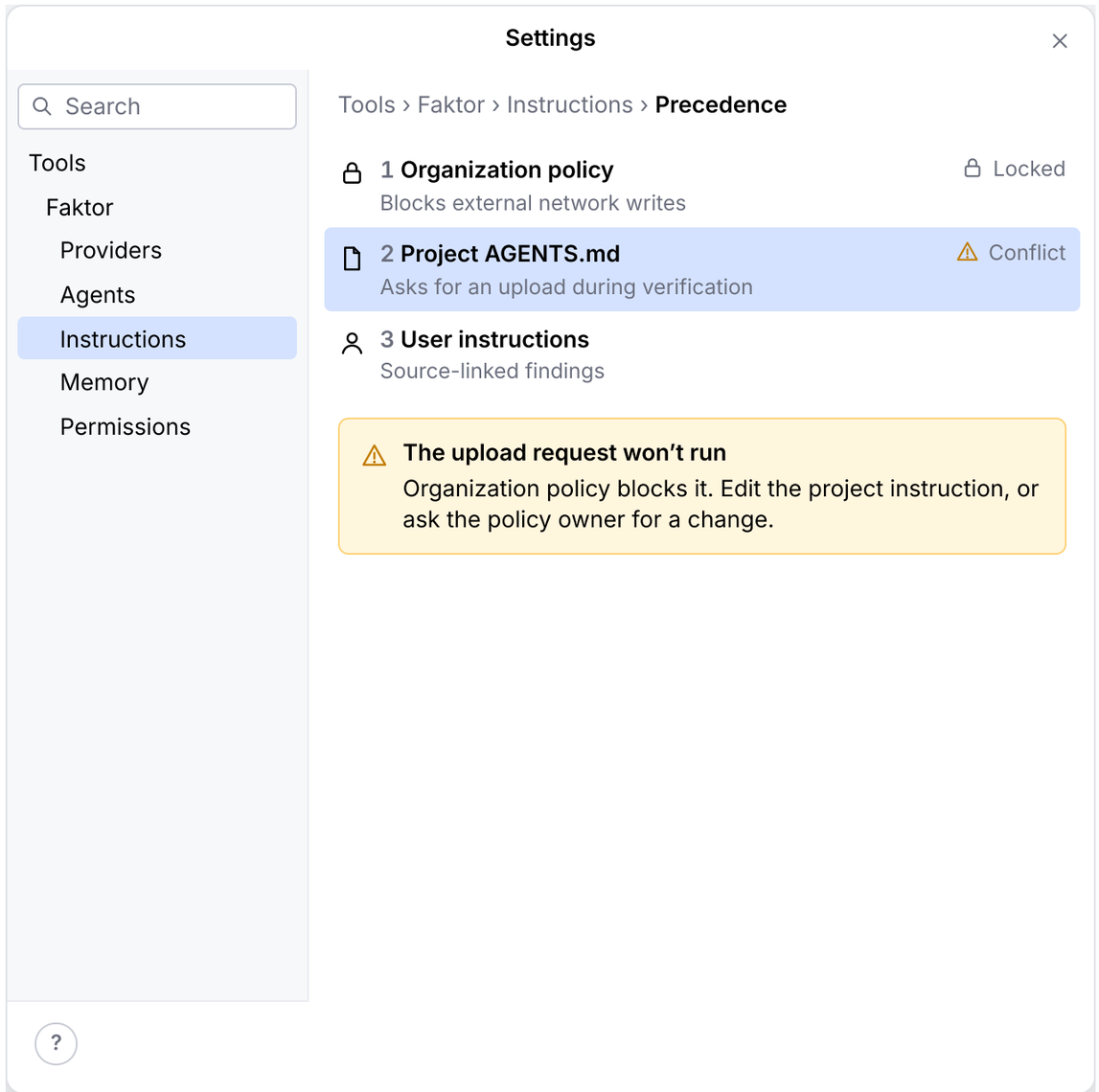
PRIMARY ACTION

Open Project Instructions · JetBrains: OK / Apply

VS CODE · DARK MODERN — EDITOR TAB · WEBVIEW SETTINGS



JETBRAINS · LIGHT — SETTINGS DIALOG · CONFIGURABLE



CONTROLS I-04 I-08 Q-12

REPLACES R7 page 056 · Expanded standard-agent surface

CONTRACT p. 136

S69 · INSTRUCTIONS AND MEMORY · BEHAVIOUR CONTRACT

Instruction precedence: contract and revision notes

Original contract retained verbatim (Expanded standard-agent surface / controls: I-04 I-08 Q-12). Design on p. 135.

What changed in revision 8

- Precedence is a numbered list; the outcome of the conflict is the headline.

Entry and return

Native Settings or the originating configuration command. Work / Inspect / History remain the only top-level clusters. Settings, Usage and Diagnostics use explicit Back. Return to the originating object, not the first row.

Scope and captured state

Application/project/connection plus effective revision; never infer task scope from an open form. Keep staged input, the effective snapshot and the acknowledged applied revision separate. Apply is atomic; Cancel preserves the previous effective configuration.

Primary action and recovery

Open project instructions. Associate validation with the exact field. A revision conflict preserves all staged values and opens the conflict resolver.

Authentication, transport and schema failures have different recovery.

Loading, empty and stale

Keep the last coherent values with a refresh state. A first read has no invented defaults. An absent capability remains explained and inspectable.

Keyboard and focus

Open on the triggering setting. Apply failure focuses the first invalid field on submission; async refresh never moves the caret. Escape cancels a transient editor and returns to its trigger.

Layout and adverse content

Align only related fields within their group. Long inputs use stacked labels. Native preferred sizes and zoom win over reference-art dimensions.

Starting point and required construction

Starting classifications: N, R. E existing operation; R runtime-backed adapter; N new contract; H host-only work. Exact per-control classifications remain in the catalog.

NEW: expose a versioned effective instruction bundle, provenance, conflict resolution and activation preview. Instruction text cannot grant execution authority.

BK01 owns validation, scope, persistence, revision comparison and acknowledgement; domain packages own parameter/capability enforcement.

Required interaction fixtures

Exercise first-read, empty/no-match, ready, refreshing/stale, error/refusal, action pending and acknowledged result. Also switch scope while the request is in flight.

Moved out of the interface

Every line of revision 7 plate copy that left the default view, and where it lives now.

"Faktor / Settings / Instructions (R7 window title)"
→ Native host chrome (VS Code view title, tabs or editor tab; JetBrains tool-window tabs, dialog title or Settings breadcrumb)

"Inspect instruction precedence / Effective instruction bundle · captured at the next admission"
→ Title and subtitle

"Managed execution policy / Locked / External network writes denied"
→ Row 1

"Project AGENTS.md / Conflict / Requests an external upload during verification"
→ Row 2

"User instructions / Require source-linked findings; no authority changes"
→ Row 3

"Effective result / The project text remains visible, but it cannot grant network authority. Edit the instruction or request a real policy change through its owner."
→ Warning banner (consequence first)

"Back to sources"
→ Back

"Open project instructions"
→ Open Project Instructions

S12 INSTRUCTIONS AND MEMORY

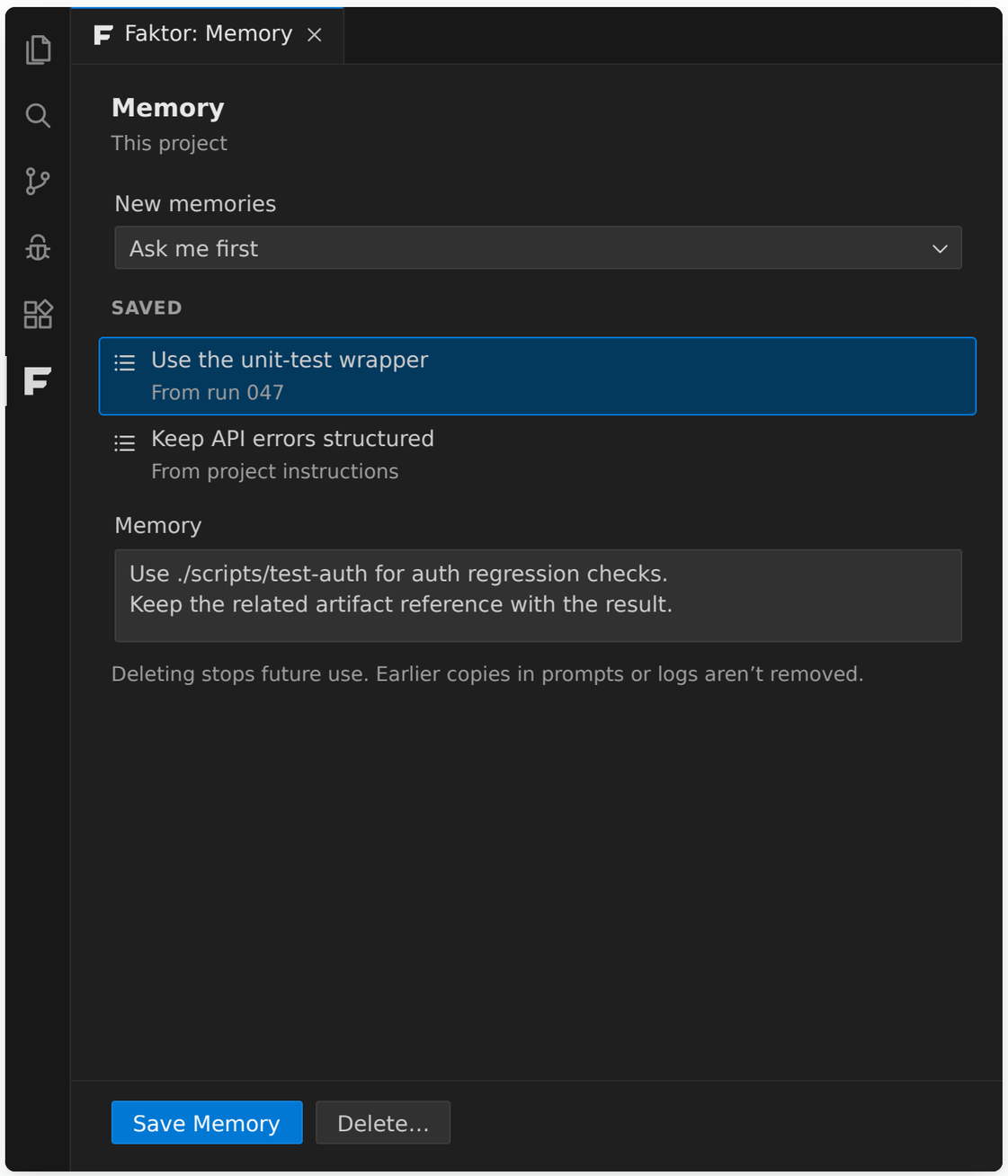
Memory

PRIMARY ACTION

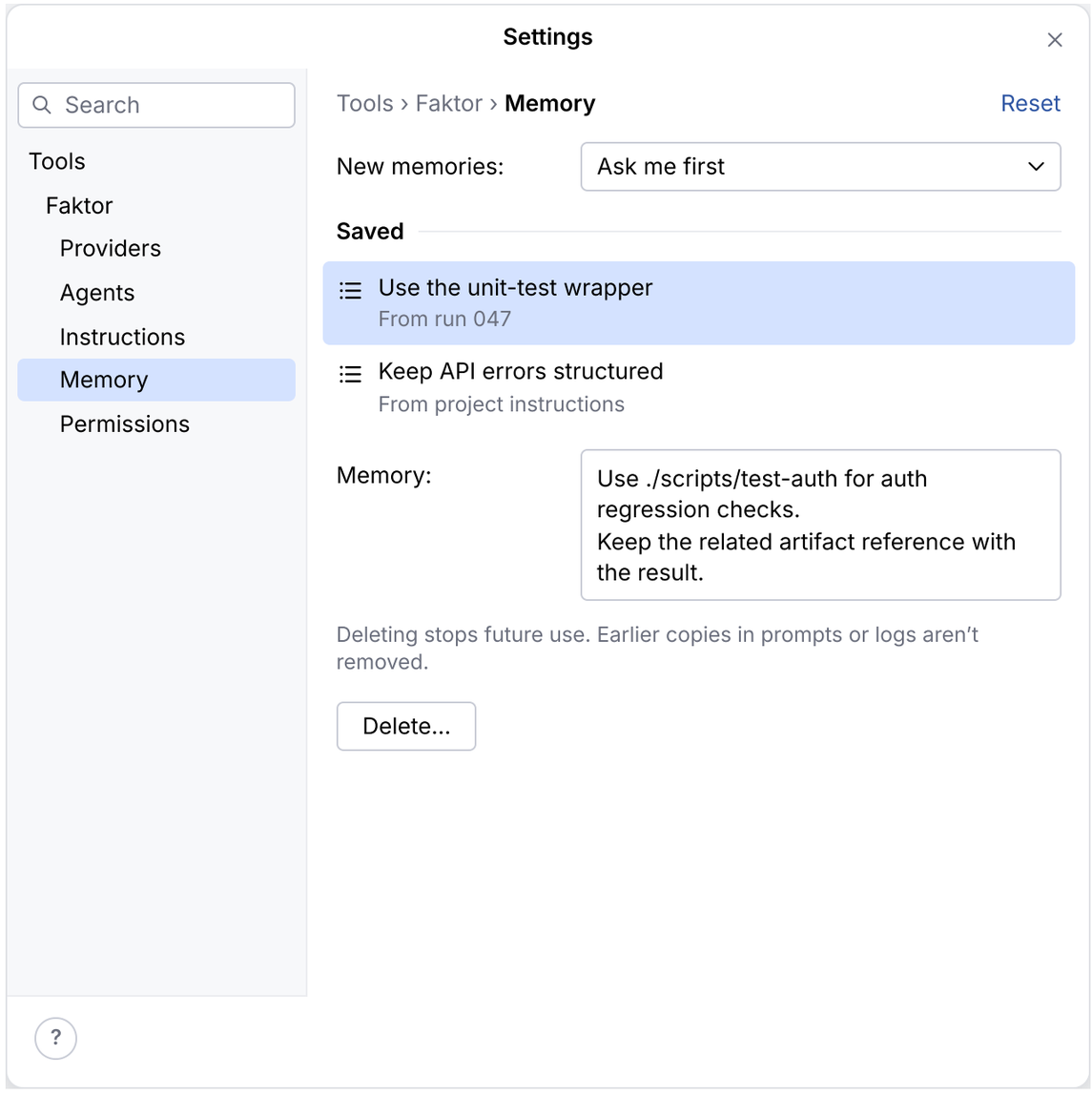
Save Memory · JetBrains: OK / Apply

Separate learned failure priors, user instructions and proposed personal/project memory. Each record has scope, source, date and deletion semantics.

VS CODE · DARK MODERN — EDITOR TAB · WEBVIEW SETTINGS



JETBRAINS · LIGHT — SETTINGS DIALOG · CONFIGURABLE



CONTROLS K-05 K-10 K-11 O-09 I-11

REPLACES R7 page 057 · Expanded standard-agent surface

CONTRACT p. 138

S12 · INSTRUCTIONS AND MEMORY · BEHAVIOUR CONTRACT

Memory: contract and revision notes

Original contract retained verbatim (Expanded standard-agent surface / controls: K-05 K-10 K-11 O-09 I-11). Design on p. 137.

What changed in revision 8

- Failure learning is not shown here, which removes the main source of confusion between the two stores.

Entry and return

Composer context picker/meter, attachment staging, memory or compaction command. Work / Inspect / History remain the only top-level clusters. Settings, Usage and Diagnostics use explicit Back. Return to the originating object, not the first row.

Scope and captured state

Draft/call receipt + exact source URI/range/version + token accounting basis; memory has origin and retention scope. Explicit inclusion acquires a versioned source snapshot. Do not replace an admitted source with the current editor or silently drop an unsupported attachment.

Primary action and recovery

Review a proposed memory diff before saving; open the originating evidence or delete the exact record after scope review.. Deleting a memory affects future use after acknowledgement, not copies in prior calls. A project cannot redirect global memory storage.

Loading, empty and stale

Measured input, draft estimate, model capacity and output/safety reserve use separate labels. null contextUsage is unavailable. Upload and compaction each retain original inputs.

Keyboard and focus

Picker arrows select sources; Enter stages the selected identity; Escape returns to the same composer caret. New usage observations do not interrupt typing.

Layout and adverse content

Wrap source labels and exact rejection reasons. Keep previews bounded with full artifact/copy actions. Do not fit a long source list between several competing scroll panes.

Starting point and required construction

Starting classifications: N, R. E existing operation; R runtime-backed adapter; N new contract; H host-only work. Exact per-control classifications remain in the catalog.

Reuse the reviewed domain operations and preserve all served fields. Build every missing native read/write adapter, capability gate and state renderer identified by the matching control/build rows.

BK04 owns accounting and compaction; BK03 owns versioned memory/customization; BK02 supplies model capabilities and BK08 owns upload handles.

Required interaction fixtures

Exercise first-read, empty/no-match, ready, refreshing/stale, error/refusal, action pending and acknowledged result. Also switch scope while the request is in flight.

Moved out of the interface

Every line of revision 7 plate copy that left the default view, and where it lives now.

"Faktor / Memory (R7 window title)"
→ Native host chrome (VS Code view title, tabs or editor tab; JetBrains tool-window tabs, dialog title or Settings breadcrumb)

"Project memory is separate from operational failure learning."
→ Failure learning lives in Efficiency (S28); rule in Specification: screen contract

"Capture policy / Manual review"
→ Field "New memories — Ask me first"

"Source: run 047 · verification artifact"
→ Row meta "From run 047"

"Source: project instructions · reviewed today"
→ Row meta; review date in Tooltip / hover help

"Selected memory"
→ Field "Memory"

"Scope / This project; Failure learning / Production ON"
→ Subtitle "This project"; failure learning on the Efficiency page

"Deleting a memory stops future use after acknowledgement. Prior prompt and artifact copies may remain."
→ Note (shortened)

"Delete selected"
→ Delete… (opens S66)

"Memory"
→ Title

"Use the unit-test wrapper / Keep API errors structured"
→ Saved rows

"Use ./scripts/test-auth for auth regression checks."
→ Memory field

"Save memory"
→ Save Memory

S66 INSTRUCTIONS AND MEMORY

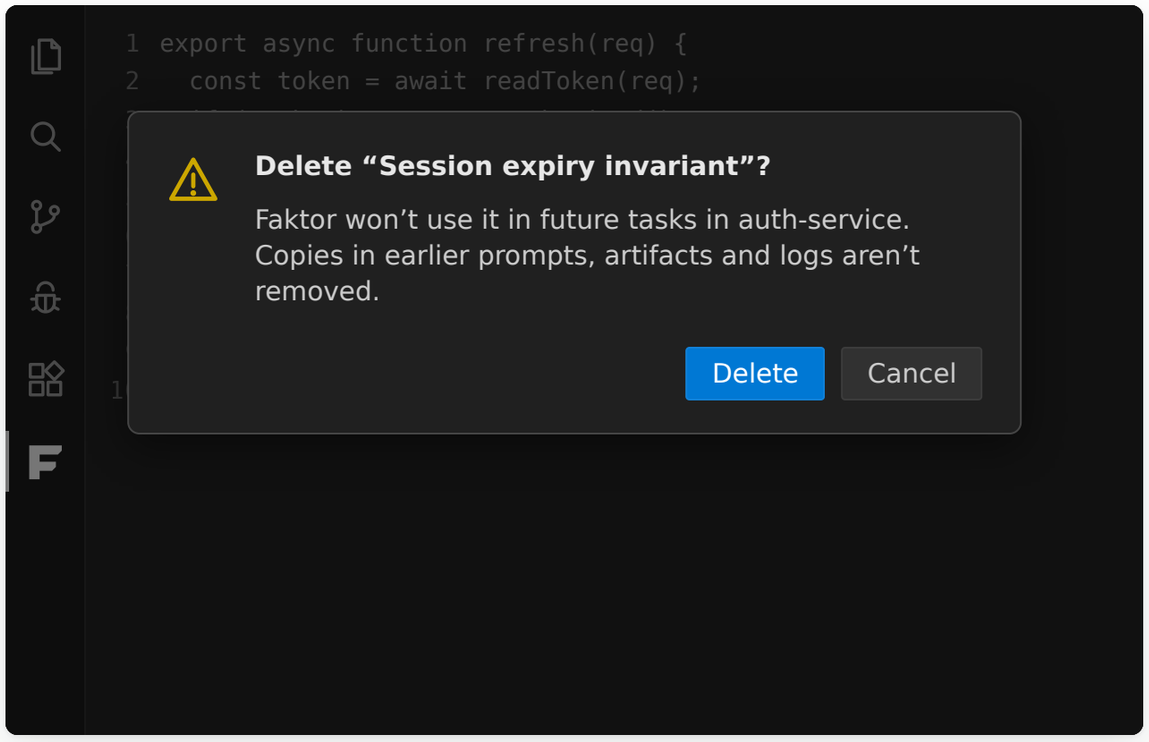
Memory deletion

PRIMARY ACTION

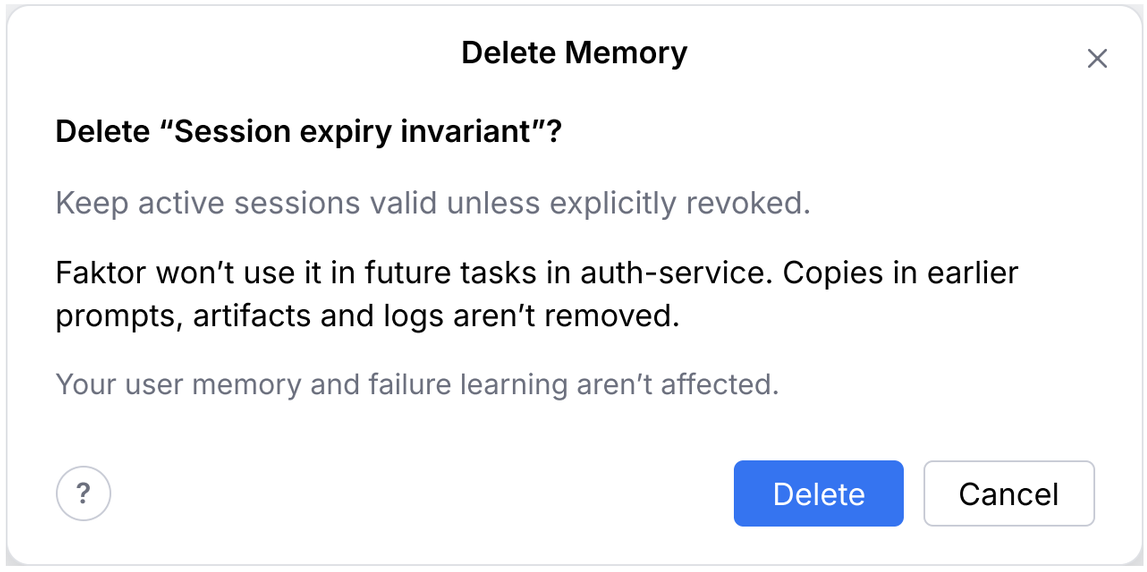
Delete

Delete one versioned memory from future retrieval while disclosing earlier prompts, artifacts and log copies.

VS CODE · DARK MODERN — MODAL MESSAGE · NATIVE



JETBRAINS · LIGHT — DIALOGWRAPPER



CONTROLS K-10 K-11

REPLACES R7 page 058 · Expanded standard-agent surface

CONTRACT p. 140

S66 · INSTRUCTIONS AND MEMORY · BEHAVIOUR CONTRACT

Memory deletion: contract and revision notes

Original contract retained verbatim (Expanded standard-agent surface / controls: K-10 K-11). Design on p. 139.

What changed in revision 8

- A native confirmation replaces a checkbox-plus-button pattern.

Entry and return

Composer context picker/meter, attachment staging, memory or compaction command. Work / Inspect / History remain the only top-level clusters. Settings, Usage and Diagnostics use explicit Back. Return to the originating object, not the first row.

Scope and captured state

Draft/call receipt + exact source URI/range/version + token accounting basis; memory has origin and retention scope. Explicit inclusion acquires a versioned source snapshot. Do not replace an admitted source with the current editor or silently drop an unsupported attachment.

Primary action and recovery

Delete project memory. Show unsupported type, oversize, upload error, capacity refusal and stale source separately. Remove only explicitly; failed compaction keeps the prior valid accounting basis.

Loading, empty and stale

Measured input, draft estimate, model capacity and output/safety reserve use separate labels. null contextUsage is unavailable. Upload and compaction each retain original inputs.

Keyboard and focus

Picker arrows select sources; Enter stages the selected identity; Escape returns to the same composer caret. New usage observations do not interrupt typing.

Layout and adverse content

Wrap source labels and exact rejection reasons. Keep previews bounded with full artifact/copy actions. Do not fit a long source list between several competing scroll panes.

Starting point and required construction

Starting classifications: N. E existing operation; R runtime-backed adapter; N new contract; H host-only work. Exact per-control classifications remain in the catalog.

NEW: add versioned project/personal memory CRUD and future-retrieval acknowledgement. Reuse no learned-failure store as generic memory.

BK04 owns accounting and compaction; BK03 owns versioned memory/customization; BK02 supplies model capabilities and BK08 owns upload handles.

Required interaction fixtures

Exercise first-read, empty/no-match, ready, refreshing/stale, error/refusal, action pending and acknowledged result. Also switch scope while the request is in flight.

Moved out of the interface

Every line of revision 7 plate copy that left the default view, and where it lives now.

"Faktor / Settings / Memory (R7 window title)"
→ Native host chrome (VS Code view title, tabs or editor tab; JetBrains tool-window tabs, dialog title or Settings breadcrumb)

"Delete this memory? / Project memory m-18 · current source version 4"
→ Question names the memory; ID and version in Tooltip / hover help

"Earlier copies can remain / Deletion removes future retrieval after acknowledgement. It does not erase earlier prompts, artifacts or logs."
→ Detail text

"Remove from future project retrieval"
→ Removed: restated by the layout or label (the dialog is the confirmation)

"This affects only m-18 in auth-service. User memory and learned failure records are separate."
→ Detail and note

"Delete project memory"
→ Delete

"Session expiry invariant"
→ Dialog question

"Keep active sessions valid unless explicitly revoked."
→ Memory text

"Cancel"
→ Cancel

PART 2 · SCREENS

2.10 Permissions and tools

9 screens. Each has a design spread for both IDEs and a contract page.

ID	SCREEN	JOB	PAGE
S13	**Permissions**	Explain effective permissions by tool, action, target and scope. Show persistent rules separately from current pending approvals and actual containment.	p. 142
S15	**Tools & MCP**	Group built-in tools and MCP servers by origin, availability and active permissions. Make discovered, enabled, healthy and authorized independently visible.	p. 144
S16	**Add MCP server**	Configure a supervised stdio server with exact command/args and explicit trust. Remote transport is offered only after capability negotiation.	p. 146
S68	**Future MCP authentication**	Specify future remote MCP authentication with exact origin, scopes, token storage and cancellation behavior.	p. 148
S17	**Skills & commands**	Discover reusable workflows without confusing installation with permission grants. Commands say whether they insert text, start a task or execute a bounded action.	p. 150
S18	**Lifecycle hook**	Edit exact native lifecycle events, executable arguments, environment, bounds and failure policy. Test under the same capability envelope.	p. 152
S67	**Hook test result**	Inspect a real bounded hook test, deadline, output and verdict without converting successful output into authority.	p. 154
S19	**Terminal**	An owned process table and bounded output coexist with the task transcript. Terminal kill, run stop and session abort are separate actions.	p. 156
S20	**Browser**	Integrate host diagnostics and explicit code actions. Browser controls require an actual advertised adapter with account, machine and network scope.	p. 158

S13 PERMISSIONS AND TOOLS

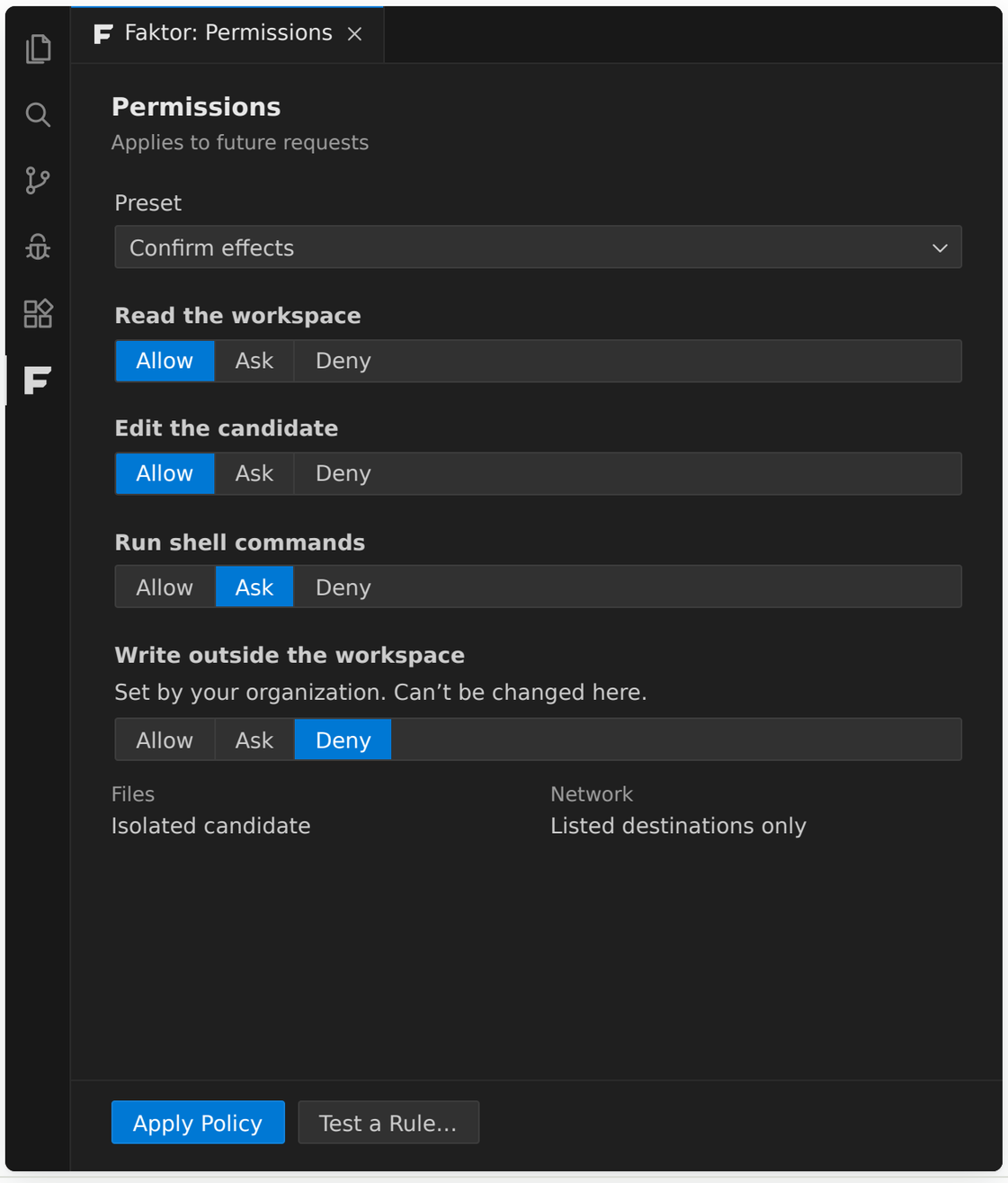
Permissions

PRIMARY ACTION

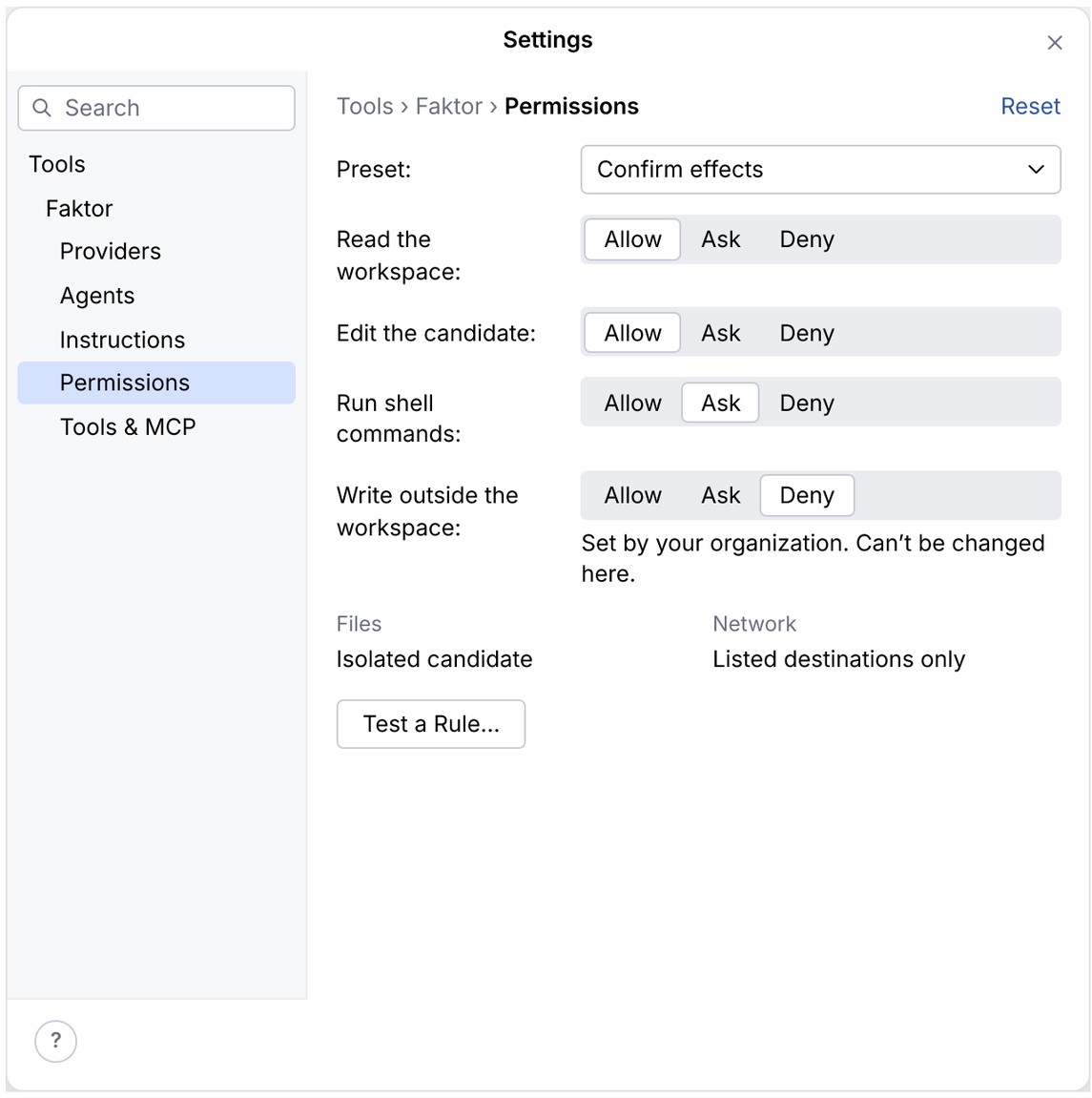
Apply Policy · JetBrains: OK / Apply

Explain effective permissions by tool, action, target and scope. Show persistent rules separately from current pending approvals and actual containment.

VS CODE · DARK MODERN — EDITOR TAB · WEBVIEW SETTINGS



JETBRAINS · LIGHT — SETTINGS DIALOG · CONFIGURABLE



CONTROLS Q-01 Q-02 Q-03 Q-04 Q-05 Q-07 Q-08 Q-09 Q-10 Q-11 Q-12 Q-13 Q-14

REPLACES R7 page 059 · Expanded standard-agent surface

CONTRACT p. 143

S13 · PERMISSIONS AND TOOLS · BEHAVIOUR CONTRACT

Permissions: contract and revision notes

Original contract retained verbatim (Expanded standard-agent surface / controls: Q-01 Q-02 Q-03 Q-04 Q-05 Q-07 Q-08 Q-09 Q-10 Q-11 Q-12 Q-13 Q-14). Design on p. 142.

What changed in revision 8

- Each permission is a labelled single-choice control; the locked one says who set it.

Entry and return

Native Settings or the originating configuration command. Work / Inspect / History remain the only top-level clusters. Settings, Usage and Diagnostics use explicit Back. Return to the originating object, not the first row.

Scope and captured state

Application/project/connection plus effective revision; never infer task scope from an open form. Keep staged input, the effective snapshot and the acknowledged applied revision separate. Apply is atomic; Cancel preserves the previous effective configuration.

Primary action and recovery

Preview a rule match against a sample operation, then apply a versioned policy to future dispatch after daemon acknowledgement.. No policy store means no Remember rule. Revocation reports future denial and already-issued effects separately; sandbox failure remains visible.

Loading, empty and stale

Keep the last coherent values with a refresh state. A first read has no invented defaults. An absent capability remains explained and inspectable.

Keyboard and focus

Open on the triggering setting. Apply failure focuses the first invalid field on submission; async refresh never moves the caret. Escape cancels a transient editor and returns to its trigger.

Layout and adverse content

Align only related fields within their group. Long inputs use stacked labels. Native preferred sizes and zoom win over reference-art dimensions.

Starting point and required construction

Starting classifications: N, R. E existing operation; R runtime-backed adapter; N new contract; H host-only work. Exact per-control classifications remain in the catalog.

Reuse the reviewed domain operations and preserve all served fields. Build every missing native read/write adapter, capability gate and state renderer identified by the matching control/build rows.

BK01 owns validation, scope, persistence, revision comparison and acknowledgement; domain packages own parameter/capability enforcement.

Required interaction fixtures

Exercise first-read, empty/no-match, ready, refreshing/stale, error/refusal, action pending and acknowledged result. Also switch scope while the request is in flight.

Moved out of the interface

Every line of revision 7 plate copy that left the default view, and where it lives now.

"Faktor / Permissions (R7 window title)"
→ Native host chrome (VS Code view title, tabs or editor tab; JetBrains tool-window tabs, dialog title or Settings breadcrumb)

"Persistent rules apply to future matching tool dispatch."
→ Subtitle "Applies to future requests"

"Policy preset"
→ Label "Preset"

"Effective tool policy"
→ Removed: restated by the layout or label

"Workspace read / Candidate edits / Shell commands / External writes"
→ Plain-language rows

"Filesystem / Isolated candidate; Network / Explicit destinations"
→ Facts "Files" and "Network"

"Managed deny remains locked / An instruction, preset or local rule cannot relax this constraint."
→ Help on the locked row

"Test rule / Apply policy"
→ Test a Rule… / Apply Policy

"Permissions"
→ Title

"Allow / Ask / Deny"
→ Segmented choices

"Confirm effects"
→ Preset value

S15 PERMISSIONS AND TOOLS

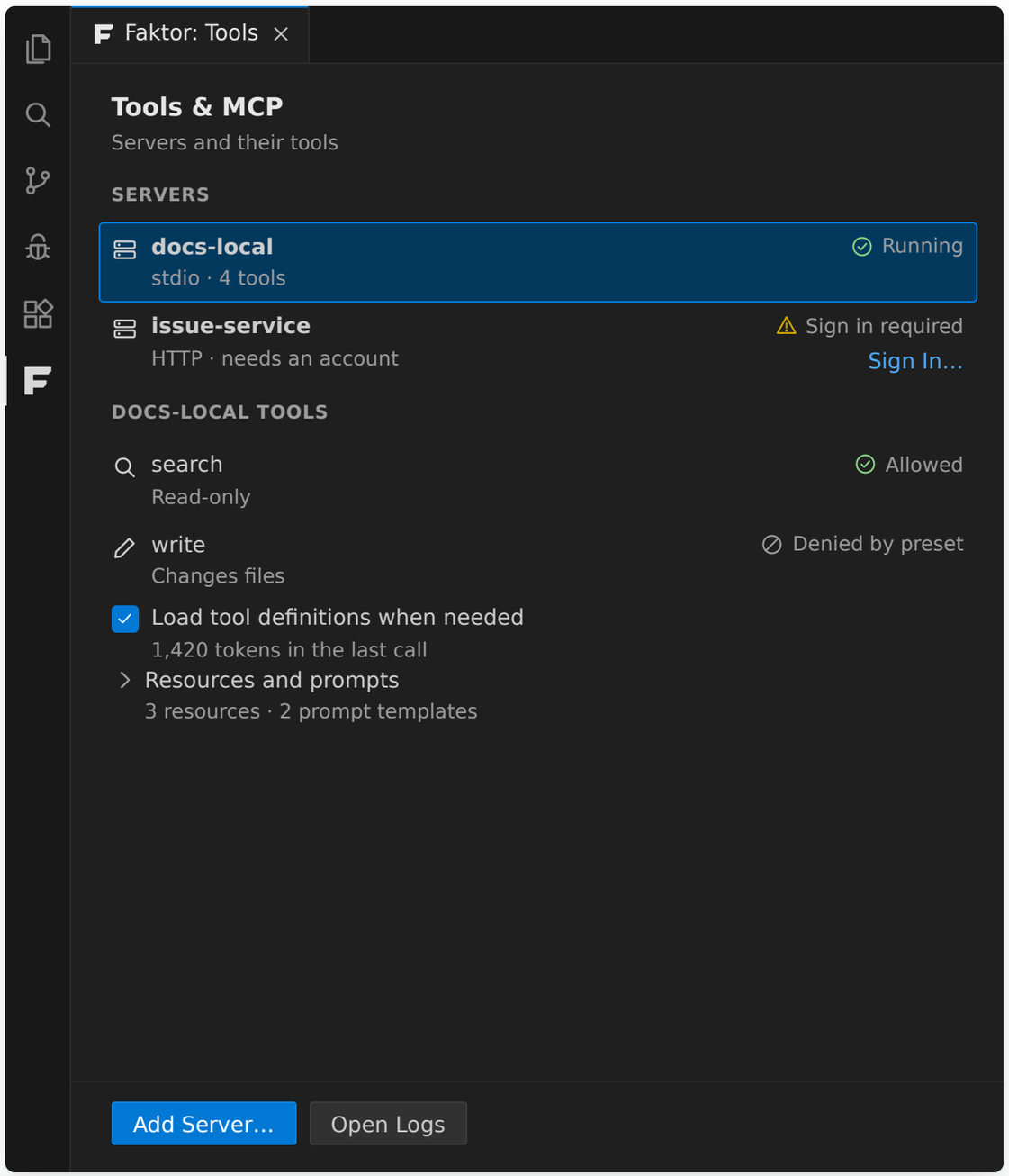
Tools & MCP

PRIMARY ACTION

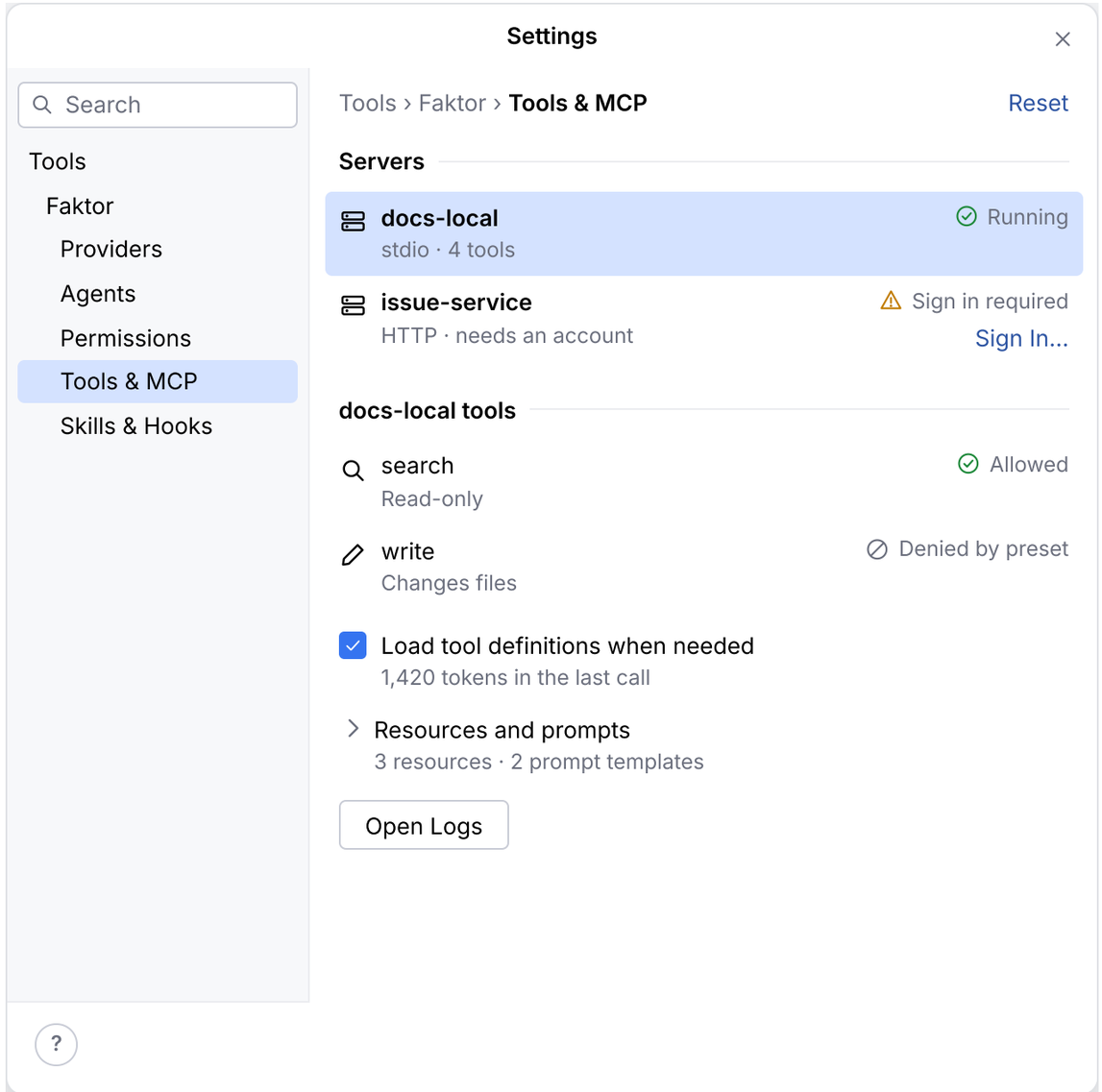
Add Server… · JetBrains: OK / Apply

Group built-in tools and MCP servers by origin, availability and active permissions. Make discovered, enabled, healthy and authorized independently visible.

VS CODE · DARK MODERN — EDITOR TAB · WEBVIEW SETTINGS



JETBRAINS · LIGHT — SETTINGS DIALOG · CONFIGURABLE



CONTROLS T-01 T-02 T-03 T-04 T-06 T-07 T-08 T-09 T-10 T-11 T-12 T-13 T-15

REPLACES R7 page 060 · Expanded standard-agent surface

CONTRACT p. 145

S15 · PERMISSIONS AND TOOLS · BEHAVIOUR CONTRACT

Tools & MCP: contract and revision notes

Original contract retained verbatim (Expanded standard-agent surface / controls: T-01 T-02 T-03 T-04 T-06 T-07 T-08 T-09 T-10 T-11 T-12 T-13 T-15). Design on p. 144.

What changed in revision 8

- Health (Running) and permission (Allowed/Denied) use different words so they can't be confused.

Entry and return

Native Settings or the originating configuration command. Work / Inspect / History remain the only top-level clusters. Settings, Usage and Diagnostics use explicit Back. Return to the originating object, not the first row.

Scope and captured state

Application/project/connection plus effective revision; never infer task scope from an open form. Keep staged input, the effective snapshot and the acknowledged applied revision separate. Apply is atomic; Cancel preserves the previous effective configuration.

Primary action and recovery

Open a server to inspect its tool schemas, permission defaults, output bounds and logs; reconnect only through owned lifecycle control.. Authentication required, failed discovery and stopped process are different states. Unknown remote effects never become Cancelled merely on disconnect.

Loading, empty and stale

Keep the last coherent values with a refresh state. A first read has no invented defaults. An absent capability remains explained and inspectable.

Keyboard and focus

Open on the triggering setting. Apply failure focuses the first invalid field on submission; async refresh never moves the caret. Escape cancels a transient editor and returns to its trigger.

Layout and adverse content

Align only related fields within their group. Long inputs use stacked labels. Native preferred sizes and zoom win over reference-art dimensions.

Starting point and required construction

Starting classifications: H, N, R. E existing operation; R runtime-backed adapter; N new contract; H host-only work. Exact per-control classifications remain in the catalog.

Reuse the reviewed domain operations and preserve all served fields. Build every missing native read/write adapter, capability gate and state renderer identified by the matching control/build rows.

BK01 owns validation, scope, persistence, revision comparison and acknowledgement; domain packages own parameter/capability enforcement.

Required interaction fixtures

Exercise first-read, empty/no-match, ready, refreshing/stale, error/refusal, action pending and acknowledged result. Also switch scope while the request is in flight.

Moved out of the interface

Every line of revision 7 plate copy that left the default view, and where it lives now.

"Faktor / Tools & MCP (R7 window title)"
→ Native host chrome (VS Code view title, tabs or editor tab; JetBrains tool-window tabs, dialog title or Settings breadcrumb)

"Connection health, enablement and permission are separate states."
→ Each row shows one state in words; rule in Specification: screen contract

"MCP servers"
→ Section "Servers"

"Ready / stdio · trusted machine configuration · 4 tools"
→ Row; trust source in Tooltip / hover help

"Sign in / HTTP · account binding required"
→ Row "Sign in required" + Sign In…

"Allow / Read-only · schema and output bounds available"
→ Row "Allowed"; bounds in Tooltip / hover help

"Deny / Effectful tool · disabled in this preset"
→ Row "Denied by preset"

"Load definitions on demand / 1,420 definition tokens observed in the current call."
→ Checkbox and help

"Resources & prompts / 3 resources · 2 prompt templates · preview before use"
→ Disclosure; preview rule in Specification: screen contract

"Tools & MCP"
→ Title

"docs-local / issue-service"
→ Server rows

"docs-local tools / search / write"
→ Tool rows

"Open logs / Add MCP server"
→ Open Logs / Add Server…

S16 PERMISSIONS AND TOOLS

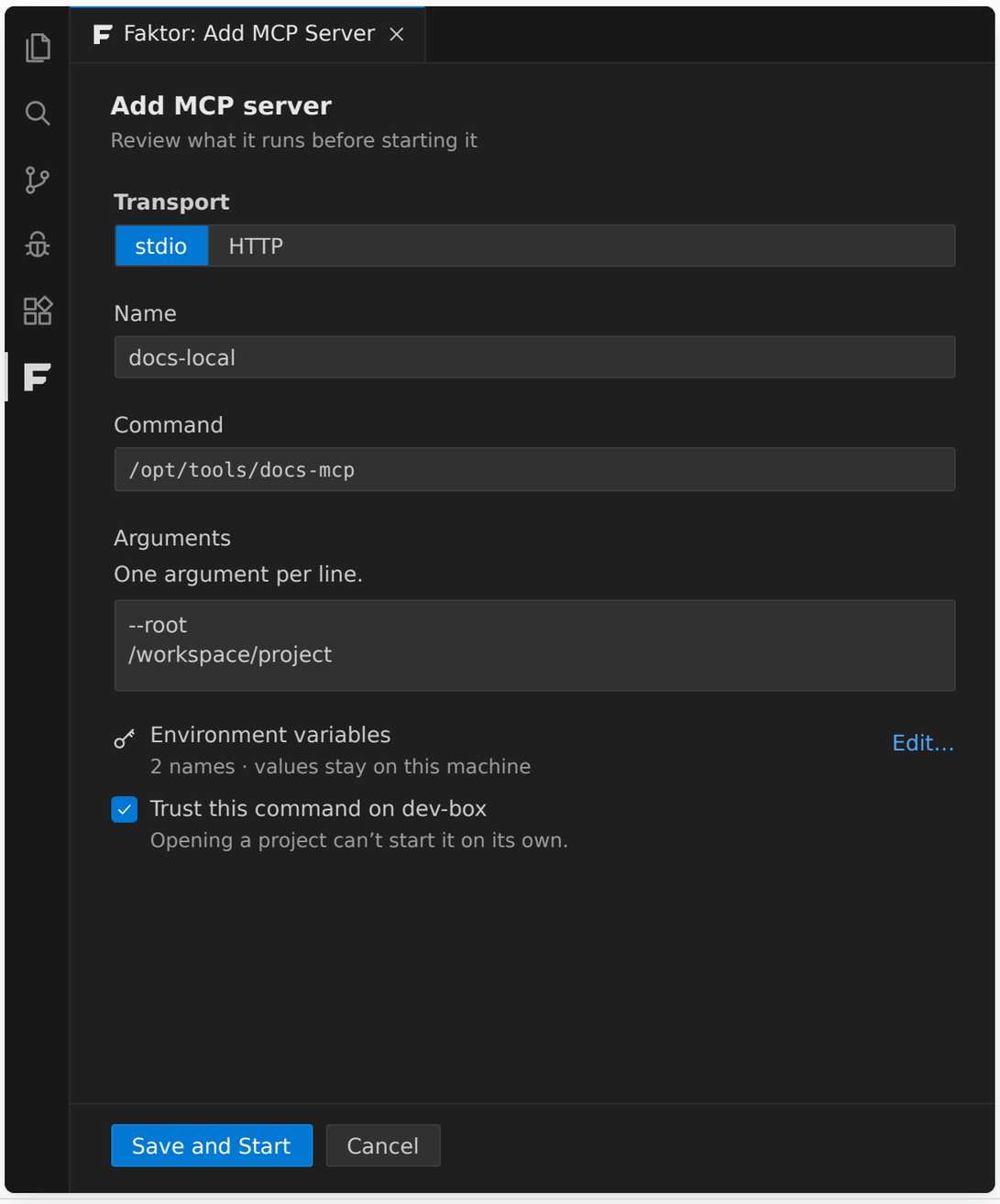
PRIMARY ACTION

Save and Start

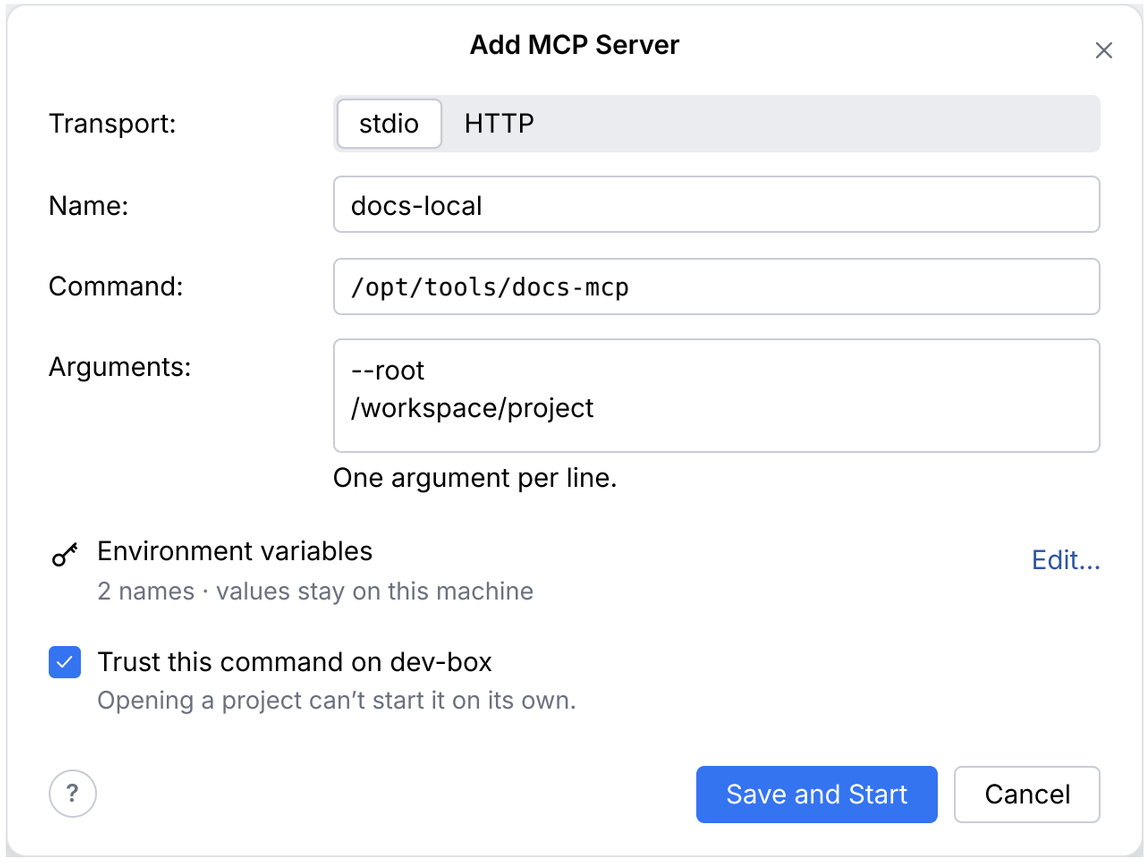
Add MCP server

Configure a supervised stdio server with exact command/args and explicit trust. Remote transport is offered only after capability negotiation.

VS CODE · DARK MODERN — EDITOR TAB · WEBVIEW



JETBRAINS · LIGHT — DIALOGWRAPPER



CONTROLS T-03 T-04 T-05 T-06 T-10 T-13 T-14 T-15

REPLACES R7 page 061 · Expanded standard-agent surface

CONTRACT p. 147

S16 · PERMISSIONS AND TOOLS · BEHAVIOUR CONTRACT

Add MCP server: contract and revision notes

Original contract retained verbatim (Expanded standard-agent surface / controls: T-03 T-04 T-05 T-06 T-10 T-13 T-14 T-15). Design on p. 146.

What changed in revision 8

- Remote transport stays visible but follows capability negotiation (contract unchanged).

Entry and return

Native Settings or the originating configuration command. Work / Inspect / History remain the only top-level clusters. Settings, Usage and Diagnostics use explicit Back. Return to the originating object, not the first row.

Scope and captured state

Application/project/connection plus effective revision; never infer task scope from an open form. Keep staged input, the effective snapshot and the acknowledged applied revision separate. Apply is atomic; Cancel preserves the previous effective configuration.

Primary action and recovery

Validate size/count bounds, review executable host and source, then save and explicitly start the trusted server.. A repository-supplied executable never starts merely because a config file exists. Invalid args or a failed start preserve the previous valid server.

Loading, empty and stale

Keep the last coherent values with a refresh state. A first read has no invented defaults. An absent capability remains explained and inspectable.

Keyboard and focus

Open on the triggering setting. Apply failure focuses the first invalid field on submission; async refresh never moves the caret. Escape cancels a transient editor and returns to its trigger.

Layout and adverse content

Align only related fields within their group. Long inputs use stacked labels. Native preferred sizes and zoom win over reference-art dimensions.

Starting point and required construction

Starting classifications: H, N, R. E existing operation; R runtime-backed adapter; N new contract; H host-only work. Exact per-control classifications remain in the catalog.

Reuse the reviewed domain operations and preserve all served fields. Build every missing native read/write adapter, capability gate and state renderer identified by the matching control/build rows.

BK01 owns validation, scope, persistence, revision comparison and acknowledgement; domain packages own parameter/capability enforcement.

Required interaction fixtures

Exercise first-read, empty/no-match, ready, refreshing/stale, error/refusal, action pending and acknowledged result. Also switch scope while the request is in flight.

Moved out of the interface

Every line of revision 7 plate copy that left the default view, and where it lives now.

"Faktor / Add MCP server (R7 window title)"
→ Native host chrome (VS Code view title, tabs or editor tab; JetBrains tool-window tabs, dialog title or Settings breadcrumb)

"Review origin and execution scope before starting a server."
→ Subtitle

"Server name"
→ Label "Name"

"Executable"
→ Label "Command"

"Each line is one argument. Limits are checked as UTF-8 bytes."
→ Help "One argument per line."; byte limits shown only as a validation error

"Environment references / 2 entries / Explicit names only. Secret values are resolved on the machine."
→ Row "Environment variables — 2 names · values stay on this machine"

"Trust this executable on dev-box / Project content cannot start it merely by being opened."
→ Checkbox and help (shortened)

"Save & start"
→ Save and Start

"Add MCP server"
→ Title

"stdio / HTTP"
→ Segmented transport

"docs-local / /opt/tools/docs-mcp / --root / /workspace/project"
→ Field values

"Cancel"
→ Cancel

S68 PERMISSIONS AND TOOLS

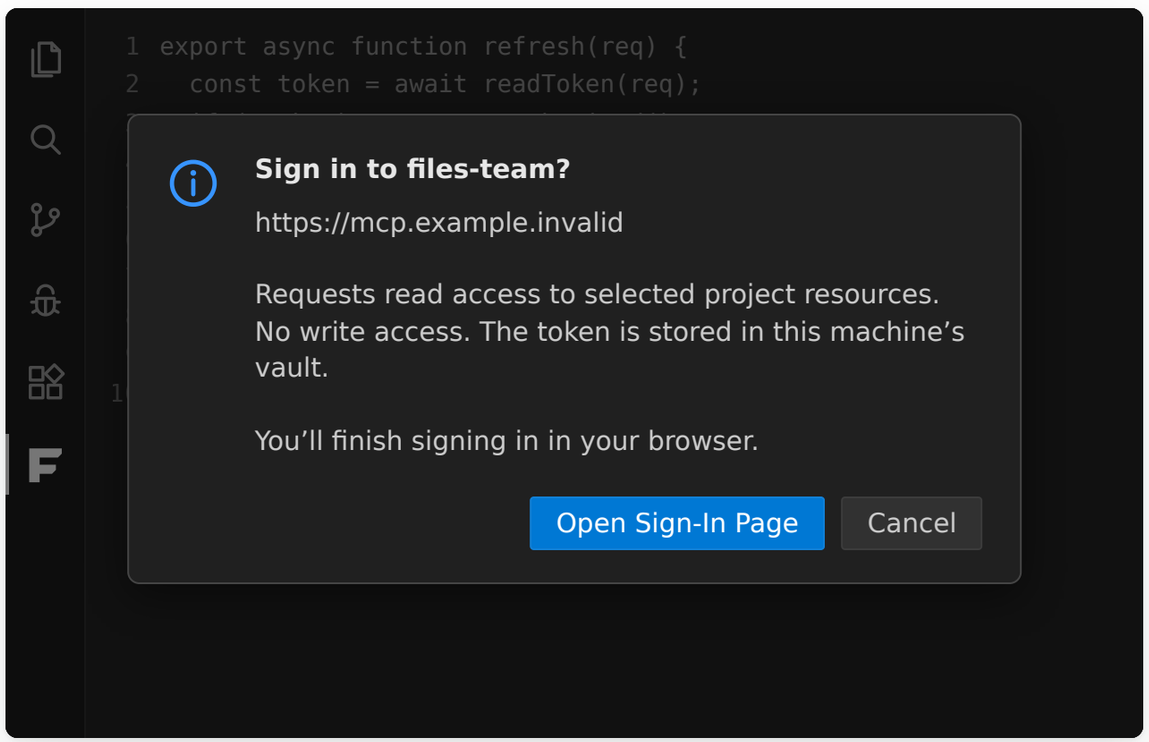
Future MCP authentication

Specify future remote MCP authentication with exact origin, scopes, token storage and cancellation behavior.

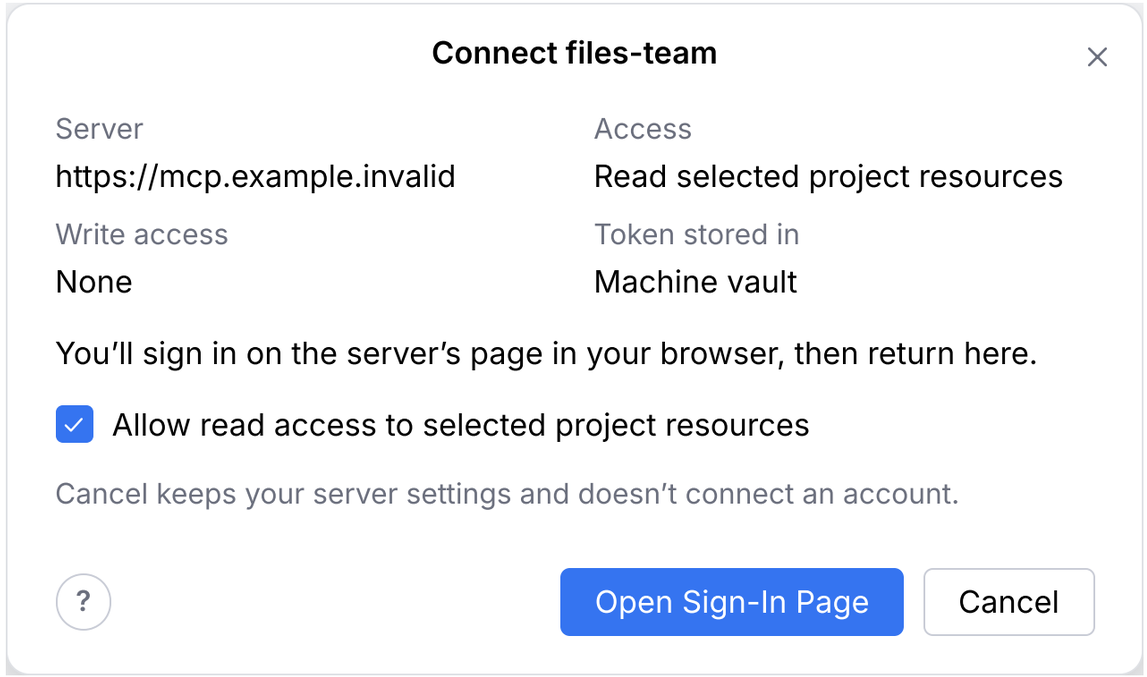
PRIMARY ACTION

Open Sign-In Page

VS CODE · DARK MODERN — MODAL MESSAGE · NATIVE



JETBRAINS · LIGHT — DIALOGWRAPPER



CONTROLS T-04 T-06

REPLACES R7 page 062 · Expanded standard-agent surface

CONTRACT p. 149

S68 · PERMISSIONS AND TOOLS · BEHAVIOUR CONTRACT

Future MCP authentication: contract and revision notes

Original contract retained verbatim (Expanded standard-agent surface / controls: T-04 T-06). Design on p. 148.

What changed in revision 8

- Exact origin and scopes stay visible: they are the decision.

Entry and return

Native Settings or the originating configuration command. Work / Inspect / History remain the only top-level clusters. Settings, Usage and Diagnostics use explicit Back. Return to the originating object, not the first row.

Scope and captured state

Application/project/connection plus effective revision; never infer task scope from an open form. Keep staged input, the effective snapshot and the acknowledged applied revision separate. Apply is atomic; Cancel preserves the previous effective configuration.

Primary action and recovery

Open sign-in page. Associate validation with the exact field. A revision conflict preserves all staged values and opens the conflict resolver. Authentication, transport and schema failures have different recovery.

Loading, empty and stale

Keep the last coherent values with a refresh state. A first read has no invented defaults. An absent capability remains explained and inspectable.

Keyboard and focus

Open on the triggering setting. Apply failure focuses the first invalid field on submission; async refresh never moves the caret. Escape cancels a transient editor and returns to its trigger.

Layout and adverse content

Align only related fields within their group. Long inputs use stacked labels. Native preferred sizes and zoom win over reference-art dimensions.

Starting point and required construction

Starting classifications: N. E existing operation; R runtime-backed adapter; N new contract; H host-only work. Exact per-control classifications remain in the catalog.

NEW: reviewed current MCP config is stdio name/command/args. Build remote discovery/transport, exact-origin OAuth, state/PKCE, secret storage and revocation before enabling this future flow.

BK01 owns validation, scope, persistence, revision comparison and acknowledgement; domain packages own parameter/capability enforcement.

Required interaction fixtures

Exercise first-read, empty/no-match, ready, refreshing/stale, error/refusal, action pending and acknowledged result. Also switch scope while the request is in flight.

Moved out of the interface

Every line of revision 7 plate copy that left the default view, and where it lives now.

"Faktor / Tools / MCP / Account (R7 window title)"
→ Native host chrome (VS Code view title, tabs or editor tab; JetBrains tool-window tabs, dialog title or Settings breadcrumb)

"Connect MCP account / files-team · sign in with the reviewed server"
→ Dialog title and question

"Requested server / https://mcp.example.invalid · exact reviewed origin"
→ Fact "Server"

"Requested scopes / Read selected project resources · no write scope"
→ Facts "Access" and "Write access"

"Credential handling / Machine vault · separate organization binding"
→ Fact "Token stored in"; binding in Tooltip / hover help

"Continue in your browser / Review the account and requested access on the server's sign-in page. Return here after signing in. Cancelling keeps the server draft and does not connect an account."
→ Two short sentences

"No write scope is requested."
→ Fact "Write access — None"

"Allow access to selected project resources"
→ Checkbox "Allow read access to selected project resources"

"Cancel / Open sign-in page"
→ Cancel / Open Sign-In Page

S17 PERMISSIONS AND TOOLS

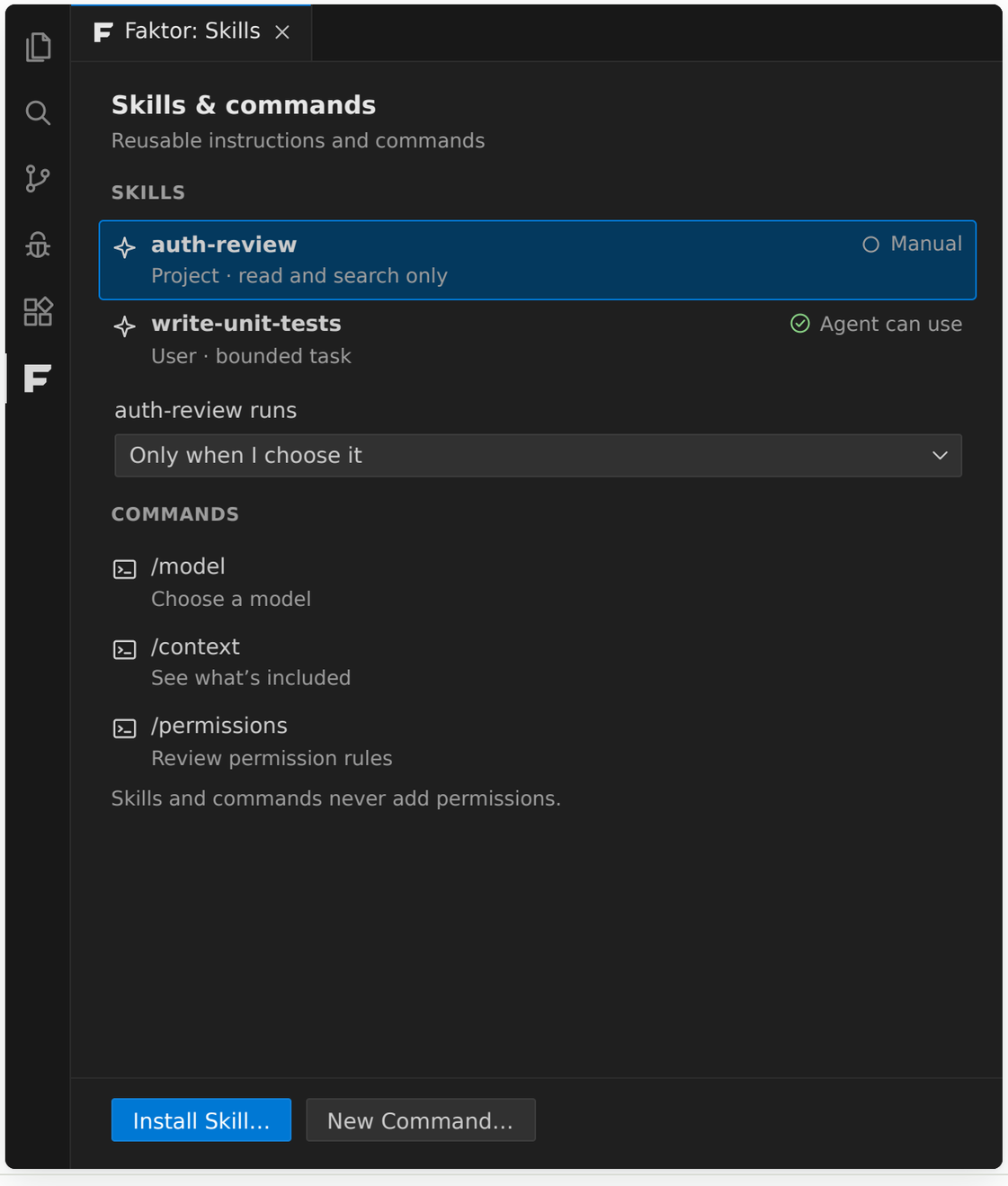
Skills & commands

PRIMARY ACTION

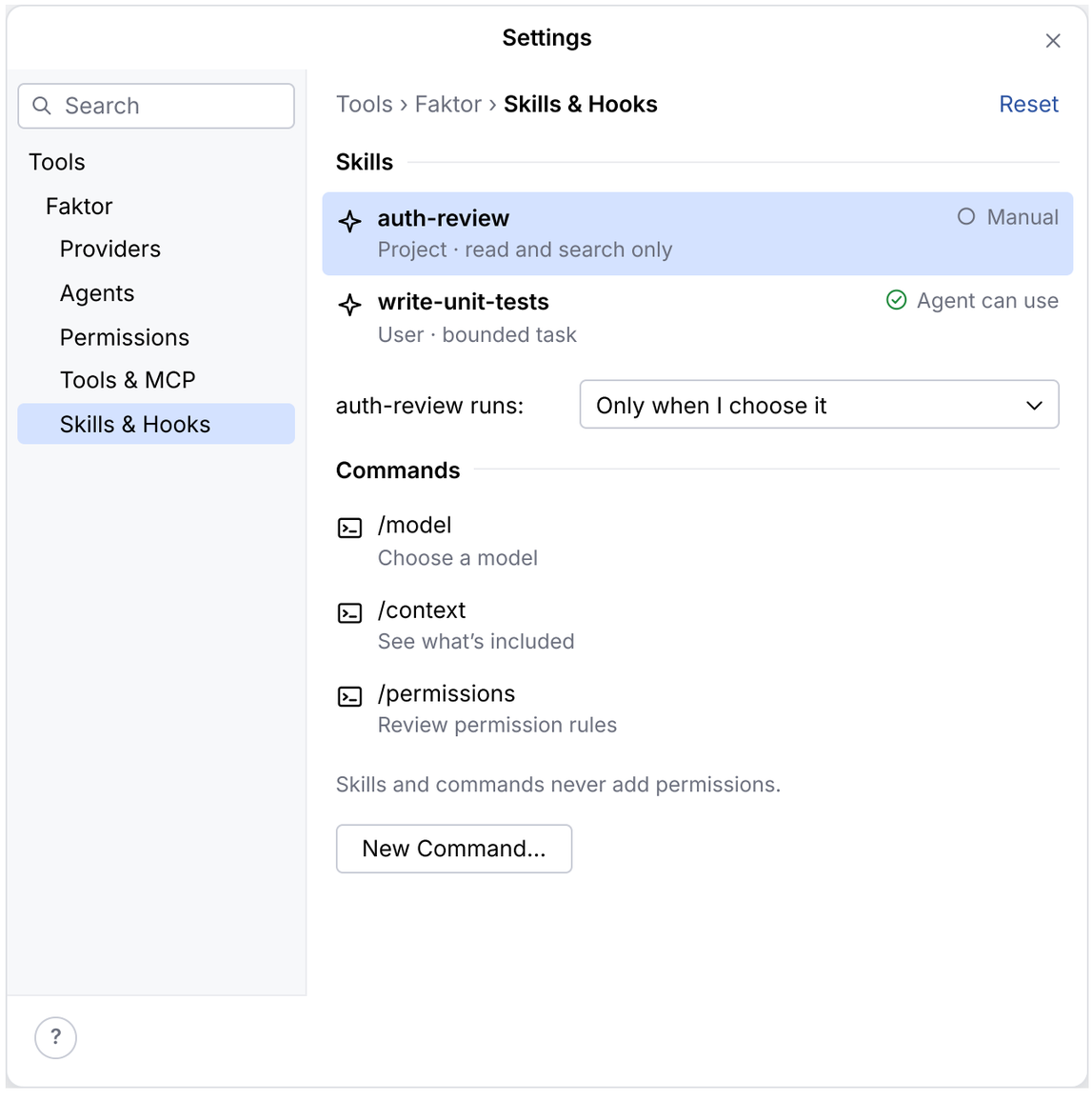
Install Skill... · JetBrains: OK / Apply

Discover reusable workflows without confusing installation with permission grants. Commands say whether they insert text, start a task or execute a bounded action.

VS CODE · DARK MODERN — EDITOR TAB · WEBVIEW SETTINGS



JETBRAINS · LIGHT — SETTINGS DIALOG · CONFIGURABLE



CONTROLS S-01 S-02 S-03 S-04 S-05 S-06 S-14 I-06

REPLACES R7 page 063 · Expanded standard-agent surface

CONTRACT p. 151

S17 · PERMISSIONS AND TOOLS · BEHAVIOUR CONTRACT

Skills & commands: contract and revision notes

Original contract retained verbatim (Expanded standard-agent surface / controls: S-01 S-02 S-03 S-04 S-05 S-06 S-14 I-06). Design on p. 150.

What changed in revision 8

- Commands are a plain list; whether each inserts text or runs a task shows on hover (Tooltip / hover help).

Entry and return

Native Settings or the originating configuration command. Work / Inspect / History remain the only top-level clusters. Settings, Usage and Diagnostics use explicit Back. Return to the originating object, not the first row.

Scope and captured state

Application/project/connection plus effective revision; never infer task scope from an open form. Keep staged input, the effective snapshot and the acknowledged applied revision separate. Apply is atomic; Cancel preserves the previous effective configuration.

Primary action and recovery

Open a skill manifest and version diff before enabling; insert a command into a draft without unexpectedly sending it.. Unavailable dependencies or a failed update keep the previous known-good version. Disabling a skill removes both discovery and agent invocation.

Loading, empty and stale

Keep the last coherent values with a refresh state. A first read has no invented defaults. An absent capability remains explained and inspectable.

Keyboard and focus

Open on the triggering setting. Apply failure focuses the first invalid field on submission; async refresh never moves the caret. Escape cancels a transient editor and returns to its trigger.

Layout and adverse content

Align only related fields within their group. Long inputs use stacked labels. Native preferred sizes and zoom win over reference-art dimensions.

Starting point and required construction

Starting classifications: H, N. E existing operation; R runtime-backed adapter; N new contract; H host-only work. Exact per-control classifications remain in the catalog.

Reuse the reviewed domain operations and preserve all served fields. Build every missing native read/write adapter, capability gate and state renderer identified by the matching control/build rows.

BK01 owns validation, scope, persistence, revision comparison and acknowledgement; domain packages own parameter/capability enforcement.

Required interaction fixtures

Exercise first-read, empty/no-match, ready, refreshing/stale, error/refusal, action pending and acknowledged result. Also switch scope while the request is in flight.

Moved out of the interface

Every line of revision 7 plate copy that left the default view, and where it lives now.

"Faktor / Skills & commands (R7 window title)"
→ Native host chrome (VS Code view title, tabs or editor tab; JetBrains tool-window tabs, dialog title or Settings breadcrumb)

"Reusable instructions keep their own origin and version."
→ Subtitle; version in Tooltip / hover help

"Installed skills"
→ Section "Skills"

"Project · version 3 · read and search only"
→ Row meta

"User · version 2 · bounded task invocation"
→ Row meta

"Selected invocation / Manual only"
→ Field "auth-review runs — Only when I choose it"

"Command palette"
→ Section "Commands"

"Enable for this project"
→ Skill row status

"Installation and invocation never grant extra tool authority."
→ Note "Skills and commands never add permissions."

"Skills & commands"
→ Title

"auth-review / write-unit-tests"
→ Skill rows

"Manual / Agent can use"
→ Statuses

"/model Choose exact model"
→ Command row "/model — Choose a model"

"/context Inspect included sources"
→ Command row "/context — See what's included"

"/permissions Review effective rules"
→ Command row "/permissions — Review permission rules"

"New command / Install skill"
→ New Command… / Install Skill…

S18 PERMISSIONS AND TOOLS

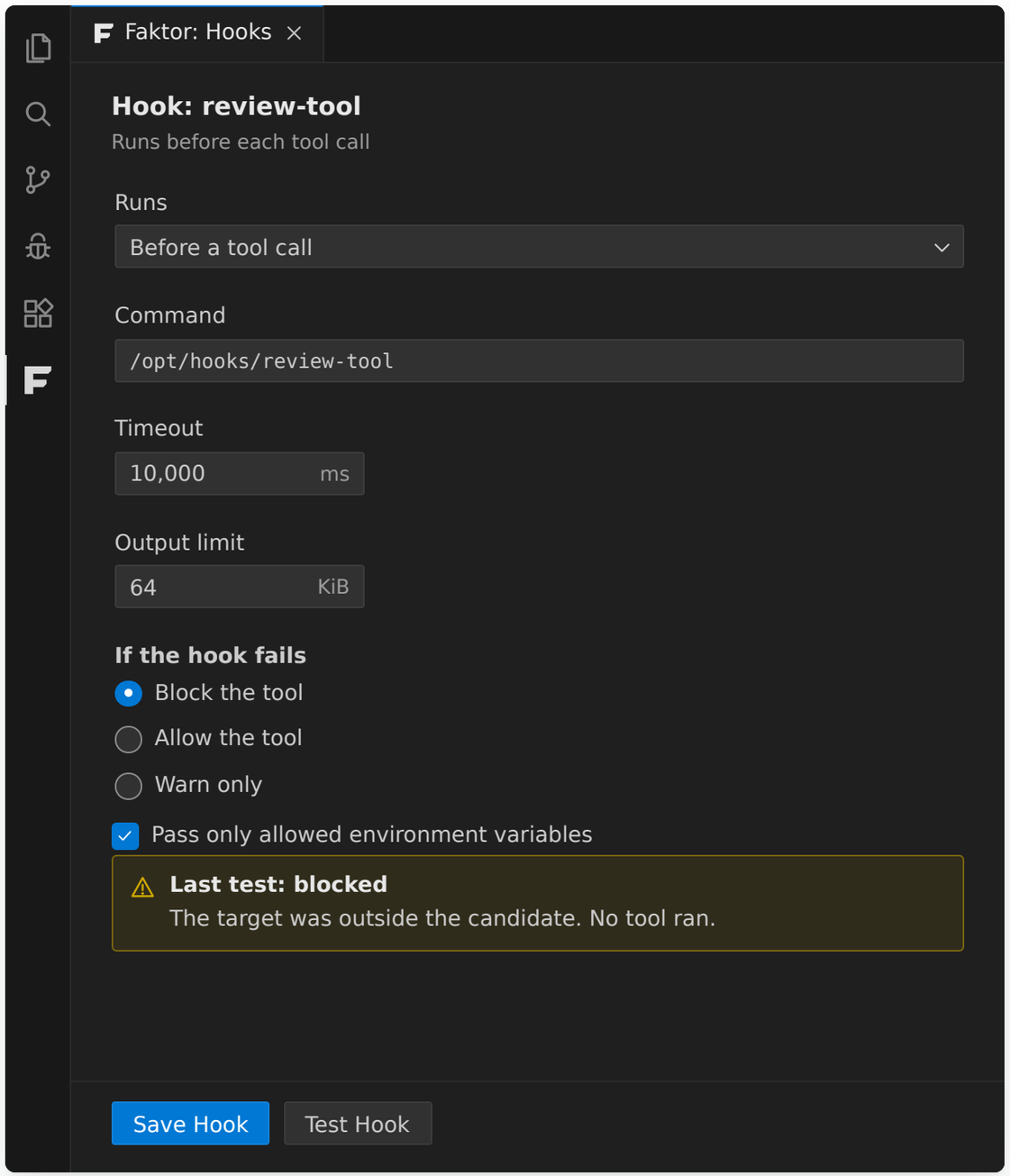
Lifecycle hook

PRIMARY ACTION

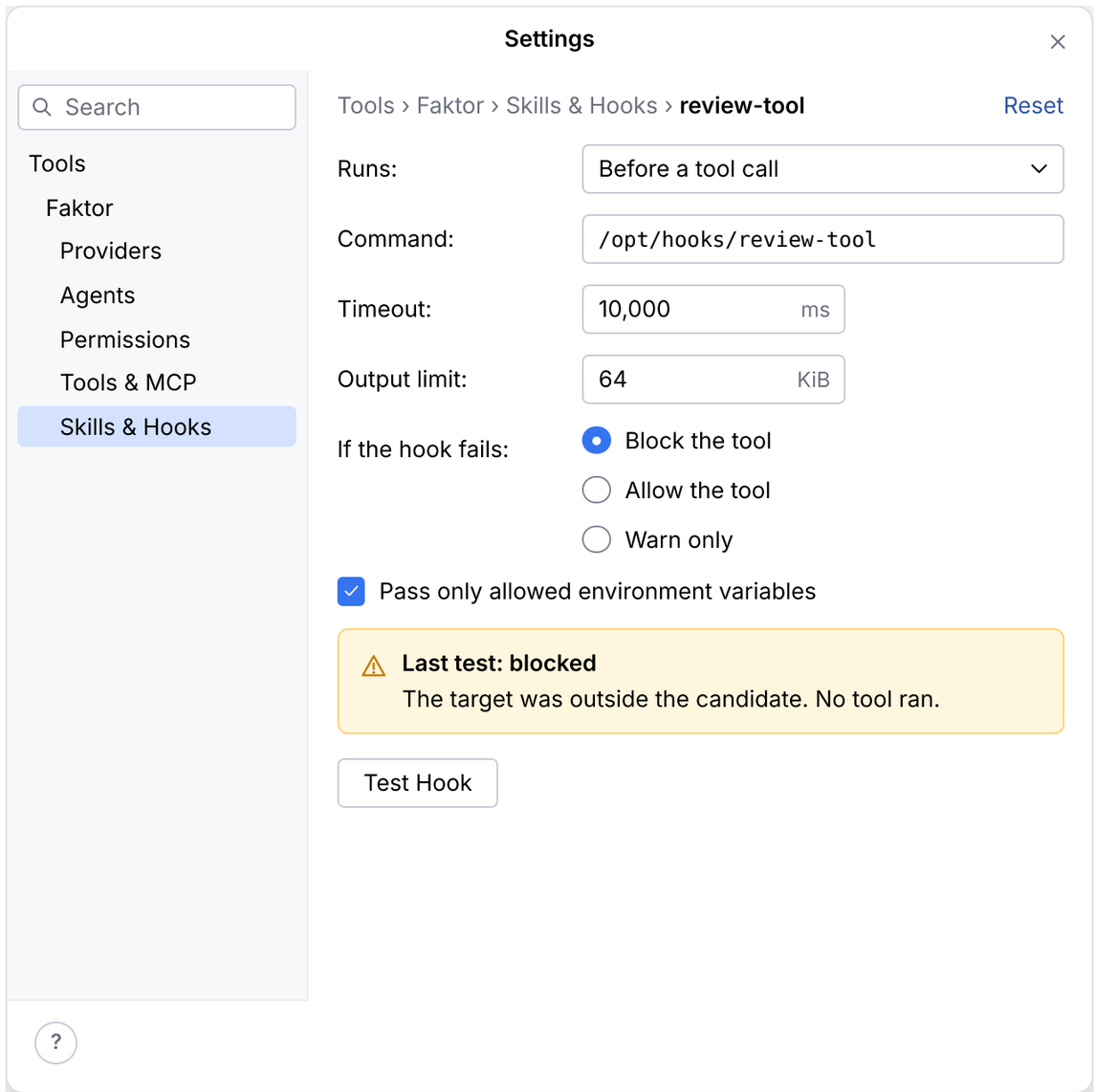
Save Hook · JetBrains: OK / Apply

Edit exact native lifecycle events, executable arguments, environment, bounds and failure policy. Test under the same capability envelope.

VS CODE · DARK MODERN — EDITOR TAB · WEBVIEW SETTINGS



JETBRAINS · LIGHT — SETTINGS DIALOG · CONFIGURABLE



CONTROLS S-07 S-08 S-09 S-10 S-11 S-12 S-13 S-14

REPLACES R7 page 064 · Expanded standard-agent surface

CONTRACT p. 153

S18 · PERMISSIONS AND TOOLS · BEHAVIOUR CONTRACT

Lifecycle hook: contract and revision notes

Original contract retained verbatim (Expanded standard-agent surface / controls: S-07 S-08 S-09 S-10 S-11 S-12 S-13 S-14). Design on p. 152.

What changed in revision 8

- Enum names (FailClosed) become outcomes the user chooses between.

Entry and return

Native Settings or the originating configuration command. Work / Inspect / History remain the only top-level clusters. Settings, Usage and Diagnostics use explicit Back. Return to the originating object, not the first row.

Scope and captured state

Application/project/connection plus effective revision; never infer task scope from an open form. Keep staged input, the effective snapshot and the acknowledged applied revision separate. Apply is atomic; Cancel preserves the previous effective configuration.

Primary action and recovery

Select an event, inspect matching inputs and run an explicitly admitted test. Save only a validated version for future events.. Deadline, nonzero exit and malformed verdict remain distinct. Partial output cannot become an Allow verdict after timeout.

Loading, empty and stale

Keep the last coherent values with a refresh state. A first read has no invented defaults. An absent capability remains explained and inspectable.

Keyboard and focus

Open on the triggering setting. Apply failure focuses the first invalid field on submission; async refresh never moves the caret. Escape cancels a transient editor and returns to its trigger.

Layout and adverse content

Align only related fields within their group. Long inputs use stacked labels. Native preferred sizes and zoom win over reference-art dimensions.

Starting point and required construction

Starting classifications: N, R. E existing operation; R runtime-backed adapter; N new contract; H host-only work. Exact per-control classifications remain in the catalog.

Reuse the reviewed domain operations and preserve all served fields. Build every missing native read/write adapter, capability gate and state renderer identified by the matching control/build rows.

BK01 owns validation, scope, persistence, revision comparison and acknowledgement; domain packages own parameter/capability enforcement.

Required interaction fixtures

Exercise first-read, empty/no-match, ready, refreshing/stale, error/refusal, action pending and acknowledged result. Also switch scope while the request is in flight.

Moved out of the interface

Every line of revision 7 plate copy that left the default view, and where it lives now.

"Faktor / Lifecycle hook (R7 window title)"
→ Native host chrome (VS Code view title, tabs or editor tab; JetBrains tool-window tabs, dialog title or Settings breadcrumb)

"Test the exact version under the real capability envelope."
→ Test Hook runs under the same permissions; rule in Specification: screen contract

"Event / PreTool"
→ Field "Runs — Before a tool call"

"Executable"
→ Label "Command"

"Deadline / 10000 ms; stdout cap / 64 KiB"
→ Timeout and Output limit

"Failure policy / FailClosed / FailOpen / Warn"
→ Plain choices: Block the tool / Allow the tool / Warn only

"Use an environment allowlist / Only explicit benign names and secret references are passed."
→ Checkbox "Pass only allowed environment variables"; detail in Tooltip / hover help

"Last test: denied / The requested target was outside the candidate scope. No production tool ran."
→ Warning banner

"Lifecycle hook"
→ Title "Hook: review-tool"

"/opt/hooks/review-tool / 10000 / 64"
→ Field values

"Test hook / Save hook"
→ Test Hook / Save Hook

S67 PERMISSIONS AND TOOLS

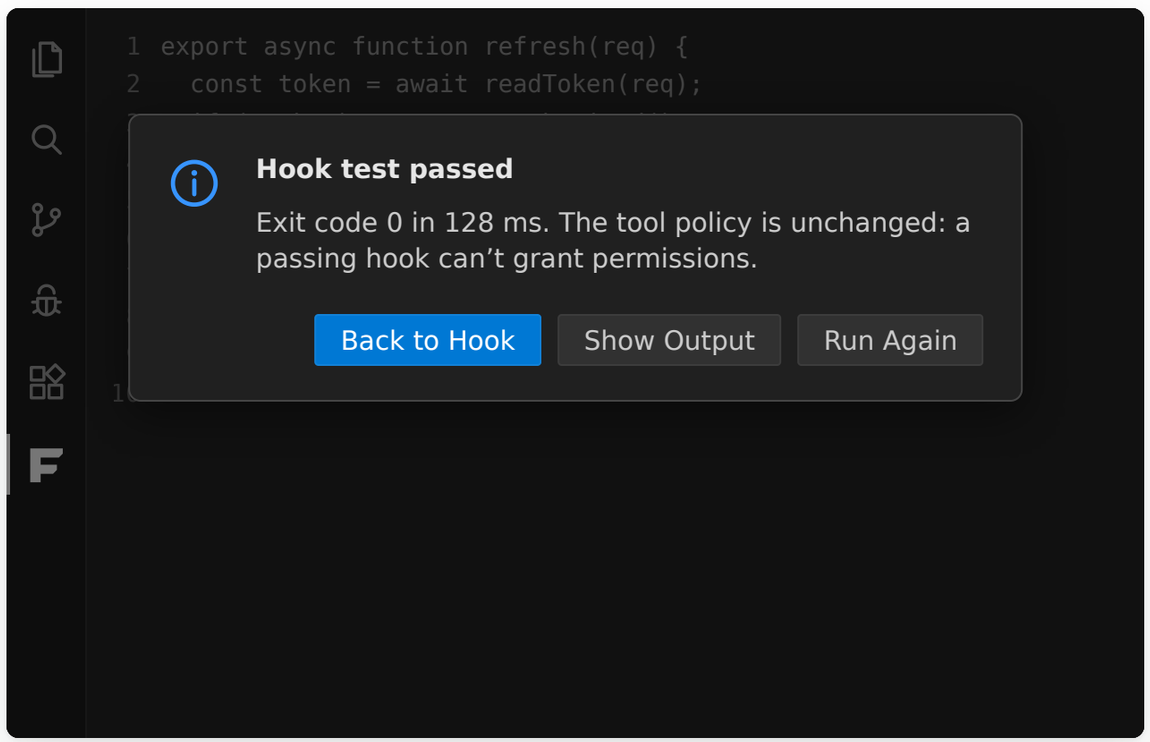
Hook test result

PRIMARY ACTION

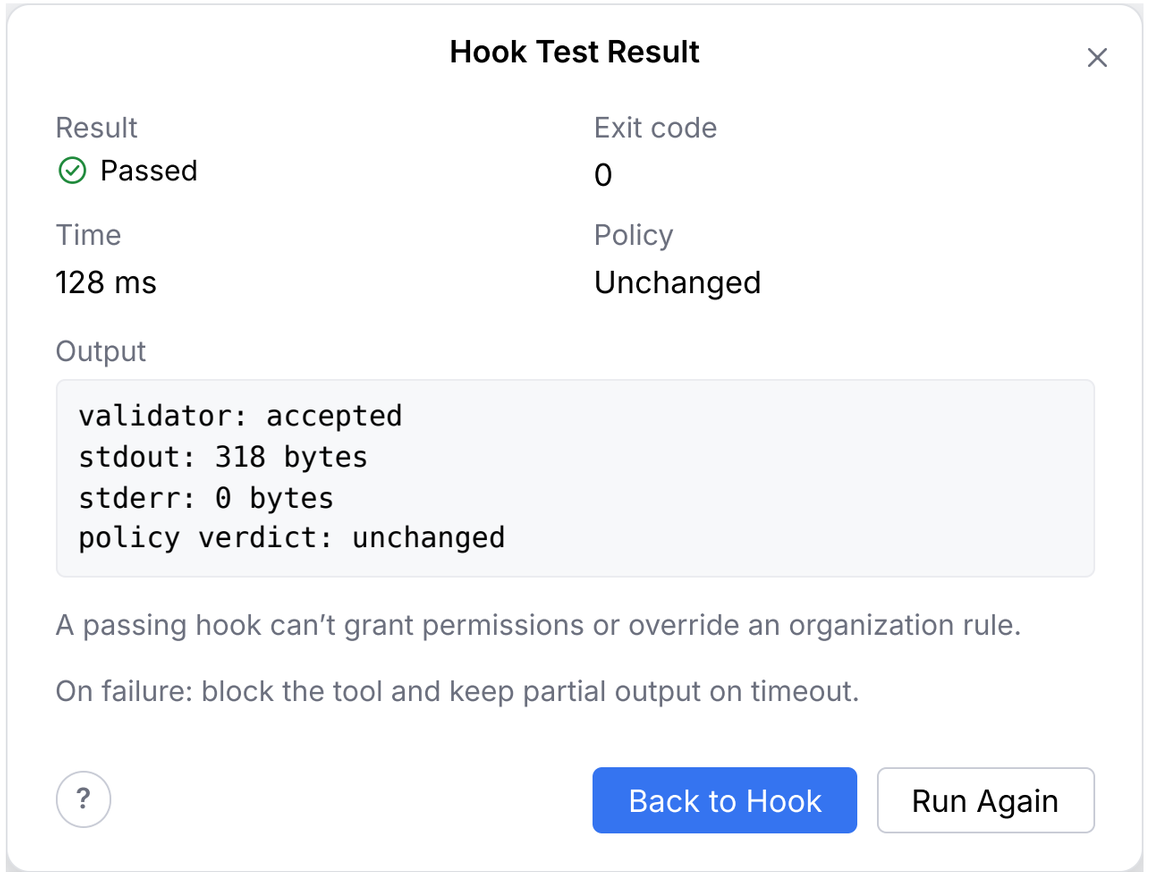
Back to Hook

Inspect a real bounded hook test, deadline, output and verdict without converting successful output into authority.

VS CODE · DARK MODERN — MODAL MESSAGE · NATIVE



JETBRAINS · LIGHT — DIALOGWRAPPER



CONTROLS S-12 S-13

REPLACES R7 page 065 · Expanded standard-agent surface

CONTRACT p. 155

S67 · PERMISSIONS AND TOOLS · BEHAVIOUR CONTRACT

Hook test result: contract and revision notes

Original contract retained verbatim (Expanded standard-agent surface / controls: S-12 S-13). Design on p. 154.

What changed in revision 8

- Shown as a result dialog over the hook page instead of a separate screen.

Entry and return

Native Settings or the originating configuration command. Work / Inspect / History remain the only top-level clusters. Settings, Usage and Diagnostics use explicit Back. Return to the originating object, not the first row.

Scope and captured state

Application/project/connection plus effective revision; never infer task scope from an open form. Keep staged input, the effective snapshot and the acknowledged applied revision separate. Apply is atomic; Cancel preserves the previous effective configuration.

Primary action and recovery

Return to hook. Associate validation with the exact field. A revision conflict preserves all staged values and opens the conflict resolver. Authentication, transport and schema failures have different recovery.

Loading, empty and stale

Keep the last coherent values with a refresh state. A first read has no invented defaults. An absent capability remains explained and inspectable.

Keyboard and focus

Open on the triggering setting. Apply failure focuses the first invalid field on submission; async refresh never moves the caret. Escape cancels a transient editor and returns to its trigger.

Layout and adverse content

Align only related fields within their group. Long inputs use stacked labels. Native preferred sizes and zoom win over reference-art dimensions.

Starting point and required construction

Starting classifications: N, R. E existing operation; R runtime-backed adapter; N new contract; H host-only work. Exact per-control classifications remain in the catalog.

NEW: expose a bounded hook-test operation using the real hook validator, environment/secret policy, deadline, output caps and typed verdict.

BK01 owns validation, scope, persistence, revision comparison and acknowledgement; domain packages own parameter/capability enforcement.

Required interaction fixtures

Exercise first-read, empty/no-match, ready, refreshing/stale, error/refusal, action pending and acknowledged result. Also switch scope while the request is in flight.

Moved out of the interface

Every line of revision 7 plate copy that left the default view, and where it lives now.

"Faktor / Settings / Hooks / Test (R7 window title)"
→ Native host chrome (VS Code view title, tabs or editor tab; JetBrains tool-window tabs, dialog title or Settings breadcrumb)

"Reviewed executable · test h-9 · same deadline and environment policy"
→ Test ID in Tooltip / hover help; conditions in Specification: screen contract

"Exit code / 0; Elapsed / 128 ms"
→ Facts

"Bounded test output"
→ Label "Output"

"Hook authority / Successful output does not grant a tool or bypass a managed deny."
→ Note (shortened)

"Failure policy / Fail-closed · retain partial output on timeout"
→ Note "On failure: block the tool…"

"Run test again / Return to hook"
→ Run Again / Back to Hook

"Hook test result"
→ Dialog title

"validator: accepted / stdout: 318 bytes / stderr: 0 bytes / policy verdict: unchanged"
→ Output block

S19 PERMISSIONS AND TOOLS

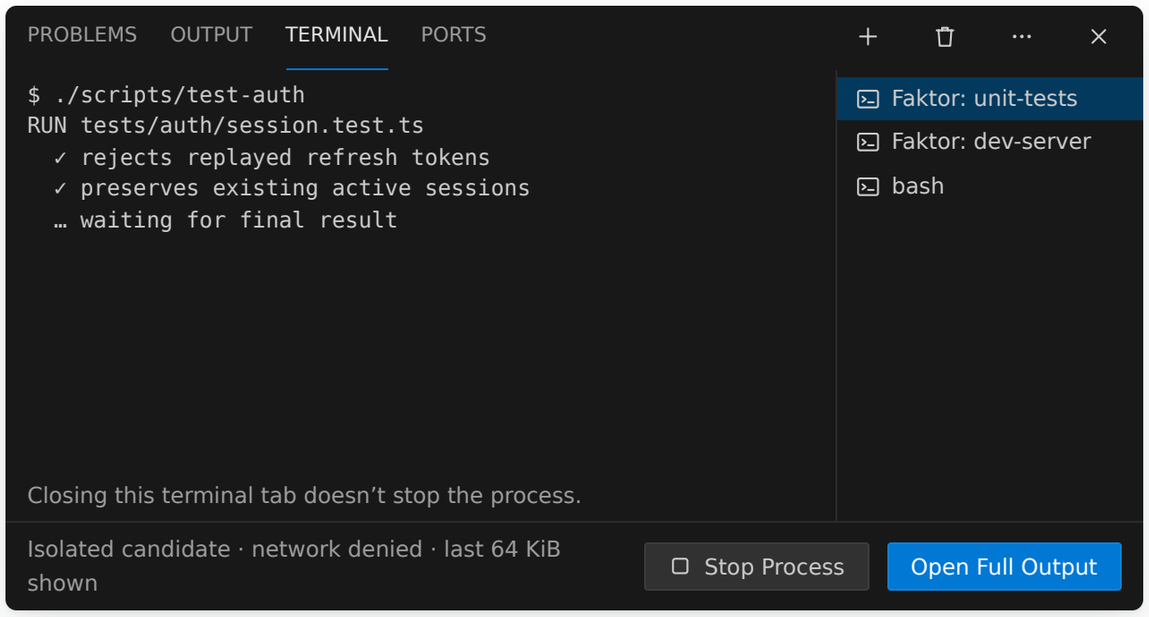
Terminal

PRIMARY ACTION

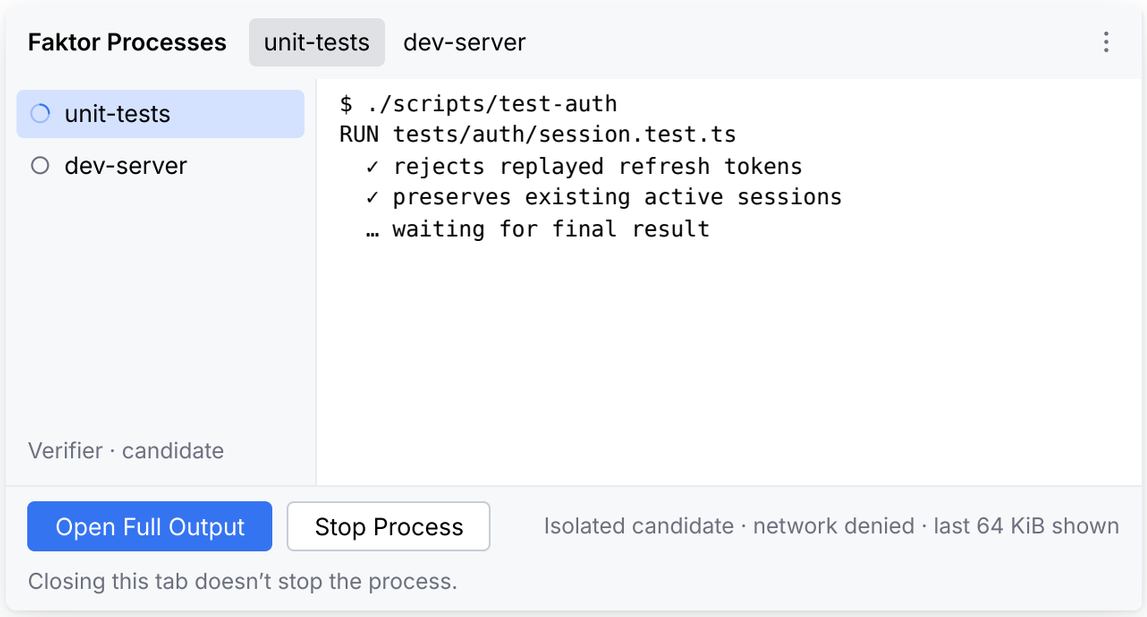
Open Full Output

An owned process table and bounded output coexist with the task transcript. Terminal kill, run stop and session abort are separate actions.

VS CODE · DARK MODERN — PANEL TERMINAL · NATIVE



JETBRAINS · LIGHT — TOOL WINDOW · CONSOLE



CONTROLS X-01 X-02 X-03 X-04 X-05 X-12 D-07 B-10

REPLACES R7 page 066 · R4-S35

CONTRACT p. 157

S19 · PERMISSIONS AND TOOLS · BEHAVIOUR CONTRACT

Terminal: contract and revision notes

Original contract retained verbatim (R4-S35 / controls: X-01 X-02 X-03 X-04 X-05 X-12 D-07 B-10). Design on p. 156.

What changed in revision 8

- VS Code: owned processes are integrated terminals in the Panel. JetBrains: a process tool window with a console.

Entry and return

Work, a captured submission receipt, or a task-specific attention item. Work / Inspect / History remain the only top-level clusters. Settings, Usage and Diagnostics use explicit Back. Return to the originating object, not the first row.

Scope and captured state

Daemon/workspace/worktree/conversation/task/request generation. Draft scope is separate from an admitted run. Capture one immutable logical start. Clear only its acknowledged draft version; preserve text typed after capture. Queue items acquire their own receipt and revision.

Primary action and recovery

Select a live process, inspect cwd/sandbox/output, then use only controls supported by its current owned handle.. A stale PID cannot authorize kill. Timeout or disconnect retains an Unknown effect state until reconciliation, with no blind replay.

Loading, empty and stale

Show the named operation and current owner. Keep the composer available for an independent draft. No fake percentages, fabricated tool steps or optimistic completion.

Keyboard and focus

Enter inserts a newline by default. The host send command respects IME composition. New approvals announce once without taking focus. Escape closes a detail; it never stops a run.

Layout and adverse content

Header, transcript scroll and composer are siblings. Collapse secondary metadata first. Keep a scoped Stop action and the active task reachable at short heights.

Starting point and required construction

Starting classifications: E, N, R. E existing operation; R runtime-backed adapter; N new contract; H host-only work. Exact per-control classifications remain in the catalog.

Reuse the reviewed domain operations and preserve all served fields. Build every missing native read/write adapter, capability gate and state renderer identified by the matching control/build rows.

BK08 owns admission, receipts, queue, control handles and reconciliation; BK05 owns authority and BK07 owns a shared exact budget ledger.

Required interaction fixtures

Exercise first-read, empty/no-match, ready, refreshing/stale, error/refusal, action pending and acknowledged result. Also switch scope while the request is in flight.

Moved out of the interface

Every line of revision 7 plate copy that left the default view, and where it lives now.

"Faktor / Inspect / Terminal (R7 window title)"
→ Native host chrome (VS Code view title, tabs or editor tab; JetBrains tool-window tabs, dialog title or Settings breadcrumb)

"Session-owned processes · dev-box"
→ Panel title "Faktor Processes" / terminal names

"unit-tests / Running / Verifier 3 · candidate/run-047"
→ Process entry and meta

"dev-server / Background / Owned background process · port shown when observed"
→ Process entry; port appears in the VS Code Ports tab when observed

"Selected output · last 64 KiB"
→ Footer note "last 64 KiB shown"

"Shell scope / Isolated candidate; Network / Denied"
→ Footer note

"Closing this output tab does not stop the process. Task stop and process stop are separate controls."
→ Note under the output; separate controls in Specification: screen contract

"Stop process / Open full output"
→ Stop Process / Open Full Output

"Terminal"
→ Panel tab

"unit-tests / dev-server"
→ Process entries

"$ ./scripts/test-auth … waiting for final result"
→ Terminal output

S20 PERMISSIONS AND TOOLS

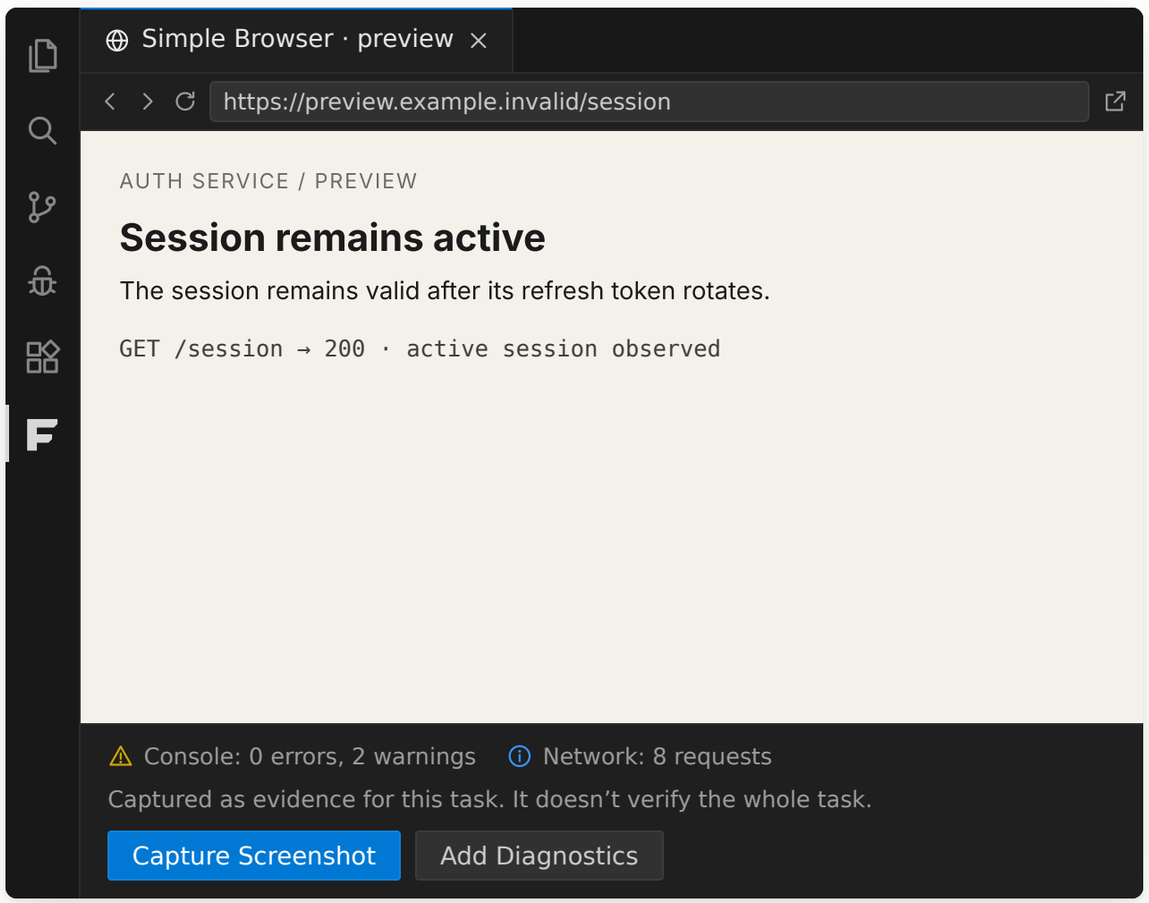
Browser

PRIMARY ACTION

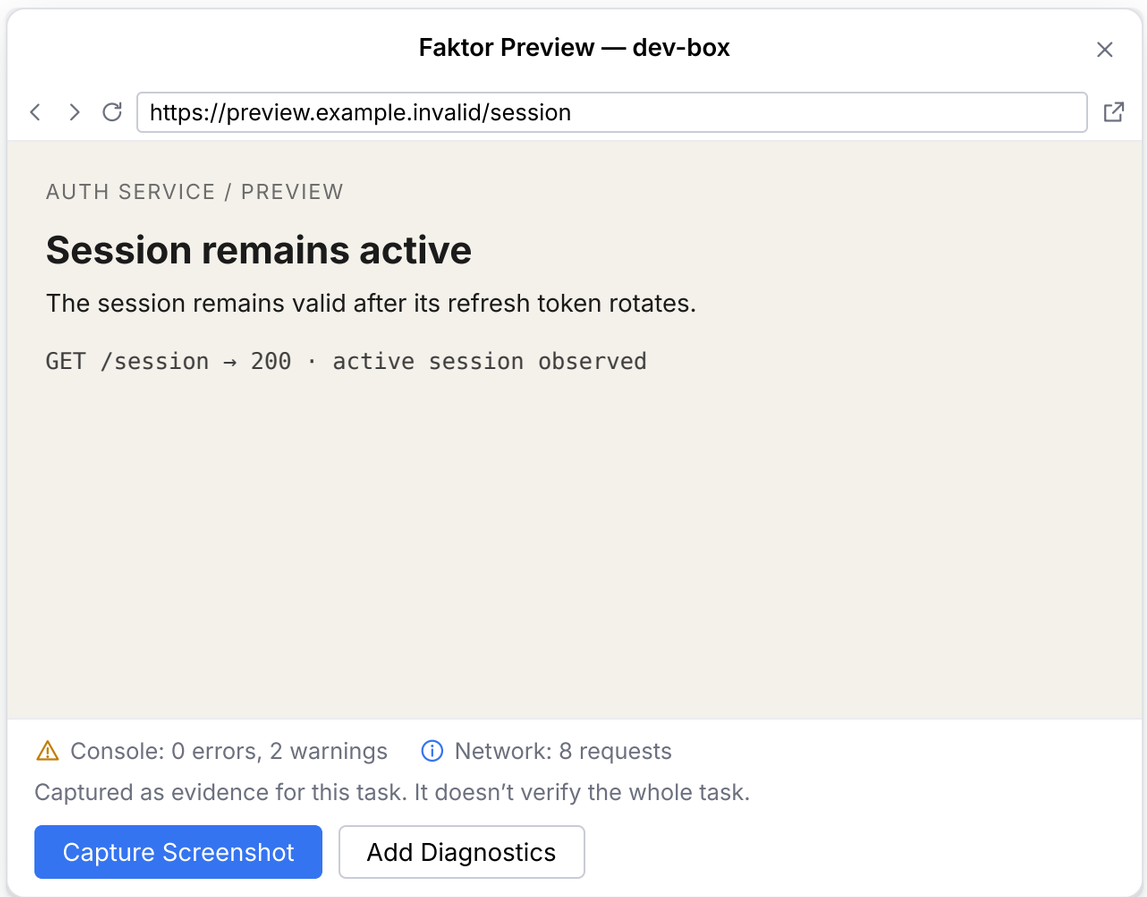
Capture Screenshot

Integrate host diagnostics and explicit code actions. Browser controls require an actual advertised adapter with account, machine and network scope.

VS CODE · DARK MODERN — SIMPLE BROWSER · NATIVE



JETBRAINS · LIGHT — JCEF PREVIEW WINDOW



CONTROLS X-06 X-07 X-08 X-09 X-10 X-11 X-13 C-03

REPLACES R7 page 067 · Expanded standard-agent surface

CONTRACT p. 159

S20 · PERMISSIONS AND TOOLS · BEHAVIOUR CONTRACT

Browser: contract and revision notes

Original contract retained verbatim (Expanded standard-agent surface / controls: X-06 X-07 X-08 X-09 X-10 X-11 X-13 C-03). Design on p. 158.

What changed in revision 8

- VS Code uses its Simple Browser editor; JetBrains uses a JCEF preview window.

Entry and return

Work, a captured submission receipt, or a task-specific attention item. Work / Inspect / History remain the only top-level clusters. Settings, Usage and Diagnostics use explicit Back. Return to the originating object, not the first row.

Scope and captured state

Daemon/workspace/worktree/conversation/task/request generation. Draft scope is separate from an admitted run. Capture one immutable logical start. Clear only its acknowledged draft version; preserve text typed after capture. Queue items acquire their own receipt and revision.

Primary action and recovery

Capture versioned selection or diagnostics into a draft; connect a browser only through an implemented adapter and scoped permission decision.. A screenshot or clean console does not prove functional correctness. Unavailable browser support is labelled as such rather than shown as an idle tab.

Loading, empty and stale

Show the named operation and current owner. Keep the composer available for an independent draft. No fake percentages, fabricated tool steps or optimistic completion.

Keyboard and focus

Enter inserts a newline by default. The host send command respects IME composition. New approvals announce once without taking focus. Escape closes a detail; it never stops a run.

Layout and adverse content

Header, transcript scroll and composer are siblings. Collapse secondary metadata first. Keep a scoped Stop action and the active task reachable at short heights.

Starting point and required construction

Starting classifications: H, N. E existing operation; R runtime-backed adapter; N new contract; H host-only work. Exact per-control classifications remain in the catalog.

Reuse the reviewed domain operations and preserve all served fields. Build every missing native read/write adapter, capability gate and state renderer identified by the matching control/build rows.

BK08 owns admission, receipts, queue, control handles and reconciliation; BK05 owns authority and BK07 owns a shared exact budget ledger.

Required interaction fixtures

Exercise first-read, empty/no-match, ready, refreshing/stale, error/refusal, action pending and acknowledged result. Also switch scope while the request is in flight.

Moved out of the interface

Every line of revision 7 plate copy that left the default view, and where it lives now.

"Faktor / Inspect / Browser (R7 window title)"
→ Native host chrome (VS Code view title, tabs or editor tab; JetBrains tool-window tabs, dialog title or Settings breadcrumb)

"Local browser on dev-box · preview account"
→ Window title "Faktor Preview — dev-box"; account in Tooltip / hover help

"Observations / Console / Network"
→ Observation bar

"Console / 0 errors · 2 warnings; Network / 8 requests observed"
→ Observation statuses

"These observations are evidence inputs. They do not establish the entire task's correctness."
→ Note (shortened)

"Add diagnostics / Capture screenshot"
→ Add Diagnostics / Capture Screenshot

"Browser"
→ Preview editor / window

"https://preview.example.invalid/session / AUTH SERVICE / PREVIEW / Session remains active"
→ Preview page

"The session remains valid after its refresh token rotates. / GET /session → 200 · active session observed"
→ Preview page

PART 2 · SCREENS

2.11 Settings and diagnostics

10 screens. Each has a design spread for both IDEs and a contract page.

ID	SCREEN	JOB	PAGE
S21	**Budgets & retries**	Show exact run limits and current accounting beside reliability controls. A soft warning is different from enforced admission refusal.	p. 161
S28	**Efficiency**	Expose current production efficiency flags and their off-switches with honest measurement, fallback and apply timing.	p. 163
S29	**General settings**	Search settings across allowed scopes and inspect each effective value, source, lock and timing. Distinguish staged, saved and runtime-applied state.	p. 165
S64	**Settings search and provenance**	Find persistent settings in the native host, inspect scope/provenance and remove an override deliberately.	p. 167
S60	**Settings revision conflict**	Resolve a rejected configuration patch against the new effective revision without losing staged edits.	p. 169
S55	**Optional control-plane account**	Connect a machine-scoped optional account with write-only secrets while local coding remains independent.	p. 171
S30	**Diagnostics**	Show readiness, version compatibility, runtime identity and bounded diagnostics with an actionable typed cause.	p. 173
S56	**Protocol-blocked recovery**	Preserve the last coherent snapshot and cursor when a typed stream refusal pauses automatic recovery.	p. 175
S31	**Workspace binding**	Display editor, execution, model, vault and candidate locations together. Remote workspaces must not blur local/hosted boundaries or path ownership.	p. 177
S58	**Narrow Work panel**	Keep a 320 px Work panel readable with stacked controls, one selected detail and an independently reachable composer.	p. 179

S21 SETTINGS AND DIAGNOSTICS

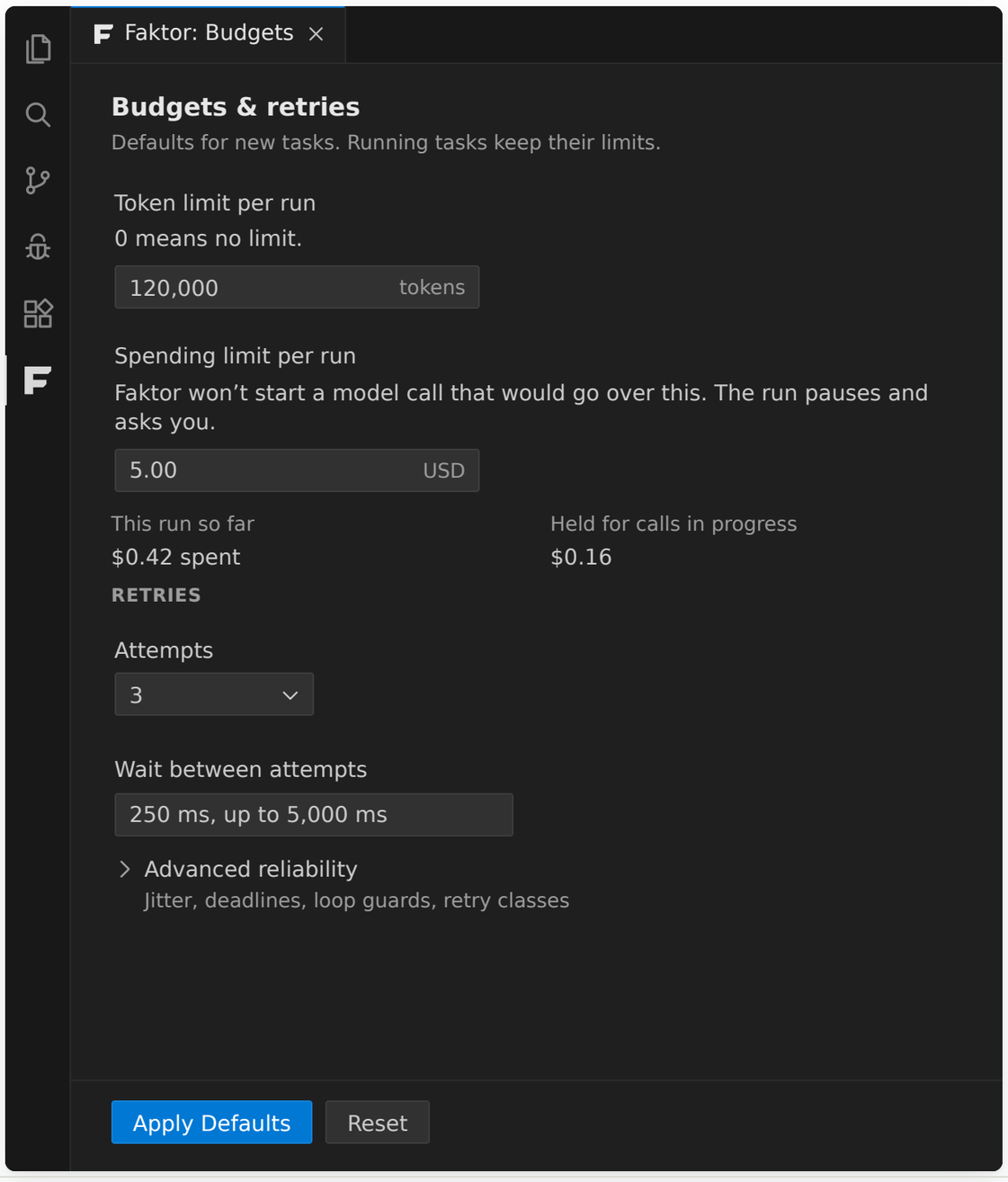
Budgets & retries

PRIMARY ACTION

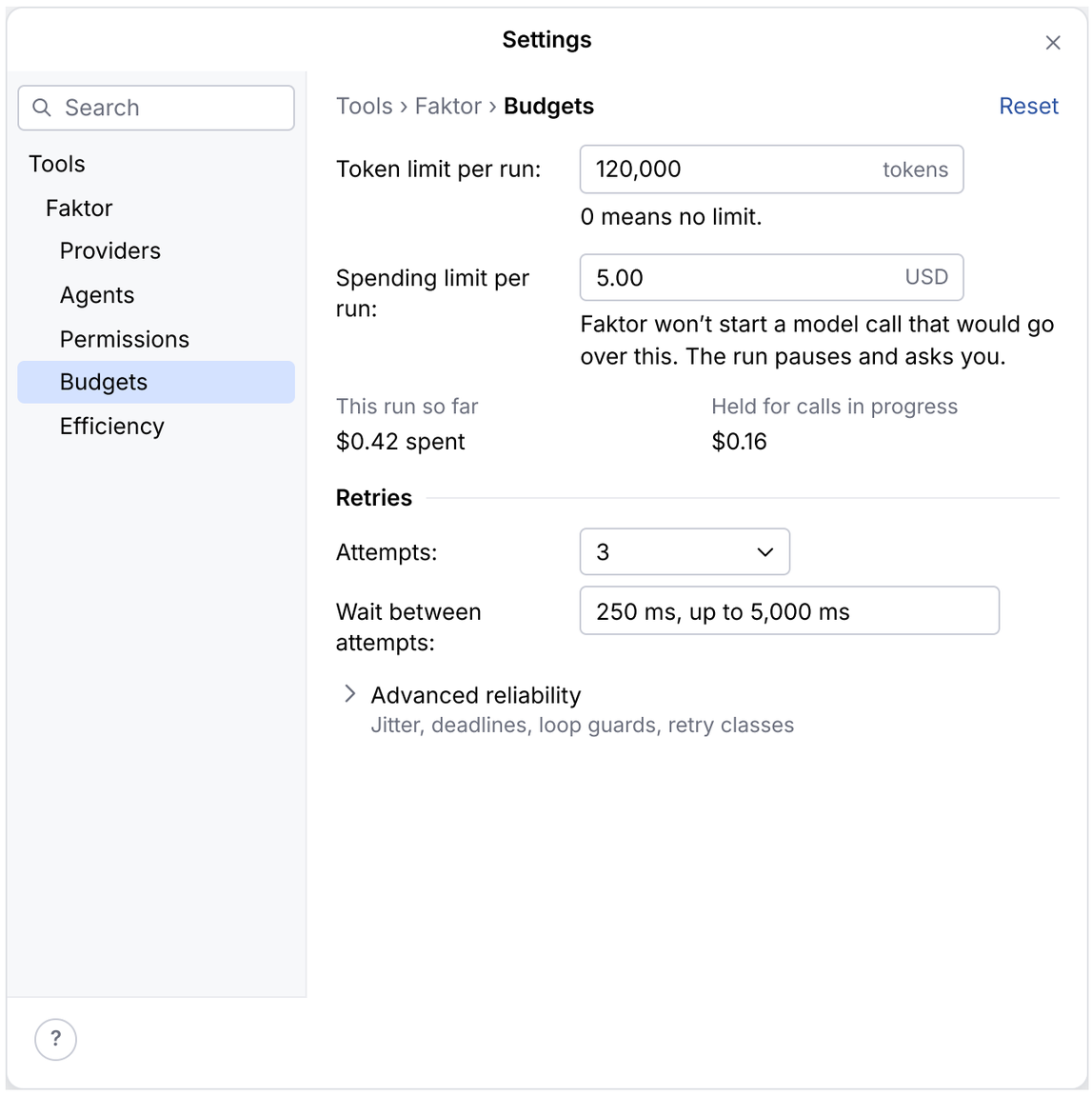
Apply Defaults · JetBrains: OK / Apply

Show exact run limits and current accounting beside reliability controls. A soft warning is different from enforced admission refusal.

VS CODE · DARK MODERN — EDITOR TAB · WEBVIEW SETTINGS



JETBRAINS · LIGHT — SETTINGS DIALOG · CONFIGURABLE



CONTROLS B-01 B-02 B-03 B-04 B-05 B-06 B-07 B-08 B-09 B-10 B-11 B-12

REPLACES R7 page 068 · Expanded standard-agent surface

CONTRACT p. 162

S21 · SETTINGS AND DIAGNOSTICS · BEHAVIOUR CONTRACT

Budgets & retries: contract and revision notes

Original contract retained verbatim (Expanded standard-agent surface / controls: B-01 B-02 B-03 B-04 B-05 B-06 B-07 B-08 B-09 B-10 B-11 B-12). Design on p. 161.

What changed in revision 8

- The spending limit now states what happens when it is reached, as requested; the micro-USD conversion lives in the B-02 contract.

Entry and return

Native Settings or the originating configuration command. Work / Inspect / History remain the only top-level clusters. Settings, Usage and Diagnostics use explicit Back. Return to the originating object, not the first row.

Scope and captured state

Application/project/connection plus effective revision; never infer task scope from an open form. Keep staged input, the effective snapshot and the acknowledged applied revision separate. Apply is atomic; Cancel preserves the previous effective configuration.

Primary action and recovery

Stage a new default or request a supported active-task allowance revision. Clearly label which captured task is affected.. Unknown price is not zero; a cap decrease cannot erase settled or reserved spend. Retry only after classifying admission and external-effect state.

Loading, empty and stale

Keep the last coherent values with a refresh state. A first read has no invented defaults. An absent capability remains explained and inspectable.

Keyboard and focus

Open on the triggering setting. Apply failure focuses the first invalid field on submission; async refresh never moves the caret. Escape cancels a transient editor and returns to its trigger.

Layout and adverse content

Align only related fields within their group. Long inputs use stacked labels. Native preferred sizes and zoom win over reference-art dimensions.

Starting point and required construction

Starting classifications: E, N, R. E existing operation; R runtime-backed adapter; N new contract; H host-only work. Exact per-control classifications remain in the catalog.

Reuse the reviewed domain operations and preserve all served fields. Build every missing native read/write adapter, capability gate and state renderer identified by the matching control/build rows.

BK01 owns validation, scope, persistence, revision comparison and acknowledgement; domain packages own parameter/capability enforcement.

Required interaction fixtures

Exercise first-read, empty/no-match, ready, refreshing/stale, error/refusal, action pending and acknowledged result. Also switch scope while the request is in flight.

Moved out of the interface

Every line of revision 7 plate copy that left the default view, and where it lives now.

"Faktor / Budgets & retries (R7 window title)"
→ Native host chrome (VS Code view title, tabs or editor tab; JetBrains tool-window tabs, dialog title or Settings breadcrumb)

"New-task defaults. The active task keeps its captured cap."
→ Subtitle "Defaults for new tasks. Running tasks keep their limits."

"Token cap"
→ Label "Token limit per run"

"0 means no limit. Use a whole non-negative token count."
→ Help "0 means no limit."; integer rule as a validation message only

"Money cap"
→ Label "Spending limit per run"

"USD 5.00 = 5,000,000 micro-USD."
→ Specification: screen contract (B-02 storage conversion) — replaced by the consequence of reaching the limit

"Settled / USD 0.42"
→ Fact "This run so far — $0.42 spent"

"Open reservations / USD 0.16"
→ Fact "Held for calls in progress — $0.16"

"Provider retries / Total attempts / 3"
→ Section "Retries", field "Attempts"

"Base delay / 250 ms; Maximum delay / 5000 ms"
→ Field "Wait between attempts — 250 ms, up to 5,000 ms"

"Advanced reliability / Jitter 0.2 · supported deadlines and loop guards · typed retry classes"
→ Disclosure

"Reset / Apply defaults"
→ Reset / Apply Defaults (JetBrains: OK/Cancel/Apply)

"120000 tokens"
→ Formatted as 120,000 tokens

"Budgets & retries"
→ Title

S28 SETTINGS AND DIAGNOSTICS

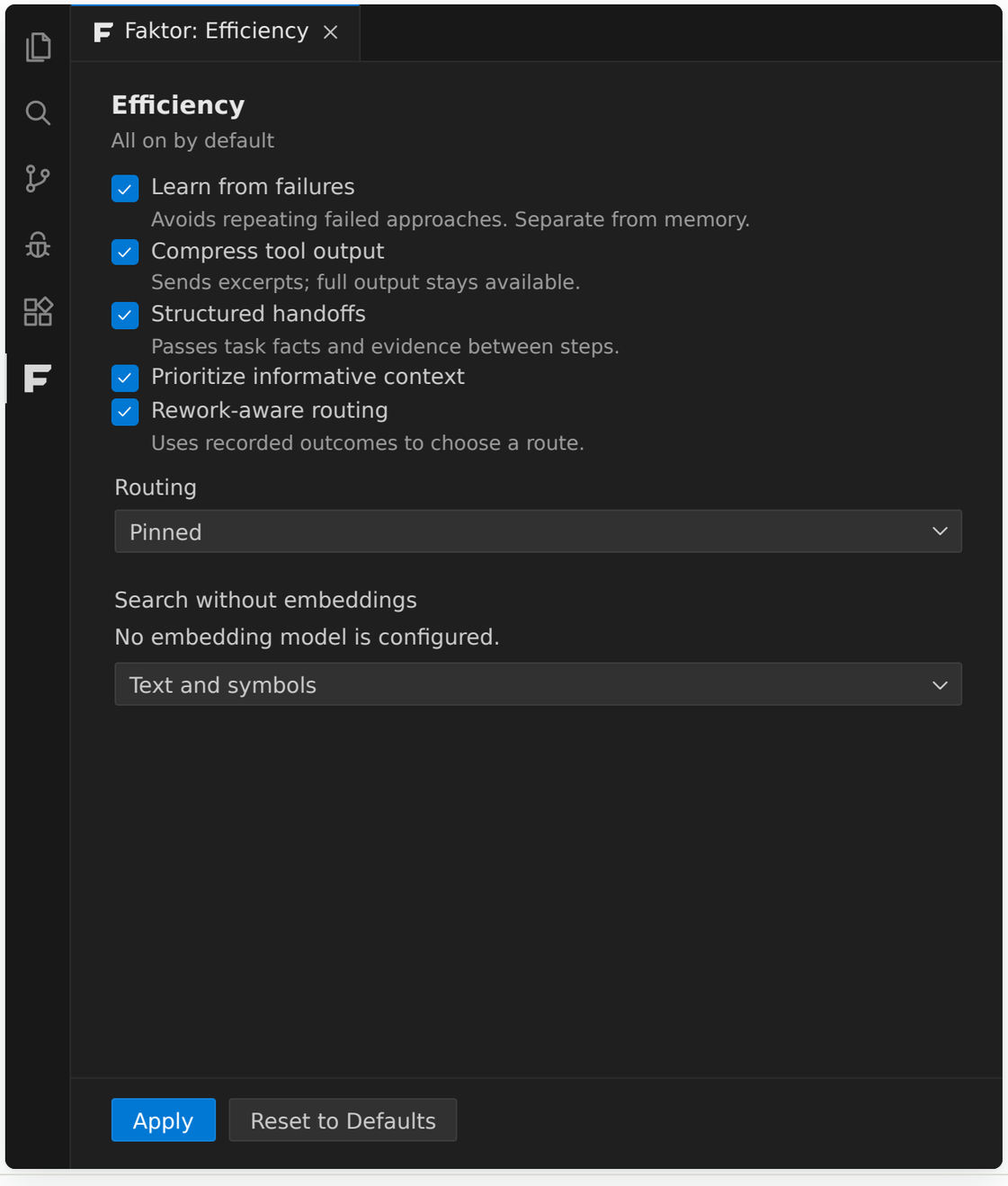
Efficiency

Expose current production efficiency flags and their off-switches with honest measurement, fallback and apply timing.

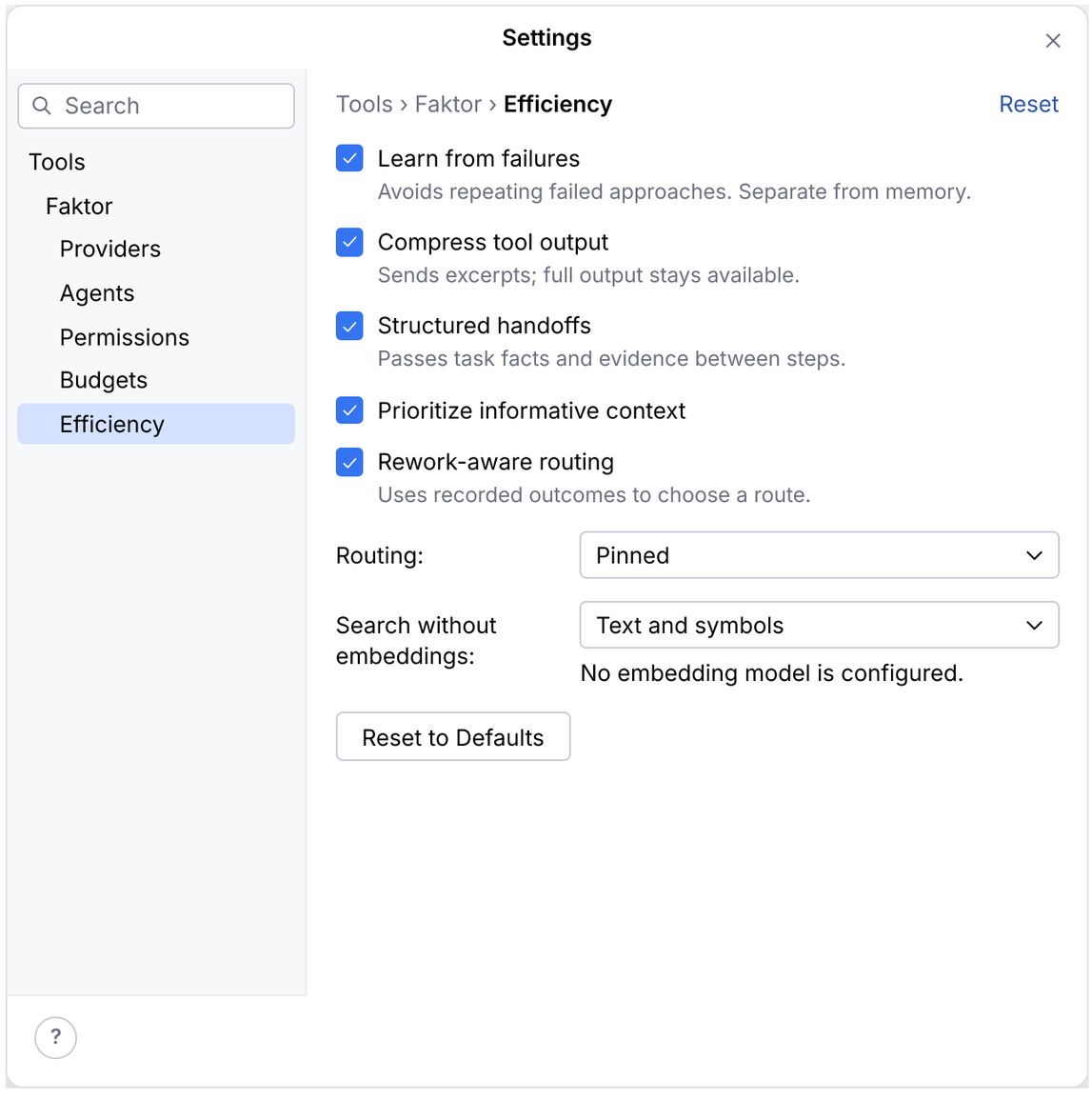
PRIMARY ACTION

Apply · JetBrains: OK / Apply

VS CODE · DARK MODERN — EDITOR TAB · WEBVIEW SETTINGS



JETBRAINS · LIGHT — SETTINGS DIALOG · CONFIGURABLE



CONTROLS M-04 M-06 M-10 K-05 K-06 K-07 K-08 K-09 K-12 C-09 C-10 C-11

REPLACES R7 page 069 · Expanded standard-agent surface

CONTRACT p. 164

S28 · SETTINGS AND DIAGNOSTICS · BEHAVIOUR CONTRACT

Efficiency: contract and revision notes

Original contract retained verbatim (Expanded standard-agent surface / controls: M-04 M-06 M-10 K-05 K-06 K-07 K-08 K-09 K-12 C-09 C-10 C-11). Design on p. 163.

What changed in revision 8

- Configuration key names leave the UI; each switch says what it does for you.

Entry and return

Native Settings or the originating configuration command. Work / Inspect / History remain the only top-level clusters. Settings, Usage and Diagnostics use explicit Back. Return to the originating object, not the first row.

Scope and captured state

Application/project/connection plus effective revision; never infer task scope from an open form. Keep staged input, the effective snapshot and the acknowledged applied revision separate. Apply is atomic; Cancel preserves the previous effective configuration.

Primary action and recovery

Preview a config diff, validate aliases and save through supported operator/native pathways. Keep provenance for each effective value.. Missing embedding support falls back to lexical/symbol retrieval. Duplicate semantic_context/information_gain aliases fail validation.

Loading, empty and stale

Keep the last coherent values with a refresh state. A first read has no invented defaults. An absent capability remains explained and inspectable.

Keyboard and focus

Open on the triggering setting. Apply failure focuses the first invalid field on submission; async refresh never moves the caret. Escape cancels a transient editor and returns to its trigger.

Layout and adverse content

Align only related fields within their group. Long inputs use stacked labels. Native preferred sizes and zoom win over reference-art dimensions.

Starting point and required construction

Starting classifications: E, N, R. E existing operation; R runtime-backed adapter; N new contract; H host-only work. Exact per-control classifications remain in the catalog.

Reuse the reviewed domain operations and preserve all served fields. Build every missing native read/write adapter, capability gate and state renderer identified by the matching control/build rows.

BK01 owns validation, scope, persistence, revision comparison and acknowledgement; domain packages own parameter/capability enforcement.

Required interaction fixtures

Exercise first-read, empty/no-match, ready, refreshing/stale, error/refusal, action pending and acknowledged result. Also switch scope while the request is in flight.

Moved out of the interface

Every line of revision 7 plate copy that left the default view, and where it lives now.

"Faktor / Efficiency (R7 window title)"
→ Native host chrome (VS Code view title, tabs or editor tab; JetBrains tool-window tabs, dialog title or Settings breadcrumb)

"All five switches below are ON in production defaults."
→ Subtitle "All on by default" and Reset to Defaults

"Failure learning / Learned operational failure priors; separate from personal memory."
→ Checkbox and help

"Compressed tool evidence / Bounded excerpts link to full durable artifacts."
→ Checkbox and help

"Typed handoff / Retain structured task facts and evidence identity."
→ Checkbox and help

"Information-gain context"
→ Checkbox "Prioritize informative context"

"semantic_context; information_gain is the same key alias."
→ Specification: screen contract (configuration key alias) — backend detail removed from the interface

"Rework-aware routing / Use recorded outcome observations without inventing scores."
→ Checkbox and help; scoring rule in Specification: screen contract

"Routing policy / Pinned"
→ Field "Routing"

"Retrieval fallback / Lexical + symbol"
→ Field "Search without embeddings — Text and symbols"

"No embedding model is configured."
→ Field help

"Reset / Apply configuration"
→ Reset to Defaults / Apply

"Efficiency"
→ Title

"Pinned"
→ Field value

S29 SETTINGS AND DIAGNOSTICS

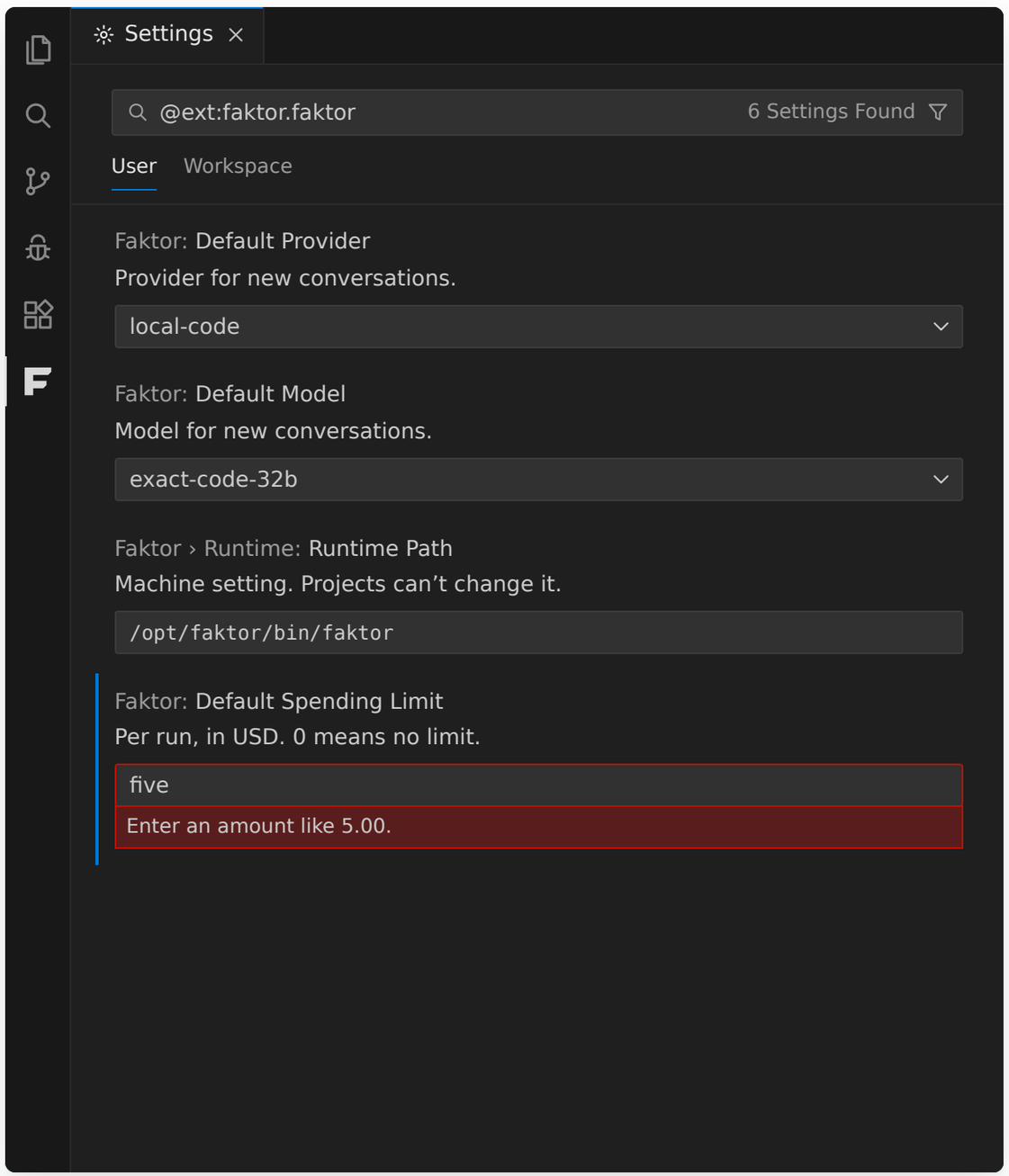
General settings

Search settings across allowed scopes and inspect each effective value, source, lock and timing. Distinguish staged, saved and runtime-applied state.

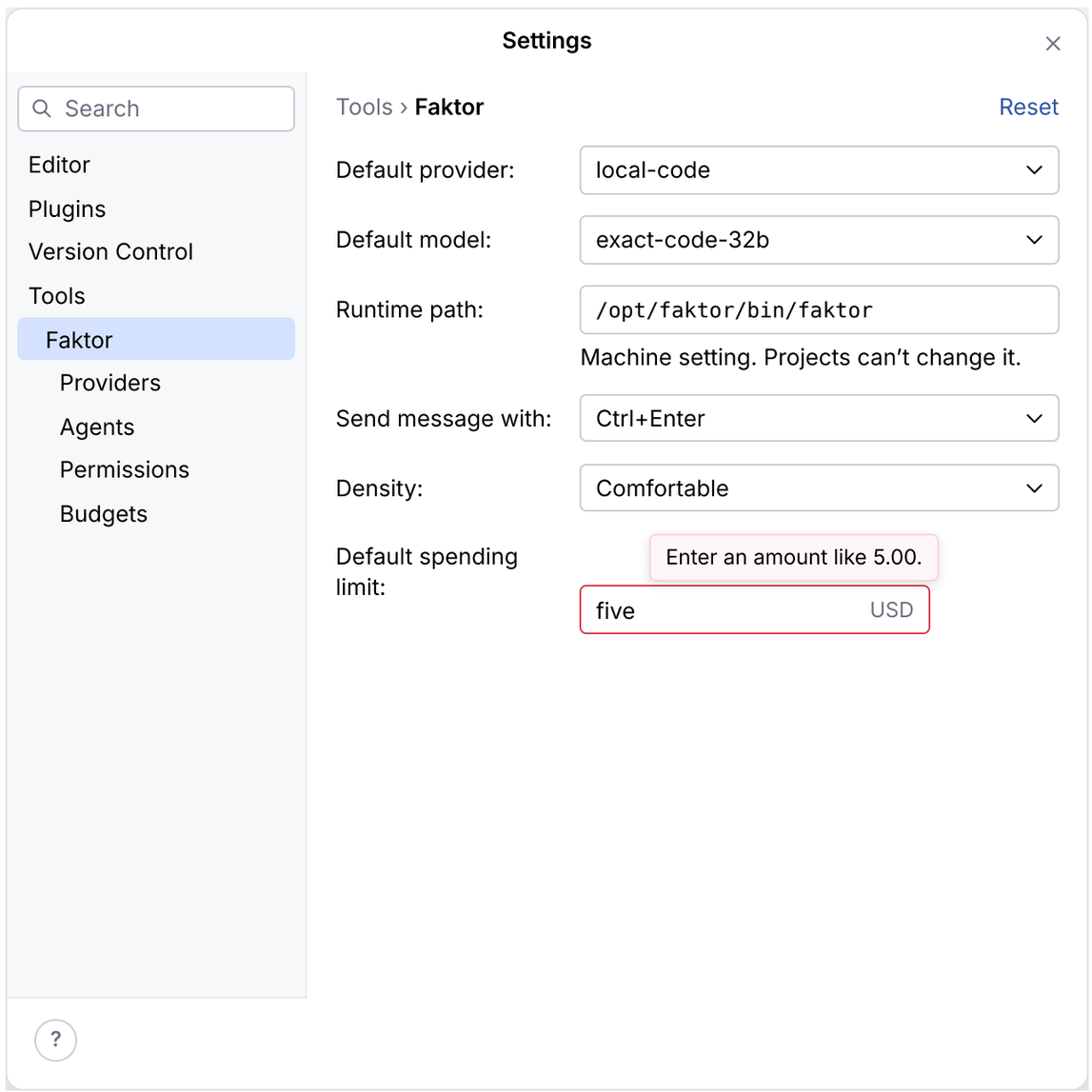
PRIMARY ACTION

VS Code: Settings saves valid values · JetBrains: OK / Apply

VS CODE · DARK MODERN — SETTINGS EDITOR · NATIVE



JETBRAINS · LIGHT — SETTINGS DIALOG · CONFIGURABLE



CONTROLS O-01 O-04 O-07 O-08 O-10 O-12 O-13 D-09 D-10 D-11 D-14

REPLACES R7 page 070 · Expanded standard-agent surface

CONTRACT p. 166

S29 · SETTINGS AND DIAGNOSTICS · BEHAVIOUR CONTRACT

General settings: contract and revision notes

Original contract retained verbatim (Expanded standard-agent surface / controls: O-01 O-04 O-07 O-08 O-10 O-12 O-13 D-09 D-10 D-11 D-14). Design on p. 165.

What changed in revision 8

- VS Code uses the native Settings editor (persistent preferences belong there); invalid values show the native validation message and are not saved.
- JetBrains shows the error tooltip on the field and, after Apply, an inline error above the buttons.

Entry and return

Native Settings or the originating configuration command. Work / Inspect / History remain the only top-level clusters. Settings, Usage and Diagnostics use explicit Back. Return to the originating object, not the first row.

Scope and captured state

Application/project/connection plus effective revision; never infer task scope from an open form. Keep staged input, the effective snapshot and the acknowledged applied revision separate. Apply is atomic; Cancel preserves the previous effective configuration.

Primary action and recovery

Apply a validated atomic patch against a known config revision, or Reset to applied values; Restore defaults edits the form until Apply.. A revision conflict opens a compare/rebase view. Machine executable paths never inherit from untrusted project settings.

Loading, empty and stale

Keep the last coherent values with a refresh state. A first read has no invented defaults. An absent capability remains explained and inspectable.

Keyboard and focus

Open on the triggering setting. Apply failure focuses the first invalid field on submission; async refresh never moves the caret. Escape cancels a transient editor and returns to its trigger.

Layout and adverse content

Align only related fields within their group. Long inputs use stacked labels. Native preferred sizes and zoom win over reference-art dimensions.

Starting point and required construction

Starting classifications: E, H, N, R. E existing operation; R runtime-backed adapter; N new contract; H host-only work. Exact per-control classifications remain in the catalog.

Reuse the reviewed domain operations and preserve all served fields. Build every missing native read/write adapter, capability gate and state renderer identified by the matching control/build rows.

BK01 owns validation, scope, persistence, revision comparison and acknowledgement; domain packages own parameter/capability enforcement.

Required interaction fixtures

Exercise first-read, empty/no-match, ready, refreshing/stale, error/refusal, action pending and acknowledged result. Also switch scope while the request is in flight.

Moved out of the interface

Every line of revision 7 plate copy that left the default view, and where it lives now.

"Faktor / General settings (R7 window title)"
→ Native host chrome (VS Code view title, tabs or editor tab; JetBrains tool-window tabs, dialog title or Settings breadcrumb)

"Application settings · one invalid staged value"
→ JetBrains: inline error above OK; VS Code: per-setting validation message

"Machine only. Project content cannot replace it."
→ Help "Machine setting. Projects can't change it."

"Send action / Ctrl/Cmd + Enter"
→ Field "Send message with"

"Default money cap"
→ Label "Default spending limit"

"Enter a non-negative decimal amount, for example 5.00."
→ Error "Enter an amount like 5.00."

"Nothing is saved until the whole form is valid. Reset restores applied values; Defaults stages a reset."
→ Specification: screen contract (atomic apply and reset semantics)

"Reset / Apply changes"
→ JetBrains: Reset link + OK/Cancel/Apply; VS Code: native Settings saves each valid value

"General settings"
→ VS Code: native Settings filtered to Faktor; JetBrains: Tools › Faktor page

"Default provider / local-code / Default model / exact-code-32b"
→ Fields

"Runtime path / /opt/faktor/bin/faktor"
→ Field

"Density / Comfortable"
→ Field (JetBrains)

"five / USD"
→ Invalid value with error

S64 SETTINGS AND DIAGNOSTICS

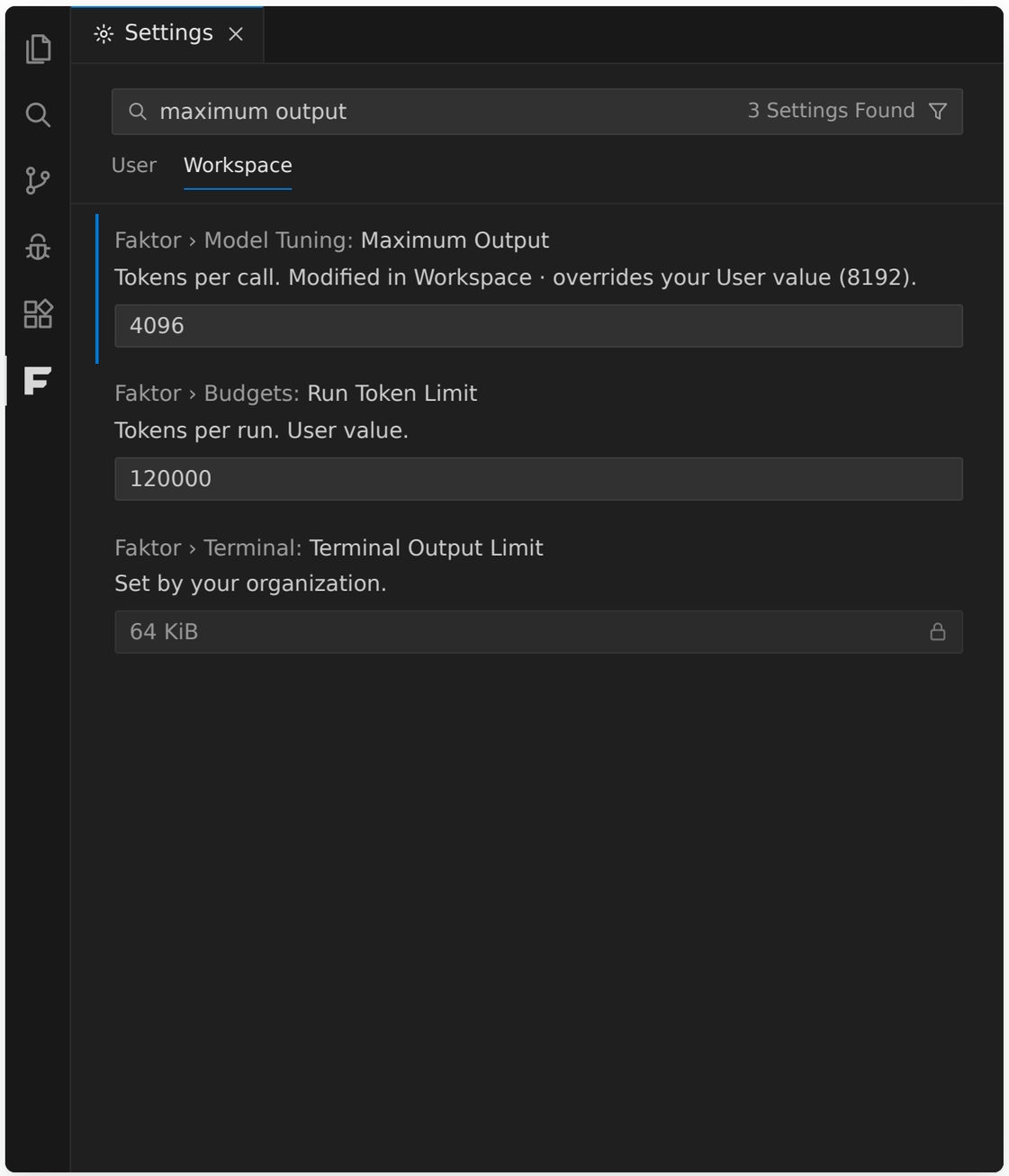
Settings search and provenance

Find persistent settings in the native host, inspect scope/provenance and remove an override deliberately.

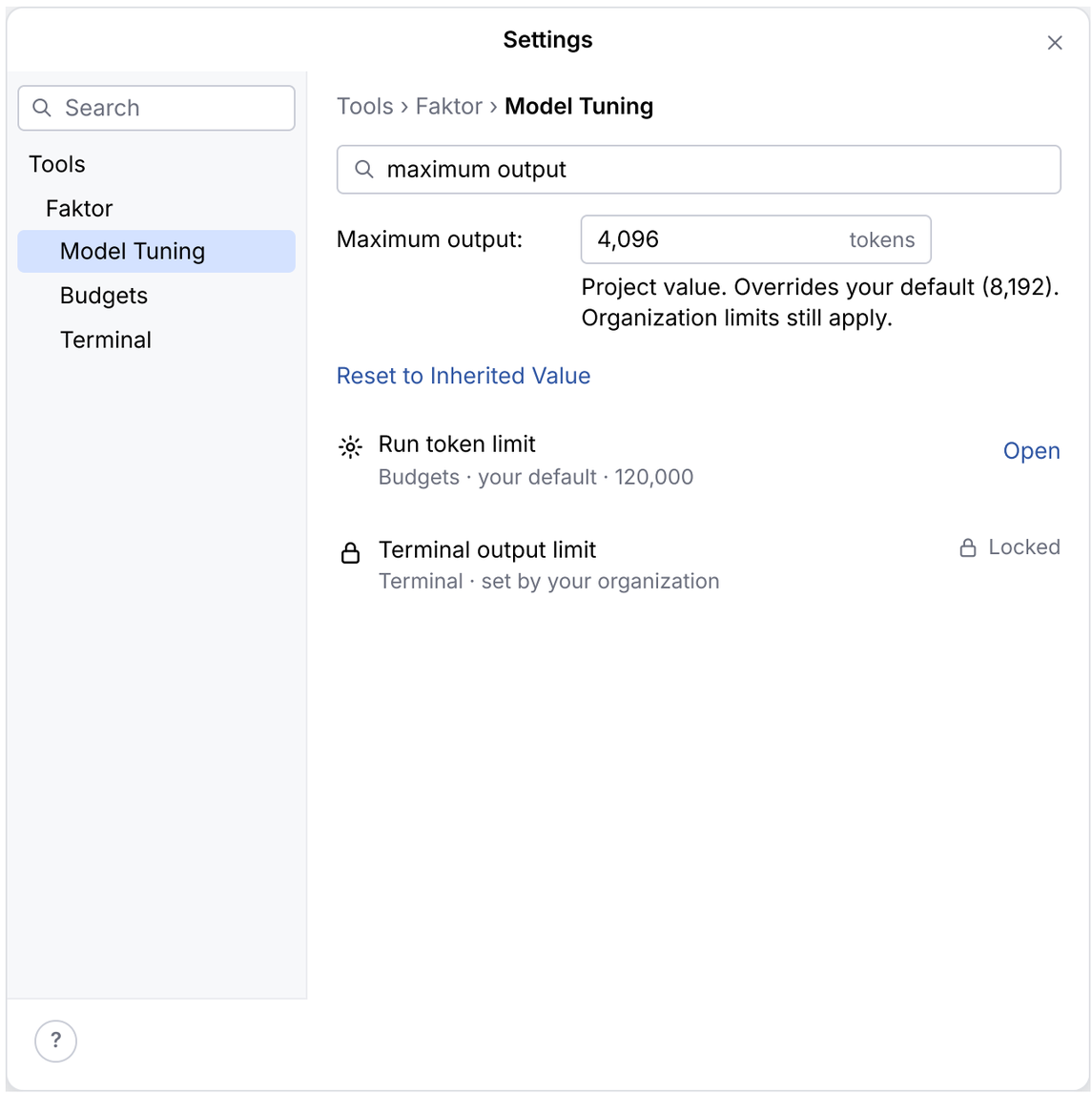
PRIMARY ACTION

VS Code: Settings saves valid values · JetBrains: OK / Apply

VS CODE · DARK MODERN — SETTINGS EDITOR · NATIVE



JETBRAINS · LIGHT — SETTINGS DIALOG · CONFIGURABLE



CONTROLS O-01 D-11

REPLACES R7 page 071 · R4-S30

CONTRACT p. 168

S64 · SETTINGS AND DIAGNOSTICS · BEHAVIOUR CONTRACT

Settings search and provenance: contract and revision notes

Original contract retained verbatim (R4-S30 / controls: O-01 D-11). Design on p. 167.

What changed in revision 8

- Provenance uses each host's own indicators (VS Code "Modified in Workspace"; JetBrains search highlighting).

Entry and return

Native Settings or the originating configuration command. Work / Inspect / History remain the only top-level clusters. Settings, Usage and Diagnostics use explicit Back. Return to the originating object, not the first row.

Scope and captured state

Application/project/connection plus effective revision; never infer task scope from an open form. Keep staged input, the effective snapshot and the acknowledged applied revision separate. Apply is atomic; Cancel preserves the previous effective configuration.

Primary action and recovery

Open model tuning. Associate validation with the exact field. A revision conflict preserves all staged values and opens the conflict resolver. Authentication, transport and schema failures have different recovery.

Loading, empty and stale

Keep the last coherent values with a refresh state. A first read has no invented defaults. An absent capability remains explained and inspectable.

Keyboard and focus

Open on the triggering setting. Apply failure focuses the first invalid field on submission; async refresh never moves the caret. Escape cancels a transient editor and returns to its trigger.

Layout and adverse content

Align only related fields within their group. Long inputs use stacked labels. Native preferred sizes and zoom win over reference-art dimensions.

Starting point and required construction

Starting classifications: H, N. E existing operation; R runtime-backed adapter; N new contract; H host-only work. Exact per-control classifications remain in the catalog.

Reuse the reviewed domain operations and preserve all served fields. Build every missing native read/write adapter, capability gate and state renderer identified by the matching control/build rows.

BK01 owns validation, scope, persistence, revision comparison and acknowledgement; domain packages own parameter/capability enforcement.

Required interaction fixtures

Exercise first-read, empty/no-match, ready, refreshing/stale, error/refusal, action pending and acknowledged result. Also switch scope while the request is in flight.

Moved out of the interface

Every line of revision 7 plate copy that left the default view, and where it lives now.

"Faktor / Settings (R7 window title)"
→ Native host chrome (VS Code view title, tabs or editor tab; JetBrains tool-window tabs, dialog title or Settings breadcrumb)

"Faktor settings / Persistent defaults use the native host settings surface."
→ The plate is the native Settings editor (VS Code) / Settings dialog (JetBrains)

"Maximum output / model tuning / Modified / Project override · 4,096 tokens · next admitted call"
→ Native "modified" indicator and help

"Run token cap / budgets / User default · 120,000 tokens · next run"
→ Search result

"Terminal output cap / tools / Machine policy · bounded output · next owned process"
→ Locked search result

"Selected setting provenance / Effective source / Project setting overrides a user default. Managed restrictions still win."
→ Help text (shortened)

"Reset to inherited / Review / Remove this override; do not write a guessed default."
→ Reset to Inherited Value (VS Code: gear menu › Reset Setting); rule in Specification: screen contract

"Clear search / Open model tuning"
→ Native search field clear / settings navigation

"maximum output"
→ Settings search text

S60 SETTINGS AND DIAGNOSTICS

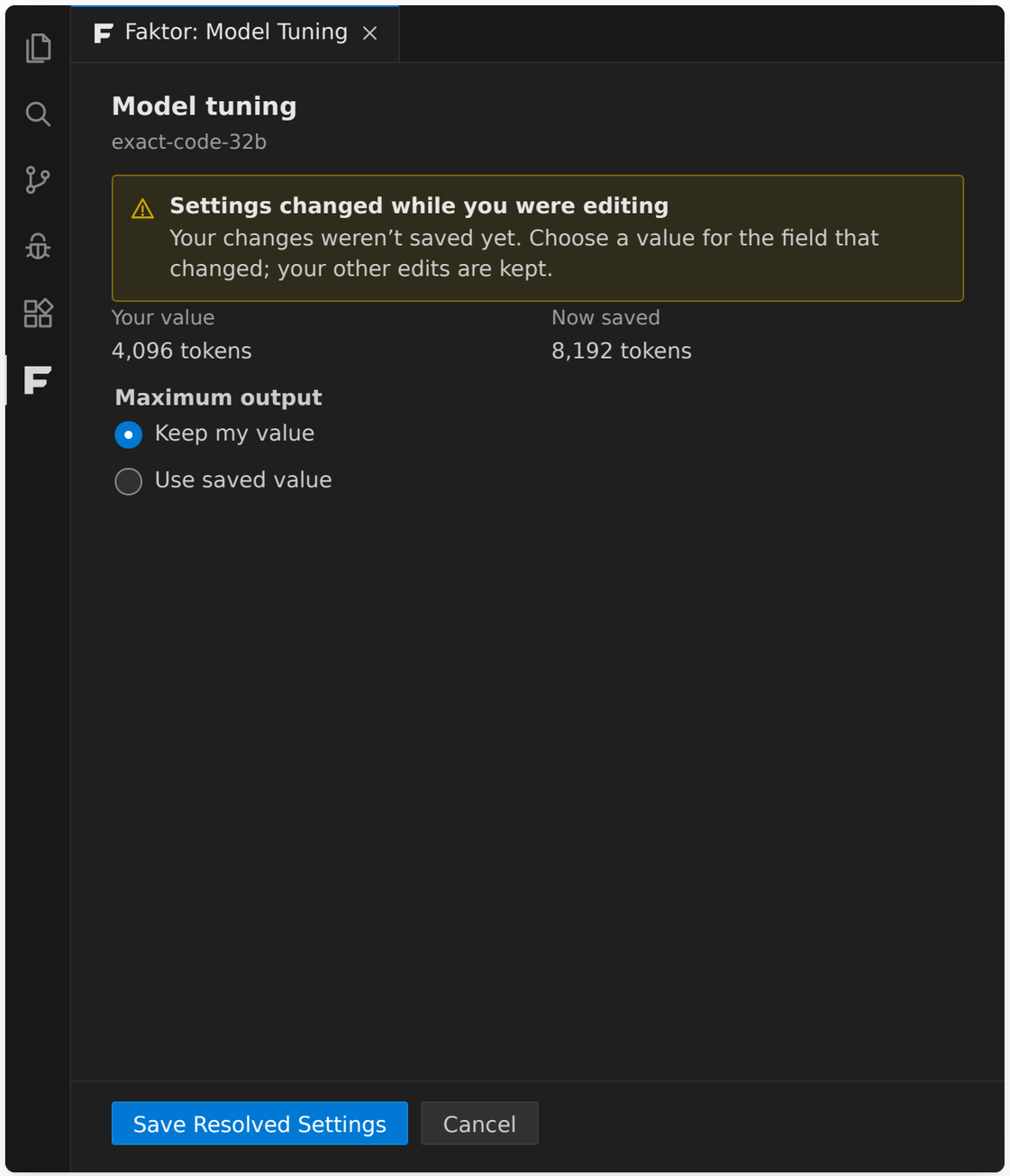
Settings revision conflict

Resolve a rejected configuration patch against the new effective revision without losing staged edits.

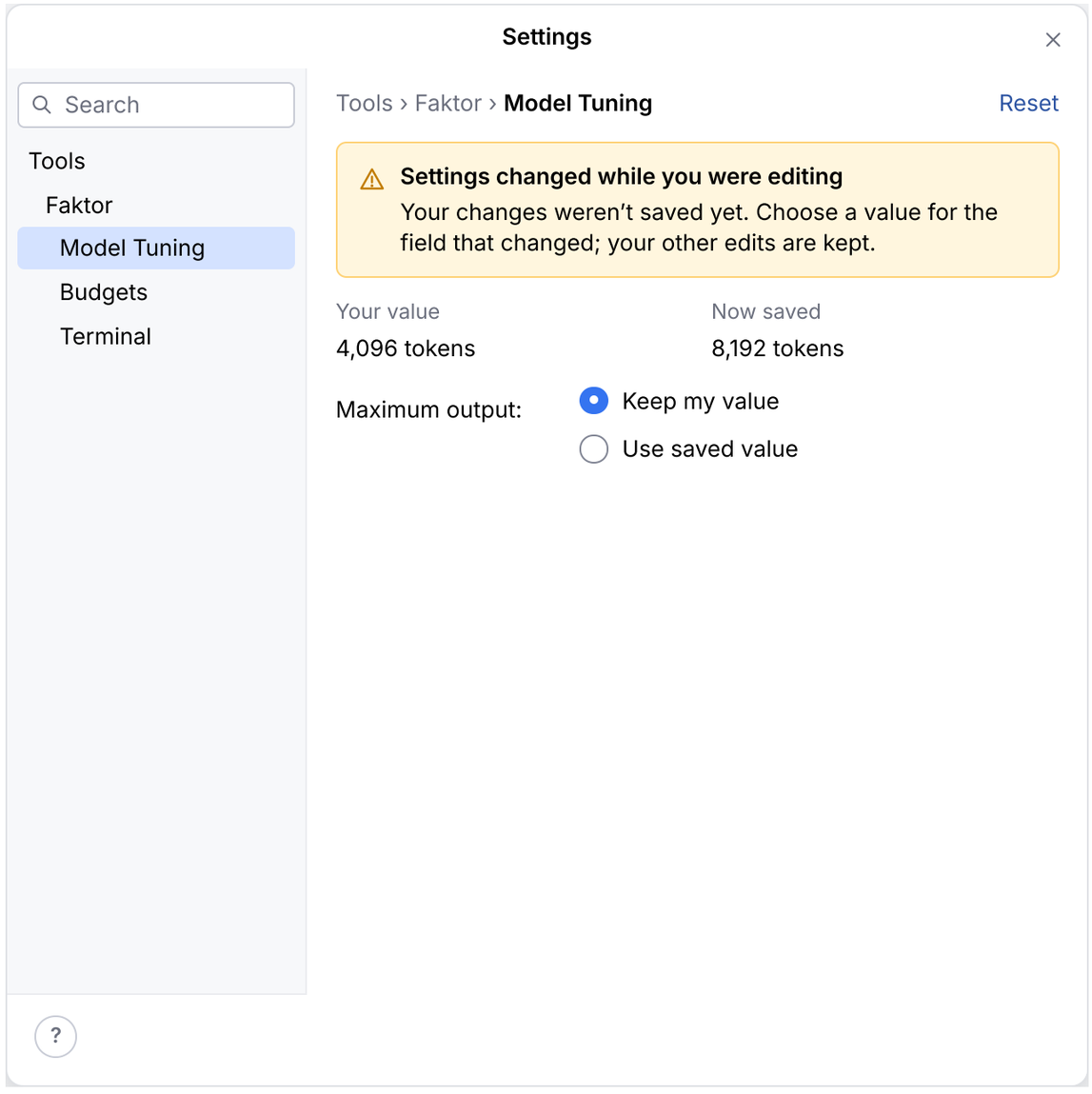
PRIMARY ACTION

Save Resolved Settings · JetBrains: OK / Apply

VS CODE · DARK MODERN — EDITOR TAB · WEBVIEW SETTINGS



JETBRAINS · LIGHT — SETTINGS DIALOG · CONFIGURABLE



CONTROLS O-01 M-12

REPLACES R7 page 072 · Expanded standard-agent surface

CONTRACT p. 170

S60 · SETTINGS AND DIAGNOSTICS · BEHAVIOUR CONTRACT

Settings revision conflict: contract and revision notes

Original contract retained verbatim (Expanded standard-agent surface / controls: O-01 M-12). Design on p. 169.

What changed in revision 8

- Conflict resolution is a single choice on the one field that changed.

Entry and return

Native Settings or the originating configuration command. Work / Inspect / History remain the only top-level clusters. Settings, Usage and Diagnostics use explicit Back. Return to the originating object, not the first row.

Scope and captured state

Application/project/connection plus effective revision; never infer task scope from an open form. Keep staged input, the effective snapshot and the acknowledged applied revision separate. Apply is atomic; Cancel preserves the previous effective configuration.

Primary action and recovery

Review resolved patch. Associate validation with the exact field. A revision conflict preserves all staged values and opens the conflict resolver.

Authentication, transport and schema failures have different recovery.

Loading, empty and stale

Keep the last coherent values with a refresh state. A first read has no invented defaults. An absent capability remains explained and inspectable.

Keyboard and focus

Open on the triggering setting. Apply failure focuses the first invalid field on submission; async refresh never moves the caret. Escape cancels a transient editor and returns to its trigger.

Layout and adverse content

Align only related fields within their group. Long inputs use stacked labels. Native preferred sizes and zoom win over reference-art dimensions.

Starting point and required construction

Starting classifications: N. E existing operation; R runtime-backed adapter; N new contract; H host-only work. Exact per-control classifications remain in the catalog.

NEW: add expected-revision compare-and-swap, atomic field validation and separate persistence/runtime-applied acknowledgements.

BK01 owns validation, scope, persistence, revision comparison and acknowledgement; domain packages own parameter/capability enforcement.

Required interaction fixtures

Exercise first-read, empty/no-match, ready, refreshing/stale, error/refusal, action pending and acknowledged result. Also switch scope while the request is in flight.

Moved out of the interface

Every line of revision 7 plate copy that left the default view, and where it lives now.

"Faktor / Settings / Conflict (R7 window title)"
→ Native host chrome (VS Code view title, tabs or editor tab; JetBrains tool-window tabs, dialog title or Settings breadcrumb)

"Settings changed elsewhere / Your patch was not applied. Effective revision is now 19."
→ Banner; revision numbers in Tooltip / hover help

"Review the conflicting field / The machine maximum-output default changed after you opened revision 18. Your unsaved values remain intact."
→ Banner text

"Your staged maximum / Effective revision 19"
→ Facts "Your value" / "Now saved"

"Unchanged fields / Preserved / Keep non-conflicting edits in the staged patch"
→ "your other edits are kept"

"Resolve with / Use effective / Keep my value"
→ Radio choice

"The service revalidates the resolved patch against revision 19. There is no silent last-write-wins update."
→ Specification: screen contract

"Review resolved patch"
→ Save Resolved Settings

"Cancel"
→ Cancel

S55 SETTINGS AND DIAGNOSTICS

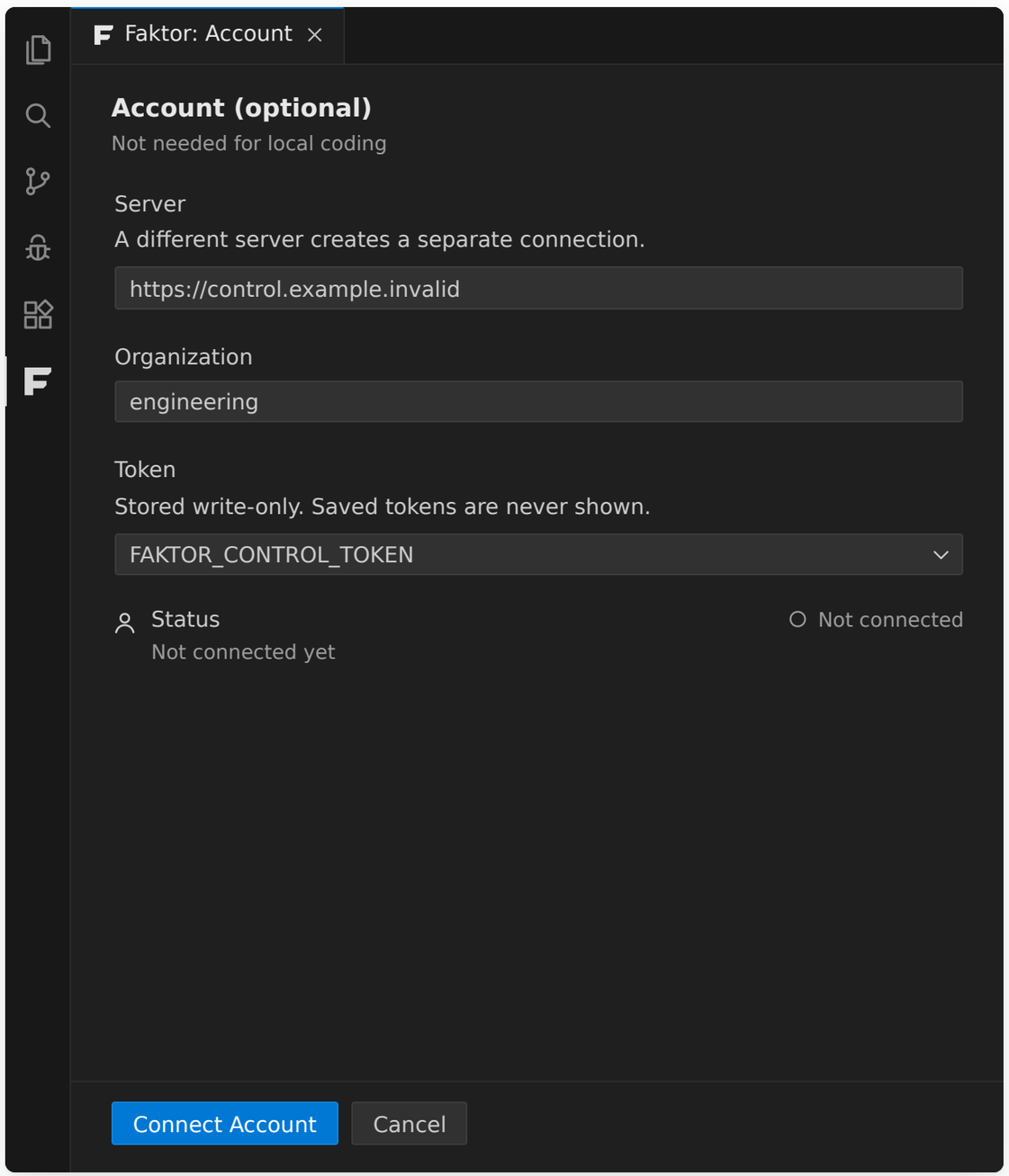
Optional control-plane account

Connect a machine-scoped optional account with write-only secrets while local coding remains independent.

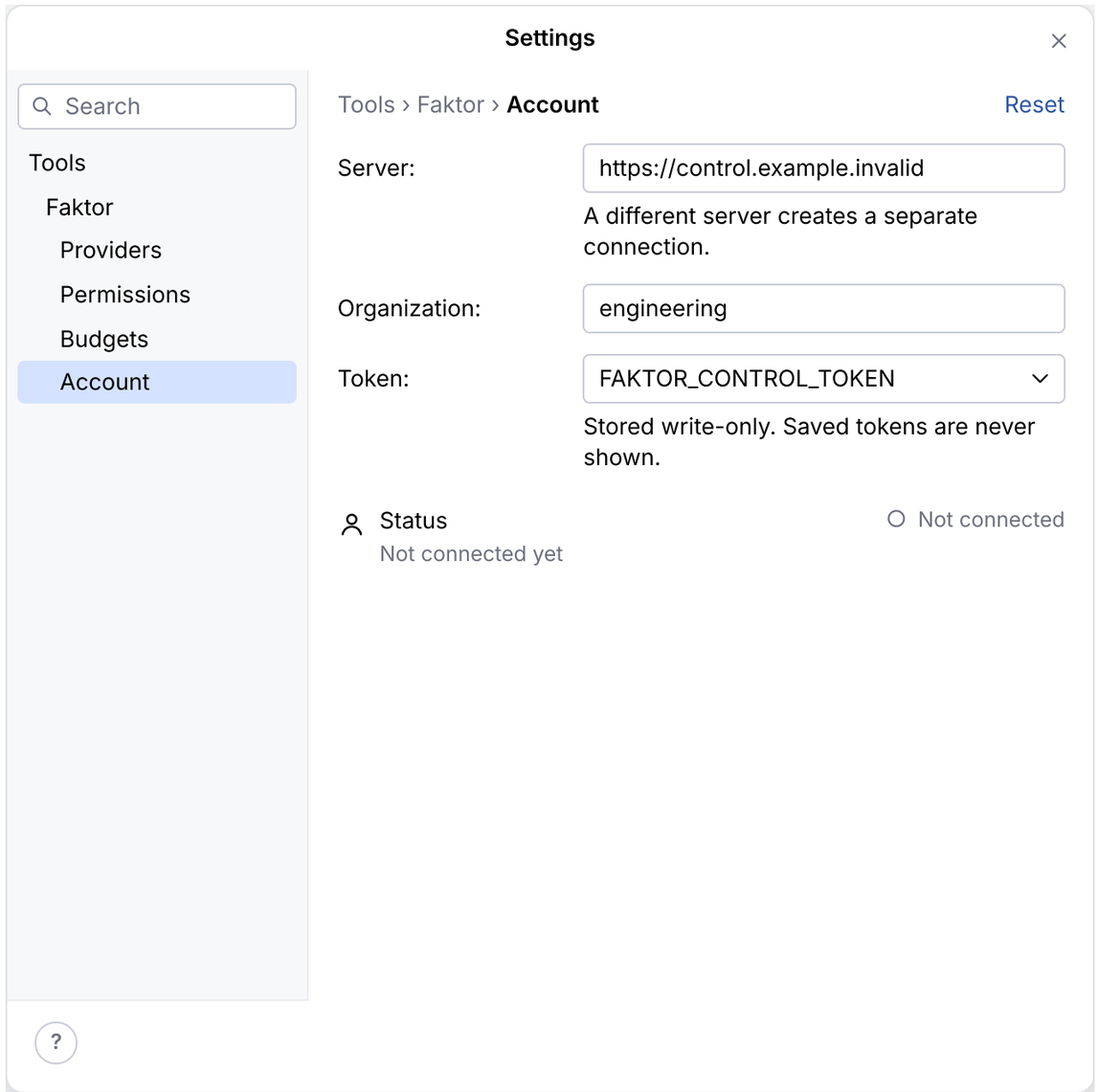
PRIMARY ACTION

Connect Account · JetBrains: OK / Apply

VS CODE · DARK MODERN — EDITOR TAB · WEBVIEW SETTINGS



JETBRAINS · LIGHT — SETTINGS DIALOG · CONFIGURABLE



CONTROLS O-07 P-04 — REPLACES R7 page 073 · R4-S31 — CONTRACT p. 172

S55 · SETTINGS AND DIAGNOSTICS · BEHAVIOUR CONTRACT

Optional control-plane account: contract and revision notes

Original contract retained verbatim (R4-S31 / controls: O-07 P-04). Design on p. 171.

What changed in revision 8

- "Optional" is in the title, so users never think an account is required.

Entry and return

Native Settings or the originating configuration command. Work / Inspect / History remain the only top-level clusters. Settings, Usage and Diagnostics use explicit Back. Return to the originating object, not the first row.

Scope and captured state

Application/project/connection plus effective revision; never infer task scope from an open form. Keep staged input, the effective snapshot and the acknowledged applied revision separate. Apply is atomic; Cancel preserves the previous effective configuration.

Primary action and recovery

Connect account. Associate validation with the exact field. A revision conflict preserves all staged values and opens the conflict resolver. Authentication, transport and schema failures have different recovery.

Loading, empty and stale

Keep the last coherent values with a refresh state. A first read has no invented defaults. An absent capability remains explained and inspectable.

Keyboard and focus

Open on the triggering setting. Apply failure focuses the first invalid field on submission; async refresh never moves the caret. Escape cancels a transient editor and returns to its trigger.

Layout and adverse content

Align only related fields within their group. Long inputs use stacked labels. Native preferred sizes and zoom win over reference-art dimensions.

Starting point and required construction

Starting classifications: E, N. E existing operation; R runtime-backed adapter; N new contract; H host-only work. Exact per-control classifications remain in the catalog.

Reuse the reviewed domain operations and preserve all served fields. Build every missing native read/write adapter, capability gate and state renderer identified by the matching control/build rows.

BK01 owns validation, scope, persistence, revision comparison and acknowledgement; domain packages own parameter/capability enforcement.

Required interaction fixtures

Exercise first-read, empty/no-match, ready, refreshing/stale, error/refusal, action pending and acknowledged result. Also switch scope while the request is in flight.

Moved out of the interface

Every line of revision 7 plate copy that left the default view, and where it lives now.

"Faktor / Settings / Account (R7 window title)"
→ Native host chrome (VS Code view title, tabs or editor tab; JetBrains tool-window tabs, dialog title or Settings breadcrumb)

"Optional control plane / Machine account binding · not required for local coding"
→ Title and subtitle

"Endpoint"
→ Label "Server"

"A different endpoint creates a different binding."
→ Help

"Credential reference"
→ Label "Token"

"Write-only secret storage. The saved secret is never shown."
→ Help

"Connection status / Not connected / No authenticated account read has completed"
→ Status row

"Removing this credential forgets the local reference. Revoking a remote session is a separate server operation."
→ Confirmation text of the Remove action (shown when connected) and Product documentation

"https://control.example.invalid / engineering / FAKTOR_CONTROL_TOKEN"
→ Field values

"Cancel / Connect account"
→ Cancel / Connect Account

S30 SETTINGS AND DIAGNOSTICS

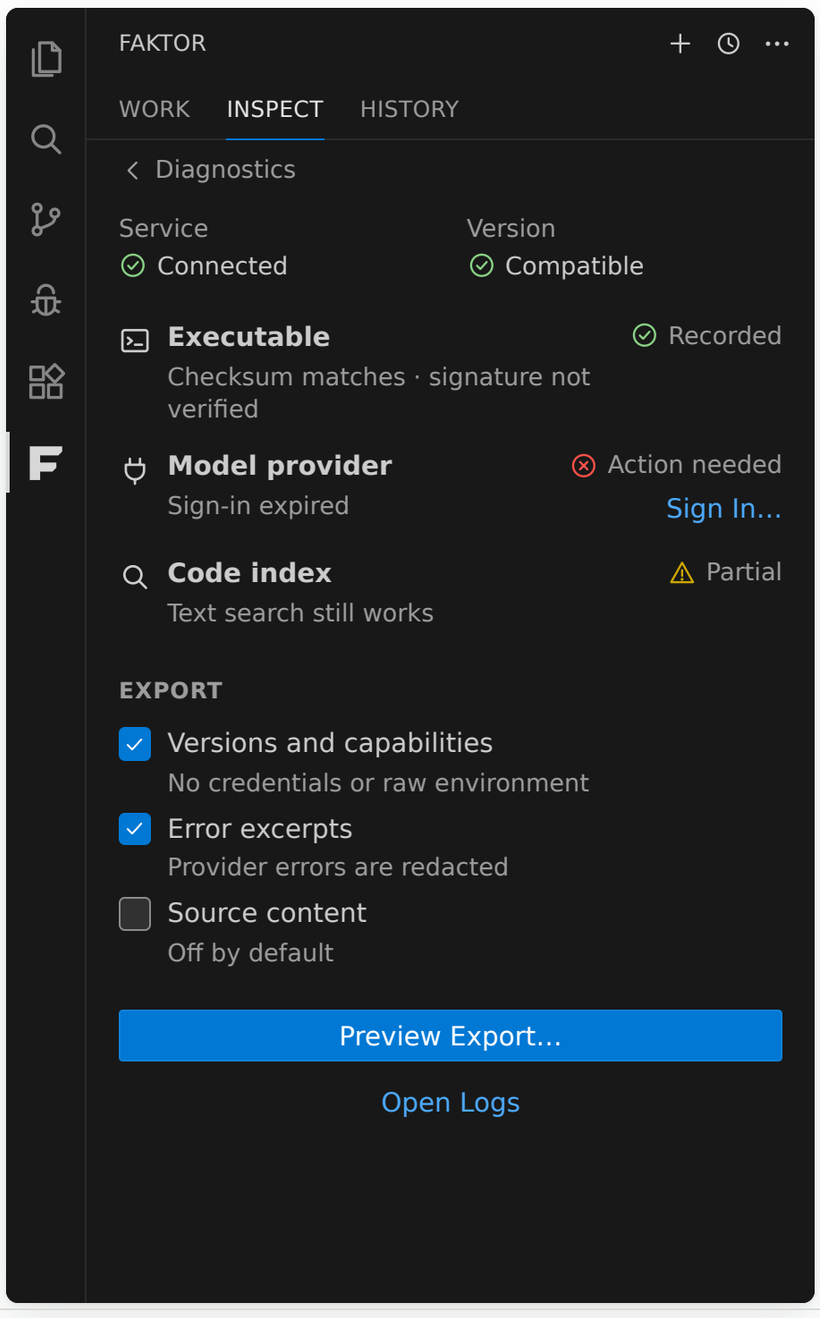
Diagnostics

Show readiness, version compatibility, runtime identity and bounded diagnostics with an actionable typed cause.

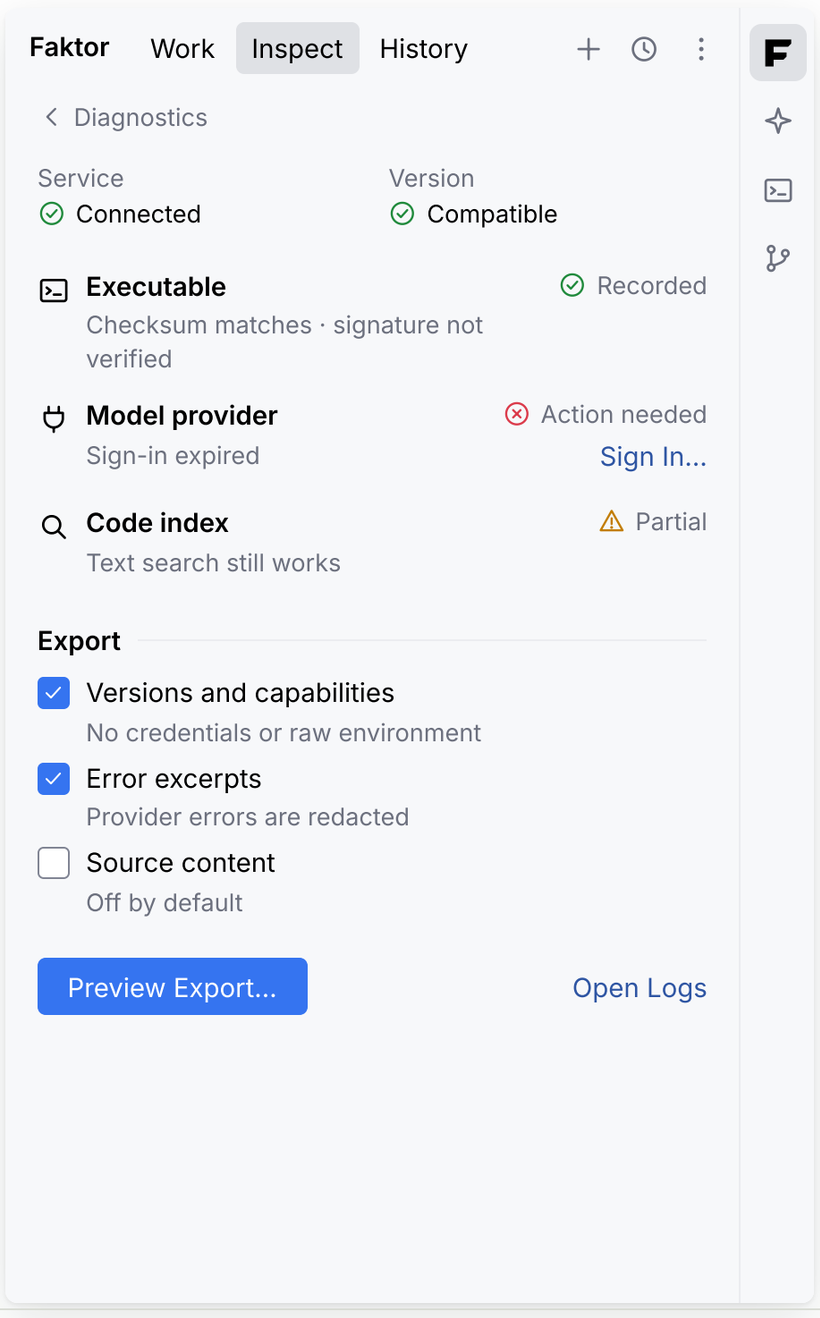
PRIMARY ACTION

Preview Export…

VS CODE · DARK MODERN — SIDEBAR VIEW · WEBVIEW



JETBRAINS · LIGHT — TOOL WINDOW · SWING / KOTLIN UI DSL



CONTROLS O-03 O-04 O-05 O-07 O-09 O-13 B-11 B-12 D-13

REPLACES R7 page 074 · R4-S32

CONTRACT p. 174

S30 · SETTINGS AND DIAGNOSTICS · BEHAVIOUR CONTRACT

Diagnostics: contract and revision notes

Original contract retained verbatim (R4-S32 / controls: O-03 O-04 O-05 O-07 O-09 O-13 B-11 B-12 D-13). Design on p. 173.

What changed in revision 8

- Every problem row carries its fix; healthy items stay quiet.

Entry and return

More / Diagnostics, the setup walkthrough, or the exact blocked operation. Work / Inspect / History remain the only top-level clusters. Settings, Usage and Diagnostics use explicit Back. Return to the originating object, not the first row.

Scope and captured state

Execution host + service instance/generation + protocol version/cursor + trust source; account bindings stay distinct. Starting, restarting and stopping the service are explicit owned operations. Rebind after recovery; opening a view is not a start command.

Primary action and recovery

Review a redacted diagnostic preview before export; explicit Start is separate from opening the tool window.. Checksum mismatch refuses start; an unrecorded exact path needs deliberate trust. A checksum match does not claim publisher-signature verification.

Loading, empty and stale

Keep last coherent data stale, retain event cursors and distinguish automatic transient reconnect from a protocol-blocked stop.

Keyboard and focus

A background disconnect never steals typing focus. Recovery opens in a native command/dialog; Back returns to the blocked task. Copy diagnostics confirms once.

Layout and adverse content

Show task-facing recovery first, technical detail behind disclosure. Wrap exact material paths; redact secrets before display and before copy/export.

Starting point and required construction

Starting classifications: E, N, R. E existing operation; R runtime-backed adapter; N new contract; H host-only work. Exact per-control classifications remain in the catalog.

Reuse the reviewed domain operations and preserve all served fields. Build every missing native read/write adapter, capability gate and state renderer identified by the matching control/build rows.

BK10 owns readiness/discovery/diagnostics; BK08 fences stream and operation owners; BK05 reports actual isolation and policy.

Required interaction fixtures

Exercise first-read, empty/no-match, ready, refreshing/stale, error/refusal, action pending and acknowledged result. Also switch scope while the request is in flight.

Moved out of the interface

Every line of revision 7 plate copy that left the default view, and where it lives now.

"Faktor / Diagnostics (R7 window title)"
→ Native host chrome (VS Code view title, tabs or editor tab; JetBrains tool-window tabs, dialog title or Settings breadcrumb)

"Last coherent snapshot retained while individual reads recover."
→ Specification: screen contract (stale-snapshot rule)

"Runtime / Connected; Protocol / Compatible"
→ Facts "Service" and "Version"

"Executable trust / Recorded / Recorded checksum matches. Publisher signature is not established."
→ Row (shortened)

"Model provider / Auth expired / Credential refresh required"
→ Row "Sign-in expired — Action needed — Sign In…"

"Index coverage / Degraded / Lexical fallback remains available"
→ Row "Code index — Partial — Text search still works"

"Diagnostic export"
→ Section "Export"

"No credential values or raw environment dump."
→ Checkbox help

"Provider errors are redacted before export."
→ Checkbox help

"Source payloads / Off by default. Preview the exact inclusion scope."
→ Checkbox and Preview Export…

"Preview diagnostics"
→ Preview Export…

"Diagnostics"
→ Breadcrumb

"Bounded error excerpts"
→ Checkbox "Error excerpts"

"Versions and capabilities / Source payloads"
→ Checkboxes

"Open logs"
→ Open Logs

S56 SETTINGS AND DIAGNOSTICS

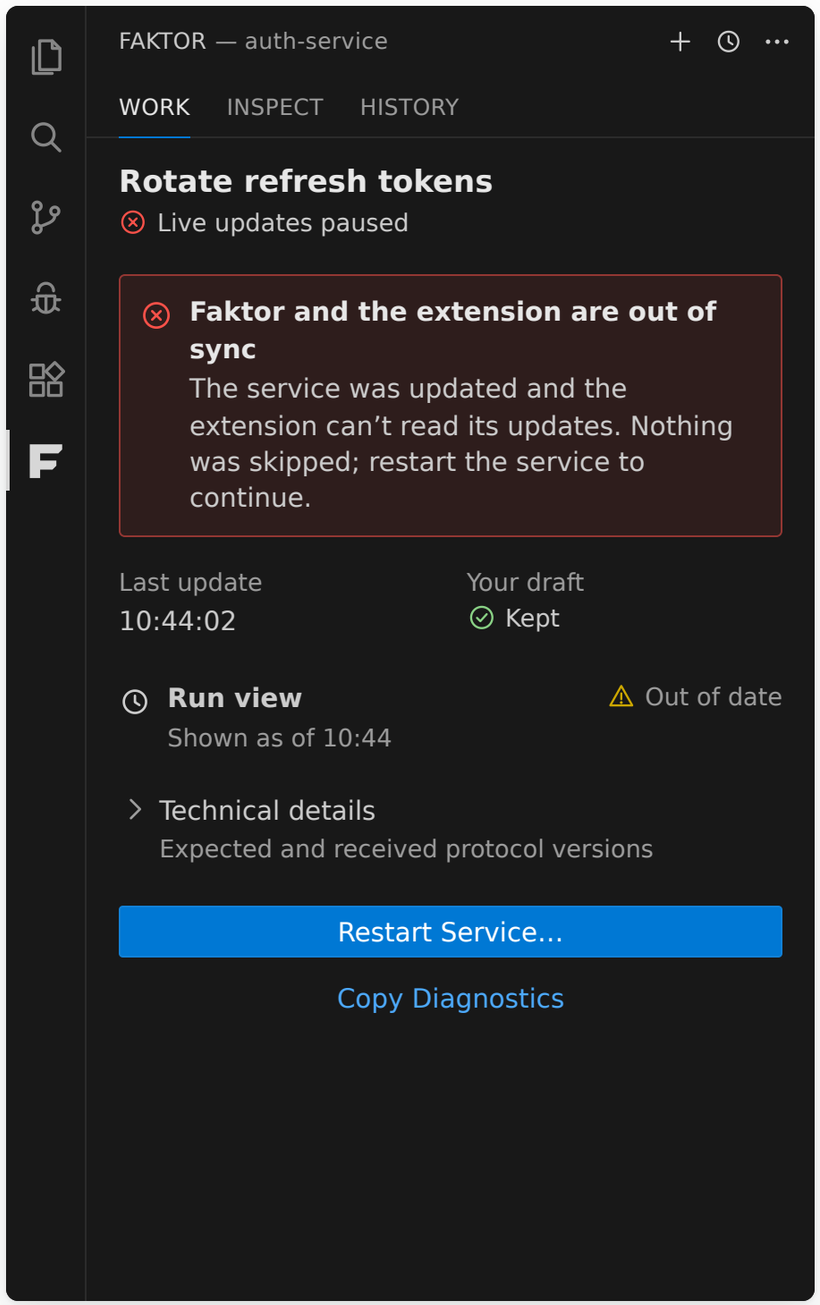
PRIMARY ACTION

Restart Service...

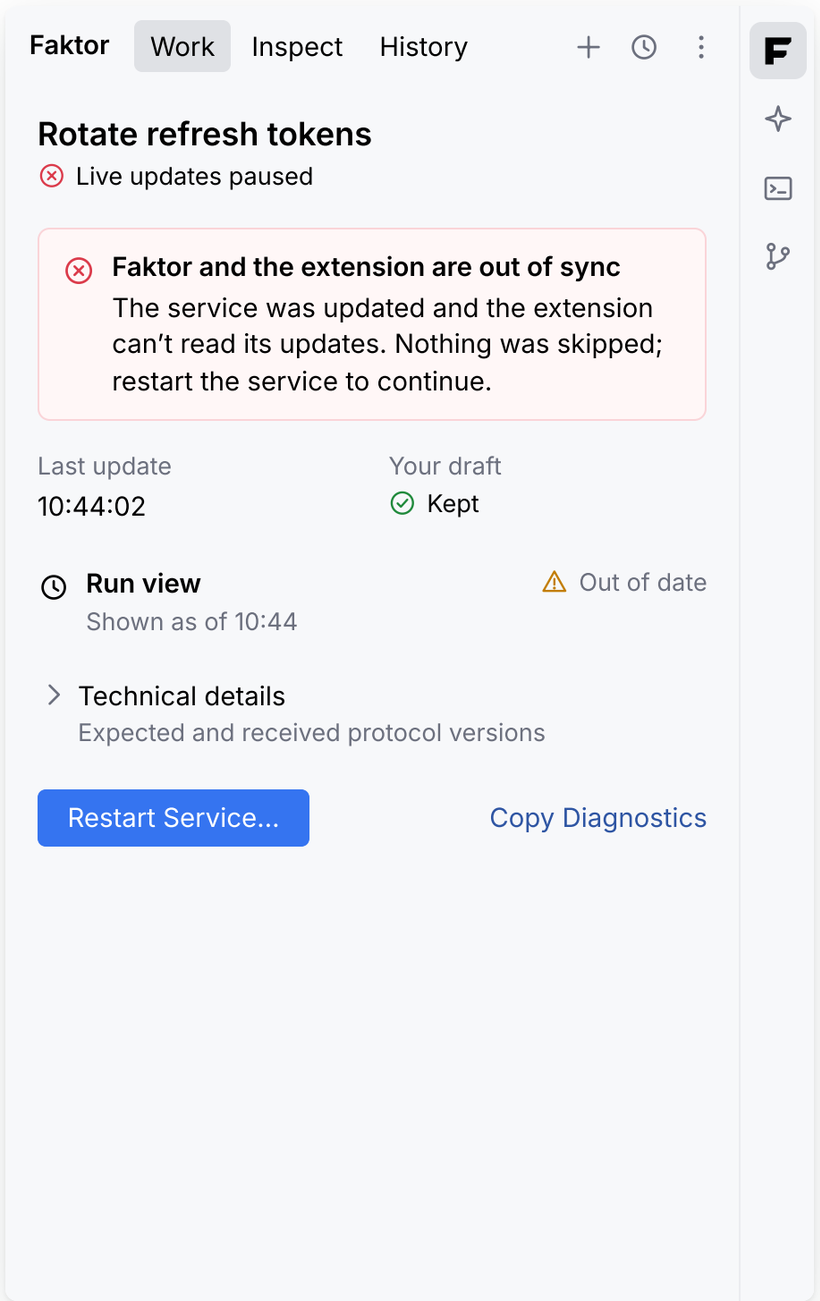
Protocol-blocked recovery

Preserve the last coherent snapshot and cursor when a typed stream refusal pauses automatic recovery.

VS CODE · DARK MODERN — SIDEBAR VIEW · WEBVIEW



JETBRAINS · LIGHT — TOOL WINDOW · SWING / KOTLIN UI DSL



CONTROLS O-04 D-13

REPLACES R7 page 075 · R4-S33

CONTRACT p. 176

S56 · SETTINGS AND DIAGNOSTICS · BEHAVIOUR CONTRACT

Protocol-blocked recovery: contract and revision notes

Original contract retained verbatim (R4-S33 / controls: O-04 D-13). Design on p. 175.

What changed in revision 8

- Explains the situation in user terms and offers one recovery action.

Entry and return

More / Diagnostics, the setup walkthrough, or the exact blocked operation. Work / Inspect / History remain the only top-level clusters. Settings, Usage and Diagnostics use explicit Back. Return to the originating object, not the first row.

Scope and captured state

Execution host + service instance/generation + protocol version/cursor + trust source; account bindings stay distinct. Starting, restarting and stopping the service are explicit owned operations. Rebind after recovery; opening a view is not a start command.

Primary action and recovery

Open service recovery. Show a bounded human reason and actionable recovery. Missing checksum, checksum mismatch, signature status, schema refusal and local/cloud boundary are separate facts.

Loading, empty and stale

Keep last coherent data stale, retain event cursors and distinguish automatic transient reconnect from a protocol-blocked stop.

Keyboard and focus

A background disconnect never steals typing focus. Recovery opens in a native command/dialog; Back returns to the blocked task. Copy diagnostics confirms once.

Layout and adverse content

Show task-facing recovery first, technical detail behind disclosure. Wrap exact material paths; redact secrets before display and before copy/export.

Starting point and required construction

Starting classifications: E. E existing operation; R runtime-backed adapter; N new contract; H host-only work. Exact per-control classifications remain in the catalog.

Reuse the reviewed domain operations and preserve all served fields. Build every missing native read/write adapter, capability gate and state renderer identified by the matching control/build rows.

BK10 owns readiness/discovery/diagnostics; BK08 fences stream and operation owners; BK05 reports actual isolation and policy.

Required interaction fixtures

Exercise first-read, empty/no-match, ready, refreshing/stale, error/refusal, action pending and acknowledged result. Also switch scope while the request is in flight.

Moved out of the interface

Every line of revision 7 plate copy that left the default view, and where it lives now.

"Faktor / Diagnostics / Live updates (R7 window title)"
→ Native host chrome (VS Code view title, tabs or editor tab; JetBrains tool-window tabs, dialog title or Settings breadcrumb)

"Live updates paused / Run 047 · last coherent event 281 · service protocol refused"
→ Status and facts; event number in Technical details

"An explicit recovery is required"
→ Primary action Restart Service…

"The stream reported an incompatible schema. Automatic reconnect is paused so events are not skipped."
→ Banner text in plain language

"Last durable cursor / 281"
→ Fact "Last update"; cursor in Technical details

"Draft and selection / Preserved"
→ Fact "Your draft — Kept"

"Last coherent run view / Stale / Clearly marked stale · no new success is inferred"
→ Row "Run view — Out of date"

"Protocol detail / Inspect / Expected version, received version and typed reason"
→ Technical details disclosure

"After compatible service recovery, refresh the snapshot and reconcile events before resuming controls."
→ Specification: screen contract (recovery sequence)

"Open service recovery"
→ Restart Service…

"Copy diagnostics"
→ Copy Diagnostics

S31 SETTINGS AND DIAGNOSTICS

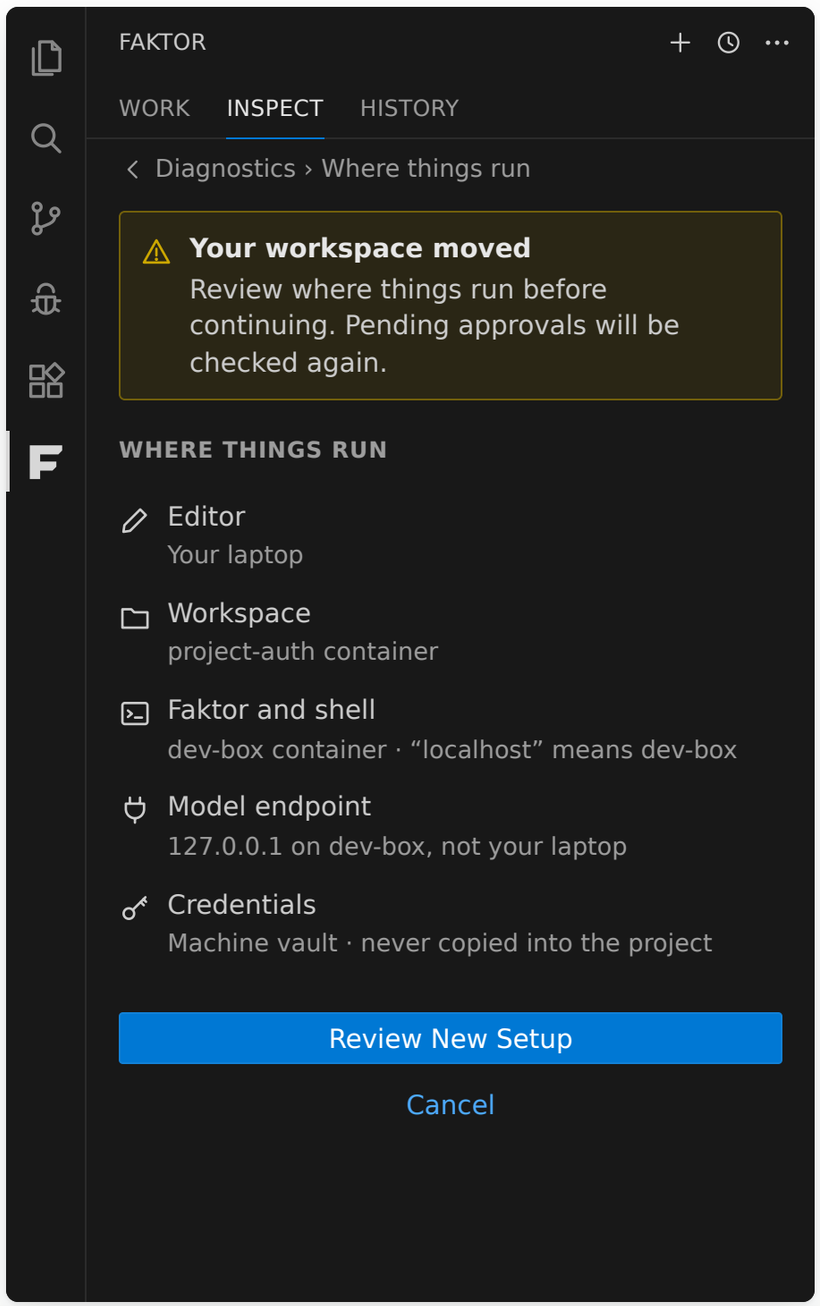
Workspace binding

PRIMARY ACTION

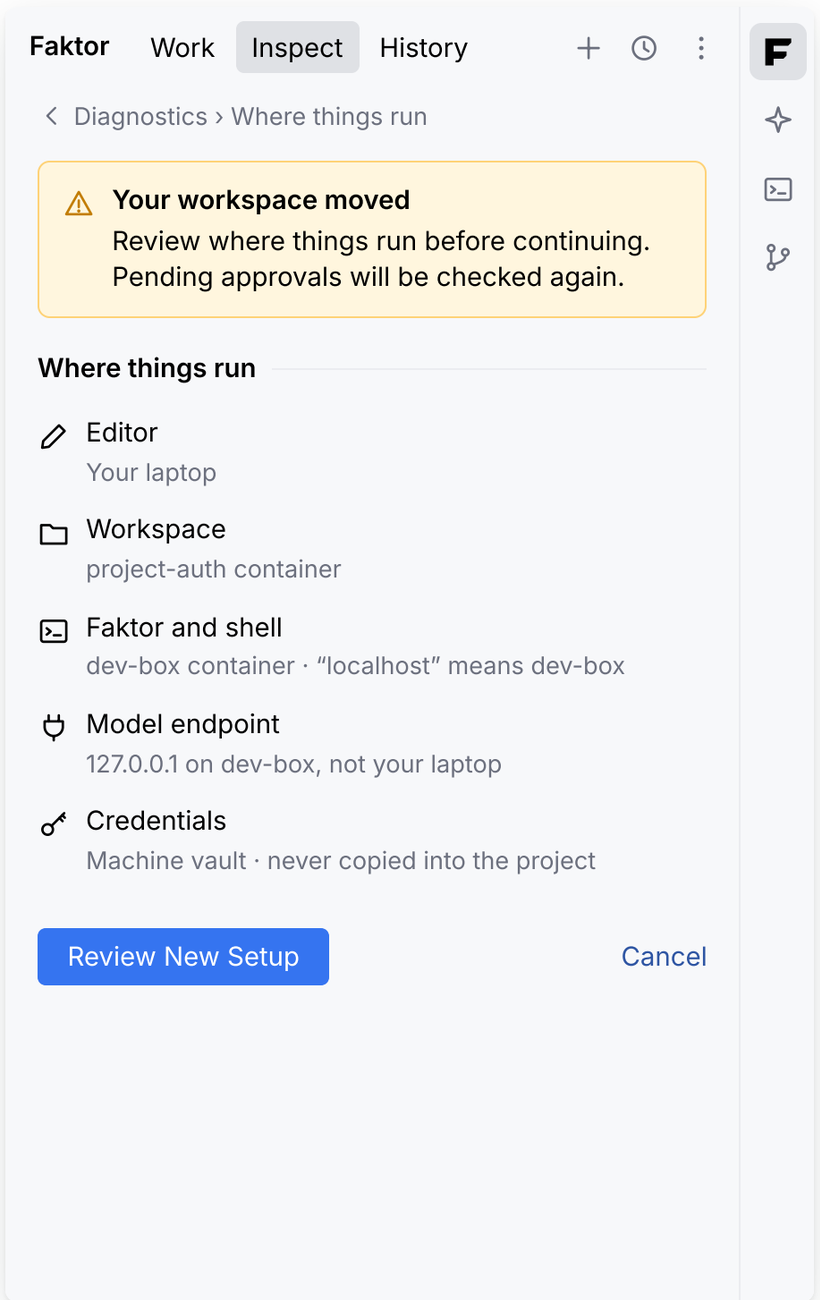
Review New Setup

Display editor, execution, model, vault and candidate locations together. Remote workspaces must not blur local/hosted boundaries or path ownership.

VS CODE · DARK MODERN — SIDEBAR VIEW · WEBVIEW



JETBRAINS · LIGHT — TOOL WINDOW · SWING / KOTLIN UI DSL



CONTROLS X-13 P-13 O-02 O-07 O-11 D-01 D-15 A-11

REPLACES R7 page 076 · Expanded standard-agent surface

CONTRACT p. 178

S31 · SETTINGS AND DIAGNOSTICS · BEHAVIOUR CONTRACT

Workspace binding: contract and revision notes

Original contract retained verbatim (Expanded standard-agent surface / controls: X-13 P-13 O-02 O-07 O-11 D-01 D-15 A-11). Design on p. 177.

What changed in revision 8

- Locality is a single list, read top to bottom from your laptop to the model.

Entry and return

More / Diagnostics, the setup walkthrough, or the exact blocked operation. Work / Inspect / History remain the only top-level clusters. Settings, Usage and Diagnostics use explicit Back. Return to the originating object, not the first row.

Scope and captured state

Execution host + service instance/generation + protocol version/cursor + trust source; account bindings stay distinct. Starting, restarting and stopping the service are explicit owned operations. Rebind after recovery; opening a view is not a start command.

Primary action and recovery

Bind a conversation to one workspace/worktree/daemon generation.

Revalidate after container, SSH or execution-host changes.. A host change invalidates process and permission handles until rebound; preserve drafts without replaying a captured effectful request elsewhere.

Loading, empty and stale

Keep last coherent data stale, retain event cursors and distinguish automatic transient reconnect from a protocol-blocked stop.

Keyboard and focus

A background disconnect never steals typing focus. Recovery opens in a native command/dialog; Back returns to the blocked task. Copy diagnostics confirms once.

Layout and adverse content

Show task-facing recovery first, technical detail behind disclosure. Wrap exact material paths; redact secrets before display and before copy/export.

Starting point and required construction

Starting classifications: E, H, N, R. E existing operation; R runtime-backed adapter; N new contract; H host-only work. Exact per-control classifications remain in the catalog.

Reuse the reviewed domain operations and preserve all served fields. Build every missing native read/write adapter, capability gate and state renderer identified by the matching control/build rows.

BK10 owns readiness/discovery/diagnostics; BK08 fences stream and operation owners; BK05 reports actual isolation and policy.

Required interaction fixtures

Exercise first-read, empty/no-match, ready, refreshing/stale, error/refusal, action pending and acknowledged result. Also switch scope while the request is in flight.

Moved out of the interface

Every line of revision 7 plate copy that left the default view, and where it lives now.

"Faktor / Workspace binding (R7 window title)"
→ Native host chrome (VS Code view title, tabs or editor tab; JetBrains tool-window tabs, dialog title or Settings breadcrumb)

"Workspace binding / Review execution locality when the remote workspace changes."
→ Title "Where things run" and banner

"Laptop · Native IDE UI and keymap"
→ Row "Editor — Your laptop"

"project-auth container · Canonical URIs and isolated candidate paths"
→ Row; path rules in Specification: screen contract

"dev-box container · Loopback URLs resolve here"
→ Row "'localhost' means dev-box"

"127.0.0.1 on dev-box · Local to execution host, not automatically the laptop"
→ Row

"Machine vault reference · Never copied into project files"
→ Row

"Rebind before resuming effects / Old process and approval handles are stale. Preserve drafts and reconcile pending operations."
→ Banner (user part); handle rules in Specification: screen contract

"Review new binding"
→ Review New Setup

"Daemon and shell"
→ Row "Faktor and shell"

"Credential binding"
→ Row "Credentials"

"Editor / Workspace / Model endpoint"
→ Rows

"Cancel"
→ Cancel

S58 SETTINGS AND DIAGNOSTICS

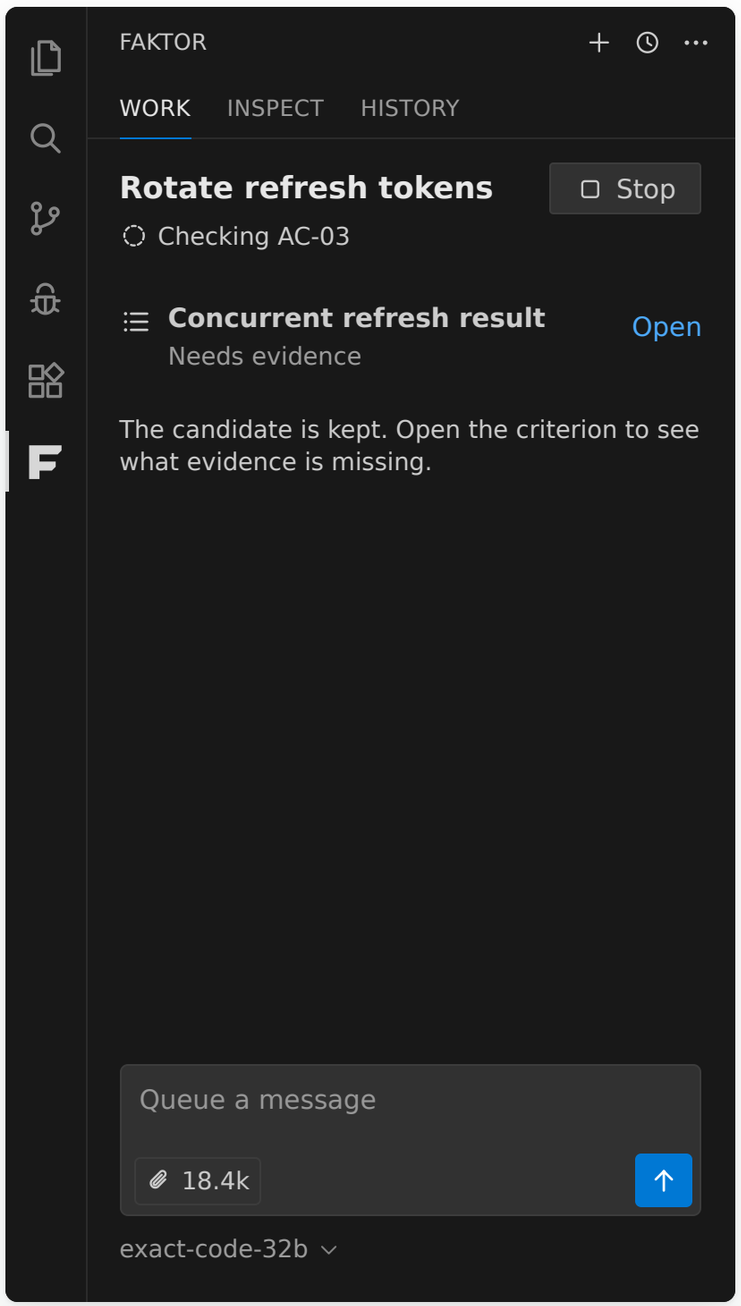
Narrow Work panel

Keep a 320 px Work panel readable with stacked controls, one selected detail and an independently reachable composer.

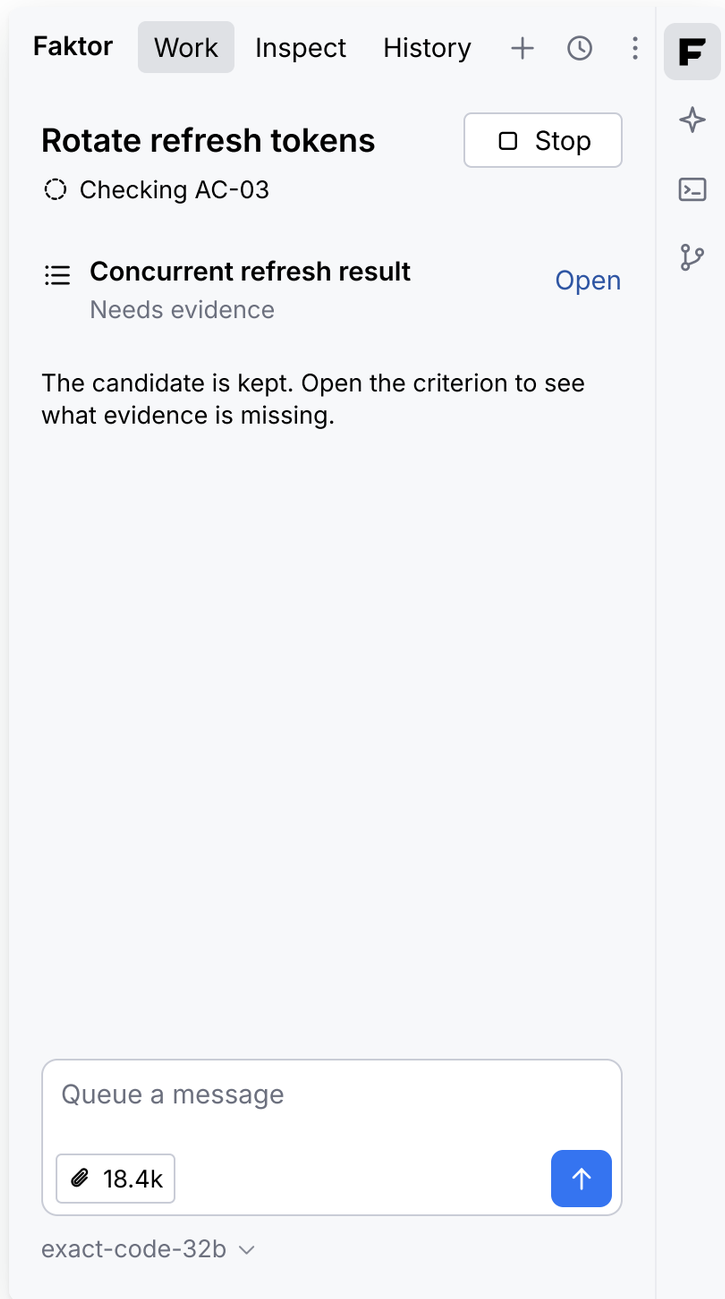
PRIMARY ACTION

Queue Message (composer)

VS CODE · DARK MODERN — SIDEBAR VIEW AT 320 PX



JETBRAINS · LIGHT — TOOL WINDOW AT 320 PX



CONTROLS D-08 D-09 D-14

REPLACES R7 page 077 · R4-S36

CONTRACT p. 180

S58 · SETTINGS AND DIAGNOSTICS · BEHAVIOUR CONTRACT

Narrow Work panel: contract and revision notes

Original contract retained verbatim (R4-S36 / controls: D-08 D-09 D-14). Design on p. 179.

What changed in revision 8

- At 320 px the header, one detail row and the composer stack; Stop stays in the header and remains reachable.

Entry and return

Work, a captured submission receipt, or a task-specific attention item. Work / Inspect / History remain the only top-level clusters. Settings, Usage and Diagnostics use explicit Back. Return to the originating object, not the first row.

Scope and captured state

Daemon/workspace/worktree/conversation/task/request generation. Draft scope is separate from an admitted run. Capture one immutable logical start. Clear only its acknowledged draft version; preserve text typed after capture. Queue items acquire their own receipt and revision.

Primary action and recovery

Queue message. Retain the captured body, options and source handles. Unknown admission/effect requires receipt reconciliation before replay. A changed body needs a different request identity.

Loading, empty and stale

Show the named operation and current owner. Keep the composer available for an independent draft. No fake percentages, fabricated tool steps or optimistic completion.

Keyboard and focus

Enter inserts a newline by default. The host send command respects IME composition. New approvals announce once without taking focus. Escape closes a detail; it never stops a run.

Layout and adverse content

Header, transcript scroll and composer are siblings. Collapse secondary metadata first. Keep a scoped Stop action and the active task reachable at short heights.

Starting point and required construction

Starting classifications: H. E existing operation; R runtime-backed adapter; N new contract; H host-only work. Exact per-control classifications remain in the catalog.

Reuse the reviewed domain operations and preserve all served fields. Build every missing native read/write adapter, capability gate and state renderer identified by the matching control/build rows.

BK08 owns admission, receipts, queue, control handles and reconciliation; BK05 owns authority and BK07 owns a shared exact budget ledger.

Required interaction fixtures

Exercise first-read, empty/no-match, ready, refreshing/stale, error/refusal, action pending and acknowledged result. Also switch scope while the request is in flight.

Moved out of the interface

Every line of revision 7 plate copy that left the default view, and where it lives now.

“Faktor / Work (R7 window title)”	→ Native host chrome (VS Code view title, tabs or editor tab; JetBrains tool-window tabs, dialog title or Settings breadcrumb)
“Run 047 · checking AC-03”	→ Header status “Checking AC-03”
“Concurrent refresh result / Pending / One criterion still needs evidence”	→ Row “Concurrent refresh result — Needs evidence — Open”
“The last candidate is retained. Open the selected criterion for its exact artifact and scope.”	→ Paragraph (shortened)
“Inspect the current criterion / Open / No second nested details pane”	→ Row action Open; one-detail rule in Specification: screen contract
“Next message / Keep existing sessions valid.”	→ Composer (queued text is shown in the Queue)
“Model / exact-code-32b”	→ Composer picker
“Stop run / Queue message”	→ Header Stop / send button
“Work”	→ Work tab
“Keep existing sessions valid.”	→ Queued message (Queue view)

PART 3

3 Validating real task flows in both IDEs

Screenshots and collision checks cannot establish this level of quality. This part defines the validation that must run in VS Code and JetBrains, with keyboard, assistive technology, zoom and narrow panels, and records what has been checked so far.

	SECTION	PAGE
3.1	Validation protocol	p. 182
3.2	End-to-end workflows W01–W16 with test records	p. 183
3.3	Alternate native themes	p. 187
3.4	Narrow panels and enlarged text	p. 194
3.5	Optical audit	p. 195
3.6	What this revision has and has not verified	p. 196

PART 3 · VALIDATION

3.1 Validation protocol

What must be done in the real IDEs before this design is called accessible or native. Status: defined, not yet executed.

Environments

HOST	BUILDS AND OS	THEMES	LAYOUTS
VS Code	Stable release current at test time; Windows 11, macOS and Ubuntu	Dark Modern, Light Modern, High Contrast Dark and Light	Sidebar 240, 320, 480 px; secondary sidebar; panel
JetBrains	IntelliJ IDEA and one other IDE on the same platform version; New UI	Light, Dark, High Contrast	Tool window docked right and bottom; floating; 320 px

Conditions

CONDITION	SETUP
Screen reader	NVDA on Windows; VoiceOver on macOS; Orca on Ubuntu (VS Code)
Keyboard only	No pointer for the whole flow; host keymaps unchanged
Zoom and text	Host zoom 100%, 150%, 200%; OS text scaling where supported
Locale	English plus one long-text locale (German) and one RTL locale (Arabic)
Adverse content	Long workspace names, paths, model IDs, multi-line errors, 50% text expansion

Procedure

- Run each of the 16 workflows (3.2) end to end in both hosts, once with a pointer and once keyboard-only.
- Repeat the approval (W06), recovery (W11) and history (W12) flows with a screen reader and record every announcement.
- At 200% zoom and at 320 px, confirm every primary, Deny and scoped Stop action remains reachable without horizontal scrolling.
- Measure button label centring in the real hosts with the same cap-centre method as 3.5 on at least 30 buttons per host and theme.
- Measure contrast of the rendered UI in each theme; resolve the host-token findings in 1.11.
- Record results in the per-workflow tables; a workflow passes only when every cell passes in both hosts.

Pass criteria

- Task completed without help and without data loss.
- Every state change is visible and announced once; nothing is announced only by colour.
- Focus is visible at all times, never trapped and never hidden behind sticky content.
- Recovery actions work and the draft survives every failure.
- Copy is understood: participants can say what the primary action will do before pressing it.

> Participants: at least five developers per host who use an agent weekly, plus at least two screen-reader users. Sessions are recorded; findings are logged against screen IDs and control IDs.

PART 3 · VALIDATION

3.2 End-to-end workflows W01–W16

Capture the request, acknowledge the effect and preserve a safe recovery path. Workflow text is retained verbatim from R7 pages 184–191; each now has a test record for both hosts.

W01

Connect and validate a custom provider

S03110 S02108 S05114

Open Providers Add; choose supported API family and execution host; enter endpoint and environment/vault reference; test auth, catalog and exact model separately; inspect capabilities and cost authority; save; pick the model; tune only offered parameters.

Recovery A failed probe preserves the staged connection and prior default. Secret values never enter shared config; unknown capability remains gated.

Complete when Provider/model pair and supported normalized tuning are captured only after acknowledgement.

	POINTER	KEYBOARD	SCREEN READER	200% / 320 PX	RECOVERY
VS Code	☐	☐	☐	☐	☐
JetBrains	☐	☐	☐	☐	☐

W02

Set up a local model without false locality

S04112 S31177 S08123

Inspect daemon host and loopback meaning; choose configured adapter; test exact endpoint; compare declared and observed allocation; configure embedder only after data-boundary review; choose retrieval fallback; capture a context preview.

Recovery Private LAN is refused under current provider policy; allow_loopback is not a bypass. Allocation/unload controls appear only when the adapter owns them.

Complete when A ready local route is explicit; observed usage remains unavailable until a durable read exists.

	POINTER	KEYBOARD	SCREEN READER	200% / 320 PX	RECOVERY
VS Code	☐	☐	☐	☐	☐
JetBrains	☐	☐	☐	☐	☐

W03

Tune a model and change it mid-session

S02108 S05114 S0140

Open model picker; inspect capability authority; choose exact provider/model; review reasoning and generation compatibility; remove invalid overrides; see Next task/New conversation timing; preserve draft; submit a new captured request.

Recovery Do not rewrite in-flight calls. Incompatible tuning stays staged; unavailable model never silently becomes another endpoint.

Complete when Each executed call records actual route and effective parameters, separately from current draft preference.

	POINTER	KEYBOARD	SCREEN READER	200% / 320 PX	RECOVERY
VS Code	☐	☐	☐	☐	☐
JetBrains	☐	☐	☐	☐	☐

W04

Move from Ask to Plan to Code

S06118 S07120 S0140

Choose enforced Ask; explore read-only; create a Plan task with named artifact scope; review the frozen plan and comments; start a new Code task with explicit contract, permissions, model and finish.

Recovery Current prompt-only Ask is not called read-only until dispatch enforcement exists. Approving a plan does not broaden tool policy.

Complete when A new Code admission binds the approved plan version and execution envelope.

	POINTER	KEYBOARD	SCREEN READER	200% / 320 PX	RECOVERY
VS Code	☐	☐	☐	☐	☐
JetBrains	☐	☐	☐	☐	☐

WORKFLOWS · CONTINUED

W05 Assemble context and resolve overflow

S09127 S08123 S10129

Mention exact sources; inspect URI/range/version; attach supported files; review automatic chips; pin required facts; compare estimated input and reserve against capacity; remove, compact or choose a capable model before capture.

Recovery Failed/oversize attachments remain visible. Missing durable usage is unavailable; estimates are labelled. No silent truncation of criteria or pending effects.

Complete when The admitted request has a versioned source manifest and valid capacity calculation.

	POINTER	KEYBOARD	SCREEN READER	200% / 320 PX	RECOVERY
VS Code	☐	☐	☐	☐	☐
JetBrains	☐	☐	☐	☐	☐

W06 Compact and inspect continuity

S10129 S08123 S3273

Open compaction plan; inspect retained facts and summary model/egress; request a new durable compact operation; preserve pending approvals and external-effect state; inspect before/after basis and measured reduction.

Recovery Failure or inadequate reduction preserves the last valid state. Collapsing old UI messages is not compaction or deletion.

Complete when Required contract, candidate identity and unresolved obligations remain reachable and active.

	POINTER	KEYBOARD	SCREEN READER	200% / 320 PX	RECOVERY
VS Code	☐	☐	☐	☐	☐
JetBrains	☐	☐	☐	☐	☐

W07 Approve once or create a narrow rule

S1442 S13142 S2246

Open attention card; inspect owner/generation, normalized tool args, cwd, target and effect; Allow once or Deny; await exact reply; optionally open a separate persistent-rule draft; test match; apply future policy after ack.

Recovery Expired or wrong-session waiters refuse. Remember is unavailable without a real store; new args require a new decision.

Complete when One exact approval resolves once; a persisted rule has named scope, version and expiry.

	POINTER	KEYBOARD	SCREEN READER	200% / 320 PX	RECOVERY
VS Code	☐	☐	☐	☐	☐
JetBrains	☐	☐	☐	☐	☐

W08 Add MCP with trust and recover auth

S16146 S15144 S13142

Choose supported transport; enter bounded spec; inspect executable source, host and environment references; review trust; save; explicitly start; observe health; inspect tool namespaces/schemas; set narrow permissions.

Recovery Remote/OAuth are not offered by the current stdio config. Auth cancellation keeps the form; disconnect does not prove effect cancellation.

Complete when Only discovered, healthy, enabled and authorized tools are offered to the correct task.

	POINTER	KEYBOARD	SCREEN READER	200% / 320 PX	RECOVERY
VS Code	☐	☐	☐	☐	☐
JetBrains	☐	☐	☐	☐	☐

WORKFLOWS · CONTINUED

W09

Install and invoke a skill safely

S17150 S11133 S13142

Inspect manifest, files, source and required capabilities; preview a version diff; choose manual/agent-selectable scope; install a validated version; resolve template variables; preview the draft and permission envelope; invoke deliberately.

Recovery Viewing a package never runs setup. Failed update retains the known-good version. Hidden/disabled cannot remain silently agent-invocable.

Complete when Versioned instructions are attributed; execution still uses effective grants.

	POINTER	KEYBOARD	SCREEN READER	200% / 320 PX	RECOVERY
VS Code	☐	☐	☐	☐	☐
JetBrains	☐	☐	☐	☐	☐

W10

Author and test a lifecycle hook

S18152 S30173 S2246

Choose a native event; configure executable/argv, allowlisted env, deadline, caps and failure policy; preview input; admit a bounded test under real capabilities; inspect verdict/exit/output; save a validated version.

Recovery Timeout, malformed verdict and nonzero exit are distinct. A fail-open choice is explicit; partial output cannot authorize after deadline.

Complete when A hook run cites exact spec version, event, envelope and bounded audit artifact.

	POINTER	KEYBOARD	SCREEN READER	200% / 320 PX	RECOVERY
VS Code	☐	☐	☐	☐	☐
JetBrains	☐	☐	☐	☐	☐

W11

Delegate and steer a bounded child

S07120 S2382 S1442

Admit a role within parent envelope; show concurrency and budget reservation; bind candidate ownership; inspect actual model and status; send a <=500-character steer or supported model control; resolve child-scoped approval; inspect typed handoff.

Recovery Queue saturation is bounded; stop/revoke does not undo issued external effects. Child failure preserves artifacts and parent obligations.

Complete when Shared budget and ownership hold; a child saying done does not establish parent verification.

	POINTER	KEYBOARD	SCREEN READER	200% / 320 PX	RECOVERY
VS Code	☐	☐	☐	☐	☐
JetBrains	☐	☐	☐	☐	☐

W12

Queue, background and interrupt work

S2246 S19156 S2691

While a task runs, choose Queue next rather than implicit interrupt; edit/remove before admission; inspect owned background process and output; deliberately stop the correct process/task/session; switch sessions with per-session drafts intact.

Recovery A closed tab is not cancellation. Unknown effect or admission is reconciled before replay. Stream loss marks stale instead of idle.

Complete when Each queue item/process/control retains owner, generation and durable result.

	POINTER	KEYBOARD	SCREEN READER	200% / 320 PX	RECOVERY
VS Code	☐	☐	☐	☐	☐
JetBrains	☐	☐	☐	☐	☐

WORKFLOWS · CONTINUED

W13

Handle rate limit, cap and unknown spend

S21161 S2246 S27105

Inspect typed 429/quota/local saturation; show known retry delay and total attempts; keep Stop available; inspect open/settled/uncertain ledger; request a supported allowance revision or end the task; reconcile any unknown admission.

Recovery Never guess reset time or zero price. A cap reduction cannot erase spend/reservations; effectful replay requires proven safety.

Complete when The final ledger retains every eligible attempted charge, including failed/cancelled work.

	POINTER	KEYBOARD	SCREEN READER	200% / 320 PX	RECOVERY
VS Code	☐	☐	☐	☐	☐
JetBrains	☐	☐	☐	☐	☐

W14

Review, restore or fork without overwriting edits

S2459 S2577 S2691

Open exact native candidate diff; review mapped criteria; choose recorded checkpoint; inspect content coverage and file preconditions; choose supported restore/fork scope; preview unrelated edits and uncovered effects; perform separate transaction if valid.

Recovery Hash-only missing blobs disable redo/diff. CAS conflict preserves user edits. Shell/network/SCM effects are not universally rewound.

Complete when A changed candidate invalidates stale proof; historical verified artifacts remain historical.

	POINTER	KEYBOARD	SCREEN READER	200% / 320 PX	RECOVERY
VS Code	☐	☐	☐	☐	☐
JetBrains	☐	☐	☐	☐	☐

W15

Finish a change with complete proof

S0140 S3331 S2459 S3273

Capture required criteria and Local/Commit/Push/PR level; implement in candidate; inspect claim artifacts; run required checks; match proof to candidate; integrate with matching identity; perform authorized finish; inspect receipts and outcome.

Recovery Missing canonical SCM, failed unit check, empty proof or mismatched landing remains pending/unverified. Session Completed is insufficient.

Complete when Required acceptance, claims, landing equality and requested finish all have explicit matching facts.

	POINTER	KEYBOARD	SCREEN READER	200% / 320 PX	RECOVERY
VS Code	☐	☐	☐	☐	☐
JetBrains	☐	☐	☐	☐	☐

W16

Recover a remote host or runtime change

S31177 S30173 S2691

Detect new daemon/workspace generation; mark old handles stale; preserve local drafts and prior projection; show execution/model/vault map; verify runtime path/readiness/protocol; rebind session and reconcile operations; resume only supported work.

Recovery A local URL on the laptop is not the daemon loopback. Do not copy credentials into project files or replay an old request on a new host.

Complete when All callbacks and controls are fenced to the new binding; unknown effects stay visible.

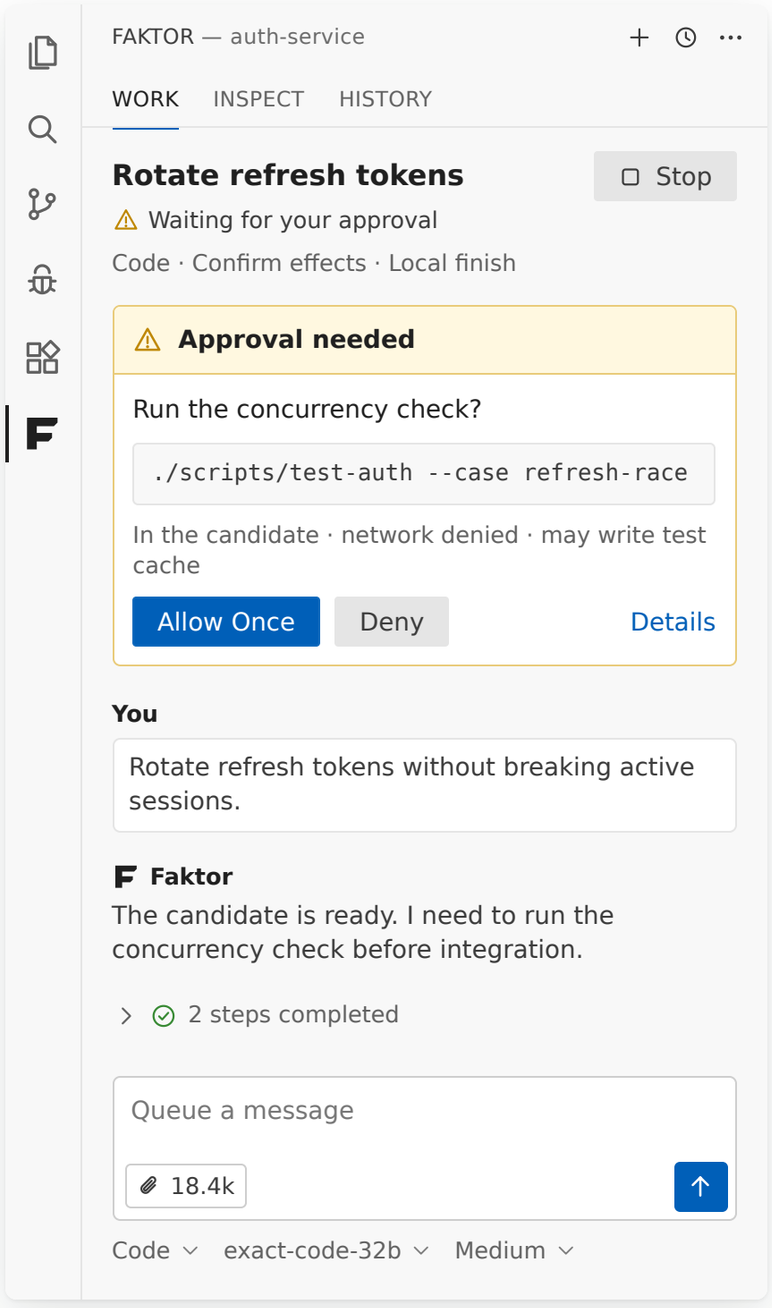
	POINTER	KEYBOARD	SCREEN READER	200% / 320 PX	RECOVERY
VS Code	☐	☐	☐	☐	☐
JetBrains	☐	☐	☐	☐	☐

3.3 ALTERNATE NATIVE THEMES · S01

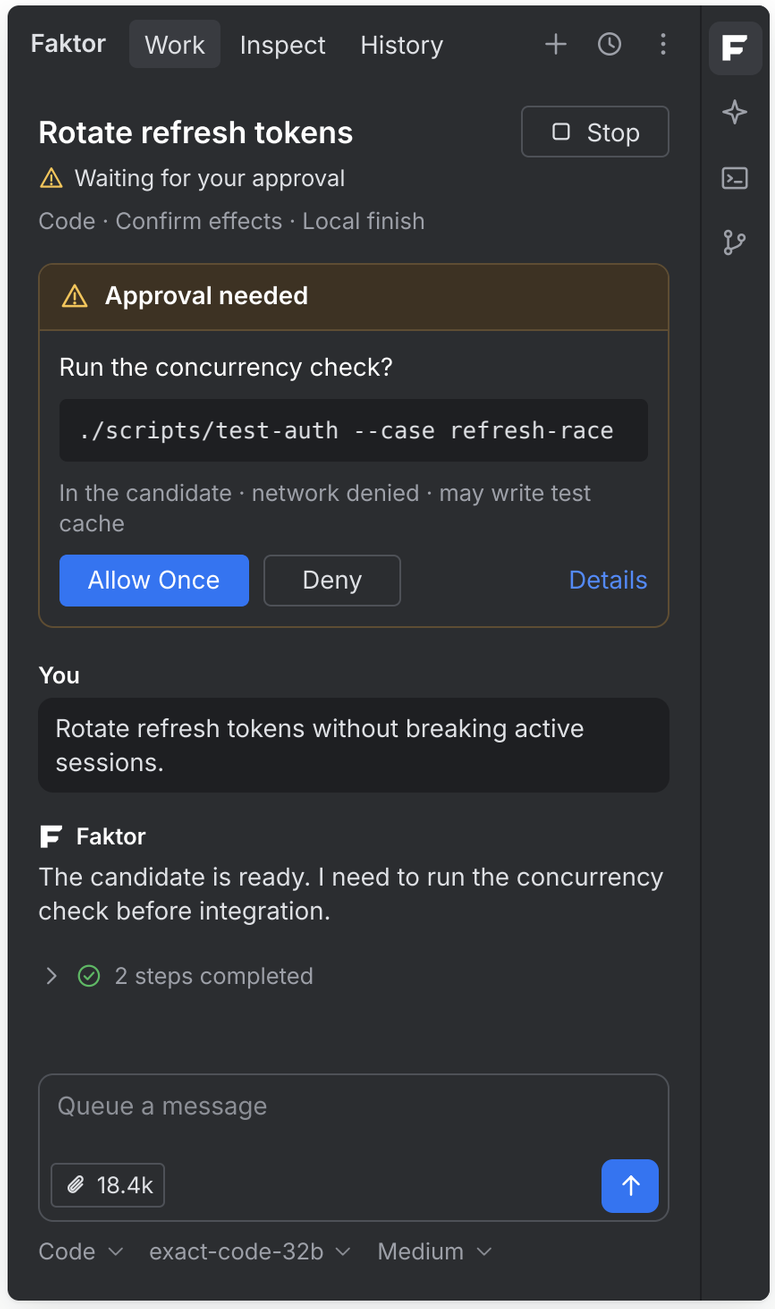
Work / alternate themes

VS Code Light Modern and JetBrains Dark use the same information and action hierarchy.

VS CODE · LIGHT MODERN — SIDEBAR VIEW · WEBVIEW



JETBRAINS · DARK — TOOL WINDOW · SWING / KOTLIN UI DSL



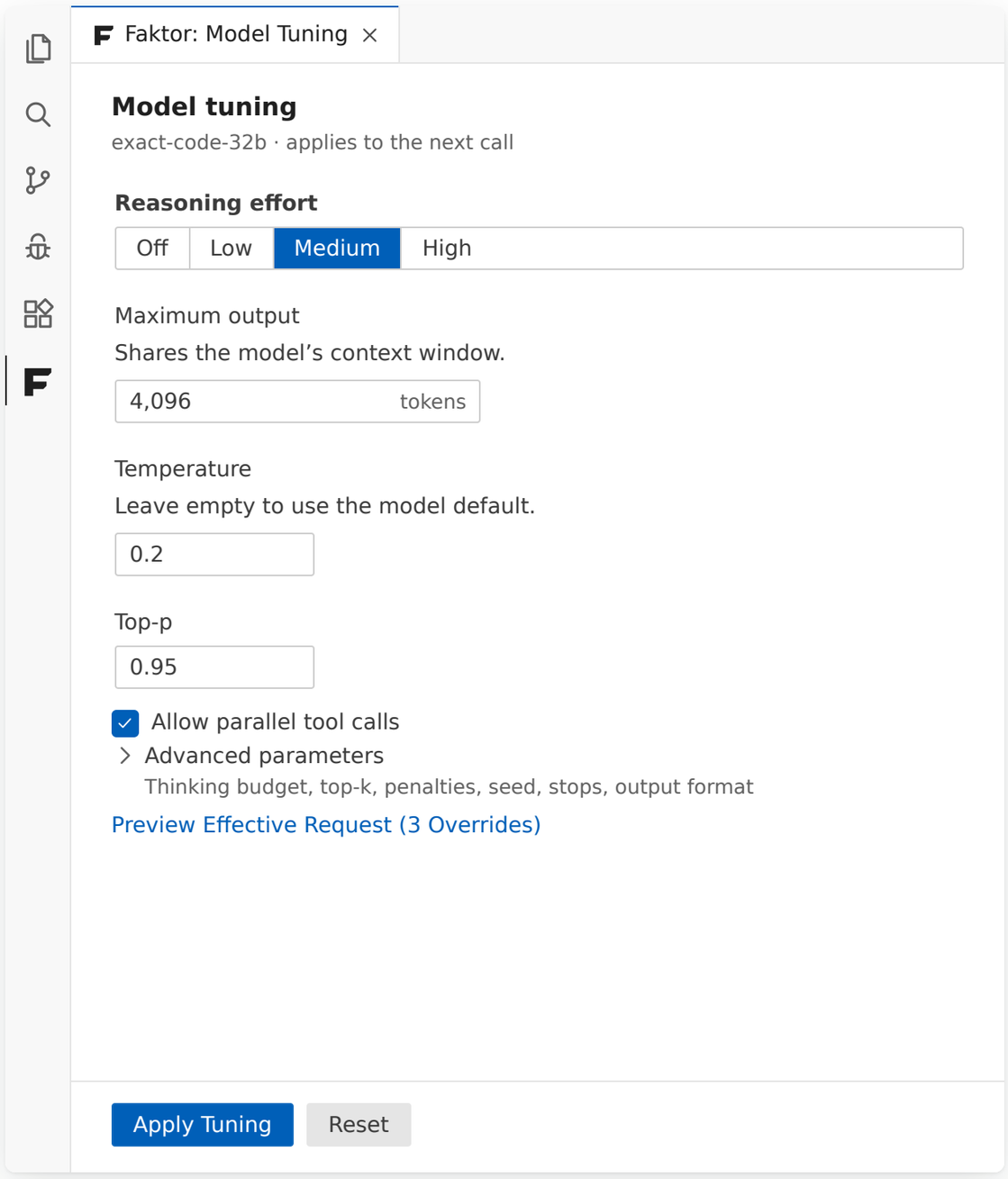
All 72 surfaces are supplied in both host themes as 288 editable vectors (R7 note, retained). Revision 8 renders every surface in four host themes in the build; high-contrast and native font/zoom integration remain required.

3.3 ALTERNATE NATIVE THEMES · S05

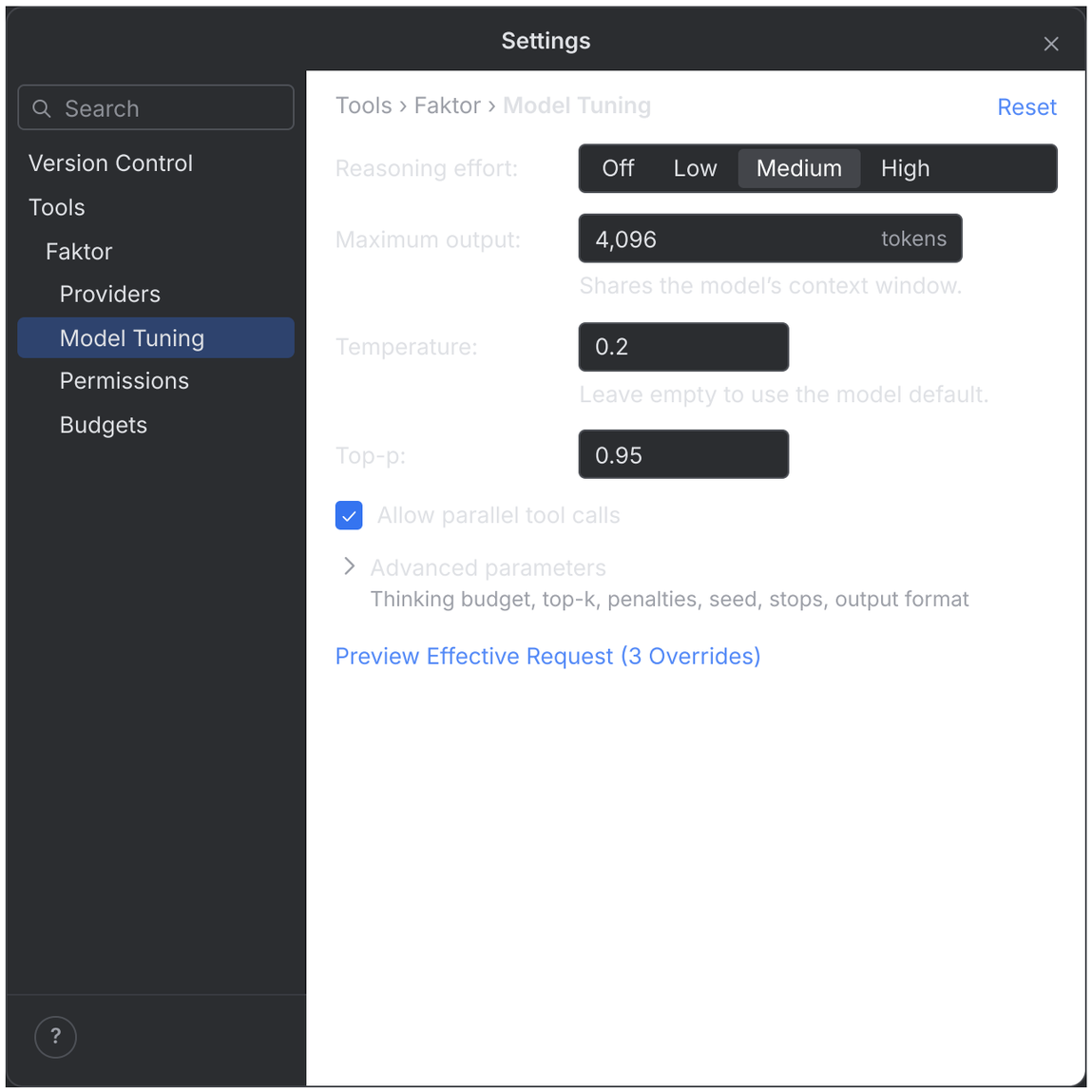
Model tuning / alternate themes

VS Code Light Modern and JetBrains Dark use the same information and action hierarchy.

VS CODE · LIGHT MODERN — EDITOR TAB · WEBVIEW SETTINGS



JETBRAINS · DARK — SETTINGS DIALOG · CONFIGURABLE



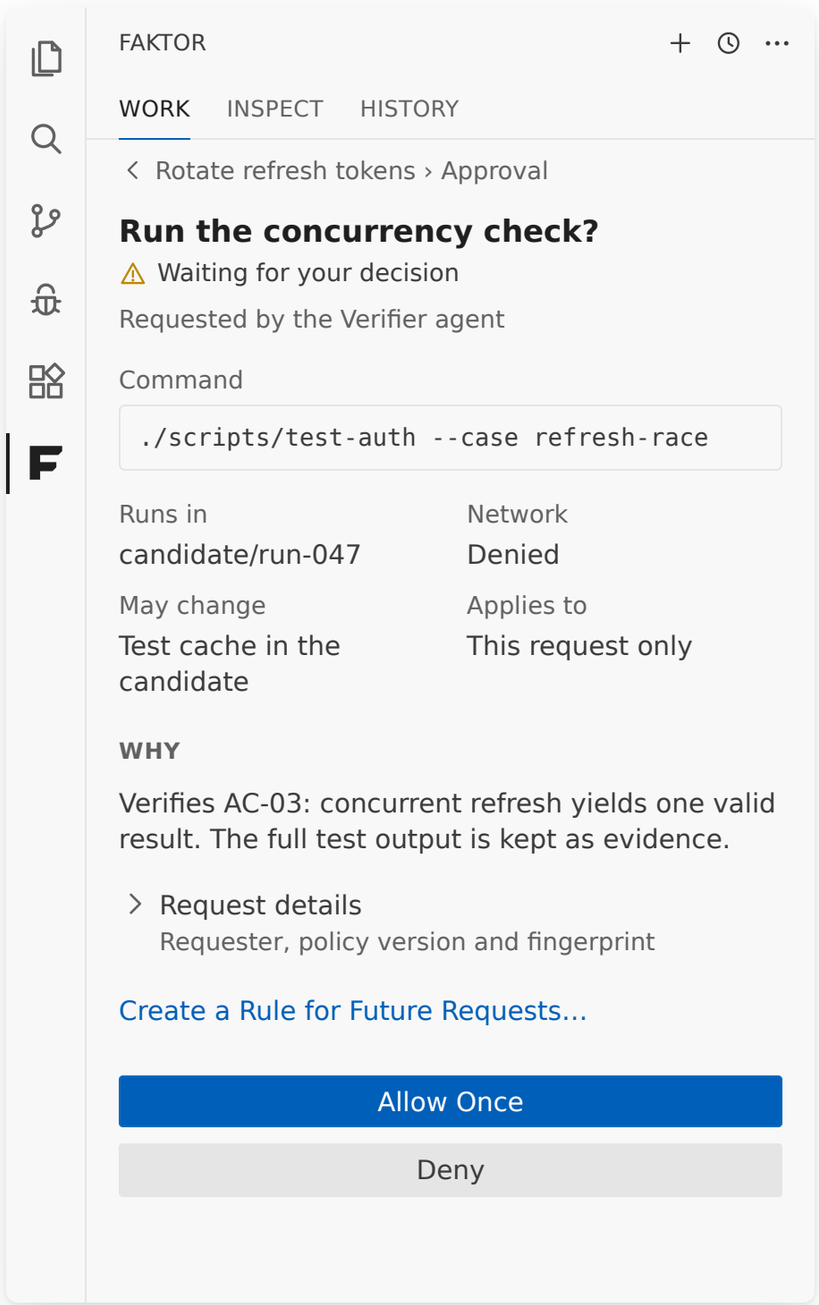
All 72 surfaces are supplied in both host themes as 288 editable vectors (R7 note, retained). Revision 8 renders every surface in four host themes in the build; high-contrast and native font/zoom integration remain required.

3.3 ALTERNATE NATIVE THEMES · S14

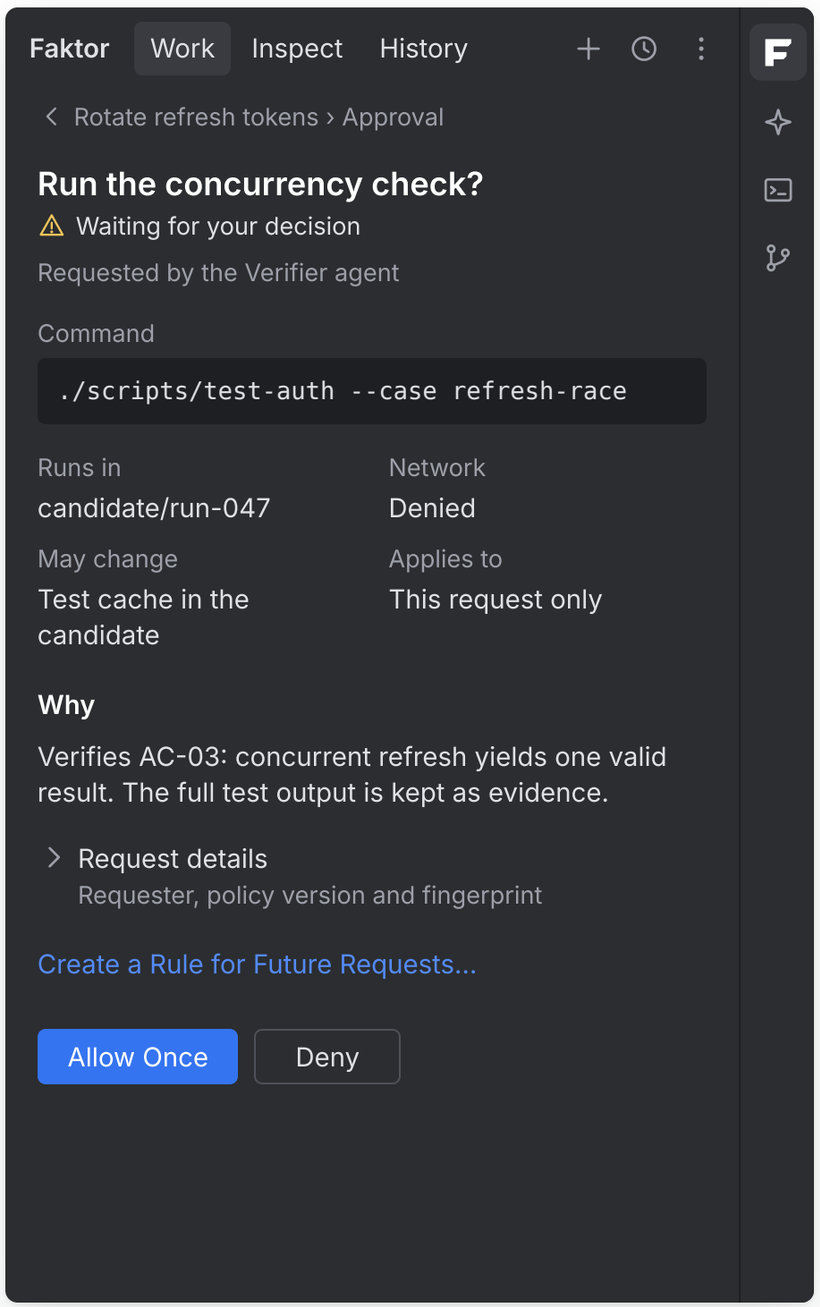
Approve an action / alternate themes

VS Code Light Modern and JetBrains Dark use the same information and action hierarchy.

VS CODE · LIGHT MODERN — SIDEBAR VIEW · WEBVIEW



JETBRAINS · DARK — TOOL WINDOW · SWING / KOTLIN UI DSL



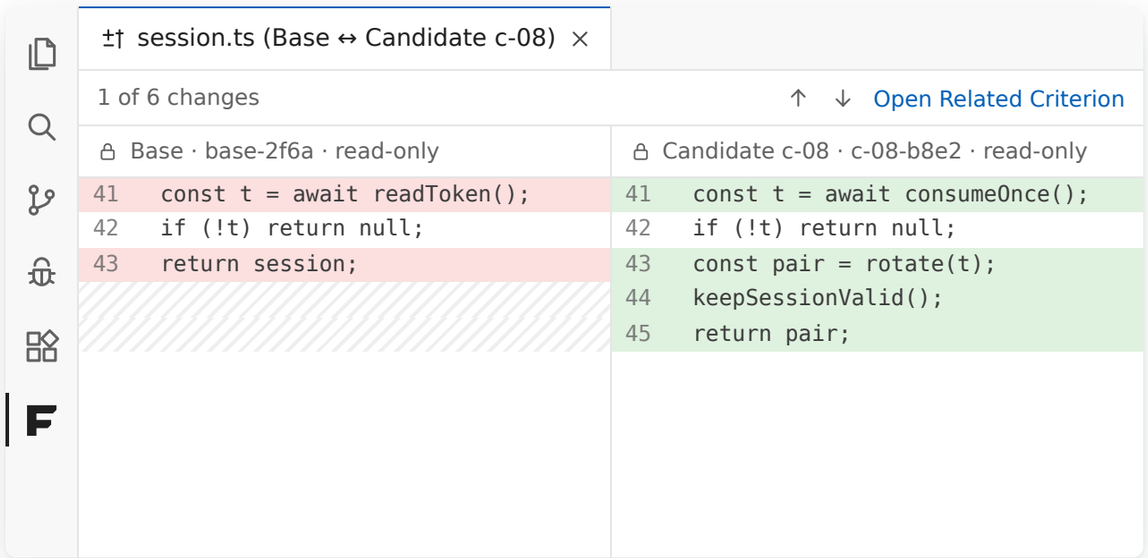
All 72 surfaces are supplied in both host themes as 288 editable vectors (R7 note, retained). Revision 8 renders every surface in four host themes in the build; high-contrast and native font/zoom integration remain required.

3.3 ALTERNATE NATIVE THEMES · S43

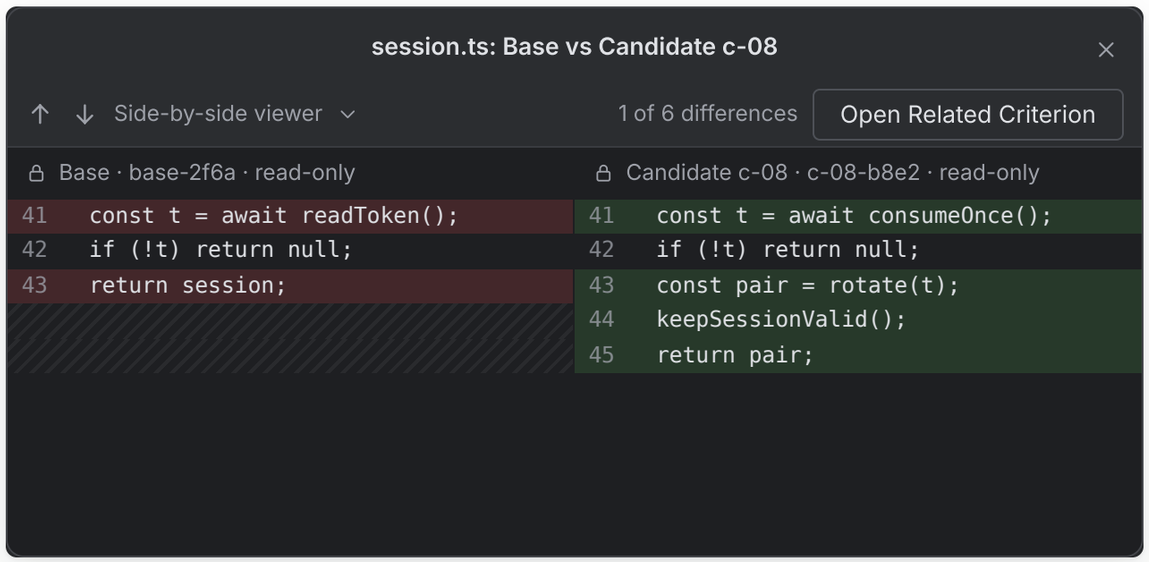
Native diff review / alternate themes

VS Code Light Modern and JetBrains Dark use the same information and action hierarchy.

VS CODE · LIGHT MODERN — DIFF EDITOR · VSCODE.DIFF



JETBRAINS · DARK — DIFF VIEWER · DIFFMANAGER



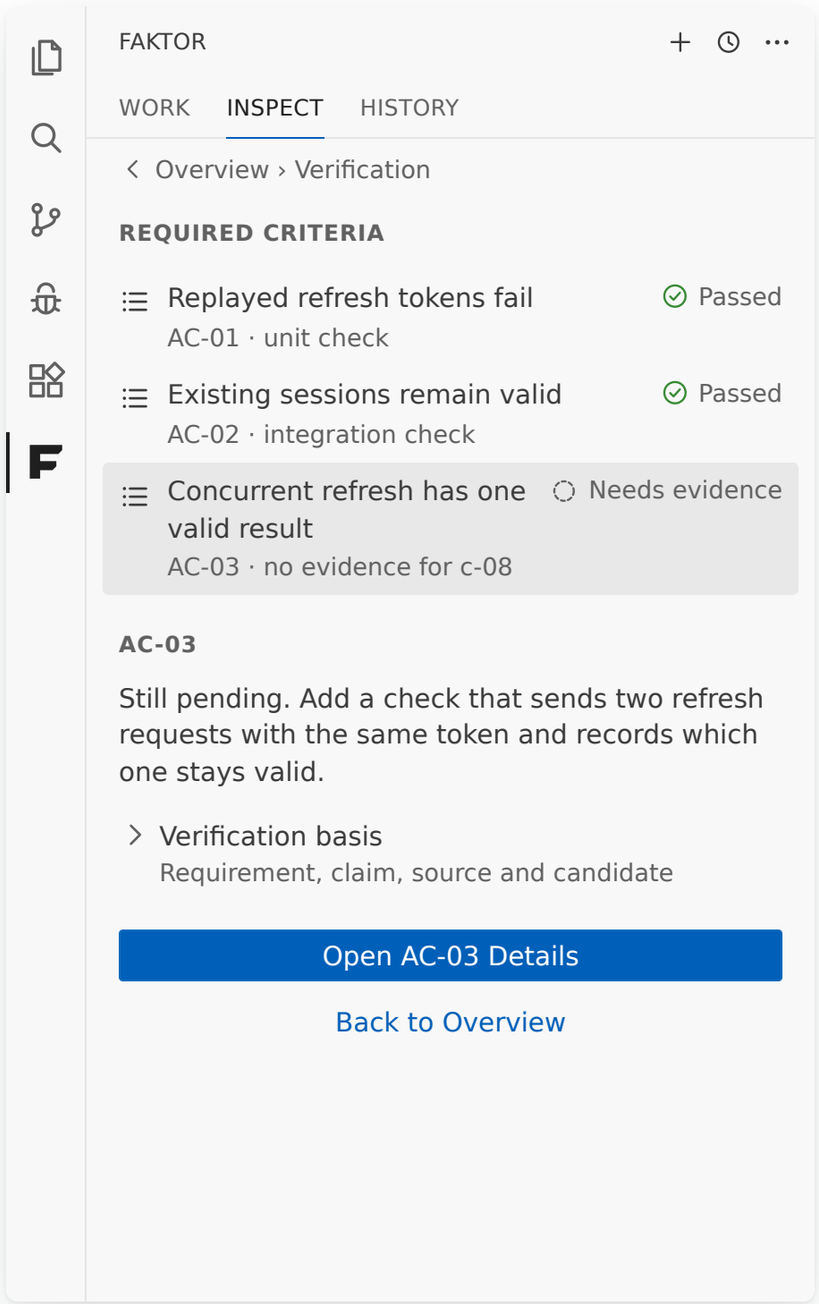
All 72 surfaces are supplied in both host themes as 288 editable vectors (R7 note, retained). Revision 8 renders every surface in four host themes in the build; high-contrast and native font/zoom integration remain required.

3.3 ALTERNATE NATIVE THEMES · S45

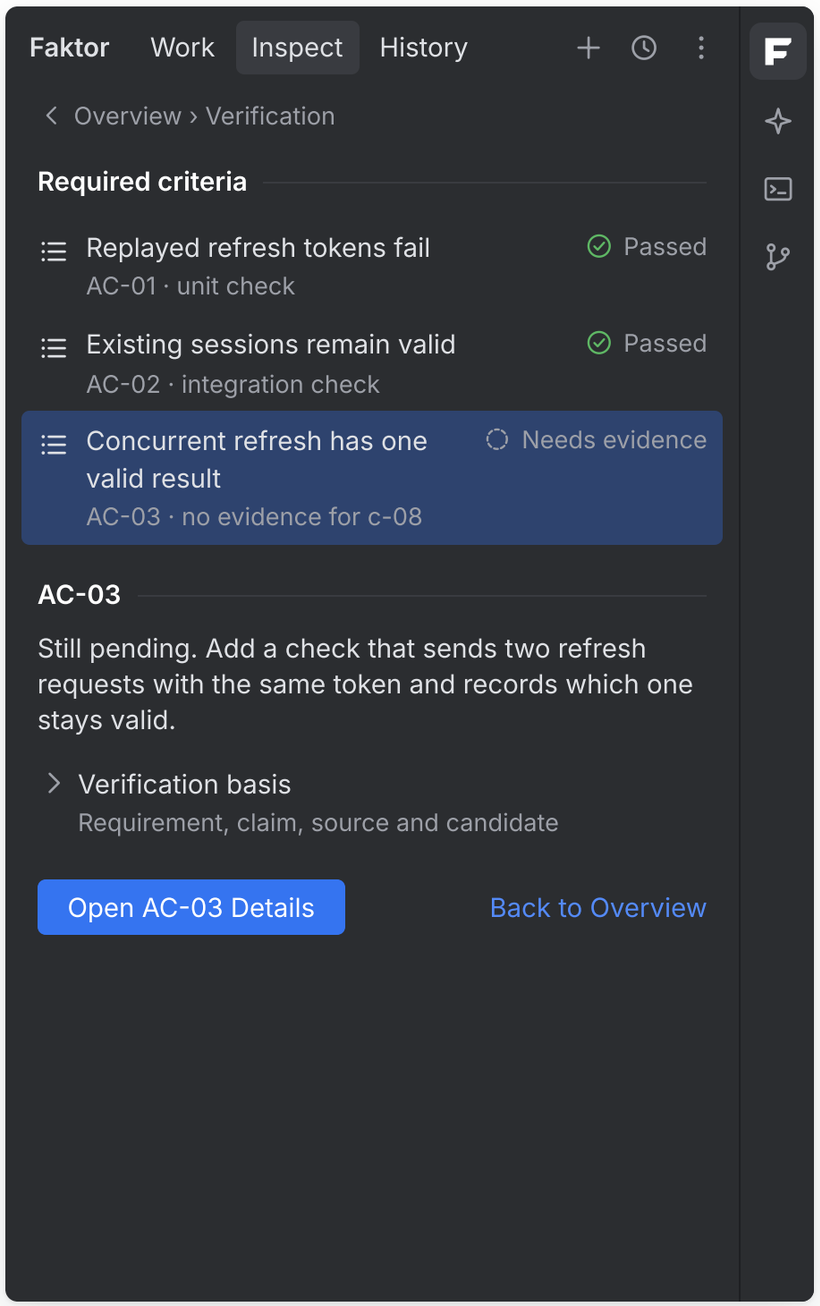
Required verification / alternate themes

VS Code Light Modern and JetBrains Dark use the same information and action hierarchy.

VS CODE · LIGHT MODERN — SIDEBAR VIEW · WEBVIEW



JETBRAINS · DARK — TOOL WINDOW · SWING / KOTLIN UI DSL



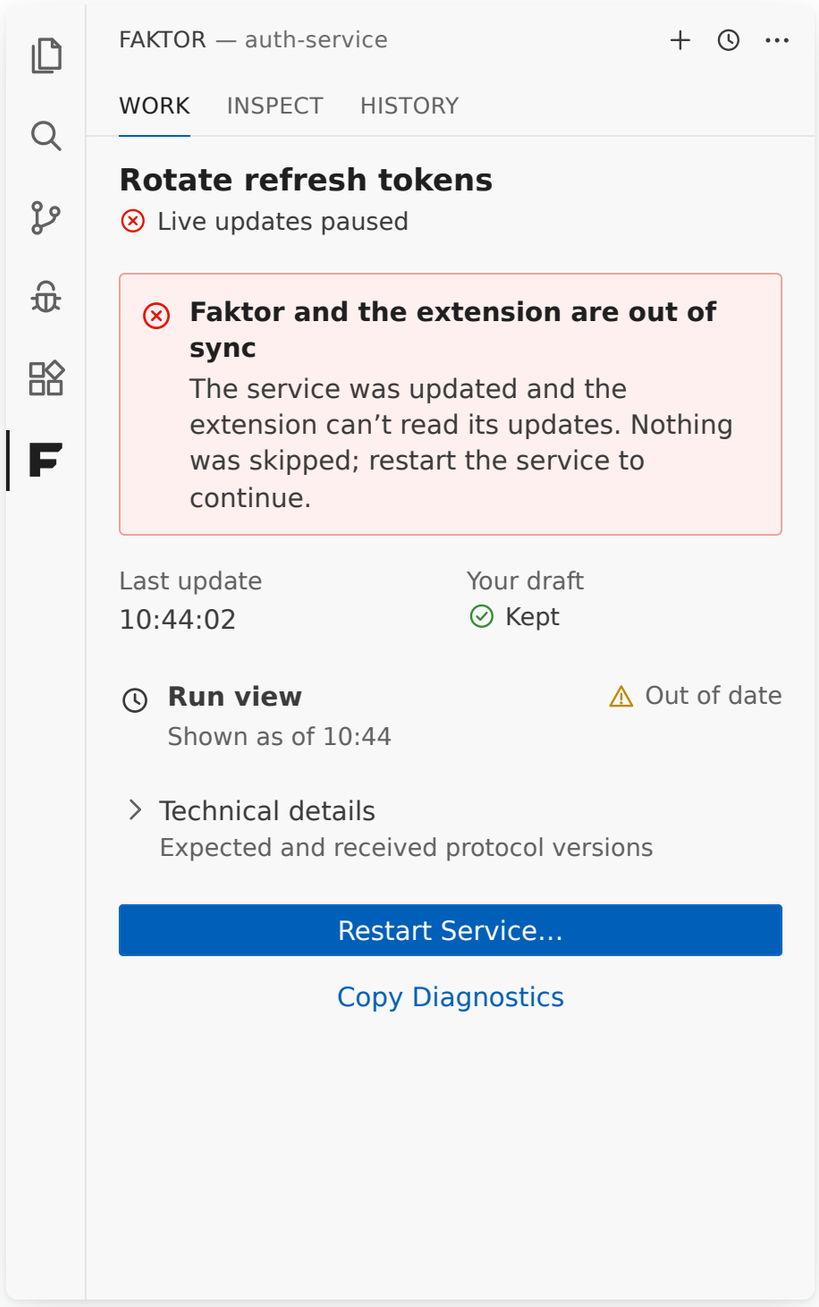
All 72 surfaces are supplied in both host themes as 288 editable vectors (R7 note, retained). Revision 8 renders every surface in four host themes in the build; high-contrast and native font/zoom integration remain required.

3.3 ALTERNATE NATIVE THEMES · S56

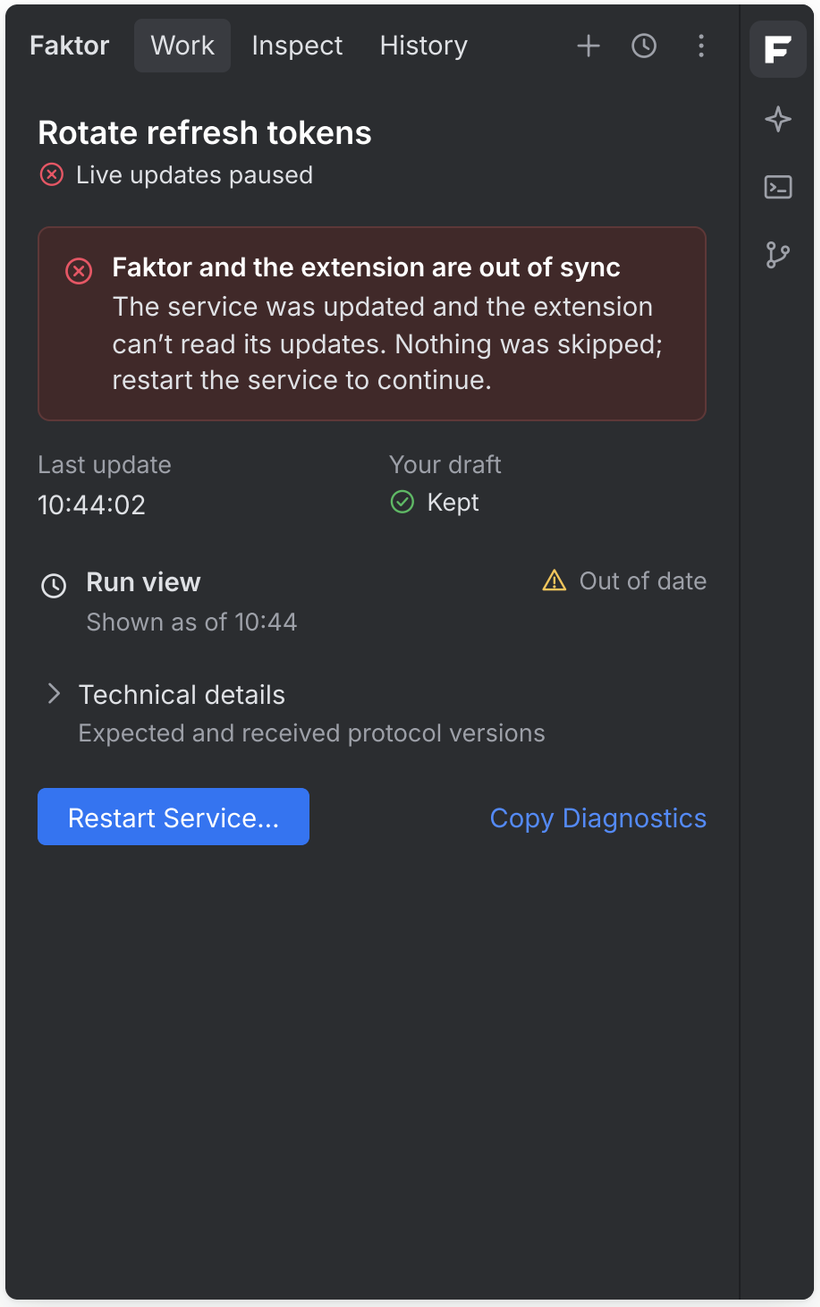
Protocol-blocked recovery / alternate themes

VS Code Light Modern and JetBrains Dark use the same information and action hierarchy.

VS CODE · LIGHT MODERN — SIDEBAR VIEW · WEBVIEW



JETBRAINS · DARK — TOOL WINDOW · SWING / KOTLIN UI DSL



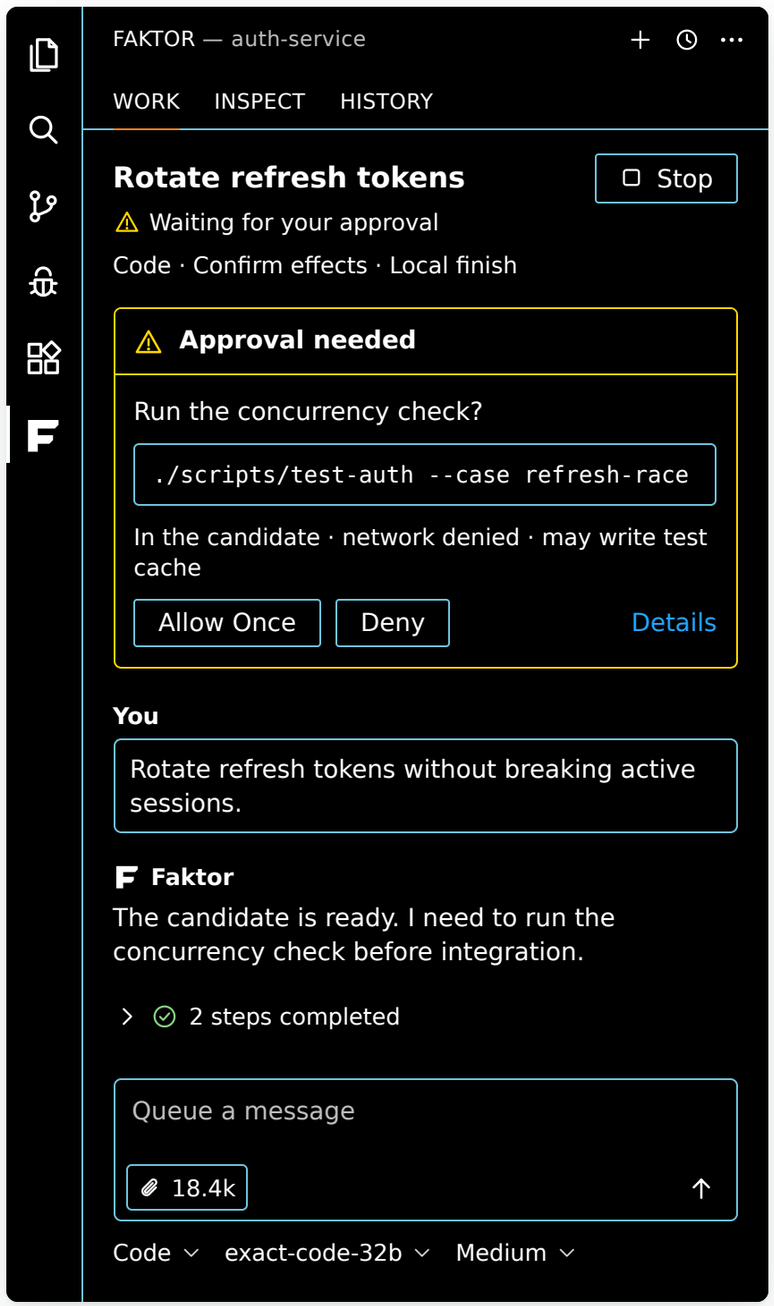
All 72 surfaces are supplied in both host themes as 288 editable vectors (R7 note, retained). Revision 8 renders every surface in four host themes in the build; high-contrast and native font/zoom integration remain required.

3.3 ALTERNATE NATIVE THEMES · S01, S14

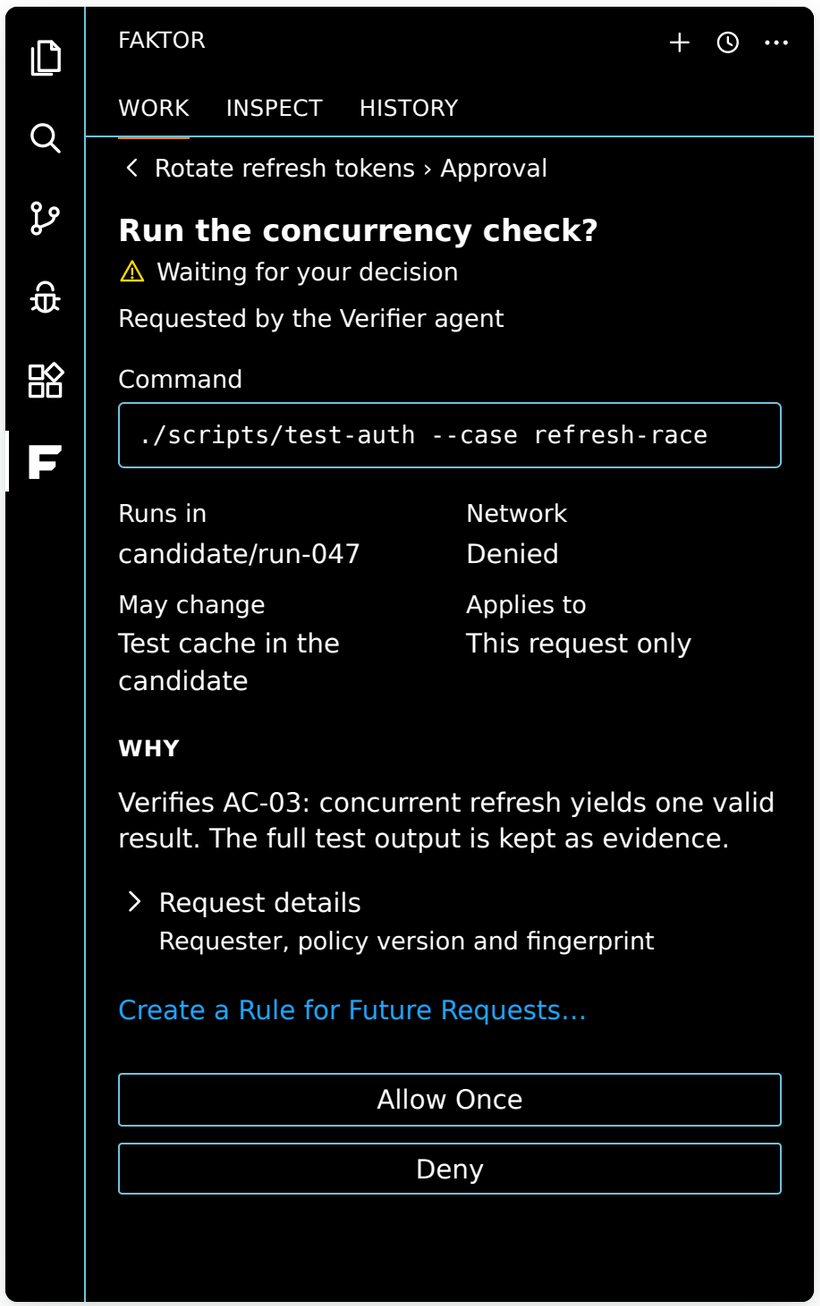
High contrast

VS Code High Contrast Dark: borders carry component boundaries; focus is a 1 px orange outline. JetBrains High Contrast must be checked in the IDE.

VS CODE · HIGH CONTRAST DARK — SIDEBAR VIEW · WEBVIEW



VS CODE · HIGH CONTRAST DARK — SIDEBAR VIEW · WEBVIEW

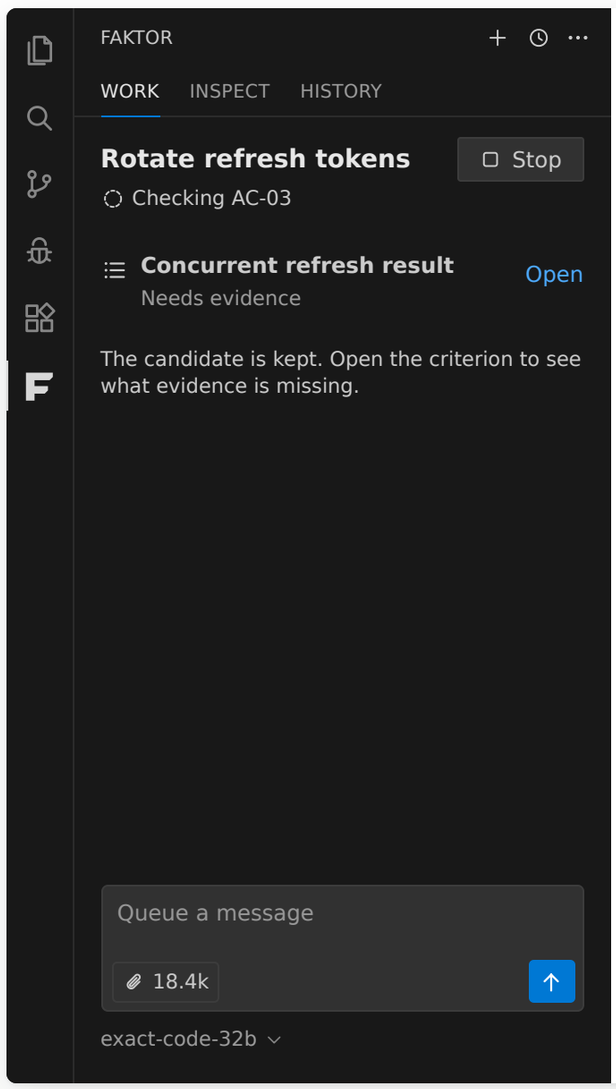


PART 3 · VALIDATION

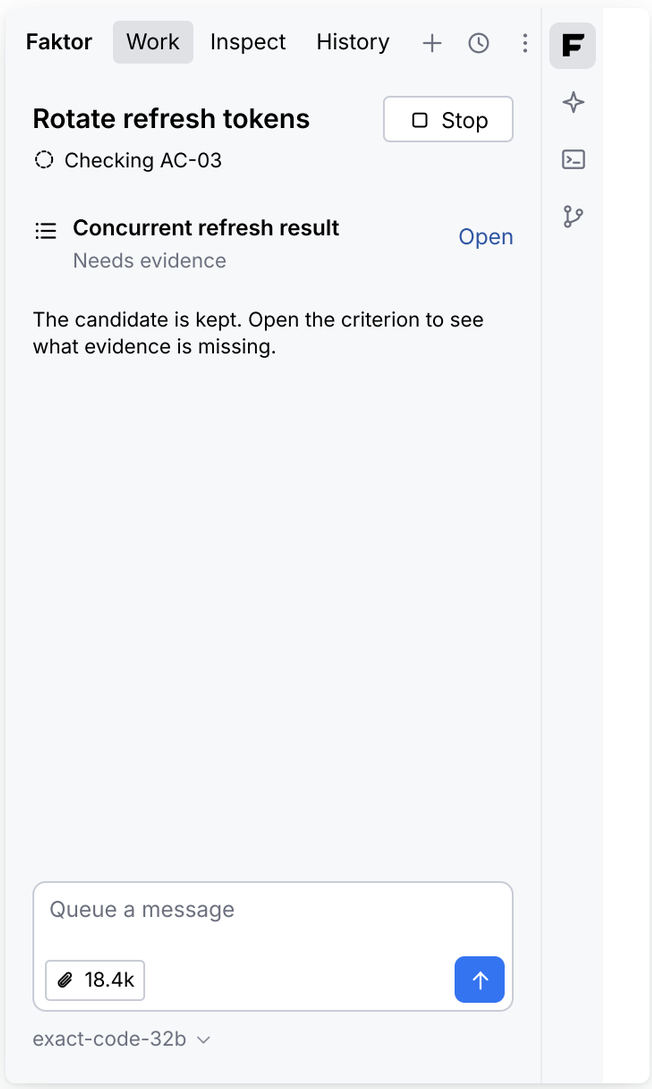
3.4 Narrow panels and enlarged text

Readable text and reachable actions at 320 px, at 200% text and with long content.

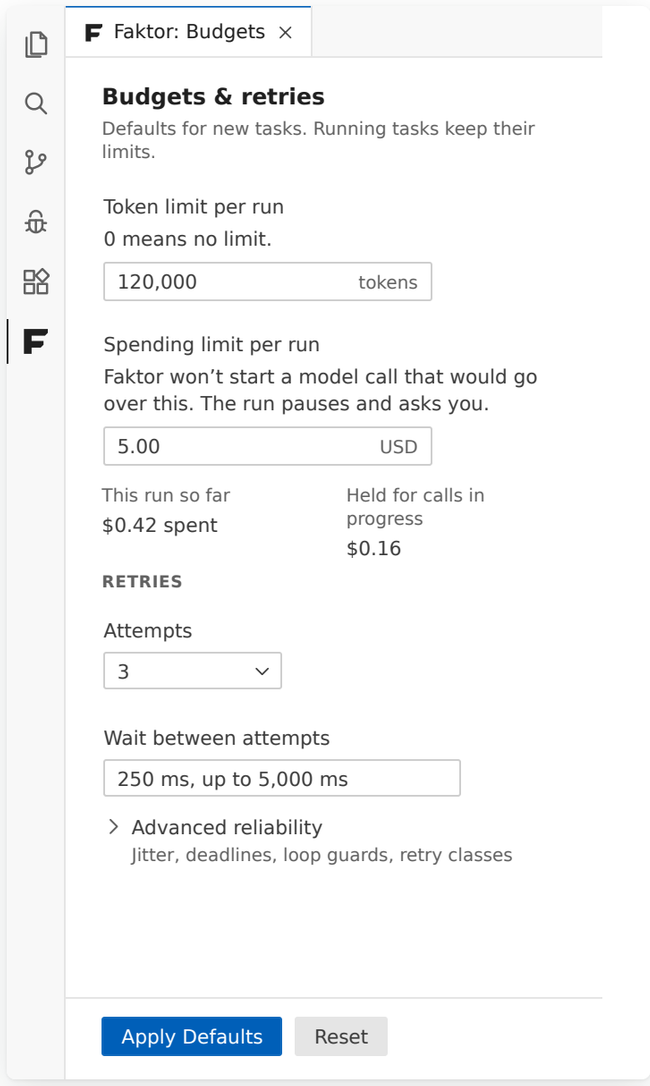
VS CODE · DARK MODERN — SIDEBAR VIEW AT 320 PX



JETBRAINS · LIGHT — TOOL WINDOW AT 320 PX



VS CODE · LIGHT MODERN
EDITOR TAB · WEBVIEW SETTINGS



- At 320 px the header, one detail row and the composer stack; Stop stays in the header.
- 200% text in a 640 px panel lays out exactly like 100% text at 320 px in a webview (CSS zoom), so the 320 px plates stand for both conditions.
- Editor-tab forms stack labels, help and controls at any width; help wraps instead of truncating.
- Paths and IDs elide in the middle and stay copyable in full.
- These are reference renderings: JetBrains Swing layouts must be checked at 200% IDE zoom in the host.

PART 3 · VALIDATION

3.5 Optical audit

Cap-height centre of every button label compared with the button midpoint.

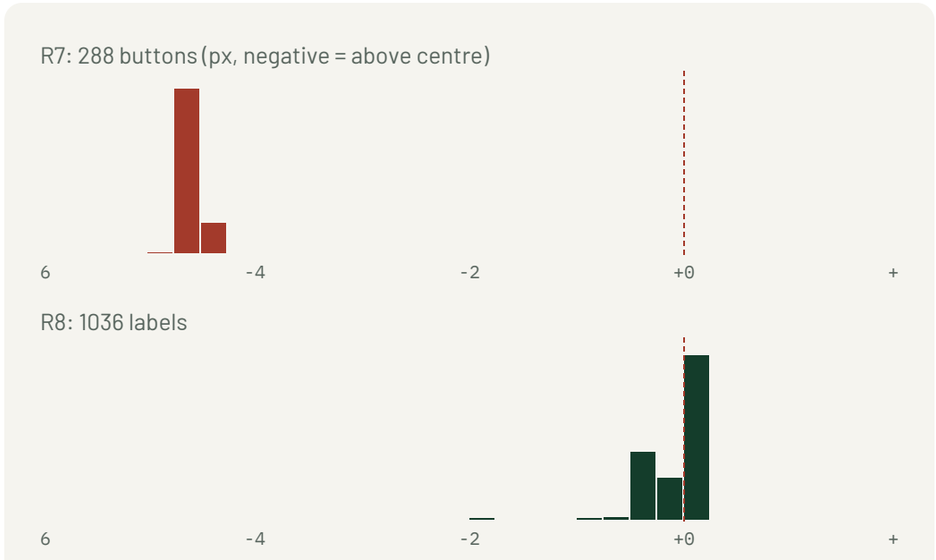
SET	LABELS	MEDIAN	MEAN	RANGE (5TH/95TH; R7 MIN/MAX)	MAX |OFFSET|
Revision 7 plates (all)	288	-4.64	-4.61	-4.41 / -4.76	4.76
JetBrains Dark	274	+0.00	-0.02	-0.33 / +0.17	2.00
JetBrains Light	274	+0.00	-0.02	-0.33 / +0.17	2.00
VS Code Dark Modern	244	-0.17	-0.14	-0.33 / +0.17	2.00
VS Code Light Modern	244	-0.17	-0.14	-0.33 / +0.17	2.00

Method

- Rasterise each plate at 3× (R7 pages at 2× from the original PDF).
- Crop the button interior, excluding the border and 7 px at each end to avoid rounded corners.
- Ink = pixels differing from the button background. Cap top = first ink row. Baseline = median of per-column lowest ink row.
- Offset = (cap top + baseline) / 2 − box centre, in logical px; negative is above centre.
- Labels measured: buttons, segmented options and tabs in all 72 screens × 4 themes.

Limits

- The renderer is Chromium with substitute fonts; real hosts use Segoe UI, SF Pro or Inter and their own rasterisers.
- Ascenders (l, d) and bowls (P) can bias the ink method by up to about 2 px; such outliers were inspected and are correctly placed.
- The same measurement must be repeated in VS Code and JetBrains builds (3.1).



PART 3 · VALIDATION

3.6 What this revision has and has not verified

Plates, measurements and checks in this document are evidence about the design, not about the shipped product.

CHECK	METHOD	RESULT	STATUS
Content retention	Every R7 plate string matched to the new plates or to a recorded relocation; R7 text sections diffed word by word against this document.	1,497 plate segments, all accounted for; 838 changes listed per screen. 5,973 text strings checked; only repeated page furniture is consolidated.	Done
Optical centring	Ink-based cap-centre measurement on rasterised labels.	R7 -4.6 px mean; R8 +0.0 px median over 1036 labels.	Done (reference renderer)
Overflow and collisions	DOM measurement of every plate in four themes.	No clipped content outside intended scroll regions.	Done (reference renderer)
Contrast	WCAG formula on plate tokens.	80 pairs; 9 host defaults below threshold (1.11).	Done; decisions open
Native conventions	Checked against VS Code UX and IntelliJ Platform UI guidelines (R8 references).	Applied in Part 1 and every spread.	Specified
Task flows in both IDEs	Protocol 3.1 and test records 3.2.	Not run. Requires implementation builds.	Not done
Assistive technology	NVDA, VoiceOver, Orca.	Not run.	Not done
Real fonts, zoom and HiDPI	In-host measurement.	Not run.	Not done

PART 4

4 Specification reference

All technical documentation from revision 7, retained in full. Original wording is unchanged; revision 8 amendments are marked where a rule changed.

	SECTION	PAGE
4.1	Detailed UI and interaction rules	p. 198
4.2	Product and behaviour specification	p. 212
4.3	Required backend and native construction	p. 226
4.4	Implementation code outlines	p. 239
4.5	All 200 control contracts	p. 257
4.6	Restored first-report content and journeys	p. 295
4.7	Original-to-revised retention map	p. 334
4.8	Acceptance A01–A40	p. 343
4.9	Source and benchmark references	p. 346

4.1 DETAILED UI AND INTERACTION RULES · R7 PAGE 078

Five consistent interaction patterns

Reference implementation requirements for both hosts; native platform conventions take precedence.

Observation

Read-only facts show their observation source and current/stale/unavailable state. They never look editable. Copy/Open is separate and keyboard reachable. A refresh has a query epoch and cannot overwrite a different scope.

Choice

Use native single/multiple selection semantics. Inherited, explicitly off, unsupported and policy-locked choices are distinct. Revalidate dependent fields when a provider/model changes; show why a choice cannot apply.

Number

Keep an explicit unit and authoritative limits. Preserve intermediate editing and distinguish unset from zero. Parse exact token/integer/money values at the trusted boundary; do not round currency through floating point.

Editor and action

An editor preserves its buffer, caret, origin and undo through validation. An action states its verb, target and scope before dispatch. Pending acknowledgement, confirmed success and effect unknown are distinct. Every control's original behavior, scope and apply timing remain authoritative; detailed interaction records are included for all 200 IDs.

4.1 DETAILED UI AND INTERACTION RULES · R7 PAGE 079

Hierarchy before decoration

Reference implementation requirements for both hosts; native platform conventions take precedence.

Three product clusters

Work owns the conversation, active task and composer. Inspect owns the selected plan, candidate, criteria, evidence and agents. History owns durable conversations. Keep Settings, Usage and Diagnostics behind their native command or More, with an explicit return. Do not promote Board, infrastructure or every tool into equal top-level tabs.

A stable task header

Show a human task name, workspace/candidate context and captured intent/authority/finish. Keep historical captured values apart from editable next-call defaults. Put service trouble in one scoped attention area. Do not repeat the same badge, title and status in three nested cards.

One selected object

Use a shallow list followed by one selected detail. An overview row opens the exact criterion, file, child or artifact. At narrow widths the detail replaces the list and gets Back. At wide widths a native adjustable splitter may expose both. Preserve the same stable selection across either layout.

Action precedence

Each surface has one named primary action that advances its current job. Neutral navigation and Copy stay secondary. Destructive or externally visible actions show their exact target. A background service refresh, optional account connection and Stop run do not compete with the task composer for visual emphasis.

> **Revision 8 amendment**
>
> Revision 8: the Work screen now has an explicit priority order: attention (approvals, questions) is pinned above the transcript; the composer is the only send surface; configuration collapses into its toolbar; usage becomes the context-chip count and the Usage inspector. See S01.

4.1 DETAILED UI AND INTERACTION RULES · R7 PAGE 080

Navigation and return paths

Reference implementation requirements for both hosts; native platform conventions take precedence.

Entry, depth and exit

Allow entry from the native command palette, a context menu, an attention item or an Inspect row. Capture a return token containing the source destination, selected entity and focus key. Closing detail returns to that exact place. Never reset to the first list row because the underlying data refreshed.

Task options are a draft editor

Task options edit goal, criteria, constraints, invariants, risks, claims, finish and run caps. Use options updates only that draft. Run task captures it. Editing options while a task runs changes a new task draft; it does not silently amend the active contract.

Native utilities

Persistent preferences belong in VS Code Settings and the JetBrains Configurable infrastructure. Use native choosers for simple selection and walkthrough/setup actions for onboarding. The report's focused settings plates specify content and grouping; they do not require a second competing settings framework inside the chat view.

External editor navigation

Open immutable candidate diffs through vscode.diff / DiffManager or equivalent platform services. Open source and artifact documents in the editor with a correct content identity. Do not use a webview button that launches an unrelated browser or replace a native diff with tiny side-by-side code in a tool window.

4.1 DETAILED UI AND INTERACTION RULES · R7 PAGE 081

Typography, spacing and content fit

Reference implementation requirements for both hosts; native platform conventions take precedence.

Measure a complete field

A field is a visible label, its control, help and validation. Wide related fields share a value edge; different groups may use different appropriate label widths. Reference art uses 24 px insets, 156 px labels, 20 px column gaps, 36 px controls, 8 px help gaps, 16 px row gaps and 24 px section gaps. Native preferred sizes and zoom remain authoritative.

Use intrinsic height

Measure wrapped text before laying out the following row. Inputs and buttons have a minimum, never a forced clipping height. A textarea has a readable minimum, grows to a bounded maximum, then scrolls its own content. Expanding an error or helper moves following content; it never overlays a sibling or the footer.

Type has semantic roles

Use native UI fonts and theme sizes. A section title is modestly stronger than body copy; a field label is clear without making every row a headline. Use monospace for code and exact technical values, not for entire descriptions. Keep line height about 1.4-1.5 for prose and 1.25-1.35 for compact labels.

Wrap meaning, preserve identity

Human labels, permission targets, error reasons and material restore paths must remain readable. A compact selector may elide a long ID only if the full value remains accessible and copyable. Code has a contained full view. Avoid a page-wide horizontal scrollbar, manual spaces for alignment and absolute positions for normal flow.

> **Revision 8 amendment**
>
> Revision 8 supersedes the fixed reference metrics in this rule (24 px insets, 156 px labels, 20 px gaps, 36 px controls, 8 px help gaps, 16 px row gaps, 24 px section gaps). Plates now inherit host metrics: VS Code 13 px UI text and 26 px controls; JetBrains Inter 13 px and 28 px controls; JetBrains label-left forms with colons, VS Code stacked settings rows. Single-line labels are trimmed to cap height and baseline so they centre optically (Part 1.2).

4.1 DETAILED UI AND INTERACTION RULES · R7 PAGE 082

Buttons, forms and validation

Reference implementation requirements for both hosts; native platform conventions take precedence.

Keep the action understandable

Label an action with its verb and affected object when ambiguity is possible: Stop child, Delete project memory, Check receipt. Avoid generic Continue if it hides a write or external effect. Do not use a status chip as a button. Keep Cancel adjacent to the primary action in native dialogs; independent Reset may remain separate.

Validate at the right moment

For simple required input, an obvious empty form may disable its primary action. In complex forms keep Apply reachable so submission can identify all problems. Do not raise a required-field error merely because the user visited it out of order. Validate structural/numeric constraints while useful; defer remote probes to explicit submit/test.

Errors remain local

On apply, mark affected fields and move focus to the first invalid field.

Associate help/error text with that same logical control. An error describes the problem and useful correction. While the user fixes it, preserve the buffer and avoid repeatedly interrupting typing. Backend validation still rejects the entire invalid patch atomically.

Pending is scoped

Freeze only actions that conflict with the pending operation. Keep the draft and navigation usable when safe. Do not optimistically paint Saved, Connected, Allowed or Stopped. A received response must match binding, request generation and expected revision before changing the visible applied state.

> **Revision 8 amendment**
>
> Revision 8: button labels are centred from font metrics (text-box trim to cap height and alphabetic baseline) instead of a fixed text offset; the reference renderer now measures a median 0.0 px offset across 1,036 labels (R7: 4.6 px high). Labels use title capitalisation on both hosts. JetBrains validation follows the platform: tooltip on the field, inline error above the buttons after submit; VS Code shows the native validation message under the input.

4.1 DETAILED UI AND INTERACTION RULES · R7 PAGE 083

Composer, admission and queue

Reference implementation requirements for both hosts; native platform conventions take precedence.

One composer

The text editor, context toolbar, model/agent controls and action row form one consistent group. They are laid out as siblings, not controls floated over text. Keep model, effort and authority available, with advanced tuning in a subordinate surface. Do not create a second prompt editor in every Inspect pane.

Captured versus editable

The active run header describes its captured contract. The composer describes the next message/task draft. Admission captures one body digest, source handles, exact model/route and caps. An acknowledgement clears only that captured draft version; text entered afterward remains intact.

Queue is explicit

While work runs, label the send action Queue message and show its queued receipt. A queue view supports edit/reorder/remove only before admission. Reordering uses a queue revision. A visible queue item is not permission to rewrite an in-flight model request or recall an already admitted effect.

Short heights and IME

Keep the composer reachable without three nested equal scroll areas.

Collapse the run header to a readable strip when needed; options open separately. Enter remains newline by default; the host send command respects keymaps and composition events. Pasting, accepting an IME candidate or closing a menu never submits accidentally.

Revision 8 amendment

Revision 8: model, effort and agent pickers sit in a quiet row under the composer; the send button carries the verb (Run Task or Queue Message) as its accessible name.

4.1 DETAILED UI AND INTERACTION RULES · R7 PAGE 084

Selections, menus, lists and trees

Reference implementation requirements for both hosts; native platform conventions take precedence.

Native component behavior

Prefer native Quick Pick, combo, list, tree and table widgets for the job. A single choice uses a single-selection component; several independent booleans use checkboxes. A radio group remains a coherent group, rather than an arbitrary two-column button grid that obscures exclusivity.

Stable row identity

Bind rows to entity IDs plus owner generation, not array positions or the currently displayed text. Refresh preserves selection and scroll anchor. If an item disappears, focus its nearest stable sibling or section heading. A sorted list never silently directs a pending action at the row that replaced the old index.

Useful density

A list row has a human title, concise supporting context and at most a small set of relevant actions. Put a status and a trailing action on separate baselines when both exist. Do not render a read-only observation as a disabled input. Keep rows large enough for focus and pointer targeting without wrapping every value into a badge.

Search the source

Transcript/history search covers the durable service data through its cursor, not just loaded DOM rows. Filters have a query epoch. Announce no matches separately from no records or read failure. Virtualize presentation while retaining accessible labels, selected keys and a way to load/search the complete underlying set.

4.1 DETAILED UI AND INTERACTION RULES · R7 PAGE 085

Feedback, streaming and attention

Reference implementation requirements for both hosts; native platform conventions take precedence.

Truthful progress

Show actual phases and named operations. Token streaming, a timer or an animated bar is not a percent-complete estimate. Distinguish queued, admitted, running, needs approval, needs input, checking, integrating and terminal run states. Engineering completion has its own proof domain.

Preserve coherent data

During transient reconnect keep the last coherent snapshot with a stale/update status. A stable schema/protocol refusal pauses automatic retry and gives explicit recovery. Preserve the durable cursor; reconcile a new snapshot and subsequent events before enabling effects. Never skip a gap and call the session current.

Respect the reader

Auto-follow only while the user is already following the tail. Preserve manual scroll position and show a return-to-latest affordance when new output arrives. Apply durable events in order; coalesce only rendering. Do not rebuild the whole transcript and lose focused buttons or selected text on each token.

Announce meaningful changes

New approvals and final outcomes announce once, politely unless immediate intervention is required. Batch token prose instead of announcing each fragment. Do not steal focus from typing for background events.

Reduced-motion mode retains static state cues and stops ornamental animation.

4.1 DETAILED UI AND INTERACTION RULES · R7 PAGE 086

Proof, evidence and finish

Reference implementation requirements for both hosts; native platform conventions take precedence.

Four separate domains

Required acceptance, named claim evidence, matching verified/landed identity, and requested finish receipts remain independently inspectable. A finished chat, a passed quick check or a recommended approach cannot fill a missing domain. The global result only becomes complete when all required current domains agree.

Criterion-to-artifact path

Each criterion exposes requiredness, claim binding, source identity, result state and exact evidence refs. Open one bounded artifact excerpt with full content available. If immutable content is absent, explain the absence. Do not compare an old claim against a convenient current file and present it as the original evidence.

Mismatch and manual edits

Changing candidate or owner checkout invalidates matching proof as appropriate. Show the old verified result as historical, not current green success. Restore and integration recheck content-addressed preconditions. Preserve unrelated user edits and uncovered external effects; no universal Undo promise is allowed.

Hierarchical finish

Local, Commit, Push and PR explicitly include preceding steps. Show native durable receipts for each requested step. Missing canonical SCM configuration blocks that step. Retry only after reconciliation and only for the blocked work; a successful arbitrary shell command does not establish a native push or PR receipt.

4.1 DETAILED UI AND INTERACTION RULES · R7 PAGE 087

Agents, comparisons and coordination

Reference implementation requirements for both hosts; native platform conventions take precedence.

Intent is not authority

Code, Ask, Plan, Debug, Review and custom presets describe intent and instruction content. They do not authorize a tool or direct owner-checkout mutation. Enforced Ask needs the runtime read-only envelope. Model/role selection, tool grants, routing and finish policy remain separate objects.

One parent envelope

Children inherit bounded authority and share parent spend/reservations. Their ceilings do not create extra money or token allowance. Show assignment, owner, wait reason and explicit queue saturation. Steering is a recorded input; stopping a child uses its current generation, never an old handle.

Compare on a real basis

Hold task, criteria and context constant when making a quality/cost comparison. Keep candidate ownership, exact models and actual routes visible. Record attempted spend from failed/cancelled eligible attempts. A winner recommendation does not automatically integrate or certify a completed engineering outcome.

Durable coordination

Keep Coordination below Agents. Posts have bounded subject/body, provenance and stable revision paging. Read acknowledgement is explicit and tied to the loaded head. Preserve a typed post through refresh. A server without the route is unavailable, never a fabricated empty discussion.

4.1 DETAILED UI AND INTERACTION RULES · R7 PAGE 088

Context, attachments and memory

Reference implementation requirements for both hosts; native platform conventions take precedence.

Accounting basis

Measured submitted input, model capacity, runtime allocation, draft estimates, output reserve and safety reserve are different values. A null contextUsage displays Unavailable with no invented percentage. A breakdown sums to its stated measured basis; output reserve is not consumed input.

Admission per source

Show source URI/range/version or upload handle, type, byte size and admission result. Unsupported/oversize/failed sources remain present with a recovery path. Switching model revalidates staged sources. Turning on all open files must show precisely what enters the next call.

Compaction preserves obligations

Preview retained facts and estimated reduction before admitting a compaction. Preserve goal, criteria, constraints, candidate/evidence identities, authority, unresolved questions and unknown effects. Failed or inadequate compaction leaves the last valid basis. Original history/artifact references remain available.

Memory deletion is bounded

Show source, origin, retrieval scope and version before edit/delete. Deletion governs future retrieval after acknowledgement; earlier prompts, artifacts or logs may retain copies. Learned failure records and personal memory are different facilities. Never reinterpret one store as the other without a real contract.

4.1 DETAILED UI AND INTERACTION RULES · R7 PAGE 089

Native host implementation

Reference implementation requirements for both hosts; native platform conventions take precedence.

VS Code

Reuse contribution points, commands, native Settings, Quick Pick, TreeView, text documents and vscode.diff where they fit. Keep one minimal Faktor view container and respect user relocation. A custom conversation webview must inherit theme/focus/font tokens and use a typed message bridge; it cannot execute trusted effects itself.

JetBrains

Use the tool-window/action APIs, Configurable, Kotlin UI DSL, JB-scaled metrics and native diff services. Preserve the user's docking, floating and keymap choices. Bind lifecycle to a proper Disposable, keep blocking I/O off EDT and publish only current-generation models on EDT.

Shared contract, independent adapters

Share semantic DTOs, validation results and state reducers at an appropriate boundary, not pixel layout code across incompatible toolkits. Each host adapts native labels, focus, selection, menus, actions and settings scopes. The effective config write path must remain one trusted service, not a second engine in each UI.

Theme and icon roles

Ship VS Code light/dark and JetBrains light/dark examples, plus high-contrast requirements. Derive text, surfaces, selection, focus and validation from host roles. Use a monochrome angular F at native tool-window sizes. Keep marketing Barlow lettering, brand ornament and promotional surfaces out of native reading controls.

> **Revision 8 amendment**
>
> Revision 8: persistent VS Code preferences use the native Settings editor (S29, S64); complex Faktor configuration that needs atomic apply opens as an editor tab. JetBrains uses Configurable pages inside Settings with OK / Cancel / Apply and a Reset link. Simple choices use Quick Pick and popup choosers (S02, S06, S09).

4.1 DETAILED UI AND INTERACTION RULES · R7 PAGE 090

Accessibility, localization and adverse content

Reference implementation requirements for both hosts; native platform conventions take precedence.

Keyboard and assistive semantics

Every interactive element has a visible name and correct role, selected/expanded/disabled state and key behavior. Native tablists, lists and trees follow their host patterns. No webview/Swing focus trap. Announce reasoned unavailable/locked state, rather than silently removing a control from navigation.

Text size and contrast

Test 100%, 150% and 200% UI/text scaling, large system fonts, both native themes and high contrast. Target WCAG AA contrast for readable text and distinguish focus/selection independently of color. Inactive controls can be visually subdued, but their reason remains readable adjacent text.

Long and translated content

Test 50% text expansion, long workspace names, URLs, file paths, IDs and multi-line errors. Use locale-aware display formatting while preserving canonical exact money/token values. Isolate code/IDs left-to-right in RTL text. Keep logical focus order and action meaning when visual direction changes.

Narrow and short

Exercise 240/320/480/720+ logical-pixel content widths and short tool-window heights. Stack fields before clipping labels. Use a selected-detail route instead of several nested panels. Avoid hiding the primary, deny or scoped stop behind an inaccessible toolbar overflow. Overflow menus need labelled equivalents and keyboard reachability.

Revision 8 amendment

Revision 8: target size follows WCAG 2.2 SC 2.5.8 (24 x 24 CSS px minimum); Faktor-drawn icon buttons and links are at least 24 px tall. Measured contrast of host default tokens is reported in Part 1.11, including host values that fall below AA and need a decision.

4.1 DETAILED UI AND INTERACTION RULES · R7 PAGE 091

Implementation checks that catch a messy UI

Reference implementation requirements for both hosts; native platform conventions take precedence.

Component fixtures

Mount every component with one-word, multi-line, empty, translated and unbreakable content. Add help, field error, global refusal, inherited/locked state and pending acknowledgement. Measure the union of its children, then assert it stays inside its content region. A passing screenshot of the shortest English string is insufficient.

Interaction fixtures

For every screen, exercise entry, selection, focus return, explicit action, pending response, success, typed failure and scope switch. Deliver a response after switching conversation and confirm it is ignored. Preserve drafts on failure. Double-click actions and stale handles must not create duplicate effects.

Streaming fixtures

Replay a long transcript, a reconnect gap, a schema refusal and an event burst. Confirm durable order, bounded logs, query epochs and manual scroll anchors. Search old messages outside the loaded window. Observe input responsiveness and assistive announcements on the actual host; reference artwork cannot certify them.

Release evidence

Run repository-prescribed checks, host integration tests and manual accessibility/zoom/theme review after implementation. Record actual measured results and failing cases. The supplied Rust/TypeScript/Kotlin files are construction outlines; they require integration with real types, ownership and platform APIs before any runtime claim.

> **Revision 8 amendment**
>
> Revision 8: reference plates and collision checks cannot certify quality. Part 3 defines task-flow validation in both IDEs with assistive technology, zoom and narrow widths; it has not been executed yet.

4.2 PRODUCT AND BEHAVIOUR SPECIFICATION · R7 PAGE 170

The product and interaction contract

The feature scope is complete. The interface makes its configuration, activity, authority and evidence readable.

Purpose	Turn an engineering objective into an inspectable change. The developer can configure the agent, understand what enters a call, control its execution and evaluate the resulting proof from one coherent workspace.
Everyday agent controls	Fast access to agent, model, supported effort, permission policy, context and budgets. Editor-sized settings cover providers, tuning, tools, customization, memory, reliability and diagnostics.
Engineering evidence	Keep isolated candidate edits, task contracts, native change review, hierarchical finish and all four completion domains: required acceptance, claims, matching integration and requested finish.
Complete construction scope	All 200 control contracts remain required. Every one maps to a screen, current-state classification, build packages and acceptance behavior. Missing backend capabilities must be built before their effects are enabled.
Implementation boundary	The visual plates are design targets, not product screenshots. Current support is pinned to the cited source revision. Code outlines explain required seams and failure handling; they are not installed plugins or a tested product patch.

F01-F40; B01-B29. Scope: Faktor IDE document only. Websites, other product identities and repositories are unchanged.

4.2 PRODUCT AND BEHAVIOUR SPECIFICATION · R7 PAGE 171

What exists today - and what does not

Source pinned to a89c00dbd612bf80048eeb4533855f0fe534954c, 17 October 2026. Do not carry forward the old document's feature assumptions.

Already in both host surfaces	Work / Inspect / History; task versus Ask entry; provider/model choice; task options and exact run caps; native change review; rich verification/outcome views; terminal and agent controls; four hierarchical finish levels Local Commit Push PR. These are redesign foundations, not new marketing claims.
Runtime concepts that need usable controls	Provider declarations and endpoint policy, model capability authority, routing, efficiency switches, compaction defaults, retries, verification policy, hook specs, MCP stdio entries, indexing status and bounded terminal output. Some are operator config rather than writable native settings. The IDE needs read/provenance, validation, persistence and apply acknowledgement.
New contracts before interactive UI	Rich provider CRUD/vault binding, per-call generation tuning, enforceable preset envelopes, durable context accounting, generalized memory, persistent per-tool policy, full MCP management/remote auth, skills management, safe rewind/fork, browser adapters and effective-settings patch/revision APIs. Add these deliberately rather than rendering optimistic controls.
Important host difference	VS Code contributes native settings and commands. JetBrains has one shared application-level settings controller with atomic validation; its persistent settings are not currently project scoped. Native host integration differs even when task semantics agree.
Evidence limits	141 selected source files were retrieved and Git-blob hash verified. The review focused on IDE declarations, settings, core config and native protocol boundaries; it is not a complete security audit or execution test of all code. Later commits may change this baseline.

F01-F07, F19-F26, F28-F30, F38-F40. Verified file inventory is retained as a design evidence summary, not bundled product source.

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Runtime facts that shape the interface

A good settings screen must reflect the actual runtime rather than fill gaps with plausible-looking values.

Ask needs enforcement	The existing Ask path submits a plain prompt without the task/finish contract. That does not itself prove read-only execution. Proposed Ask and Plan presets must restrict offered tools and enforce the same envelope at dispatch.
Context usage is unavailable	The current projection explicitly leaves contextUsage null because there is no durable context-usage read API in this revision. Show measured usage as Unavailable. A draft estimate can still be displayed with its method and basis; it cannot be labelled live utilization.
Efficiency defaults are production ON	failure_learning, ccr, typed_handoff, semantic_context and rework_routing are all enabled by production_defaults(). A type-level test default or stale root comment must not reverse their displayed defaults. information_gain is an alias for semantic_context, not a sixth toggle.
Isolation and networking remain real boundaries	Mutation is shadow-only; direct-owner mode was removed. allow_loopback has adapter-specific defaults and does not permit private LAN destinations. A new LAN policy needs a scoped backend implementation, not a UI workaround.
Generation and rewind need more than labels	A ReasoningMode enum does not prove a writable generation API. A checkpoint hash does not prove stored content for diff/redo. Disable the unsupported operation with its concrete missing prerequisite.
Local deletion and revocation differ	Removing an IDE-stored credential is not remote account revocation. Closing a tool window is not process cancellation. Hiding history is not data deletion. Every action names the exact scope and acknowledgement it can deliver.

Source baseline: a89c00dbd612 / F05, F07-F12, F14-F17, F21, F31.

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Competitive baseline, translated into Faktor

Current primary product documentation was reviewed on 17 October 2026. This is a design comparison, not a claim of identical host implementations.

Model and provider controls	Kilo documents model selection, supported reasoning variants and custom model/endpoint configuration. Claude documents model/effort configuration and policy. Faktor needs fast selection plus a full, capability-driven connection and generation editor. Do not copy discontinued Kilo profiles or assume every model accepts every sampling option.
Authority and customization	Current Kilo exposes per-tool Allow/Ask/Deny and distinct agents. Claude separates permissions, settings, subagents, memory, skills and hooks. Faktor needs equally discoverable control surfaces while retaining its own native schemas and capability enforcement.
Operational visibility	Current agent workflows expose interruption, queueing, background work, context, history and tool state. Faktor should make the current owner, actual model, pending request, process, cost basis and recovery action visible throughout a session.
Rewind and history expectations	Kilo snapshots and Claude checkpointing have defined coverage and restore limitations. Faktor must implement and disclose its own CAS/content boundary. A title filter is not transcript search, and a conversation fork is not a file-system rollback.
Host behavior matters	VS Code guidance favors its settings, commands, themes and editors; IntelliJ guidance favors native tool windows, actions and Configurable. Shared semantics belong in a common state/protocol contract; placement, keymaps and widgets belong in host adapters.

B01-B09, B11-B20, B23-B29. Dedicated benchmark references support each principle; no foreign source code or brand assets are reused.

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Design decisions that make the interface usable

The familiar coding-agent controls are visible; specialist engineering detail remains one deliberate step away.

Fast control row	Agent and provider/model form the first row above the composer. Effort and permission policy form the quieter second row. Context and token/money caps sit below the draft. Tools and connection health open from the toolbar menu, with relevant failures kept in attention. At narrow widths, controls wrap in priority order without hiding authority or Run/Stop.
A real configuration center	The gear opens native settings or an editor-sized configuration route with deep links to the exact category. Avoid ten tiny popovers in the sidebar. Search spans labels, descriptions, source and effective value; modified, overridden, locked and unavailable filters are available.
Progressive disclosure with a complete model	Basic controls show good defaults and meaningful inheritance. Advanced fields expose validation, source and reset. Unsupported relevant controls explain the missing capability; irrelevant ones stay out of normal menus. No form is a decorative promise of future runtime behavior.
Always explain waiting	Running, queued, model thinking, tool active, approval needed, child waiting, retry delay, provider quota, disconnected and unknown effect have different labels and actions. A pending approval is discoverable from Work, Inspect, session history and the host attention badge.
Keep editing and evidence connected	Changed files open in native diffs. Criteria, claim artifacts, candidate identity, verification and finish are linked from the same scoped run. Long details open in the editor area; the compact tool window stays useful while coding.

Screens S01-S32; control groups M-O. Host guidance: B23-B25. All new routes in this document are proposed unless marked E.

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Information architecture and entry points

One conversational workspace, three durable destinations and one searchable configuration system.

Work	Conversation selector and workspace breadcrumb; transcript; active run/attention strip; composer; agent/model/effort/authority; context and caps; task options. New conversation, session manager, settings and overflow are available from the native action toolbar.
Inspect	Overview / Plan / Changes / Verification / Agents, with Tools and Terminal details reached contextually. Each detail keeps session, task, candidate and observation basis visible. Wide diff, context and call detail use native editor tabs; Back restores the prior selection.
History	Conversation title filter and paged results, current-project/worktree scope, engineering outcome, pending attention and available lifecycle actions. Open-transcript Find is separate. Indexed cross-history content search is advertised only after a real read contract exists.
Configuration categories	General; Providers; Models; Model tuning; Agents; Context & retrieval; Compaction & memory; Instructions & rules; Permissions & sandbox; Tools & MCP; Skills & hooks; Terminal & editor tools; Budgets & reliability; Efficiency; Privacy & diagnostics. Search aliases link related categories without duplicating storage.
Command discovery	Register native commands and keymap entries for show/focus Faktor, new conversation, send, stop task, pick model, inspect context, open permissions, find transcript, open settings and diagnostics. Proposed slash aliases are discoverability routes, not invented current CLI commands.

F02, F04, F24, F30; D-01–D-15, O-01, S-05. Category names are display IA, not current config keys.

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Settings contract: requested, effective, applied

Every writable field needs a complete state model. A green Save toast is not proof that the next model call uses the new value.

Field descriptor	Logical ID; type; unit; allowed range/enum; unset representation; default authority; allowed scopes; secret classification; required capabilities; dependency rules; restart/apply timing. Catalog IDs such as G-01 are design identifiers, not existing TOML or API fields.
Resolved snapshot	Return requested value, effective value, winning source, configuration revision, policy lock/reason, capability authority and last acknowledgement. Show Unavailable separately from inherited/unset. A versioned snapshot belongs to the specific daemon and workspace binding.
Scope and precedence	Proposed ordinary preference order: captured task override, permitted project/workspace preference, user/application preference, daemon default. Managed constraints restrict all of them. A field may exclude scopes entirely: binary paths, install roots and secret bindings remain machine-only. JetBrains project preferences need new work.
Transactional editing	Keep a staged draft separate from the applied snapshot. Validate the whole patch, resolve dependencies and show a diff. Commit atomically against an expected revision. Conflict keeps the draft and opens compare/rebase; partial persistence is forbidden. Runtime acknowledgement advances the displayed effective state.
Reset and timing	Reset restores applied values; Restore defaults stages removal of overrides until Apply. Saved but restart-required is labelled. Running requests retain their captured configuration; edits say Next call, Next task, New conversation or Restart required. Never update an active task by changing next-draft defaults.

F01, F05, F07, F30; O-01, O-12, M-12, G-14, I-11. The generalized resolver/patch contract is proposed; existing host atomic validation is retained.

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Protocol additions before enabling the new UX

Build typed contracts around pure core decisions and bounded host adapters. Names below describe capabilities, not existing endpoint routes.

Configuration read and atomic patch	Versioned descriptors and effective snapshot; expected-revision patch; full validation report; secret-reference binding; explicit restart/apply acknowledgement. Never add arbitrary UI fields to strict existing task DTOs: unknown fields currently fail.
Generation and context	Model-adapter parameter descriptors and normalized effective call receipt; versioned context manifest with exact accounting basis, observed input, output reserve and unavailable reasons; manual compaction operation with cancellation, retained-facts manifest and measured result.
Customization and authority	Preset/rule/memory/skill CRUD with version, source and allowed scope; persistent policy simulation and enforcement; hook management over exact native specs; tool/server lifecycle and auth contracts. Prompt instructions cannot substitute for dispatcher policy.
Owned operations and recovery	Session/worktree/daemon generation on callbacks; bounded operation IDs and queues; durable admission lookup; exact approval generation; tool/process effect state; restore preconditions and content coverage; reconciliation before replay. Every response states whether it is terminal or retryable.
UI-only work first	Favorites, disclosure state, density, transcript Find, attributed editor-context picking, command registration and navigation can be implemented in host adapters with bounded local persistence. Host-only state must not claim runtime application or fabricate durable session metadata.

F07, F15, F19-F20, F25, F32, F34, F38-F39; all N and H catalog entries. Backend evolution must remain additive and version-negotiated.

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Context, locality and authority need different displays

A single “context” number cannot answer what was sent, where it was sent or what the agent was allowed to do.

Account for the window	Show capacity authority, observed runtime allocation, estimated draft input, actual submitted input, output reserve, safety reserve and estimated remaining headroom as different fields. Breakdown categories must share a basis and sum correctly. For the screen sample: 18,400 + 4,096 + 2,048 + 8,224 = 32,768.
Attribute every inclusion	Files, selections, tabs, diagnostics, terminal excerpts, diffs, MCP content, retrieved chunks and past chats have source URI, range/version, byte size and inclusion reason. A visible chip explains auto inclusion and allows removal before capture. A filename alone is not a content attachment.
Surface the data boundary	Show editor host, daemon/shell host, model endpoint, embedding endpoint, vault and worker location. “Local” is relative to execution host. An Economy fallback may cross an egress boundary only if policy allows it and the capture discloses the change.
Do not confuse retrieval with enforcement	Search ignores reduce discovery; they do not automatically prevent an authorized shell read. AGENTS.md, skill text and MCP responses are attributed instructions/data, not authority to alter grants. Per-tool dispatch and canonical path/network checks enforce the boundary.
Keep proof across transformations	Compaction, compression, cache use, rework routing and typed handoff cannot erase criteria, pending permission/effect state or candidate identity. A model summary does not re-verify stale artifacts. The UI links full durable evidence behind bounded excerpts.

F07–F11, F17–F18, F23, F27, F31; C-01–C-14, K-01–K-12, O-02. Diagram-free accounting intentionally uses exact numbers and source labels.

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Run lifecycle and truthful status vocabulary

Use explicit state and a next action. Do not infer backend state from animation, elapsed time or a disabled Send button.

Draft validating admitted	Draft is local and editable. Validation reports invalid field/attachment/capability/authority without losing text. Admission freezes the body and UUID; a durable receipt establishes ownership. A response timeout enters Admission unknown until lookup/reconciliation.
Queued active waiting	Queued says owner, capacity/budget reason and whether it can be edited or removed. Active names phase: model, tool, verification or integration. Waiting names approval, child, input, rate limit, connectivity or required proof. Stop remains scoped and discoverable.
Stopping cancelled or effect unknown	A Stop click becomes a pending control request, not an instant terminal fact. Show what acknowledged cancellation and what may already have happened externally. Unknown effect persists through disconnect and requires reconciliation; do not offer automatic repeat.
Terminal conversation versus engineering outcome	Conversation completed, task failed, task cancelled and task stopped are run states. Verified/unverified/pending engineering outcome is derived from required acceptance, claim evidence, matching integration and requested finish. Completed chat text cannot resolve missing evidence.
Connection and freshness overlay	Connected/fresh, connected/stale, reconnecting, stopped, incompatible and refused are independent from task state. Keep the last coherent projection with its age. Never blank to an idle state or merge events from another session after a switch.

F07, F15, F19, F25, F32, F34; B-11-B-12, D-01, D-05, V-09. Transitions must be driven by durable receipts or typed observations.

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Interaction patterns for every screen

The same rules apply to providers, models, permissions, hooks and runtime controls.

Empty, loading, stale and error	An empty collection explains how to add or discover an item. Loading has a cancellable/bounded request, not placeholder values. Stale keeps the last valid data with age. Error names the typed cause, affected scope and safe recovery action; field errors keep edits.
Optimism has limits	Expansion, filtering, favorites and draft editing may be immediate. Saved settings, Connected, Allowed, Cancelled, Restored, Verified and PR created require their corresponding acknowledgements. Duplicate clicks reuse or fence the exact pending operation rather than issue a second effect.
High-impact operation preview	Deleting connection/config/history, remembering a broad rule, running a hook test, restoring files and publishing SCM actions explain exact target, scope, retained data and external effects. Do not add ritual confirmation to harmless navigation or read-only inspection.
Focus and keyboard continuity	Opening a sheet or picker moves focus to its labelled first control. Escape cancels staged popup work and restores the invoker. Form errors focus a summary with links to fields. Dialog close, session switch and Back never discard an unsaved draft silently.
Cancellation and partial results	Keep completed tool artifacts, observed spend and changed-file identity after failure or cancellation. Bounded outputs link to full artifacts. The interface never treats truncated output as complete evidence or a timeout as a confirmed remote rollback.

F05, F07, F15, F19, F25-F26, F32-F34; D-08-D-14. Detailed field rules follow in the 200-control catalog.

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Two native implementations, one semantic contract

The 66 visual plates show 33 distinct states in each editor. Host differences are deliberate rather than theme-only variants.

VS Code placement	Retain the Faktor activity container and relocatable view. Work, Inspect and History live in the view; settings routes use native settings/deep links or editor-sized detail. QuickPick is used for model/session/command selection when its limitations fit the task; rich previews use an editor/detail route.
VS Code implementation map	package.json owns settings, commands, menus and keybindings. webview.ts owns rendering only through typed view models. nativeClient.ts carries validated protocol DTOs; taskStart.ts captures draft/admission; lifecycleFence.ts rejects stale callbacks; changeReview.ts opens native diffs; refreshScheduler.ts coalesces rendering without dropping durable facts.
JetBrains placement	Respect the movable/dockable Faktor tool window. Use native ActionSystem toolbar, lists, dialogs and editor diff. Configuration enters Settings/Preferences through Configurable; the shared application settings controller remains the validation/persistence authority for both native and fallback surfaces.
JetBrains implementation map	FaktorChatPanel owns orchestration and composition; SettingsPanel/Configurable use the shared model/controller. NativeProtocol carries strict DTOs. Dedicated Attachments, History, Agents, Terminal, Outcomes and Changes panels consume the same scoped state. Use UIManager/JBUI metrics and native focus handling.
Shared semantic core	One descriptor schema, state vocabulary, effective-value contract, unit conversion, policy/capability validation and source manifest. Provider quirks stay in adapters. Host implementations share test fixtures and behavior, not copied DOM or a second policy engine.

F01-F07, F19-F26, F28, F30, F32, F34, F38-F40; B23-B25, B29. Proposed adapter changes require product implementation.

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Responsive, accessible and calm under load

The compact sidebar must remain useful on a small laptop and with larger text, keyboard navigation or a screen reader.

Responsive behavior

At 240-319 px, controls become a labelled vertical stack and long values wrap; the primary Send/Stop and attention remain visible. At 320-479 px, two compact rows wrap chips. At >=480 px, show full configuration summary. Wide settings and diffs use editor/dialog space, never a horizontal sidebar scroll trap.

Typography and targets

IDE text inherits host UI/editor fonts and zoom. Minimum interactive icons have accessible names; dense native toolbar targets use host metrics and visible focus, while primary composer actions use 36 px where host metrics permit where space allows. Document typography is brand typography, not a request to replace host fonts.

Keyboard and assistive technology

Use normal Tab/Shift+Tab traversal and native lists/menu semantics. Expose labels, selected state, validation errors, row relations and action scope. Announce phase/approval/error changes deliberately rather than every token; keep the full transcript navigable. IME composition, multiline paste and Enter never unexpectedly submit.

Contrast and motion

Use host semantic tokens, focus colors and high-contrast modes. Pair color with text/icon/shape for all status. No auto-scrolling after the user scrolls up, no blinking progress, no decorative motion. Reduced-motion settings remove nonessential transitions.

Bounded rendering and search

Maintain keyed windowed transcript rendering, anchored scroll and follow-tail state. Page older content and search beyond rendered rows. Coalesce high-frequency paint around host frame scheduling while preserving durable event order. Cap terminal/log buffers and link artifacts; do not set invented performance certifications.

F02, F04, F26, F32; D-08-D-14. Acceptance: 200% text, 240 px panel, keyboard-only, screen reader, high contrast and long-running stream fixtures.

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Implementation waves and release boundaries

Deliver coherent workflows in dependency order. A screen is ready only when its values, effects and recovery are real.

Wave 1 - expose and unify existing capability	Native command/keymap/navigation work; provider/model capability sheet; exact budget UX; task options/finish parity; durable history; native diff; terminal/agents/attention; source-correct defaults; searchable applied settings. Add host-only favorites, context source picking and transcript Find. Avoid adding inert placeholder features.
Wave 2 - configuration and model control	Versioned descriptors/effective snapshot/atomic patch, provider CRUD and secure credential handoff, adapter parameter mapping, normalized request receipt, enforceable agent presets and persistent per-tool policy. Complete model/permission/budget workflows before surfacing advanced sampling knobs.
Wave 3 - context and customization	Durable context manifest/accounting, compaction operations, attributed memory/rules, MCP management with supervised stdio first, hook editor/test, skills lifecycle and bounded command templates. Add remote MCP/OAuth only behind actual transport/auth capabilities.
Wave 4 - advanced recovery and scale	Content-aware restore/fork/redo, remote-workspace binding, bounded parallel session manager, browser adapter, model comparison and deeper outcome analysis. Each requires explicit scope, identity, capacity/ledger and effect reconciliation.
Definition of ready	Both hosts pass the same behavioral fixtures for the delivered slice. Required controls have keyboard access, help, defaults, validation, source, apply timing, unavailable/loading/error/stale states and safe recovery. No fabricated usage or unsupported setting can appear active. Publish only claims backed by actual product tests.

The catalog assigns E/R/N/H per control. These are implementation classifications, not evidence that a proposed feature has already shipped.

4.3 REQUIRED BACKEND AND NATIVE CONSTRUCTION · R7 PAGE 192

Build the missing features - every UX is a requirement

Missing capabilities are assumed to be built. Unavailable states describe honest interim behavior, not permission to omit the feature.

The implementation obligation

All 200 control IDs have a construction row in Build_Map.csv. Each row names current status, required behavior, default, allowed scope, apply timing, screens, backend packages, host work and verification. Every screen references those same IDs; nothing in the visual design is left as an unowned placeholder.

Existing features need complete UX too

E entries reuse current operations and add the specified presentation, state and recovery. R entries require a native adapter/read-write contract where only runtime config exists. N entries require new backend work. H entries require bounded host implementation. These labels explain the starting point, not release priority or optionality.

Backend-first gates, parallel host work

Implement pure decision types and store/adapter contracts first; host layouts can develop against versioned fixture snapshots. Enable effects only after server capability negotiation and end-to-end acceptance. A mock must never become the production source of model usage, permissions, saved settings or completion.

Outlines are a construction guide

The handoff includes original Rust, TypeScript, Kotlin and CSS outlines, schema/spacing tokens, fixtures and a build playbook. They name the seams, ownership and tricky branches. They are deliberately unintegrated reference code; actual repository types, async primitives and error enums must be reused or extended rather than duplicated.

4.3 REQUIRED BACKEND AND NATIVE CONSTRUCTION · R7 PAGE 193

Backend work packages and ownership

Use domain boundaries already present in Faktor. Keep pure decisions inward and provider/platform effects in adapters.

BK01-BK03: configuration, models and customization	BK01 supplies descriptors, effective snapshots, atomic patches and secret references. BK02 supplies provider lifecycle, capability/parameter mapping and call receipts. BK03 supplies trusted versioned presets/rules/memory/skills and deterministic instruction manifests. Rich UI depends on all three agreeing on IDs, scope and version.
BK04-BK06: context, authority and tools	BK04 persists exact context/accounting and compaction. BK05 owns effective policy, simulation, enforcement and truthful sandbox status. BK06 owns tool/MCP/hook lifecycle, auth and bounded process effects. A visible control never broadens authority just because it is enabled or installed.
BK07-BK10: execution and proof	BK07 owns shared money/token reservations, routing/retries/efficiency reads and changes. BK08 owns session/queue/process/remote identity. BK09 owns review/restore/fork/evidence/finish. BK10 owns runtime trust, readiness, diagnostics and optional services. Cross-package operations use durable IDs and typed receipts.
UI01-UI03: native delivery	UI01 and UI02 are host adapters over the same semantic fixtures. UI03 is the spacing/text/overflow system and visual acceptance gate. Use adapters to fit each host; do not move policy or ledger decisions into a webview or Swing event listener.

4.3 REQUIRED BACKEND AND NATIVE CONSTRUCTION · R7 PAGE 194

Configuration service: types, validation and apply

Reference: contracts/configuration.rs. Add a typed native configuration service; do not loosen existing strict task-run bodies.

Pure patch decision	Take current revision, descriptors, policy, capability snapshot and proposed changes. Validate field types/units/ranges, permitted scope, secret-reference kind and cross-field dependencies. Return a complete error list or an immutable validated patch. Unknown keys fail; defaults resolve on the server.
Transactional persistence	Compare expected revision inside the store transaction; persist all changes plus the new revision or none. Return saved revision and apply state per field. A separate runtime/config reload acknowledgement advances effective state. Persisted-but-not-applied is a first-class state.
Secret handoff	Define write-only secret binding at the machine-owned vault/daemon boundary. Return opaque reference and verification state only. Store no secret values in normal snapshots, request previews, logs, shared project files or config export. Rotation has a new binding generation; deletion and remote revoke are distinct.
Host integration tip	Do not teach each widget precedence. Read the descriptor/snapshot once, create a staged form model and send one validated patch. Render source, lock and timing beside the field. Keep the draft on conflict and show a compare view; never overwrite another editor's changes silently.

4.3 REQUIRED BACKEND AND NATIVE CONSTRUCTION · R7 PAGE 195

Providers and model tuning: a real parameter contract

Reference: contracts/generation.rs. Implement all G controls through adapter-supported parameter descriptors and normalized receipts.

Provider lifecycle	Create/update/test/delete a stable connection ID with API family, endpoint and secret reference. Test stages are reachability, authentication, model listing and selected capability probes; each has a bounded deadline and distinct result. A probe may cost usage and must not be described as a coding benchmark.
Parameter descriptors	Serve supported enums/ranges, unset/default semantics, mutual constraints and apply timing for reasoning, thinking/output budgets, temperature, top-p/k, penalties, seed, stops, parallel tools, output schema, tier/cache and streaming. Do not infer support from provider marketing names.
Call compilation	Validate requested values against exact capability authority and enforced policy; normalize in the provider adapter; capture effective parameters and actual route with the admitted call. If an option is unsupported or unmapped, reject before dispatch rather than silently ignore it.
Local runtime extension	Add allocation, keep-alive and unload only to adapters that can observe/apply them with ownership checks. LAN access needs a new destination-scoped authenticated policy. Keep existing loopback defaults and forbidden destination classes intact until that policy is implemented.

4.3 REQUIRED BACKEND AND NATIVE CONSTRUCTION · R7 PAGE 196

Context and compaction: durable accounting, not guessed meters

Reference: contracts/context.rs. Build the currently missing context read model before labelling any value measured.

Persist the manifest	At context compilation record source ID/URI/version/hash, inclusion cause, category, byte/token basis and omission reason; bind it to call/task/model/capability revision. Keep estimates versioned and separate from observed provider counts. Include capacity authority, allocation and output/safety reserves.
Validate accounting	Sum nonoverlapping categories with checked integers. Reject input + output reserve + safety reserve above the admitted window; do not hide-truncate task facts. If tokenizer/accounting is unknown, expose unknown basis and apply the runtime's conservative admission policy.
Compaction operation	Introduce an owned operation with expected context/task version, previewed retained-facts set and model route. Preserve contract, permissions, candidate identity, pending effects and source references. A cancellation/failure leaves the prior basis. Persist measured result and summary version only after successful commit.
UI rendering tip	Use a labelled segmented bar only when categories share the same basis. Put numbers in text, keep the legend aligned and separate consumed input from reserved capacity. Estimate and Measured are explicit adjacent labels, never a tooltip-only caveat.

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Permissions, tools and hooks: one enforcement boundary

References: contracts/authority.rs and contracts/tools.rs. The UI previews policy; the dispatcher enforces it.

Shared evaluator	Represent normalized tool/action/target, preset grant, persistent policy, managed constraints and expiry. A pure evaluator returns Allow, Ask or Deny with rule/version/reason. The simulator calls the same function. Canonical path, shell command and network normalization happen in trusted adapters, not string widgets.
One-shot waiter	A response includes exact session/request/generation and captured action fingerprint. Consume once transactionally. Expired/wrong-session/changed-action/duplicate replies return typed results. Remember decision creates a separate versioned rule and does not bless unrelated pending operations.
Supervised extensions	MCP first needs bounded stdio CRUD, environment references, start/stop/discovery/health/logs and toolset identity. Add remote transport/OAuth as a later negotiated extension, not a property forced into today's McpEntry. New tools resolve policy by stable namespace.
Hooks and customization	Hook tests use real capability, environment allowlist, deadline and output caps. Reuse native event/failure enums. Skills/rules/presets carry trusted origin/version and cannot change grants by instruction text. Memory CRUD includes provenance, retention and acknowledged future-use deletion.

4.3 REQUIRED BACKEND AND NATIVE CONSTRUCTION · R7 PAGE 198

Sessions and recovery: own every asynchronous effect

Reference: contracts/operations.rs. Add any missing queue/search/fork/remote operations with explicit identity and capacity.

Binding and idempotence	Bind UI state and requests to daemon/workspace/worktree/session/task generation. Capture one immutable submission body and UUID. Same key/same body may recover its receipt; changed body conflicts. Reconnect looks up receipt and cursor before considering replay.
Queue, children and process owners	Create bounded queues with owner/capacity/cancel state. Child admission reserves within parent budget and grants edit ownership explicitly. Terminal controls require a live owned handle, not just PID. Closing an output tab is host UI state only.
Effect-aware control	Stop/pause/background/model/steer actions advertise actual support and return accepted/settled/refused/unknown effect. Keep stopped model, owned process and remote side effect distinct. Do not “complete” an effect by disabling a button or dropping a subscription.
History and remote binding	Title filter, paged transcript Find and indexed cross-session recall need separate read contracts. Rename/archive/fork require durable metadata. Remote rebind invalidates old handles, preserves drafts and requires URI/vault/model locality review before resuming effects.

4.3 REQUIRED BACKEND AND NATIVE CONSTRUCTION · R7 PAGE 199

Review, restore, money and proof: preserve exact identity

References: contracts/review.rs and contracts/budget.rs. Never infer reversible effects or financial facts from presentation state.

Content-aware restore	Record optional content handles alongside before/after hashes and disclose file/effect coverage. Preview expected-current hashes and target content; enforce all CAS preconditions and restore transactionally. Missing content or unrelated edits refuse safely. Fork conversation lineage separately from file restore.
Candidate and verification	Tie review comments, required criteria, claim artifacts, checks, integration and requested finish to exact candidate identity. Any changed/partially selected candidate invalidates stale evidence. Retain historical proof; do not recast it as proof for the new tree.
Exact budgets	Parse decimal USD into checked micro-USD without float arithmetic; validate nonnegative integer tokens. Reserve against the shared parent envelope, settle/refund with durable ledger identity and preserve uncertain charges. A revision cannot erase existing obligations or assume a missing price is zero.
Outcome reads	VCSC uses all eligible attempted spend divided by verified landed changes for an explicit cohort/period. No verified denominator means unavailable. Successful-run cost, local estimates, provider quota, list rates and observed charges are separate named fields.

4.3 REQUIRED BACKEND AND NATIVE CONSTRUCTION · R7 PAGE 200

VS Code implementation outline

Reference: hosts/vscode/controller.ts and layout.css. Use the current plugin architecture; the outline is framework-neutral.

State and rendering	Separate immutable scoped state, staged settings/draft state and local disclosure/scroll state. Render pure view models; commands carry current binding and expected revision. A stale async result is discarded before it can mutate the selected session. Never interpolate untrusted HTML into a webview.
Native entry points	Register command IDs/keybindings/menus in package.json. Use native settings for simple contributed preferences, QuickPick for one-dimensional choices and native diff for review. Use an editor-sized advanced form when multi-field validation and staged atomic Apply are necessary; do not force that into a tiny picker.
Safe layout	Use host font/color/focus tokens, minmax(0,1fr), min-width:0 on grid/flex children, bounded text wrapping, visible helper/error slots and independent scrolling. The composer is a sibling of the transcript, not an absolutely positioned overlay. Do not rely on fixed height or magic negative margins.
Lifecycle and performance	Dispose subscriptions on view/session changes; reject late results by binding; page data and coalesce paint. Keep durable event application separate from render scheduling. Preserve manual scroll and focus during streaming; never rerender the entire form and erase a user's in-progress input.

4.3 REQUIRED BACKEND AND NATIVE CONSTRUCTION · R7 PAGE 201

JetBrains implementation outline

Reference: hosts/jetbrains/FaktorController.kt. Render native components and keep network/process effects off the EDT.

Model/controller separation	Use typed immutable snapshots and one shared settings controller. Read/validate/commit asynchronously; publish only current binding results on EDT. Disposable scopes cancel observations when the tool window/project closes. Do not hold a project reference in a global callback.
Native layout	Use UI DSL/FormBuilder or equivalent native layout, JBUI-scaled insets, UIManager fonts/colors, ActionSystem and native diff. Prefer preferred/minimum sizes and weighted rows over setBounds or hardcoded Swing pixels. Labels use labelFor; validation uses the platform's error treatment.
Settings semantics	Configurable.isModified compares staged and applied values. apply validates all before one write; reset restores applied values; Restore defaults stages defaults until apply. Current application scope remains explicit; new project settings need a separate allowed-scope service rather than writing application keys per project.
Focus and task controls	Restore the invoking component after modal close; maintain selected list row and text caret during snapshots. Disable only the pending operation's action until acknowledgement. Run stop, process stop, approval and model controls retain exact session/generation scope.

4.3 REQUIRED BACKEND AND NATIVE CONSTRUCTION · R7 PAGE 202

Spacing system and text layout rules

UI03 is a release requirement. Use semantic host-scaled tokens rather than individual developer guesses.

The shared grid	Use a 4 logical-pixel base. The wide surface has 24 px insets, a 156 px label column, 20 px column gutter, 36 px controls, 8 px helper gap and 16 px between complete field rows. Section gaps are 24 px. Compact host surfaces may adapt insets deliberately; they do not move individual controls off-grid.
Type and intrinsic height	Inherit host UI fonts, size and zoom. Body line-height is 1.4-1.5; labels 1.25-1.35. Controls grow when text or native metrics require it. A field row is label/control plus its helper or error plus the shared row gap. Never clip a fixed single-line box to make the mockup fit.
Responsive widths	At less than 560 logical px, stack labels above values. Use min-width:0 and wrap paths, IDs, descriptions and error text deliberately. Ellipsis is allowed only for compact selectors that retain their complete accessible/copy value. Material command and restore targets remain fully inspectable.
Content and actions	Lay out header, minmax(0,1fr) scroll content and intrinsic footer as siblings. The composer and action bar never overlay messages or form rows. Errors expand their own field. Reserve attention for blockers and effect scope, not decorative badges or implementation jargon.

4.3 REQUIRED BACKEND AND NATIVE CONSTRUCTION · R7 PAGE 203

Microcopy, overflow and state implementation tips

The wording and empty/error states are designed behavior, not polish left to the end.

Field text	Use sentence-case labels and explicit units: Maximum output (tokens), Total attempts, Run cap (USD). Helper text explains consequence or timing in one or two short sentences. Validation says what is wrong and how to fix it. Keep the user's invalid input; do not coerce or silently round.
Button text and feedback	Use verbs with scope: Allow once, Stop task, Stop process, Apply changes, Open native diff, Review restore. During acknowledgement show Applying… or Stopping… beside the same action. Connected, Saved, Applied, Verified and Restored require their different receipts; a toast never replaces persistent effective state.
Overflow and attention	Long URLs, model IDs, rule paths and localized text are fixture inputs. At narrow widths stack rather than shrink text. Tooltips supplement visible labels; they never carry the only error, capability limitation or dangerous effect scope. Badge counts and status icons always have accessible names.
Visual review gate	Review every screen with empty/large/invalid/stale data; 240/320/480 px tool windows; 200% text; long names; keyboard/IME; both host themes/high contrast; and running stream. Capture reference snapshots at fixed fixtures; reject overlap, hidden actions, off-grid helper text or mismatched state before merge.

4.3 REQUIRED BACKEND AND NATIVE CONSTRUCTION · R7 PAGE 204

Build, integrate and verify the full specification

The handoff is designed to become tracked implementation work, with every UX connected to a contract and acceptance condition.

Turn the map into work items

Create work items from Build_Map.csv by package and dependency, preserving each control ID and its screen links. Include the full required behavior, allowed scope, timing and verification row. E/R/N/H does not authorize omitting UX; negotiate the backend version and then complete both host adapters.

Reuse native code and tests

Read current AGENTS.md and repository contribution commands before editing the product. Extend existing typed DTOs/config/owners rather than duplicate policy. Add focused pure-core validation/state-machine tests, adapter contract fixtures and host integration/screenshot cases for the delivered behavior.

Reference outlines and fixtures

contracts/*.rs, hosts/vscode/controller.ts, hosts/vscode/layout.css, hosts/jetbrains/FaktorController.kt, spacing-tokens.json and ux-fixtures.json demonstrate the intended seams. They are original design reference code, not installed plugins or a drop-in patch. Integrate them into real workspace types and run the repository's checks.

Acceptance and traceability

The 40 scenarios in this document cover the major failure boundaries; every one of the 200 build rows additionally requires its own default, validation, apply, fence and recovery verification. Native visual plates are reference fixtures, not evidence that runtime behavior is already implemented.

4.4 IMPLEMENTATION CODE OUTLINES · R7 PAGE 205

Atomic configuration

contracts/configuration.rs / original unintegrated reference code

```rust
pub struct Snapshot { pub revision: u64, pub values: BTreeMap<String, Value> }
pub struct ValidatedPatch(Patch); // Only validation constructs this token.
pub enum ApplyState { Effective, NextCall, NextTask, NewSession, RestartRequired, Failed(String) }
pub struct CommitReceipt {
    pub saved_revision: u64, pub runtime_revision: Option<u64>,
    pub field_states: BTreeMap<String, ApplyState>,
}
pub trait DomainValidation {
    fn validate(&self, descriptor: &Descriptor, change: &Change) -> Vec<FieldError>;
    fn cross_validate(&self, snapshot: &Snapshot, patch: &Patch) -> Vec<FieldError>;
}
pub fn validate_patch(
    snapshot: &Snapshot, descriptors: &BTreeMap<String, Descriptor>,
    patch: Patch, domains: &impl DomainValidation,
) -> Result<ValidatedPatch, Vec<FieldError>> {
    let mut errors = Vec::new(); let mut seen = BTreeSet::new();
    if patch.expected_revision != snapshot.revision {
        errors.push(FieldError { key: "revision".into(), message: "Refresh and rebase the draft" });
    }
    for change in &patch.changes {
        let Some(d) = descriptors.get(&change.key) else {
            errors.push(FieldError { key: change.key.clone(), message: "Unknown setting" }); continue;
        };
        if !seen.insert((change.key.clone(), change.scope.clone())) ||
            !d.writable || !d.allowed_scopes.contains(&change.scope) {
            errors.push(FieldError { key: change.key.clone(), message: "Duplicate, locked or invalid scope"
});
        }
        if d.secret_reference_only && !matches!(change.value, Value::SecretReference(_) | Value::Unset) {
            errors.push(FieldError { key: change.key.clone(), message: "Use an opaque secret reference" });
        }
        errors.extend(domains.validate(d, change));
```

Build intent

Validate the full staged patch, compare the expected revision inside one transaction, then acknowledge runtime application separately.

Reuse the actual repository types, async owners and adapters. These outlines are a construction guide, not a compiled product patch.

The handoff contains complete reference files, per-control build mapping and required integration checks.

4.4 IMPLEMENTATION CODE OUTLINES · R7 PAGE 206

Generation parameters

contracts/generation.rs / original unintegrated reference code

```rust
pub struct ModelDescriptor {
    pub provider_id: String, pub model_id: String, pub authority: Authority,
    pub parameters: BTreeMap<String, ParameterDescriptor>,
}
pub struct RequestDraft { pub values: BTreeMap<String, Parameter> }
pub struct EffectiveRequest {
    pub provider_id: String, pub model_id: String,
    pub parameters: BTreeMap<String, Parameter>, pub capability_revision: u64,
}
pub fn compile(
    model: &ModelDescriptor, draft: RequestDraft, revision: u64,
) -> Result<EffectiveRequest, Vec<String>> {
    let mut errors = Vec::new();
    for (key, value) in &draft.values {
        let Some(spec) = model.parameters.get(key) else {
            errors.push(format!("{key}: unsupported parameter")); continue;
        };
        if spec.requires_exact_authority && !matches!(model.authority, Authority::KnownExact) {
            errors.push(format!("{key}: exact capability authority required"));
        }
        let valid = match (&spec.kind, value) {
            (Kind::Number { min, max }, Parameter::Number(v)) => v.is_finite() && v >= min && v <= max,
            (Kind::Integer { min, max }, Parameter::Integer(v)) => v >= min && v <= max,
            (Kind::Choices(choices), Parameter::Choice(v)) => choices.contains(v),
            (Kind::Toggle, Parameter::Toggle(_)) => true, _ => false,
        };
        if !valid { errors.push(format!("{key}: invalid type or value")); }
    }
    // Also call adapter cross-field validation: context/output/thinking,
    // sampling restrictions, structured tools, stops and enforced policy.
    if !errors.is_empty() { return Err(errors); }
    Ok(EffectiveRequest { provider_id: model.provider_id.clone(), model_id: model.model_id.clone(),
```

Build intent

Only compile parameters offered by the selected adapter and capability authority. Unsupported or unmapped fields fail before dispatch.

Reuse the actual repository types, async owners and adapters. These outlines are a construction guide, not a compiled product patch.

The handoff contains complete reference files, per-control build mapping and required integration checks.

4.4 IMPLEMENTATION CODE OUTLINES · R7 PAGE 207

Context and compaction

contracts/context.rs / original unintegrated reference code

```rust
pub enum Basis { Estimated { method: String, version: String }, Observed { receipt: String }, Unavailable(String) }
pub struct Source {
    pub id: String, pub uri: String, pub version: String, pub content_hash: String,
    pub category: String, pub tokens: Option<u64>, pub inclusion_reason: String,
}
pub struct Manifest {
    pub revision: u64, pub call_id: String, pub model_id: String,
    pub capacity: u64, pub input: Option<u64>, pub output_reserve: u64,
    pub safety_reserve: u64, pub basis: Basis, pub sources: Vec<Source>,
}
pub fn headroom(m: &Manifest) -> Result<Option<u64>, &'static str> {
    let Some(input) = m.input else { return Ok(None); };
    let total = input.checked_add(m.output_reserve)
        .and_then(|v| v.checked_add(m.safety_reserve)).ok_or("Accounting overflow")?;
    m.capacity.checked_sub(total).map(Some).ok_or("Context exceeds the admitted capacity")
}
pub fn sum_categories(values: &[u64]) -> Result<u64, &'static str> {
    values.iter().try_fold(0_u64, |sum, value| sum.checked_add(*value).ok_or("Category overflow"))
}
pub struct RequiredFacts {
    pub contract_version: String, pub candidate_id: String, pub policy_version: String,
    pub pending_operations: Vec<String>, pub evidence_refs: Vec<String>,
}
pub struct CompactionPlan {
    pub operation_id: String, pub expected_context_revision: u64,
    pub retained: RequiredFacts, pub model_route: String,
}
pub enum CompactionOutcome {
    Applied { new_revision: u64, measured_before: u64, measured_after: u64, summary_ref: String },
    Refused(String), Cancelled, Failed(String), EffectUnknown,
```

Build intent

Persist one versioned accounting basis and source manifest. Unknown usage remains unavailable. A failed compaction keeps the prior basis.

Reuse the actual repository types, async owners and adapters. These outlines are a construction guide, not a compiled product patch.

The handoff contains complete reference files, per-control build mapping and required integration checks.

4.4 IMPLEMENTATION CODE OUTLINES · R7 PAGE 208

Policy and approvals

contracts/authority.rs / original unintegrated reference code

```
pub struct Action { pub tool_id: String, pub canonical_target: String, pub effect: String }
pub struct Rule { pub tool_id: String, pub target: String, pub decision: Decision }
pub struct Policy { pub revision: u64, pub managed_denies: Vec<Rule>, pub rules: Vec<Rule> }
pub struct Verdict { pub decision: Decision, pub revision: u64, pub reason: &'static str }
fn matches(rule: &Rule, action: &Action) -> bool {
    // Exact-match example ONLY. Real typed matchers must be normalized server-side.
    rule.tool_id == action.tool_id && rule.target == action.canonical_target
}
pub fn evaluate(policy: &Policy, action: &Action, envelope_allows: bool) -> Verdict {
    if !envelope_allows || policy.managed_denies.iter().any(|r| matches(r, action)) {
        return Verdict { decision: Decision::Deny, revision: policy.revision, reason: "Enforced constraint" };
    }
    let decision = policy.rules.iter().find(|r| matches(r, action))
        .map(|r| r.decision.clone()).unwrap_or(Decision::Ask);
    Verdict { decision, revision: policy.revision, reason: "Effective rule or unmatched Ask" }
}
pub struct Approval {
    pub session: String, pub generation: u64, pub request: String,
    pub action_fingerprint: String, pub consumed: bool, pub expired: bool,
}
pub struct Reply {
    pub session: String, pub generation: u64, pub request: String, pub action_fingerprint: String,
}
pub fn consume(waiter: &mut Approval, reply: &Reply) -> Result<(), &'static str> {
    if waiter.expired || waiter.consumed { return Err("Expired or already resolved"); }
    if waiter.session != reply.session || waiter.generation != reply.generation ||
        waiter.request != reply.request || waiter.action_fingerprint != reply.action_fingerprint {
        return Err("Wrong owner, generation or changed action");
    }
    waiter.consumed = true; Ok(()) // Perform atomically in the actual owner/store.
}
```

Build intent

The simulator and runtime share a trusted evaluator. Consume one exact waiter generation; persistent rules use a separate versioned operation.

Reuse the actual repository types, async owners and adapters. These outlines are a construction guide, not a compiled product patch.

The handoff contains complete reference files, per-control build mapping and required integration checks.

4.4 IMPLEMENTATION CODE OUTLINES · R7 PAGE 209

MCP and hooks

contracts/tools.rs / original unintegrated reference code

```rust
//! Supervised MCP and native hooks. Remote/OAuth requires a separate capability contract.
pub struct StdioServer { pub name: String, pub command: String, pub args: Vec<String> }
pub fn validate_stdio(existing_count: usize, s: &StdioServer) -> Result<(), &'static str> {
    if existing_count >= 8 { return Err("At most 8 servers"); }
    if s.name.is_empty() || s.name.len() > 128 { return Err("Name must fit 128 UTF-8 bytes"); }
    if s.command.is_empty() || s.command.len() > 512 { return Err("Command must fit 512 bytes"); }
    if s.args.len() > 32 || s.args.iter().any(|a| a.len() > 512) { return Err("Argument bounds exceeded"); }
    Ok(())
}
pub enum Health { Stopped, Connecting, Ready, AuthenticationRequired, Failed(String), Stale }
pub enum HookEvent {
    SessionStart, SessionResume, TaskStart, PreModel, PostModel, PreTool, PostTool,
    ToolError, PreEdit, PostEdit, PreCommit, SubagentStart, SubagentStop,
    AgentError, AgentStop, TaskComplete, SessionEnd,
}
pub enum FailurePolicy { FailClosed, FailOpen, Warn }
pub enum HookOutcome { Verdict(String), TimedOut, NonZero(i32), Malformed, Cancelled, EffectUnknown }
pub struct HookBounds { pub deadline_ms: u64, pub stdout_bytes: u64, pub stderr_bytes: u64 }
pub struct HookSpec {
    pub id: String, pub event: HookEvent, pub executable: String, pub args: Vec<String>,
    pub environment_allowlist: Vec<String>, pub bounds: HookBounds,
    pub failure_policy: FailurePolicy, pub capability_scope: String,
}
// Required adapters: trusted config store; vault refs; owned process lifecycle;
// bounded discovery/output/log pages; policy evaluator; actual protocol handshake.
// A server becomes Ready only after observed handshake, not after config save.
// A timeout ignores partial stdout verdicts; explicit failure policy is resolved separately.
// Hook testing never emits a real production lifecycle event merely to preview input.
```

Build intent

Build supervised stdio lifecycle first; add remote transport and auth through explicit capabilities. Hook deadlines and output bounds remain enforced.

Reuse the actual repository types, async owners and adapters. These outlines are a construction guide, not a compiled product patch.

The handoff contains complete reference files, per-control build mapping and required integration checks.

4.4 IMPLEMENTATION CODE OUTLINES · R7 PAGE 210

Customization trust

contracts/customization.rs / original unintegrated reference code

```rust
pub enum Origin { BuiltIn, User, Project, Managed }
pub enum Invocation { Manual, AgentSelectable, Disabled }
pub struct VersionedContent {
    pub id: String, pub revision: u64, pub origin: Origin, pub source_uri: String,
    pub content_hash: String, pub trusted: bool, pub text: String,
}
pub struct Preset {
    pub content: VersionedContent, pub allowed_tools: BTreeSet<String>,
    pub child_roles: BTreeSet<String>, pub maximum_children: usize,
}
pub struct Skill {
    pub content: VersionedContent, pub invocation: Invocation,
    pub required_capabilities: BTreeSet<String>,
}
pub fn may_invoke(skill: &Skill, agent_selected: bool, granted: &BTreeSet<String>) -> Result<(), &'static str> {
    if !skill.content.trusted { return Err("Review source and trust first"); }
    if matches!(skill.invocation, Invocation::Disabled) ||
        (agent_selected && matches!(skill.invocation, Invocation::Manual)) { return Err("Invocation disabled"); }
    if !skill.required_capabilities.is_subset(granted) { return Err("Missing required capability grants"); }
    Ok(())
}
pub struct Memory {
    pub content: VersionedContent, pub source_receipt: String, pub created_at: String,
    pub retention_policy: String, pub project_id: Option<String>,
}
pub struct DeletionReceipt {
    pub record_id: String, pub stopped_future_use_at_revision: u64,
    pub retained_prior_artifact_refs: Vec<String>,
}
// Build list/read/diff/save/delete services with expected revision and allowed scope.
```

Build intent

Version the source, origin and scope. A skill or preset remains instruction content and cannot grant itself more execution authority.

Reuse the actual repository types, async owners and adapters. These outlines are a construction guide, not a compiled product patch.

The handoff contains complete reference files, per-control build mapping and required integration checks.

4.4 IMPLEMENTATION CODE OUTLINES · R7 PAGE 211

Budgets and ledger

contracts/budget.rs / original unintegrated reference code

```rust
//! Exact micro-USD conversion and shared ledger envelope. No floating-point currency.
pub fn parse_usd(s: &str) -> Result<u64, &'static str> {
    let mut parts = s.split('.'); let whole = parts.next().ok_or("Missing amount")?;
    let fraction = parts.next().unwrap_or("");
    if parts.next().is_some() || whole.is_empty() || fraction.len() > 6 ||
        !whole.bytes().all(|b| b.is_ascii_digit()) || !fraction.bytes().all(|b| b.is_ascii_digit()) {
        return Err("Use a non-negative decimal with at most six fractional digits");
    }
    let dollars: u64 = whole.parse().map_err(|_| "Amount overflow")?;
    let micros: u64 = if fraction.is_empty() { 0 } else {
        fraction.parse::<u64>().map_err(|_| "Invalid fraction")?
            .checked_mul(10_u64.pow(6 - fraction.len() as u32)).ok_or("Amount overflow")?
    };
    dollars.checked_mul(1_000_000).and_then(|v| v.checked_add(micros)).ok_or("Amount overflow")
}
pub struct Envelope { pub cap: Option<u64>, pub settled: u64, pub open: u64, pub uncertain: u64 }
pub fn may_reserve(e: &Envelope, additional: u64) -> Result<bool, &'static str> {
    let obligations = e.settled.checked_add(e.open).and_then(|v| v.checked_add(e.uncertain))
        .and_then(|v| v.checked_add(additional)).ok_or("Ledger overflow")?;
    Ok(e.cap.map(|cap| obligations <= cap).unwrap_or(true))
}
pub fn vcsc(attempted_micro_usd: u64, verified_changes: u64) -> Option<f64> {
    if verified_changes == 0 { None } else { Some(attempted_micro_usd as f64 / verified_changes as f64 / 1_000_000.0) }
}
// VCSC floating-point presentation is NOT the authoritative monetary ledger.
// Existing config's 0 = no limit maps to Option::None only at a validated boundary.
// Reservations need atomic parent/child ownership and idempotent settlement/refund.
// Unknown price is a distinct state and cannot enter the ledger as zero.
```

Build intent

Parse monetary values exactly, preserve unknown pricing and enforce one atomic parent/child reservation envelope.

Reuse the actual repository types, async owners and adapters. These outlines are a construction guide, not a compiled product patch.

The handoff contains complete reference files, per-control build mapping and required integration checks.

4.4 IMPLEMENTATION CODE OUTLINES · R7 PAGE 212

Owned operations

contracts/operations.rs / original unintegrated reference code

```
//! Every effect has durable identity and a current owner. Original design outline.
#[derive(Clone, Debug, PartialEq, Eq)]
pub struct Binding {
    pub daemon: String, pub workspace: String, pub worktree: String,
    pub session: String, pub generation: u64,
}
pub enum EffectState { NotIssued, Admitted, InFlight, Confirmed, CancelledAcknowledged, Unknown }
pub struct CapturedSubmission { pub id: String, pub body_hash: [u8; 32], pub body: Vec<u8> }
pub struct Receipt { pub submission_id: String, pub body_hash: [u8; 32], pub run_id: String }
pub enum Admission { New, Existing(String), Conflict }
pub fn reconcile(c: &CapturedSubmission, existing: Option<&Receipt>) -> Admission {
    match existing {
        None => Admission::New,
        Some(r) if r.submission_id == c.id && r.body_hash == c.body_hash =>
Admission::Existing(r.run_id.clone()),
        Some(_) => Admission::Conflict,
    }
}
pub struct OwnedProcess { pub binding: Binding, pub process_id: String, pub handle_generation: u64, pub live:
bool }
pub fn may_control(p: &OwnedProcess, current: &Binding, handle_generation: u64) -> bool {
    p.live && p.binding == *current && p.handle_generation == handle_generation
}
pub struct QueuedItem { pub binding: Binding, pub submission: CapturedSubmission, pub admitted: bool }
// The actual admission store atomically enforces unique submission ID + body identity.
// A lookup saying None is not permission to replay an already-issued external effect.
// Add bounded queue metadata/read/edit/cancel with owned receipts and explicit timing.
// Stop returns accepted then settled/refused/unknown; UI waits for the real state.
// Rebinding invalidates old handles but preserves drafts and unresolved effect records.
```

Build intent

Every request, queue item, process and control carries a current owner. Reconcile receipts and uncertain effects before replay.

Reuse the actual repository types, async owners and adapters. These outlines are a construction guide, not a compiled product patch.

The handoff contains complete reference files, per-control build mapping and required integration checks.

4.4 IMPLEMENTATION CODE OUTLINES · R7 PAGE 213

Restore and proof

contracts/review.rs / original unintegrated reference code

```
//! File restore and conversation fork are separate owned transactions.
pub struct FileRestore {
    pub uri: String, pub expected_current_hash: String,
    pub target_hash: String, pub target_content_ref: Option<String>,
}
pub fn validate_restore(files: &[FileRestore], observed: &[(String, String)]) -> Result<(), &'static str> {
    for file in files {
        if file.target_content_ref.is_none() { return Err("Target content is unavailable"); }
        let current = observed.iter().find(|(uri, _)| uri == &file.uri).map(|(_, hash)| hash);
        if current != Some(&file.expected_current_hash) { return Err("Files changed; preserve edits and rebase"); }
    }
    Ok(()) // Recheck all preconditions INSIDE actual restore transaction.
}
pub enum Finish { Local, Commit, Push, PullRequest }
pub struct Proof {
    pub candidate: String, pub verified_candidate: Option<String>, pub landed_candidate: Option<String>,
    pub required_total: usize, pub required_passed: usize,
    pub claims_complete: bool, pub finish_receipt_matches: bool,
}
pub fn blockers(p: &Proof) -> Vec<&'static str> {
    let mut reasons = Vec::new();
    if p.required_total == 0 || p.required_passed != p.required_total { reasons.push("Acceptance proof incomplete"); }
    if !p.claims_complete { reasons.push("Claim evidence incomplete"); }
    if p.verified_candidate.as_ref() != Some(&p.candidate) || p.landed_candidate.as_ref() != Some(&p.candidate) {
        reasons.push("Verification/integration identity mismatch");
    }
    if !p.finish_receipt_matches { reasons.push("Requested finish pending"); }
    reasons
```

Build intent

Recheck every content precondition at commit. Matching candidate evidence, integration and finish remain separate requirements.

Reuse the actual repository types, async owners and adapters. These outlines are a construction guide, not a compiled product patch.

The handoff contains complete reference files, per-control build mapping and required integration checks.

4.4 IMPLEMENTATION CODE OUTLINES · R7 PAGE 214

Runtime diagnostics

contracts/diagnostics.rs / original unintegrated reference code

```rust
//! Diagnostics are allowlisted structured fields, not a raw environment dump.
use std::collections::BTreeMap;
pub struct RuntimeStatus {
    pub executable_path: String, pub version: String, pub protocol_compatible: bool,
    pub recorded_checksum_matches: Option<bool>, pub publisher_signature_verified: Option<bool>,
    pub readiness: BTreeMap<String, String>, pub observed_at: String,
}
pub struct DiagnosticSelection {
    pub versions: bool, pub capabilities: bool, pub bounded_errors: bool,
    pub source_payloads_explicitly_selected: bool,
}
pub struct DiagnosticPreview { pub fields: BTreeMap<String, String>, pub omitted_categories: Vec<String> }
pub fn basic_preview(status: &RuntimeStatus) -> DiagnosticPreview {
    let mut fields = BTreeMap::new();
    fields.insert("version".into(), status.version.clone());
    fields.insert("protocol_compatible".into(), status.protocol_compatible.to_string());
    fields.insert("observed_at".into(), status.observed_at.clone());
    DiagnosticPreview { fields, omitted_categories: vec!["secret values".into(), "source payloads".into()] }
}
// Build explicit diagnostic field redactors and byte caps before serializing.
// Provider errors can embed request secrets: do not copy raw error strings blindly.
// Recorded checksum match and publisher signature verification are separate facts.
// Optional account/workers/updater unavailable states never block legitimate local work.
```

Build intent

Allowlist diagnostic fields and redact provider errors. Recorded checksum and publisher signature are different facts.

Reuse the actual repository types, async owners and adapters. These outlines are a construction guide, not a compiled product patch.

The handoff contains complete reference files, per-control build mapping and required integration checks.

4.4 IMPLEMENTATION CODE OUTLINES · R7 PAGE 215

VS Code controller

hosts/vscode/controller.ts / original unintegrated reference code

```
/** Original framework-neutral outline; integrate with the current plugin's typed DTOs. */
export type Binding = Readonly<{ daemon: string; workspace: string; worktree: string; session: string; generation: number }>;
export type Snapshot = Readonly<{ binding: Binding; revision: number; phase: string; observedAt: string }>;
export type FieldError = Readonly<{ id: string; message: string }>;
export interface Api { read(binding: Binding, signal: AbortSignal): Promise<Snapshot>; }
export interface View { render(snapshot: Snapshot): void; showError(message: string): void; }
const same = (a: Binding, b: Binding) => a.daemon === b.daemon && a.workspace === b.workspace &&
  a.worktree === b.worktree && a.session === b.session && a.generation === b.generation;
export class Controller {
  private binding?: Binding;
  private pending?: AbortController;
  private requestEpoch = 0;
  constructor(private readonly api: Api, private readonly view: View) {}
  bind(binding: Binding): void {
    this.pending?.abort(); this.binding = binding; this.requestEpoch++;
    // Draft/scroll/disclosure state is keyed by Binding, not one global current session.
  }
  async refresh(): Promise<void> {
    if (!this.binding) return;
    this.pending?.abort(); const captured = this.binding; const epoch = ++this.requestEpoch;
    const abort = new AbortController(); this.pending = abort;
    try {
      const next = await this.api.read(captured, abort.signal);
      if (abort.signal.aborted || epoch !== this.requestEpoch || !this.binding ||
          !same(captured, this.binding) || !same(next.binding, captured)) return;
      this.view.render(next); // Coalesce paint; never discard durable event application.
    } catch (error) {
      if (abort.signal.aborted || epoch !== this.requestEpoch) return;
      this.view.showError('Refresh failed. Keep the last coherent view and reconcile.');
    }
```

Build intent

Use typed scoped snapshots and dispose stale requests. Keep rendering, staged edits and durable command effects separate.

Reuse the actual repository types, async owners and adapters. These outlines are a construction guide, not a compiled product patch.

The handoff contains complete reference files, per-control build mapping and required integration checks.

4.4 IMPLEMENTATION CODE OUTLINES · R7 PAGE 216

JetBrains controller

hosts/jetbrains/FaktorController.kt / original unintegrated reference code

```
data class Binding(val daemon: String, val workspace: String, val worktree: String,
                   val session: String, val generation: Long)
data class Snapshot(val binding: Binding, val revision: Long, val phase: String)
fun interface CancelHandle { fun cancel() }
interface UiDispatcher { fun post(block: () -> Unit) } // Implement with the platform EDT.
interface SnapshotApi {
    fun read(binding: Binding, result: (Result<Snapshot>) -> Unit): CancelHandle
}
interface FaktorView { fun render(snapshot: Snapshot); fun showStaleError(message: String) }
class FaktorController(private val api: SnapshotApi, private val ui: UiDispatcher,
                       private val view: FaktorView) {
    // Controller methods run on EDT; result callbacks may arrive from any thread.
    private var binding: Binding? = null
    private var pending: CancelHandle? = null
    private var epoch = 0L
    private var disposed = false
    fun bind(next: Binding) {
        check(!disposed); pending?.cancel(); pending = null; epoch += 1; binding = next
    }
    fun refresh() {
        val captured = binding ?: return
        pending?.cancel(); epoch += 1; val requestedEpoch = epoch
        pending = api.read(captured) { result ->
            ui.post {
                if (disposed || requestedEpoch != epoch || binding != captured) return@post
                result.fold(
                    onSuccess = { if (it.binding == captured) view.render(it) },
                    onFailure = { view.showStaleError("Refresh failed; last valid state retained.") }
                )
            }
        }
```

Build intent

Publish current-generation snapshots on EDT and use one shared validation/persistence model for native and fallback settings.

Reuse the actual repository types, async owners and adapters. These outlines are a construction guide, not a compiled product patch.

The handoff contains complete reference files, per-control build mapping and required integration checks.

4.4 IMPLEMENTATION CODE OUTLINES · R7 PAGE 217

Layout implementation

hosts/vscode/layout.css / original unintegrated reference code

```
/* Revision 7. Inherit the actual host font, theme and zoom. */
:root {
  --panel-inset: 24px;
  --field-row-gap: 16px;
  --field-column-gap: 20px;
  --helper-gap: 8px;
  --section-gap: 24px;
  --control-height: 36px;
  color: var(--vscode-foreground);
  font: var(--vscode-font-size)/1.45 var(--vscode-font-family);
}
* { box-sizing: border-box; }
html, body, #root { margin: 0; height: 100%; min-width: 0; }
.faktor-shell {
  container-type: inline-size;
  container-name: faktor;
  display: grid;
  grid-template-rows: auto minmax(0, 1fr) auto;
  height: 100%;
  min-height: 0;
}
.content-scroll { min-height: 0; min-width: 0; overflow: auto; }
.panel { padding: var(--panel-inset); min-width: 0; }
.section + .section { margin-top: var(--section-gap); }
.field {
  display: grid;
  grid-template-columns: 156px minmax(0, 1fr);
  column-gap: var(--field-column-gap);
  row-gap: var(--helper-gap);
  margin-bottom: var(--field-row-gap);
}
.field > * { min-width: 0; }
.field label { align-self: center; line-height: 1.3; }
.field-control { width: 100%; min-height: var(--control-height); }
```

Build intent

Match the visual grid with semantic tokens, min-width:0, intrinsic rows, a scrollable content region and a separately reserved footer.

Reuse the actual repository types, async owners and adapters. These outlines are a construction guide, not a compiled product patch.

The handoff contains complete reference files, per-control build mapping and required integration checks.

4.4 IMPLEMENTATION CODE OUTLINES · R7 PAGE 218

Scoped interaction ownership

hosts/vscode/interaction.ts / original unintegrated reference code

```
}
export class ViewOwner {
  private epoch = 0;
  private pending?: AbortController;
  constructor(private binding: Binding) {}
  rebind(next: Binding): void {
    this.pending?.abort(); this.binding = next; this.epoch++;
    // Keep drafts, selection and scroll in a map keyed by their original binding.
  }
  async read<T>(load: (b: Binding, signal: AbortSignal) => Promise<T>, paint: (v: T) => void): Promise<void>
{
    this.pending?.abort();
    const controller = this.pending = new AbortController();
    const captured = this.binding, epoch = ++this.epoch;
    try {
      const value = await load(captured, controller.signal);
      if (controller.signal.aborted || epoch !== this.epoch || !sameBinding(captured, this.binding)) return;
      paint(value); // Publish a coherent model; do not infer durable command success here.
    } catch (error) {
      if (controller.signal.aborted || epoch !== this.epoch) return;
      throw error; // Project the typed error; keep last coherent data explicitly stale.
    }
  }
  dispose(): void { this.pending?.abort(); this.epoch++; }
}
export function clearOnlyCapturedDraft(current: Draft, captured: CapturedStart): Draft {
  return current.version === captured.draft.version
    ? { version: current.version + 1, text: '', sourceHandles: [] }
    : current;
}
```

Build intent

Fence each response by its full binding and request epoch. Preserve later draft edits when acknowledging an older captured submission.

Reuse the actual repository types, async owners and adapters. These outlines are a construction guide, not a compiled product patch.

The handoff contains complete reference files, per-control build mapping and required integration checks.

4.4 IMPLEMENTATION CODE OUTLINES · R7 PAGE 219

Keyboard and focus preservation

hosts/vscode/interaction.ts / original unintegrated reference code

```
  if (composing || event.isComposing || event.keyCode === 229) return false;
  return event.key === 'Enter' && !event.shiftKey && !event.altKey &&
    (mac ? event.metaKey && !event.ctrlKey : event.ctrlKey && !event.metaKey);
}
export function restoreFocus(root: HTMLElement, wanted: FocusKey, fallback: FocusKey): void {
  const key = (v: FocusKey) => JSON.stringify(v);
  const nodes = [...root.querySelectorAll<HTMLElement>('[data-focus-key]')];
  const node = nodes.find(n => n.dataset.focusKey === key(wanted)) ??
    nodes.find(n => n.dataset.focusKey === key(fallback));
  node?.focus({ preventScroll: true });
  // The caller chooses a nearest stable sibling/heading, not a guessed array index.
}
// Admission: reuse the same UUID and captured digest after ambiguous delivery.
// Never put credentials in this webview module. The extension owns trusted effects.
// Stop run / abort turn / stop process are separate typed commands and receipts.
```

Build intent

Use native send keymaps with IME guards, stable focus keys and an explicit fallback when an entity disappears.

Reuse the actual repository types, async owners and adapters. These outlines are a construction guide, not a compiled product patch.

The handoff contains complete reference files, per-control build mapping and required integration checks.

4.4 IMPLEMENTATION CODE OUTLINES · R7 PAGE 220

Immutable native diff integration

hosts/vscode/immutable-diff.ts / original unintegrated reference code

```
  uri(s: Snapshot): vscode.Uri {
    // URI identity includes the full immutable candidate/hash/path tuple.
    const uri = vscode.Uri.from({ scheme: this.scheme, path: '/' + s.path,
      query: new URLSearchParams({ candidate: s.candidate, hash: s.contentHash }).toString() });
    this.snapshots.set(uri.toString(), s); return uri;
  }
  async provideTextDocumentContent(uri: vscode.Uri, token: vscode.CancellationToken): Promise<string> {
    const snapshot = this.snapshots.get(uri.toString());
    if (!snapshot) throw new Error('The snapshot identity is unavailable.');
    const controller = new AbortController(); this.reads.add(controller);
    const cancellation = token.onCancellationRequested(() => controller.abort());
    if (token.isCancellationRequested) controller.abort();
    try {
      const text = await this.read(snapshot, controller.signal);
      if (controller.signal.aborted) throw new vscode.CancellationError();
      if (text === null) throw new Error('Immutable content is unavailable for this snapshot.');
      return text; // Never fall back to a convenient live working-tree file.
    } finally { cancellation.dispose(); this.reads.delete(controller); }
  }
  async open(base: Snapshot, candidate: Snapshot): Promise<void> {
    await vscode.commands.executeCommand('vscode.diff', this.uri(base), this.uri(candidate),
      `${candidate.path} - candidate review`, { preview: true });
  }
  dispose(): void {
    for (const controller of this.reads) controller.abort();
    this.reads.clear(); this.snapshots.clear(); this.registration.dispose();
  }
}
// Enforce the immutable read and hash match at the service boundary too.
// Keep provider content bounded/cached by immutable identity using the real repository policy.
```

Build intent

Content identity includes candidate, hash and path. Missing immutable content is an unavailable read, never a fallback to live owner files.

Reuse the actual repository types, async owners and adapters. These outlines are a construction guide, not a compiled product patch.

The handoff contains complete reference files, per-control build mapping and required integration checks.

4.4 IMPLEMENTATION CODE OUTLINES · R7 PAGE 221

Native JetBrains form layout

hosts/jetbrains/FaktorForm.kt / original unintegrated reference code

```
class FaktorForm(
  private val owner: Disposable,
  private val model: StagedTuning,
  private val gateway: TuningGateway,
  private val expectedRevision: Long,
) {
  val component: DialogPanel = panel {
    group("Next-call tuning") {
      row("Maximum output:") {
        textField().bindText(model::maximumOutput).align(AlignX.FILL)
          .comment("Tokens. The model context and adapter limits are validated on apply.")
      }
      row("Run money cap:") {
        textField().bindText(model::moneyCap).align(AlignX.FILL)
          .comment("USD. Zero means no limit; an empty inherited value is different.")
      }
    }
  }
  fun isModified(): Boolean = component.isModified()
  fun reset() = component.reset()
  fun apply() {
    component.apply()
    try {
      gateway.validateAndCommit(expectedRevision, model.copy())
    } catch (failure: IllegalArgumentException) {
      // Replace with typed field/refusal/revision-conflict projection from the real service.
      throw ConfigurationException(failure.message ?: "Review the entered values.")
    }
  }
  fun preferredFocus(): JComponent? = component.preferredFocusedComponent
```

Build intent

Reuse platform Settings/UI DSL and native size hints. Schedule real service work off EDT and project typed field/revision errors back to the current form.

Reuse the actual repository types, async owners and adapters. These outlines are a construction guide, not a compiled product patch.

The handoff contains complete reference files, per-control build mapping and required integration checks.

4.4 IMPLEMENTATION CODE OUTLINES · R7 PAGE 222

Typed inspection states

contracts/inspection.rs / original unintegrated reference code

```rust
// Construction outline: use actual repository IDs, proof reducers and native DTOs.
pub struct InspectionBinding {
    pub task: String,
    pub candidate: String,
    pub snapshot_version: u64,
}
pub enum ReadState<T> {
    NotRead,
    Loading,
    Current(T),
    Stale { previous: Option<T>, reason: String },
    Unavailable { code: String, reason: String },
}
pub enum CriterionResult {
    Pending,
    Passed { evidence: Vec<String> },
    Failed { evidence: Vec<String> },
    Refused { reason: String },
    Unavailable { reason: String },
}
pub struct CriterionView {
    pub id: String,
    pub required: bool,
    pub claim: String,
    pub binding: InspectionBinding,
    pub result: CriterionResult,
}
pub struct InspectionRequest {
    pub binding: InspectionBinding,
    pub selected_entity: Option<String>,
    pub request_epoch: u64,
```

Build intent

Keep missing, stale, refused and failed evidence distinct. Retain claim, candidate and artifact provenance; reuse the actual trusted proof reducer.

Reuse the actual repository types, async owners and adapters. These outlines are a construction guide, not a compiled product patch.

The handoff contains complete reference files, per-control build mapping and required integration checks.

4.5 ALL 200 CONTROL CONTRACTS · R7 PAGES 223–226 · 14 CONTROLS

M · Models and routing

Keep the selected model, actual routed model and capability provenance separate. Make selection reachable from the composer in one gesture; show advanced configuration in an editor-sized settings surface.

E existing / R runtime-backed native work / N new contract to build / H host UX. Every control remains required.

CONTROL / WIDGET	DEFAULT / SCOPE / APPLIES	REQUIRED BEHAVIOUR / BUILD / PATTERN
M-01 **Default provider** Served-instance picker	**DEFAULT** Daemon catalog choice **SCOPE** User/application; draft override **APPLIES** New conversation	Validate provider plus model together. If unavailable, keep the previous selection and offer Refresh; never silently use another endpoint. E / Build BK01 UI01 UI02 UI03 BK02 BK07 UI pattern: choice / full interaction: M-01 in Control_UX_Contracts.csv
M-02 **Default model** Searchable model picker	**DEFAULT** Served catalog default **SCOPE** User/application; conversation **APPLIES** New conversation or supported next task	Show full model ID, provider and current availability. A change requiring a new session offers Create new conversation with the draft preserved. E / Build BK01 UI01 UI02 UI03 BK02 BK07 UI pattern: choice / full interaction: M-02 in Control_UX_Contracts.csv
M-03 **Favorites and recent models** Star and recent list	**DEFAULT** None; local MRU **SCOPE** User **APPLIES** Immediate UI	Favorites refer to provider plus exact model ID. Removed models remain labelled unavailable until deliberately removed. H / Build UI01 UI02 UI03 UI pattern: choice / full interaction: M-03 in Control_UX_Contracts.csv
M-04 **Routing policy** Pinned / Economy picker	**DEFAULT** Show effective daemon policy **SCOPE** Daemon; captured task **APPLIES** Next admitted task	Pinned must never fall back silently; Economy explains selected and substituted model. Expose the actual route separately from the requested preference. R / Build BK01 UI01 UI02 UI03 BK02 BK07 UI pattern: choice / full interaction: M-04 in Control_UX_Contracts.csv
M-05 **Per-role model** Planner / coder / verifier / summarizer map	**DEFAULT** Inherit selected model **SCOPE** Agent preset; task **APPLIES** Next child or model call after ack	Resolve exact model capabilities and shared budget for each role. A role model is not permission to create a child agent. N / Build BK01 UI01 UI02 UI03 BK02 BK07 UI pattern: choice / full interaction: M-05 in Control_UX_Contracts.csv
M-06 **Fallback chain** Ordered allowed candidates	**DEFAULT** No added fallback **SCOPE** Task/daemon policy **APPLIES** Next routing decision	Explain trigger, price/data-boundary change and quality floor. Do not route local-only work to a hosted endpoint. Exhaustion is a visible failure. N / Build BK01 UI01 UI02 UI03 BK02 BK07 UI pattern: choice / full interaction: M-06 in Control_UX_Contracts.csv

M · MODELS AND ROUTING · CONTINUED

CONTROL / WIDGET	DEFAULT / SCOPE / APPLIES	REQUIRED BEHAVIOUR / BUILD / PATTERN
M-07 **Model capability sheet** Read-only badges and details	**DEFAULT** Served capability authority **SCOPE** Provider/model **APPLIES** On catalog refresh	Show context, output, tools, parallel tools, reasoning, thinking, vision, documents, streaming and structured output. Unknown, generic declaration and exact-model authority remain distinct. R / Build BK01 UI01 UI02 UI03 BK02 BK07 UI pattern: observation / full interaction: M-07 in Control_UX_Contracts.csv
M-08 **Catalog refresh and probe** Refresh / probe exact model	**DEFAULT** No automatic destructive probe **SCOPE** Daemon instance **APPLIES** On explicit action	Show last observation time and probe failure. A context allocation probe does not certify tool-call or coding reliability. R / Build BK01 UI01 UI02 UI03 BK02 BK07 UI pattern: action / full interaction: M-08 in Control_UX_Contracts.csv
M-09 **Context-window override** Positive integer, capability-bounded	**DEFAULT** Served effective limit **SCOPE** Model declaration; operator **APPLIES** After config validation and ack	Lower limits may constrain a model; larger declarations need supported authority. Show configured versus runtime-observed limits and reject impossible reservations. N / Build BK01 UI01 UI02 UI03 BK02 BK07 UI pattern: number / full interaction: M-09 in Control_UX_Contracts.csv
M-10 **Quality floors** Advanced declared thresholds	**DEFAULT** Effective daemon floors **SCOPE** Operator/task policy **APPLIES** Next route	Show user-declared versus observed quality and sample counts. Never relabel a manually entered score as a measured benchmark. R / Build BK01 UI01 UI02 UI03 BK02 BK07 UI pattern: number / full interaction: M-10 in Control_UX_Contracts.csv
M-11 **Actual model per call** Read-only call identity	**DEFAULT** No call yet **SCOPE** Turn/call **APPLIES** On durable observation	Display provider, exact model, route reason and role on expanded calls. A composer preference must not masquerade as the model that executed. R / Build BK01 UI01 UI02 UI03 BK02 BK07 UI pattern: observation / full interaction: M-11 in Control_UX_Contracts.csv
M-12 **Switch timing** Next task / supported next turn	**DEFAULT** Current call unchanged **SCOPE** Draft/session **APPLIES** After explicit acknowledgement	Changing a picker never rewrites an in-flight request. Unsupported mid-session changes offer a new session; show when the new value actually applies. N / Build BK01 UI01 UI02 UI03 BK02 BK07 UI pattern: choice / full interaction: M-12 in Control_UX_Contracts.csv
M-13 **Cost and limit details** Read-only price/limit sheet	**DEFAULT** Unavailable if not served **SCOPE** Model/provider **APPLIES** At quote or observation	Separate list/declared/observed price and quota. Cache prices may be absent; missing rates do not become zero. No hard-coded model price recommendation. R / Build BK01 UI01 UI02 UI03 BK02 BK07 UI pattern: observation / full interaction: M-13 in Control_UX_Contracts.csv

M · MODELS AND ROUTING · CONTINUED

CONTROL / WIDGET	DEFAULT / SCOPE / APPLIES	REQUIRED BEHAVIOUR / BUILD / PATTERN
M-14 **Model comparison** Bounded side-by-side candidate run	**DEFAULT** Off **SCOPE** Task **APPLIES** New comparison admission	Name all candidate models, shared cap and evidence basis before running. Compare verified outcomes and attempted spend; keep candidate identity and user choice explicit. N / Build BK01 UI01 UI02 UI03 BK02 BK07 UI pattern: action / full interaction: M-14 in Control_UX_Contracts.csv

4.5 ALL 200 CONTROL CONTRACTS · R7 PAGES 227–230 · 15 CONTROLS

P · Providers, credentials and local inference

Connections are endpoint and credential bindings; models are served capabilities; agent presets are instructions and permissions. These three objects must not be conflated.

E existing / R runtime-backed native work / N new contract to build / H host UX. Every control remains required.

CONTROL / WIDGET	DEFAULT / SCOPE / APPLIES	REQUIRED BEHAVIOUR / BUILD / PATTERN
P-01 **Connection label and instance ID** Label + immutable unique ID	**DEFAULT** User supplied label **SCOPE** Machine/daemon **APPLIES** After save	Show readable label and reveal exact ID. Validate uniqueness before storing. Renaming a label must not move credentials or alias a different endpoint. N / Build BK01 UI01 UI02 UI03 BK02 BK10 UI pattern: editor / full interaction: P-01 in Control_UX_Contracts.csv
P-02 **Provider/API family** Capability-supported family picker	**DEFAULT** Detected suggestion, user confirmed **SCOPE** Connection **APPLIES** After save/restart if required	Distinguish OpenAI Chat, Responses and other actually supported adapters. Never infer a remote API family from a marketing model name. R / Build BK01 UI01 UI02 UI03 BK02 BK10 UI pattern: choice / full interaction: P-02 in Control_UX_Contracts.csv
P-03 **Base URL** Validated endpoint field	**DEFAULT** Adapter default, shown explicitly **SCOPE** Machine/connection **APPLIES** After validation and connection ack	Show which machine resolves the URL. Reject invalid schemes, userinfo and blocked destinations; keep refusal inline without exposing credential fragments. R / Build BK01 UI01 UI02 UI03 BK02 BK10 UI pattern: editor / full interaction: P-03 in Control_UX_Contracts.csv
P-04 **Credential reference** Secret-store or environment-name selector	**DEFAULT** No credential value **SCOPE** Machine secret store **APPLIES** On next authorized request	Current provider config uses environment references. Direct IDE provider-key entry requires a secure daemon handoff contract; never put key values in shared settings or exports. N / Build BK01 UI01 UI02 UI03 BK02 BK10 UI pattern: choice / full interaction: P-04 in Control_UX_Contracts.csv
P-05 **Custom headers** Bounded name + secret-reference rows	**DEFAULT** None **SCOPE** Connection, secret values in vault **APPLIES** After save	Current adapter helper bounds: 32 headers, names 128 bytes, values 8192 bytes. Native editor persistence and vault binding still need implementation. Mask auth-shaped values. N / Build BK01 UI01 UI02 UI03 BK02 BK10 UI pattern: editor / full interaction: P-05 in Control_UX_Contracts.csv
P-06 **Models on custom endpoint** Fetch models / add exact ID	**DEFAULT** Served or explicitly declared IDs **SCOPE** Connection **APPLIES** After catalog reload	Display live, exact declaration or generic declaration provenance. Failed discovery does not erase explicit IDs or falsely validate tool calling. R / Build BK01 UI01 UI02 UI03 BK02 BK10 UI pattern: action / full interaction: P-06 in Control_UX_Contracts.csv

P · PROVIDERS, CREDENTIALS AND LOCAL INFERENCE · CONTINUED

CONTROL / WIDGET	DEFAULT / SCOPE / APPLIES	REQUIRED BEHAVIOUR / BUILD / PATTERN
P-07 **Local engine choice** Ollama / compatible endpoint	**DEFAULT** Existing configured local adapter **SCOPE** Connection **APPLIES** After readiness	Expose the protocol being used, not a promise of support for every inference server. LM Studio/vLLM are endpoint examples only when that compatible path is configured. R / Build BK01 UI01 UI02 UI03 BK02 BK10 UI pattern: choice / full interaction: P-07 in Control_UX_Contracts.csv
P-08 **Loopback permission** Explicit local-endpoint opt-in	**DEFAULT** Adapter default; show actual value **SCOPE** Machine/connection **APPLIES** After validated config reload	Ollama defaults to permitted loopback; custom OpenAI uses explicit opt-in. This switch does not authorize private LAN, link-local or metadata destinations. R / Build BK01 UI01 UI02 UI03 BK02 BK10 UI pattern: choice / full interaction: P-08 in Control_UX_Contracts.csv
P-09 **Private LAN inference** Trusted destination policy editor	**DEFAULT** Not available under current provider rule **SCOPE** Machine/operator policy **APPLIES** Only after new backend policy exists	A factory-LAN address cannot be enabled by allow_loopback. Show the current typed refusal; scoped authenticated LAN policy is new work, not a hidden workaround toggle. N / Build BK01 UI01 UI02 UI03 BK02 BK10 UI pattern: editor / full interaction: P-09 in Control_UX_Contracts.csv
P-10 **Connection test** Test auth / list / minimal capability probe	**DEFAULT** No request until selected **SCOPE** Connection **APPLIES** On explicit action	Show stages and target; note if the chosen probe incurs provider usage. Auth OK, model listed and model capable are separate results. N / Build BK01 UI01 UI02 UI03 BK02 BK10 UI pattern: action / full interaction: P-10 in Control_UX_Contracts.csv
P-11 **Transport timeout** Positive duration within adapter range	**DEFAULT** Effective adapter policy **SCOPE** Connection **APPLIES** Next request after ack	Show connect, first-byte and total deadlines where supported. Do not retry a request whose external effect or streamed admission is unknown. N / Build BK01 UI01 UI02 UI03 BK02 BK10 UI pattern: number / full interaction: P-11 in Control_UX_Contracts.csv
P-12 **Proxy, TLS and certificates** Managed options with validation	**DEFAULT** System/daemon policy **SCOPE** Machine/operator **APPLIES** After reload	Show effective proxy and certificate source without secrets. Do not provide a convenient disable-TLS-verification checkbox as an ordinary setting. N / Build BK01 UI01 UI02 UI03 BK02 BK10 UI pattern: editor / full interaction: P-12 in Control_UX_Contracts.csv
P-13 **Remote execution location** Read-only host and locality badges	**DEFAULT** Current daemon host **SCOPE** Connection/session **APPLIES** On binding	Local means local to the daemon, not necessarily the laptop. SSH/container sessions show endpoint reachability, vault location and file-resolution host. H / Build UI01 UI02 UI03 UI pattern: observation / full interaction: P-13 in Control_UX_Contracts.csv

P · PROVIDERS, CREDENTIALS AND LOCAL INFERENCE · CONTINUED

CONTROL / WIDGET	DEFAULT / SCOPE / APPLIES	REQUIRED BEHAVIOUR / BUILD / PATTERN
P-14 **Reconnect, rotate and delete** Explicit connection lifecycle actions	**DEFAULT** Keep active binding until replaced **SCOPE** Machine/connection **APPLIES** After acknowledgement	Rotation is write-only. Delete shows affected drafts and sessions; active requests retain their captured credential generation, with clear failure if revoked. N / Build BK01 UI01 UI02 UI03 BK02 BK10 UI pattern: action / full interaction: P-14 in Control_UX_Contracts.csv
P-15 **Local allocation and keep-alive** Adapter-offered context allocation / keep-alive / unload	**DEFAULT** Inherit runtime policy **SCOPE** Local connection/operator **APPLIES** After explicit runtime acknowledgement	Show configured versus observed allocation. Offer only supported engine controls; unloading needs ownership and active-use checks. No generic GPU or thread sliders without a real engine contract. N / Build BK01 UI01 UI02 UI03 BK02 BK10 UI pattern: editor / full interaction: P-15 in Control_UX_Contracts.csv

4.5 ALL 200 CONTROL CONTRACTS · R7 PAGES 231–234 · 14 CONTROLS

G · Reasoning and generation

ReasoningMode and capability flags in core do not establish a writable per-turn settings API. Every parameter below needs capability, policy and effective-value acknowledgement before it is enabled.

E existing / R runtime-backed native work / N new contract to build / H host UX. Every control remains required.

CONTROL / WIDGET	DEFAULT / SCOPE / APPLIES	REQUIRED BEHAVIOUR / BUILD / PATTERN
G-01 **Reasoning effort** Served enum; Off/Low/Medium/High when mapped	**DEFAULT** Inherit model/daemon **SCOPE** Model default; draft/task override **APPLIES** Next admitted call	Use the adapter-offered values. Current core enum has four levels; do not expose Max or a made-up numeric range for every model. N / Build BK01 UI01 UI02 UI03 BK02 UI pattern: choice / full interaction: G-01 in Control_UX_Contracts.csv
G-02 **Thinking token budget** Integer bounded by model capability	**DEFAULT** Provider/model default **SCOPE** Model/task **APPLIES** Next call	Keep thinking budget separate from total output reserve. Hide if the adapter cannot control it; explain any shared reasoning/output accounting. N / Build BK01 UI01 UI02 UI03 BK02 UI pattern: number / full interaction: G-02 in Control_UX_Contracts.csv
G-03 **Maximum output** Positive token limit	**DEFAULT** Served maxOutput **SCOPE** Model/task **APPLIES** Next call	Validate against context and output capabilities. Increasing output reserve reduces space for input; warn before discarding or compacting selected context. N / Build BK01 UI01 UI02 UI03 BK02 UI pattern: number / full interaction: G-03 in Control_UX_Contracts.csv
G-04 **Temperature** Adapter-offered numeric range	**DEFAULT** Provider default; unset stays unset **SCOPE** Model/task **APPLIES** Next call	Unsupported models show Not supported rather than a nonfunctional slider. Default reset removes the override; never send a null or arbitrary zero accidentally. N / Build BK01 UI01 UI02 UI03 BK02 UI pattern: number / full interaction: G-04 in Control_UX_Contracts.csv
G-05 **Top-p** Adapter-supported probability value	**DEFAULT** Unset **SCOPE** Model/task **APPLIES** Next call	Validate the exact adapter range. Warn on mutually restrictive sampling choices; preserve provider defaults unless the user sets a value. N / Build BK01 UI01 UI02 UI03 BK02 UI pattern: number / full interaction: G-05 in Control_UX_Contracts.csv
G-06 **Top-k** Adapter-supported integer	**DEFAULT** Unset **SCOPE** Model/task **APPLIES** Next call	Only show when the selected adapter accepts it. A local runtime option is not automatically a hosted-model capability. N / Build BK01 UI01 UI02 UI03 BK02 UI pattern: number / full interaction: G-06 in Control_UX_Contracts.csv

G · REASONING AND GENERATION · CONTINUED

CONTROL / WIDGET	DEFAULT / SCOPE / APPLIES	REQUIRED BEHAVIOUR / BUILD / PATTERN
G-07 **Frequency/presence penalties** Separate supported values	**DEFAULT** Unset **SCOPE** Model/task **APPLIES** Next call	Render independent labels, ranges and reset actions. Unsupported request fields must be rejected before submission, not silently ignored. N / Build BK01 UI01 UI02 UI03 BK02 UI pattern: number / full interaction: G-07 in Control_UX_Contracts.csv
G-08 **Seed and determinism** Optional integer + caveat	**DEFAULT** Unset **SCOPE** Model/task **APPLIES** Next call	A seed is best-effort provider behavior, not reproducibility proof. Show whether the provider acknowledged it and retain it in the captured request summary. N / Build BK01 UI01 UI02 UI03 BK02 UI pattern: number / full interaction: G-08 in Control_UX_Contracts.csv
G-09 **Stop sequences** Bounded list of strings	**DEFAULT** None **SCOPE** Model/task **APPLIES** Next call	Warn when stops can truncate structured tool output. Do not allow an advanced text setting to corrupt native tool-call framing. N / Build BK01 UI01 UI02 UI03 BK02 UI pattern: editor / full interaction: G-09 in Control_UX_Contracts.csv
G-10 **Parallel model tool calls** Capability-gated switch and cap	**DEFAULT** Effective supported policy **SCOPE** Agent/task **APPLIES** Next model call	Needs both model support and runtime-safe scheduling. Concurrency must not bypass shared budget, capability grants, cancellation or edit ownership. N / Build BK01 UI01 UI02 UI03 BK02 UI pattern: choice / full interaction: G-10 in Control_UX_Contracts.csv
G-11 **Output format/schema** Text / supported structured schema	**DEFAULT** Normal agent protocol **SCOPE** Task **APPLIES** Next supported operation	Validate schema size and compatibility. Do not replace Faktor internal tool envelopes with user-supplied formatting or claim JSON merely because prose looks structured. N / Build BK01 UI01 UI02 UI03 BK02 UI pattern: editor / full interaction: G-11 in Control_UX_Contracts.csv
G-12 **Service tier and cache policy** Adapter-supported options	**DEFAULT** Provider default **SCOPE** Connection/task policy **APPLIES** Next call	Display actual tier and cache-write/read basis. A cache toggle must not claim that a prefix hit occurred; observe that separately. N / Build BK01 UI01 UI02 UI03 BK02 UI pattern: choice / full interaction: G-12 in Control_UX_Contracts.csv
G-13 **Stream responses** Supported on/off preference	**DEFAULT** Stream when supported **SCOPE** User/model **APPLIES** Next call	Preserve stop, stale-state and transcript semantics when disabled. A quiet model call is not an idle task; show request phase without fake progress. N / Build BK01 UI01 UI02 UI03 BK02 UI pattern: choice / full interaction: G-13 in Control_UX_Contracts.csv

G · REASONING AND GENERATION · CONTINUED

CONTROL / WIDGET	DEFAULT / SCOPE / APPLIES	REQUIRED BEHAVIOUR / BUILD / PATTERN
G-14 **Effective request preview** Read-only normalized summary	**DEFAULT** No request captured yet **SCOPE** Draft and immutable call receipt **APPLIES** Before admission / after ack	List requested, effective and unsupported fields with source and timing. Do not dump hidden system secrets or claim to expose private model reasoning. N / Build BK01 UI01 UI02 UI03 BK02 UI pattern: observation / full interaction: G-14 in Control_UX_Contracts.csv

4.5 ALL 200 CONTROL CONTRACTS · R7 PAGES 235–237 · 12 CONTROLS

A · Agent presets and delegation

Separate task intent, model selection, permissions, routing and finish policy. A named agent is an explicit preset over those objects, not a new personality label that secretly broadens authority.

E existing / R runtime-backed native work / N new contract to build / H host UX. Every control remains required.

CONTROL / WIDGET	DEFAULT / SCOPE / APPLIES	REQUIRED BEHAVIOUR / BUILD / PATTERN
A-01 **Agent preset picker** Code / Ask / Plan / Debug / Review / Custom	**DEFAULT** Code as proposed initial preset **SCOPE** User default; draft **APPLIES** Before admission	Describe scope and tool envelope in the picker. Existing Task/Ask controls are not equivalent to enforced read-only Ask; the stricter preset needs backend enforcement. N / Build BK01 UI01 UI02 UI03 BK03 BK05 BK08 UI pattern: choice / full interaction: A-01 in Control_UX_Contracts.csv
A-02 **Ask policy** Read-only enforced envelope	**DEFAULT** No writes or effectful tools **SCOPE** Preset/task **APPLIES** At task admission	Allow exploration and answers; filter tools and deny writes at dispatch. Current plain prompt/Ask route alone does not prove read-only operation. N / Build BK01 UI01 UI02 UI03 BK03 BK05 BK08 UI pattern: observation / full interaction: A-02 in Control_UX_Contracts.csv
A-03 **Plan policy** Exploration + named plan artifact writes	**DEFAULT** No code edits **SCOPE** Preset/task **APPLIES** At admission	Show plan file and review comments. User approval admits a new Code task with a frozen plan version; it does not mutate the planning run in place. N / Build BK01 UI01 UI02 UI03 BK03 BK05 BK08 UI pattern: choice / full interaction: A-03 in Control_UX_Contracts.csv
A-04 **Custom role instructions** Editor with preview and source	**DEFAULT** Inherit baseline instructions **SCOPE** User/project preset **APPLIES** Next task	Preview the merged instruction bundle and token estimate. Prompt text cannot widen a capability grant or override managed policy. N / Build BK01 UI01 UI02 UI03 BK03 BK05 BK08 UI pattern: editor / full interaction: A-04 in Control_UX_Contracts.csv
A-05 **Tool allowlist per preset** Grouped searchable selector	**DEFAULT** Effective preset envelope **SCOPE** Preset/task **APPLIES** At admission	Show inherited denies and exact MCP namespaces. A newly discovered tool remains unavailable until policy resolution allows it. N / Build BK01 UI01 UI02 UI03 BK03 BK05 BK08 UI pattern: choice / full interaction: A-05 in Control_UX_Contracts.csv
A-06 **Role model and reasoning** Inherited model + supported overrides	**DEFAULT** Inherit **SCOPE** Preset/task/child **APPLIES** Next admitted child/call	Show fallback and pricing boundary. Existing child model control is preserved; richer preset defaults need explicit configuration and adapter mapping. R / Build BK01 UI01 UI02 UI03 BK03 BK05 BK08 UI pattern: choice / full interaction: A-06 in Control_UX_Contracts.csv

A · AGENT PRESETS AND DELEGATION · CONTINUED

CONTROL / WIDGET	DEFAULT / SCOPE / APPLIES	REQUIRED BEHAVIOUR / BUILD / PATTERN
A-07 **Delegation policy** Off / ask / permitted bounded delegation	**DEFAULT** Effective runtime policy **SCOPE** Task **APPLIES** Before new child admission	Show max depth, parent budget share and allowed child roles. Disabling delegation prevents new children; it does not pretend to stop already-running children. N / Build BK01 UI01 UI02 UI03 BK03 BK05 BK08 UI pattern: choice / full interaction: A-07 in Control_UX_Contracts.csv
A-08 **Child concurrency limit** Integer within served capacity	**DEFAULT** Effective daemon limit **SCOPE** Task/operator **APPLIES** Next child admission	Display waiting versus active children. Saturation creates an owned bounded queue or a typed refusal; never unbounded spawning. N / Build BK01 UI01 UI02 UI03 BK03 BK05 BK08 UI pattern: number / full interaction: A-08 in Control_UX_Contracts.csv
A-09 **Child budget allocation** Shared cap plus optional child ceiling	**DEFAULT** Shared parent envelope **SCOPE** Task/child **APPLIES** On grant acknowledgement	A child never gets an independent unlimited wallet. Preserve open reservations and uncertain spend when reducing an allowance. R / Build BK01 UI01 UI02 UI03 BK03 BK05 BK08 UI pattern: number / full interaction: A-09 in Control_UX_Contracts.csv
A-10 **Steer, pause and stop child** Contextual lifecycle actions	**DEFAULT** No action until selected **SCOPE** Child run **APPLIES** After durable control ack	Existing steer/model/agent controls are reused where served. Steer text retains the current 500-character bound; unsupported pause is disabled, not simulated. E / Build BK01 UI01 UI02 UI03 BK03 BK05 BK08 UI pattern: action / full interaction: A-10 in Control_UX_Contracts.csv
A-11 **Worktree/edit ownership** Read-only ownership + explicit assignment	**DEFAULT** Isolated candidate **SCOPE** Task/child **APPLIES** Before edit ownership grant	Keep parallel agents from overwriting the same candidate. Shared-file contention shows owner, conflict and handoff; agents do not acquire arbitrary owner-checkout write access. R / Build BK01 UI01 UI02 UI03 BK03 BK05 BK08 UI pattern: observation / full interaction: A-11 in Control_UX_Contracts.csv
A-12 **Results and handoff schema** Typed summary/criteria/artifact fields	**DEFAULT** Existing task evidence where served **SCOPE** Child to parent **APPLIES** On child completion	Show what was requested, returned, verified and still missing. Child text saying done is not the parent task completion basis. R / Build BK01 UI01 UI02 UI03 BK03 BK05 BK08 UI pattern: editor / full interaction: A-12 in Control_UX_Contracts.csv

4.5 ALL 200 CONTROL CONTRACTS · R7 PAGES 238–241 · 14 CONTROLS

C · Context, attachments and retrieval

Make context an inspectable input budget. Separate model capacity, runtime allocation, estimated input, actual submitted input, output reserve and safety reserve. A null contextUsage value is unavailable, not 0%.

E existing / R runtime-backed native work / N new contract to build / H host UX. Every control remains required.

CONTROL / WIDGET	DEFAULT / SCOPE / APPLIES	REQUIRED BEHAVIOUR / BUILD / PATTERN
C-01 **Context meter** Used / reserved / available display	**DEFAULT** Unavailable until authoritative read **SCOPE** Call/session **APPLIES** On measured snapshot	Current projection contextUsage is null. Add a durable versioned read contract before showing real percentages; keep draft estimates explicitly labelled. N / Build BK01 UI01 UI02 UI03 BK04 BK07 UI pattern: observation / full interaction: C-01 in Control_UX_Contracts.csv
C-02 **Context breakdown** System / rules / tools / history / files / retrieval	**DEFAULT** Measured or estimated categories **SCOPE** Draft/call **APPLIES** On preview / after call	Categories must be mutually accountable and sum to the same basis. Output reserve and safety margin are not consumed input tokens. N / Build BK01 UI01 UI02 UI03 BK04 BK07 UI pattern: observation / full interaction: C-02 in Control_UX_Contracts.csv
C-03 **Mention picker** @file / folder / selection / diagnostics / terminal / diff / URL / past chat	**DEFAULT** Nothing silently attached **SCOPE** Draft **APPLIES** On explicit selection	Show host URI, range, size and include/exclude state. A reference from another worktree resolves there, never against a global current directory. H / Build UI01 UI02 UI03 UI pattern: choice / full interaction: C-03 in Control_UX_Contracts.csv
C-04 **Current editor context** Off / selection / active file / explicit tabs	**DEFAULT** Proposed: selection only with visible chip **SCOPE** User/workspace default **APPLIES** At draft capture	Display every automatic inclusion. A tab title is not its contents; enabling all open files must show the added input and file scope. N / Build BK01 UI01 UI02 UI03 BK04 BK07 UI pattern: choice / full interaction: C-04 in Control_UX_Contracts.csv
C-05 **Attachment staging** Add / remove / retry per file	**DEFAULT** Existing typed upload behavior **SCOPE** Session/draft **APPLIES** On upload and admission	Preserve current MIME/byte gates from the served exact model. Unsupported image/document attachments remain visible with a reason, never silently dropped. E / Build BK01 UI01 UI02 UI03 BK04 BK07 UI pattern: action / full interaction: C-05 in Control_UX_Contracts.csv
C-06 **Clipboard and large paste** Text/image admission with preview	**DEFAULT** Respect model capability **SCOPE** Draft **APPLIES** On paste / send	Collapse large text behind a preview while preserving full bounded content. Show an invalid image or oversize payload without clearing the rest of the draft. E / Build BK01 UI01 UI02 UI03 BK04 BK07 UI pattern: editor / full interaction: C-06 in Control_UX_Contracts.csv

C · CONTEXT, ATTACHMENTS AND RETRIEVAL · CONTINUED

CONTROL / WIDGET	DEFAULT / SCOPE / APPLIES	REQUIRED BEHAVIOUR / BUILD / PATTERN
C-07 **Pin/exclude context items** Per-item controls	**DEFAULT** No implicit pinned content **SCOPE** Draft/task **APPLIES** Next captured request	Pinning keeps priority, not permission to exceed a hard window. Explain when an item cannot fit and require a deliberate trim or alternate model. N / Build BK01 UI01 UI02 UI03 BK04 BK07 UI pattern: choice / full interaction: C-07 in Control_UX_Contracts.csv
C-08 **Ignore and protected patterns** Scope-aware ignore editor	**DEFAULT** Existing tool/security rules remain authoritative **SCOPE** Project/user policy **APPLIES** After policy ack	Separate search exclusions from security denies. A retrieval ignore file is not a guarantee that an authorized shell cannot read a file. N / Build BK01 UI01 UI02 UI03 BK04 BK07 UI pattern: editor / full interaction: C-08 in Control_UX_Contracts.csv
C-09 **Retrieval strategy** Lexical / symbol / semantic / hybrid when served	**DEFAULT** Effective available strategy **SCOPE** Daemon/task **APPLIES** Next retrieval	No configured embedder means lexical/symbol fallback. Never label the index semantic merely because the UI contains that option. R / Build BK01 UI01 UI02 UI03 BK04 BK07 UI pattern: choice / full interaction: C-09 in Control_UX_Contracts.csv
C-10 **Embedding provider/model** Served provider + model + embedding policy	**DEFAULT** No embedder when absent **SCOPE** Daemon/operator **APPLIES** After validated reload/reindex	Model changes invalidate incompatible vectors and show rebuild state. Credentials and data egress are disclosed before embedding a codebase. R / Build BK01 UI01 UI02 UI03 BK04 BK07 UI pattern: choice / full interaction: C-10 in Control_UX_Contracts.csv
C-11 **Index coverage and freshness** Read-only coverage + reindex action if served	**DEFAULT** Current index snapshot **SCOPE** Workspace/index **APPLIES** On status refresh	Show indexed, ignored, stale and failed paths and commit basis. Existing status/coverage endpoints are reused; do not infer completion from a background spinner. E / Build BK01 UI01 UI02 UI03 BK04 BK07 UI pattern: observation / full interaction: C-11 in Control_UX_Contracts.csv
C-12 **Context capacity guard** Preview and resolve overflow	**DEFAULT** Block an over-cap request **SCOPE** Draft/call **APPLIES** Before admission	Offer remove, shorten, compact or choose a capable model. No hidden truncation of acceptance criteria, permissions or captured engineering obligations. N / Build BK01 UI01 UI02 UI03 BK04 BK07 UI pattern: observation / full interaction: C-12 in Control_UX_Contracts.csv
C-13 **Source preview and provenance** Open included excerpt/source	**DEFAULT** Read-only **SCOPE** Context item **APPLIES** On selection	Show exact version/range and how it entered context. Treat tool output, fetched pages and repository instructions as data, not authority to change policy. H / Build UI01 UI02 UI03 UI pattern: observation / full interaction: C-13 in Control_UX_Contracts.csv

C · CONTEXT, ATTACHMENTS AND RETRIEVAL · CONTINUED

CONTROL / WIDGET	DEFAULT / SCOPE / APPLIES	REQUIRED BEHAVIOUR / BUILD / PATTERN
C-14 **Context export** Redacted manifest and bounded excerpt export	**DEFAULT** Exclude credentials and secret payloads **SCOPE** User/session **APPLIES** On explicit export	Include IDs, hashes, basis and omission reasons. Mark estimates; do not export a guessed hidden prompt or private model chain of thought. N / Build BK01 UI01 UI02 UI03 BK04 BK07 UI pattern: action / full interaction: C-14 in Control_UX_Contracts.csv

4.5 ALL 200 CONTROL CONTRACTS · R7 PAGES 242–245 · 12 CONTROLS

K · Compaction, memory and efficiency

Preserve task intent, required evidence, permissions, candidate identity and pending operations across compaction. Operational learned failure facts are not the same as free-form personal memory.

E existing / R runtime-backed native work / N new contract to build / H host UX. Every control remains required.

CONTROL / WIDGET	DEFAULT / SCOPE / APPLIES	REQUIRED BEHAVIOUR / BUILD / PATTERN
K-01 **Automatic compaction threshold** Validated usage fraction	**DEFAULT** Daemon default 0.65 **SCOPE** Daemon; future task override **APPLIES** Next compaction decision	Use the daemon usage denominator. Show effective threshold and read source; do not reuse a graphical estimated context ratio as authoritative runtime usage. R / Build BK01 UI01 UI02 UI03 BK03 BK04 BK07 UI pattern: number / full interaction: K-01 in Control_UX_Contracts.csv
K-02 **Compaction model** Served model choice	**DEFAULT** Inherit when absent **SCOPE** Daemon/operator **APPLIES** Next compaction	Show egress and cost implications and required capabilities. An unavailable compactor preserves the last valid state and reports failure. R / Build BK01 UI01 UI02 UI03 BK03 BK04 BK07 UI pattern: choice / full interaction: K-02 in Control_UX_Contracts.csv
K-03 **Compact now** Preview / compact / cancel before dispatch	**DEFAULT** No automatic manual action **SCOPE** Session **APPLIES** After durable operation ack	Requires an explicit new native operation contract. Show preserved facts, removed material and reduction; a failed minimum reduction cannot be labelled successful. N / Build BK01 UI01 UI02 UI03 BK03 BK04 BK07 UI pattern: action / full interaction: K-03 in Control_UX_Contracts.csv
K-04 **Post-compaction inspection** Before/after manifest + task facts	**DEFAULT** Latest successful compaction **SCOPE** Session **APPLIES** After completion	Retain immutable original history by reference, summary version and stale evidence flags. Re-expanding UI text does not restore model context automatically. N / Build BK01 UI01 UI02 UI03 BK03 BK04 BK07 UI pattern: observation / full interaction: K-04 in Control_UX_Contracts.csv
K-05 **Failure learning** On/off with provenance	**DEFAULT** Production ON **SCOPE** Daemon/operator **APPLIES** After config reload	Expose the real efficiency.failure_learning off-switch. Separate learned failure priors from personal memories and show disabled/fallback status. R / Build BK01 UI01 UI02 UI03 BK03 BK04 BK07 UI pattern: choice / full interaction: K-05 in Control_UX_Contracts.csv
K-06 **Compressed tool evidence** On/off with archive links	**DEFAULT** Production ON **SCOPE** Daemon/operator **APPLIES** After config reload	Expose efficiency.ccr. Disabling it does not remove bounded-output limits; show the durable full artifact and bounded model excerpt separately. R / Build BK01 UI01 UI02 UI03 BK03 BK04 BK07 UI pattern: choice / full interaction: K-06 in Control_UX_Contracts.csv

K · COMPACTION, MEMORY AND EFFICIENCY · CONTINUED

CONTROL / WIDGET	DEFAULT / SCOPE / APPLIES	REQUIRED BEHAVIOUR / BUILD / PATTERN
K-07 **Typed handoff** On/off with retained-facts preview	**DEFAULT** Production ON **SCOPE** Daemon/operator **APPLIES** After config reload	Expose efficiency.typed_handoff. Changing it never erases the task ledger; describe the current projection basis and loss of structured handoff benefits. R / Build BK01 UI01 UI02 UI03 BK03 BK04 BK07 UI pattern: choice / full interaction: K-07 in Control_UX_Contracts.csv
K-08 **Information-gain context** On/off	**DEFAULT** Production ON **SCOPE** Daemon/operator **APPLIES** After config reload	Canonical key semantic_context; information_gain is an alias, not another switch. Duplicate aliases must fail validation instead of silently winning. R / Build BK01 UI01 UI02 UI03 BK03 BK04 BK07 UI pattern: choice / full interaction: K-08 in Control_UX_Contracts.csv
K-09 **Rework-aware routing** On/off + outcome sample count	**DEFAULT** Production ON **SCOPE** Daemon/operator **APPLIES** After config reload	Expose efficiency.rework_routing and distinguish prior declarations from verified outcome observations. No measured savings claim without a matching denominator. R / Build BK01 UI01 UI02 UI03 BK03 BK04 BK07 UI pattern: choice / full interaction: K-09 in Control_UX_Contracts.csv
K-10 **Personal/project memory** Off / manual / controlled auto capture	**DEFAULT** Proposed: manual only **SCOPE** User/project; separate stores **APPLIES** Next request after ack	New memory authority needs list/read/edit/delete, source, timestamp and retention. Do not reinterpret current failure-learning records as general auto-memory. N / Build BK01 UI01 UI02 UI03 BK03 BK04 BK07 UI pattern: choice / full interaction: K-10 in Control_UX_Contracts.csv
K-11 **Memory review and deletion** Diff before save; delete with scope	**DEFAULT** No cross-project sharing **SCOPE** Memory record **APPLIES** After durable save/delete ack	Deleting a memory does not remove earlier prompt copies. Show downstream retention scope; prevent a project from selecting an arbitrary global memory location. N / Build BK01 UI01 UI02 UI03 BK03 BK04 BK07 UI pattern: action / full interaction: K-11 in Control_UX_Contracts.csv
K-12 **Efficiency effect display** Observed counters and basis	**DEFAULT** Unavailable when not recorded **SCOPE** Session/workspace **APPLIES** After measurement	Show compression reduction, cache observations and routing outcomes separately. No fabricated percent saved or zero-cost assumption for locally hosted inference. N / Build BK01 UI01 UI02 UI03 BK03 BK04 BK07 UI pattern: observation / full interaction: K-12 in Control_UX_Contracts.csv

4.5 ALL 200 CONTROL CONTRACTS · R7 PAGES 246–248 · 11 CONTROLS

I · Instructions, rules and prompt composition

Show the effective instruction bundle and its source. Instruction precedence is a product contract; it must never be confused with enforced filesystem, shell or network policy.

E existing / R runtime-backed native work / N new contract to build / H host UX. Every control remains required.

CONTROL / WIDGET	DEFAULT / SCOPE / APPLIES	REQUIRED BEHAVIOUR / BUILD / PATTERN
I-01 **User instructions** Markdown editor and preview	**DEFAULT** Effective daemon baseline **SCOPE** User/daemon **APPLIES** Next request after ack	Keep unsaved text local. Show token estimate and source; do not claim a new native instructions API is already present. R / Build BK01 UI01 UI02 UI03 BK03 UI pattern: editor / full interaction: I-01 in Control_UX_Contracts.csv
I-02 **Project instructions** AGENTS.md source view/editor	**DEFAULT** Discover only supported trusted sources **SCOPE** Project **APPLIES** Next request after reload/ack	Display path, trust and active workspace. The repository guide is not proof that the runtime discovers every nested AGENTS.md file today. N / Build BK01 UI01 UI02 UI03 BK03 UI pattern: editor / full interaction: I-02 in Control_UX_Contracts.csv
I-03 **Scoped rules** Rule editor: path scope + trigger	**DEFAULT** No new rule enabled silently **SCOPE** Project/user **APPLIES** Next matching task	Validate glob and test against sample paths. Explain always/manual/path-matched activation and show the matching source in the request preview. N / Build BK01 UI01 UI02 UI03 BK03 UI pattern: editor / full interaction: I-03 in Control_UX_Contracts.csv
I-04 **Effective order** Read-only winning-value/merged-bundle inspector	**DEFAULT** Backend-served resolution **SCOPE** Task/session **APPLIES** On config snapshot	Separate enforced managed constraints from instruction text. Proposed ordinary preference order is task, project, user, daemon default; fields may restrict allowed scopes. N / Build BK01 UI01 UI02 UI03 BK03 UI pattern: observation / full interaction: I-04 in Control_UX_Contracts.csv
I-05 **Rule enable and disable** Toggle with policy lock	**DEFAULT** Existing effective set **SCOPE** Rule/source **APPLIES** After acknowledged reload	Show inactive, invalid, disabled and locked states. Disabling one file must not disable an unrelated rule with a matching display name. N / Build BK01 UI01 UI02 UI03 BK03 UI pattern: choice / full interaction: I-05 in Control_UX_Contracts.csv
I-06 **Prompt templates** Named template + variables	**DEFAULT** None **SCOPE** User/project **APPLIES** On insert into draft	Template insertion does not execute tools or send a prompt. Required variables remain visibly unresolved until filled; preserve the original draft on cancel. H / Build UI01 UI02 UI03 UI pattern: editor / full interaction: I-06 in Control_UX_Contracts.csv

I · INSTRUCTIONS, RULES AND PROMPT COMPOSITION · CONTINUED

CONTROL / WIDGET	DEFAULT / SCOPE / APPLIES	REQUIRED BEHAVIOUR / BUILD / PATTERN
I-07 **Instruction conflict display** Explain conflicting sources	**DEFAULT** No silent conflict resolution in UI **SCOPE** Effective config **APPLIES** On validation	Show winner and reason, with links to editable sources. A locked policy cannot be overridden by appending a more forceful instruction. N / Build BK01 UI01 UI02 UI03 BK03 UI pattern: observation / full interaction: I-07 in Control_UX_Contracts.csv
I-08 **Import rules or presets** Preview, map and validate	**DEFAULT** No import executed automatically **SCOPE** User/project **APPLIES** After explicit import save	Import supported semantics only. Kilo/Claude config files are not native Faktor formats; unsupported keys remain a visible migration report. N / Build BK01 UI01 UI02 UI03 BK03 UI pattern: action / full interaction: I-08 in Control_UX_Contracts.csv
I-09 **System prompt inspection** Redacted effective summary	**DEFAULT** No secret/system dump **SCOPE** Task/call **APPLIES** Before admission when available	Show user-editable instructions and loaded-source manifest. Do not claim to reveal hidden provider instructions or protected model internals. N / Build BK01 UI01 UI02 UI03 BK03 UI pattern: observation / full interaction: I-09 in Control_UX_Contracts.csv
I-10 **Output style** Concise / explanatory / custom	**DEFAULT** Project/user preference **SCOPE** User/preset **APPLIES** Next request	A communication style affects prose, not tool grants, verification policy or data egress. Show it in the effective prompt preview without occupying the main toolbar. N / Build BK01 UI01 UI02 UI03 BK03 UI pattern: choice / full interaction: I-10 in Control_UX_Contracts.csv
I-11 **Instruction version binding** Read-only version/hash receipt	**DEFAULT** Captured at admission **SCOPE** Task/call **APPLIES** At durable admission	Running work retains its captured bundle. Editing a file shows Next task or Restart required; it must not silently rewrite the instructions of an existing run. N / Build BK01 UI01 UI02 UI03 BK03 UI pattern: observation / full interaction: I-11 in Control_UX_Contracts.csv

4.5 ALL 200 CONTROL CONTRACTS · R7 PAGES 249–252 · 14 CONTROLS

Q · Permissions, autonomy and sandbox

Make the authority level visible beside the composer. The model cannot grant itself powers. A session-owned one-shot reply is different from a persistent rule, and approval is different from OS isolation.

E existing / R runtime-backed native work / N new contract to build / H host UX. Every control remains required.

CONTROL / WIDGET	DEFAULT / SCOPE / APPLIES	REQUIRED BEHAVIOUR / BUILD / PATTERN
Q-01 **Permission preset** Inspect / confirm edits / project auto / custom	**DEFAULT** Proposed: inspect/read and ask before effects **SCOPE** Task/user within managed bounds **APPLIES** At admission; revocation acknowledged	Explain the exact tool categories affected. Avoid an ambiguous full-auto switch; full privilege is an explicit advanced policy, never inferred from Code mode. N / Build BK01 UI01 UI02 UI03 BK05 UI pattern: choice / full interaction: Q-01 in Control_UX_Contracts.csv
Q-02 **Per-tool rule** Allow / Ask / Deny	**DEFAULT** Ask when no authorized rule applies **SCOPE** User/project/managed **APPLIES** After policy reload/ack	Persistent rules require a real policy store and dispatcher enforcement. Current permission-reply API alone cannot save an Always allow rule. N / Build BK01 UI01 UI02 UI03 BK05 UI pattern: choice / full interaction: Q-02 in Control_UX_Contracts.csv
Q-03 **File scope matcher** Canonical roots + validated patterns	**DEFAULT** Captured workspace/candidate **SCOPE** Rule/task **APPLIES** At each dispatch	Distinguish read, create, edit and delete. Normalize symlinks and worktree ownership server-side; a pretty glob is not a containment guarantee. N / Build BK01 UI01 UI02 UI03 BK05 UI pattern: editor / full interaction: Q-03 in Control_UX_Contracts.csv
Q-04 **Command rule matcher** Structured executable/argument policy	**DEFAULT** Ask for unclassified commands **SCOPE** Rule/task **APPLIES** At each dispatch	Preview the exact normalized command. Shell injection/composition and interpreter scripts must not be approved through a misleading prefix match. N / Build BK01 UI01 UI02 UI03 BK05 UI pattern: editor / full interaction: Q-04 in Control_UX_Contracts.csv
Q-05 **Network and external effects** Destination + action policy	**DEFAULT** Effective daemon deny/allow rules **SCOPE** Operator/task **APPLIES** At dispatch	Read-only HTTP, authenticated API writes, uploads and messages have different effects. Show destination class and data boundary, not only a tool name. R / Build BK01 UI01 UI02 UI03 BK05 UI pattern: editor / full interaction: Q-05 in Control_UX_Contracts.csv
Q-06 **One-shot approval card** Allow once / Deny + request details	**DEFAULT** Pending current request **SCOPE** Session/request generation **APPLIES** After exact reply ack	Preserve existing target/reason and stale/wrong-session handling. Disable duplicate reply buttons until a durable or typed outcome arrives. E / Build BK01 UI01 UI02 UI03 BK05 UI pattern: action / full interaction: Q-06 in Control_UX_Contracts.csv

Q · PERMISSIONS, AUTONOMY AND SANDBOX · CONTINUED

CONTROL / WIDGET	DEFAULT / SCOPE / APPLIES	REQUIRED BEHAVIOUR / BUILD / PATTERN
Q-07 **Remember this decision** Preview persistent rule, then save	**DEFAULT** Off **SCOPE** Named policy scope **APPLIES** Future matching dispatch only	A remembered rule cannot pre-authorize another pending ID. Show match scope and expiry; unavailable persistence leaves Allow once as the only approval option. N / Build BK01 UI01 UI02 UI03 BK05 UI pattern: action / full interaction: Q-07 in Control_UX_Contracts.csv
Q-08 **Rule evaluation preview** Test tool/action against policy	**DEFAULT** Read-only simulator **SCOPE** Rule draft **APPLIES** Before save	Show which normalized rule wins and why. Simulator output is advisory until the runtime returns the same versioned effective policy. N / Build BK01 UI01 UI02 UI03 BK05 UI pattern: action / full interaction: Q-08 in Control_UX_Contracts.csv
Q-09 **Expiry and revocation** Timed/session/task scope	**DEFAULT** One request for one-shot approval **SCOPE** Grant/rule **APPLIES** After revocation ack	Revoke prevents future effects and reports active operations separately. Do not imply an already-issued remote effect can be undone by revoking a local grant. N / Build BK01 UI01 UI02 UI03 BK05 UI pattern: choice / full interaction: Q-09 in Control_UX_Contracts.csv
Q-10 **Sandbox status** Read-only mode and guarantees	**DEFAULT** Actual served platform state **SCOPE** Daemon host **APPLIES** On readiness/status refresh	Show filesystem, shell and network guarantees separately. Network-capable fallback is not a fully isolated shell. A failed setup stays a real blocker. R / Build BK01 UI01 UI02 UI03 BK05 UI pattern: observation / full interaction: Q-10 in Control_UX_Contracts.csv
Q-11 **Sandbox setup choice** Native guidance/host-controlled flow	**DEFAULT** Current fail-closed default **SCOPE** Machine/operator **APPLIES** After durable choice and restart if required	Reuse current faktor sandbox choose guidance until a real native setup API exists. Do not change shell security through an unacknowledged webview toggle. R / Build BK01 UI01 UI02 UI03 BK05 UI pattern: action / full interaction: Q-11 in Control_UX_Contracts.csv
Q-12 **Managed policy locks** Read-only lock + source + contact route	**DEFAULT** No managed assumptions when disabled **SCOPE** Organization/machine **APPLIES** On effective-config read	A locked setting cannot be relaxed by user/project defaults. Local coding remains available when the optional control plane is disabled. R / Build BK01 UI01 UI02 UI03 BK05 UI pattern: observation / full interaction: Q-12 in Control_UX_Contracts.csv
Q-13 **Audit decision history** Bounded searchable decision log	**DEFAULT** Only recorded decisions **SCOPE** Task/organization when enabled **APPLIES** After durable event	Show actor, scope, time, target and outcome; redact payload secrets. Permission requested, allowed, dispatched and effect confirmed remain different facts. R / Build BK01 UI01 UI02 UI03 BK05 UI pattern: observation / full interaction: Q-13 in Control_UX_Contracts.csv

Q · PERMISSIONS, AUTONOMY AND SANDBOX · CONTINUED

CONTROL / WIDGET	DEFAULT / SCOPE / APPLIES	REQUIRED BEHAVIOUR / BUILD / PATTERN
Q-14 **Prompt-injection boundary** Source labels and policy refusal display	**DEFAULT** Untrusted retrieved content **SCOPE** Tool/source item **APPLIES** On dispatch and review	A file, MCP result or webpage cannot instruct the UI to elevate authority. Show why an instruction was treated as data without blocking ordinary legitimate content. N / Build BK01 UI01 UI02 UI03 BK05 UI pattern: observation / full interaction: Q-14 in Control_UX_Contracts.csv

4.5 ALL 200 CONTROL CONTRACTS · R7 PAGES 253–256 · 15 CONTROLS

T · Tool catalog and MCP

A tool can be installed, discovered, enabled, granted, healthy, busy or unavailable. These are separate states. Current McpEntry exposes stdio name/command/args; remote transport and OAuth must be treated as new contracts.

E existing / R runtime-backed native work / N new contract to build / H host UX. Every control remains required.

CONTROL / WIDGET	DEFAULT / SCOPE / APPLIES	REQUIRED BEHAVIOUR / BUILD / PATTERN
T-01 **Tool catalog** Search grouped built-in/MCP tools	**DEFAULT** Current served tools only **SCOPE** Daemon/preset **APPLIES** On tool discovery	Show namespace, description, input schema, effect class and availability reason. An enabled tool can still be denied by active policy. N / Build BK01 UI01 UI02 UI03 BK05 BK06 UI pattern: observation / full interaction: T-01 in Control_UX_Contracts.csv
T-02 **Tool enablement** Enable/disable within preset envelope	**DEFAULT** Inherited effective set **SCOPE** Preset/task **APPLIES** After toolset ack	Changing visibility never broadens grants. Disabled tools stop being offered on future calls; in-flight calls remain listed and separately cancellable. N / Build BK01 UI01 UI02 UI03 BK05 BK06 UI pattern: choice / full interaction: T-02 in Control_UX_Contracts.csv
T-03 **Add stdio MCP server** Name, executable and argument rows	**DEFAULT** No added server **SCOPE** Machine/daemon trusted scope **APPLIES** After validation and supervised start	Match current bounds: at most 8 servers; name 128 bytes, command 512 bytes, 32 args of 512 bytes. Native CRUD is still required. R / Build BK01 UI01 UI02 UI03 BK05 BK06 UI pattern: editor / full interaction: T-03 in Control_UX_Contracts.csv
T-04 **Remote MCP transport** Streamable HTTP when supported	**DEFAULT** Unavailable in reviewed McpEntry config **SCOPE** Connection/operator **APPLIES** After new transport capability	Do not display current stdio config as an HTTP/OAuth implementation. Validate destination, authentication and reconnect semantics before enabling. N / Build BK01 UI01 UI02 UI03 BK05 BK06 UI pattern: choice / full interaction: T-04 in Control_UX_Contracts.csv
T-05 **MCP environment and secrets** Explicit names/references and preview	**DEFAULT** No inherited secret expansion **SCOPE** Machine/server **APPLIES** Next supervised start	A new configuration/handoff contract must define allowed environment values. Never paste a bearer token into a shared project JSON field. N / Build BK01 UI01 UI02 UI03 BK05 BK06 UI pattern: editor / full interaction: T-05 in Control_UX_Contracts.csv
T-06 **MCP OAuth lifecycle** Connect / reconnect / revoke	**DEFAULT** Signed out **SCOPE** Machine/server account **APPLIES** After auth callback/revoke	Show account and server binding. Cancellation preserves the draft; failed auth does not become Connected. Current reviewed MCP config has no OAuth fields. N / Build BK01 UI01 UI02 UI03 BK05 BK06 UI pattern: action / full interaction: T-06 in Control_UX_Contracts.csv

T · TOOL CATALOG AND MCP · CONTINUED

CONTROL / WIDGET	DEFAULT / SCOPE / APPLIES	REQUIRED BEHAVIOUR / BUILD / PATTERN
T-07 **Server health** Connecting / ready / auth / failed / stopped	**DEFAULT** Observed state or unavailable **SCOPE** Server instance **APPLIES** On heartbeat/observation	Show last success and bounded error detail. Installed does not mean connected, and connection does not certify tool behavior. N / Build BK01 UI01 UI02 UI03 BK05 BK06 UI pattern: observation / full interaction: T-07 in Control_UX_Contracts.csv
T-08 **Per-MCP-tool permissions** Namespaced Allow/Ask/Deny	**DEFAULT** Inherited policy; ask for effects **SCOPE** Rule/preset **APPLIES** At dispatch	Use server instance plus stable tool name, not a display label. Newly discovered effectful tools do not inherit a careless wildcard from another server. N / Build BK01 UI01 UI02 UI03 BK05 BK06 UI pattern: choice / full interaction: T-08 in Control_UX_Contracts.csv
T-09 **Tool argument inspector** Schema and normalized argument preview	**DEFAULT** Collapsed inline summary **SCOPE** Call **APPLIES** Before dispatch/after result	Redact secrets but show the exact target and material changes. Arguments changed after approval require a new authority decision. H / Build UI01 UI02 UI03 UI pattern: observation / full interaction: T-09 in Control_UX_Contracts.csv
T-10 **Timeout and output bounds** Validated duration and byte caps	**DEFAULT** Effective runtime bounds **SCOPE** Tool/server policy **APPLIES** Next operation	Show truncation and full artifact reference. A timeout marks whether an external effect is unknown; retry is not automatic for effectful tools. R / Build BK01 UI01 UI02 UI03 BK05 BK06 UI pattern: number / full interaction: T-10 in Control_UX_Contracts.csv
T-11 **Tool-definition loading** Deferred / pinned when supported	**DEFAULT** Effective runtime strategy **SCOPE** Session/preset **APPLIES** Next context compile	Show definition/token cost when measurable. Installing many tools must not quietly consume the whole context window; unsupported deferred loading is explicit. N / Build BK01 UI01 UI02 UI03 BK05 BK06 UI pattern: choice / full interaction: T-11 in Control_UX_Contracts.csv
T-12 **MCP resources and prompts** Catalog + selected-item preview	**DEFAULT** None selected **SCOPE** Session/draft **APPLIES** On explicit read/invoke	Resources, templates, prompts and tools have different lifecycles. Treat server-supplied instructions as attributed content, not unconditional policy. N / Build BK01 UI01 UI02 UI03 BK05 BK06 UI pattern: observation / full interaction: T-12 in Control_UX_Contracts.csv
T-13 **Restart, logs and remove** Bounded logs and owned lifecycle actions	**DEFAULT** Keep current server until action **SCOPE** Server instance **APPLIES** After owner ack	Restart reconciles pending calls and process ownership. Removing a config never turns an unknown remote effect into cancelled. R / Build BK01 UI01 UI02 UI03 BK05 BK06 UI pattern: action / full interaction: T-13 in Control_UX_Contracts.csv

T · TOOL CATALOG AND MCP · CONTINUED

CONTROL / WIDGET	DEFAULT / SCOPE / APPLIES	REQUIRED BEHAVIOUR / BUILD / PATTERN
T-14 **Shared/project server trust** Review executable, origin and scope	**DEFAULT** Not executable merely from repo config **SCOPE** Project/operator **APPLIES** After explicit trusted installation	A checked-out repository cannot silently run arbitrary stdio servers. Show executable provenance and which user/machine granted the start authority. N / Build BK01 UI01 UI02 UI03 BK05 BK06 UI pattern: action / full interaction: T-14 in Control_UX_Contracts.csv
T-15 **Tool diagnostics/export** Redacted connection and schema bundle	**DEFAULT** Exclude credential values **SCOPE** User/server **APPLIES** On explicit export	Include versions, toolset hash and bounded failure context. Copying diagnostics must never export environment secrets or private tool-result payloads by default. H / Build UI01 UI02 UI03 UI pattern: action / full interaction: T-15 in Control_UX_Contracts.csv

4.5 ALL 200 CONTROL CONTRACTS · R7 PAGES 257–260 · 14 CONTROLS

S · Skills, commands and hooks

Skills are reusable instruction packages; commands are entry points; hooks are lifecycle executables. None of the three is a shortcut around the runtime capability envelope.

E existing / R runtime-backed native work / N new contract to build / H host UX. Every control remains required.

CONTROL / WIDGET	DEFAULT / SCOPE / APPLIES	REQUIRED BEHAVIOUR / BUILD / PATTERN
S-01 **Skill library** Search installed trusted skills	**DEFAULT** No fabricated marketplace inventory **SCOPE** User/project/preset **APPLIES** On discovery	Show source, version, description, invocation policy and required capabilities. Current source review does not establish a complete skill-management API. N / Build BK01 UI01 UI02 UI03 BK03 BK06 UI pattern: observation / full interaction: S-01 in Control_UX_Contracts.csv
S-02 **Skill invocation policy** Manual / agent-selectable / disabled	**DEFAULT** Proposed: manual for effectful workflows **SCOPE** Skill/preset **APPLIES** Next invocation	Visibility and authority are distinct. A hidden skill must not remain silently invocable through a guessed command alias. N / Build BK01 UI01 UI02 UI03 BK03 BK06 UI pattern: choice / full interaction: S-02 in Control_UX_Contracts.csv
S-03 **Skill details and preview** Manifest, files and declared capabilities	**DEFAULT** Read-only before installation **SCOPE** Skill source **APPLIES** On open	Show exactly which instructions and scripts would load. Do not run setup or install dependencies merely to display a package. N / Build BK01 UI01 UI02 UI03 BK03 BK06 UI pattern: observation / full interaction: S-03 in Control_UX_Contracts.csv
S-04 **Install/update/remove skill** Preview version diff, then apply	**DEFAULT** No silent project installation **SCOPE** User/project trusted store **APPLIES** After validated install ack	Keep prior known-good version on failed update. Removing a skill removes future discovery without pretending to erase earlier context copies. N / Build BK01 UI01 UI02 UI03 BK03 BK06 UI pattern: action / full interaction: S-04 in Control_UX_Contracts.csv
S-05 **Slash-command palette** Filterable commands with scope and effect	**DEFAULT** Documented available commands **SCOPE** Host/session **APPLIES** At invocation	Commands insert or act according to a visible label. Proposed names in this document are UX routes, not existing CLI commands. H / Build UI01 UI02 UI03 UI pattern: action / full interaction: S-05 in Control_UX_Contracts.csv
S-06 **Custom commands/templates** Editor and dry insertion preview	**DEFAULT** None **SCOPE** User/project **APPLIES** On explicit invocation	Validate arguments and show whether a command is prompt-only or executable. Do not interpret pasted command-looking text as an automatic shell action. N / Build BK01 UI01 UI02 UI03 BK03 BK06 UI pattern: editor / full interaction: S-06 in Control_UX_Contracts.csv

S · SKILLS, COMMANDS AND HOOKS · CONTINUED

CONTROL / WIDGET	DEFAULT / SCOPE / APPLIES	REQUIRED BEHAVIOUR / BUILD / PATTERN
S-07 **Hook event selector** Current 17-event enum	**DEFAULT** No new hook installed **SCOPE** Daemon/approved package **APPLIES** On config apply	Use exact native HookEvent names; do not copy Claude-specific events into Faktor config. Expose only events the running daemon supports. R / Build BK01 UI01 UI02 UI03 BK03 BK06 UI pattern: choice / full interaction: S-07 in Control_UX_Contracts.csv
S-08 **Hook executable and args** Executable reference + bounded argv	**DEFAULT** User supplied **SCOPE** Machine/trusted hook spec **APPLIES** After validation before spawn	Show executable host, working directory and version/source. A project hook cannot silently replace the machine runtime or expand its capability set. R / Build BK01 UI01 UI02 UI03 BK03 BK06 UI pattern: editor / full interaction: S-08 in Control_UX_Contracts.csv
S-09 **Hook match filter** Tool/event/path predicates when supported	**DEFAULT** Event set only unless extended **SCOPE** Hook spec **APPLIES** Next matching event	Preview matching events. A richer matcher is a new schema feature if the current HookSpec cannot express it. N / Build BK01 UI01 UI02 UI03 BK03 BK06 UI pattern: editor / full interaction: S-09 in Control_UX_Contracts.csv
S-10 **Hook environment** Explicit allowlist + secret references	**DEFAULT** Native allowlist true **SCOPE** Machine/hook spec **APPLIES** Next hook run	Show benign passthrough and explicit entries. Changing the switch must not dump the entire daemon environment into the child process. R / Build BK01 UI01 UI02 UI03 BK03 BK06 UI pattern: editor / full interaction: S-10 in Control_UX_Contracts.csv
S-11 **Hook deadline/output caps** Bounded deadline/stdout/stderr	**DEFAULT** Effective native hook spec **SCOPE** Hook spec **APPLIES** Next run	On expiry, owned process tree is stopped and partial output cannot masquerade as an Allow verdict. Preserve a bounded audit excerpt. R / Build BK01 UI01 UI02 UI03 BK03 BK06 UI pattern: number / full interaction: S-11 in Control_UX_Contracts.csv
S-12 **Hook failure policy** FailClosed / FailOpen / Warn	**DEFAULT** Explicit per-hook policy **SCOPE** Operator/hook **APPLIES** Next run	Explain what an operation will do after failure. FailOpen is a policy choice, not an error swallowed by the UI; managed constraints still apply. R / Build BK01 UI01 UI02 UI03 BK03 BK06 UI pattern: choice / full interaction: S-12 in Control_UX_Contracts.csv
S-13 **Hook test and audit** Dry-test inputs + recorded verdicts	**DEFAULT** No production event triggered by preview **SCOPE** Hook/test session **APPLIES** After explicit test admission	Test under the real capability/deadline envelope. Show Allow/Deny/Warn/metadata changes, exit state and exact tested version. N / Build BK01 UI01 UI02 UI03 BK03 BK06 UI pattern: action / full interaction: S-13 in Control_UX_Contracts.csv

S · SKILLS, COMMANDS AND HOOKS · CONTINUED

CONTROL / WIDGET	DEFAULT / SCOPE / APPLIES	REQUIRED BEHAVIOUR / BUILD / PATTERN
S-14 **Packaged customization trust** Origin, update and conflict review	**DEFAULT** Untrusted until reviewed **SCOPE** User/project/operator **APPLIES** Before enabling executable content	Keep rule/skill/hook installation distinct from permission grants. Lock managed versions where policy requires and explain unsigned/unverified provenance honestly. N / Build BK01 UI01 UI02 UI03 BK03 BK06 UI pattern: action / full interaction: S-14 in Control_UX_Contracts.csv

4.5 ALL 200 CONTROL CONTRACTS · R7 PAGES 261–264 · 13 CONTROLS

X · Terminal, browser and language tooling

Keep long-running processes owned and visible. A browser, terminal or language server is a concrete adapter capability, not an icon that implies a tool exists.

E existing / R runtime-backed native work / N new contract to build / H host UX. Every control remains required.

CONTROL / WIDGET	DEFAULT / SCOPE / APPLIES	REQUIRED BEHAVIOUR / BUILD / PATTERN
X-01 **Terminal process list** Session-owned processes and states	**DEFAULT** Existing native process view **SCOPE** Session/task/agent **APPLIES** On event/snapshot	Show owner, working directory, age, process state and last output. A PID without a live owned handle is stale, not proof that the process is controllable. E / Build BK01 UI01 UI02 UI03 BK05 BK06 BK08 UI pattern: observation / full interaction: X-01 in Control_UX_Contracts.csv
X-02 **Shell/profile and cwd** Served shell choice + workspace URI	**DEFAULT** Daemon policy and candidate cwd **SCOPE** Machine/task **APPLIES** Before spawn	Show which machine/container owns the shell. Do not use a global current directory or run in the owner checkout because the editor selection changed. R / Build BK01 UI01 UI02 UI03 BK05 BK06 BK08 UI pattern: choice / full interaction: X-02 in Control_UX_Contracts.csv
X-03 **Terminal output detail** Bounded head/tail + full artifact	**DEFAULT** Current bounded service view **SCOPE** Process/output artifact **APPLIES** On output page	Show truncation, byte count and cursor. Do not append unlimited logs to the chat DOM or model context. E / Build BK01 UI01 UI02 UI03 BK05 BK06 BK08 UI pattern: observation / full interaction: X-03 in Control_UX_Contracts.csv
X-04 **Run in background** Owned background lifecycle action	**DEFAULT** Foreground unless configured/selected **SCOPE** Process/task **APPLIES** After background ack	Clarify that closing a UI tab does not terminate an owned process. Show dependencies and whether the parent waits for the result. N / Build BK01 UI01 UI02 UI03 BK05 BK06 BK08 UI pattern: action / full interaction: X-04 in Control_UX_Contracts.csv
X-05 **Stop, input and reconcile** Controls gated by owned handle	**DEFAULT** No action without valid owner **SCOPE** Process/session **APPLIES** After exact lifecycle ack	Keep terminal kill, run cancellation and session abort separate. Unknown process effect requires reconciliation, not a blind restart. E / Build BK01 UI01 UI02 UI03 BK05 BK06 BK08 UI pattern: action / full interaction: X-05 in Control_UX_Contracts.csv
X-06 **Browser connection** Adapter selector and auth/status	**DEFAULT** Unavailable until advertised **SCOPE** Machine/session **APPLIES** After explicit connect	Browser automation needs a real tool adapter and permission envelope. Show local/remote browser and session scope; no product promise from a mock preview. N / Build BK01 UI01 UI02 UI03 BK05 BK06 BK08 UI pattern: choice / full interaction: X-06 in Control_UX_Contracts.csv

X · TERMINAL, BROWSER AND LANGUAGE TOOLING · CONTINUED

CONTROL / WIDGET	DEFAULT / SCOPE / APPLIES	REQUIRED BEHAVIOUR / BUILD / PATTERN
X-07 **Preview URL and network scope** Validated URL + effective boundary	**DEFAULT** No automatic arbitrary navigation **SCOPE** Task/browser **APPLIES** At navigation admission	Expose target host and credentials/account context. Do not equate a successfully opened page with verified application behavior. N / Build BK01 UI01 UI02 UI03 BK05 BK06 BK08 UI pattern: editor / full interaction: X-07 in Control_UX_Contracts.csv
X-08 **Screenshot/artifact capture** Explicit capture and attachment	**DEFAULT** No automatic cross-context sharing **SCOPE** Task/artifact **APPLIES** After capture	Show origin and capture time; control whether screenshots enter model context. Keep secret-bearing pages out of shared exports by default. N / Build BK01 UI01 UI02 UI03 BK05 BK06 BK08 UI pattern: action / full interaction: X-08 in Control_UX_Contracts.csv
X-09 **Browser console/network observations** Bounded issue table	**DEFAULT** Unavailable without adapter events **SCOPE** Task/browser **APPLIES** On observation	A clean console is one observation, not total correctness. Filter by severity while preserving the recorded underlying results. N / Build BK01 UI01 UI02 UI03 BK05 BK06 BK08 UI pattern: observation / full interaction: X-09 in Control_UX_Contracts.csv
X-10 **Language-server diagnostics** Native Problems integration	**DEFAULT** Current host diagnostics when included **SCOPE** Workspace/draft/task **APPLIES** On explicit context capture	Show file, range and diagnostic version. Stale diagnostics are not evidence that the current candidate fails or passes. H / Build UI01 UI02 UI03 UI pattern: observation / full interaction: X-10 in Control_UX_Contracts.csv
X-11 **Inline code actions** Explain / fix / test / add to context	**DEFAULT** No action automatically selected **SCOPE** Editor selection **APPLIES** On explicit action	Capture file URI, range and version. Fix opens a bounded task draft; it does not silently edit or overwrite a running conversation. H / Build UI01 UI02 UI03 UI pattern: action / full interaction: X-11 in Control_UX_Contracts.csv
X-12 **Terminal command approval display** Command, cwd, env names, bounds	**DEFAULT** Effective permission policy **SCOPE** Operation **APPLIES** Before execution	Keep secret values masked but material executable/arguments visible. Show sandbox guarantees and whether output will be attached to the model. R / Build BK01 UI01 UI02 UI03 BK05 BK06 BK08 UI pattern: observation / full interaction: X-12 in Control_UX_Contracts.csv
X-13 **SSH, WSL and container binding** Read-only execution/context/vault map	**DEFAULT** Actual active host binding **SCOPE** Workspace/session **APPLIES** On binding/reconnect	A local URL resolves on the execution host. Enforce URI/path mapping and invalidate stale handles after a remote-host change without losing drafts. N / Build BK01 UI01 UI02 UI03 BK05 BK06 BK08 UI pattern: observation / full interaction: X-13 in Control_UX_Contracts.csv

4.5 ALL 200 CONTROL CONTRACTS · R7 PAGES 265–268 · 12 CONTROLS

B · Budgets, retries and reliability

A budget is an enforced envelope, not a cost badge. Show requested cap, effective shared allowance, reserved, settled and uncertain amounts with explicit units and provenance.

E existing / R runtime-backed native work / N new contract to build / H host UX. Every control remains required.

CONTROL / WIDGET	DEFAULT / SCOPE / APPLIES	REQUIRED BEHAVIOUR / BUILD / PATTERN
B-01 **Run token cap** Exact non-negative integer	**DEFAULT** Existing 0 = no limit **SCOPE** User default; per-run override **APPLIES** Captured at admission	Show No limit rather than raw zero in the normal UI. Reject decimals, negatives, overflow and malformed pasted numbers before posting; preserve the draft. E / Build BK01 UI01 UI02 UI03 BK07 BK08 UI pattern: number / full interaction: B-01 in Control_UX_Contracts.csv
B-02 **Run money cap** Exact currency input plus advanced micro-USD	**DEFAULT** Existing 0 = no limit **SCOPE** User default; per-run override **APPLIES** Captured at admission	Convert exactly using decimal arithmetic; never binary floating-point rounding. Example: USD 5.00 = 5000000 micro-USD, not 5 tokens or 5000000 dollars. E / Build BK01 UI01 UI02 UI03 BK07 BK08 UI pattern: number / full interaction: B-02 in Control_UX_Contracts.csv
B-03 **Cap warning and enforcement** Soft warning + hard-refusal states	**DEFAULT** Effective backend policy **SCOPE** Task **APPLIES** Before reservation/dispatch	Warn on configured threshold where supported. Unknown price cannot be assumed to fit; expose refusal or explicitly documented bounded pricing policy. N / Build BK01 UI01 UI02 UI03 BK07 BK08 UI pattern: observation / full interaction: B-03 in Control_UX_Contracts.csv
B-04 **Reservation display** Open / settled / refunded / uncertain	**DEFAULT** Only durable observed values **SCOPE** Task/session **APPLIES** After ledger observation	Open reservations are not final charges. Provider-reported and computed spend are different fields; uncertainty survives retries and cancellation. E / Build BK01 UI01 UI02 UI03 BK07 BK08 UI pattern: observation / full interaction: B-04 in Control_UX_Contracts.csv
B-05 **Budget revision** Explicit new allowance request	**DEFAULT** No silent active-run edit **SCOPE** Task/policy **APPLIES** After durable revision ack	A lower cap cannot erase settled spend or existing reservations. Show when a new ceiling blocks future work; revised draft defaults do not alter captured tasks. N / Build BK01 UI01 UI02 UI03 BK07 BK08 UI pattern: action / full interaction: B-05 in Control_UX_Contracts.csv
B-06 **Provider retry attempts** Integer 1..8	**DEFAULT** Daemon 3 total attempts **SCOPE** Daemon/operator **APPLIES** Next eligible request	Label total attempts, not retries, to avoid off-by-one cost expectations. State-aware replay must never repeat a request after durable effect/stream admission. R / Build BK01 UI01 UI02 UI03 BK07 BK08 UI pattern: number / full interaction: B-06 in Control_UX_Contracts.csv

B · BUDGETS, RETRIES AND RELIABILITY · CONTINUED

CONTROL / WIDGET	DEFAULT / SCOPE / APPLIES	REQUIRED BEHAVIOUR / BUILD / PATTERN
B-07 **Backoff and jitter** Base/max durations and fraction	**DEFAULT** 250 ms / 5000 ms / 0.2 **SCOPE** Daemon/operator **APPLIES** Next eligible request	Base 1..60000 ms; max >= base and <=120000 ms; jitter 0..1. Show retry class and countdown, with Stop available. R / Build BK01 UI01 UI02 UI03 BK07 BK08 UI pattern: number / full interaction: B-07 in Control_UX_Contracts.csv
B-08 **Rate limit recovery** Retry-After/queue state + cancel	**DEFAULT** Effective retry policy **SCOPE** Provider/operation **APPLIES** On typed limit	Distinguish provider rate limit, account quota and local queue saturation. Do not advertise a reset time that the provider did not supply. N / Build BK01 UI01 UI02 UI03 BK07 BK08 UI pattern: action / full interaction: B-08 in Control_UX_Contracts.csv
B-09 **Loop/step/consecutive-failur e limit** Served limits only	**DEFAULT** Effective daemon bound **SCOPE** Task/preset **APPLIES** At admission or policy ack	Expose actual loop guards without inventing an active max-turns knob. If not supported, show Planned and keep bounded backend behavior authoritative. N / Build BK01 UI01 UI02 UI03 BK07 BK08 UI pattern: number / full interaction: B-09 in Control_UX_Contracts.csv
B-10 **Task/operation deadline** Explicit duration and cancellation policy	**DEFAULT** Served owner deadline **SCOPE** Task/operation **APPLIES** At admission	Show what expires: model call, tool, task or session. Timeout is not necessarily cancellation acknowledged or remote effect rolled back. R / Build BK01 UI01 UI02 UI03 BK07 BK08 UI pattern: number / full interaction: B-10 in Control_UX_Contracts.csv
B-11 **Connectivity and auth failure** Retry/connect/update actions by typed cause	**DEFAULT** Keep coherent snapshot **SCOPE** Connection/session **APPLIES** On classified error	401, 429, timeout, unavailable provider and stale daemon need different actions. No generic Retry button that can duplicate an admitted effectful task. E / Build BK01 UI01 UI02 UI03 BK07 BK08 UI pattern: action / full interaction: B-11 in Control_UX_Contracts.csv
B-12 **Uncertain admission recovery** Receipt lookup / same-key retry when safe	**DEFAULT** Original immutable submission **SCOPE** Prompt/task **APPLIES** After durable reconciliation	Same UUID with same captured body may recover its receipt; changing model, attachments or budget creates a different request and must not reuse that key. E / Build BK01 UI01 UI02 UI03 BK07 BK08 UI pattern: action / full interaction: B-12 in Control_UX_Contracts.csv

4.5 ALL 200 CONTROL CONTRACTS · R7 PAGES 269–272 · 12 CONTROLS

V · Changes, checkpoints, verification and finish

Keep Faktor engineering proof visible while adding familiar review/undo controls. Review, restore, verify, integrate and publish are separate operations with separate authority.

E existing / R runtime-backed native work / N new contract to build / H host UX. Every control remains required.

CONTROL / WIDGET	DEFAULT / SCOPE / APPLIES	REQUIRED BEHAVIOUR / BUILD / PATTERN
V-01 **Candidate changes view** File tree + unified/split native diff	**DEFAULT** Current immutable candidate **SCOPE** Task/candidate **APPLIES** On served change basis	Preserve current native diff integration. Display base/candidate hashes and scope; editor navigation must never compare against a different session's files. E / Build BK01 UI01 UI02 UI03 BK09 UI pattern: observation / full interaction: V-01 in Control_UX_Contracts.csv
V-02 **Review comments and next task** Line comments + bounded follow-up	**DEFAULT** No edit from merely reading **SCOPE** Review/task draft **APPLIES** On explicit submit	Tie comments to exact file and diff version. Stale comments prompt rebase or clarification; a follow-up task gets its own captured contract. N / Build BK01 UI01 UI02 UI03 BK09 UI pattern: editor / full interaction: V-02 in Control_UX_Contracts.csv
V-03 **Hunk/file acceptance** Per-hunk review plus selected-change policy	**DEFAULT** Read-only review unless supported **SCOPE** Candidate **APPLIES** After explicit supported action	Review marks do not integrate files. Partial landing needs new proof for the changed candidate; never treat old full-candidate evidence as proof for a cherry-picked subset. N / Build BK01 UI01 UI02 UI03 BK09 UI pattern: choice / full interaction: V-03 in Control_UX_Contracts.csv
V-04 **Checkpoint timeline** Stored before/after identity and coverage	**DEFAULT** Only actual recorded checkpoints **SCOPE** Session/candidate **APPLIES** On checkpoint read	Hash-only rows may lack content for redo/diff. Show unavailable rather than promising a universal undo; no checkpoint is invented from a completed message. R / Build BK01 UI01 UI02 UI03 BK09 UI pattern: observation / full interaction: V-04 in Control_UX_Contracts.csv
V-05 **Restore preview** Code / conversation / both only when supported	**DEFAULT** No destructive default selected **SCOPE** Checkpoint/branch **APPLIES** Before a separate restore transaction	Show affected and untracked/uncovered files, external effects and unrelated user edits. Exact content preconditions must be enforced by the backend. N / Build BK01 UI01 UI02 UI03 BK09 UI pattern: action / full interaction: V-05 in Control_UX_Contracts.csv
V-06 **Fork and redo** New branch lineage + supported redo	**DEFAULT** Preserve current session **SCOPE** Session/checkpoint **APPLIES** After durable fork/restore ack	Conversation branching and file restoration are different. Missing after blobs disable redo; a fork must not erase subsequent work or duplicate remote side effects. N / Build BK01 UI01 UI02 UI03 BK09 UI pattern: action / full interaction: V-06 in Control_UX_Contracts.csv

V · CHANGES, CHECKPOINTS, VERIFICATION AND FINISH · CONTINUED

CONTROL / WIDGET	DEFAULT / SCOPE / APPLIES	REQUIRED BEHAVIOUR / BUILD / PATTERN
V-07 **Acceptance criteria and contract** Ordered criteria + constraints/invariants/risks/clai ms	**DEFAULT** Existing per-task fields **SCOPE** Task draft **APPLIES** Captured at admission	Keep required status and claim scope visible. Show a versioned contract; editing the next draft cannot change criteria that an active candidate was evaluated against. E / Build BK01 UI01 UI02 UI03 BK09 UI pattern: editor / full interaction: V-07 in Control_UX_Contracts.csv
V-08 **Verification policy** Quick/unit/full and supported external provider	**DEFAULT** 60 s quick / 600 s unit / full background **SCOPE** Daemon/task policy **APPLIES** After supported config reload	quick_max_s=0 disables service and leaves mutating work Unverified. Advanced/optional provider availability is explicit; native checks remain independently useful. R / Build BK01 UI01 UI02 UI03 BK09 UI pattern: choice / full interaction: V-08 in Control_UX_Contracts.csv
V-09 **Completion basis** Acceptance / claims / matching landing / finish	**DEFAULT** Pending until all required facts exist **SCOPE** Task/candidate **APPLIES** On durable proof/landing	Separate every boundary; missing required result stays pending. Session Completed or an empty passing required set cannot independently establish verified engineering success. E / Build BK01 UI01 UI02 UI03 BK09 UI pattern: observation / full interaction: V-09 in Control_UX_Contracts.csv
V-10 **Finish level** Local / commit / push / PR, hierarchical	**DEFAULT** Local **SCOPE** Task draft **APPLIES** Captured at admission	Both current hosts now expose matching four levels. State implied earlier actions; selecting PR requires actual SCM configuration and does not bypass verification. E / Build BK01 UI01 UI02 UI03 BK09 UI pattern: choice / full interaction: V-10 in Control_UX_Contracts.csv
V-11 **SCM target and privacy** Remote/base/branch/draft PR summary	**DEFAULT** Effective configured target **SCOPE** Task/operator **APPLIES** Before captured finish	Show repo, target branch and access scope. Missing canonical SCM is a blocker; a successful arbitrary shell command is not a native PR receipt. R / Build BK01 UI01 UI02 UI03 BK09 UI pattern: choice / full interaction: V-11 in Control_UX_Contracts.csv
V-12 **Conflict and manual-edit recovery** Inspect / rebase new candidate / preserve edits	**DEFAULT** No overwrite of unrelated work **SCOPE** Candidate/owner checkout **APPLIES** On compare-and-swap failure	Show changed basis and new verification obligations. Restoring or editing invalidates stale proof; retain the previous verified artifact as historical evidence. E / Build BK01 UI01 UI02 UI03 BK09 UI pattern: action / full interaction: V-12 in Control_UX_Contracts.csv

4.5 ALL 200 CONTROL CONTRACTS · R7 PAGES 273–277 · 15 CONTROLS

D · Sessions, displays and accessibility

Keep Work compact without hiding standard agent controls. Display preferences are host-owned; session operations remain durable, scoped and capability-gated.

E existing / R runtime-backed native work / N new contract to build / H host UX. Every control remains required.

CONTROL / WIDGET	DEFAULT / SCOPE / APPLIES	REQUIRED BEHAVIOUR / BUILD / PATTERN
D-01 **Conversation identity and selector** Title + workspace/worktree + active run	**DEFAULT** Existing durable conversation **SCOPE** Session/host **APPLIES** On deliberate switch	Preserve per-session draft, scroll anchor and tool view. Late callbacks from the previous session cannot overwrite the newly selected session. E / Build BK01 UI01 UI02 UI03 BK08 BK09 UI pattern: choice / full interaction: D-01 in Control_UX_Contracts.csv
D-02 **Title/history search** Explicit Search conversation titles	**DEFAULT** Existing title filter **SCOPE** Selected project/session list **APPLIES** On query	Show loaded/list scope and paging. Do not claim full-text conversation search from a title-only filter. Distinguish no matches from history unavailable. E / Build BK01 UI01 UI02 UI03 BK08 BK09 UI pattern: editor / full interaction: D-02 in Control_UX_Contracts.csv
D-03 **Open-transcript search** Find with next/previous, case and regex options	**DEFAULT** Off **SCOPE** Open conversation **APPLIES** On host action	Search the full paged transcript, including not-yet-rendered messages. A virtualized screen subset is not the full search universe. H / Build UI01 UI02 UI03 BK08 UI pattern: editor / full interaction: D-03 in Control_UX_Contracts.csv
D-04 **Rename, pin and archive** Durable session metadata actions	**DEFAULT** Current state **SCOPE** Session **APPLIES** After metadata ack	Pinning may be host-local if labelled. Rename/archive needs real metadata support; never just change the displayed text and lose it on refresh. N / Build BK01 UI01 UI02 UI03 BK08 BK09 UI pattern: action / full interaction: D-04 in Control_UX_Contracts.csv
D-05 **Queue and steer main run** Visible queue with edit/remove/next-step timing	**DEFAULT** Do not interrupt implicitly **SCOPE** Session/prompt queue **APPLIES** After admission or queue edit ack	Use existing prompt admission/queue semantics where available. Separate send next, interrupt then send, and a draft not yet admitted; show queue ownership and cap. R / Build BK01 UI01 UI02 UI03 BK08 BK09 UI pattern: editor / full interaction: D-05 in Control_UX_Contracts.csv
D-06 **Fork, export and delete** Scoped lifecycle/export actions	**DEFAULT** No deletion selected **SCOPE** Session **APPLIES** After appropriate durable/host action	Export a bounded redacted transcript plus metadata. Delete states what records, artifacts and forks remain; a running task needs a real stop/retain decision. N / Build BK01 UI01 UI02 UI03 BK08 BK09 UI pattern: action / full interaction: D-06 in Control_UX_Contracts.csv

D · SESSIONS, DISPLAYS AND ACCESSIBILITY · CONTINUED

CONTROL / WIDGET	DEFAULT / SCOPE / APPLIES	REQUIRED BEHAVIOUR / BUILD / PATTERN
D-07 **Live tool timeline** Running/queued/waiting/failed/unknown parts	**DEFAULT** Compact tool summaries **SCOPE** Session/operation **APPLIES** On durable events	Show model calls, tool results, approvals, cost and changed-file badges. Preserve original timestamps and output artifacts; do not invent a percent-complete bar. E / Build BK01 UI01 UI02 UI03 BK08 BK09 UI pattern: observation / full interaction: D-07 in Control_UX_Contracts.csv
D-08 **Focus/verbose display** Compact / balanced / detailed	**DEFAULT** Proposed: balanced **SCOPE** User **APPLIES** Immediate UI	Collapsing tool detail does not hide pending approval, failures, missing proof or current task checklist. Expansion is local UI state, not new model context. H / Build UI01 UI02 UI03 UI pattern: choice / full interaction: D-08 in Control_UX_Contracts.csv
D-09 **Theme, zoom and density** Inherit host; optional density	**DEFAULT** Host theme/font/zoom **SCOPE** User/host **APPLIES** Immediate UI	Respect VS Code tokens and JetBrains UIManager/JBUI metrics. Marketing fonts stay on document/website surfaces, not IDE reading controls. H / Build UI01 UI02 UI03 UI pattern: choice / full interaction: D-09 in Control_UX_Contracts.csv
D-10 **Composer send behavior** Enter inserts newline; host send action	**DEFAULT** Current Ctrl/Cmd+Enter pattern **SCOPE** User/host **APPLIES** Immediate UI	Offer discoverable send settings without stealing standard Tab/Shift+Tab focus. IME composition and paste must never accidentally admit a task. H / Build UI01 UI02 UI03 UI pattern: choice / full interaction: D-10 in Control_UX_Contracts.csv
D-11 **Keymap/commands and discovery** Native command registration and searchable help	**DEFAULT** Host keymap **SCOPE** User/host **APPLIES** Immediate UI	Show shortcuts in tooltips and menus; allow remap. Proposed /model, /context and /permissions are aliases only after implemented, not current command claims. H / Build UI01 UI02 UI03 UI pattern: action / full interaction: D-11 in Control_UX_Contracts.csv
D-12 **Notifications and attention** Permission/failure/completion categories	**DEFAULT** Local notices; no unsolicited external sends **SCOPE** User **APPLIES** Immediate UI	Never cover the editor for routine progress. Pending questions name the child/session; alerts remain available to screen readers without announcing every streamed token. H / Build UI01 UI02 UI03 UI pattern: choice / full interaction: D-12 in Control_UX_Contracts.csv
D-13 **Performance and transcript window** Paged history, anchored viewport, bounded buffers	**DEFAULT** Existing host windowing/coalescing **SCOPE** Host/session **APPLIES** On render/page	Preserve follow-tail versus manual scroll. Recover offscreen matches through paging; do not drop durable events to make animation look smoother. E / Build BK01 UI01 UI02 UI03 BK08 BK09 UI pattern: observation / full interaction: D-13 in Control_UX_Contracts.csv

D · SESSIONS, DISPLAYS AND ACCESSIBILITY · CONTINUED

CONTROL / WIDGET	DEFAULT / SCOPE / APPLIES	REQUIRED BEHAVIOUR / BUILD / PATTERN
D-14 **Accessibility and reduced motion** Native semantics + audit/help view	**DEFAULT** Host accessibility preferences **SCOPE** User/host **APPLIES** Immediate UI	Keyboard reaches every control; visible focus and labels survive 240 px panels and 200% text. Use names/shapes with color, editable plain-text tables and deliberate live-region summaries. H / Build UI01 UI02 UI03 UI pattern: observation / full interaction: D-14 in Control_UX_Contracts.csv
D-15 **Parallel conversations and worktrees** Session manager with owned tabs and active-run badges	**DEFAULT** Existing selected conversation; bounded proposed parallelism **SCOPE** Host/session/worktree **APPLIES** After supported admission or switch ack	Show which worktree, daemon and run owns each tab. A tab switch does not stop work; creating another active task requires explicit capacity and budget admission. Never share drafts or changed-file identities across worktrees. N / Build BK01 UI01 UI02 UI03 BK08 BK09 UI pattern: observation / full interaction: D-15 in Control_UX_Contracts.csv

4.5 ALL 200 CONTROL CONTRACTS · R7 PAGES 278–281 · 13 CONTROLS

O · Privacy, diagnostics, optional cloud and distribution

Keep the local agent independent of cloud services and of the other company products. Native verification and coding remain useful without Kiwi-CI, Tangerine or a control-plane account.

E existing / R runtime-backed native work / N new contract to build / H host UX. Every control remains required.

CONTROL / WIDGET	DEFAULT / SCOPE / APPLIES	REQUIRED BEHAVIOUR / BUILD / PATTERN
O-01 **Effective-settings search** Search all settings and explain winner	**DEFAULT** All effective readable values **SCOPE** Allowed scopes only **APPLIES** On snapshot refresh	Each row shows label, source, scope, timing, reset and lock. Hide secrets; compare changes before save and keep invalid edits without partial persistence. N / Build BK01 UI01 UI02 UI03 BK07 BK10 UI pattern: editor / full interaction: O-01 in Control_UX_Contracts.csv
O-02 **Local/hosted data boundary** Endpoint/egress summary and policy	**DEFAULT** Actual configured route **SCOPE** Task/connection **APPLIES** Before capture and route change	Local to the service is not automatically local to the user. A hosted fallback must show its data boundary; turning cloud features off does not disable legitimate local work. R / Build BK01 UI01 UI02 UI03 BK07 BK10 UI pattern: observation / full interaction: O-02 in Control_UX_Contracts.csv
O-03 **Diagnostic export** Preview and redacted bounded bundle	**DEFAULT** Local save only **SCOPE** User/session **APPLIES** On explicit action	Include versions, route/capability state, ownership and bounded errors; exclude credential values, source payloads and private screenshots unless deliberately selected. E / Build BK01 UI01 UI02 UI03 BK07 BK10 UI pattern: action / full interaction: O-03 in Control_UX_Contracts.csv
O-04 **Runtime location and trust** Version/path/launcher/checksum details	**DEFAULT** Current discovery policy **SCOPE** Machine/application only **APPLIES** At explicit service start	Project settings cannot change the executable path. Recorded checksum match is not signature verification; mismatch refuses start and unrecorded paths need exact-path confirmation. E / Build BK01 UI01 UI02 UI03 BK07 BK10 UI pattern: observation / full interaction: O-04 in Control_UX_Contracts.csv
O-05 **Start/stop/update lifecycle** Explicit actions and readiness stages	**DEFAULT** No start from mere view opening **SCOPE** Machine/daemon **APPLIES** After lifecycle ack	Show starting, ready, incompatible, stopped and refused separately. Signed updater availability is conditional; an update action must preserve active-run ownership and rollback state. E / Build BK01 UI01 UI02 UI03 BK07 BK10 UI pattern: action / full interaction: O-05 in Control_UX_Contracts.csv
O-06 **Usage/pricing/account view** Observed counts and quota basis	**DEFAULT** Unavailable rather than zero **SCOPE** Local/session/org when enabled **APPLIES** On served usage/account read	Separate API charges, local operating estimates, subscription quota and reservations. Provider quotas are only shown when supplied; no fake balance on a local model. R / Build BK01 UI01 UI02 UI03 BK07 BK10 UI pattern: observation / full interaction: O-06 in Control_UX_Contracts.csv

O · PRIVACY, DIAGNOSTICS, OPTIONAL CLOUD AND DISTRIBUTION · CONTINUED

CONTROL / WIDGET	DEFAULT / SCOPE / APPLIES	REQUIRED BEHAVIOUR / BUILD / PATTERN
O-07 **Optional control-plane sign-in** Endpoint + organization + scoped secret-store flow	**DEFAULT** Disabled/empty endpoint **SCOPE** Machine/user account **APPLIES** After verified auth/revoke	VS Code has a guided path; JetBrains currently includes manual credential storage. Removing a local stored credential is distinct from revoking a remote session. E / Build BK01 UI01 UI02 UI03 BK07 BK10 UI pattern: editor / full interaction: O-07 in Control_UX_Contracts.csv
O-08 **Managed configuration** Lock/provenance inspector and allowed overrides	**DEFAULT** Disabled when enterprise not enabled **SCOPE** Organization/operator **APPLIES** After effective policy ack	Reuse current enterprise effective-config concepts only when enabled. Do not imply arbitrary local preference keys are already enforced by that service. R / Build BK01 UI01 UI02 UI03 BK07 BK10 UI pattern: observation / full interaction: O-08 in Control_UX_Contracts.csv
O-09 **Retention and cleanup** Bounded policy and dry-run list	**DEFAULT** Current stored retention policy **SCOPE** User/operator **APPLIES** After acknowledged policy/job	Show sessions, snapshots, artifacts, logs and learned facts separately. Never delete active-owned records or silently equate hiding chat with deleting retained files. R / Build BK01 UI01 UI02 UI03 BK07 BK10 UI pattern: editor / full interaction: O-09 in Control_UX_Contracts.csv
O-10 **Telemetry and feedback** Explicit categories and payload preview	**DEFAULT** Proposed: no content telemetry by default **SCOPE** User/managed policy **APPLIES** After acknowledged choice	Local UX counters and uploaded content are different. Opt-in/out must show endpoint and scope; no unsupported compliance or confidentiality badge. N / Build BK01 UI01 UI02 UI03 BK07 BK10 UI pattern: choice / full interaction: O-10 in Control_UX_Contracts.csv
O-11 **Worker placement** Local / served remote worker	**DEFAULT** Local unless configured **SCOPE** Task/operator **APPLIES** At placement admission	Workers are optional. Show job generation, machine, lease and egress boundary; the IDE cannot invent a remote capacity or silently move local work to cloud. R / Build BK01 UI01 UI02 UI03 BK07 BK10 UI pattern: choice / full interaction: O-11 in Control_UX_Contracts.csv
O-12 **Import/export configuration** Redacted schema-validated bundle	**DEFAULT** No secrets or executable auto-start **SCOPE** User/operator **APPLIES** After preview and apply	Report unsupported keys, policy conflicts, required restart and missing secret references. Never import machine executable paths from an untrusted project. N / Build BK01 UI01 UI02 UI03 BK07 BK10 UI pattern: action / full interaction: O-12 in Control_UX_Contracts.csv
O-13 **Feature/capability status** Available / disabled / unsupported / needs development	**DEFAULT** Actual daemon and host support **SCOPE** Daemon/host **APPLIES** On version/capability handshake	Keep unavailable controls explainable and relevant, not a sea of inert toggles. The design catalog is complete; shipped menus include only backed capabilities. N / Build BK01 UI01 UI02 UI03 BK07 BK10 UI pattern: observation / full interaction: O-13 in Control_UX_Contracts.csv

4.6 RESTORED FIRST-REPORT CONTENT · CONTENT PRESERVATION · R7 PAGE 282

What has been restored

Retention is checked against the original content, not only against control counts.

The first full IDE report

The complete 12 October report had 36 screen contracts, 34 named states, 16 end-to-end journeys, 14 dated source conflicts, 10 file-level implementation targets, keyboard/focus rules and validation plans. These are restored below and preserved exactly as structured data and as the original 105-page PDF in the handoff.

The second full IDE report

All 200 controls, defaults, scopes, apply timing, detailed behavior, source references, 16 standard-agent workflows, backend/native build requirements, 40 acceptance scenarios and original code outlines are retained. The original 183-page PDF and complete Markdown remain in the dated source archive.

Identity and website

All 42 pages of the independent Faktor identity report and all 57 pages of the full Faktor website PDF are retained as clearly dated appendices. The approved angular F and fixed wordmark stay unchanged. KiwiCaptcha remains a separate product with its own identity, website and files.

How conflicting dates work

Historical source descriptions are labelled 12 October and pinned to their original commit. The current reviewed design baseline remains a89c00dbd612. Explicit corrections supersede older current-state assertions; the originals remain available for audit. No unsupported feature is silently dropped from the final construction scope.

4.6 RESTORED FIRST-REPORT CONTENT · RESTORED FIRST-REPORT CONTRACTS · R7 PAGES 283, 284, 285, 286, 287, 288, 289, 290, 291

Original screen coverage / fully restored

Dated source statements below are preserved as history; use the explicit 17 October corrections for current support.

R4-S01 / First open / no project

Restored design: S34.

Job: Locate the workspace and explicit start path.

VS Code: Work empty surface

JetBrains: Work empty surface

12 October source: Current stopped/workspace state exists; dedicated preflight checklist is proposed.

Original actions: Open project / Start service / Setup guidance

Original classification: proposed_ui_existing_backend. States: T00, T01, T02.

Trace: V02, V03, V06, V15, V16, V18.

R4-S02 / Service discovery and trust

Restored design: S35.

Job: Show the exact app-level executable and trust basis.

VS Code: Native host dialog + Diagnostics

JetBrains: Native dialog + application chooser

12 October source: Current explicit discovery, unverified confirm and mismatch refusal exist. VS Code has settings guidance rather than a registered Locate chooser.

Original actions: Locate daemon… / Cancel / Technical details

Original classification: proposed_host_integration. States: T02, T03, T04. Trace: V07, V17, V38.

R4-S03 / Ready conversation

Restored design: S36.

Job: Make task entry the primary job.

VS Code: Work

JetBrains: Work

12 October source: Current conversation/current run/composer exists in both.

Original actions: Run task / Ask / Attach / Run options

Original classification: presentation_refinement. States: T06. Trace: V03, V04, V18.

R4-S04 / Task options

Restored design: S33.

Job: Capture criteria, invariants, risks, finish and budget without losing the draft.

VS Code: Composer disclosure or compact options sheet

JetBrains: Scrollable native task options sheet

12 October source: Contract/finish options exist; per-run budget sheet and normalized finish selector are proposed.

Original actions: Apply options / Cancel

Original classification: proposed_ui_existing_backend. States: T06, T07. Trace: V02, V04, V09, V14, V18.

R4-S05 / Attachments

Restored design: S37.

Job: Show admitted files/documents/images with model-specific limits.

VS Code: Composer chips + attachment detail

JetBrains: Composer chips + Attachments panel

12 October source: Existing host-owned bytes/IDs and model-capability admission.

Original actions: Choose files / Remove / Clear / Retry

Original classification: presentation_refinement. States: T06, T08. Trace: V04, V09, V35.

R4-S06 / Submission in flight

Restored design: S38.

Job: Represent one captured logical start.

VS Code: Composer + notice

JetBrains: Composer + notice

12 October source: Both have single-flight capture and retained submission identity.

Original actions: Starting… / Review draft only

Original classification: presentation_refinement. States: T07. Trace: V04, V09, V18, V44.

ORIGINAL SCREEN COVERAGE / FULLY RESTORED · CONTINUED

R4-S07 / Ambiguous start / retry

Restored design: S39.

Job: Keep the draft, options and successfully uploaded refs.

VS Code: Work notice

JetBrains: Work notice

12 October source: Retry semantics are present. Structured human error cards are current VS Code, proposed parity in JetBrains.

Original actions: Review and retry / Technical details

Original classification: proposed_ui_existing_backend. States: T08. Trace: V08, V09, V18.

R4-S08 / Active run

Restored design: S01.

Job: Show goal, phase, candidate and the relevant next action.

VS Code: Current run strip in Work

JetBrains: Current run strip in Work

12 October source: Current run and cancel are present; richer strip is a presentation proposal.

Original actions: Inspect / Stop current run

Original classification: presentation_refinement. States: T09, T12, T33. Trace: V03, V06, V18, V53.

R4-S09 / Contextual approval

Restored design: S14.

Job: Tie Allow/Deny to the exact request.

VS Code: Inline above composer or attention strip

JetBrains: Work attention card + Inspect Agents/Approvals

12 October source: VS Code has inline cards; JetBrains approvals are under Agents. Shared inline Work placement is proposed for JetBrains.

Original actions: Allow / Deny / Details

Original classification: proposed_ui_existing_backend. States: T10. Trace: V04, V27, V28.

R4-S10 / Needs input / blocker

Restored design: S40.

Job: Explain what is owed and where to act.

VS Code: Work attention + Inspect Overview

JetBrains: Work attention + Inspect Overview

12 October source: Blocker state/actions exist; clearer routing and human copy are proposed.

Original actions: Resolve item / Inspect related agent

Original classification: presentation_refinement. States: T11, T19, T22. Trace: V01, V28, V18.

R4-S11 / Inspect overview

Restored design: S41.

Job: Summarize run, scope, criteria, candidate, integration, finish and cost.

VS Code: Inspect Overview proposed subview

JetBrains: Inspect Overview

12 October source: Current cockpit/Overview exist; expose four status domains distinctly.

Original actions: Open verification / Review changes / Plan

Original classification: presentation_refinement. States: T12, T13, T18, T21, T29.

Trace: V10, V24, V53.

R4-S12 / Plan

Restored design: S42.

Job: Show dependencies and blocked work without database vocabulary.

VS Code: Inspect Plan proposed subview

JetBrains: Inspect Plan

12 October source: Existing structured step/agent models; proposed compact list-first rendering.

Original actions: Open step / Inspect blockers

Original classification: presentation_refinement. States: T09, T11, T13. Trace: V10, V23, V24.

ORIGINAL SCREEN COVERAGE / FULLY RESTORED · CONTINUED

R4-S13 / Changes list

Restored design: S24.

Job: Associate changed paths with candidate and evidence.

VS Code: Inspect Changes proposed subview

JetBrains: Inspect Changes

12 October source: JetBrains currently lists paths and refs; both need native editor diff navigation.

Original actions: Review selected change / Open evidence

Original classification: proposed_host_integration. States: T17, T18. Trace: V10, V12, V21.

R4-S14 / Native diff review

Restored design: S43.

Job: Compare immutable before/candidate content in the host editor.

VS Code: Native editor diff with Faktor scope toolbar

JetBrains: Native diff editor via platform adapter

12 October source: No current native diff/open-file action was found in inspected Changes/host dispatch. Requires host adapter plus immutable content seam.

Original actions: Next change / Previous change / Open claim / Return to run

Original classification: proposed_host_integration. States: T17, T18, T19. Trace: V06, V12, V21, V18.

R4-S15 / Integration conflict

Restored design: S44.

Job: Keep independently changed files and show the stale basis.

VS Code: Changes conflict detail + native diff

JetBrains: Changes conflict detail + native diff

12 October source: Proof/integration conflict count exists; direct rebase/apply controls require backend capability.

Original actions: Inspect conflict / Recheck current basis

Original classification: proposed_backend_extension. States: T19, T23. Trace: V12, V24, V47.

R4-S16 / Verification criteria

Restored design: S45.

Job: Show each requirement, verdict, basis and evidence.

VS Code: Inspect Verification proposed subview

JetBrains: Inspect Verification

12 October source: Current criterion proof models and detail are implemented.

Original actions: View evidence / Filter required / Show all

Original classification: presentation_refinement. States: T13, T14, T15, T16. Trace: V04, V10, V22, V24.

R4-S17 / Evidence detail

Restored design: S46.

Job: Open the result behind a claim without changing scope.

VS Code: Inline detail or editor artifact preview

JetBrains: Verification Evidence navigator

12 October source: Existing bounded retrieval and text preview. Native external artifact opening is proposed.

Original actions: Return to criterion / View full artifact if available

Original classification: proposed_host_integration. States: T13, T14, T15. Trace: V04, V12, V25.

R4-S18 / Full proof and mismatch

Restored design: S47.

Job: Show current proof chain, identity match, review and missing members.

VS Code: Inspect Verification proof detail

JetBrains: Inspect Verification / Plan proof detail

12 October source: Both clients consume strict taskProof; more readable evidence summary proposed.

Original actions: Inspect mismatch / Refresh proof / Advanced identities

Original classification: presentation_refinement. States: T15, T23, T24. Trace: V10, V12, V23, V24.

ORIGINAL SCREEN COVERAGE / FULLY RESTORED · CONTINUED

R4-S19 / Finish actions

Restored design: S48.

Job: Separate verified candidate, landing and requested publication.

VS Code: Inspect Overview/Verification finish section

JetBrains: Inspect Overview/Plan finish section

12 October source: Durable finish rows and native SCM gate exist; normalized four-level selector is a design change.

Original actions: View requested action / Setup SCM when missing

Original classification: presentation_refinement. States: T18, T21, T22. Trace: V10, V24, V47.

R4-S20 / Verified result

Restored design: S65.

Job: Show the reviewed result with its actual completion basis.

VS Code: Work result + Inspect proof

JetBrains: Work result + Inspect proof

12 October source: Strict proof state is served; avoid using generic done/session status as proof.

Original actions: Review result / New task

Original classification: presentation_refinement. States: T24. Trace: V10, V24, V47.

R4-S21 / Cancelled or failed result

Restored design: S49.

Job: Retain useful work and name the terminal outcome.

VS Code: Work result + Inspect

JetBrains: Work result + Inspect

12 October source: Current cancellation/failure routes and records exist.

Original actions: Inspect recorded work / New task

Original classification: presentation_refinement. States: T25, T26. Trace: V08, V18, V44.

R4-S22 / Agents roster

Restored design: S23.

Job: Show human role, work state, dependency and applicable controls.

VS Code: Inspect Agents proposed subview

JetBrains: Inspect Agents/Roster

12 October source: Existing roster and pause/resume/cancel/retry/steer/model/budget controls.

Original actions: Open agent / Applicable per-agent actions

Original classification: presentation_refinement. States: T09, T11, T10. Trace: V04, V10, V29, V19.

R4-S23 / Agent detail / steer

Restored design: S50.

Job: Scope the message/control to one agent and preserve input on failure.

VS Code: Inspect Agents detail + native input

JetBrains: Inspect Agents detail + native input

12 October source: Agent controls and native dialogs exist; structured drawer refinement proposed.

Original actions: Send steer / Resume / Cancel / Technical details

Original classification: presentation_refinement. States: T09, T11. Trace: V06, V29, V18.

R4-S24 / Compare approaches

Restored design: S51.

Job: Compare identical criteria candidates before choosing a recommendation.

VS Code: Inspect Plan comparison

JetBrains: Inspect Plan comparison

12 October source: Existing bounded 2-4 candidate tournament and recommend/abort routes.

Original actions: Recommend winner when settled / Abort comparison

Original classification: presentation_refinement. States: T12, T13, T14, T17. Trace: V10, V30, V19.

ORIGINAL SCREEN COVERAGE / FULLY RESTORED · CONTINUED

R4-S25 / Coordination

Restored design: S52.

Job: Make agent communication contextual and provenance visible.

VS Code: Inspect Agents/Coordination proposed grouping

JetBrains: Inspect Agents/Coordination

12 October source: Existing board read/post/paging. No instruction-authority promotion.

Original actions: Post / Read / Load older / Advanced

Original classification: presentation_refinement. States: T09. Trace: V04, V31, V19.

R4-S26 / Prior conversations

Restored design: S26.

Job: Find and reopen actual durable work.

VS Code: History proposed visible destination

JetBrains: History existing destination

12 October source: Native session listing exists in VS Code host; visible selector absent. JetBrains already searches/opens.

Original actions: Open conversation / Search / Refresh

Original classification: proposed_ui_existing_backend. States: T31. Trace: V01, V06, V12, V26.

R4-S27 / Run outcome analytics

Restored design: S53.

Job: Read actual all-attempt cost and routing cohorts.

VS Code: History / Outcomes proposed

JetBrains: History / Outcomes proposed

12 October source: Native outcomes endpoint/schema exists; inspected host services do not currently wire this dedicated UI.

Original actions: Open outcome / Filter cohort / Explain metric

Original classification: proposed_ui_existing_backend. States: T29. Trace: V44, V45, V57.

R4-S28 / Usage and quota

Restored design: S54.

Job: Show current-period metering distinct from engineering outcome cost.

VS Code: Overflow Usage

JetBrains: More Usage

12 October source: Existing billing UI and quota/credits routes; role-aware actions.

Original actions: Refresh usage / Previous / Next / Grant credits if permitted

Original classification: presentation_refinement. States: T29, T30. Trace: V10, V33, V19.

R4-S29 / Model and provider selection

Restored design: S02.

Job: Choose from the advertised catalog and show active selection.

VS Code: Native settings/quick pick; composer chip

JetBrains: Settings or native chooser; composer chip

12 October source: Current new-session provider defaults; JetBrains selection exists. Proposed per-run chooser must respect API/model capability.

Original actions: Select model / Refresh catalog

Original classification: proposed_ui_existing_backend. States: T06, T15. Trace: V02, V06, V32, V44.

R4-S30 / Settings

Restored design: S64.

Job: Separate machine service settings, run defaults and optional control plane.

VS Code: Native VS Code Settings scoped correctly

JetBrains: Native Settings/Configurable proposed; current utility panel

12 October source: Machine/application service path exists; new JetBrains platform Configurable integration is proposed.

Original actions: Apply changed configuration / Cancel / Restore default

Original classification: proposed_host_integration. States: T02, T29, T30. Trace: V02, V07, V17, V32.

ORIGINAL SCREEN COVERAGE / FULLY RESTORED · CONTINUED

R4-S31 / Optional control-plane credentials

Restored design: S55.

Job: Keep local work usable and store secrets outside settings.

VS Code: Native sign-in command and SecretStorage

JetBrains: Settings utility and PasswordSafe

12 October source: Current scoped credential stores and sign-out behavior. No default cloud endpoint.

Original actions: Store credential / Sign out / Back

Original classification: presentation_refinement. States: T30. Trace: V06, V55, V16, V32.

R4-S32 / Diagnostics

Restored design: S30.

Job: Expose service/readiness/stream/index and exact advanced details.

VS Code: Overflow Diagnostics

JetBrains: More Diagnostics

12 October source: Existing service/stream and recovery actions; add clearer readiness and source details.

Original actions: Reconnect / Refresh / Restart service

Original classification: presentation_refinement. States: T27, T28, T33. Trace: V03, V13, V34, V39, V44.

R4-S33 / Protocol blocked recovery

Restored design: S56.

Job: Retain last good cursor and show why a fresh read is needed.

VS Code: Inline Live updates paused card

JetBrains: Work attention + Diagnostics

12 October source: VS Code inline recovery exists; JetBrains stream/Diagnostics exists. No skip-event action.

Original actions: Reconnect after repair / Refresh from snapshot / Technical details

Original classification: presentation_refinement. States: T28. Trace: V13, V19, V39.

R4-S34 / Sandbox platform setup

Restored design: S57.

Job: Show explicit fail-closed choice and trade-off guidance.

VS Code: Diagnostics setup guidance

JetBrains: Diagnostics setup guidance

12 October source: Current CLI durable choice exists, not a complete native IDE grant workflow.

Original actions: Open setup guidance / Keep current fail-closed state

Original classification: proposed_ui_existing_backend. States: T32. Trace: V46.

R4-S35 / Owned command output

Restored design: S19.

Job: Inspect command, owner, exit/result and bounded output.

VS Code: Inspect Overview command details

JetBrains: Inspect Overview Terminal detail

12 October source: Current tool excerpts; JetBrains terminal spawn/output/events exist.

Original actions: Show output / Load more evidence

Original classification: presentation_refinement. States: T09, T14, T28. Trace: V04, V36, V19.

R4-S36 / Narrow tool window

Restored design: S58.

Job: Keep labels/actions readable at 240px without horizontal action bars.

VS Code: Work/Inspect single column

JetBrains: Work/Inspect single column

12 October source: Current reflow helpers exist; JetBrains toolbar horizontal-scroll fallback remains. Proposed overflow avoids that fallback.

Original actions: Primary contextual action / Overflow secondary actions

Original classification: presentation_refinement. States: T06, T10, T13. Trace: V05, V20, V18.

4.6 RESTORED FIRST-REPORT CONTENT · RESTORED STATE & MICROCOPY CONTRACT · R7 PAGES 292, 293, 294, 295, 296, 297

Every named operational state

The label, condition, supporting copy and action are a single semantic contract.

R4-T00 / Workspace unavailable

Condition: No project

Visible copy: Open a project to choose where this run works.

Actions: Open project

May claim verified completion: No.

Semantic tone: none. Sources: V06, V16.

R4-T01 / Faktor is stopped

Condition: Stopped

Visible copy: Start the service when you are ready.

Actions: Start service

May claim verified completion: No.

Semantic tone: neutral. Sources: V06, V16, V19.

R4-T02 / Faktor could not find its service

Condition: Missing binary

Visible copy: Locate an application-level executable or install the service.

Actions: Locate daemon…

May claim verified completion: No.

Semantic tone: attention. Sources: V07, V17, V38.

R4-T03 / Checksum record unavailable

Condition: Unverified binary

Visible copy: Confirm the exact executable before starting.

Actions: Start anyway / Cancel

May claim verified completion: No.

Semantic tone: attention. Sources: V07, V17, V38.

R4-T04 / The service executable does not match its record

Condition: Binary refused

Visible copy: Choose another executable or repair the installation.

Actions: Locate daemon…

May claim verified completion: No.

Semantic tone: failure. Sources: V07, V37, V38.

R4-T05 / Starting Faktor

Condition: Starting

Visible copy: Resolving the confirmed executable and waiting for readiness.

Actions: Cancel startup

May claim verified completion: No.

Semantic tone: active. Sources: V06, V19, V44.

R4-T06 / Ready

Condition: Ready

Visible copy: Describe a task or ask a question.

Actions: Run task

May claim verified completion: No.

Semantic tone: neutral. Sources: V03, V18.

R4-T07 / Starting the run

Condition: Submitting

Visible copy: Your captured draft is being admitted once.

Actions: Wait

May claim verified completion: No.

Semantic tone: active. Sources: V04, V09, V18.

EVERY NAMED OPERATIONAL STATE · CONTINUED

R4-T08 / Start response unavailable

Condition: Start ambiguous

Visible copy: The draft was kept. Retry checks the same logical submission.

Actions: Review and retry

May claim verified completion: No.

Semantic tone: attention. Sources: V06, V09, V18, V44.

R4-T09 / Running

Condition: Working

Visible copy: The candidate is separate from your workspace.

Actions: Inspect / Stop current run

May claim verified completion: No.

Semantic tone: active. Sources: V01, V03, V18.

R4-T10 / Faktor needs your approval

Condition: Approval

Visible copy: Review the action, target and stated reason.

Actions: Allow / Deny

May claim verified completion: No.

Semantic tone: attention. Sources: V04, V27.

R4-T11 / Needs your input

Condition: Needs input

Visible copy: The run is waiting for a decision or clarification.

Actions: Reply in composer

May claim verified completion: No.

Semantic tone: attention. Sources: V01, V18, V28.

R4-T12 / Checking the change

Condition: Verifying

Visible copy: Required results remain attached to this candidate.

Actions: Inspect verification

May claim verified completion: No.

Semantic tone: active. Sources: V42, V49.

R4-T13 / Completion pending

Condition: Required result missing

Visible copy: One required result is missing.

Actions: View missing result

May claim verified completion: No.

Semantic tone: attention. Sources: V48, V10, V24.

R4-T14 / Required check failed

Condition: Required check failed

Visible copy: Inspect the result and repair the candidate.

Actions: View evidence

May claim verified completion: No.

Semantic tone: failure. Sources: V42, V48, V49.

R4-T15 / Verification unavailable

Condition: Verification unavailable

Visible copy: No verdict is served for this claim.

Actions: Inspect reason / Refresh

May claim verified completion: No.

Semantic tone: attention. Sources: V49, V10, V24.

EVERY NAMED OPERATIONAL STATE · CONTINUED

R4-T16 / Verification refused

Condition: Verification refused

Visible copy: A policy or contract prevents this check.

Actions: Inspect reason

May claim verified completion: No.

Semantic tone: attention. Sources: V49.

R4-T17 / Candidate checks passed

Condition: Candidate verified

Visible copy: The candidate’s required checks passed; integration and finish actions remain separate.

Actions: Review changes

May claim verified completion: No.

Semantic tone: partial_success. Sources: V12, V24, V47.

R4-T18 / Not integrated

Condition: Integration pending

Visible copy: The verified candidate has not landed.

Actions: Review changes

May claim verified completion: No.

Semantic tone: attention. Sources: V12, V24, V47.

R4-T19 / Workspace changed

Condition: Integration conflict

Visible copy: The current workspace no longer matches the integration basis.

Actions: Inspect conflict

May claim verified completion: No.

Semantic tone: failure. Sources: V12, V24.

R4-T20 / Integrating the candidate

Condition: Integration active

Visible copy: An integration transaction is in progress.

Actions: Inspect transaction

May claim verified completion: No.

Semantic tone: active. Sources: V12, V47.

R4-T21 / Finish action pending

Condition: Finish pending

Visible copy: A requested action lacks a succeeded durable result.

Actions: View finish actions

May claim verified completion: No.

Semantic tone: attention. Sources: V10, V24, V47.

R4-T22 / Pull request integration unavailable

Condition: PR not configured

Visible copy: Configure the native SCM integration; no PR success is inferred.

Actions: Open setup guidance

May claim verified completion: No.

Semantic tone: attention. Sources: V47.

R4-T23 / The result needs verification

Condition: Proof mismatch

Visible copy: The landed state differs from the verified candidate.

Actions: Inspect mismatch

May claim verified completion: No.

Semantic tone: failure. Sources: V10, V12, V24.

EVERY NAMED OPERATIONAL STATE · CONTINUED

R4-T24 / Verified complete

Condition: Verified complete

Visible copy: The current proof summary supports completion and requested finish actions.

Actions: Review result / New task

May claim verified completion: Yes, only on matching current proof.

Semantic tone: success. Sources: V10, V24, V47.

R4-T25 / Run cancelled

Condition: Cancelled

Visible copy: Cancellation is a terminal run outcome, not verification success.

Actions: Review recorded work / New task

May claim verified completion: No.

Semantic tone: neutral. Sources: V18, V19, V44.

R4-T26 / Run failed

Condition: Failed

Visible copy: The result and reason remain available for inspection.

Actions: Inspect result

May claim verified completion: No.

Semantic tone: failure. Sources: V08, V18, V44.

R4-T27 / Reconnecting live updates

Condition: Transient reconnect

Visible copy: The last valid journal position is kept.

Actions: Wait / Refresh

May claim verified completion: No.

Semantic tone: attention. Sources: V13, V39.

R4-T28 / Live updates paused

Condition: Protocol blocked

Visible copy: The client cannot read the next durable event. Nothing is skipped.

Actions: Reconnect / Refresh from snapshot

May claim verified completion: No.

Semantic tone: attention. Sources: V13, V39.

R4-T29 / Cost unavailable

Condition: Cost unavailable

Visible copy: The current authority has not supplied a cost value.

Actions: View details

May claim verified completion: No.

Semantic tone: neutral. Sources: V10, V24, V33.

R4-T30 / Billing is disabled locally

Condition: Billing disabled

Visible copy: This surface has no active billing data.

Actions: Back

May claim verified completion: No.

Semantic tone: neutral. Sources: V10, V33.

R4-T31 / History unavailable

Condition: History unavailable

Visible copy: The conversation list could not be read.

Actions: Refresh history

May claim verified completion: No.

Semantic tone: attention. Sources: V26.

EVERY NAMED OPERATIONAL STATE · CONTINUED

R4-T32 / Shell execution needs a decision

Condition: Sandbox setup

Visible copy: The secure default remains fail-closed on this platform.

Actions: Open setup guidance

May claim verified completion: No.

Semantic tone: attention. Sources: V46.

R4-T33 / Repository context is partial

Condition: Partial repository index

Visible copy: The index is rebuilding or capped; no full coverage is implied.

Actions: Inspect coverage

May claim verified completion: No.

Semantic tone: attention. Sources: V06, V18, V34.

4.6 RESTORED FIRST-REPORT CONTENT · RESTORED END-TO-END JOURNEYS · R7 PAGE 298

R4-F01 / First launch and explicit service start

All original steps and guards retained. Current designs preserve the same task and proof boundary.

Entry

Open Faktor with a project.

Interaction

1. Show stopped state without starting a process.
2. Choose Start service.
3. Resolve machine/application binary and show source/path.
4. If missing, open setup/Locate flow.
5. If checksum record is absent, require the existing exact-path native confirmation.
6. If checksum mismatches or record is malformed, refuse startup.
7. Wait for startup/readiness; create/reopen workspace-bound conversation.

Required guard

Opening the view never implicitly grants binary or shell execution. Bare checksum match is not signature verification.

Exit and trace

Ready or explicit refused/stopped state.

Original screens: S01, S02, S03, S32.

Current designs: S34, S35, S36, S30.

Sources: V06, V07, V16, V17, V19, V38, V44.

4.6 RESTORED FIRST-REPORT CONTENT · RESTORED END-TO-END JOURNEYS · R7 PAGE 299

R4-F02 / Prepare a task

All original steps and guards retained. Current designs preserve the same task and proof boundary.

Entry

Ready Work composer.

Interaction

1. Type goal; example chips fill and focus without sending.
2. Choose Task mode where the proposed mode selector exists.
3. Expand Run options only if needed.
4. Enter line-based contract rows, select finish level and review model/budget.
5. Validate visible input bounds without silently dropping authored text.

Required guard

One composer and one scoped draft. Ask path never inherits Task finish actions.

Exit and trace

Ready-to-submit captured task draft.

Original screens: S03, S04, S29.

Current designs: S36, S33, S02.

Sources: V03, V04, V09, V14, V18.

4.6 RESTORED FIRST-REPORT CONTENT · RESTORED END-TO-END JOURNEYS · R7 PAGE 300

R4-F03 / Attach context

All original steps and guards retained. Current designs preserve the same task and proof boundary.

Entry

Draft with a selected model.

Interaction

1. Choose files or paste an image.
2. Show type, filename, size and admission state.
3. Use workspace-relative files for repository context.
4. Use advertised image/document capability and limits for binary delivery.
5. Refuse unsupported or oversized attachments before partial admission.
6. Remove or retry explicitly; keep successful upload references on start failure.

Required guard

Model catalog authority supplies limits; missing capability uses conservative bounds. No fabricated vision/document capability.

Exit and trace

Admitted attachment set or a specific retained-draft refusal.

Original screens: S05, S07.

Current designs: S37, S39.

Sources: V09, V35, V44.

4.6 RESTORED FIRST-REPORT CONTENT · RESTORED END-TO-END JOURNEYS · R7 PAGE 301

R4-F04 / Admit one run and handle ambiguous response

All original steps and guards retained. Current designs preserve the same task and proof boundary.

Entry

Run task click or submit chord.

Interaction

1. Capture immutable goal/options/attachments/session.
2. Claim single-flight before posting.
3. Keep editable draft separate from captured submission.
4. Reuse logical identity for a retry of the same ambiguous request.
5. On durable acceptance clear only the captured unchanged draft/options.
6. On failure keep draft and uploaded refs; focus the useful field.

Required guard

A double click or Ctrl/Cmd+Enter plus click must not start two runs. Session changes invalidate capture rather than redirecting it.

Exit and trace

One admitted run or retryable/refused state.

Original screens: S06, S07, S08.

Current designs: S38, S39, S01.

Sources: V04, V09, V18, V44.

4.6 RESTORED FIRST-REPORT CONTENT · RESTORED END-TO-END JOURNEYS · R7 PAGE 302

R4-F05 / Follow work without losing the editor

All original steps and guards retained. Current designs preserve the same task and proof boundary.

Entry

Active run.

Interaction

1. Show current goal/phase and candidate boundary in the run strip.
2. Append activity to the conversation.
3. Respect manual transcript scroll; offer jump-to-latest without forced jumps.
4. Open Inspect to plan/evidence as needed.
5. Return to Work preserving composer and selection.

Required guard

Activity volume and plan-step completion are not verification. No fictional percent complete.

Exit and trace

Continued run, input need or verification stage.

Original screens: S08, S11, S12.

Current designs: S01, S41, S42.

Sources: V03, V04, V18, V23.

R4-F06 / Decide a permission request

All original steps and guards retained. Current designs preserve the same task and proof boundary.

Entry

One exact pending request.

Interaction

1. Show requester/action/target and authority-provided reason.
2. Open details for capability, IDs and original payload.
3. Allow or Deny applies to that request only.
4. Disable duplicate reply until the result is known.
5. Re-read current pending list.
6. On expired/already-resolved or wrong-session refusal, preserve reason and offer refresh; never silently retry.

Required guard

No inferred purpose such as Needed to verify if the source did not supply it. No Allow always control without policy support.

Exit and trace

Recorded decision or explicit refused/unavailable state.

Original screens: S09, S10, S22.

Current designs: S14, S40, S23.

Sources: V04, V06, V27, V28.

4.6 RESTORED FIRST-REPORT CONTENT · RESTORED END-TO-END JOURNEYS · R7 PAGE 304

R4-F07 / Inspect a failed or missing claim

All original steps and guards retained. Current designs preserve the same task and proof boundary.

Entry

Required failure, pending result or unavailable verdict.

Interaction

1. Open criterion by its stable identity.
2. Show requirement/origin/verdict/basis and candidate.
3. Retrieve evidence attached to that claim.
4. Show truncated preview honestly and full artifact action only when available.
5. Return to the same criterion.
6. Keep failed/pending/unavailable state until new evidence authority changes it.

Required guard

Fail, Pending, Unavailable and Refused stay distinct. An unrelated passing test cannot close the claim.

Exit and trace

Useful evidence or explicit retrieval failure.

Original screens: S16, S17, S18.

Current designs: S45, S46, S47.

Sources: V04, V10, V12, V22, V24, V25, V49.

4.6 RESTORED FIRST-REPORT CONTENT · RESTORED END-TO-END JOURNEYS · R7 PAGE 305

R4-F08 / Review candidate changes in the native editor

All original steps and guards retained. Current designs preserve the same task and proof boundary.

Entry

Candidate changes available.

Interaction

1. Select a changed path under a named candidate.
2. Open immutable before/candidate sides using a proposed host diff adapter.
3. Show source basis and candidate scope near the diff.
4. Navigate files using the host editor.
5. Open related criteria/evidence.
6. If immutable content is unavailable, show diff unavailable rather than compare unrelated live files.

Required guard

Native diff action and immutable-content read seam are proposed. No blanket Accept all or UI-only bypass of integration authority.

Exit and trace

Reviewable diff or explicit missing-basis state.

Original screens: S13, S14, S15.

Current designs: S24, S43, S44.

Sources: V06, V12, V18, V21, V24.

4.6 RESTORED FIRST-REPORT CONTENT · RESTORED END-TO-END JOURNEYS · R7 PAGE 306

R4-F09 / Verify, integrate and finish

All original steps and guards retained. Current designs preserve the same task and proof boundary.

Entry

Verification results and candidate state.

Interaction

1. Show required-check acceptance separately from task proof.
2. Review candidate and any integration conflicts.
3. Let the authoritative runtime integration path operate against its recorded basis.
4. Show landed/verified equality and transaction status.
5. Show each requested commit/push/PR step and its durable outcome.
6. Declare Verified complete only from current proofState verified.

Required guard

Candidate pass is not landed completion. A skipped/missing finish row is not success; native PR requires configured SCM reconciliation. Direct integrate/rebase controls are proposed only where an actual capability seam exists.

Exit and trace

Verified complete or explicit unresolved blocker/mismatch.

Original screens: S18, S19, S20.

Current designs: S47, S48, S65.

Sources: V10, V12, V24, V47, V48, V49.

4.6 RESTORED FIRST-REPORT CONTENT · RESTORED END-TO-END JOURNEYS · R7 PAGE 307

R4-F10 / Stop a run without erasing evidence

All original steps and guards retained. Current designs preserve the same task and proof boundary.

Entry

Current active run.

Interaction

1. Stop current run addresses the displayed run, not an unrelated child or the entire service.
2. Acknowledge cancellation request, not terminal success.
3. Wait for durable cancelled/failed/completed result.
4. Keep conversation, candidate/evidence and draft.
5. Offer review or a new task.

Required guard

Run cancellation, turn abort, child cancellation and service stop are different controls with different scope.

Exit and trace

Terminal run result or still-pending cancellation.

Original screens: S08, S21, S22.

Current designs: S01, S49, S23.

Sources: V06, V18, V19, V44.

4.6 RESTORED FIRST-REPORT CONTENT · RESTORED END-TO-END JOURNEYS · R7 PAGE 308

R4-F11 / Reconnect and recover

All original steps and guards retained. Current designs preserve the same task and proof boundary.

Entry

Disconnected or protocol-blocked stream.

Interaction

1. Distinguish transient retry from a stable unreadable event.
2. Keep the last valid cursor and current view marked stale as needed.
3. Reconnect after the cause is repaired, or refresh authoritative snapshot.
4. Do not advance through an unreadable durable frame.
5. After service restart, preserve session/cursor according to the host service seam.
6. Revalidate proof and cost reads under the new authority.

Required guard

Heartbeat never certifies work or advances the durable cursor. No skip-event button and no blind re-run of unknown external effects.

Exit and trace

Fresh live/read state or explicit ongoing block.

Original screens: S32, S33, S18.

Current designs: S30, S56, S47.

Sources: V13, V19, V39, V44.

4.6 RESTORED FIRST-REPORT CONTENT · RESTORED END-TO-END JOURNEYS · R7 PAGE 309

R4-F12 / Resume prior work

All original steps and guards retained. Current designs preserve the same task and proof boundary.

Entry

History.

Interaction

1. Search title and read actual date/state.
2. Open the selected durable conversation.
3. Rebind workspace/model and refresh authority-bound views.
4. Load messages with proper paging and then live updates.
5. Restore local unsent draft only to the same conversation/workspace identity.
6. Present Completed session state separately from verified run evidence.

Required guard

VS Code visible History is new UI work. Session listing is capped; do not label it all history unless pagination/cap semantics support that.

Exit and trace

Opened conversation with current proof or explicit unavailable fields.

Original screens: S26, S03, S11.

Current designs: S26, S36, S41.

Sources: V06, V12, V26, V44.

4.6 RESTORED FIRST-REPORT CONTENT · RESTORED END-TO-END JOURNEYS · R7 PAGE 310

R4-F13 / Read outcome cost honestly

All original steps and guards retained. Current designs preserve the same task and proof boundary.

Entry

Outcomes or Usage utility.

Interaction

1. Choose whether reading run outcome economics or commercial period usage.
2. Show current metrics with sample/cohort/count and truncation.
3. Distinguish successful-run cost from all-attempt VCSC.
4. Keep local-only, local-final, local-first and fallback cohorts distinct.
5. For no successful denominator display No verified changes yet; preserve numeric zero in data without calling success free.
6. For unavailable reads show no monetary value and a reason.

Required guard

Unmeasured zero recorder fields are not savings claims. A hosted step is not automatically a frontier escalation.

Exit and trace

An interpretable recorded cohort or explicit empty/unavailable state.

Original screens: S27, S28.

Current designs: S53, S54.

Sources: V10, V33, V45, V57.

4.6 RESTORED FIRST-REPORT CONTENT · RESTORED END-TO-END JOURNEYS · R7 PAGE 311

R4-F14 / Configure without changing product independence

All original steps and guards retained. Current designs preserve the same task and proof boundary.

Entry

Settings or model chooser.

Interaction

1. Separate per-run defaults from machine service configuration and optional cloud control plane.
2. Choose models from the advertised catalog.
3. Keep service binary/data/install/args in machine/application scope.
4. Store credentials only in SecretStorage/PasswordSafe.
5. Leave empty cloud endpoint disabled rather than substitute a vendor URL.
6. Explain applies to next conversation/run when current model should not change retroactively.

Required guard

No required Kiwi/Tangerine/KiwiCaptcha step. No project-selected executable and no silent scope escalation.

Exit and trace

Validated configuration or retained form with refusal.

Original screens: S29, S30, S31.

Current designs: S02, S64, S55.

Sources: V02, V06, V07, V17, V32, V43.

4.6 RESTORED FIRST-REPORT CONTENT · RESTORED END-TO-END JOURNEYS · R7 PAGE 312

R4-F15 / Compare candidate approaches

All original steps and guards retained. Current designs preserve the same task and proof boundary.

Entry

Plan / Compare approaches.

Interaction

1. Use one identical criterion set for all candidates.
2. Start bounded 2-4 candidates when budget/policy permits.
3. Show cost/verification/review and unresolved results per candidate.
4. Enable recommendation only after every candidate is settled.
5. Show recommended winner and the separate integration path.
6. Abort names the comparison scope, preserving recorded results.

Required guard

A winner recommendation is not automatic landing or proof. No invented cheaper-by-percent claim.

Exit and trace

Recorded recommendation, abort or unresolved comparison.

Original screens: S24, S19.

Current designs: S51, S48.

Sources: V10, V30, V19.

4.6 RESTORED FIRST-REPORT CONTENT · RESTORED END-TO-END JOURNEYS · R7 PAGE 313

R4-F16 / Resolve platform shell setup

All original steps and guards retained. Current designs preserve the same task and proof boundary.

Entry

Shell/verification fail-closed platform refusal.

Interaction

1. Show that current fail-closed policy blocked the command.
2. Explain absence of an OS network-isolation backend.
3. Open exact CLI/config setup guidance.
4. Preserve the secure default until the operator explicitly changes it.
5. If a future native grant control is built, record/display the exact weaker-network trade-off and authority response.

Required guard

Current backend choice is made by faktor sandbox choose or explicit config; a new IDE write route is not currently established.

Exit and trace

Guidance only, or explicitly recorded operator choice from real configuration.

Original screens: S34, S32.

Current designs: S57, S30.

Sources: V46.

4.6 RESTORED FIRST-REPORT CONTENT · SOURCE CONFLICT RECONCILIATION · R7 PAGES 314, 315, 316, 317

Dated findings and their current disposition

The original findings remain visible; their older implementation-status assertions are not silently reused.

R4-D01 / Service discovery and implicit startup changed

Prior: Repository-relative executable discovery and optional activation auto-start were present in prior sources.

Current: Application/machine settings, environment or signed install layout only; explicit action starts the process.

Design: Add precise stopped/missing/unverified/refused/start/readiness screens. Do not show workspace target build fallback.

17 October disposition: Keep explicit start and machine/application executable authority. Opening the view is not service activation.

Historical trace: V06, V07, V16, V17, V38, V51.

R4-D02 / VCSC changed

Prior: Prior definition measured spend only over verified successful runs.

Current: VCSC includes eligible spend across all attempted work, failed and cancelled included, divided by verified changes. Successful-run cost remains distinct.

Design: Retain old definition as dated history and add current correction; use updated field names.

17 October disposition: Use all eligible attempted spend, including eligible failed/cancelled attempts, divided by verified landed changes; zero denominator is unavailable.

Historical trace: V45, V57.

R4-D03 / Local route economics changed

Prior: One local-substitution rate counted runs with any local step.

Current: Local-only, local-final, local-first success and local-to-hosted fallback are separate cohorts.

Design: A local-then-hosted repair is fallback, not successful purely local substitution. Explain denominators.

17 October disposition: Separate observed local route, actual hosted fallback, unknown pricing and unmeasured local operating cost.

Historical trace: V45, V57.

R4-D04 / New explicit platform shell choice

Prior: No first_run.rs workflow in the prior baseline.

Current: Fail-closed default and a durable explicit network-capable grant with weaker guarantee.

Design: Setup guidance is a real user need. Do not promise all platform shells are OS isolated.

17 October disposition: Reuse the real fail-closed/native first-run shell choice. A new native editor must not imply stronger isolation than the platform establishes.

Historical trace: V46.

R4-D05 / JetBrains README route/stream prose is stale

Current conflict: README routing table lists older /session/create, /session/list and /api/session events; stream paragraph says heartbeats advance and oversized events skip. Current clients use native routes, do not advance heartbeat cursor and block malformed durable events.

Design: Use actual current NativeClient/NativeEventStream paths and recovery semantics; preserve README excerpt only as historical/source-conflict note.

17 October disposition: Use the pinned source routes and durable stream behavior, not stale README prose. Keep exact protocol refusal and cursor recovery.

Historical trace: V52, V39, V40, V44.

R4-D06 / PR execution prose is stale

Current conflict: Older certification paragraph describes PR command execution only; current runner and README use canonical typed native SCM reconciliation and a missing-SCM blocker.

Design: Show requested PR blocked when configuration is absent; generic command success is not native PR proof.

17 October disposition: A PR request uses native SCM configuration and durable step reconciliation; arbitrary shell success does not replace it.

Historical trace: V47, V50, V58.

R4-D07 / Finish controls differ between hosts

Current conflict: VS Code four hierarchical radio levels; JetBrains three independent booleans.

Design: Proposed shared four-level presentation explicitly maps implicit earlier steps; label as a design change and test normalized envelope.

17 October disposition: The reviewed 17 October source has four finish levels in both hosts. The older JetBrains checkbox difference is historical.

Historical trace: V04, V18.

R4-D08 / History product vocabulary exceeds VS Code visible UI

Current conflict: Terminology describes Work/Inspect/History; current VS Code manifest/webview exposes single Chat with Inspect disclosure, no visible History switcher.

Design: Add History UI and host session-switch action rather than depict it as existing.

17 October disposition: The reviewed 17 October source has Work, Inspect and History in both hosts. Visible History is no longer an unbuilt basic feature.

Historical trace: V01, V02, V03, V06, V12, V26.

R4-D09 / Native diff review is absent from current inspected Changes surface

Current conflict: Changed file list and evidence refs exist; native editor diff opening and immutable content navigation are not wired in inspected host dispatch/ChangesPanel.

Design: Provide explicit host adapter/immutable read seam; prototype diff screens remain proposed.

17 October disposition: The reviewed 17 October source has native diff opening in both hosts. Preserve immutable-content correctness and its unavailable state.

Historical trace: V06, V12, V18, V21.

R4-D10 / Checksum wording conflates two authorities

Current conflict: Bare .sha256 matching code performs hash comparison while some strings call it a signed checksum record. The install bootstrap separately verifies signed activation/digest.

Design: Use Matches recorded checksum / No checksum record / Checksum mismatch for bare files. Keep signed install verification wording scoped to launcher.

17 October disposition: Recorded checksum, missing checksum and publisher-signature verification are separate authorities and labels.

Historical trace: V07, V38, V51.

R4-D11 / Some machine-scope claims are broader than visible reads

Current conflict: VS Code package declares startupTimeout/control-plane coordinates machine scoped; extension uses generic config reads for those fields while binary/data/install/args go through daemonSetting inspect refusal.

Design: Use one machine-scoped read/validation adapter consistently before presenting guaranteed refusal parity. Review setting descriptions against actual runtime reads.

17 October disposition: Keep machine/application authority explicit; native project-scoped preference editing needs a real new scope adapter where absent.

Historical trace: V02, V06, V07.

R4-D12 / JetBrains evidence references can lose buttons in one row

Current conflict: VerificationPanel.criterionCard adds every evidence button at BorderLayout.WEST, so later buttons replace earlier placements.

Design: Use a wrapping action row or evidence list; retain every served reference and original identity.

17 October disposition: Use intrinsic evidence rows, wrapping action groups and one selected detail. Never distribute evidence buttons by manual spaces.

Historical trace: V22.

R4-D13 / JetBrains overview can collapse missing proof/cost into weaker wording

Current conflict: Overview verification uses not verified; state fallback may tint completed/done positive without full proof; missing spend says No spend recorded yet.

Design: Preserve proof unavailable as explicit current state and distinguish absent/unavailable/known zero money. Neutral completed session/run without verified proof.

17 October disposition: Do not collapse missing proof, unknown cost, pending finish or identity mismatch into a generic Done or zero-cost presentation.

Historical trace: V53, V24.

DATED FINDINGS AND THEIR CURRENT DISPOSITION · CONTINUED

R4-D14 / Approvals purpose is partly inferred

Current conflict: VS Code generic card says Needed to verify based on a target even when original payload may not state that purpose.

Design: Show authority-provided reason if present; otherwise Action requested for this run, plus exact payload.

17 October disposition: Show the served reason when present. Otherwise say Action requested for this run with the exact payload; never infer a reassuring purpose.

Historical trace: V04, V27.

4.6 RESTORED FIRST-REPORT CONTENT · RESTORED CONSTRUCTION TARGETS · R7 PAGES 318, 319, 320, 321, 322

File-level implementation targets

Full original file targets remain searchable in the editable specification.

R4-I01 / Shared Work/Inspect/History shell, VS Code visible History and host-specific adapters

Priority: P1

VS Code: apps/vscode/src/webview.ts, apps/vscode/media/chat.js, apps/vscode/media/chat.css, apps/vscode/src/extension.ts, apps/vscode/package.json JetBrains:

apps/jetbrains/frontend/src/main/kotlin/dev/faktor/frontend/FaktorChatPanel.kt, apps/jetbrains/frontend/src/main/kotlin/dev/faktor/frontend/FaktorToolWindowFa ctory.kt, apps/jetbrains/frontend/src/main/kotlin/dev/faktor/frontend/HistoryPanel.kt

Authority/backend: Existing session/read/run endpoints; add host open/switch action and scoped state handling.

12 October classification: proposed_ui_existing_backend.

Current disposition: integrate with the reviewed native features and required BK/UI packages; this older classification is historical.

R4-I02 / Normalize task options, hierarchical finish levels and draft capture

Priority: P1

VS Code: apps/vscode/src/taskStart.ts, apps/vscode/media/chat.js, apps/vscode/src/contract.ts JetBrains:

apps/jetbrains/frontend/src/main/kotlin/dev/faktor/frontend/FaktorChatPanel.kt, apps/jetbrains/shared/src/main/kotlin/dev/faktor/shared/NativeProtocol.kt

Authority/backend: Existing strict task admission and NativeCompletionContract; UI changes must map exact booleans.

12 October classification: proposed_ui_existing_backend.

Current disposition: integrate with the reviewed native features and required BK/UI packages; this older classification is historical.

R4-I03 / Native diff review with immutable source basis

Priority: P1

VS Code: apps/vscode/src/extension.ts, apps/vscode/src/nativeClient.ts, apps/vscode/media/chat.js JetBrains:

apps/jetbrains/frontend/src/main/kotlin/dev/faktor/frontend/ChangesPanel.kt, apps/jetbrains/frontend/src/main/kotlin/dev/faktor/frontend/FaktorToolWindowFa ctory.kt, apps/jetbrains/frontend/src/main/kotlin/dev/faktor/frontend/FaktorFrontendServi ce.kt, apps/jetbrains/backend/src/main/kotlin/dev/faktor/backend/NativeClient.kt

Authority/backend: Need authenticated bounded before/candidate content or artifact read contract; do not use unrelated live contents. New adapter files can isolate host-specific native diff APIs.

12 October classification: proposed_host_integration.

Current disposition: integrate with the reviewed native features and required BK/UI packages; this older classification is historical.

R4-I04 / Separate proof/acceptance/integration/finish and preserve unavailable fields

Priority: P0

VS Code: apps/vscode/src/cockpit.ts, apps/vscode/src/state.ts, apps/vscode/src/nativeClient.ts, apps/vscode/media/chat.js JetBrains:

apps/jetbrains/frontend/src/main/kotlin/dev/faktor/frontend/TaskTreeModel.kt, apps/jetbrains/frontend/src/main/kotlin/dev/faktor/frontend/OverviewPanel.kt, apps/jetbrains/frontend/src/main/kotlin/dev/faktor/frontend/VerificationPanel.kt, apps/jetbrains/frontend/src/main/kotlin/dev/faktor/frontend/TaskTreePanel.kt

Authority/backend: Existing task proof, completion steps and integration transaction fields.

12 October classification: presentation_refinement.

Current disposition: integrate with the reviewed native features and required BK/UI packages; this older classification is historical.

FILE-LEVEL IMPLEMENTATION TARGETS · CONTINUED

R4-I05 / Contextual approvals, exact reason and no duplicate replies

Priority: P1

VS Code: apps/vscode/media/chat.js, apps/vscode/src/extension.ts, apps/vscode/src/errors.ts JetBrains:

apps/jetbrains/frontend/src/main/kotlin/dev/faktor/frontend/PermissionsPanel.kt, apps/jetbrains/frontend/src/main/kotlin/dev/faktor/frontend/BlockersPanel.kt, apps/jetbrains/frontend/src/main/kotlin/dev/faktor/frontend/FaktorChatPanel.kt

Authority/backend: Existing session-scoped permission reply with typed expired/wrong-session refusals.

12 October classification: proposed_ui_existing_backend.

Current disposition: integrate with the reviewed native features and required BK/UI packages; this older classification is historical.

R4-I06 / Human error/recovery parity, readiness and startup trust flows

Priority: P1

VS Code: apps/vscode/src/errors.ts, apps/vscode/src/daemon.ts, apps/vscode/src/extension.ts, apps/vscode/src/eventStream.ts JetBrains:

apps/jetbrains/frontend/src/main/kotlin/dev/faktor/frontend/FaktorDaemonSettin gs.kt, apps/jetbrains/frontend/src/main/kotlin/dev/faktor/frontend/DiagnosticsPanel.kt, apps/jetbrains/frontend/src/main/kotlin/dev/faktor/frontend/FaktorChatPanel.kt, apps/jetbrains/frontend/src/main/kotlin/dev/faktor/frontend/FaktorFrontendServi ce.kt, apps/jetbrains/backend/src/main/kotlin/dev/faktor/backend/DaemonBinary.kt, apps/jetbrains/backend/src/main/kotlin/dev/faktor/backend/NativeEventStream.k t

Authority/backend: Existing native health/ready; exact confirmation, typed refusal and cursor preservation. Native UI prompts must run on host UI thread; service/network work off it.

12 October classification: proposed_host_integration.

Current disposition: integrate with the reviewed native features and required BK/UI packages; this older classification is historical.

R4-I07 / Current VCSC/local cohort outcome analytics; no unavailable-as-zero

Priority: P1

VS Code: apps/vscode/src/nativeClient.ts, apps/vscode/src/cockpit.ts, apps/vscode/src/state.ts, apps/vscode/src/money.ts JetBrains:

apps/jetbrains/backend/src/main/kotlin/dev/faktor/backend/NativeClient.kt, apps/jetbrains/frontend/src/main/kotlin/dev/faktor/frontend/FaktorFrontendServi ce.kt, apps/jetbrains/frontend/src/main/kotlin/dev/faktor/frontend/HistoryPanel.kt, apps/jetbrains/frontend/src/main/kotlin/dev/faktor/frontend/UsagePanel.kt, apps/jetbrains/shared/src/main/kotlin/dev/faktor/shared/NativeProtocol.kt

Authority/backend: docs/outcome-ledger.md; crates/agent/src/outcome_ledger.rs; crates/protocol/schema/faktor-protocol.schema.json

12 October classification: proposed_ui_existing_backend.

Current disposition: integrate with the reviewed native features and required BK/UI packages; this older classification is historical.

R4-I08 / Native settings scope and optional control-plane separation

Priority: P1

VS Code: apps/vscode/package.json, apps/vscode/src/extension.ts, apps/vscode/src/daemon.ts, apps/vscode/src/controlPlaneAuth.ts JetBrains:

apps/jetbrains/frontend/src/main/kotlin/dev/faktor/frontend/SettingsPanel.kt, apps/jetbrains/frontend/src/main/kotlin/dev/faktor/frontend/FaktorDaemonSettin gs.kt, apps/jetbrains/frontend/src/main/kotlin/dev/faktor/frontend/FaktorToolWindowFa ctory.kt, apps/jetbrains/frontend/src/main/resources/META-INF/plugin.xml

Authority/backend: Current default-empty cloud endpoint; machine binary config; secret storage separate. Proposed platform Configurable/action adapter.

12 October classification: proposed_host_integration.

Current disposition: integrate with the reviewed native features and required BK/UI packages; this older classification is historical.

FILE-LEVEL IMPLEMENTATION TARGETS · CONTINUED

R4-I09 / Narrow-width/focus/semantic evidence component repairs

Priority: P1

VS Code: apps/vscode/media/chat.css, apps/vscode/media/chat.js, apps/vscode/src/webview.ts JetBrains:

apps/jetbrains/frontend/src/main/kotlin/dev/faktor/frontend/PanelSupport.kt, apps/jetbrains/frontend/src/main/kotlin/dev/faktor/frontend/FaktorChatPanel.kt, apps/jetbrains/frontend/src/main/kotlin/dev/faktor/frontend/VerificationPanel.kt, apps/jetbrains/frontend/src/main/kotlin/dev/faktor/frontend/EvidenceNavigatorP anel.kt, apps/jetbrains/frontend/src/main/kotlin/dev/faktor/frontend/HistoryPanel.kt

Authority/backend: No backend change; avoid state inference or evidence loss during redraw.

12 October classification: presentation_refinement.

Current disposition: integrate with the reviewed native features and required BK/UI packages; this older classification is historical.

R4-I10 / Sandbox first-run guidance and capability-controlled future grant UI

Priority: P2

VS Code: apps/vscode/src/errors.ts, apps/vscode/src/extension.ts, apps/vscode/media/chat.js JetBrains:

apps/jetbrains/frontend/src/main/kotlin/dev/faktor/frontend/DiagnosticsPanel.kt, apps/jetbrains/frontend/src/main/kotlin/dev/faktor/frontend/FaktorChatPanel.kt, apps/jetbrains/frontend/src/main/kotlin/dev/faktor/frontend/FaktorFrontendServi ce.kt

Authority/backend: crates/cli/src/config/first_run.rs

12 October classification: guidance_existing_backend; native choice write control deferred until capability is established.

Current disposition: integrate with the reviewed native features and required BK/UI packages; this older classification is historical.

4.6 RESTORED FIRST-REPORT CONTENT · DATED SOURCE INVENTORY · R7 PAGES 323, 324

Original host inventory

Preserved verbatim as a dated baseline. The 17 October current-source corrections take precedence.

VSCODE / current / 12 October

Placement: Faktor activity-bar container; one custom webview view faktor.chat named Chat. Users may relocate the container/view.

Navigation: Conversation/current run/composer are primary. Inspect and diagnostics overflow disclose the rest. There is no current visible History destination.

Composer: One task goal, Run task, New task, attachment chips, Finish when done radios and Acceptance & contract disclosure. Ctrl/Cmd+Enter sends; Enter inserts newline.

Finish: Local / Commit / Push / PR hierarchy: later choices explicitly include earlier actions.

Utilities: Native settings and commands plus webview Diagnostics; status-bar entry opens Faktor.

Theme: VS Code foreground/sidebar/button/focus/editor tokens and host fonts; static measured verdict-pair exception.

VSCODE / proposed / 12 October

Placement: Retain one Faktor container and one default Work surface. Honor moves to Secondary Sidebar or Panel; do not reopen or force a particular side.

Navigation: A compact Work / Inspect / History switcher; Inspect subviews Overview / Plan / Changes / Verification / Agents, with same scope breadcrumb.

Composer: Run task primary, Ask as an explicit alternate mode when useful; one composer. Task options disclose criteria, contract, finish, attachments and per-run budget.

Finish: Keep four hierarchical levels with their actual included actions described.

Utilities: Native Settings for persistent configuration; settings/model quick picks for workspace-aware choices; Diagnostics/Usage via overflow.

Additions status: History navigation, full Inspect subgrouping, native diff integration, per-run budget sheet and run-outcomes view are proposals, not current shipped UI.

JETBRAINS / current / 12 October

Placement: Plugin toolWindow id Faktor is anchored right in plugin.xml; content is FaktorChatPanel, native Swing frontend.

Navigation: Work / Inspect / History already exist. Inspect: Overview / Plan /

Changes / Verification / Agents. Agents: Roster / Approvals / Coordination.

Composer: One input, Run task primary, Ask secondary, Abort, Run options disclosure. Acceptance criteria are comma-separated. Four contract fields are line-based.

Finish: Three independent Commit / Push / Create PR checkboxes; not the VS Code hierarchical radio-level UI.

Utilities: Toolbar New session/Refresh/Settings/More. Settings, Usage and Diagnostics replace clusters with an explicit Back action.

Theme: UIManager-derived reading fonts/surfaces; PanelSupport provides a theme-adjusted teal accent and rounded 8/6 px cards/controls.

JETBRAINS / proposed / 12 October

Placement: Retain right tool window as default and respect docking/floating/moving. Native host action toolbar; use editor area for wide native diffs.

Navigation: Keep the three clusters; make Inspect drill-down shallower and replace stacked equal splits with one selected detail surface.

Composer: One composer, clear Task/Ask mode, run summary strip and wrapped chips. Task options open in a bounded scrollable host sheet; label any new mode control as design addition.

Finish: Unify with four hierarchical levels; map to NativeCompletionContract explicitly. This changes current presentation and must be implemented deliberately.

Utilities: Use platform Settings/Configurable and action APIs where possible; current standalone Swing controls remain a fallback, not a second product workflow.

Additions status: Native diff opening, normalized finish selector, IDE keymap registrations, richer outcome history and adapter-driven configuration are proposed integrations.

4.6 RESTORED FIRST-REPORT CONTENT · RESTORED RESPONSIVE CONTRACT · R7 PAGE 325

Original responsive requirements

These requirements remain mandatory alongside the refined intrinsic-layout system.

240-319 px content

Single column. Name/current phase with Inspect and overflow; top clusters as a compact row. Labels above fields, full-width evidence and finish rows. No fixed key column.

Actions: Primary and stop/deny remain visible; secondary utilities in a labeled overflow.

Diff: Open in native editor, never squeeze side-by-side content into tool window.

320-479 px content

Compact run strip and wrapped metadata; one selected Inspect subview; options scroll inside a bounded area.

Actions: Primary row wraps; two action rows maximum before overflow.

480-719 px content

Useful label/value rows, evidence summary plus selected detail, single-column transcript; tab/subview navigation text visible.

Actions: Frequent contextual actions exposed, advanced controls disclosed.

720+ px panel/floating area

Optional plan/list master-detail with user-adjustable splitter. Work conversation remains focal.

Actions: More room never introduces unrelated dashboards.

Zoom

Reflow at 100%, 150% and 200% IDE UI scale plus large user reading fonts. No clipped titles, raw hashes as primary labels or horizontal main-page action toolbar.

Short height

Keep composer reachable; collapse current-run readout to a summary strip, open full options as a scrollable sheet; do not stack three equal nested scroll panes.

Paths and code

Human labels/paths wrap. Code/diff may use explicit contained horizontal scrolling with copy/full view; never ellipsize the sole meaning.

Brand boundary

Native host reading/editor fonts and focus/theme tokens take precedence over marketing Barlow/Ochre. The fixed angular F identifies the product; no circular/koru ornament.

4.6 RESTORED FIRST-REPORT CONTENT · RESTORED INTERACTION INVARIANTS · R7 PAGES 326, 327

Keyboard, focus and assistive behavior

All original focus rules remain, including IME, lost-item focus and bounded live announcements.

Current / vscode / 12 October

Enter inserts newline; Ctrl/Cmd+Enter starts task with IME composition guard. Native disclosure/controls are focusable and async transcript/approval rebuild tries to preserve exact focused item.

Current / jetbrains / 12 October

Ctrl+Enter in input starts task. Default Swing tabs/lists/controls have native navigation; app-level IDE keymap registration for full proposed workflow is not present in current plugin.xml.

Open Faktor

User-remappable host command/action; avoid an assumed unclaimed global shortcut.

Send current task

Ctrl+Enter on Windows/Linux; Cmd+Enter on macOS if platform keymap permits. Enter remains newline; IME composition never submits.

Move between clusters

Host tab semantics/arrow keys while tablist focused; one active panel and stable selection.

Move between plan/history/evidence rows

Arrow navigation; Enter opens selected detail; Space activates check/radio where native component expects it.

Close transient detail/options

Escape returns to exact trigger; not a run cancellation shortcut.

Stop run

Explicit scoped command/action with visible label; never overload Escape.

Focus contract

Async data refresh preserves the same logical row/button, not just the same text label.

A new approval announces without moving focus away from typing.

Opening a detail moves focus to its heading or first useful control.

Closing returns to its original claim/path/request/trigger.

If an item disappears, focus goes to its nearest stable sibling or section heading.

Submitting clears only acknowledged captured draft; failure focuses the retained draft/invalid field.

Background output never replaces current selection or repeatedly re-announces the transcript.

Accessibility contract

Text+glyph for every semantic state.

Theme-derived text/focus tokens; no color-only success or failure.

Accurate labels, roles, expanded/selected/disabled states.

Native keyboard traversal; no webview/Swing focus trap.

Reduced motion keeps static state cues; screen-reader mode avoids ornamental animation.

New requests and final outcomes announced once; streaming prose uses bounded polite updates.

4.6 RESTORED FIRST-REPORT CONTENT · RESTORED ACCEPTANCE REQUIREMENTS · R7 PAGE 328

Original validation plan

These are required integration scenarios; no product-runtime certification is claimed by this design artifact.

Submission

Double click/chord race admits one logical run.

Captured draft survives ambiguous response and retry uses same identity. Typed invalid input refuses without silently dropping contract/attachments. Session switch cannot redirect old draft.

Proof

No verified complete with missing required check, unavailable claim, stale snapshot or incomplete finish.

A zero-required-check Pass reducer does not automatically create proof. Failed proof read replaces stale verified presentation with unavailable.

Money

Known zero, absent value, unavailable read and no denominator are visually distinct.

VCSC includes failed/cancelled eligible attempt spend.

Local-to-hosted fallback never counted as purely local success.

Large microUSD strings remain exact; display no scientific notation.

Approvals

Every Allow/Deny targets exact session/request.

Duplicate reply is gated.

Expired/wrong-session refusal remains visible and is not retried implicitly. No fabricated purpose or persistent policy control.

Recovery

Transient retry differs from stable protocol block.

Cursor never passes malformed durable event.

Restart preserves authority scope correctly and stale proof/cost is not reused.

No blind repeat of unknown external action.

Host UI

Both hosts have light/dark/high-contrast and large-font specimens.

240/320/420/720+ content widths keep all decisions reachable.

Keyboard/IME/focus restoration and screen-reader text states are meaningful.

Native diff opens proper immutable basis; unavailable basis never opens misleading comparison.

Opening tool window does not launch the service.

Delivery claims

Each new control labeled as proposed design unless actually present.

Current-source facts pinned and prior material preserved with dated addendum.

Generated/prototype IDE scenes are not described as runtime screenshots or certification.

Manual mobile hardware testing is not reintroduced.

4.6 RESTORED FIRST-REPORT CONTENT · RESTORED ILLUSTRATIVE-STATE CONTRACT · R7 PAGE 329

Unresolved and completed examples stay distinct

The incomplete refresh-token example must not turn green merely because its illustration advances.

Label

Design prototype - proposed interface. Illustrative data; no live workspace/model/service connection.

Required unresolved example

{ "goal": "Rotate refresh tokens without breaking active sessions.", "criteria": [{ "label": "Replayed refresh tokens fail", "state": "Pass" }, { "label": "Existing sessions remain valid", "state": "Pass" }, { "label": "Concurrent refresh yields one valid result", "state": "Needs evidence" }], "summary": "2 criteria passed · 1 needs evidence", "candidate": "Isolated candidate", "integration": "Not integrated", "finish": "Not started", "cost": "Unavailable in this illustration", "global_verified_complete": false }

Sample complete example

{ "label": "Separate illustrative completed run", "proof": "Verified complete", "integration": "Landed state matches verified state", "finish": "Only requested durable succeeded steps shown", "copy_warning": "Do not morph unresolved example into completion without explicitly changing the served evidence/state in the illustration." }

Scene mapping

ready: R4-S03 / current S36 task_options: R4-S04 / current S33 active_run: R4-S08 / current S01 approval: R4-S09 / current S14 plan: R4-S12 / current S42 diff_review: R4-S14 / current S43 evidence: R4-S16 / current S45 completion: R4-S19 / current S48 error_retry: R4-S07 / current S39 history: R4-S26 / current S26 settings: R4-S30 / current S64 diagnostics: R4-S32 / current S30

4.7 ORIGINAL-TO-REVISED RETENTION MAP · ORIGINAL-TO-REVISED COVERAGE · R7 PAGES 330, 331, 332, 333, 334, 335

Content retention map

Stable source namespaces keep old screen IDs distinct from the redesigned screen IDs. Revision 8 resolves every destination to a page in this document.

R4-S01 / First open / no project

Destination: R7 page 006 → R8 p. 22 / S34 Restored as a dedicated current design and full dated source contract

R4-S02 / Service discovery and trust

Destination: R7 page 007 → R8 p. 25 / S35 Restored as a dedicated current design and full dated source contract

R4-S03 / Ready conversation

Destination: R7 page 009 → R8 p. 29 / S36 Restored as a dedicated current design and full dated source contract

R4-S04 / Task options

Destination: R7 page 010 → R8 p. 31 / S33 Restored as a dedicated current design and full dated source contract

R4-S05 / Attachments

Destination: R7 page 011 → R8 p. 33 / S37 Restored as a dedicated current design and full dated source contract

R4-S06 / Submission in flight

Destination: R7 page 012 → R8 p. 35 / S38 Restored as a dedicated current design and full dated source contract

R4-S07 / Ambiguous start / retry

Destination: R7 page 013 → R8 p. 37 / S39 Restored as a dedicated current design and full dated source contract

R4-S08 / Active run

Destination: R7 page 014 → R8 p. 40 / S01 Restored as a dedicated current design and full dated source contract

R4-S09 / Contextual approval

Destination: R7 page 015 → R8 p. 42 / S14 Restored as a dedicated current design and full dated source contract

R4-S10 / Needs input / blocker

Destination: R7 page 016 → R8 p. 44 / S40 Restored as a dedicated current design and full dated source contract

R4-S11 / Inspect overview

Destination: R7 page 020 → R8 p. 55 / S41 Restored as a dedicated current design and full dated source contract

R4-S12 / Plan

Destination: R7 page 021 → R8 p. 57 / S42 Restored as a dedicated current design and full dated source contract

R4-S13 / Changes list

Destination: R7 page 022 → R8 p. 59 / S24 Restored as a dedicated current design and full dated source contract

R4-S14 / Native diff review

Destination: R7 page 023 → R8 p. 61 / S43 Restored as a dedicated current design and full dated source contract

R4-S15 / Integration conflict

Destination: R7 page 027 → R8 p. 69 / S44 Restored as a dedicated current design and full dated source contract

R4-S16 / Verification criteria

Destination: R7 page 024 → R8 p. 63 / S45 Restored as a dedicated current design and full dated source contract

R4-S17 / Evidence detail

Destination: R7 page 025 → R8 p. 65 / S46 Restored as a dedicated current design and full dated source contract

R4-S18 / Full proof and mismatch

Destination: R7 page 026 → R8 p. 67 / S47 Restored as a dedicated current design and full dated source contract

CONTENT RETENTION MAP · CONTINUED

R4-S19 / Finish actions

Destination: R7 page 028 → R8 p. 71 / S48 Restored as a dedicated current design and full dated source contract

R4-S20 / Verified result

Destination: R7 page 030 → R8 p. 75 / S65 Restored as a dedicated current design and full dated source contract

R4-S21 / Cancelled or failed result

Destination: R7 page 032 → R8 p. 79 / S49 Restored as a dedicated current design and full dated source contract

R4-S22 / Agents roster

Destination: R7 page 033 → R8 p. 82 / S23 Restored as a dedicated current design and full dated source contract

R4-S23 / Agent detail / steer

Destination: R7 page 034 → R8 p. 84 / S50 Restored as a dedicated current design and full dated source contract

R4-S24 / Compare approaches

Destination: R7 page 035 → R8 p. 86 / S51 Restored as a dedicated current design and full dated source contract

R4-S25 / Coordination

Destination: R7 page 036 → R8 p. 88 / S52 Restored as a dedicated current design and full dated source contract

R4-S26 / Prior conversations

Destination: R7 page 037 → R8 p. 91 / S26 Restored as a dedicated current design and full dated source contract

R4-S27 / Run outcome analytics

Destination: R7 page 041 → R8 p. 101 / S53 Restored as a dedicated current design and full dated source contract

R4-S28 / Usage and quota

Destination: R7 page 042 → R8 p. 103 / S54 Restored as a dedicated current design and full dated source contract

R4-S29 / Model and provider selection

Destination: R7 page 044 → R8 p. 108 / S02 Restored as a dedicated current design and full dated source contract

R4-S30 / Settings

Destination: R7 page 071 → R8 p. 167 / S64 Restored as a dedicated current design and full dated source contract

R4-S31 / Optional control-plane credentials

Destination: R7 page 073 → R8 p. 171 / S55 Restored as a dedicated current design and full dated source contract

R4-S32 / Diagnostics

Destination: R7 page 074 → R8 p. 173 / S30 Restored as a dedicated current design and full dated source contract

R4-S33 / Protocol blocked recovery

Destination: R7 page 075 → R8 p. 175 / S56 Restored as a dedicated current design and full dated source contract

R4-S34 / Sandbox platform setup

Destination: R7 page 008 → R8 p. 27 / S57 Restored as a dedicated current design and full dated source contract

R4-S35 / Owned command output

Destination: R7 page 066 → R8 p. 156 / S19 Restored as a dedicated current design and full dated source contract

R4-S36 / Narrow tool window

Destination: R7 page 077 → R8 p. 179 / S58 Restored as a dedicated current design and full dated source contract

R4/state_inventory / state inventory

Destination: R7 page 292 → R8 p. 302 / restored-state_inventory Complete original values retained; current-state corrections are explicit

R4/flows / flows

Destination: R7 page 298 → R8 p. 307 / restored-flow-F01 Complete original values retained; current-state corrections are explicit

CONTENT RETENTION MAP · CONTINUED

R4/source_changes_and_conflicts / source changes and conflicts

Destination: R7 page 314 → R8 p. 323 / restored-source_changes_and_conflicts Complete original values retained; current-state corrections are explicit

R4/implementation_targets / implementation targets

Destination: R7 page 318 → R8 p. 326 / restored-implementation_targets Complete original values retained; current-state corrections are explicit

R4/validation_plan / validation plan

Destination: R7 page 328 → R8 p. 332 / restored-validation_plan Complete original values retained; current-state corrections are explicit

R4/responsive / responsive

Destination: R7 page 325 → R8 p. 330 / restored-responsive Complete original values retained; current-state corrections are explicit

R4/keyboard_and_accessibility / keyboard and accessibility

Destination: R7 page 326 → R8 p. 331 / restored-keyboard_and_accessibility Complete original values retained; current-state corrections are explicit

R4/prototype_contract / prototype contract

Destination: R7 page 329 → R8 p. 333 / restored-prototype_contract Complete original values retained; current-state corrections are explicit

R4/host_inventory / host inventory

Destination: R7 page 323 → R8 p. 329 / restored-host_inventory Complete original values retained; current-state corrections are explicit

R4/sources / sources

Destination: references Complete original values retained; current-state corrections are explicit

R4/official_sources / official sources

Destination: references Complete original values retained; current-state corrections are explicit

R5/M / Models and routing

Destination: R7 page 223 → R8 p. 257 / catalog-M 14 controls, all original default/scope/timing/behavior/source values retained

R5/P / Providers, credentials and local inference

Destination: R7 page 227 → R8 p. 260 / catalog-P 15 controls, all original default/scope/timing/behavior/source values retained

R5/G / Reasoning and generation

Destination: R7 page 231 → R8 p. 263 / catalog-G 14 controls, all original default/scope/timing/behavior/source values retained

R5/A / Agent presets and delegation

Destination: R7 page 235 → R8 p. 266 / catalog-A 12 controls, all original default/scope/timing/behavior/source values retained

R5/C / Context, attachments and retrieval

Destination: R7 page 238 → R8 p. 268 / catalog-C 14 controls, all original default/scope/timing/behavior/source values retained

R5/K / Compaction, memory and efficiency

Destination: R7 page 242 → R8 p. 271 / catalog-K 12 controls, all original default/scope/timing/behavior/source values retained

R5/I / Instructions, rules and prompt composition

Destination: R7 page 246 → R8 p. 273 / catalog-I 11 controls, all original default/scope/timing/behavior/source values retained

R5/Q / Permissions, autonomy and sandbox

Destination: R7 page 249 → R8 p. 275 / catalog-Q 14 controls, all original default/scope/timing/behavior/source values retained

R5/T / Tool catalog and MCP

Destination: R7 page 253 → R8 p. 278 / catalog-T 15 controls, all original default/scope/timing/behavior/source values retained

R5/S / Skills, commands and hooks

Destination: R7 page 257 → R8 p. 281 / catalog-S 14 controls, all original default/scope/timing/behavior/source values retained

R5/X / Terminal, browser and language tooling

Destination: R7 page 261 → R8 p. 284 / catalog-X 13 controls, all original default/scope/timing/behavior/source values retained

CONTENT RETENTION MAP · CONTINUED

R5/B / Budgets, retries and reliability

Destination: R7 page 265 → R8 p. 286 / catalog-B 12 controls, all original default/scope/timing/behavior/source values retained

R5/V / Changes, checkpoints, verification and finish

Destination: R7 page 269 → R8 p. 288 / catalog-V 12 controls, all original default/scope/timing/behavior/source values retained

R5/D / Sessions, displays and accessibility

Destination: R7 page 273 → R8 p. 290 / catalog-D 15 controls, all original default/scope/timing/behavior/source values retained

R5/O / Privacy, diagnostics, optional cloud and distribution

Destination: R7 page 278 → R8 p. 293 / catalog-O 13 controls, all original default/scope/timing/behavior/source values retained

R5/narrative / narrative

Destination: R7 page 170 → R8 p. 212 / brief Complete second-report content retained; reference code files retained in full

R5/flows / flows

Destination: R7 page 184 → R8 p. 183 / flows Complete second-report content retained; reference code files retained in full

R5/construction / construction

Destination: R7 page 192 → R8 p. 226 / build-mandate Complete second-report content retained; reference code files retained in full

R5/packages / packages

Destination: R7 page 193 → R8 p. 227 / backend-map Complete second-report content retained; reference code files retained in full

R5/acceptance / acceptance

Destination: acceptance Complete second-report content retained; reference code files retained in full

Faktor/identity / Independent visual identity

Destination: retained-identity All 42 original pages retained in the PDF and exact source archive

Faktor/website / Complete independent website

Destination: retained-identity All 57 original pages retained after the identity appendix, with exact source archive

4.7 RETENTION MAP · PAGE INDEX

Every revision 7 page, located in revision 8

All 451 pages of revision 7 map to a page in this document. Screen designs map to their redesigned spread; contracts to the screen's contract page; appendices to their framed original.

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4.8 ACCEPTANCE SCENARIOS · R7 PAGES 336–339

Acceptance A01 – A40

Behavioral requirements for product integration. These scenarios were not executed against a Faktor release.

A01 **Provider/model atomicity**

An unavailable provider/model pair is rejected; neither default changes and the draft survives.

A02 **Capability authority**

Unknown model authority disables gated tools/vision/schema; manual labels do not certify support.

A03 **Generation compatibility**

Changing model revalidates all overrides; unsupported parameters cannot leak into strict DTOs.

A04 **Unset versus zero**

Reset removes the tuning override; temperature 0 and absent temperature remain distinct.

A05 **Active-call capture**

Editing effort/model/budget does not rewrite an in-flight request or its historical receipt.

A06 **Secret storage**

Provider/MCP/hook key values are absent from shared config, diagnostics, logs and exported bundles.

A07 **Loopback versus LAN**

Ollama default and custom-provider opt-in differ; private LAN remains refused under current policy.

A08 **Remote locality**

A loopback URL resolves on the daemon host; UI, workspace, vault and model locations are explicit.

A09 **Ask enforcement**

Ask cannot dispatch a write/effectful tool even if prompt text requests it; prompt-only mode is labelled.

A10 **Preset and policy**

A custom agent, rule, skill or hook cannot widen managed authority or edit owner checkout directly.

A11 **Context null**

contextUsage=null renders Unavailable, never 0%, complete or a fabricated green utilization bar.

A12 **Context arithmetic**

All estimate categories sum to one basis; output/safety reserve are not counted as consumed input.

A13 **Source identity**

Selection, diagnostic and file attachments retain URI/range/version across worktree/session switches.

A14 **Attachment failure**

Unsupported/oversize/stale attachment remains visible, blocks invalid admission and preserves text.

ACCEPTANCE · CONTINUED

A15 **Compaction continuity**

Failed/minimal-reduction compaction does not advance success; criteria and pending effects survive.

A16 **Memory deletion**

Future-use deletion is acknowledged; earlier prompt/artifact copies are disclosed as retained.

A17 **Approval exactly once**

Duplicate clicks cannot reply twice; stale/wrong-session/generation waiters refuse visibly.

A18 **Persistent rule boundary**

Allow once cannot silently persist; remembered rules require policy save/version acknowledgement.

A19 **Command normalization**

Compound shell/interpreter/symlink targets cannot exploit a decorative prefix/glob approval.

A20 **Sandbox truth**

Failed setup blocks affected actions; fallback network access is not labelled complete isolation.

A21 **MCP bounds**

Server count and UTF-8 byte/argument limits fail before persistence; prior valid config stays intact.

A22 **MCP transport/auth**

Reviewed stdio config never appears as implemented remote/OAuth; cancelled auth is not Connected.

A23 **Project executable trust**

Opening a repository or skill manifest does not execute MCP servers, hooks, setup or installs.

A24 **Hooks and verdict**

Deadline/nonzero/malformed output differ; partial stdout after timeout cannot authorize a tool.

A25 **Child budgets**

All child reservations share parent envelope; saturation is bounded and ownership contention is visible.

A26 **Steer bounds**

Empty and >500-character steer are rejected without truncation; stale child controls are fenced.

A27 **Owned process**

Stale PID cannot authorize input/kill; process stop, task stop and session abort remain separate.

A28 **Unknown effect**

Disconnect/timeout does not mark an external effect cancelled or offer blind automatic replay.

A29 **Admission idempotence**

Same UUID + same captured body recovers receipt; altered body conflicts and needs a new request.

A30 **Budget exactness**

USD 5.00 becomes exactly 5000000 micro-USD; negatives/fractions/overflow fail token/micro-unit input.

ACCEPTANCE · CONTINUED

A31 **Unknown pricing**

Missing rate/quota stays unavailable; cancelled/failed attempted spend remains in eligible accounting.

A32 **Retry policy**

3 total attempts is not 3 retries; range/backoff validation and effect-state replay rules hold.

A33 **Checkpoint coverage**

Missing after blobs disable redo/diff; uncovered shell/network/SCM effects are named before restore.

A34 **Manual-edit CAS**

Restore/integration refuses changed preconditions and preserves unrelated user edits and prior artifacts.

A35 **Engineering completion**

Missing required acceptance/claims/matching landing/finish remains pending despite Completed chat.

A36 **VCSC denominator**

All eligible attempted spend / verified landed changes; zero verified denominator is unavailable.

A37 **Atomic settings**

Invalid patch writes nothing; revision conflict retains draft; Reset and Defaults have native semantics.

A38 **Session/host fencing**

Late callbacks and old handles never populate or control a newly selected session/worktree/daemon.

A39 **Accessible compact view**

240 px + 200% text, keyboard, IME, high contrast and screen reader retain all essential controls.

A40 **Long stream and full search**

Windowing/coalescing preserve event order/scroll; Find reaches unloaded history and logs stay bounded.

Every control additionally needs default, validation, atomic apply, timing, ownership-fence and recovery checks from Build_Map.csv.

4.9 EVIDENCE · R7 PAGES 340–351 · 141 REFERENCES

Source and benchmark references

F/B: 17 October source and benchmark / R4: dated 12 October source / U: native UX guidance reviewed 17 October. Sources support the pinned current-state boundaries. UI and construction requirements are authored design targets.

F01 **VS Code settings and commands**
apps/vscode/package.json / line 117

F03 **VS Code native change review**
apps/vscode/src/changeReview.ts / line 33

F05 **JetBrains application-level settings**
apps/jetbrains/frontend/src/main/kotlin/dev/faktor/frontend/FaktorSettingsModel.kt / line 45

F07 **Native API and current feature boundary**
docs/native-protocol.md / line 129

F09 **Model capabilities, reasoning and pricing authority**
crates/core/src/model.rs / line 30

F11 **Core config, compaction and retries**
crates/cli/src/config/core.rs / line 10

F02 **VS Code complete UI declarations**
apps/vscode/src/webview.ts / line 294

F04 **JetBrains task/ask, finish and work surface**
apps/jetbrains/frontend/src/main/kotlin/dev/faktor/frontend/FaktorChatPanel.kt / line 83

F06 **JetBrains provider/model selection**
apps/jetbrains/frontend/src/main/kotlin/dev/faktor/frontend/SettingsPanel.kt/ line 25

F08 **Provider configuration and endpoint policy**
crates/cli/src/config/provider.rs / line 31

F10 **Production efficiency flags**
crates/cli/src/config/efficiency.rs / line 74

F12 **Current MCP configuration and limits**
crates/cli/src/config/mcp.rs / line 7

F13 **Hooks and lifecycle contract**
crates/hooks/src/lib.rs / line 34

F15 **Session-owned permission replies**
crates/server/src/permission.rs / line 58

F17 **Isolated mutation policy**
crates/cli/src/config/tasks.rs / line 21

F19 **IDE usage and financial state validation**
apps/vscode/src/nativeClient.ts / line 2682

F21 **IDE credentials and endpoint scope**
apps/jetbrains/frontend/src/main/kotlin/dev/faktor/frontend/ControlPlaneCredentials.kt / line 18

F23 **Typed attachments and validation**
apps/jetbrains/frontend/src/main/kotlin/dev/faktor/frontend/AttachmentsPanel.kt / line 742

F14 **Content-aware checkpoint boundary**
crates/session/src/checkpoints.rs / line 18

F16 **Verification policy and optional provider**
crates/cli/src/config/verification.rs / line 12

F18 **Semantic embedding configuration**
crates/cli/src/config/embedding.rs / line 14

F20 **Agent steering and bound**
apps/vscode/src/steer.ts / line 8

F22 **Optional enterprise settings**
crates/cli/src/config/enterprise.rs / line 27

F24 **Durable title-filtered history**
apps/jetbrains/frontend/src/main/kotlin/dev/faktor/frontend/HistoryPanel.kt /line 73

F25 **Session-switch ownership fencing**
apps/vscode/src/lifecycleFence.ts / line 9

F27 **Context allocation math**
crates/context/src/budget.rs / line 9

F29 **Completion configuration**
crates/cli/src/config/completion.rs / line 21

F31 **Shell and network policy**
crates/cli/src/config/sandbox.rs / line 49

F33 **IDE source authority and architectural constraints**
AGENTS.md / line 6

F35 **Updater configuration**
crates/cli/src/config/updater.rs / line 38

F26 **Bounded native terminal view**
apps/jetbrains/frontend/src/main/kotlin/dev/faktor/frontend/TerminalPanel.kt/ line 30

F28 **Agents and contextual controls**
apps/jetbrains/frontend/src/main/kotlin/dev/faktor/frontend/AgentsPanel.kt /line 22

F30 **Platform settings integration**
apps/jetbrains/frontend/src/main/kotlin/dev/faktor/frontend/FaktorConfigurable.kt / line 58

F32 **Streaming and rendering scheduler**
apps/vscode/src/refreshScheduler.ts / line 13

F34 **Run admission and captured draft**
apps/vscode/src/taskStart.ts / line 37

F36 **Optional worker placement**
crates/cli/src/config/workers.rs / line 35

F37 **Exact provider-header limits and secrecy**
crates/provider/src/config.rs / line 28

F39 **Current native Kotlin protocol client**
apps/jetbrains/shared/src/main/kotlin/dev/faktor/shared/NativeProtocol.kt /line 2

B01 **Kilo current extension migration**
kilo.ai/docs/code-with-ai/platforms/vscode/whats-new

B03 **Kilo custom models**
kilo.ai/docs/code-with-ai/agents/custom-models

B05 **Kilo agent selection**
kilo.ai/docs/code-with-ai/agents/using-agents

B07 **Kilo snapshots**
kilo.ai/docs/code-with-ai/features/checkpoints

F38 **Current client methods and strict DTOs**
apps/vscode/src/nativeClient.ts / line 4168

F40 **Current engineering-outcome presentation**
apps/jetbrains/frontend/src/main/kotlin/dev/faktor/frontend/OutcomesPanel.kt/ line 119

B02 **Kilo model selection**
kilo.ai/docs/code-with-ai/agents/model-selection

B04 **Kilo permissions**
kilo.ai/docs/getting-started/settings/auto-approving-actions

B06 **Kilo context mentions**
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B08 **Kilo session history**
kilo.ai/docs/code-with-ai/agents/session-history

SOURCE AND BENCHMARK REFERENCES · CONTINUED

B09 **Kilo MCP management**
kilo.ai/docs/automate/mcp/using-in-kilo-code

B11 **Claude configuration precedence**
code.claude.com/docs/en/settings

B13 **Claude interactive controls**
code.claude.com/docs/en/interactive-mode

B15 **Claude MCP integration**
code.claude.com/docs/en/mcp

B17 **Claude checkpoint limits**
code.claude.com/docs/en/checkpointing

B19 **Claude hooks**
code.claude.com/docs/en/hooks

B10 **Claude Code IDE workflow**
code.claude.com/docs/en/vs-code

B12 **Claude model configuration**
code.claude.com/docs/en/model-config

B14 **Claude permission configuration**
code.claude.com/docs/en/permissions

B16 **Claude subagents**
code.claude.com/docs/en/sub-agents

B18 **Claude memory**
code.claude.com/docs/en/memory

B20 **Claude skills**
code.claude.com/docs/en/skills

B21 **Claude cost management**
code.claude.com/docs/en/costs

B23 **VS Code UX guidance**
code.visualstudio.com/api/ux-guidelines/overview

B25 **IntelliJ tool windows**
plugins.jetbrains.com/docs/intellij/tool-windows.html

B27 **Cursor rules**
cursor.com/docs/context/rules

B29 **Claude JetBrains integration**
code.claude.com/docs/en/jetbrains

R4-V02 **package.json / dated source**
apps/vscode/package.json

B22 **Claude sandbox**
code.claude.com/docs/en/sandboxing

B24 **IntelliJ settings**
plugins.jetbrains.com/docs/intellij/settings.html

B26 **Copilot host feature matrix**
docs.github.com/en/copilot/reference/copilot-feature-matrix

B28 **VS Code MCP management**
code.visualstudio.com/docs/agent-customization/mcp-servers

R4-V01 **ui-terminology.md / dated source**
docs/ui-terminology.md

R4-V03 **webview.ts / dated source**
apps/vscode/src/webview.ts

R4-V04 **chat.js / dated source**
apps/vscode/media/chat.js

R4-V06 **extension.ts / dated source**
apps/vscode/src/extension.ts

R4-V08 **errors.ts / dated source**
apps/vscode/src/errors.ts

R4-V10 **cockpit.ts / dated source**
apps/vscode/src/cockpit.ts

R4-V12 **nativeClient.ts / dated source**
apps/vscode/src/nativeClient.ts

R4-V14 **contract.ts / dated source**
apps/vscode/src/contract.ts

R4-V05 **chat.css / dated source**
apps/vscode/media/chat.css

R4-V07 **daemon.ts / dated source**
apps/vscode/src/daemon.ts

R4-V09 **taskStart.ts / dated source**
apps/vscode/src/taskStart.ts

R4-V11 **state.ts / dated source**
apps/vscode/src/state.ts

R4-V13 **eventStream.ts / dated source**
apps/vscode/src/eventStream.ts

R4-V15 **plugin.xml / dated source**
apps/jetbrains/frontend/src/main/resources/META-INF/plugin.xml

R4-V16 **FaktorToolWindowFactory.kt / dated source**
apps/jetbrains/frontend/src/main/kotlin/dev/faktor/frontend/FaktorToolWindowFactory.kt

R4-V18 **FaktorChatPanel.kt / dated source**
apps/jetbrains/frontend/src/main/kotlin/dev/faktor/frontend/FaktorChatPanel.kt

R4-V20 **PanelSupport.kt / dated source**
apps/jetbrains/frontend/src/main/kotlin/dev/faktor/frontend/PanelSupport.kt

R4-V22 **VerificationPanel.kt / dated source**
apps/jetbrains/frontend/src/main/kotlin/dev/faktor/frontend/VerificationPanel.kt

R4-V24 **TaskTreeModel.kt / dated source**
apps/jetbrains/frontend/src/main/kotlin/dev/faktor/frontend/TaskTreeModel.kt

R4-V26 **HistoryPanel.kt / dated source**
apps/jetbrains/frontend/src/main/kotlin/dev/faktor/frontend/HistoryPanel.kt

R4-V17 **FaktorDaemonSettings.kt / dated source**
apps/jetbrains/frontend/src/main/kotlin/dev/faktor/frontend/FaktorDaemonSettings.kt

R4-V19 **FaktorFrontendService.kt / dated source**
apps/jetbrains/frontend/src/main/kotlin/dev/faktor/frontend/FaktorFrontendService.kt

R4-V21 **ChangesPanel.kt / dated source**
apps/jetbrains/frontend/src/main/kotlin/dev/faktor/frontend/ChangesPanel.kt

R4-V23 **TaskTreePanel.kt / dated source**
apps/jetbrains/frontend/src/main/kotlin/dev/faktor/frontend/TaskTreePanel.kt

R4-V25 **EvidenceNavigatorPanel.kt / dated source**
apps/jetbrains/frontend/src/main/kotlin/dev/faktor/frontend/EvidenceNavigatorPanel.kt

R4-V27 **PermissionsPanel.kt / dated source**
apps/jetbrains/frontend/src/main/kotlin/dev/faktor/frontend/PermissionsPanel.kt

R4-V28 **BlockersPanel.kt / dated source**
apps/jetbrains/frontend/src/main/kotlin/dev/faktor/frontend/BlockersPanel.kt

R4-V30 **TournamentPanel.kt / dated source**
apps/jetbrains/frontend/src/main/kotlin/dev/faktor/frontend/TournamentPanel.kt

R4-V32 **SettingsPanel.kt / dated source**
apps/jetbrains/frontend/src/main/kotlin/dev/faktor/frontend/SettingsPanel.kt

R4-V34 **DiagnosticsPanel.kt / dated source**
apps/jetbrains/frontend/src/main/kotlin/dev/faktor/frontend/DiagnosticsPanel.kt

R4-V36 **TerminalPanel.kt / dated source**
apps/jetbrains/frontend/src/main/kotlin/dev/faktor/frontend/TerminalPanel.kt

SOURCE AND BENCHMARK REFERENCES · CONTINUED

R4-V38 **DaemonBinary.kt / dated source**
apps/jetbrains/backend/src/main/kotlin/dev/faktor/backend/DaemonBinary.kt

R4-V29 **AgentsPanel.kt / dated source**
apps/jetbrains/frontend/src/main/kotlin/dev/faktor/frontend/AgentsPanel.kt

R4-V31 **BoardPanel.kt / dated source**
apps/jetbrains/frontend/src/main/kotlin/dev/faktor/frontend/BoardPanel.kt

R4-V33 **UsagePanel.kt / dated source**
apps/jetbrains/frontend/src/main/kotlin/dev/faktor/frontend/UsagePanel.kt

R4-V35 **AttachmentsPanel.kt / dated source**
apps/jetbrains/frontend/src/main/kotlin/dev/faktor/frontend/AttachmentsPanel.kt

R4-V37 **BackendProcessManager.kt / dated source**
apps/jetbrains/backend/src/main/kotlin/dev/faktor/backend/BackendProcessManager.kt

R4-V39 **NativeEventStream.kt / dated source**
apps/jetbrains/backend/src/main/kotlin/dev/faktor/backend/NativeEventStream.kt

R4-V40 **NativeClient.kt / dated source**
apps/jetbrains/backend/src/main/kotlin/dev/faktor/backend/NativeClient.kt

R4-V42 **completion_proof.md / dated source**
docs/specs/completion_proof.md

R4-V44 **native-protocol.md / dated source**
docs/native-protocol.md

R4-V46 **first_run.rs / dated source**
crates/cli/src/config/first_run.rs

R4-V48 **lib.rs / dated source**
crates/verify/src/lib.rs

R4-V50 **README.md / dated source**
README.md

R4-V41 **NativeProtocol.kt / dated source**
apps/jetbrains/shared/src/main/kotlin/dev/faktor/shared/NativeProtocol.kt

R4-V43 **autonomous-engineering.md / dated source**
docs/specs/autonomous-engineering.md

R4-V45 **outcome-ledger.md / dated source**
docs/outcome-ledger.md

R4-V47 **completion_steps.rs / dated source**
crates/orchestrator/src/completion_steps.rs

R4-V49 **provider.rs / dated source**
crates/verify/src/provider.rs

R4-V51 **README.md / dated source**
apps/README.md

R4-V52 **README.md / dated source**
apps/jetbrains/README.md

R4-V54 **money.ts / dated source**
apps/vscode/src/money.ts

R4-V56 **branding.md / dated source**
docs/specs/branding.md

R4-V58 **certification.md / dated source**
docs/certification.md

R4-O02 **VS Code Sidebars UX Guidelines**
code.visualstudio.com/api/ux-guidelines/sidebars

R4-O04 **VS Code Webview API - accessibility**
code.visualstudio.com/api/extension-guides/webview

R4-V53 **OverviewPanel.kt / dated source**
apps/jetbrains/frontend/src/main/kotlin/dev/faktor/frontend/OverviewPanel.kt

R4-V55 **controlPlaneAuth.ts / dated source**
apps/vscode/src/controlPlaneAuth.ts

R4-V57 **outcome_ledger.rs / dated source**
crates/agent/src/outcome_ledger.rs

R4-001 **VS Code Views UX Guidelines**
code.visualstudio.com/api/ux-guidelines/views

R4-003 **VS Code Webviews UX Guidelines**
code.visualstudio.com/api/ux-guidelines/webviews

R4-005 **JetBrains Tool Window UI Guidelines**
plugins.jetbrains.com/docs/intellij/tool-window.html

R4-006 **JetBrains Accessibility**
plugins.jetbrains.com/docs/intellij/accessibility.html

U01 **VS Code / native extension UX**
code.visualstudio.com/api/ux-guidelines/overview

U03 **VS Code / custom webview boundaries**
code.visualstudio.com/api/ux-guidelines/webviews

U05 **JetBrains / tool-window hierarchy**
plugins.jetbrains.com/docs/intellij/tool-window.html

U07 **JetBrains / Kotlin UI DSL**
plugins.jetbrains.com/docs/intellij/kotlin-ui-dsl-version-2.html

R4-007 **JetBrains Working with Icons**
plugins.jetbrains.com/docs/intellij/icons.html

U02 **VS Code / views and relocation**
code.visualstudio.com/api/ux-guidelines/views

U04 **JetBrains / related-control layout**
plugins.jetbrains.com/docs/intellij/layout.html

U06 **JetBrains / validation timing**
plugins.jetbrains.com/docs/intellij/validation-errors.html

4.9 EVIDENCE · REVISION 8 ADDITIONS

Revision 8 guidance references

Native-host and accessibility guidance consulted on 17 October 2026 for this revision. Quoted rules are summarised in Part 1.

R8-01 **VS Code UX Guidelines: overview**
code.visualstudio.com/api/ux-guidelines/overview

R8-02 **VS Code UX Guidelines: sidebars**
code.visualstudio.com/api/ux-guidelines/sidebars

R8-03 **VS Code UX Guidelines: webviews**
code.visualstudio.com/api/ux-guidelines/webviews

R8-04 **IntelliJ Platform UI Guidelines: button**
plugins.jetbrains.com/docs/intellij/button.html

R8-05 **IntelliJ Platform UI Guidelines: validation errors**
plugins.jetbrains.com/docs/intellij/validation-errors.html

R8-06 **IntelliJ Platform UI Guidelines: inline help text**
plugins.jetbrains.com/docs/intellij/inline-help-text.html

R8-07 **IntelliJ Platform UI Guidelines: input field**
plugins.jetbrains.com/docs/intellij/input-field.html

R8-08 **IntelliJ Platform UI Guidelines: groups of controls**
plugins.jetbrains.com/docs/intellij/groups-of-controls.html

R8-09 **IntelliJ IDEA Help: New UI (Inter UI font)**
jetbrains.com/help/idea/new-ui.html

R8-10 **WCAG 2.2 (contrast 1.4.3, 1.4.11; reflow 1.4.10; resize 1.4.4; focus 2.4.7, 2.4.11; target 2.5.8, 2.5.5; name/role/value 4.1.2; status 4.1.3)**
w3.org/TR/WCAG22/

R8-11 **Nielsen Norman Group: 10 usability heuristics for user interface design**
nngroup.com/articles/ten-usability-heuristics/

R8-12 **CSS Inline Layout Module Level 3: text-box-trim and text-box-edge**
w3.org/TR/css-inline-3/

PART 5 · APPENDICES · R7 PAGE 352

Retained independent identity and website

Original content is retained in full, with dated scope rather than silently trimmed or rewritten.

Independent Faktor visual identity

The following 42 pages retain the complete 12 October identity report: product analysis, positioning, family relationship, approved angular F, custom wordmark, logo suite, clear space, pixel proofs, palette, typography, layout, voice, applications, website architecture, rollout, research and reusable briefs.

Complete Faktor website

The next 57 pages retain the full website report, including product/approach, evidence, source setup, outcome economics, editor design, task contracts, brand, FAQ, source notes and historical references. These pages keep their original date and local pagination.

Which facts govern implementation

The main revised specification and its pinned 17 October source support supersede old current-state implementation labels. Historical pages remain complete; the approved identity and product independence remain in force. The handoff contains exact source PDFs and hashes.

Related products remain separate

KiwiCaptcha owns its identity, widget and website. Kiwi-CI and Tangerine remain optional independent products with defined interoperability. No sibling logo is embedded into the Faktor mark, and no account or sibling backend becomes a prerequisite for native Faktor operation.

How these pages are reproduced

Pages 353–451 of revision 7 are reproduced unchanged and in order, scaled into a frame so their own dates, page numbers and design stay intact. They are dated source documents for the product identity and website, not IDE interface plates, so revision 8 does not redraw them.

	DOCUMENT	DATE	LENGTH	STARTS
A	**Faktor Expanded, edition 4**	12 October 2026	8 pages	p. 351
B	**Faktor visual identity v3**	11 October 2026	34 pages	p. 359
C	**The complete Faktor website**	Website capture	57 pages	p. 393

faktor

Independent product identity / edition 4 / 12 October 2026

Faktor Expanded.

Complete editor design for VS Code and JetBrains, a fully developed independent website, and the complete earlier identity system.

> The complete 11 October identity guide follows this new edition. Its original pages, analysis, generated presentations, vector construction, typography, applications and export rules are retained as dated reference.

Bel Consulting / Kiwi-CI product family
Current product source: 16e8bd4cf2832affbb6a2e9e9281beee7527ac4d
Each product keeps its own mark, palette, typography, website and download suite.

Faktor / edition 4

The identity stays precise.

The Faktor mark is a solid angular F. There is no circle, ring, curl, spiral or koru. The custom wordmark is fixed vector lettering. Barlow Semi Condensed carries the marketing headings, Barlow supports reading, and IBM Plex Mono carries technical labels.

The relation to Kiwi-CI comes through disciplined spacing, restrained company attribution, robust monochrome assets and a common standard of clarity. Faktor retains its warm evidence accent and architectural geometry.

Inside the editors, inherit native host fonts, user editor fonts, native focus behavior and theme tokens. Marketing typography should not replace the developer's configured reading environment.

FaktorExpanded. 2

Faktor / edition 4

Two hosts. A coherent work model.

Work retains conversation, current run and composer. Inspect explains plan, candidate, changes, verification and agent observations. History preserves project-scoped conversation and outcomes.

JetBrains already exposes Work, Inspect and History. VS Code currently exposes Chat with an Inspect disclosure. A visible VS Code History destination, shared switcher, richer outcome views and native side-by-side diff adapters are proposed additions.

JetBrains currently has independent Commit, Push and Create PR checkboxes. The normalized four-level finish selector is a proposed presentation change and needs explicit contract mapping.

The separate UI/UX book contains 36 screen specifications, 34 states, 16 flows and all 56 host/theme visual specimens. The website contains a local interactive 14-state design prototype.

Faktor / edition 4

Completion has four distinct boundaries.

Required-check acceptance, named claim and candidate evidence, verified/landed identity and integration, and the requested finish actions each need their own recorded basis. A completed conversation or passing subset of checks does not establish an accepted engineering outcome.

The pending visual specimen deliberately has a missing concurrent-refresh result. It remains pending. The accepted visual specimen is a separate illustrative candidate with complete evidence and successful integration.

Current VCSC is all eligible attempted spend, including failed and cancelled attempts, divided by verified changes. A successful run's cost is a separate field. Missing cost and unrecorded savings are unavailable, not zero.

Faktor / edition 4

Start deliberately. Recover coherently.

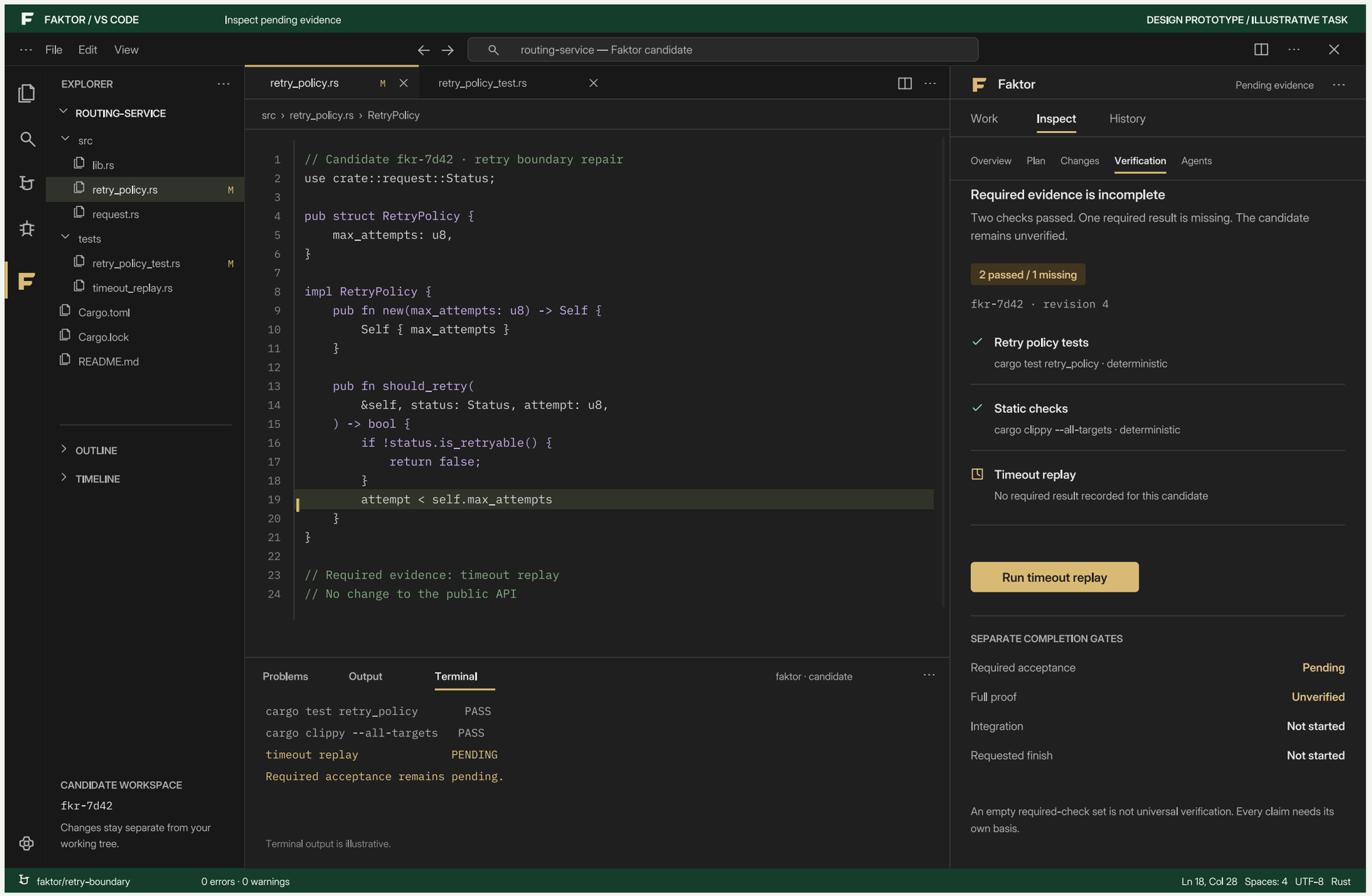
Opening the view does not start the service. Resolve machine/application/environment or signed-install layout discovery. Do not fall back to a project repository target binary.

An executable without a recorded checksum needs an exact-path native Start anyway / Cancel confirmation. A checksum mismatch or malformed checksum is a hard refusal. A match to a recorded checksum is not a signature-verification claim.

On malformed durable event frames, retain the last coherent snapshot and do not advance the cursor beyond unseen events. Preserve task draft, finish selection, attachments and captured contract; retry or refresh must not admit duplicate runs.

vscode / full visual specimen / proposed layout

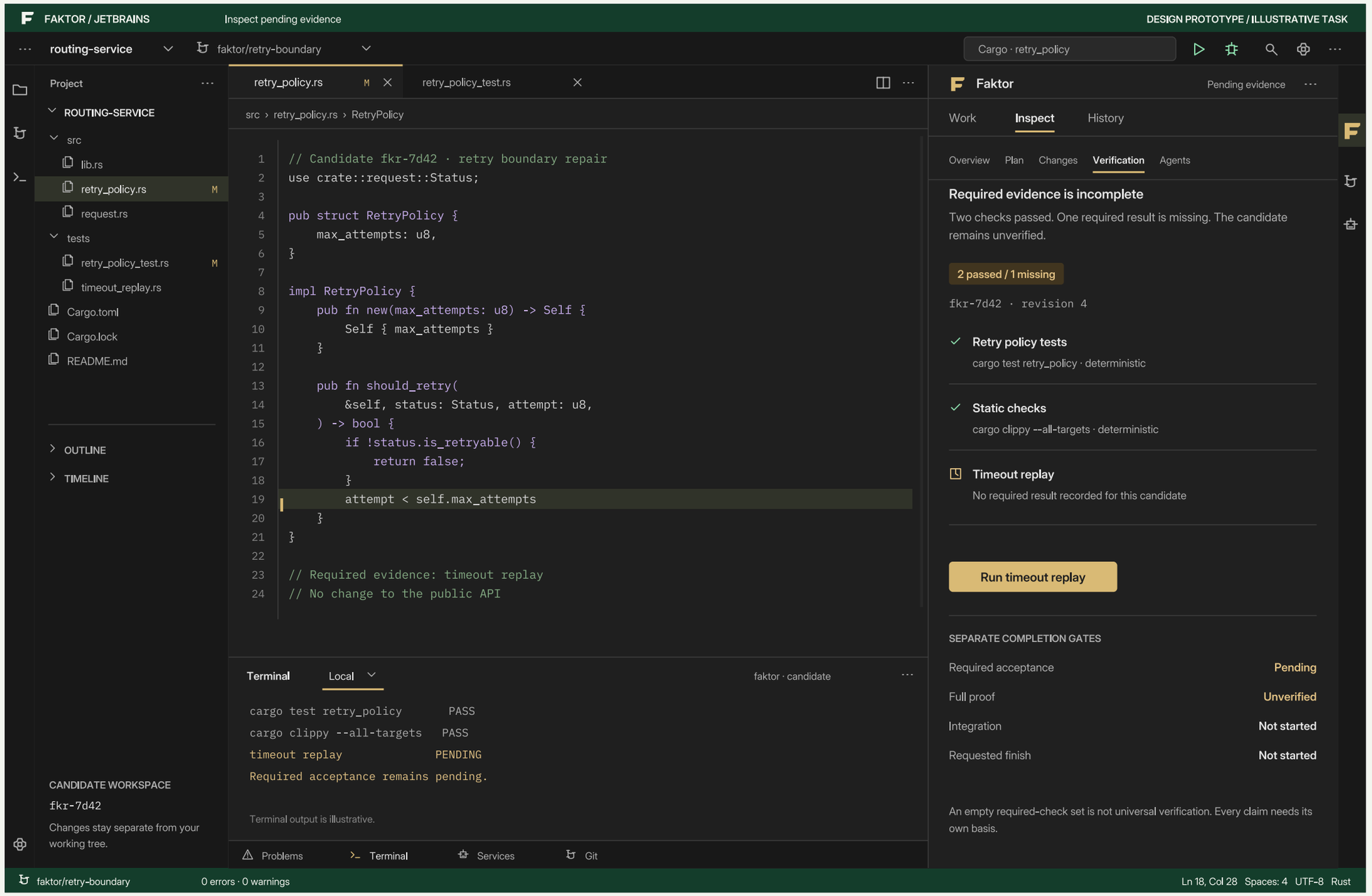
Missing evidence remains pending.



Illustrative task and repository. The complete two-host light/dark atlas is in Faktor_IDE_UI_UX.pdf.

jetbrains / full visual specimen / proposed layout

Missing evidence remains pending.



Illustrative task and repository. The complete two-host light/dark atlas is in Faktor_IDE_UI_UX.pdf.

Complete deliverables / retained source

Every part stays available.

https://faktor.windyclock5.chatgpt.site

- Faktor_Website.pdf - every current website page and the complete dated prior website.
- Faktor_Visual_Identity.pdf - this current addition and every original identity guide page.
- Faktor_Logo_Kit.zip - the complete original logo suite, fonts, previews, identity specifications and new source-grounded design handoff.
- Faktor_IDE_UI_UX.pdf - complete editor specification and all 56 host/theme visual specimens.

Read the current edition first. The following original guide is retained without removing or rewriting its pages.
Earlier product-source claims are dated to their original baseline; current corrections above and the new website take precedence.

Source review 12 October 2026 / 16e8bd4cf2832affbb6a2e9e9281beee7527ac4d

BEL CONSULTING / INDEPENDENT PRODUCT

Faktor / Visual identity

faktor

Make the change. Show the evidence.

Product analysis. Fixed logo artwork. An individual type system. Reusable applications. A dedicated website.

RESEARCH / LETTERING / LOGO / IDENTITY / INTERFACE / WEBSITE

PRODUCT ANALYSIS

Understand the product before the mark.

The identity starts with the work people need to complete and the evidence the source actually supports.

Primary job

Turn an engineering objective into a change the developer can inspect. Plan the work, edit an isolated candidate, evaluate required criteria and preserve the basis for completion.

Product boundary

A standalone native Rust engineering agent with IDE and CLI surfaces. Native intelligence and verification form the baseline. Optional services enrich neutral provider interfaces.

Why it deserves its own identity

The developer is evaluating a change, its constraints and its completion chain. A broad AI-platform story hides that concrete job. The brand should make the work, scope and evidence legible.

What the design should express

Directed action, stable boundaries and an explicit finish. An angular initial identifies the product. Narrow editorial headings organize the explanation. Unresolved evidence remains visible.

Study scope: selected pinned source and implementation review, then design synthesis. Normative specifications are distinguished from observed behavior; this is not a new runtime implementation or audit.

SOURCE-TO-VISUAL RATIONALE / 1

Architecture becomes design decisions.

The source establishes behavior. The visual consequence is an explicit design interpretation.

Source behavior	Design consequence	Sources
A task includes goals, invariants, risks and proof obligations.	Use a visible engineering brief with separate labeled fields rather than a vague dashboard.	F02 / F11 / F12
Change planning is dependency-aware and bounded.	Numbered work, orthogonal connectors, stable columns and finite diagrams.	F02 / F06
Candidate editing is separate from integration.	Make candidate, verified state and landing visibly distinct.	F01 / F06 / F15
Edits and snapshots respect content identity.	Definite edges, stable alignment and clipped terminals; no ornamental fluidity.	F06
Required checks and provider results have explicit unresolved outcomes.	Text plus static glyph and distinct marker. Brand ochre is not success.	F08 / F09 / F13 / F14
The runtime uses durable state and journals.	Calm stable interfaces and motion that acknowledges an actual change.	F01 / F06

SOURCE-TO-VISUAL RATIONALE / 2

The detail carries the distinction.

Product-specific choices continue beyond the symbol and into layout, voice and state meaning.

Source behavior	Design consequence	Sources
Finish actions are user-selected and completion-gated.	A labeled local / commit / push / PR strip, with missing status visible.	F12 / F16
The outcome ledger records actual economics and verification basis.	Show the record and metric definitions without fabricated numeric claims.	F05
IDE surfaces belong to Faktor and inherit the host.	One bare F for both hosts; marketing type stays outside reading controls.	F01 / F12 / F13
Faktor is standalone.	Own root site, identity pack, metadata and CTA. Family endorsement is secondary.	F02 / F07 / F09

The complete mapped review, precise claim inventory and source inspection scope are supplied separately in specifications/product-content.json and product-strategy.md.

PRODUCT CLAIMS AND BOUNDARIES

Trust has conditions. Keep them visible.

Strong positioning follows the supported contract and leaves unresolved conditions intact.

Check acceptance

Required results reduce to Pass, Fail or Pending. A missing required result stays Pending. Optional checks do not determine that reducer. An empty required set can pass without establishing a verified change.

Claim evidence

Evidence answers a named claim about a candidate snapshot. A response for a different claim or state cannot silently certify the requested change. Unavailable is no verdict; refusal is a distinct outcome.

Integration and finish

Keep run, verified and landed identities explicit. Requested local, commit, push and native pull-request actions have durable status. A missing or skipped step is not a success.

The ledger definition

A verified change requires accepted and integrated with a deterministic-checks final basis. Preserve cost, rework and outcome definitions. Publish no invented savings, completion rate or price.

Current native SCM reconciliation controls PR copy. Older certification prose includes a stale command-only description; the review follows the newer README and completion runner.

AUDIENCE JOBS AND POSITIONING

Give each reader the proof they need.

The same identity can support more than one depth of information without merging distinct product experiences.

Reader	Job	What the design makes clear
Developer implementing a feature	Translate an objective into changes, checks and a reviewable result.	One worked brief paired with criteria, candidate and evidence.
Maintainer changing a legacy system	Keep existing behavior and constraints visible during change.	Separate invariants and risks; show one unresolved row.
Reviewer	Find the relevant candidate and evidence behind a claim.	Evidence rows and Inspect, not an undifferentiated green banner.
Engineer repairing a failure	Understand the failed check and next action.	Named result, counterexample and repair step.
Engineering lead	Assess measured cost, rework and accepted outcomes.	Ledger fields, explicit basis and definitions; no invented savings.
Developer selecting a provider	Keep model choice separate from engineering workflow.	Configured model as secondary metadata; provider documentation.

FAMILY ARCHITECTURE

Related by discipline. Individually owned.

Kiwi CI is the reference for restraint and family foundations; neither product is a renamed Kiwi CI.



KIWI CI / APPROVED FAMILY REFERENCE



FAKTOR / DISTINCT FIXED ARTWORK

Shared foundations

Ink, Forest, Paper and Sage; substantial one-color silhouettes; generous clear space; restrained technical metadata; IBM Plex Mono; precise product language.

Individual character

Cut F, ochre accent, custom clipped f/k/t lettering, Barlow editorial type and orthogonal graphic grammar. No circle, curl or koru in the Faktor symbol.

Independent delivery

An individual logo suite, font system, guide, source study, downloads and root website. Bel Consulting endorses the product. No sibling installation is required.

The family resemblance is a design interpretation based on the approved Kiwi CI artwork. A common body font is not required; exact outlined logo lettering remains separate from UI typography.

INDIVIDUAL LOGO PRESENTATION

Faktor / Primary symbol

Native geometry is authoritative; the generated presentation demonstrates the visual direction.



Cut F

One solid initial. A 16-unit upright stem, 14-unit arms and 6-unit clipped tips. The upper arm is longer than the middle. Its path uses only M, V, H, L and Z commands.

No circular motif

The symbol has no circle, curl, koru, arc or curve. The clipped tips suggest directed work and a clear finish; that meaning comes from use, not a claim of universal correctness.

INDIVIDUAL LIGHT-LOGO PRESENTATION

Faktor / Primary signature

Fixed logo lettering, preserved proportions and generous space on Paper.



Production file: assets/faktor-lockup-ink.svg. The exact SVG is an outlined master with no live font dependency. Generated pixels are presentation artwork rather than a vector source.

INDIVIDUAL DARK-LOGO PRESENTATION

Faktor / Reverse signature

Use the product accent on Ink, or the supplied white master when the context requires it.



Production file: assets/faktor-lockup-accent.svg. Soft ochre has strong contrast on Ink. Avoid shadows, glow, gradients and an altered mark-to-wordmark relationship.

WORDMARK CONSTRUCTION

Lettering with its own character.

The logo is fixed artwork. The website font explains the voice without replacing that artwork.

faktor

Bespoke f, k and t

The f and t repeat the clipped-arm detail of the symbol. The k has a dedicated angular contour. Their widths and terminal cuts are drawn as path artwork rather than stock UI type.

A readable word rhythm

Barlow Semi Condensed 600 informs a, o and r. Pair-specific optical adjustments preserve the word’s density. The o is a letter counter, not a circular product emblem.

The OFL font files remain unmodified. This is edited logo artwork derived partly from licensed outlines; it is not a new full typeface. Exact positions and contours are supplied in lettering.json and SVG.

LOCKUPS AND REPRODUCTION

A complete suite, with specific uses.

Use the primary horizontal signature whenever the full name can be read comfortably.



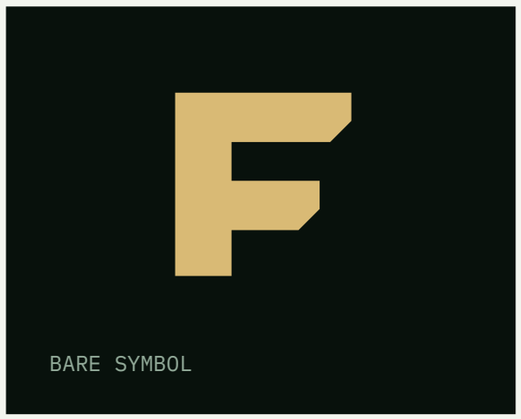
PRIMARY / HORIZONTAL



SECONDARY / STACKED



WORDMARK / KNOWN CONTEXT



BARE SYMBOL



APP CONTAINER

Every symbol, lockup, wordmark and stacked master has currentColor, Ink, white and accent variants. An app tile is a platform container; it is not an extra enclosing shape in the bare mark.

CLEAR SPACE AND MINIMUM SIZE

Space and scale preserve recognition.

Use visible artwork to measure the exclusion zone; empty viewBox padding does not count as clear space.



Use	Size	Rule
Bare symbol	24 px preferred	One-color, high-contrast context.
Tiny icon	16 px supplied	Use the proofed icon; check the target slot.
Horizontal lockup	160 px wide	Keep the complete name readable.
Raster export	2× / 3× output	Prefer SVG when accepted.
Print	Process-specific proof	Convert in the target production profile.

Keep H/4 clear space on all sides of the visible artwork. H is the visible symbol height in the selected signature. Preserve the approved symbol-to-wordmark gap.

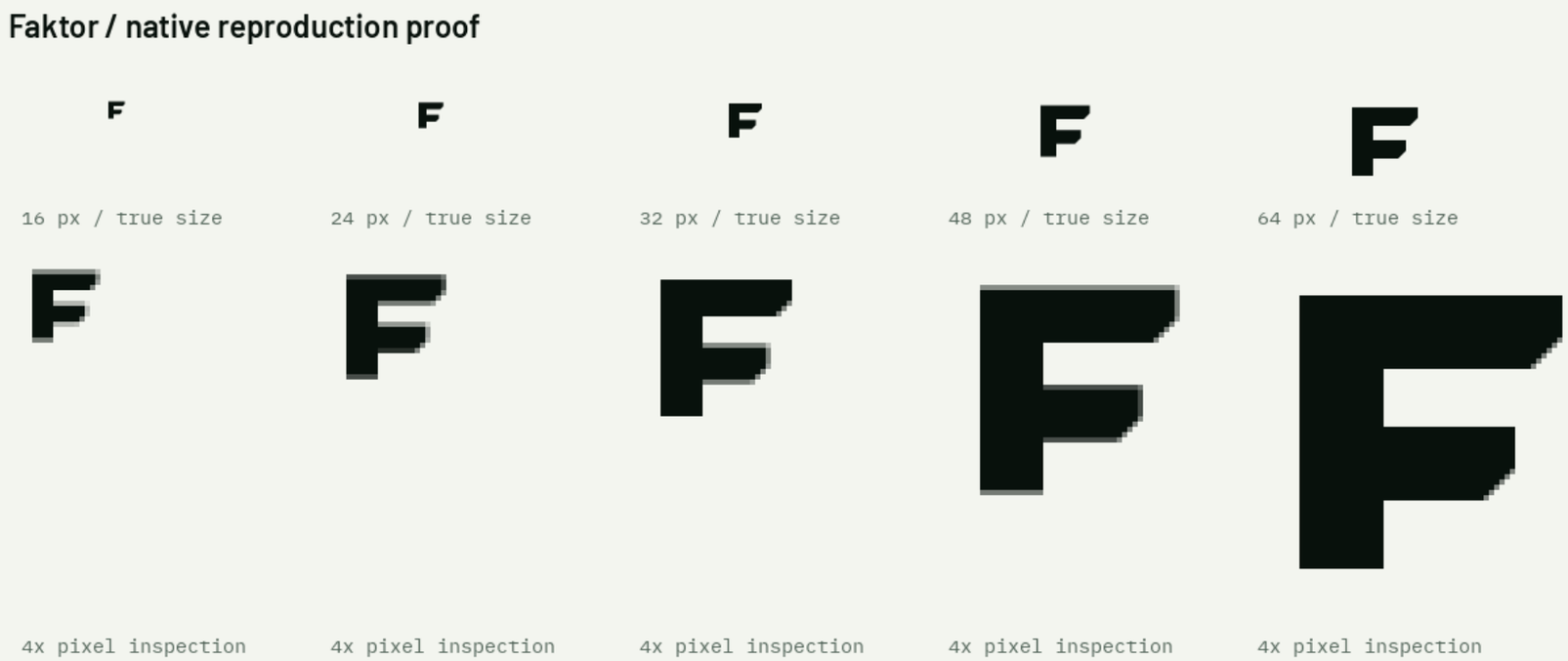
Keep the master intact

No stretching, rotation, letter-spacing changes, cropping, outlining, gradients, shadows or added ornament. On photography, use a quiet solid backing.

NATIVE PIXEL REPRODUCTION PROOFS

The tiny icon is a real constraint.

True-size samples and 4× pixel enlargement at 16, 24, 32, 48 and 64 px.



faktor

One-color native master

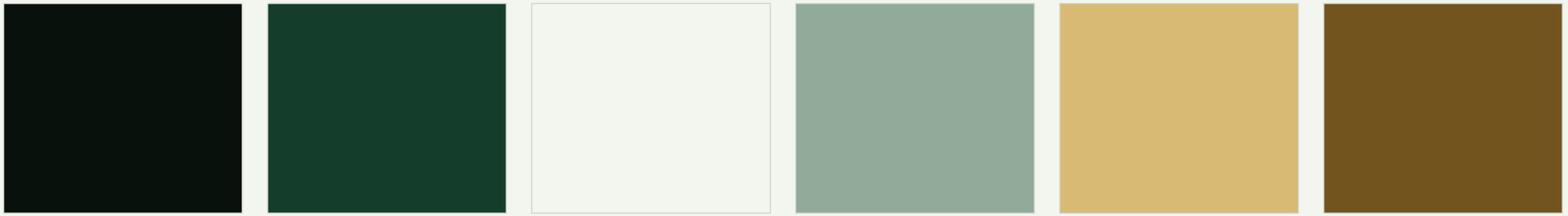
Fixed outlines / transparent exports

These proofs use the native SVG and supplied raster output. 24 px remains the preferred bare-symbol size; the 16 px coil has less interior separation. Physical print and every display condition still need local proofing.

COLOR SYSTEM AND CONTRAST

A palette with clear responsibilities.

Family foundations plus an individual accent. Brand colors and outcome semantics remain distinct.



Ink	Forest	Paper	Sage	Ochre	Ochre Deep
#08110C	#143D2B	#F3F5EF	#92AA99	#D9BA75	#72541E

Foreground / background	Contrast	Use
Ink / Paper	17.45:1	Normal text
Forest / Paper	11.04:1	Normal text
Muted / Paper	5.66:1	Normal text
Sage / Ink	7.69:1	Normal text
Ochre / Ink	10.25:1	Normal text
Ochre Deep / Paper	6.37:1	Normal text
Ochre / Paper	1.7:1	Decorative accent only

Normal text target: 4.5:1; essential non-text graphics: 3:1. Logos have a text-contrast exemption, but reproduction still benefits from strong contrast. Never rely on color alone for state meaning.

INDEPENDENT TYPE SYSTEM

Typography is part of the identity.

The display and reading fonts are now specific to this product; technical metadata connects the family.

Make the change. Show the evidence.

Readable product language. Deliberate detail.

`01 / status: pending / scope: local`

Role	Typeface and scale	Behavior
Display heading	Barlow Semi Condensed 600/700 / 56-108 px	Short, concrete heading; controlled line length.
Section heading	Barlow Semi Condensed / 38-66 px	One clear proposition; editorial rhythm.
Reading / controls	Barlow / 19-25 px	Open leading; visible labels; natural wrapping.
Technical metadata	IBM Plex Mono / 10-14 px	Short IDs, source context, states and code.
Product host	inherit IDE	IDE / widget keeps its existing reading environment.

Self-hosted WOFF and TTF files plus OFL notices are supplied. The logo's fixed paths are independent of font loading. Do not add marketing font requests to the production widget or repaint the IDE.

LAYOUT, GRAPHICS AND MOTION

Build a visual grammar around the work.

A few repeatable choices make the identity useful across pages, documentation and interfaces.

Engineering-sheet composition

Thin rules, stable columns, annotations and one focal example. Use a 4 px unit and 8 px base. Square or 2 px corners and restrained clipped terminals echo the F without a border around everything.

Graphic language

Use orthogonal or stepped traces to explain work order. Keep candidate boundaries, criteria and evidence visible. Avoid neural meshes, AI sparkles and an invented robot mascot.

Interface density

Marketing headings can be large; IDE typography remains native. Conversation, current run and composer lead. Inspect holds detail. A monospace CLI must remain useful in one color.

Motion with meaning

Reveal detail or acknowledge a deliberate interaction. Do not animate pending checks into apparent completion. No invented percentage. Reduced-motion mode preserves the same state and labels.

VOICE AND COPY RULES

Write what the state actually means.

Direct language, bounded claims and recovery that names a useful next action.

Situation	Use	Avoid
Campaign	Make the change. Show the evidence.	Vague claims of intelligence.
Pending	One required result is missing. Completion stays pending.	A global green Verified banner.
Failure	Required check failed. Inspect the result.	A cheerful finish that hides failure.
Unavailable	Verification unavailable. No verdict for this claim.	Turning absence into success.
Mismatch	Landed state differs from the verified candidate.	Using stale evidence for new code.
Cost	Recorded spend, rework and final basis.	Invented savings or public prices.

The brand can be confident while naming scope. Source and integration documentation are honest calls to action; no fabricated customers, customer logos, reviews, performance figures or release availability are included.

INTERFACE STATE CONTRACT

One change. Four distinct conditions.

Required-check acceptance, claim evidence, integration and requested finish actions should be visible together.

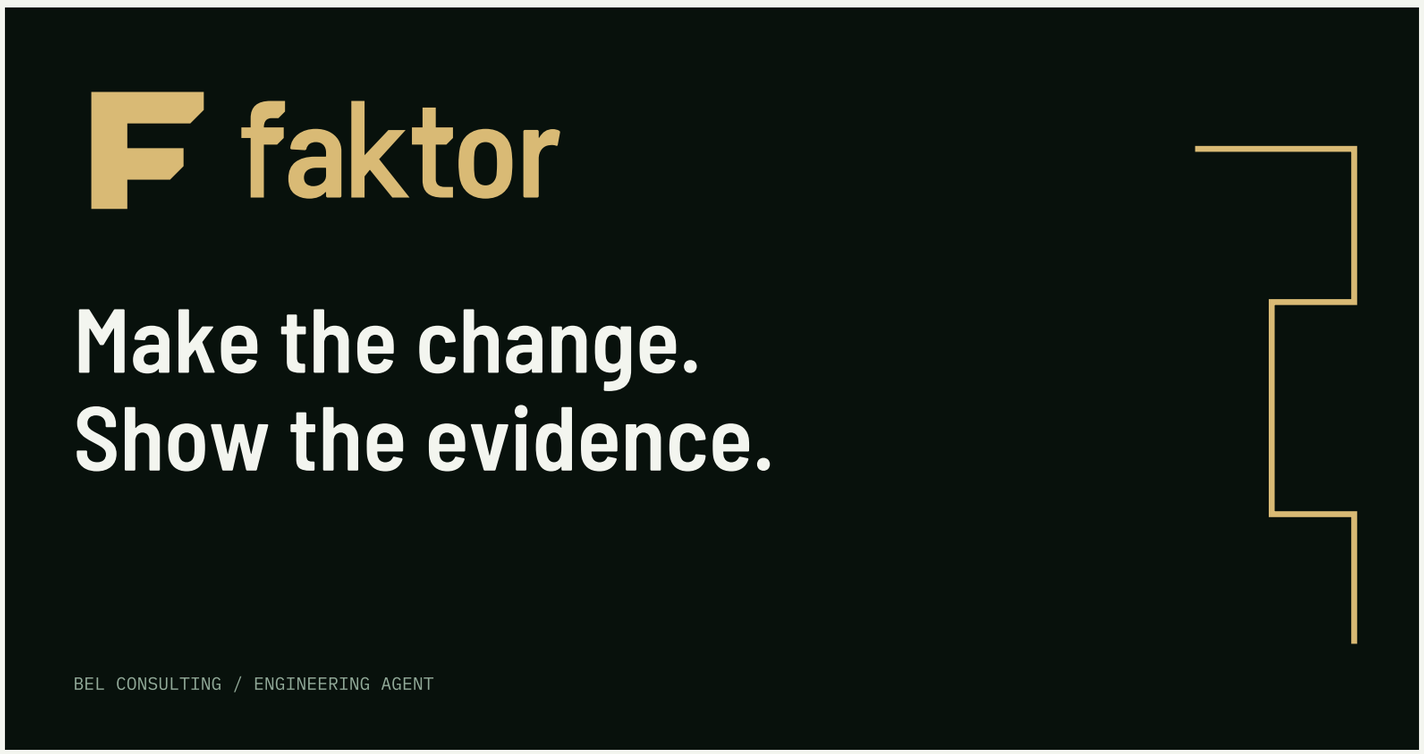
Layer	Visible states	UI consequence
Required acceptance	Pass / Fail / Pending	Text plus check, cross or unresolved glyph. Missing required result stays pending.
Claim evidence	Passed / Failed / Unavailable / Refused	Show claim and candidate. Unavailable carries no verdict.
Candidate / integration	Edited / verified / landed identities	Expose snapshot equality and mismatch. Recheck a different landed state.
Finish actions	Requested / running / succeeded / blocked	A skipped or absent step cannot silently count as succeeded.
Outcome record	Accepted / integrated / final basis	Only the defined verified-change basis enters that ledger count.

Success light #145C39 / dark #68DC93. Pending light #72541E / dark #D9BA75. Failure light #9D3131 / dark #FFB3AE. Every treatment includes text and a shape.

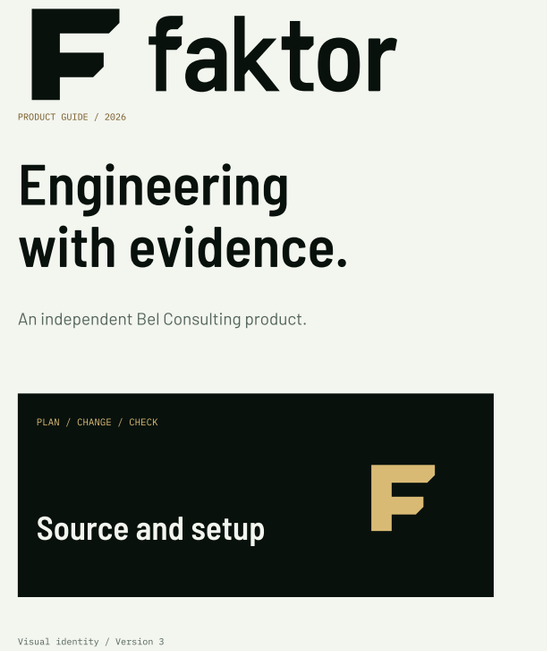
REUSABLE ASSET SYSTEM

Applications built from exact masters.

Social card, document cover and application icon are native vector artwork with fixed font outlines.



SOCIAL / 1200 × 630



DOCUMENT / 900 × 1200



App sizes 16-256 px and multi-resolution ICO. Use the bare currentColor symbol for monochrome activity-bar or toolbar slots.

Native logo and text outlines keep these applications reproducible. RGB values need a destination print profile and a physical proof for print use.

NATIVE IDENTITY BOARD

The exact visual system in one view.

This board uses the supplied SVG masters, exact palette and actual font outlines.

FAKTOR / VISUAL IDENTITY

faktor

BEL CONSULTING / INDEPENDENT PRODUCT

Make the change.
Show the evidence.

PALETTE / SHARED FOUNDATIONS, OWN ACCENT

Ink	Forest	Paper	Sage	Ochre	Ochre Deep
#08110C	#143D2B	#F3F5EF	#92AA99	#D9BA75	#72541E

TYPOGRAPHY / OWN VOICE

Barlow Semi Condensed

Character in the lettering.
Readable product language. Exact labels.

IBM PLEX MONO / TECHNICAL METADATA

PRODUCT WEBSITE / NATIVE DESIGN SPECIMEN

faktor

Make the change.
Show the evidence.

View the source

COMPONENT / ILLUSTRATIVE STATE

CANDIDATE / LOCAL WORKSPACE

2 checks passed / 1 pending

Concurrent refresh / needs evidence

COMPLETION PENDING

APP / ICON

FIXED OUTLINED WORDMARK

faktor

H/4 clear space. One-color masters. No effects.

Source review / logo suite / applications / own website

NATIVE MASTER / VERSION 3.0

The native board governs exact reproduction. Editable vector artwork and a high-resolution raster export are included in the assets folder.

GENERATED IDENTITY-BOARD PRESENTATION

The identity as an application direction.

Concept applications explore the design language. Native files govern contours, type, color and product copy.

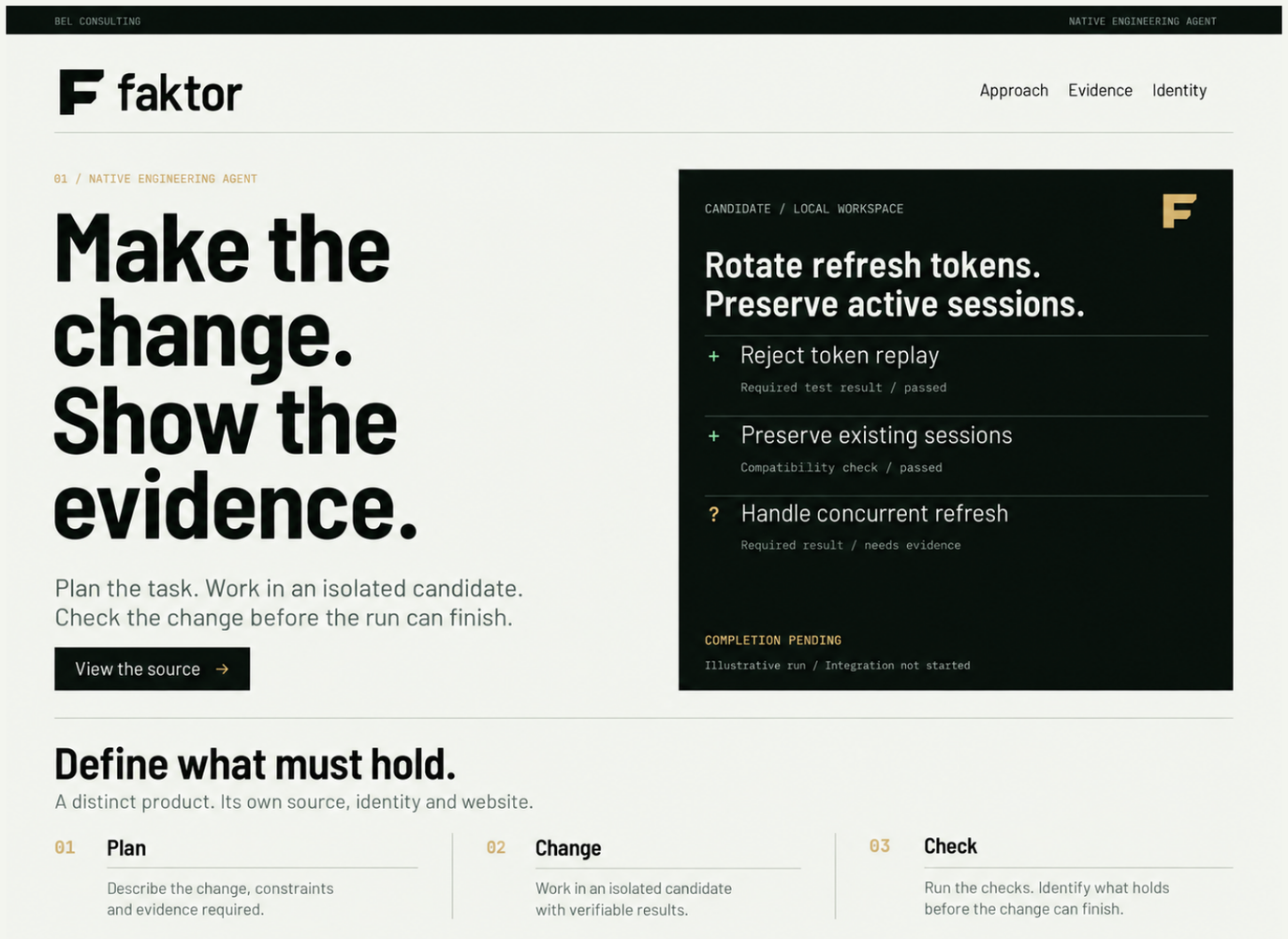


Generated application sketches are not exact product screenshots or performance claims. Use the native board, tokens, source-grounded copy and implemented website for production decisions.

INDIVIDUAL WEBSITE VISUAL PRESENTATION

Faktor / Website design

The concept establishes the product's own composition and typography.



This is a visual direction specimen. The implemented multi-page site uses native logo artwork, local fonts and accessible local interactions. It is a separate origin and project.

IMPLEMENTED INFORMATION ARCHITECTURE

A complete, independent website.

Distinct page structures, useful product depth and appropriate source-based next steps.

Page	Job	Implemented content
Home	Explain the engineering work	Seven editorial sections, candidate/evidence example, Work / Inspect / History tabs, source setup and FAQ.
Approach	Expose the completion chain	Intent, planning, candidate isolation, check acceptance, named claims, integration, finish actions and ledger definition.
Identity	Provide the complete brand	Light/reverse native masters, palette, typography, exact board and guide/kit/token downloads.

Interaction

Native buttons switch illustrative tabs with arrow/Home/End keyboard behavior. Copy command handles failure. Theme is stored locally. No agent, model request or repository mutation runs on the marketing site.

https://faktor.windyclock5.chatgpt.site

IMPLEMENTATION HANDOFF

Roll out the artwork at its real source.

This package supplies brand files and websites. The product repositories were reviewed, not modified.

Target	Action
apps/vscode/media/faktor.svg	Replace with bare angular F, preserve currentColor/theme semantics.
apps/jetbrains/frontend/src/main/resources/META-INF/faktor.svg	Replace terminal-like icon with the same bare angular F, match host constraints.
docs/ and repository headers	Use custom wordmark/lockup and industrial editorial templates; preserve product naming and source links.
release metadata/app assets	Use exact supplied master and appropriate app tile; respect branding scan and licensing.

Preserve the host

Use the bare currentColor Cut F consistently in both IDEs. Keep native theme, reading/editor fonts and focus tokens. Preserve Work / Inspect / History and the conversation-first surface.

The target map is a concrete handoff, not a claim that these source edits or release checks were performed in this branding task.

DELIVERABLES AND VERIFICATION

Everything is supplied in its own project.

The guide and kit are standalone; neither contains a sibling product's reusable logo assets.

Folder / file	Contents	Authority
assets/	21 native SVGs; transparent logo PNGs; app sizes; favicon; social, cover and identity board.	Native path artwork
fonts/	Own display/body fonts, IBM Plex Mono, local web formats and OFL license notices.	Unmodified licensed binaries
specifications/	Tokens, geometry, lettering, 4,100+ word strategy, source map, copy and rollout decisions.	Pinned review and explicit design interpretation
briefs/ + previews/	Five generation briefs, five new presentations, native logo and website specimens.	Concept direction; native artwork wins
qa/	Contrast, icon proofs, SVG structure, website checks and rendered PDF review.	Recorded verification scope
Faktor_Visual_Identity.pdf	Complete analysis, standards, applications and reference appendices.	This document

Website validation checks local files, anchors, labels, assets and JavaScript plus local interaction behavior. Managed browser QA was unavailable. PDF pages and native reproduction proofs were rendered and visually inspected.

PINNED RESEARCH INDEX

Product sources / 1

Repository sbelakho2/faktor / c43f97451d68 / source baseline

F01

README.md

Full README; product, setup, runtime and status claims.

F02

docs/specs/autonomous-engineering.md

Full normative specification and descriptive implementation status; normative requirements are not automatically treated as shipped cla

F03

docs/specs/completion_proof.md

Full normative completion specification and implementation notes.

F04

docs/ui-terminology.md

Full shared user-facing terminology.

F05

docs/outcome-ledger.md

Full ledger description and metrics semantics.

F06

docs/architecture.md

Relevant architecture sections sampled: product boundary, state machine, context, providers, editing, snapshots, supervision and schedu

F07

crates/semantic/src/lib.rs

Provider abstraction, validation and native fallback contract; full small module inspected.

F08

crates/verify/src/lib.rs

Project/check derivation and required-only acceptance inspected; tests read as semantics, not executed.

F09

crates/verify/src/provider.rs

Typed provider envelope, candidate/claim binding, outcome semantics and response validation sampled.

F10

crates/verify/src/review.rs

Independent completion-review package description and bounds inspected; no whole-file security audit.

F11

crates/agent/src/runtime/contract_compiler.rs

Full scanner/compiler wiring inspected; bounded and best-effort behavior distinguished from normative intent.

F12

apps/vscode/src/webview.ts

Full webview host and product HTML inspected.

Exact source URLs are linked. Inspection is scoped; long files may have targeted sections reviewed. Complete ranges/hashes where recorded are in product-content.json.

PINNED RESEARCH INDEX

Product sources / 2

Repository sbelakho2/faktor / c43f97451d68 / source baseline

F13

apps/vscode/media/chat.css

Theme inheritance, hierarchy, focus, verification state treatment and reduced-motion rules inspected.

F14

apps/vscode/media/chat.js

Relevant criterion rendering, missing-data handling, durable completion presentation and task finish controls sampled.

F15

docs/certification.md

Completion gate and one-read proof-chain documentation sampled.

F16

crates/orchestrator/src/completion_steps.rs

Current ordered-step contract, native typed SCM reconciliation and configuration blocker inspected; much of the larger file is not audi

F17

docs/native-protocol.md

Task start and outcome endpoints inspected; native protocol boundary cross-checked.

F18

docs/specs/branding.md

Full product naming and scan requirements inspected.

F19

apps/vscode/media/faktor.svg

Existing small activity icon inspected as a rollout target.

F20

apps/jetbrains/frontend/src/main/resources/META-INF/faktor.svg

Existing JetBrains icon inspected as a rollout target.

Exact source URLs are linked. Inspection is scoped; long files may have targeted sections reviewed. Complete ranges/hashes where recorded are in product-content.json.

DESIGN REFERENCES AND FAMILY BENCHMARK

Identification, reproduction and context.

Primary design sources inform the reasoning. They do not certify a mark or prescribe a universal formula.

Paul Rand / Logos, Flags, and Escutcheons

Identification and meaning established through use.

Pentagram / Slack identity

Reproduction across sizes, surfaces and a coherent family.

Chermayeff & Geismar & Haviv / NBC

Simple, distinctive silhouette and purposeful recognition.

W3C / Contrast minimum and non-text contrast

Readable text; state meaning beyond color alone.

Jeremy Tribby / Barlow

Official type source; OFL notices are supplied.

Kiwi CI benchmark: 14-page guide, individual primary/symbol/dark presentations, identity board, analysis, applications and full generation briefs. This package matches that scope and adds native masters and its own multi-page website.

REUSABLE BRIEF APPENDIX

Primary symbol / Full generation brief

Reference: previews/native-light-symbol.png / Native SVG and tokens remain authoritative.

Create a primary standalone symbol presentation for Faktor. Reproduce the attached exact symbol master, alone and optically centered, crisp Ink #08110C on warm Paper #F3F5EF with generous clear space. The symbol is a solid upright angular F, with a longer top arm, shorter middle arm and clipped arm tips. All edges are straight. It must have NO circle, curl, spiral, Koru, curves, round container, extra emblem, AI sparkle or decorative stroke. No words or caption, no palette or framing, no mockup props, shadow, gradient, texture or 3D. Flat high-resolution vector-like graphic, square 1000 by 1000 composition. This product is standalone within the Kiwi CI family.

Use the supplied local reference artwork when regenerating. A generated presentation cannot redefine exact logo geometry, lettering, state semantics, color values or implemented site behavior.

REUSABLE BRIEF APPENDIX

Primary horizontal logo / Full generation brief

Reference: previews/native-light-logo.png / Native SVG and tokens remain authoritative.

Create an exceptionally refined primary logo presentation for FAKTOR, an independent native engineering agent from Bel Consulting. This is a revision of an existing logo and must reproduce the FIRST attached master artwork faithfully: solid angular Cut F followed by the custom lowercase faktor lettering. Do not invent a new font. The f and t have clipped arm terminals and the k is bespoke; the other letters have an engineered semi-condensed rhythm. Maintain their forms, proportions and optical spacing from that master. NO circle, koru, curl, spiral, rounded enclosure, extra emblem, shield, AI sparkle, bot, generic gradient or 3D bevel anywhere in the Faktor symbol. The second attachment is the approved sibling Kiwi CI family reference: use it only for restraint, common Ink/Forest/Paper palette, substantial one-color mark, and quiet presentation quality. Do NOT borrow its bird or its wordmark. Faktor must be separate. Produce a beautifully balanced flat graphic logo proof with the entire horizontal lockup large, crisp near-black Ink #08110C on warm Paper #F3F5EF, generous clear space, centered optically. The cut F is the existing exact master, not a rounded F. Logo lettering should stay exactly like the first attached master and should look intentional, crafted, not fashionable geometric SaaS type. No text other than the logo; no caption, no palette chips, no rule, no mockup props. Single flat solid color, vector-like sharp edges. Landscape 1400 by 850 composition, high resolution, no texture.

Use the supplied local reference artwork when regenerating. A generated presentation cannot redefine exact logo geometry, lettering, state semantics, color values or implemented site behavior.

REUSABLE BRIEF APPENDIX

Dark horizontal logo / Full generation brief

Reference: previews/native-dark-logo.png / Native SVG and tokens remain authoritative.

Create the dark-mode logo presentation for Faktor. Reproduce the attached exact master artwork faithfully, including the fixed letterforms and spacing. Place this entire horizontal logo alone, optically centered, with generous clear space on a flat Ink #08110C background. Artwork color is Ochre #D9BA75. Its solid angular Cut F and bespoke f/k/t letters have clipped straight terminals. Never add circles, koru, spiral or an enclosing emblem. Match the supplied artwork proportions. No extra text, texture, gradient, shadows, bevels, background illustration, props, light source or ornament. Pure crisp one-color flat logo proof, landscape 1400 by 850. This is a sibling of Kiwi CI with its own independent identity; the attached master is authoritative.

Use the supplied local reference artwork when regenerating. A generated presentation cannot redefine exact logo geometry, lettering, state semantics, color values or implemented site behavior.

REUSABLE BRIEF APPENDIX

Visual identity board / Full generation brief

Reference: previews/native-light-logo.png / Native SVG and tokens remain authoritative.

Create a polished, extensive visual identity board for FAKTOR, a standalone native engineering agent from Bel Consulting. Use the attached FIRST exact new custom logo master faithfully: solid angular Cut F and lowercase faktor lettering with bespoke clipped f, k and t, engineered semi-condensed letter rhythm. NO circle, Koru, spiral or curl in the Faktor symbol. Do not replace the custom wordmark with generic Inter. The second attachment is approved Kiwi CI family direction: shared Ink #08110C, Forest #143D2B, Paper #F3F5EF, Sage #92AA99 and disciplined technical tone; no bird for Faktor. Faktor's own accent is Ochre #D9BA75 with Deep Ochre #72541E for dark text on Paper. Use Barlow Semi Condensed style display typography and Barlow body, IBM Plex Mono-like small metadata. Create an art-directed design board with clear editorial composition and generous whitespace, landscape 1600 by 1100. Main area: dominant precise logo, restrained palette specimens with correct labels/hex, type specimen 'Make the change. Show the evidence.' Small lower application area: independent Faktor website composition with narrow large heading and orthogonal engineering-sheet illustration; source/documentation cover, app icon; dark IDE-like Work / Inspect / History product example. Product facts: it plans engineering work, edits in an isolated candidate workspace and retains an evidence basis. Example state clearly says '2 checks passed / 1 pending' with unresolved evidence, not a blanket verified banner. State colors distinct from ochre brand accent. Sibling name belongs only in tiny company footer. Native vector-like logos must remain undistorted. Graphic grammar: thin rules, square or 2px corners, clipped terminal details, orthogonal marks; no generic equal card grid, no neon purple, AI sparkles, robots, fake performance statistics, invented clients, effects or gradient. The board should look like a crafted independent product identity, related to the family through palette and clarity. Select only essential legible short labels; no dense fake text. All elements flat unless a single lightly angled document cover mockup helps demonstrate application.

Use the supplied local reference artwork when regenerating. A generated presentation cannot redefine exact logo geometry, lettering, state semantics, color values or implemented site behavior.

REUSABLE BRIEF APPENDIX

Website visual direction / Full generation brief

Reference: previews/native-website.png / Native SVG and tokens remain authoritative.

Create a carefully art-directed WEBSITE DESIGN presentation for Faktor, an independent native engineering agent from Bel Consulting. This is an edit of the attached native website design specimen. Preserve the precise primary logo (solid angular Cut F and custom lowercase factor spelled faktor), new type character, warm Paper #F3F5EF, Ink #08110C, Ochre #D9BA75 and restrained thin-rule editorial layout. The factor logo contains no circles, koru, curl or spiral. The new lettering has clipped f/t and bespoke k; do not substitute Inter or generic geometric AI type. Barlow Semi Condensed 600/700-style headline, Barlow body, Plex Mono technical metadata. Follow the attached layout: small company strip; logo left and Approach / Evidence / Identity navigation right; large narrow 'Make the change. Show the evidence.' headline on the left; on the right a dark engineering sheet for 'Rotate refresh tokens. Preserve active sessions.' with two passed criteria and one missing concurrent-refresh result, clearly 'Completion pending'; no global verified badge. Keep illustration labeled Illustrative run, integration not started. A real View the source CTA belongs below the exact supporting copy. Below the hero, show one editorial row 'Define what must hold.' and the beginning of three numbered Plan / Change / Check columns, not a generic card grid. Refine spacing and pixel-level alignment while retaining this distinct composition and typography. No fake testimonials, invented customers, pricing, human imagery, neural mesh, gradient, 3D device mockup, heavy blur or dashboard clutter. Render as a crisp flat desktop website design, landscape 1400 by 1020; this is a visual concept, not a screenshot of a deployed browser. Native graphic clarity, confident specific lettering, independently branded but in the Kiwi CI family.

Use the supplied local reference artwork when regenerating. A generated presentation cannot redefine exact logo geometry, lettering, state semantics, color values or implemented site behavior.

faktor

Complete website / edition 4 / 12 October 2026

Faktor
The full website.

Every current page, complete documentation, identity system and dated prior website content in a paginated edition.

01 / Make the change. Show the evidence.
02 / Define the finish. Then make the change.
03 / Source to workspace
04 / A result belongs to a change
05 / Count the attempts. Keep the basis.
06 / The workflow stays yours.
07 / Useful work starts with a bounded request.
08 / Faktor visual identity
09 / Current edition and complete prior reference
10 / Work, inspect and finish deliberately.

Live website: https://faktor.windyclock5.chatgpt.site
Current source: 16e8bd4cf2832affbb6a2e9e9281beee7527ac4d
Local interactive examples are design simulations. PDFs show static states; live pages retain the controls.

Bel Consulting

Native engineering agent

faktor

Approach In your editor Documentation Identity

01 / Native engineering agent

Make the change. Show the evidence.

Faktor plans the task, works in an isolated candidate workspace, and checks what changed before the run can finish.

View the source ↗

See the engineering loop

VS CODE / JETBRAINS / CLI · NATIVE RUST RUNTIME

CANDIDATE / LOCAL WORKSPACE

Rotate refresh tokens. Preserve active sessions.

✓ Reject token replay
Required test result / passed

✓ Preserve existing sessions
Required compatibility check / passed

◇ Handle concurrent refresh
Required result / needs evidence

COMPLETION PENDING 02 / 03

Illustrative run Integration not started

02 / Define what must hold

A task needs more than a prompt.

Keep the goal, acceptance criteria, invariants and proof obligations together. The plan orders the work; the checks evaluate the change.

01 / PLAN

Give the work conditions.

Translate the objective into ordered tasks and observable criteria. Keep the risks and dependencies in view.

02 / CHANGE

Work in a candidate.

Use an isolated workspace with transactions and checkpoints. Separate editing from integration.

03 / CHECK

Keep the basis visible.

Connect required checks and named claims to the candidate they actually evaluated.

Explore the approach →

03 / A result belongs to a claim

See what passed. See what is missing.

Inspect connects criteria, candidate state, check results and review. A missing required result keeps completion pending.

Illustrative run / Rotate refresh tokens without breaking active sessions

Criterion	State	Evidence basis
Replayed refresh tokens fail	√ Passed	Required test result
Existing sessions remain valid	√ Passed	Required compatibility check
Concurrent refresh yields one valid result	◇ Needs evidence	Required result not recorded

2 criteria passed / 1 needs evidence / Candidate not integrated

04 / Done has conditions

Choose where the run stops.

Leave the changes locally, commit, push or open a pull request. Requested finish actions are recorded and checked as part of completion.

01	02	03	04
Local change	Commit	Push	Pull request

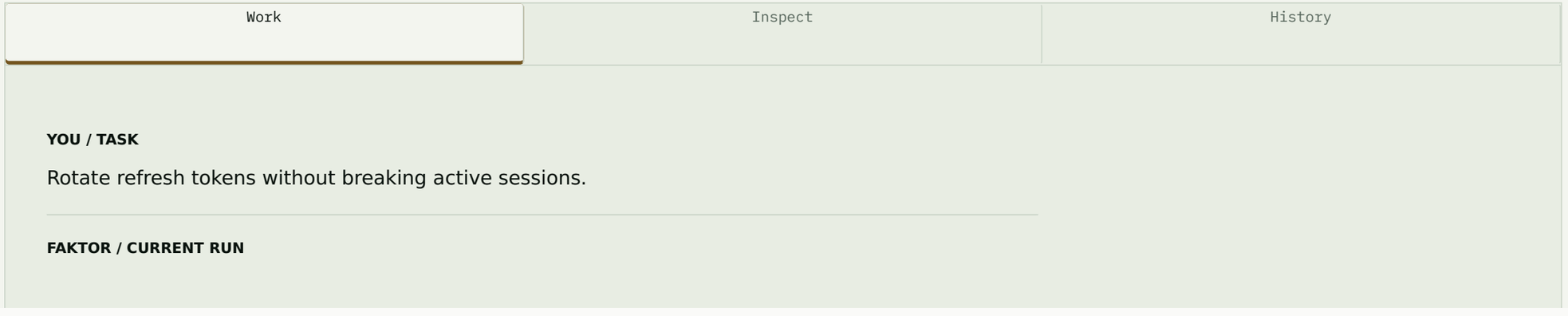
Native pull requests use the configured SCM integration. Missing configuration stays visible as a blocker.

Expanded editor flow / design proposal

05 / In your editor

Work stays in front.

Conversation, current run, composer. Open Inspect for the plan and evidence. History keeps durable outcomes within reach.



The candidate is ready for review. Two required criteria passed. The concurrent refresh result is still missing.

Add a constraint or inspect the evidence...

Illustrative interface. Tabs switch examples locally.

06 / The run leaves a record

Spend. Rework. Outcome.

Verified cost / VCSC

All eligible attempted spend / verified changes.

First-pass result

Keep initial verification and repair attempts distinct.

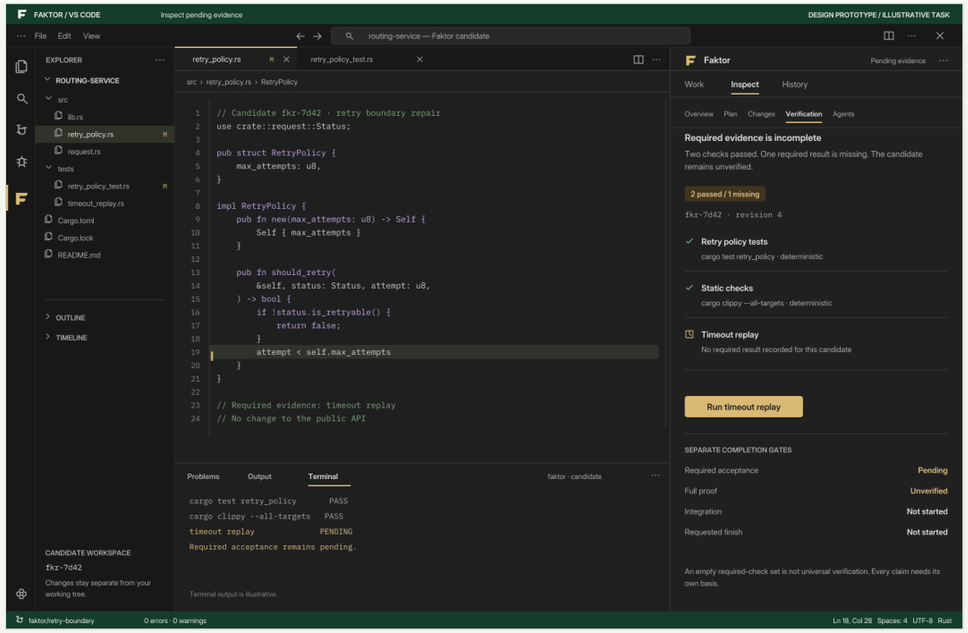
Final basis

Record acceptance, integration and the verification basis together.

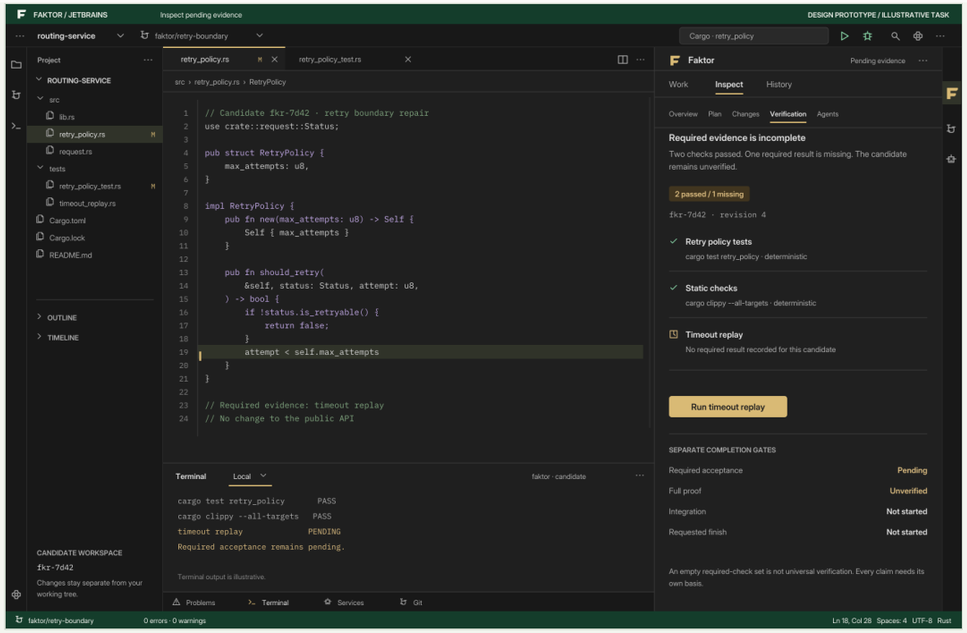
A full editor experience

One task. Two native hosts.

Explore the complete proposed Faktor experience in VS Code and JetBrains: setup, planning, permissions, evidence, diff review, recovery and history.



VS CODE / EVIDENCE PENDING / DESIGN PROTOTYPE



JETBRAINS / EVIDENCE PENDING / DESIGN PROTOTYPE

Explore both editor flows → Download the UI/UX specification

Concrete engineering work

Choose a task worth reviewing.

01

Preserve behavior during a migration.

Define compatibility criteria and invariants before a session, schema or API change. The evidence remains attached to the candidate that was checked.

02

Repair a failure with a visible basis.

Start with the failed check or counterexample. Keep the repair scoped and inspect the candidate before choosing an integration or finish action.

03

Work with deliberate model choice.

Keep model configuration separate from the engineering contract. Review native, local or hosted steps through their recorded outcomes and spend.

Explore task patterns →

07 / Start in your workspace

Source, setup and the runtime.

Explore the native runtime and its IDE surfaces. Start with the source workspace and its setup guide.

Read setup ↗

SOURCE-WORKSPACE COMMAND

```
cargo run -p faktor-cli -- run --data-dir /tmp/faktor-demo "explain this repo"
```

Copy command

VS Code: Faktor: Open
JetBrains: Faktor tool window

Useful boundaries

Before you begin.

Does Faktor require another Bel Consulting product?

No. Native intelligence and verification form the baseline. Optional providers can enrich neutral interfaces.

Does a passing check mean the whole change is verified?

A result has a scope. Inspect the required criteria, named claims, candidate identity, integration state and final basis together. An empty set of required checks can pass without establishing a verified change.

What stays in the IDE?

The product uses the host's theme and typography. Work, Inspect and History are shared vocabulary across its native surfaces.

faktor

An engineering agent for changes you can inspect.

Source & documentation ↗

Identity & downloads

Research & prior edition

An independent Bel Consulting product.
Also in the family: KiwiCaptcha · Kiwi CI

Bel Consulting Native engineering agent

faktor

Approach In your editor Documentation Identity

Approach / Engineering loop

Define the finish. Then make the change.

A goal, a candidate and a visible evidence basis. Each part does a different job.

01 / INTENT

Compile the task.

Keep the goal, criteria, invariants, risks and proof obligations together before work starts.

02 / PLAN

Order the work.

A task graph records dependencies and makes the plan inspectable.

03 / CANDIDATE

Separate the edit.

Transactions, checkpoints and isolated candidates bound changes before integration.

The completion chain

A green result has a scope.

Required-check acceptance, claim evidence, integration and requested finish actions stay distinct. A result for an earlier candidate cannot certify a different landed state.

1. **Required acceptance.** Every required result must be present and pass. Missing evidence stays Pending; a failure stays Fail.
2. **Claim evidence.** Name the claim and the candidate snapshot. Unavailable means no verdict; refusal is a separate outcome.
3. **Integration.** Keep verified and landed identities visible. Recheck a state that differs from the verified candidate.
4. **Finish actions.** Local changes, commit, push and native pull request are user choices with durable outcomes.

The outcome ledger counts a verified change when it is accepted and integrated with a deterministic-checks final basis. A passing empty check set does not establish that condition.

One workflow in the host.

Work keeps conversation and the current task in front. Inspect exposes the plan, agents and evidence. History holds outcomes. IDE surfaces inherit their editor’s typography and theme.

Start from the source.

The source workspace includes the native Rust runtime, CLI, VS Code and JetBrains surfaces. Optional providers do not become product dependencies.

Open the repository ↗ | Explore the example

Explore the contract

A complete route through the work.

- Evidence and completion ↗
- Outcome and cost definitions ↗
- Runtime and providers ↗
- VS Code and JetBrains design ↗

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Approach In your editor Documentation Identity

Documentation / Source to workspace

Start with a workspace. Keep the finish explicit.

Runtime setup, editor configuration, task criteria, evidence and recorded outcomes in one practical guide.

ON THIS PAGE

Source quickstart
Editor setup
Task contract
Completion
Recovery

Source quickstart

Build and run from the Faktor source workspace. These commands are source-development entry points; use the repository's current installation notes for packaged releases.

```
cargo build --workspace
```

Copy build command

```
cargo run -p faktor-cli -- doctor
```

Copy doctor command

```
cargo run -p faktor-cli -- run --data-dir /tmp/faktor-demo "explain this repo"
```

Copy run command

Choose the host.

VS Code opens Faktor through its command and webview. JetBrains uses a right-anchored tool window. Keep binary discovery, daemon transport and project binding understandable before the first run.

The expanded design introduces a consistent Work / Inspect / History structure while retaining host-specific behavior. The current VS Code History surface and common finish selector are proposed additions; JetBrains already has the three top-level views.

Open the complete editor design →

Give the task conditions.

Record the goal, acceptance criteria, invariants, risks and requested finish action. An isolated candidate separates editing from integration. Context and permissions stay scoped to the task.

Completion has a chain.

Required acceptance, claim evidence, candidate identity, integration and requested finish actions are distinct. Missing data stays visible. An empty required-check set can pass without establishing a verified change.

Read the evidence contract →

Recover without losing the task.

Preserve the draft, attached context and finish choice when a start fails. Explain daemon, provider, verification and SCM blockers separately. A reconnect must not silently skip malformed durable frames or advance beyond unseen events.

Runtime and provider configuration →

Reviewed source: Faktor README at 16e8bd4cf283. Detailed source and proposed-control mapping are supplied in the UI/UX specification.

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Start with a workspace.Keep the finish explicit. 4

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Evidence / Claim, candidate and result

A result belongs to a change.

Review what was requested, which candidate was checked and what can still prevent completion.

The completion chain

Four questions before the finish.

01

Did the required criteria pass?

Pass requires present, passing required results. A missing result is Pending; a failed required check is Fail. An empty set is not a complete proof of the change.

02

What claim did the evidence answer?

Keep the named claim, source and candidate snapshot visible. Unavailable provides no verdict. A refusal is not an observation of correctness.

03

Is this the state that will land?

Expose verified and landed identities. A different landed snapshot needs its own evidence basis; a result for earlier code cannot certify later edits.

04

Did the requested actions succeed?

Local changes, commit, push and native pull request carry distinct requested outcomes. Missing SCM configuration, skipped work and unsuccessful steps stay explicit.

Illustrative refresh-token task

Keep the unresolved result.

Criterion	Required result	Next useful action
Reject replayed refresh tokens	Passed / test result	Inspect the recorded basis.
Preserve active sessions	Passed / compatibility check	Review the candidate identity.
Concurrent refresh yields one valid result	Missing / Pending	Run or acquire the required result.

A result belongsto a change. 2

Illustrative task. No live run or verification request is made by this website.

From a result to an outcome

The final basis stays in the record.

The verified-change definition requires accepted and integrated with a deterministic-checks final basis. Metrics retain this boundary instead of counting every green check as a successful engineering change.

Explore outcome and cost definitions → Inspect the interface design →

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Source & documentation ↗

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Outcomes / Measured records

Count the attempts. Keep the basis.

Cost and performance mean more when failed and cancelled work remains in the denominator's context.

Current source definitions

Three different measurements.

Measure	Definition	Reading rule
Verified Cost per Successful Change	All eligible attempted spend divided by verified changes. Failed and cancelled attempted spend remains eligible under the ledger's rules.	Keep period and cohort together; an absent denominator is not a zero-cost success.
Successful-run cost	Spend associated with successful runs.	A separate measure from VCSC; do not silently substitute it.

Measure	Definition	Reading rule
Recorded final basis	Accepted / integrated / deterministic-checks basis.	Keep the source-defined verified-change boundary explicit.

Local execution cohorts

Keep the comparison like for like.

01

Local-only

Runs whose model steps stayed local.

02

Local-final

Runs with a local final path under the current ledger definition.

03

Local-first with fallback

Runs beginning locally and using a hosted fallback. Keep escalation and spend visible.

> The current recorder does not establish avoided hosted inference, verification process spend or frontier-escalation savings. Those fields are honest zeros at this source baseline, rather than measured savings claims.

What the interface should show

A run record you can inspect.

Field	Presentation
Outcome and final basis	Accepted, integrated and verification basis in separate labeled rows.
Spend	Recorded values where available; “Cost unavailable” when not available.
Rework and first-pass result	Initial verification and repair attempts remain distinct.
Eligible attempted spend	Available in outcome context, not hidden because a run failed.
Cohort and period	Visible alongside aggregate comparisons.

Current source: outcome-ledger.md. The prior edition’s metric wording is retained as dated reference.

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Bel Consulting Native engineering agent

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Approach In your editor Documentation Identity

Runtime / Provider configuration

The workflow stays yours. The model is a choice.

Connect the runtime deliberately, then keep model and optional service configuration separate from the engineering contract.

Before the first run

Make readiness legible.

01

Locate the binary.

Show the configured path, discovery result and executable readiness. A bare checksum match means it matches the recorded checksum; it is different from a signed launcher’s signature verification.

02

Bind the project and daemon.

Explain project scope, transport and autostart state. A disconnected daemon is a connectivity condition, not a failed verification verdict.

03

Configure the chosen model.

Keep local or hosted provider details in settings. Secrets remain masked, and diagnostic copies redact credentials.

04

Inspect optional adapters.

Optional intelligence and verification providers enrich neutral interfaces. The native baseline does not acquire a sibling-product dependency.

Where the editor puts the detail

Keep Work uncluttered.

Work

Conversation, current run, scoped requests and composer. A short configuration status belongs beside the task; raw URLs, cursors and transport details belong in diagnostics.

Settings and diagnostics

Binary discovery, daemon lifecycle, provider readiness and redacted support details. Name the failed seam and the next safe recovery action.

Explore the configuration and diagnostics states →

---

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An engineering agent for changes you can inspect.

Source & documentation ↗

Identity & downloads

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Bel Consulting Native engineering agent

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Approach In your editor Documentation Identity

Task patterns / Reviewable engineering

Useful work starts with a bounded request.

Choose observable conditions and the stop point before handing the task to an engineering agent.

Migration

Rotate refresh tokens without breaking sessions.

Contract field	Example
Goal	Rotate refresh tokens.
Acceptance	Reject replay; preserve sessions; constrain concurrent refresh behavior.
Invariant	Active sessions continue under the stated compatibility conditions.

Contract field	Example
Candidate	Isolated workspace and snapshot identity.
Finish choice	Leave changes locally for review.

> Illustrative brief from the design study. The website does not run this migration or claim its results.

Failure repair

Start with the counterexample.

01

Attach the failing basis.

Keep the relevant test or observed failure alongside the task.

02

Constrain the repair.

State preserved behavior, affected scope and the requested verification.

03

Review before landing.

Inspect changed files, unrelated changes and the evidence for the candidate being integrated.

Budget-aware work

Keep spend and rework observable.

Recorded spend, attempted outcomes and model cohorts help evaluate an engineering workflow. They do not establish unmeasured token savings or guarantee that a cheaper model produces the same result.

Read the current outcome definitions →

---

faktor

An engineering agent for changes you can inspect.

Source & documentation ↗

Identity & downloads

Research & prior edition

An independent Bel Consulting product.
Also in the family: KiwiCaptcha · Kiwi CI

Bel Consulting — Native engineering agent

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Approach · In your editor · Documentation · Identity

Faktor / Visual identity

Built for a clear finish.

Clipped terminals. A clear initial. Deliberate evidence.

The fixed logo masters



INK / PAPER — Download SVG



OCHRE / INK — Download SVG

Fixed outlined artwork. Preserve letter spacing and H/4 clear space. Preferred symbol size: 24 px; tiny icons supplied separately.

Palette / Family foundations, own accent

Ink, Paper and Ochre.



Ink	Forest	Paper	Sage	Ochre	Ochre Deep
#08110C	#143D2B	#F3F5EF	#92AA99	#D9BA75	#72541E

Use the soft accent on Ink; the deeper accent for readable text on Paper. State colors keep their own semantic meaning.

Typography / Its own voice

Make the change. Show the evidence.

Barlow Semi Condensed for headings.
Barlow for the reading surface.

Built for aclear finish.

2

IBM Plex Mono for technical metadata. The logo is fixed outlined artwork; host interfaces retain their own typography.

The identity in use

A complete application system.



Built for aclear finish.

3

Application concept. Use the native SVG masters for production artwork.

Downloads

Everything in its own project.

Logo suite, source analysis, usage rules, design tokens, font licenses, applications and generation briefs.

Visual identity guide	PDF
Complete identity kit	ZIP
Standalone symbol	SVG
Stacked logo	SVG
Application icon	SVG
Design tokens	JSON

Current application package

The websites, in PDF form.

Complete website edition	PDF / ALL PRODUCT PAGES
All editor visual assets	SVG + PNG / BOTH HOSTS / BOTH THEMES

VS Code & JetBrains UI/UX

PDF / FLOWS, STATES AND HANDOFF

Retained prior content

DATED RESEARCH & WEBSITE

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Source & documentation ↗

Identity & downloads

Research & prior edition

An independent Bel Consulting product.
Also in the family: KiwiCaptcha · Kiwi CI

Built for aclear finish. 5

Bel Consulting Native engineering agent

faktor

Approach In your editor Documentation Identity

Research / Retained content

Current edition. Complete prior reference.

All earlier identity work and website content remain available, with their original source baselines clearly dated.

Current baseline / 12 October 2026

Faktor at 16e8bd4cf283

Cost definitions now distinguish all eligible attempted spend, successful-run cost and local execution cohorts. VS Code and JetBrains have different implemented surfaces; the complete shared UI is explicitly a design extension.

Current source baseline: 16e8bd4cf2832affbb6a2e9e9281beee7527ac4d
Prior research baseline: c43f97451d68b2d8e4884e39c3b475d2737a8b27

Prior website / 11 October 2026

The earlier edition, retained.

Index / Prior edition ↗

Approach / Prior edition ↗

Brand / Prior edition ↗

The expanded identity guide retains every original page, image, construction rule, source analysis and generation brief. Dated corrections introduce the new edition.

faktor

An engineering agent for changes you can inspect.

Source & documentation ↗

Identity & downloads

Research & prior edition

An independent Bel Consulting product.
Also in the family: KiwiCaptcha · Kiwi CI

Bel Consulting Native engineering agent

faktor

Approach In your editor Documentation Identity

Faktor / VS Code & JetBrains

Work, inspect and finish deliberately.

A complete editor design: task capture, scoped permissions, candidate review, evidence, recovery and durable outcomes.

Interactive editor design

Switch the host, theme and flow state, or use the controls inside the tool view.

> Local design prototype. No executable starts, repository changes, model requests, permissions or SCM actions occur here. Proposed controls and current-source differences are documented below.

Editor VS Code ˅ **Host theme** Dark ˅ **Flow state** Evidence pending ˅

routing-service — Visual Studio Code — DESIGN PROTOTYPE / ILLUSTRATIVE TASK

PROJECT / ROUTING-SERVICE

- routing-service
  - src
    - auth
      - session.rs
      - session_test.rs
  - tests
  - Cargo.toml
  - README.md

session.rs | session_test.rs

routing-service › src › auth › session.rs

```
1  use crate::auth::RefreshToken;
2
3  pub fn rotate_session(token: RefreshToken) {
4      // Candidate change / illustrative code
5      validate_session(&token);
6      reject_replayed_token(&token);
7
8      // Required concurrent-refresh result is missing
9      let next = issue_rotated_token(&token);
10     preserve_active_session(&token, &next);
11     record_rotation(&next);
12 }
```

TERMINAL / ILLUSTRATIVE OUTPUT

```
Required test: replay rejection / passed
Compatibility: active sessions / passed
Concurrent refresh result / missing
Completion remains pending.
```

Faktor — Settings

Work | Inspect | History

Plan | Agents | Evidence | Diff

Completion pending

Two required criteria passed. One required result is missing.

✓ Reject token replay
Required test result / passed

✓ Preserve active sessions
Compatibility check / passed

◇ Concurrent refresh yields one valid result
Required result / missing

Candidate	candidate-08 / illustrative
Integration	Not started
Final basis	Not available

Review changed files | Open criteria

Evidence references

Replay test result

Session compatibility result

Concurrent-refresh result not recorded

routing-service / candidate workspace — Evidence pending

Work, inspectand finish deliberately. 2

Completion remains pending.

Next design state → | Reset specimen | **Complete editor asset kit ↓** | Complete UI/UX PDF ↓

Host-specific design

Familiar in each editor.

Area	VS Code	JetBrains
Placement	Movable view/container and webview. Current source has Chat and Inspect disclosure.	Right-anchored Faktor tool window with Work / Inspect / History.
Shared navigation	A proposed Work / Inspect / History shell, including a visible History addition.	Retains the existing top-level views and makes their content more coherent.
Finish choice	Hierarchical finish levels already exist.	Current independent checkboxes become a proposed common four-level selector.
Diff review	Proposed native side-by-side diff adapter.	Proposed native DiffManager adapter.
Typography and theme	Inherit VS Code font, editor settings, focus and semantic tokens.	Inherit JetBrains UI font, editor settings and native theme.
Native confirmation	Keep exact-path service-start confirmation and fail-closed checks.	Keep exact-path service-start confirmation and fail-closed checks.

Thirty-six specified screens

The complete flow, beyond the hero.

01

Capture and preserve the task.

Project readiness, binary discovery, native start consent, typed task criteria, context attachments and captured finish choices.

02

Work with bounded authority.

Named current work, stop conditions and target/reason/scope permission review. No blanket grant is implied.

03

Inspect the evaluated candidate.

Plan, agents, all evidence references, review/verification, changed files and matching verified/landed identities.

04

Recover coherently.

Ambiguous submission, reconnect, malformed durable frame, unavailable proof, missing cost and SCM blockers retain useful state.

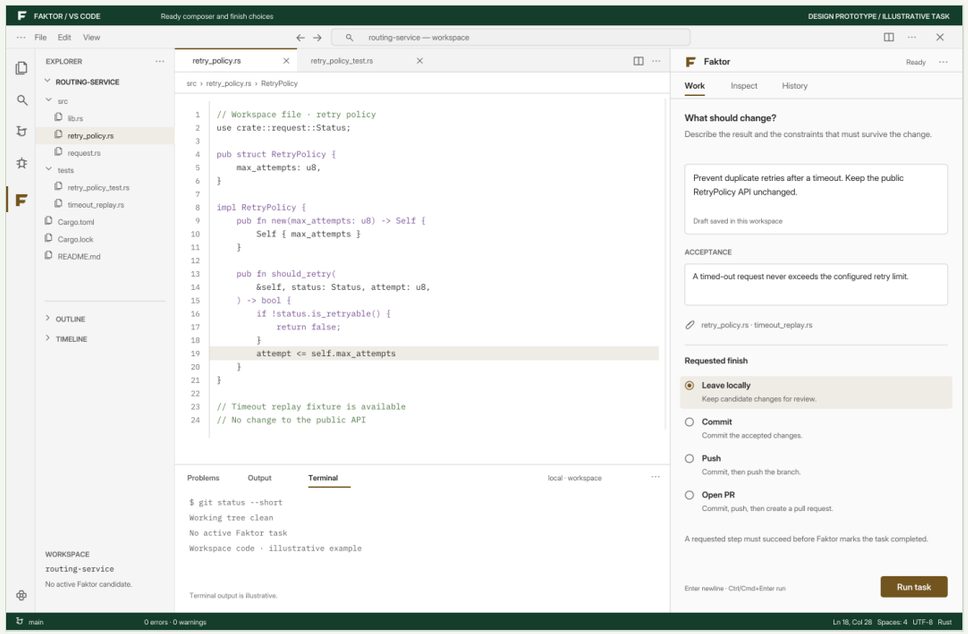
05

Record the outcome.

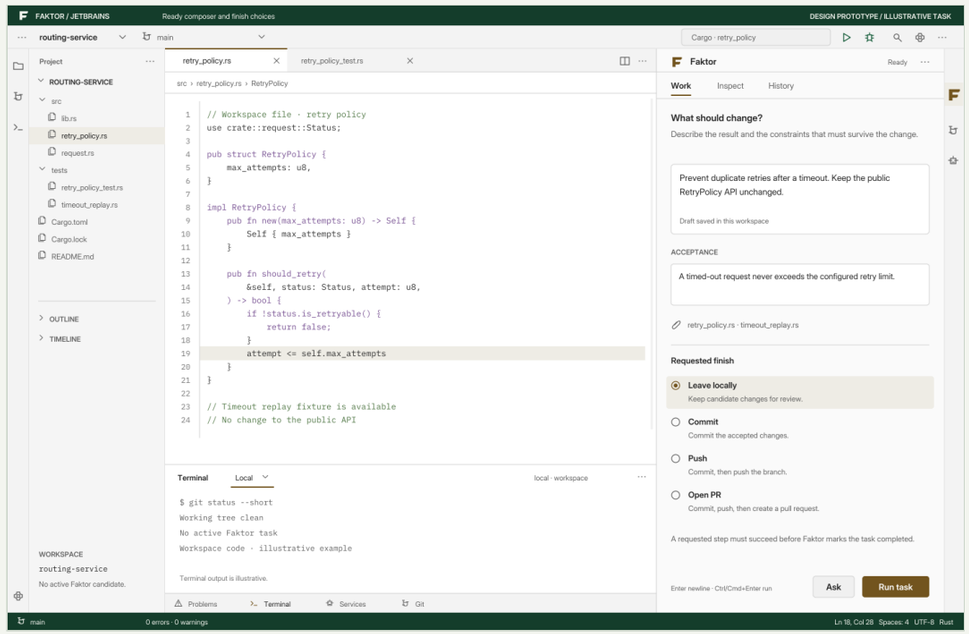
History separates conversation completion, engineering acceptance, integration, requested steps, final basis and recorded cost.

Reproduction in both themes

Native layout specimens.



VS CODE / LIGHT / COMPOSER



JETBRAINS / LIGHT / COMPOSER

The complete kit contains 56 native SVG and PNG screen assets: 14 core states × 2 hosts × 2 themes. The standalone specification maps 36 screens, 34 states and 16 end-to-end flows to the current source and proposed work.

Keyboard and recovery

Small details that keep the work usable.

Condition	Interaction contract
Keyboard	Roving main tabs, visible focus, Ctrl/Cmd + Enter scoped to the composer, Escape for confirmation recovery.
Narrow pane	Wrap paths and key/value rows, stack composer actions, move only secondary controls to accessible overflow.
Async refresh	Retain item identity and coherent snapshot; do not steal focus or skip an unprocessed durable frame.
Submission	One admitted logical request; keep the captured draft, finish and context when the response is ambiguous.
Unavailable information	Show unavailable proof/cost explicitly. Unavailable cost is not zero.
Diagnostics	Mask credentials; copied support details are redacted.

Implementation handoff

Build the shared contract in native adapters.

The source study distinguishes presentation refinements, UI work using existing backend support, new host integrations and backend extensions. The artifact is a complete design and handoff, not a claim that these extension changes are deployed.

Download the complete specification

Runtime and editor setup

faktor

An engineering agent for changes you can inspect.

Source & documentation ↗

Identity & downloads

Research & prior edition

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Work, inspectand finish deliberately. 7

Bel Consulting Native engineering agent

faktor

Approach Evidence Identity

Prior edition / 11 October 2026 Retained design and pinned research. Current source and expanded experience are in the updated edition. Open current website →

Approach / Engineering loop

Define the finish. Then make the change.

A goal, a candidate and a visible evidence basis. Each part does a different job.

01 / INTENT

Compile the task.

Keep the goal, criteria, invariants, risks and proof obligations together before work starts.

02 / PLAN

Order the work.

A task graph records dependencies and makes the plan inspectable.

03 / CANDIDATE

Separate the edit.

Transactions, checkpoints and isolated candidates bound changes before integration.

The completion chain

A green result has a scope.

Required-check acceptance, claim evidence, integration and requested finish actions stay distinct. A result for an earlier candidate cannot certify a different landed state.

1. **Required acceptance.** Every required result must be present and pass. Missing evidence stays Pending; a failure stays Fail.
2. **Claim evidence.** Name the claim and the candidate snapshot. Unavailable means no verdict; refusal is a separate outcome.
3. **Integration.** Keep verified and landed identities visible. Recheck a state that differs from the verified candidate.
4. **Finish actions.** Local changes, commit, push and native pull request are user choices with durable outcomes.

The outcome ledger counts a verified change when it is accepted and integrated with a deterministic-checks final basis. A passing empty check set does not establish that condition.

One workflow in the host.

Work keeps conversation and the current task in front. Inspect exposes the plan, agents and evidence. History holds outcomes. IDE surfaces inherit their editor’s typography and theme.

Start from the source.

The source workspace includes the native Rust runtime, CLI, VS Code and JetBrains surfaces. Optional providers do not become product dependencies.

Open the repository ↗ | Explore the example

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Source & documentation ↗

Logo & design assets

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Define the finish.Then make the change. 3

Bel Consulting Native engineering agent

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Approach Evidence Identity

Prior edition / 11 October 2026 Retained design and pinned research. Current source and expanded experience are in the updated edition. Open current website →

Faktor / Visual identity

Built for a clear finish.

Clipped terminals. A clear initial. Deliberate evidence.

The fixed logo masters



INK / PAPER Download SVG



OCHRE / INK Download SVG

Fixed outlined artwork. Preserve letter spacing and H/4 clear space. Preferred symbol size: 24 px; tiny icons supplied separately.

Palette / Family foundations, own accent

Ink, Paper and Ochre.

Ink	Forest	Paper	Sage	Ochre	Ochre Deep
#08110C	#143D2B	#F3F5EF	#92AA99	#D9BA75	#72541E

Use the soft accent on Ink; the deeper accent for readable text on Paper. State colors keep their own semantic meaning.

Typography / Its own voice

Built for aclear finish. 2

Make the change. Show the evidence.

Barlow Semi Condensed for headings.
Barlow for the reading surface.

IBM Plex Mono for technical metadata. The logo is fixed outlined artwork; host interfaces retain their own typography.

The identity in use

A complete application system.

FAKTOR / VISUAL IDENTITY

faktor

BEL CONSULTING / INDEPENDENT PRODUCT

Make the change.
Show the evidence.

PALETTE / SHARED FOUNDATIONS, OWN ACCENT

Ink	Forest	Paper	Sage	Ochre	Ochre Deep
#08110C	#143D2B	#F3F5EF	#92AA99	#D9BA75	#72541E

TYPOGRAPHY / OWN VOICE

Barlow Semi Condensed

Character in the lettering.

Readable product language. Exact labels.

IBM PLEX MONO / TECHNICAL METADATA

PRODUCT WEBSITE / NATIVE DESIGN SPECIMEN

faktor

Make the change.
Show the evidence.

View the source

COMPONENT / ILLUSTRATIVE STATE

CANDIDATE / LOCAL WORKSPACE

2 checks passed / 1 pending

Concurrent refresh / needs evidence

COMPLETION PENDING

APP / ICON

FIXED OUTLINED WORDMARK

faktor

H/4 clear space. One-color masters. No effects.

Source review / logo suite / applications / own website

NATIVE MASTER / VERSION 3.0

Application concept. Use the native SVG masters for production artwork.

Downloads

Everything in its own project.

Logo suite, source analysis, usage rules, design tokens, font licenses, applications and generation briefs.

Visual identity guide	PDF
Complete design kit	ZIP
Standalone symbol	SVG
Stacked logo	SVG
Application icon	SVG
Design tokens	JSON

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Source & documentation ↗

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Bel Consulting

Native engineering agent

faktor

Approach Evidence Identity

Prior edition / 11 October 2026 Retained design and pinned research. Current source and expanded experience are in the updated edition. Open current website →

01 / Native engineering agent

Make the change. Show the evidence.

Faktor plans the task, works in an isolated candidate workspace, and checks what changed before the run can finish.

View the source ↗

See the engineering loop

CANDIDATE / LOCAL WORKSPACE

Rotate refresh tokens. Preserve active sessions.

✓ Reject token replay
Required test result / passed

✓ Preserve existing sessions
Required compatibility check / passed

◇ Handle concurrent refresh
Required result / needs evidence

COMPLETION PENDING 02 / 03

Illustrative run Integration not started

VS CODE / JETBRAINS / CLI · NATIVE RUST RUNTIME

02 / Define what must hold

A task needs more than a prompt.

Keep the goal, acceptance criteria, invariants and proof obligations together. The plan orders the work; the checks evaluate the change.

01 / PLAN

Give the work conditions.

Translate the objective into ordered tasks and observable criteria. Keep the risks and dependencies in view.

02 / CHANGE

Work in a candidate.

Use an isolated workspace with transactions and checkpoints. Separate editing from integration.

03 / CHECK

Keep the basis visible.

Connect required checks and named claims to the candidate they actually evaluated.

Explore the approach →

03 / A result belongs to a claim

See what passed. See what is missing.

Inspect connects criteria, candidate state, check results and review. A missing required result keeps completion pending.

Illustrative run / Rotate refresh tokens without breaking active sessions

Criterion	State	Evidence basis
Replayed refresh tokens fail	√ Passed	Required test result
Existing sessions remain valid	√ Passed	Required compatibility check
Concurrent refresh yields one valid result	◇ Needs evidence	Required result not recorded

2 criteria passed / 1 needs evidence / Candidate not integrated

04 / Done has conditions

Choose where the run stops.

Leave the changes locally, commit, push or open a pull request. Requested finish actions are recorded and checked as part of completion.

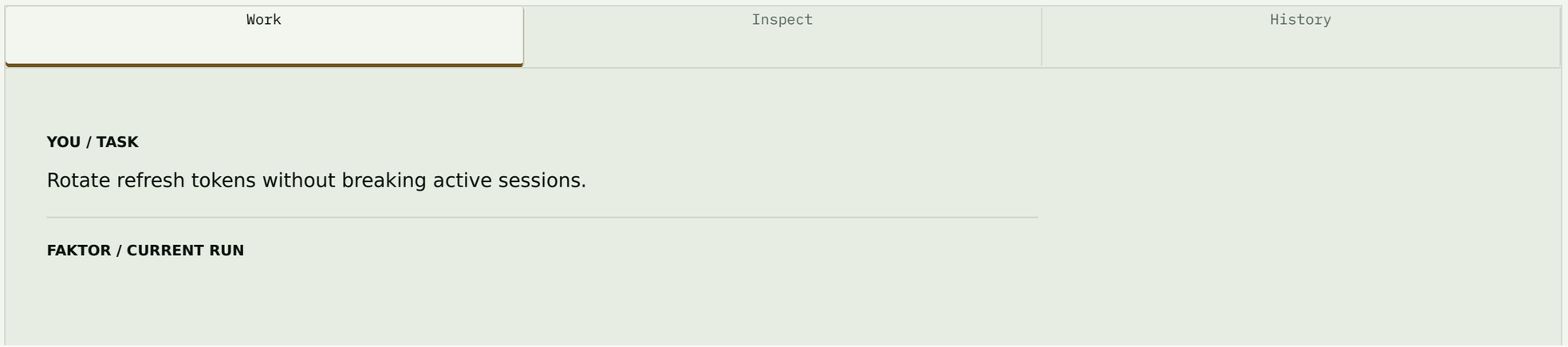
01	02	03	04
Local change	Commit	Push	Pull request

Native pull requests use the configured SCM integration. Missing configuration stays visible as a blocker.

05 / In your editor

Work stays in front.

Conversation, current run, composer. Open Inspect for the plan and evidence. History keeps durable outcomes within reach.



The candidate is ready for review. Two required criteria passed. The concurrent refresh result is still missing.

Add a constraint or inspect the evidence...

Illustrative interface. Tabs switch examples locally.

06 / The run leaves a record

Spend. Rework. Outcome.

Verified cost

Recorded model and verification spend per successful change.

First-pass result

Keep initial verification and repair attempts distinct.

Final basis

Record acceptance, integration and the verification basis together.

07 / Start in your workspace

Source, setup and the runtime.

Explore the native runtime and its IDE surfaces. Start with the source workspace and its setup guide.

Read setup ↗

SOURCE-WORKSPACE COMMAND

```
cargo run -p faktor-cli -- run --data-dir /tmp/faktor-demo "explain this repo"
```

Copy command

VS Code: Faktor: Open
JetBrains: Faktor tool window

Useful boundaries

Before you begin.

Does Faktor require another Bel Consulting product?

No. Native intelligence and verification form the baseline. Optional providers can enrich neutral interfaces.

Does a passing check mean the whole change is verified?

A result has a scope. Inspect the required criteria, named claims, candidate identity, integration state and final basis together. An empty set of required checks can pass without establishing a verified change.

What stays in the IDE?

The product uses the host's theme and typography. Work, Inspect and History are shared vocabulary across its native surfaces.

faktor

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Source & documentation ↗

Logo & design assets

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